

OPERATIONS MANAGEMENT

RETOPEMAN1



**CENTRE FOR
DISTANCE AND
ONLINE EDUCATION**

This book is only for NCDOE

NMIMS Centre for Distance and Online Education

Director, NCDOE: Dr Priya Mary Mathew

Advisory Committee:

Dr Sneha Utekar
Deputy Registrar
NCDOE

Dr Vikas Gaundare
Assistant Professor
NCDOE

Dr Brinda Sampat
Director-CIQA
NCDOE

Author: B. Mahadevan, Indian Institute of Management Bangalore

First Reviewer: Prof. Girish Parameswaran, Academic Associate

Second Reviewer: Dr. Praful More, Assistant Professor

General Editors (Language): Dr Urmila Rai

Although the author and publisher have made every effort to ensure that the information in this book was correct at the time of editing and printing, the author and publisher do not assume and hereby disclaim any liability to any party for any loss or damage arising out of the use of this book caused by errors or omissions, whether such errors or omissions result from negligence, accident or any other cause. Further, names, pictures, images, characters, businesses, places, events and incidents are either the products of the author's imagination or used in a fictitious manner. Any resemblance to actual persons, living or dead or actual events is purely coincidental and do not intend to hurt sentiments of any individual, community, sector religion.

In case of binding mistake, misprints or missing pages, etc., the publisher's entire liability and your exclusive remedy is the replacement of this book within a reasonable time of purchase by a similar edition/reprint of the book.

Copyright © 2025 Pearson India Education Services Pvt. Ltd

Copyright © NMIMS NCDOE 2025

This book is sold subject to the condition that it shall not, by way of trade or otherwise, be lent, resold, hired out, or otherwise circulated without the publisher's prior written consent in any form of binding or cover other than that in which it is published and without a similar condition including this condition being imposed on the subsequent purchaser and without limiting the rights under copyright reserved above, no part of this publication may be reproduced, stored in or introduced into a retrieval system, or transmitted in any form or by any means (electronic, mechanical, photocopying, recording or otherwise), without the prior written permission of both the copyright owner and the publisher of this book.

ISBN 978-93-671-3001-8

First Impression

Published by Pearson India Education Services Pvt. Ltd, CIN: U72200TN2005PTC057128.

Head Office: 1st Floor, Berger Tower, Plot No. C-001A/2, Sector 16B,
Noida - 201 301, Uttar Pradesh, India.

Registered Office: Featherlite, 'The Address' 5th Floor, Survey No 203/10B,
200 Ft MMRD Road, Zamin Pallavaram, Chennai – 600 044.

Website: in.pearson.com, Email: companysecretary.india@pearson.com

This ISBN is customised for NCDOE

CONTENTS

CHAPTER NO.	CHAPTER NAME	PAGE NO.
1	Operations Management: Trends and Issues	01
2	Location Decision	31
3	Plant Layout: Concept and Types	68
4	Plant Layout: Design	105
5	Product and Service Design	137
6	Productivity and Production	155
7	Manufacturing Economics: Foundation Concepts	201
8	Manufacturing Economics: Planning Systems	229
9	Inventory Management: Concepts	255
10	Inventory Classification and Planning Systems	284
11	Total Quality Management	313
12	Theory of Constraints	351

OPERATIONS MANAGEMENT

CURRICULUM

- 1. Operations Management: Trends and Issues** – Introduction to Operations management, Manufacturing and service sector trends in India, Operations as a key Functional Area, Operations Management: A systems Perspective, Operations Management Functions, Challenges in Operations management, Current Priorities for Operations Management.
- 2. Location Decision** – Globalization of Operations, Factors Affecting Location Decisions, Location Planning Methods, Other Issues in Location Planning.
- 3. Plant Layout: Concept and Types** – Determinants of Process Characteristics in Operations, Types of Processes and Operations Systems, The Process–Product Matrix, Layout Planning, Types of Layouts
- 4. Plant Layout: Design** – Performance measure for Layout Design, Design of Process layout, Design of Product layout, Design of Group Technology Layouts, One Worker Multiple Machine Layout, Technology issue in Process layout.
- 5. Product and service design** – New Product development, Manufacturability and value engineering, Quality function development, Process-Chain-Network Analysis.
- 6. Productivity and Production** – Planning Hierarchies in Operations, Aggregate operations Planning, The need for aggregate Operations planning, Alternatives for managing demands, Alternatives for managing supply, Basic Strategies for aggregate Operations planning, Aggregate Operations planning methods, Operations research tools for Operations planning, Master operations scheduling.
- 7. Manufacturing Economics: Foundation Concepts** – Dependent demand Attributes, Planning a Framework: The basic building blocks.
- 8. Manufacturing Economics: Planning Systems** – MRP Logic, Using the MRP system, Capacity requirements planning, Distribution requirement planning, Manufacturing Resources planning, Enterprise resource planning, Resource planning in services.
- 9. Inventory Management: Concepts** – Understanding Inventory, Inventory planning for independent demand Items, Types of Inventory, Inventory costs, Inventory control for Deterministic demand items, Handling Uncertainty in Demand.

- 10. Inventory Classification and Planning systems:** Introduction to Inventory Planning Systems, Inventory control systems, Selective control of Inventory, Inventory planning for single period demand, Other issues in Inventory planning and control.
- 11. Total Quality Management** – The Quality resolution, Quality Gurus, Definition of Quality, Total quality management, Quality management tools, Enhancing total quality management through 5s and waste reduction, Quality certification and awards, and design of quality assurance systems.
- 12. Theory of Constraints** – The needs for scheduling, Scheduling – Alternative terms, The loading of machines, The scheduling context, Scheduling of flow shops, Scheduling of Job shops, Input-Output controls, Operations control issues in Mass production systems, Operations planning and control based on the theory of constraints.

1

CHAPTER

OPERATIONS MANAGEMENT: TRENDS AND ISSUES

- 1.1 Introduction to Operations Management
- 1.2 Manufacturing and Service Sector Trends in India
- 1.3 Services as a Part of Operations Management
 - 1.3.1 Intangibility
 - 1.3.2 Heterogeneity
 - 1.3.3 Simultaneous Production and Consumption
 - 1.3.4 Perishability
- 1.4 Operations as a Key Functional Area
- 1.5 Operations Management: A Systems Perspective
- 1.6 Operations Management Functions
 - 1.6.1 Design Versus Operational Control Issues
 - 1.6.2 Long-term Versus Short-term Issues
- 1.7 Challenges in Operations Management
 - 1.7.1 Competitive Pressure due to Economic Reforms
 - 1.7.2 Growing Customer Expectations
 - 1.7.3 Technological Developments
 - 1.7.4 Environmental Issues
- 1.8 Current Priorities for Operations Management
- 1.9 Summary
- 1.10 Review Questions
- 1.11 Net-Wise Exercises
- 1.12 Multiple-choice Questions

NOTES

NARAYANA HEALTH (NH): A JOURNEY THAT BEGAN WITH WINING THE HEART OF INDIA

India's healthcare expenses are growing at a CAGR of 10-12% with roughly 60% of the expenses going towards private healthcare. India faces 31% of the global deaths caused due to cardio-vascular diseases. Based on a WHO sponsored global burden of disease study, the standardized cardio-vascular deaths in India is 272 for a million population, compared to a global average of 235. There are only 5500 cardiologists available in India i.e. roughly a cardiologist for a population of 30,000 people. There is a huge shortage of doctors and paramedical staff in this country. To complicate the matter even more, 70% of the doctors are in urban areas, whereas 70% of the people live in rural areas. This appears to be a serious problem of lack of adequate healthcare for citizens. However, a closer look at the story of Narayana Health (NH) suggests that mass healthcare cover is not about money but about planning, passion, and compassion. Operations management can work at its best to address such problems.

NH journey first started with a 300-bed hospital started by Dr. Devi Shetty on the outskirts of Bangalore city in 2001 that began providing heart surgeries to patients at significantly low costs. Currently, the group has 20 hospitals across India with 30+ specialties and a capacity of 6373 beds. As of Q3 2022-23, their capital cost of 40 lakhs per operational bed is one of the lowest among similar specialty hospitals. As of 2021, NH does 14000 cardiac surgeries per year. Despite the huge volume, they command responsiveness of 98.5% of the patients receiving doctor consultation within two hours of arrival at the hospital. By 2022, their employee strength had grown to 18,500 including 3970 doctors. They are rapidly expanding their footprint across the country. Managing such a massive expansion requires strategic thinking, the creation of new facilities in a cost-effective manner, and operating them on a day-to-day basis with superior planning. All these issues belong to the domain of Operations management.

Many of the operational aspects of the NH network are governed by the vision that nobody should go back from the hospital for want of money. A heart surgery package at NH could be anywhere between ₹75,000 and ₹150,000 compared to a typical cost of ₹300,000 in other hospitals. Achieving such dramatic cost reduction calls for employing several methods. These include input cost reduction, benefiting from process innovations and scale economies, and bringing innovative ideas in design of facilities and resource deployment. Further,

productivity improvements would be very crucial and an ongoing activity in such a system. In NH, we find several examples illustrating these.

Due to volumes and supplier development efforts, NH is able to buy a pair of glove at ₹4.50 as against ₹9.50 by directly sourcing it from Malaysia. Many of the expensive equipment such as MRI scan machine are installed by vendors on a pay-per-use model rather than an outright purchase. This brings down the cost of an MRI scan and also increases its utilization. Hospitals are located on the outskirts of the cities to bring down the capital costs. The Mysore cardiac hospital was built using pre-fabricated materials and has no air-conditioning, except in operation theatres, thereby bringing down the cost of the facility dramatically low.

Another strategy to bring down cost is to increase the volume of patient flow. To achieve this, NH has diversified into 30+ specialties (such as eye care, orthopedics, neuroscience, oncology and so-on). Furthermore, NH recruits the best doctors and pays them higher than average salaries in the market. However, they are convinced to work extra during the day. For the doctors at NH, 20 years ago, the plan was to do five surgeries per day with a 150 bed capacity, which itself felt high. However, currently with 6373 beds, they perform over 30 surgeries a day. As a result, nearly 10-12% of all cardiac surgeries performed in India are performed as a part of the NH network.

Narayana Health is an example of a service system. Like NH, every other service organizations as well as manufacturing organizations face similar set of issues, although in varying degrees. As illustrated in the case of NH, these issues can be addressed by applying several tools and techniques, collectively known as Operations Management.

Source: Based on Babu, V. (2012), "Pulse on the future", *Business World*, March 5, 2012, pp 40–45; Chaki, D. (2013), "Straight from the heart", *Business India*, Sep. 30–Oct. 13, 2013, pp 61–63.

NOTES

1.1 INTRODUCTION TO OPERATIONS MANAGEMENT

Manufacturing, service, and agriculture are the major economic activities in any country. In India, manufacturing and service together constitute nearly 84 per cent of the gross domestic product (GDP) at constant prices. In recent years, growth in these sectors has largely been the engine for Indian economic growth. Moreover the growth of service sector has steadily increased from 40 percent in 1996 to over 54 percent in 2021. In view of

NOTES

their contribution to the country's GDP, management of manufacturing and service operations are important economic activities. Significant improvements in productivity and cost savings can be achieved through operations management—a discipline that focuses on activities that relate to the planning and control of operations in manufacturing and service organizations. Efficient operations management can also have a positive impact on the overall health of the economy.

A manufacturing firm essentially engages in converting a variety of inputs into products that are useful for individuals and organizations. For example, a manufacturer of machine tools employs several production workers, buys raw material and components from various suppliers, and manufactures machine tools in a factory using an extensive array of manufacturing facilities. In this case, the factory encompasses a large number of interrelated conversion processes for the transformation of raw material into the final product, that is, the machine tool.

A service organization, on the other hand, responds to the requirements of customers and satisfies their needs through a service delivery process. Service organizations leave an impression in the minds of their customers through their service delivery. Typical examples of such organizations include management consultancies, automobile garages, hotels, hospitals and banks. A service organization may not always make use of material inputs and may not always produce products that are used by the customer. For instance, a law firm providing legal consultancy to its clients may not provide material inputs to the system or produce material output. Instead, the input and output are informational and experiential in nature. However, in the case of a service system like an automobile garage, a restaurant or a health care system, there are material inputs and material outputs (in the form of products consumed by the customers, as in the case of a restaurant). Despite this difference, service systems also have a conversion process that utilizes resources and delivers useful outputs from the system.

An *operations system* is defined as one in which several activities are performed to transform a set of inputs into a useful output using a transformation process. These inputs and outputs can be tangible, in the form of raw materials and physical products, or intangible, in the form of information and experiences. Viewed in this manner, manufacturing and service systems could be broadly classified as operations systems. **Operations management** is a systematic approach to addressing issues in the transformation process that converts inputs into useful, revenue-generating outputs. Four aspects of this definition merit closer attention:

Operations management is a systematic approach to addressing issues in the transformation process that converts inputs into useful, revenue-generating outputs.

- ⇒ *Operations management is a systematic approach.* It involves understanding the nature of issues and problems to be studied; establishing measures of performance; collecting relevant data; using scientific tools, techniques, and solution methodologies for analysis; and developing effective as well as efficient solutions to the problem at hand. Therefore, for successful operations management, the focus should be on developing a set of tools and techniques to analyse the problems faced within an operations system.
- ⇒ *Operations management involves addressing various issues that an organization faces.* These issues vary markedly in terms of the time frame, the nature of the problem, and the commitment of the required resources. Simple problems include deciding how to re-route jobs when a machine breaks down on a shop floor, or how to handle a surge in demand in a service system. On the other hand, decisions such as where to locate the plant, what capacity to build in the system, and what types of products and services to offer to the customers require greater commitment of resources and time. Operations management provides alternative methodologies to address such wide-ranging issues in an organization.
- ⇒ *Transformation processes are central to operations systems.* The transformation process ensures that inputs are converted into useful outputs. Therefore, the focus of operations management is to address the design, planning, and operational control of the transformation process.
- ⇒ *The goal of operations management is to ensure that the organization is able to keep costs to a minimum and obtain revenue in excess of costs through careful planning and control of operations.* An appropriate performance evaluation system is required for this. Therefore, operations management also involves the development of performance evaluation systems and methods through which the operating system can make improvements to meet targeted performance measures.

Since manufacturing and service organizations form the operations system, it helps to understand the prevalent trends and the status of manufacturing and service organizations in India before we dwell on the various elements of operations management.

1.2 MANUFACTURING AND SERVICE SECTOR TRENDS IN INDIA

The index of industrial production (IIP) is a measure of the growth in the manufacturing sector. The Centre for Monitoring Indian Economy (CMIE) collects data on several macroeconomic indicators, including the IIP. Table 1.1 shows some of the salient aspects of the manufacturing and service sectors of the Indian economy. It is evident from Table 1.1 that the corporate sector has been going through tough times. There has not been much growth in sales

NOTES

NOTES

TABLE 1.1 SALIENT ASPECTS OF THE CORPORATE SECTOR OF THE INDIAN ECONOMY

Index of Industrial Performance	FY2018–19	FY2019–20	FY2020–21	FY2021–22
Manufacturing	3.83	–1.42	–9.56	11.75
Capital Goods	2.74	–13.96	–18.69	16.97
Consumer Goods	4.65	–3.85	–7.33	6.63
Intermediate Goods	0.87	9.08	–9.44	15.44
Corporate Sector Performance				
Sales–Manufacturing	14.0	–2.9	–2.0	32.4
Sales – Non–financial Services	13.7	2.2	–1.7	26.0
Sales – Financial Services	10.5	12.4	7.0	3.6
Operating Expense	14.93	–1.68	–3.49	32.30
Compensation to employees	10.6	8.7	3.6	17.4
Profit after tax (PAT)	40.9	–43.9	161.5	58.6
No of companies	34,811	34,063	32,837	18,715

All numbers in the table represent growth % over the previous year.

Data compiled from Economic Outlook of Centre for Monitoring Indian Economy (CMIE) using their time series data available at <http://economicoutlook.cmie.com>

revenue. However, there has been a steady increase in operating expenses and compensation to the employees. This has resulted in a situation where the profit after tax (PAT) has been progressively and significantly shrinking in the last three years. The service sector companies have been experiencing this more significantly than the manufacturing sector companies as evident from the table. On the whole, this points to the importance of deploying sound operations management practices in both manufacturing and service sectors of the Indian economy.

Over the years, there has been an increase in raw-material consumption. This perhaps indicates that manufacturing organizations are increasingly buying components and semi-processed items from suppliers. Furthermore, there has been a gradual increase in the cost of material and labour, which are primary inputs in the manufacturing system. Such an increase in the input cost is likely to put greater pressure on firms to cut down waste and improve productivity to remain competitive in the market. Therefore, the focus areas of operations management are likely to be in the areas of better supplier management, elimination of waste from the system, and improvement in overall productivity. Several sectors of the industry have been focusing on some of these initiatives.

Globally, India is emerging as an important manufacturing base and is competing closely with China in attracting several

multinational companies to set up their manufacturing plants. Several studies point to emerging opportunities for Indian manufacturing to grow and attain a global presence. India has a unique advantage in the form of abundant low-cost labour and technical manpower. The Global Manufacturing Competitiveness Index 2013 released by Deloitte rated India at number four, next to China, Germany, and the US among 38 countries. It predicted India to reach the second position in the next 5 years. The report suggested that India's rich talent pool of scientists, researchers, and engineers as well as its large, well-educated English-speaking workforce and democratic regime would make it an attractive destination for manufacturers. Global manufacturing executives increasingly view India as a place where they can design, develop, and manufacture innovative products for sale in local as well as in global markets. Therefore, Indian manufacturing firms can, on the one hand, exploit the low-cost advantage as an entry strategy to capture global markets and, on the other develop, unique capabilities to position as an active participant in the global value chain.

The examples discussed here underscore the need for use of operations management to remain competitive in business and tap the emerging opportunities in the global arena. Effective operations management also requires a greater understanding of the various activities pursued under its banner and the critical linkages between operations and other aspects of business.

1.3 SERVICES AS A PART OF OPERATIONS MANAGEMENT

The service sector encompasses a wide spectrum of activities in every country. The growth of the service sector in India during 2002 to 2007 has been very significant. Table 1.2 provides some details on the share of some sectors of services in the overall GDP. The central government began taxing three services in 1994–95. This has grown steadily, and as of 2012–13, the number of services taxed has gone up to 119. During this period, both the number of assesseees and the service tax revenue has been growing very rapidly. According to the Central Board of Excise and Customs (CBEC), between 1994–95 and 2012–13 the tax revenue has grown from ₹4.1 billion to ₹1.325 trillion. These figures indicate the growing importance of services in the Indian economy and the need to apply management practices to plan and control operations in the service sector.

Although services and manufacturing are classified as separate sectors in a macroeconomic sense, from the perspective of operations management, this separation is artificial. In operations management, a “pure product” and a “pure service” are just two ends of the spectrum, and not separate entities. In reality, a vast majority of operations share a continuum of services and products. Therefore, most of the principles, tools and techniques of operations management apply to both these sectors. This product–service continuum is illustrated in Figure 1.1.

NOTES

NOTES

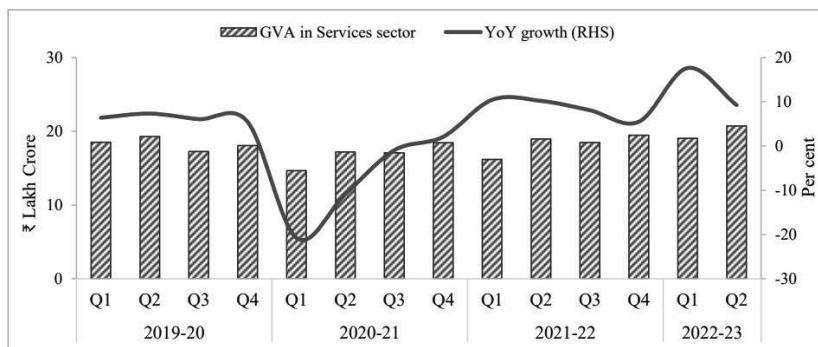
TABLE 1.2 SERVICE SECTORS IN INDIA: SHARE OF GDP IN PERCENTAGE

Index of Industrial Performance	2018–19	2019–20	2020–21
Total Services (Excluding construction)	7.7	5.5	–8.8
Trade, hotels, transport, communication & services related to broadcasting	7.7	3.6	–21.41
Financial, real estate & professional services	6.8	4.6	–0.82
Public administration, defence & other services	9.4	10.0	–3.68

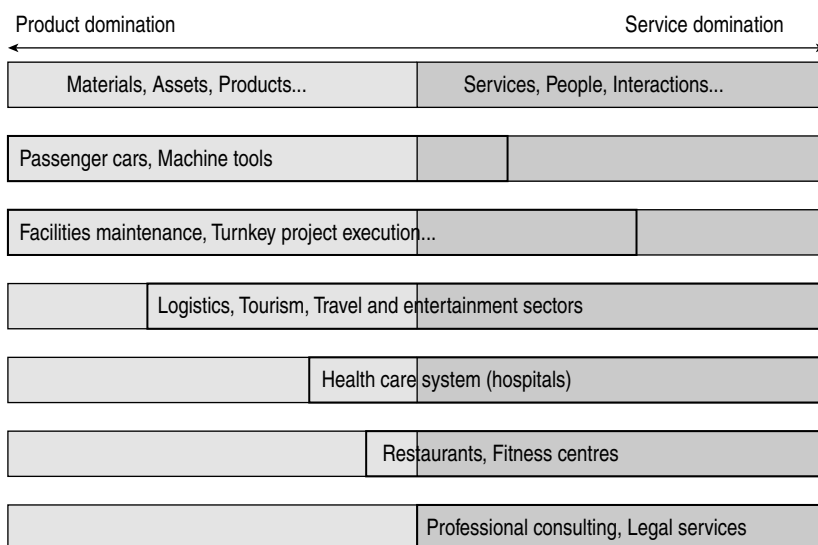
Advanced Estimate

* Compiled from Chapter 10 on Services Sector in Economic Survey 2012–13, Government of India, Ministry of Finance, Economic Division. For more details see <http://indiabudget.nic.in>

Note: We are not having the old level of break-down anymore. So an alternative is to replace the table with a graph extracted from Economic Survey in the last budget. The graph is given below.



Source: NSO, MOSPI

**Figure 1.1** The service–product continuum

Services such as management consulting, health spas, and education have dominant service attributes. They form one end of the spectrum. Similarly, manufacturing and supply of machine tools, gadgets, and consumables have a dominant product attribute, and they form the other end of the spectrum. However, several businesses share both service and product attributes. Take the case of automobiles. There is a product attribute in it since it involves the physical structure of the passenger car. On the other hand, there is also the experiential component of using the car, which forms a significant part of the product. This is the service component. Similarly, in the case of a restaurant, the food items share both product and service attributes. There are certain important differences between services and manufacturing. Let us take a brief look at each of these differences.

1.3.1 Intangibility

Fundamentally, services differ from manufacturing with respect to tangibility. Because services are experiences rather than objects, they cannot be touched, tasted or felt as in the case of objects. At the most, the recipients of services can form an opinion (based on some personal assessment) about the quality of the service offered. This has important implications for defining and assessing the quality of the service.

On the other hand, in a product-oriented operation, the product is defined by certain attributes and the customer faces less ambiguity with respect to the product, its attributes, and its performance. This is because the customer can touch and feel the product and make his/her own assessment of the product.

Fundamentally, services differ from manufacturing with respect to tangibility, heterogeneity, simultaneity, and perishability.

1.3.2 Heterogeneity

The second differentiating aspect of services is the high degree of heterogeneity associated with them. Since the experiential component is dominant in a service, it is likely that no two services are exactly alike. The differences are attributed to the differences in the service receivers (customers), the service providers, and other parameters of the service delivery system. Therefore, a dentist attending to two consecutive patients having identical ailments may provide more or less the same type of service. Nevertheless, the two patients may have different perceptions of the quality of the service and may have different satisfaction levels. Moreover, the time spent by the dentist in both the cases could vary greatly. High heterogeneity results in high variability in the operations system performance and the need to factor them into the planning and control of operations.

NOTES

N O T E S

High heterogeneity results in high variability in the operations system performance.

1.3.3 Simultaneous Production and Consumption

More often, services occur in the presence of the customer, who may also be involved at the time the service is produced for his/her consumption. In the example of the dentist and the patient, the doctor and the patient are in the system together to produce and consume the service. This holds for education, entertainment, travel, tourism, and hotel services as well. In the case of manufacturing, however, most goods are produced at some point in time and distributed to the customer later. This difference has implications for the design, planning, and control of service operations, as the degree of customer contact in a service delivery system is likely to be high.

VIDEO INSIGHTS 1.1

Operations management in service organizations have a different set of issues to address. In order to develop an understanding of the peculiar issues pertaining to operations management in service organizations, a peep into service systems will be a valuable exercise. Hospitality and Healthcare sectors of services face unique challenges arising out of people-intensive service delivery, where the customer involvement is also high. To understand the issues in these sectors, find the video links (Video Insights) in the Instructor Resources or Student Resources under section **Downloadable Resources** (<http://www.pearsoned.co.in/BMahadevan/>) subsections **Bonus Material**.

1.3.4 Perishability

Services are perishable. This implies that they cannot be inventoried. Thirty minutes of a doctor's consulting expertise today cannot be stored for future use, reused, or returned in a future period. Similarly, the treatment for an acute headache likely to be faced by a patient sometime in the future cannot be stored in advance and reused at that time. The possibility of inventorying the supply and using it at a later time is very common in a manufacturing system. The implication for operations is that service systems require methods that work without inventories.

The characteristics of tangibility, heterogeneity, simultaneity, and perishability apply to "pure services," which are more experiential in nature. Furthermore, businesses with dominant service characteristics will have these features dominating over the other

Services cannot be inventoried.

product-related features. For example, in the case of a fast-food joint, there is both the service aspect and the product aspect. A customer having a plate of *idli* with *sambar* experiences all the four attributes discussed here when viewed from the service angle. However, there is also a product angle to the restaurant. For instance, viewing *idli* as a product may enable the restaurant owner to inventory it for future use (later in the day).

Despite these differences between products and services, from an operations management perspective, there are similarities between these two categories. Table 1.3 lists some of the salient differences and similarities between a manufacturing and a service system. The differences, as we have already discussed, are mainly due to the four distinguishing features of services. The similarities, on the other hand, are due to the basic need to manage operations. Design of operations requires decisions on capacities, products, and services to be offered. Operational planning and control requires that suppliers are managed, operations properly planned and scheduled, and demand matched with supply. These issues are common to both manufacturing and services, though the tools and techniques used to handle these issues may be specific to manufacturing and services.

TABLE 1.3 A COMPARISON OF MANUFACTURING AND SERVICE ORGANIZATIONS

Manufacturing Organizations	Service Organizations
Differences	
Physical, durable product	Intangible, perishable product
Output can be inventoried	Output cannot be inventoried
Low customer contact	High customer contact
Long response time	Short response time
Regional, national, international markets	Local markets
Large facilities	Small facilities
Capital intensive	Labour intensive
Quality easily measured	Quality not easily measured
Similarities	
Is concerned about quality, productivity and timely response to its customers	
Must make choices about capacity, location, layout	
Has suppliers to deal with	
Has to plan operations, schedules and resources	
Must balance capacity with demand by a careful choice of resources	
Has to make an estimate of demand	

NOTES

OPERATIONS MANAGEMENT IN SERVICE SYSTEMS

NOTES

Intangibility, heterogeneity, perishability, and simultaneous production and consumption are the factors that differentiate service systems from manufacturing systems. Despite these differences, many of the operations management tools that were initially developed for manufacturing organizations are applicable to services as well. A study on the Ahmedabad Regional Passport Office (ARPO) illustrates this fact.

In March 2001, the regional passport officer of the ARPO was concerned about the long lead time taken for processing passport applications and issuing passports. The estimated lead time was 145 days at that time, so there was always a long queue of applicants waiting at the ARPO, either to know the status of the application or to submit the application form. To improve the response time, the existing process needed to be studied and redesigned. The situation called for the application of the principles of operations management to a service setting.

A detailed study of the process was undertaken to investigate the reasons for the long lead time. Excessive delays and long lead times often point to operations management issues pertaining to capacity, process, procedures, and use of technology and people. Process mapping, estimating the time for each step in the process, and assessment of capacity imbalances are important steps in studying issues related to long lead times in any operations system. In the case of the ARPO, this data was collected. Although the ARPO had a sanctioned staff of 103, the staff strength was 57 in January 2002. It augmented the capacity by employing 40 casual workers. It received about 235,000 applications in a year. Detailed studies were initiated to understand the various processes involved in issuing passports. See Figure 1.2 for an illustration of the process followed for passport application and approval.

Detailed discussions with the officers revealed several opportunities for improving the existing situation. The activities at the ARPO were labour-intensive, and the organizational rules and procedures were highly centralized. Further, the discussions revealed that there was scope for automation of some of the activities. In the present scenario, it took a lot of time to access information on the application status, and Web-based services could reduce this time. The uneven load on police stations and the need to modify and simplify certain rules were some other issues that merited attention. Independent of the study, some administrative and procedural changes were already underway at the ARPO. Attention to all the issues that were highlighted during the study through better capacity management could help reduce the lead time in processing passports.

The data collected during the study was subjected to detailed analysis, and alternative methods were identified for

the design and operational control of the operations at the ARPO. These include the use of waiting line models to explore alternative ways of redesigning the service delivery system; process redesigning to eliminate waste; improvements in productivity and the discovery of hidden capacity in the system; the identification of a new system for tracking operational performance, benchmarking and improvement; and the use of workload analysis and capacity planning methodologies. We shall discuss these aspects of operations management in the forthcoming chapters.

Source: Adapted from N. Ravichandran and D. Bahuguna, "Rule Bound Government Agency to Customer Centric Service Facility: Can Indian Passport Offices Make The Leap?" *IIMB Management Review* 18, no. 1 (2006): 59–66.

NOTES

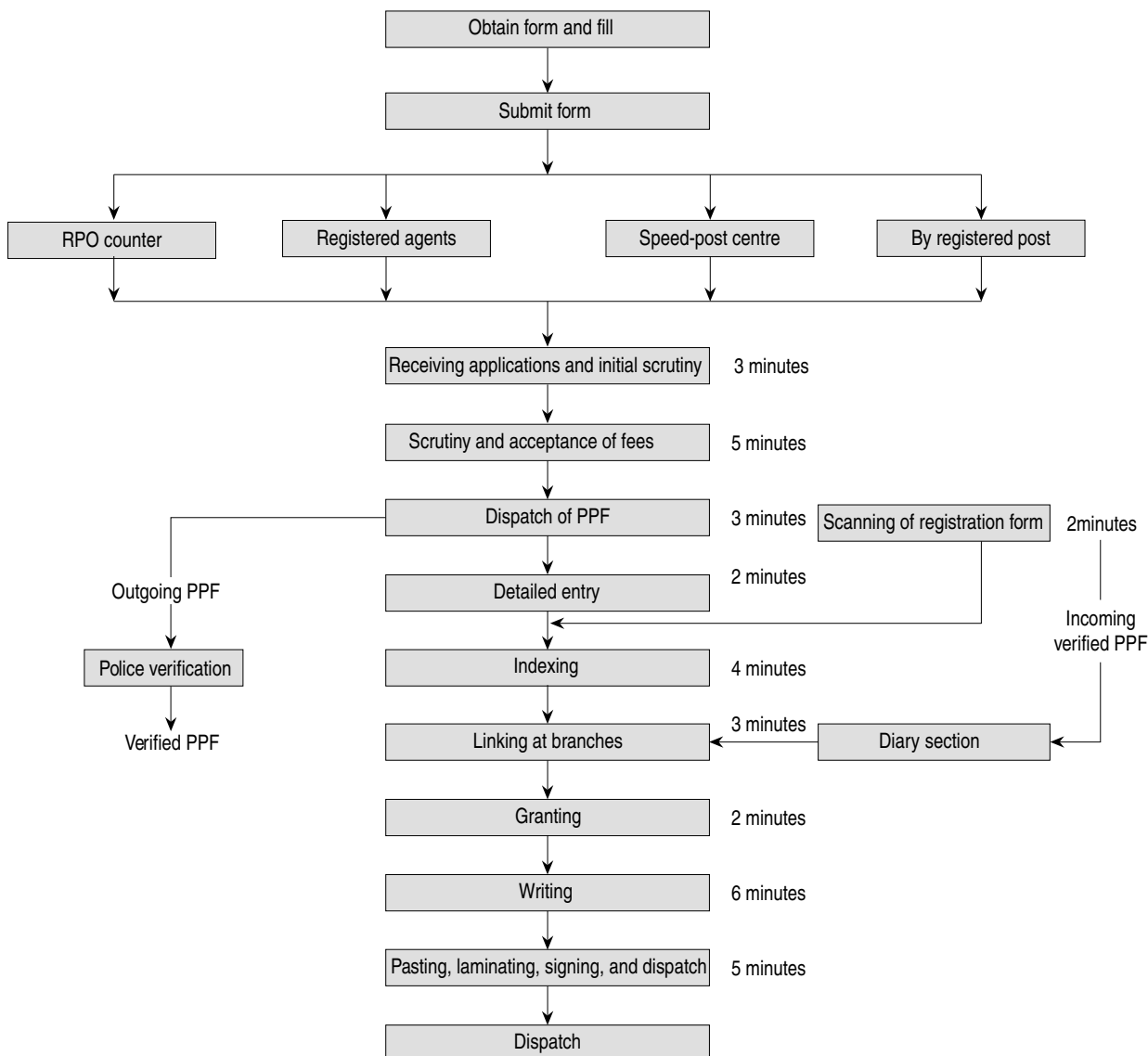


Figure 1.2 Process-flow diagram for application processing

NOTES

1.4 OPERATIONS AS A KEY FUNCTIONAL AREA

Before we understand the various sub-functions in operations in detail, it would help to appreciate the role of operations in an organization and its relationship with other functional areas of business. Irrespective of the activities and the type of business it is involved in, every organization has a few important activities. These include operations, marketing, finance, and human resources management. *Operations management*, as we have already seen, deals with the management of the conversion process in an organization. The *marketing* function is responsible for understanding the requirements of customers, creating a demand for the products and services produced, and satisfying customer requirements by delivering the right products and services to customers at the right time. Both operational and marketing activities require estimates of financial needs, tapping the market for funds, and management of working capital. These activities broadly constitute the *finance* function. Every organization employs a number of people who have varied skills, backgrounds and work requirements. Managing the workforce and addressing a host of issues related to them is another important requirement in an organization. The *human resources management* function deals with such issues. These four areas form the basic functions of any organization, as depicted in Figure 1.3.

As shown in the figure, the four functions have mutual interactions among them. The decisions made in each of these functional areas could form an important input in another functional area. For example, organizations typically begin their yearly plan with the marketing function making an estimate of the next year's sales. This input forms the basis for production planning in the operations area of business. Depending on the production plans, procurement planning is done and all these factors lead to a certain estimate of the fund requirements. This forms an important input for the finance function. At the time these plans are executed, the interactions between these functions are even greater. The human resource management function influences the productive capacity of labour available in real time. The actual production of goods and services influences the marketing activities to be undertaken and the quantity and timing of available funds from sales. Such interactions are common in most organizations.

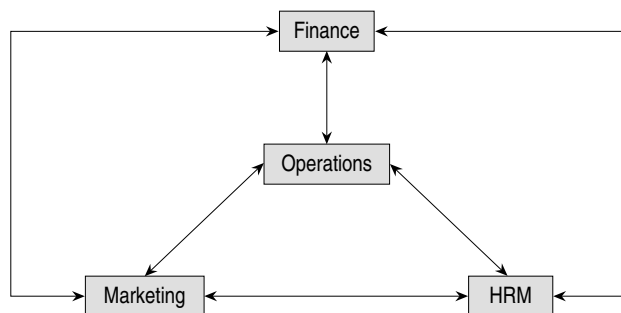


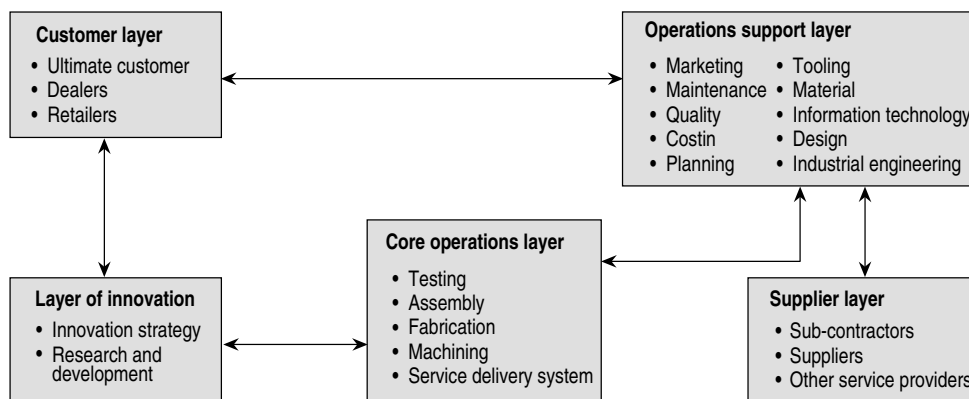
Figure 1.3 The basic functions of an organization

In every manufacturing or service organization, there are several sub-functions within the core operations function. Figure 1.4 provides a broader view of the nature of activities pursued within the core operations function and its linkages with other functions, both within and outside the organization. As shown in the figure, the operations function in an organization has five layers: (i) the customer layer, (ii) the core operations layer, (iii) the operations support layer, (iv) the innovation layer, and (v) the supplier layer. The *customer layer* consists of the end customer and various others in the distribution chain, including retailers and dealers. Within an organization, there are three layers. The *core operations layer* represents the manufacturing set-up in the case of a manufacturing organization. This may consist of fabrication, machining, assembly and final inspection. In the case of a service organization, the service delivery system consisting of the physical facilities and personnel forms the core operations layer. In every organization, there are other activities that directly interact with the core operations layer and provide a variety of support services. These form the *operations support layer*. This typically includes areas such as marketing, quality, design, planning, costing, information technology, purchase and materials, stores and maintenance. The third layer internal to an organization is the *innovation layer*. Every organization needs to devise strategies to remain competitive in the market. The strategy formulation process drives the research and development activities in an organization and enables the organization to bring new products and services to its customers.

The last layer, that is, the *supplier layer*, is external to the organization. Consisting of a network of suppliers, sub-contractors and other specialized service providers, this layer collectively represents the inbound logistics of the organization.

The customer layer interacts with the marketing function, and the inputs from the marketing function are crucial for planning operations. The supplier layer interacts with the materials or

NOTES



Source: B. Mahadevan, *The New Manufacturing Architecture* (New Delhi: Tata McGraw Hill, 1999).

Figure 1.4 Operations function and its linkages in an organization

NOTES

procurement function of an organization. The procurement function, in turn, interacts with the core operations layer and ensures timely availability of the various materials required. The customer layer also provides critical inputs to the layer of innovation, which forms the basis for the strategy formulation process. Based on the strategies made, several changes are made to the core operations layer. Therefore, there is a critical linkage between the core operations layer and the innovation layer.

The basic inputs in an operating system are labour, material, and capital.

1.5 OPERATIONS MANAGEMENT: A SYSTEMS PERSPECTIVE

A systems perspective facilitates a comprehensive understanding of the various aspects of operations management. A systems perspective essentially involves identifying the input, the output, and the processing and feedback mechanisms in a system. Figure 1.5 presents a systems perspective of the operations management function.

The basic inputs in an operating system are labour, material and capital. The availability of skilled labour is an essential requirement as it is not possible to operate the system satisfactorily and deliver good-quality goods and services without adequate supply. The second input in operations is material. All manufacturing organizations process raw materials and convert them into useful products. Materials are required in several service systems as well. A healthcare system such as a multi-specialty hospital requires medical consumables such as cotton, disposable syringes, and other equipment. Similarly, a hotel requires a variety of raw material for preparation of food.

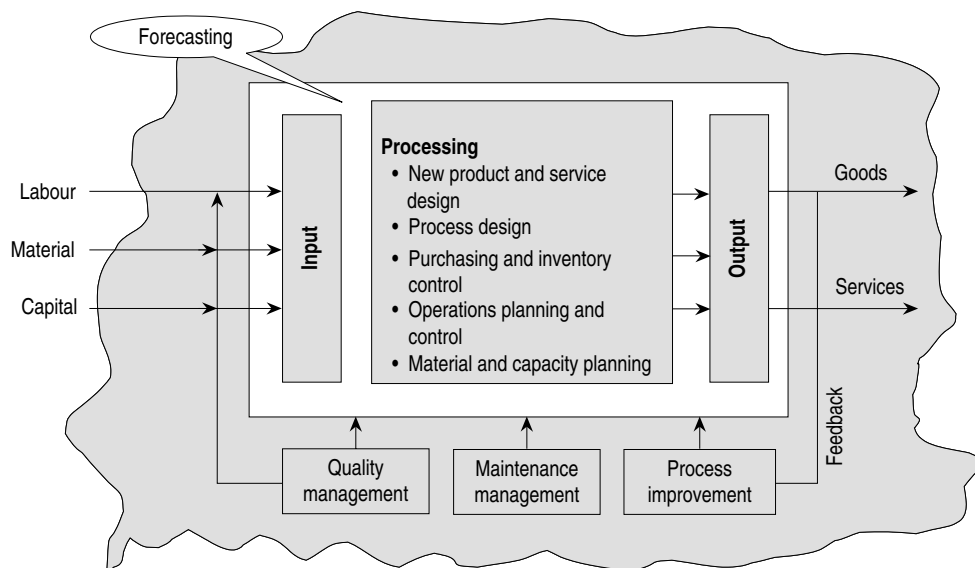


FIGURE 1.5 Operations management: a systems perspective

Therefore, material forms a critical input in operations. The third important input in operations is capital. Capital is required to buy resources such as plants and machinery, to pay labour, and for investing in materials that are being processed into finished goods.

Processing includes the various activities that an operating system undertakes to convert the raw material into useful products for customers. The conversion process adds value to the product and enables the organization to sell it in the market. In the case of service organizations, the conversion process involves committing the various resources at its disposal to deliver useful services to the customer. Irrespective of whether it is a manufacturing or a service organization, there are a set of activities to be performed in the conversion process. These activities are presented in the processing section of Figure 1.5.

Processing includes the various activities that an operating system undertakes to convert the raw material into useful products for the customers.

Product and process designs are an integral part of the conversion process. Every operations system must make important decisions with respect to the type of products and services to be offered. Furthermore, once the right mix of products and services are identified, appropriate processes for manufacturing and delivering them to the customers need to be identified. This involves deciding on the type of technology to use, the type of machines to be used in the conversion process, and the exact method of creating the products and services. These steps are known as product and process design. For example, when the low-cost airline Spice Jet started its operations, it had to make these choices. It chose to fly to only three locations connecting western India to Delhi. It had to decide on the type of aircraft to use, the in-flight services to offer, and various other aspects, such as ticketing and reservation, airport logistics, and

VIDEO INSIGHTS 1.2

The best way to understand the need for operations management and the issues to be addressed in a manufacturing system is to make a factory visit. A factory visit gives us a bird's eye view of various aspects of managing the operations. The Honda factory in the USA and Mahindra's XUV 500 factory in India provide a good introduction to the concepts of operations management in manufacturing organizations. To know more about this, find the video links (Video Insights) in the Instructor Resources or Student Resources under section **Downloadable Resources** (<http://www.pearsoned.co.in/BMahadevan/>) subsections **Bonus Material**.

NOTES

NOTES

A planning system will help the airline in route planning, rerouting aircraft, and short-term capacity management.

passenger interfaces at the airport and outside. In manufacturing, a similar exercise can be visualized for Hyundai, when it started to manufacture passenger cars in India in 1998.

Once the product and process designs are finalized, an operations system must focus on ensuring that the demand for products and services is met. At this stage, the organization may also require operations planning to ensure the availability of adequate material and capacity to meet the targeted production and service delivery. This is known as operations planning. The operations system also requires methods for real-time control of operations as there may be several deviations from the plan. For example, there might be a breakdown of machinery or rejection of a lot of material supplied by a vendor. Alternatively, some customer might change his/her order quantity or cancel the order altogether. The operations control activity takes care of these changing requirements in an operations system. In the airline example, the issues relate to the pricing of tickets, capacity planning, and the handling of tickets cancellations and schedule delays. A planning system will help the airline in route planning, rerouting aircraft, and short-term capacity management.

A fast-food restaurant may provide various types of food for breakfast, as per its process and product design. In several manufacturing organizations, services are also offered in the form of after-sales support and warranty.

Another important activity in the conversion process pertains to ensuring adequate supply of materials for the operations system. This requires that the suppliers of various materials are identified and relationships established with them. With such an arrangement, it will be possible to place orders for material with the suppliers and receive them within the stipulated time. This activity is labelled as purchasing and inventory control in Figure 1.5. From an operations management perspective, the processing part of the system primarily focuses on identifying the nature of resources required, planning for material and capacity, and ensuring that changes in real time are addressed.

An important aspect of operations management is estimating demand. This is done through *forecasting*.

The output of an operations system consists of goods and services. An organization that manufactures passenger cars will provide many variants of the passenger car. On the other hand, a fast-food restaurant may provide various types of food for breakfast, as per its process and product design. In several

manufacturing organizations, services are also offered in the form of after-sales support and warranty. In addition to these tangible outputs, it is inevitable that an operations system will leave certain impressions and value judgments in the minds of its customers and stakeholders. For instance, a manufacturer of copper products such as Hindustan Copper will not only supply copper products to its customers, but also create an impression in the mind of the customer. For instance, its customers may feel that though the quality of the product is good, it is costly in comparison with its competition. People living in the vicinity of its factory may form opinions about the company on matters related to environmental pollution. The labour practices adopted by the company are likely to influence its prospective employees.

In recent times, organizations have begun to understand their role as a responsible entity in society. Pollution of the atmosphere creates greater problems leading to global warming. Industries such as leather processing and petrochemicals tend to generate large amounts of effluents. Sustainability is another area of major concern. In a country like India, businesses and society compete for limited natural resources such as water. In the state of Kerala, the local society protested against Coca-Cola for drawing water from the locality for their bottling plant. In such situations, the government and judiciary bodies such as the Supreme Court play an active role and intervene to ensure that these issues are resolved. To avoid disrupting the balance in society, an operations system must take such factors into consideration while configuring its input–processing–output elements.

It is useful to understand how the operations system considers these factors and reacts to them. In any system, the feedback loop serves the purpose of identifying the deviation paths and highlighting the areas that need immediate correction. In an operations system, there are three feedback loops available for making corrective measures. These include quality management, maintenance management, and process improvement. These operations management activities provide checkpoints to identify the areas requiring improvement and ensure that corrective measures are indeed taken.

The demand for the goods and services offered by an organization cannot be entirely controlled by the organization. At best, an organization can influence the demand to an extent. From a systems perspective, the demand is an exogenous variable. However, planning for production, capacity and material clearly depend on the demand that the operations system is likely to face in the upcoming periods. Therefore, an important aspect of operations management is to estimate demand. This is done through what is known as *forecasting*.

NOTES

NOTES

TABLE 1.4 Operations Management Functions

Design Issues	Operational Control Issues
Product and service design	Forecasting
Process design	Operations planning and control
Quality management	Supply chain management
Location and layout of facilities	Maintenance management
Capacity planning	Continuous improvement of operations

1.6 OPERATIONS MANAGEMENT FUNCTIONS

From the description of the various aspects of operations management in the systems perspective, one can distil some of the important functions of operations management. These functions are carried out irrespective of whether the organization is in manufacturing or services. Table 1.4 lists these functions.

The important functions can be understood by: (a) categorizing them under design issues and control issues, as shown in Table 1.4; and (b) classifying them as long-term issues and short-term issues, based on the planning horizon.

1.6.1 Design Versus Operational Control Issues

An examination of the left-hand column in Table 1.4 shows that the operations management decisions listed in that column pertain to the design of an operations system. **Design issues** relate to the configuration of the operations system and provide an overall framework under which the operations system will function. The operations management functions pertaining to design include designing products, services and processes, setting up a quality assurance system, deciding on the layout and location of facilities, and capacity planning. Design issues in operations management establish the overall constraints under which the operations system functions. A few examples will help clarify this characteristic feature of design issues in operations management.

For example, once the capacity of the resources to be used in the system is decided, it sets limits for the actual use of the system in operation. If Maruti Suzuki India Limited installed an engine plant that has the capacity to produce 200,000 engines annually, then, in actual operation, it can produce no more than 200,000 engines. Similarly, if a multi-specialty hospital is located in Pune with a

Design issues in operations management establish the overall constraints under which the operations system functions.

200-bed facility, it is understood that it will largely serve residents in the vicinity of Pune up to a capacity of 200 in-patients at any time. Setting policy guidelines and procedural details for various aspects of the operations system is also a design activity. A typical example is the setting up of a quality management system that establishes rules and policies with respect to what is acceptable and what is not.

Operations management decisions listed in the right-hand column of Table 1.4 share a different characteristic. They all address the vital issue of planning and operations control. Once the design choices are exercised, operations management amounts to putting the available resources to best use and handling various issues as and when they arise. The available capacity could be better utilized through production planning and by scheduling operations so that idle time is minimized. Furthermore, required capacity and material could be estimated and made available through purchasing and scheduling procedures. All these constitute operational control decisions in operations management.

Every issue addressed in design is inevitably addressed again in operational control. However, the context differs in the two cases. Capacity planning as a design issue lends itself to a different type of analysis compared to capacity planning in operational control. In the case of the latter, the objective is to match the requirement to the available capacity, whereas in the case of the former, the objective is to find out how much capacity is required for meeting targeted business plans. Moreover, design issues often turn out to be strategic in nature, while operational control issues are tactical in nature. Strategic decisions frequently involve large capital outlay and are made with critical inputs from an operations strategy process. Such decisions are made by the top management to improve the competitiveness of the organization. Operational control issues, on the other hand, are tactical, repetitive and routine in nature. Lower-level operations managers often make such decisions.

1.6.2 Long-term Versus Short-term Issues

Another useful approach to understanding the various operations management functions is to view them from the planning horizon perspective. Certain operations decisions are made once every five to ten years. For example, the decision to locate a manufacturing or service delivery facility is made as and when a new facility is to be introduced. Similarly, product line decisions and capacity augmentation decisions are made once every three to five years. Typically, a majority of design decisions are made with a planning horizon of five to ten years. Such decisions usually require a long lead time, multiple levels of decision-making

NOTES

Every issue addressed in design is inevitably addressed again in operational control.

NOTES

and huge capital outlay. Therefore, they tend to be made at less frequent intervals.

Some other decisions are made in fixed cycles of one year. Every organization makes a business plan coinciding with its financial year, wherein specific targets for sales are established. The annual business planning exercise leads to aggregate operations planning. Once the aggregate operations planning is done, master scheduling and material and capacity requirements planning are done for the next quarter or three months. These are medium-term decisions. Finally, several operations management decisions are made for a short run of a week or less. These decisions include detailed scheduling of operations, quality management and control, and reacting to disruptions and changes in plans.

1.7 CHALLENGES IN OPERATIONS MANAGEMENT

Challenges in operations management arise mostly from developments that create a need for new, and, at times, more efficient system of operations management. The economic reforms in the 1990s opened up the Indian markets to overseas players, which posed new challenges for domestic companies. With time, there have been more changes in the form of rising customer expectations, technological developments, and growing awareness about environmental issues. Each of these developments needs to be addressed with appropriate changes in the strategy, design and implementation of operations management practices.

1.7.1 Competitive Pressure due to Economic Reforms

It is more than 20 years since the economic liberalization and globalization policies of the Government of India came into force. During these years, the Indian corporate sector has learnt what exactly the melting pot of global competition means. Due to integration of the Indian economy with the global economy, the Indian companies are required to benchmark the manufacturing sector against the best in the world to enhance their competitiveness. Companies that are able to meet the stiff competition are able to discover the unprecedented opportunity for growth as they discover new markets and customers in other parts of the world. Competitive pressure manifests in terms of three major challenges:

Falling prices There has been a continuous fall in prices of branded, engineered consumer products as well as industrial products. For example, the on road price of a Hyundai Santro GLS model in 2000 was ₹440,000/- and the price of a Santro Xing GLS model in April 2014 was about ₹480,000/-. Over the last 15 years, the annual growth rate in price of several white goods such as refrigerators, washing machines, and ovens have been far less than inflation rate. Similarly, industrial buyers constantly enforce annual cost reduction targets for their suppliers. Therefore, learning to profitably operate in an era of falling prices is a challenge.

Shrinking delivery quote Several years back it was not uncommon to book a vehicle (such as a car or a scooter) and wait for taking the delivery. The car manufacturer may not be able to satisfy the customer in terms of the colour choice. However, due to economic reforms, these issues have been pushed to the background. Today, responding to the requirements of the market, rather quickly, is very important. Otherwise, the customers will switch to other suppliers who are willing to respond to them fast. Several service firms such as insurance providers and banks are expected to provide a high level of responsiveness. Manufacturing companies are also constantly facing the challenge of quickly reacting to their customer needs. For example, textile manufacturers are expected to cut their lead time from order placement to final delivery down to 2 months. Otherwise, large retailers such as Wal-Mart, Gap, and Tesco are not interested in placing orders on them. Therefore, developing new methods of shrinking lead time is another challenge for the companies.

Build-to-order requirements Customers increasingly demand several variations of products and services. They also demand flexibility to change their requirements until the point of service and product delivery. Systems that allow them to customize, configure, and visualize their own version of products and services are more in demand than a standard off-the-shelf version with a few pre-determined variations. In order to respond to these emerging requirements, a firm need to develop build-to-order capabilities.

1.7.2 Growing Customer Expectations

Look at the number of tariff plans and options offered by mobile service providers such as Airtel and Vodafone these days. They provide multiple postpaid and prepaid options that appear to satisfy every customer category in the market. This is a good indication of the growing customer expectations and the ability of organizations to respond to them. Such developments have been greatly facilitated by the economic liberalization process that we discussed earlier. A decade ago, customers would not have made any fuss about the services provided by the Department of Post and Telegraphs, simply because they had no other choice. The situation was similar in most sectors, whether a customer wanted to travel by air, get a phone connection, or buy a car. Today, however, they have a variety of choices in of these situations. Many courier services can substitute India Post. Several private airlines and telecom companies offer multiple versions of services. Many different models of cars, starting from basic models such as the Maruti 800 and the Omni to high-end cars such as the Toyota Camry are available today.

Exposure to such developments increases customer awareness, enabling them to demand better-quality products and services. They prefer more variety, greater levels of customization,

NOTES

NOTES

and the ability to have it in discrete quantities and without waiting too much for the product or service they want. This means that manufacturing organizations and service providers constantly need to invest in understanding changing customer tastes and provide goods and services by aligning various aspects of the operations. Organizations need to develop the capability to bring out newer product and service versions that are not only better but also cost-effective. This introduces new challenges for operations management.

1.7.3 Technological Developments

In the last few years, technological developments have changed the way businesses in many sectors operate. Take, for example, the banking sector in our country. With the advent of automated teller machines (ATMs) and the Internet, the face of banking is undergoing a sea change. Customers need not visit a bank branch to draw cash or to check their account balance. These transactions can be carried out at an ATM. Drafts and cheques can be replaced with electronic payment gateways and fund transfer mechanisms. Similar facilities are now available in other sectors. For example, consider the process involved in buying a train ticket. By visiting a Web site like <http://www.irctc.co.in>, a customer can accomplish all the required tasks pertaining to ticket booking and cancellation at leisure.

The changes that manufacturing organizations are going through are no less than service organizations. In this age of electronic markets, a manufacturing organization can procure goods and services by organizing a reverse auction on the Internet. In a matter of 3 to 4 hours, the best price for a component and the supplier willing to provide the component at a desired quality can be located. Similarly, a team of design personnel from across different geographical locations can participate in new-product development using technological tools.

From these examples, we can appreciate that today's businesses are constantly challenged by the rapid technological advancements. These advancements tend to have a dramatic impact on the competitiveness of businesses. They force structural changes in a sector of the industry and trigger several changes in the operational practices of firms.

1.7.4 Environmental Issues

Growing industrialization at one level indicates the economic progress of a nation. However, at another level, it introduces new concerns and challenges in operations. In the recent past, there has been growing concern regarding the impact of business activities on the environment. These include concerns regarding the depletion of natural resources and the waste generated from production systems and

end-of-life products. Increasingly, firms are under pressure to take responsibility of restoring, sustaining, and expanding the planet's ecosystem instead of merely exploiting it.

When the Government of India announced a scheme for special economic zones (SEZs), it generated controversies and social concerns. In several such cases, local communities were reluctant to part with fertile land, which was then used for industrial purposes. This has implications for the location of manufacturing and service facilities.

Consider the case of passenger cars. In a country like India, 12 out of 1000 people have a passenger car. In the United States, this figure is 512 per 1000 people. While this could be an indicator for the passenger-car industry to set its growth projections, it also indicates the challenges arising out of availability of petrol and diesel and increased emissions. Unless the issue of emissions is addressed and alternative fuels and non-conventional technologies are invented, the sector will not grow to meet its potential. Design of products and operational practices must take these environmental concerns into consideration.

Growing urbanization creates societal problems arising out of scarcity of available resources and generation of solid wastes. With high population density in cities, the consumption of energy and water in countries like India is on the rise. Such a situation requires better practices and newer methods of addressing these requirements using better operational practices. These days, due to the increased use of electronic gadgets, the amount of e-waste being generated is also on the increase. Similarly, the use of plastics and bio-non-degradable items also pose challenges to businesses and society.

Businesses need to refine their operations management practices to be able to address these requirements. Operations management practices must address environmental concerns in order to ensure a sustainable world.

1.8 CURRENT PRIORITIES FOR OPERATIONS MANAGEMENT

Each of the challenges identified in the previous discussion clearly points to new priorities for operations management. Economic reforms and liberalization have established the importance of the customer/market. Organizations need to relate the operations system to the customer/market. Otherwise, they may find it difficult to survive in the long run. As customer aspirations grow, organizations must find ways and means of dealing with the proliferation of variety. Operations management practices need to be developed for this purpose. One way in which organizations can meet growing customer aspirations is by getting into a continuous learning mode. In the future, addressing emerging environmental concerns through better operations practices will become a necessity. Some such priorities for present-day organizations are discussed in some detail here:

NOTES

NOTES

- ⇒ *Relate the operations system to customer/market requirements.* Choices in manufacturing and services cannot be made on the basis of internal conveniences, as was the practice in the pre-reforms era. All the choices made in an organization demand that the customer is central to the design, whether it concerns performance measures, manufacturing and service-delivery system design, supplier management, or product development. If the customer has difficulty using a product or service, there should be methods for understanding the difficulty and rectifying the mistakes. This calls for better quality management practices in every organization.
- ⇒ *Acquire capabilities to tolerate product/service proliferation.* Customers are likely to demand more choices and value-offerings from manufacturing and service organizations. In order to remain competitive, every organization needs to understand customer requirements and incorporate them in new product development initiatives. New procedures and systems should link the customer layer to other layers of an organization (see Figure 1.4). The development of new products and models often becomes necessary for organizations. However, large product/service variations often result in a spectrum of offerings ranging from “low-volume, high-variety” to “high-volume, low-variety.” Operations management practices need to address these requirements and enable managers to handle the complexities arising out of variety.
- ⇒ *Develop systems and procedures that promote learning.* Continuous improvements in factors affecting competition, such as quality, delivery, cost, and responsiveness, are very important for every organization. These improvements do not take place in isolation or by mere accident. Instead, improvements happen out of continuous learning on the part of the employees in an organization. A good operations management practice in an organization will help the organization set up a continuous improvement methodology. New systems and procedures that help employees to learn need to be put in place.
- ⇒ *Develop green manufacturing practices.* Green manufacturing practices are operational practices that ensure that products are manufactured in a sustainable manner by conserving scarce natural resources and minimizing pollution and other negative impacts on society and nature.

Environmental imbalances are of great concern as they threaten the society and its sustainability. Non-governmental organizations (NGOs) and interest groups such as Greenpeace, international bodies such as the World Trade Organization (WTO), and government and regulatory bodies impose stringent regulations on organizations to reduce the adverse effects of manufacturing activities on the environment. Policy makers have brought in new legislations that put regulatory pressures on businesses

as a means to address environmental concerns. Notable among these are the EU directives on compulsory product-take-back at the end of life of a product, The Netherlands National Environmental Plan, and the package recycling and product take-back laws in Germany. In Germany, the Commercial and Industrial Waste Avoidance and Management Act holds producers responsible for end-of-life disposition, recovery, and reuse of their products. Therefore, it is inevitable for every organization to develop green manufacturing practices.

In an era where human beings are facing serious environmental concerns like global warming and depletion of natural resources, organizations should reduce the total environmental burden by moving beyond mere “greening” and take ownership of the environmental impacts associated with the total life cycle of the product. This essentially calls for fundamental changes in the underlying product and process design, taking into account all possible effects a product might have throughout its life on the environment, both within the firm and beyond its boundaries. The use of clean technologies is another important requirement for maintaining a balanced environment.

NOTES

1.9 SUMMARY

- *Operations management* is a systematic approach to addressing issues in the transformation process that converts inputs into useful, revenue-building outputs.
- Globally, India is emerging as an important manufacturing base and is closely competing with China in attracting several multinational companies to set up their manufacturing plants in India. Several recent studies point to emerging opportunities for Indian manufacturing to grow and attain a global presence.
- From an operations management perspective, a “pure product” and a “pure service” are just two ends of a spectrum. In reality, a vast majority of operations share the continuum of products and services.
- Despite several important differences between products and services, from an operations management perspective, there are several similarities between the two settings.
- Every organization has five layers that make up its value chain. The physical arrangement of these five layers depicts the structural aspects of an operations system.
- The decision context in operations management can be broadly classified as *design issues* and *operations control issues*. Moreover, there are *long-term* and *short-term decisions* involved in operations management.
- Some of the challenges faced by operations firms include the need to address increased competition due to economic liberalization and globalization, growing aspirations of customers, rapid technological advances, and emerging environmental concerns.

1.10 REVIEW QUESTIONS

1. What is an operations system? Give some examples of operations systems.
2. What do you understand by the term “operations management”?
3. What are the major operations management issues that manufacturing organizations face in India?
4. How would you define a service system? Give some examples to support your definition.
5. How important is it to apply formal management principles to service operations?
6. “There is nothing like a service system or a manufacturing system. In reality, there is a continuum between these two extremes.” Comment on this statement.
7. “Services are very different from manufacturing. Therefore, it is not appropriate to use the same set of principles for managing operations in manufacturing and service organizations.” Do you agree with this statement? Give reasons in support of your argument.
8. What do you understand by the term *systems perspective*? Use the example of operations management to explain.
9. What is the role of operations in an organization? What are the other functions in an organization? Are these functions independent of one another? Give some examples to support your argument.
10. What are the various functions of operations and how are they linked to other parts of an organization?
11. Distinguish between the following terms:
 - (a) Service attributes and product attributes
 - (b) Manufacturing organizations and service organizations
 - (c) Design functions and operational control functions in operations
 - (d) Long-term functions and short-term functions in operations
12. Identify three emerging environmental concerns and their impact on operations management. What changes do you

1.11 NET-WISE EXERCISES

envisage in operations management to address these concerns?

1. Visit the following Web sites to understand the current trends and the growth in India's manufacturing sector:
 - <http://indiabudget.nic.in/budget2013-2014/es2012-13/echap-09.pdf>
 - <http://nmcc.nic.in/default.aspx> (examine the contents under various heads)
 - http://nmcc.nic.in/pdf/strategy_paper_0306.pdf (go through the report on a strategy for the manufacturing sector)

Now, answer the following questions:

- (a) What is your understanding of the status of the manufacturing sector in the country?
 - (b) What are the key steps that the National Manufacturing Competitiveness Program proposes to improve the competitiveness of manufacturing?
2. Visit <http://www.servicetax.gov.in/to> know more about the growth of service tax in India. Look up http://fieo.org/view_section.php?id=0,23&lang=0 to know more about the status of the Indian service sector and its export potential. Furthermore, there are several important links listed in this Web site. Look them up and prepare a report on the status of the service sector and its export potential by answering the following questions:

- (a) Comment on the growth of the Indian services sector and its exports during the last ten years.
 - (b) Is India a dominant player in global trade for services? Does it export more services? What types of services are exported from India?
3. In the future, all manufacturing and service organizations would be required to adhere to sustainable practices. Therefore, every operations manager must develop a good understanding of the issues and challenges that they are likely to face and possible actions that they can take to address these concerns. Visit the following URLs and study in detail how environmental issues are being addressed by organizations:
 - <http://www.ecocycle.org/>
 - www8.hp.com/us/en/hp-information/global-citizenship/reporting.html
 Prepare a report to answer the following questions:
 - (a) What are the implications of the mandatory requirement of "product take-back"? How will it affect the operations management practices of manufacturing organizations? Can you identify two areas on which product take-back will have significant impact and show how they will be affected?
 - (b) How does Hewlett-Packard address environmental concerns? Identify three interesting aspects of their policy and show how an operations manager could implement these policies in his/her organization.

1.12 MULTIPLE-CHOICE QUESTIONS

- 1-1. Which are the three major economic activities in any country?
 - A. Manufacturing, Service and Agriculture
 - B. Foreign Direct Investments (FDI), Agriculture and Manufacturing
 - C. Agriculture, IT sector and Telecommunication
 - D. None of the above
- 1-2. What does IIP, CMIE and PAT stands for?
 - A. Index of Industrial Production, Centre for Monitoring Indian Economy, Profit After Tax
 - B. International Index of Production, Centralized Monitoring of Indian Economy, Profit Associated with Tax
 - C. Indian Index of Production, Centre for Measuring Indian Economy, Profit Aligned with Tax
 - D. None of the above
- 1-3. Services are always:
 - A. Inventoried
 - B. Perishable
 - C. Reusable
 - D. Returnable
- 1-4. What are the key functional areas in any Organization?
 - A. Warehousing, Logistics, Procurement and Sales
 - B. Housekeeping, Supervising, Material Control and Tools Management
 - C. Finance, Operations, Marketing and HRM
 - D. Routine Checking, Medications, Oxygen Cylinder Fixing and Reporting
- 1-5. How many layers are there in an operational function in any organization?

A. 4	B. 6
C. 5	D. 7
- 1-6. What an organization has to do to achieve the targeted production and service delivery?
 - A. Structural planning
 - B. Financial planning
 - C. Functional planning
 - D. Operational planning
- 1-7. The wholesome activity that an operating system uses to convert the raw materials into useful products is termed as:
 - A. Processing
 - B. Progressing
 - C. Conversion process
 - D. None of the above
- 1-8. A very crucial and important aspect of Operations Management is:
 - A. Reaching the desired targeted production
 - B. Estimating the demand
 - C. Ensuring smooth operations on the floor
 - D. All the above
- 1-9. What includes the various activities that an operating system undertakes to convert the raw material into useful products for the customers?
 - A. Processing
 - B. Forecasting
 - C. Quality management
 - D. Maintenance management
- 1-10. In operations management what establishes the overall constraints under which the operations system functions?
 - A. Design issues
 - B. Long term issues
 - C. Short term issues
 - D. Control issues

Answer Keys: 1-1. A 1-2. C 1-3. B 1-4. C 1-5. C 1-6. D 1-7. A 1-8. B 1-9. A 1-10. A

2

CHAPTER

LOCATION DECISION

2.1	Globalization of Operations
2.1.1	Regulatory Issues
2.1.2	Factor Advantages
2.1.3	Expanding Markets in Developing Countries
2.2	Factors Affecting Location Decisions
2.2.1	Market-related Issues
2.2.2	Cost-related Issues
2.2.3	Regulatory and Policy Issues
2.2.4	Other Issues
2.3	Location Planning Methods
2.3.1	Location Factor Rating
2.3.2	The Centre-of-gravity Method
2.3.3	The Load–Distance Method
2.3.4	The Transportation Model
2.4	Other Issues in Location Planning
2.5	Summary
2.6	Formula Review Questions
2.7	Review Questions
2.8	Problems
2.9	Net-Wise Exercises
2.10	Case Study
2.11	Multiple-choice Questions

N O T E S

AKSHAYA PATRA: THE NATION-WIDE MID-DAY MEAL SCHEME OF ISKCON

Mid-day meal scheme (MDMS) is a mechanism to address the socio-economic objectives of boosting enrolment, retention, participation, and nutritional status of children at schools. The Akshaya Patra Foundation (TAPF) is a non-profit organization that operates MDMS program for about 2 million children in over 22,000 schools across 15 states and 2 union territories of India. The vision of TAPF is 'no child in India shall be deprived of education because of hunger'. From the humble beginning of delivering lunch to 1500 children in 2000, it has grown to delivering lunch for 2 million children by 2022.

Amongst many problems to be solved by TAPF, one of the fundamental problems is the location of the kitchens. A centralized kitchen may reduce the capital investments and help them benefit from economies of scale. On the other hand, it may increase the distribution cost as well as the time to reach the schools, which are the demand points. Loading food into vans must start at about 8 am each day, and the vans must complete their delivery schedule before 12.30 pm. Since schools break for lunch at 1 pm, any delay in delivery could result in students going back to their afternoon sessions with an empty stomach. Therefore, maintaining a strict cooking to consumption time is critical for TAPF logistics. This will put realistic limits to centralizing the kitchen.

The other option is to decentralize the kitchen. While this may improve the responsiveness of the delivery network and bring down the distribution cost, food preparation costs may go up as too much of capacity need to be built. These factors determine the location decision for TAPF.

The TAPF operating model involves setting up cooking infrastructure in a city that can cater to the demands of a number of rural schools in the surrounding areas using delivery vans. The capacity of the kitchen and the size of the delivery fleet are determined based on the estimated demand in a region. The maximum available time window for 'cooking to consumption' is about 6 hours. In the city of Bangalore there are two kitchens, one in Vasanthapura and the other in Hare Krishna Hills, Rajaji Nagar. The Vasanthapura kitchen caters to about 630 schools in the surrounding areas. TAPF has located 64 centralized kitchens spread across 15 states and 2 union territories to serve nearly 1.8 million children. These kitchens have a daily cooking capacity in the range of 50,000 to 185,000 meals.

As is evident from the above example, location planning calls for a systematic approach to collect data on demand points, quantum of demand, transportation costs, etc., and uses this for solving the location problem. Other factors such as strict window for delivery also need to be included in the analysis of the location problem. We shall look at these issues in some detail in this chapter.

Source: Mahadevan, B., Sivakumar, S., Dinesh Kumar, D. and Ganeshram, K. (2013). 'Redesigning Mid-day meal logistics of AkshayaPatra Foundation: OR at work in feeding hungry school children', *Interfaces*, 43 (6), pp 530 – 546.

NOTES

The controversy surrounding the Tata Nano project in the Singur special economic zone (SEZ) in West Bengal highlights the importance of location decisions for an organization. Ratan Tata's ambitious objective of building the world's cheapest car became a victim of a tussle between the CPI(M) leadership and the Trinamool Congress. For the CPI(M), the Nano project could have been a symbol of the new spirit of industrialization in the state. This was important because, over the years, West Bengal had been losing out to new hubs of corporate excellence such as Bangalore and Hyderabad. For Trinamool Congress leader Mamata Banerjee, however, it was an opportunity for gaining the goodwill of the farmers as she felt that they had been given a raw deal when their farmland had been acquired. Thus, location decisions seem to have a political dimension in a country like India due to its growing population and the paucity of land.

Location decisions are an integral part of designing a supply chain for an organization as it determines the flow of materials from the raw material suppliers to the factories, and finally to the customers. Location decisions affect the overall profitability of a firm. This is because the cost of manufacturing and the total cost of logistics and distribution are direct fallouts of location decisions. For instance, locating manufacturing facilities far away from the market could raise the logistics and distribution costs. On the other hand, locating manufacturing facilities in regions with high factor costs may make the firm unattractive as the cost of manufacturing may turn out to be high. Moreover, once location decisions are made, they are likely to stay longer as huge investments are made. Therefore, location decisions are strategic in nature.

Location decisions pertain to the choice of appropriate geographical sites for locating various manufacturing and/or service facilities of an organization. In reality, a firm may have to locate several operating units to cater to the requirements of the market. At one extreme, it is possible to have a single location in which all facilities could be located. Products are manufactured in this facility

NOTES

and sent to the markets using a distribution system. Companies in the aerospace and defence sectors, such as Boeing and Airbus, are examples of this category. Another example is the printer-manufacturing facility of Hewlett-Packard. At the other extreme, many facilities are located in multiple markets so that it becomes easy to access customers. Automobile manufacturers such as Ford and Toyota are examples of this category. Essel Propack, the Indian manufacturer of tubes for toothpastes, creams, etc., has factories in all continents except Australia. Both the extremes may prove inappropriate in most situations. However, they highlight the basic trade-off involved in solving the location puzzle.

Location decisions pertain to the choice of appropriate geographical sites for locating the various manufacturing and service facilities of an organization.

Locating facilities in regions that offer attractive cost advantages is one aspect of the trade-off. It may be expensive to set up a distribution system that is both efficient and responsive (it leads to an increase in transportation costs and the costs of coordinating and communicating with the supplier about products and availability). On the other hand, locating facilities very close to the markets provides the advantages of quicker responses and better customer service.

Operations managers need to evaluate intermediate solutions to arrive at an optimal location of facilities. Location planning essentially addresses this issue and provides the operations manager the needed tools and techniques to study the location problem. Location decisions affect other functional areas of business as well. Choice of locations is an important input to marketing as the management needs to plan their marketing strategy in relation to the location of facilities vis-à-vis the customers. In this chapter, we shall look at these issues in some detail and understand how operations managers could make decisions with respect to location of the organization facilities.

2.1 GLOBALIZATION OF OPERATIONS

Location issues have become more prominent in recent years due to the increased pace of economic reforms in several countries and the consequent globalization of markets. The globalization of markets opens up new opportunities to multinational corporations on issues related to location, which they have not faced before. For instance, ABB decided to identify factories that could produce world-class products at internationally competitive prices. Moreover, the factories needed to have high levels of technical capability and domain expertise. This exercise resulted in selecting plants at Vadodara and Nasik for circuit breakers of ranges above 72.5 kV. Similarly, their motor division in Faridabad, the only one available east of the Suez Canal, is capable of making variable drive motors for global markets.

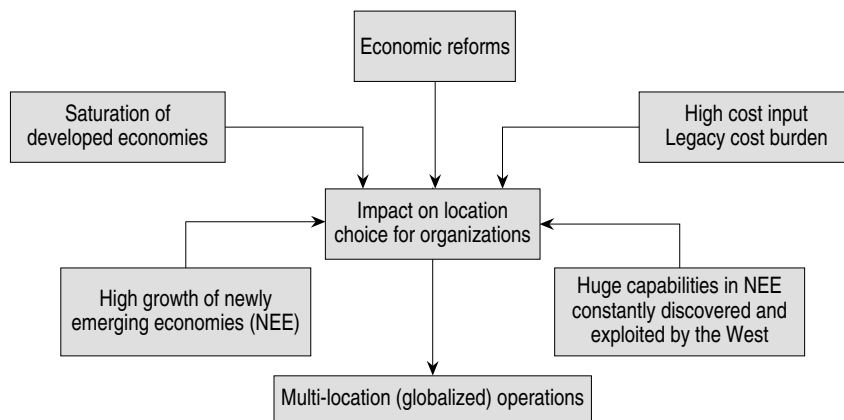


Figure 2.1 Impact of globalization on location decisions of organizations

These examples amplify the close relationship between globalizing operations and new dimensions to the location of manufacturing facilities. Therefore, we shall examine in detail the factors that have caused globalization of operations and how these influence the location decisions of firms. Figure 2.1 presents a schematic representation of the impact of globalization on the location decisions of organizations.

2.1.1 Regulatory Issues

The most significant factor that drives globalization is the ongoing economic and regulatory reforms in several developing countries. Beginning from 1991, we embarked on a set of regulatory changes that made India much more attractive in terms of locating a manufacturing facility. Two events have been broadly responsible for this. First is the reduction in customs and excise tariffs and a move towards the single-point value-added tax (VAT) regime. The other is the delicensing of several sectors of industry and the progressive removal of the cap on foreign direct investment (FDI). Further, there has been a progressive simplification of the procedural aspects of setting up operations here and running them on a day-to-day basis.

These changes have several implications for the location choices of multinational firms. The removal of entry barriers and reduction in the cost of manufacturing due to tariff reductions will make India an attractive destination to which manufacturing bases can be shifted. In the case of multinational firms with manufacturing facilities already located in India, these facilities will be attractive candidates for further development and growth.

The removal of entry barriers and reduction in the cost of manufacturing due to tariff reductions will make India an attractive destination to which manufacturing bases can be shifted.

Another relevant point for location planning with respect to regulatory issues is the emergence of regional trading blocs. A trading

NOTES

NOTES

bloc is essentially a group of geographically separated nations that provide advantageous access to markets, manufacturing facilities, and technologies within the member countries. If required, it could also have a common currency, as in the case of the European Union. Such an arrangement provides multiple options for locations for an organization operating in such markets. India is a member of two trading blocs, the South Asian Association for Regional Cooperation (SAARC) and the Association of Southeast Asian Nations (ASEAN).

2.1.2 Factor Advantages

The globalization of manufacturing is also triggered by factor advantages that an organization can enjoy by operating in specific locations. Generally, developed countries in Europe and in the United States are characterized by the high cost of labour. On the other hand, developing countries offer significant advantages to a firm due to the availability of cheap labour. Therefore, in location planning, these advantages are likely to be considered. The recent shifting of manufacturing bases to China and India and the large-scale shifting of BPO activities to India are primarily related to the factor cost advantages that these countries offer to firms in the western world. For a similar reason, the bulk of semiconductor and electronic appliances manufacturing has been shifting to countries such as South Korea, Taiwan and Malaysia.

Factor advantages may encourage a firm to globalize its operations.

Factor advantages also accrue to a firm on account of the availability of skilled labour and other resources required in plenty for manufacturing. The other resources may be power and water (as in the case of the chemical processing industry) or the availability of technical infrastructure in the form of well-developed ancillary industries. Essentially, when these resources are available in plenty, firms develop confidence that the location of their facilities will benefit them in the long run and enable them to make further enhancements in the product offering without much difficulty. Moreover, the plentiful availability of these resources will introduce a healthy competition among them, paving the way for cost reduction and the overall improvement of quality and delivery reliability. These factors may significantly influence the location decisions of a firm and encourage the firm to globalize its operations.

2.1.3 Expanding Markets in Developing Countries

Another phenomenon that promotes the globalization of operations and provides more alternatives for location decisions is the growth rate of the economy in developed and developing countries. Developing countries such as India and China are growing at an annual rate of six per cent and above. In contrast, most developed countries witness very little growth, amounting to less than two

per cent. Let us consider the automobile industry. In the United States, for every 1000 individuals, nearly 797 own a car. On the other hand, in India, only 41 out of 1000 own a car. This merely illustrates the untapped potential for market growth in developing countries. Moreover, developing countries (primarily in Asia) have very high populations. In several of these countries, the middle-income group is larger than the entire population of most developed countries. Clearly, such a scenario means that there are expanding markets in developing countries, and the desire of multinational firms has been to capture a significant market share in these regions. Therefore, several firms have been considering new locations in these markets for their manufacturing facilities in recent years.

The factors that drive the globalization of operations are aptly summarized by the study on world competitiveness rating by the Geneva-based World Economic Forum. Figure 2.2 illustrates the various factors that make a country (location) attractive for firms. The model proposes three tiers of competitiveness. At the topmost level, there are certain factors that make a country attractive. These include the quality of judicial, economic, and financial institutions, openness to international trade and finance, and the role of its government in creating an environment conducive for business. At the second level, the sectoral competitiveness determines how attractive a location is. This level broadly covers issues of quality and

NOTES

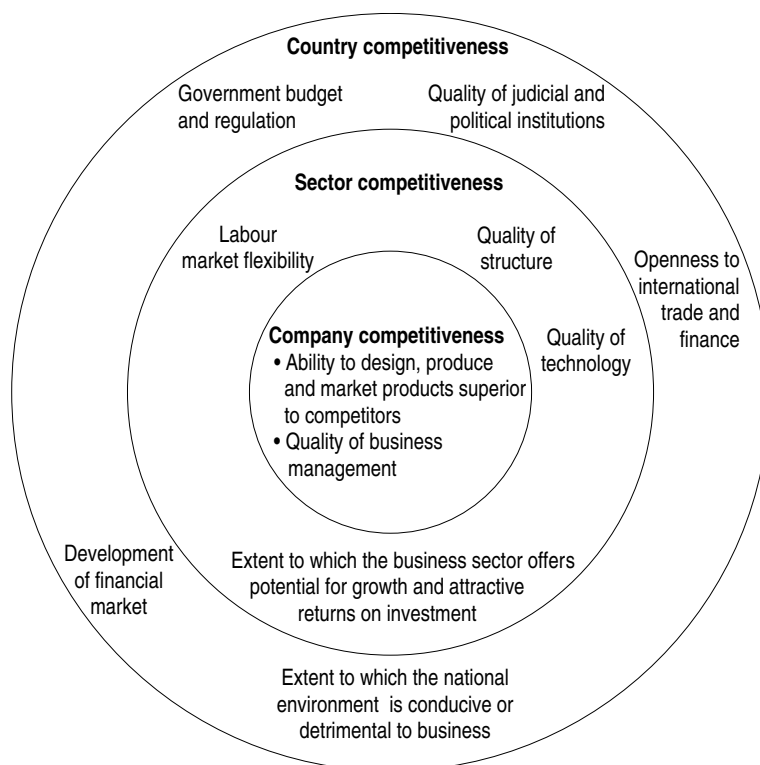


Figure 2.2 A three-tier model for assessing the competitiveness of a location

NOTES

the availability of manpower and associated infrastructure for the sector. At the lowest level, firm-level issues contribute to the attractiveness of the location. This basically pertains to the ability of the firm to compete effectively in the market by operating in a particular location.

Location planning in a globalized scenario also introduces additional complexities that managers had not faced earlier. Foremost is the multilingual and multicultural setting of the various locations. Managing an operating system in these introduces costs that are unknown *a priori* and therefore difficult to factor into a location decision-making framework. There are cultural dimensions and regional practices that account for productivity differences between locations. Complexities in communication and coordination arising out of a multilingual set-up also contribute to additional costs and loss of productivity. Moreover, unknown laws, regulations, and unfamiliar trade practices result in further differences between what is assumed about a location and the reality.

Despite these complexities, more and more organizations see merit in considering alternative locations. Developing and third-world countries are seen to be increasingly attractive options and more and more organizations have been making an effort to shift their manufacturing bases to these locations. The underlying logic behind this appears to be that of identifying the most significant factors from a large number of potential ones and using a methodology for studying the impact of these factors on the overall attractiveness of these locations.

2.2 FACTORS AFFECTING LOCATION DECISIONS

From the previous discussion, it is clear that there are several factors that affect the location decision. Therefore, a location planning exercise requires three steps:

1. Identify the set of factors that could influence the location decision.
2. Establish the relative importance of these measures for the location decision.
3. Develop a methodology to assess the impact of these factors.

We shall dwell on these three issues in some detail.

We frequently hear that multinationals seem to be investing more in China compared to India. We also know that huge manufacturing bases are set up in China to manufacture a variety of consumer goods such as toys and textiles and intermediary low-value-added engineering goods and components. Does that mean that China is a more attractive location compared to India? If so, why is it so? In order to understand this, let us look at some results from the study conducted by the World Bank on doing business in 2005. Table 2.1 presents a comparative study of India and China on some chosen parameters.

TABLE 2.1 DOING BUSINESS IN 2014: A COMPARISON OF INDIA AND CHINA

Parameters	China	India
Ease of doing Business (score)	77.9	71.0
Starting a business (score)	94.1	81.6
Starting a business – procedures required (score)	85.0	47.1
Starting a business – cost (% of income as per capita – score)	99.5	96.4
Starting a business – Time (days)	8.6	17.5
Registering property (score)	81.0	47.6
Enforcing contracts – (score)	80.9	41.2
Enforcing contracts cost (% of claims – score)	81.9	65.2
Enforcing contracts – Total Time (days)	496.3	1445

Source: World Bank. 2013. Doing Business 2014: Understanding Regulations for Small and Medium-Size Enterprises. Washington, DC: World Bank Group. DOI: 10.1596/978-0-8213-9984-2. License: Creative Commons Attribution CC BY 3.0.

From a cursory analysis of the table, one may come to the conclusion that it is relatively easier to do business in China compared to India. This is because there are fewer procedures related to starting or closing business in China. Furthermore, it is easier to enforce contracts. We also notice that the costs associated with running the business, closing, or enforcing the contracts are much lesser in China. It is evident from this example that one can develop a set of factors that affect the location decisions of organizations and use it as a basis for the site selection.

In order to study the factors that affect a location decision in an organization, it is useful to categorize them and understand how each of these are relevant to the location problem. Table 2.2 shows a four-part classification of the various factors that affect the location decision.

2.2.1 Market-related Issues

The first set of factors in Table 2.2 comprise *market-related* issues. The existence of a market for the firm's products and services significantly influences the location decision. In general, it is very desirable to locate the operating system close to either the market or the source of raw material and other critical inputs for the system. Locating systems far away from these will often make the system unviable due to large transportation and distribution costs arising out of bringing the product to the market where it is ultimately consumed. Availability of raw material and other inputs also significantly affect the overall cost of the system, and hence the profitability. In India, successive governments have been trying to woo investors to backward and underdeveloped regions by providing tax rebates and other sops. In spite of that, firms have not been attracted to these places. This is due to the market-related issues of location planning

NOTES

NOTES

TABLE 2.2 FACTORS AFFECTING LOCATION DECISIONS

Market-related Issues	Cost-related Issues
Market for products and services	Factor costs of inputs
Raw material availability	Transportation costs
Number and proximity of suppliers	Taxes and other tariff issues
Availability of skilled labour	Cost of manufacture/service
Quality of infrastructure	Currency and exchange rate fluctuations
Demand–supply gap	
Nature of competition	
Regulatory and Policy Issues	Other Issues
Government and economic stability	Culture
Quality of legal and other institutions	Climate
Trading blocks and trading agreements	Quality of life

and the absence of infrastructure facilities. The importance of market-related issues is much more in the case of a service organization.

The other related issue is the size of the market. A large market provides several advantages to an organization. It has higher demand for products and services. It also assures ample supply of key human resources for the operating system. Large markets are very attractive locations, as we currently see in the case of countries such as China and India. Furthermore, the nature of competition also determines the suitability of a location. If a market is already saturated and there are already several established firms competing for the market share, it requires considerable effort to successfully establish operations in such a market.

Typically, logistics and distribution costs are often considered for location analysis because these costs are direct, tangible, and easy to measure and analyse.

2.2.2 Cost-related Issues

Cost-related issues capture the desirability or otherwise of competing locations on the basis of the cost of operating the system in alternative locations. Typically, logistics and distribution costs are often considered for the analysis simply because these costs are direct, tangible, and easy to measure and analyse. However, depending on the availability of data and sophistication of the analysis, other costs, such as wages or input factor costs and tax and other tariff benefits could also be included. Our earlier discussion on the hesitation of firms to locate their operating systems in far-off and underdeveloped regions could be easily justified by taking a cost-oriented approach to location planning.

ARAVIND EYE HOSPITAL'S APPROACH TO LOCATION OF HEALTHCARE DELIVERY SYSTEMS

NOTES

Research shows that permanent eye care facilities in rural areas motivate people to seek earlier treatment for vision problems, allowing them to reintegrate back into the workforce instead of becoming visually impaired. In order to address the needs of the rural population, the location of healthcare delivery services requires a well thought out strategy so that the target segment is reached in the most effective manner. Being a service system, reach is an important aspect of location planning. Aravind Eye Hospital has a three tier approach for location planning.

At the first level, it has full-fledged hospitals in 14 locations in South India. The hospitals provide high quality and affordable eye-care services to the rich and poor alike, yet are financially self-supporting. They have well equipped specialty clinics with comprehensive support facilities. In the year ending March 2020, 4.6 million out-patient visits and 520,000 surgeries were performed in these hospitals.

At the second level, Aravind Eye Care has setup vision centres to provide eye care for the needy population located at the primary level. The vision centres offer innovative Internet-based Information Technology (IT) that allows patients in rural areas to be remotely diagnosed by ophthalmologists at the base hospital. Via high-speed wireless video-conferencing, doctors can consult with hundreds of rural patients per day, providing high quality eye care while eliminating the need for patients to travel to hospital (unless more advanced treatment is needed). By March 2022, Aravind has established more than 91 IT enabled vision centers providing telemedicine facilities in various districts of Tamilnadu. Each vision center will cover a population of about 45,000 to 50,000.

Community eye clinics offer a permanent access point for comprehensive primary (non-surgical) eye care, in an underserved suburban or semi-urban setting less than one hour drive (30 to 60 kilometres) from the base hospital. The community eye centre model is intended to replace regional eye camps and to manage primary eye care in order to enable the base hospital to focus on secondary and tertiary care. The main difference between a community eye centre and vision centre (besides its bigger size of about 1000 square feet) is the presence of a full-time ophthalmologist in the former set up. The staff consists of one doctor, five paramedics (2 senior ophthalmic assistants, and 3 junior), and one paramedically trained receptionist. The services offered include lab services (urine, blood sugar), optical shop, medical shop, and treatment follow-up.

Source: <http://www.aravind.org/ClinicalServices.aspx>. Last accessed on 20 May 2014.

NOTES

2.2.3 Regulatory and Policy Issues

So far, we have looked at factors that apply to a limited geographical reach (such as a country). However, when we analyse the feasibility of locations cutting across countries, other factors enter into the analysis. First among them is the *regulatory and policy climate* of local governments. The quality of the legal and judicial institutions in that area is one major element in decision making. For example, countries that offer very little or no protection for intellectual property will be least attractive for firms. In such a setting, the imitation of original research and interesting ideas will be rampant and a vibrant gray market will thrive. On the other hand, good-quality governance, availability of free markets, public finances, robust financial institutions and free access to markets will increase the attractiveness of the location.

2.2.4 Other Issues

Finally, *other issues* that influence a location decision include cultural, linguistic, and climatic aspects, along with the overall quality of life. These factors provide a certain assurance to firms that operating systems can work without much difficulty, as the key input for the system, in the form of human resources, will be readily available.

Any location planning decision needs to take these factors into consideration. Depending on the scope of the location planning and the type of modelling of the problem, location planners can judiciously make use of these factors for selection from among competing candidates.

2.3 LOCATION PLANNING METHODS

Generally, location decisions could be one of the two: *one facility–multiple candidates* and *multiple facilities–multiple candidates*. In the former, the decision boils down to selecting one location out of a group for a single facility. For example, a firm may want to choose one location out of several alternatives to build its factory. In the process of selecting the ideal location, one can suitably incorporate any of the factors discussed so far and assess the attractiveness of each location on their basis. We shall discuss two methods used to perform this exercise.

Generally, location decisions could be one of the two: one facility–multiple candidates and multiple facilities– multiple candidates.

The second type is one of selecting k locations out of the n alternatives available. A typical example could be a firm wanting to locate four regional warehouses to serve the demand throughout the country. Therefore, it may have to choose four locations out of the 10 alternatives available to it. Some sort of network flow modelling is involved in this case to decide on an appropriate set of locations.

In addition to choosing k locations, the location planner must also decide how these locations will serve multiple markets that are spread out geographically. One of the popular network models is the transportation method, which could address this issue.

2.3.1 Location Factor Rating

One of the simplest methods to compare alternative sites using the factors identified to be relevant is to use a factor rating method. **Factor rating** is a simple methodology to assess the attractiveness of each potential location. This method involves four steps in which the relevant factors are identified, their relative importance established, the performance of each location in each factor assessed, and, finally, all this information is combined to rank the locations. The advantage of this method is its ability to incorporate any factor into the analysis as long as the decision maker can assess its relative importance. It is also simple to compute and comprehend. However, the limitation of this method is that it can be used only for initial screening and broad-level ranking. A detailed cost-based analysis may be desirable to assess the quantum of benefit or impact of each candidate location before a final decision is made on the location.

Factor rating is a simple methodology to assess the attractiveness of each potential location.

The four-step process of identifying an appropriate location is as follows:

1. Identify and list all the relevant factors for the location decision. We have already discussed some of the factors to be considered. One can choose several factors from the list and add a few more that are specific to the context.
2. Establish the relative importance of each factor in the final decision. The easiest method to establish the relative importance of each factor is to rate each factor on a scale of 0 to 100. Once all the factors are rated, these ratings can be normalized to obtain the relative weights. A simple example of normalization is provided in Table 2.3 for a hypothetical case of six factors.

TABLE 2.3 AN EXAMPLE OF NORMALIZATION

Factor	Rating	Normalized
1	30	$30/300 = 0.10$
2	60	$60/300 = 0.20$
3	35	$35/300 = 0.12$
4	80	$80/300 = 0.27$
5	45	$45/300 = 0.15$
6	50	$50/300 = 0.17$
Total	300	

NOTES

SPECIAL ECONOMIC ZONES: A POLICY ANGLE TO LOCATION PLANNING

NOTES

India was one of the first in Asia to recognize the effectiveness of the export processing zone (EPZ) model in promoting exports, with Asia's first EPZ set-up in Kandla in 1965. With a view to overcome the shortcomings experienced on account of the multiplicity of controls and clearances, the absence of world-class infrastructure and an unstable fiscal regime, and also to attract larger foreign investments in India, the Special Economic Zones Policy was announced in April 2000. This policy intended to make SEZs an engine for economic growth, supported by quality infrastructure complemented by an attractive fiscal package with the minimum possible regulations at both central and state levels. The Special Economic Zones Act, 2005, was passed by the parliament in May 2005 and presidential assent was given on 23 June, 2005.

The SEZ rules provide for different minimum land requirements for different classes of SEZs. Every SEZ is divided into a processing area where the SEZ units alone would come up and the non-processing area where the supporting infrastructure was to be created. The total land requirement for SEZs for which formal approvals have already been granted until May 2009 is approximately 67,680 hectares. The land for the 270 notified SEZs where operations have since commenced is approximately over 31,405 hectares. Units located in SEZs get several incentives and benefits. These range from income tax concessions, service tax and sales tax exemptions, and duty free import.

Out of the 531 formal approvals given till date, 174 approvals are for sector-specific and multi-product SEZs for manufacture of textiles and apparels, leather footwear, automobile components, engineering, etc., which would involve labour-intensive manufacturing. SEZs are going to lead to the creation of employment for a large number of unemployed rural youth.

The Nokia and Flextronics electronics and hardware SEZs in Sriperumbudur are already providing employment to 14,577 and 1,058 people respectively. The Hyderabad Gems SEZ for jewellery manufacturing in Hyderabad has already employed 2,145 people, a majority of whom are from landless families, after providing training to them. They have a projected direct employment for about 2,267 people. The Apache SEZ being set up in Andhra Pradesh will employ 20,000 people to manufacture 1,000,000 pairs of shoes every month. They currently employ 5,536 people. Brandix Apparels, a Sri Lankan FDI project would provide employment to 60,000 workers over a period of 3 years. Even in the services sector, 12.5 million m² of space is expected in the IT/ITES SEZs, which, as per the NASSCOM standards, translates into 1.25 million jobs. Therefore, it is expected that the establishment of SEZs would lead to faster growth of labour-intensive manufacturing and services in the country.

From these statistics, one can conclude that the benefits derived from SEZs include employment generation, increased exports, and the development of quality infrastructure. The benefits derived from the multiplier effect of the investments and additional economic activity in the SEZs and the employment generated could far outweigh the tax exemptions and the losses on account of land acquisition.

Source: <http://sezindia.nic.in/HTMLS/about.htm>, last accessed on 20 May, 2014.

3. Rate the performance of each candidate location using a rating mechanism.
4. Compute a total score for each location, based on its performance against each factor, and rank them in the decreasing order of the score.

2.3.2 The Centre-of-gravity Method

In the location factor rating method, we utilized a simple weighted scheme to assess the attractiveness of each location. However, it is possible to utilize more direct measures for evaluating each location. Sometimes, location planners tend to use a distance measure to evaluate the impact of a proposed location. Such an approach has merit when the dominant requirement is proximity to the market for products (or for raw materials). By locating itself too far away, an organization may end up spending considerably on logistics and distribution. Therefore, assessing the distance of a site vis-à-vis the reference market could be a better method for selecting appropriate locations.

In the **centre-of-gravity method**, all the demand points (or the supply points, if the raw material is supplied from several locations) are represented in a Cartesian coordinate system. Each demand (or supply) point will also have weights, which indicates the quantum of the demand (or the supply) per unit time. In this context, it is possible to identify the centre of gravity of the various demand (or supply) points. Locating the new facility at this point will be most appropriate.

The **centre-of-gravity method** merely indicates the ideal location in the grid map that would ensure that the weighted distances travelled on the whole is minimum.

The centre-of-gravity method merely indicates the ideal location on the grid map that would ensure that the weighted distance travelled on the whole is minimal. However, location planners may have to include other practical considerations before making a final decision. For instance, the coordinates indicated by the centre-of-gravity method may point to an infeasible site to set up a plant. In some cases, it may be very expensive to acquire the land and develop the site. There could be regulatory constraints in locating a factory (as in the case of having a factory in a residential area). Therefore, one method to solve this problem could be to evaluate multiple sites in the area and select one among them.

NOTES

NOTES

EXAMPLE 2.1

A manufacturer of garments is actively considering five alternative locations for setting up its factory. The locations vary in terms of their advantages to the firm. Hence, the firm requires a method of identifying the most appropriate location. Based on a survey of its senior executives, the firm has arrived at six factors to be considered for final site selection. The ratings of each factor on a scale of 1 to 100 provide this information. Furthermore, based on the detailed analysis of both the qualitative and quantitative data available for each of the locations, the ratings of the locations against each factor have also been arrived at (on a scale of 0 to 100). Using this information (see Tables 2.4 and 2.5), obtain a ranking of the alternative locations.

TABLE 2.4 FACTOR RATINGS

Factors	Rating
Availability of infrastructure	90
Size of the market	60
Industrial relations climate	50
Tax benefits and concessions	30
Availability of cheap labour	30
Nearness to port	65

Solution

Establishing the Relative Importance Through Normalization

The first step in the solution is to establish the relative importance of each factor using a normalization procedure. The sum of all ratings is 325. Therefore, by dividing each factor rating by 325 one can obtain the relative weight of the factors. The workings are presented in Table 2.6.

TABLE 2.5 RATING OF EACH LOCATION AGAINST EACH FACTOR

Factors	Location 1	Location 2	Location 3	Location 4	Location 5
Availability of infrastructure	20	40	60	35	55
Size of the market	30	30	40	60	80
Industrial relations climate	80	30	50	60	50
Tax benefits and concessions	80	20	10	20	20
Availability of cheap labour	70	70	45	50	50
Nearness to port	20	40	90	50	60

Computing the Performance of Each Location

TABLE 2.6 RELATIVE WEIGHT OF EACH FACTOR

Factors	Rating	Relative Weights
Availability of infrastructure	90	0.28
Size of the market	60	0.18
Industrial relations climate	50	0.15
Tax benefits and concessions	30	0.09
Availability of cheap labour	30	0.09
Nearness to port	65	0.20
Sum of all factor ratings	325	1.00

Since the normalized weight of each factor is available, one can compute how each location fares by weighing the rating of the location against each factor with the weight for the factor. The computation for Location 3 is as follows:

$$\text{Overall rating for Location 3} = (60 \times 0.28) + (40 \times 0.18) + (50 \times 0.15) + (10 \times 0.09) + (45 \times 0.09) + (90 \times 0.20) = 54.45$$

The computation for all the other locations is done in the same manner. Based on these overall scores, the locations are finally ranked. Table 2.7 has all the relevant computations.

On the basis of this computation, we find that Location 5 is the best, followed by Location 3. It will be useful to shortlist these two locations and perform additional analysis by collecting more information, such as costs and distances, in order to finally select the most appropriate location. Therefore, it is evident that the location factor analysis method provides a mechanism for a rough-cut and quick evaluation of several sites before identifying two or three probable sites for detailed analysis and final selection.

NOTES

TABLE 2.7 RANKING OF THE FIVE LOCATIONS

Factors	Relative Weights	Location 1	Location 2	Location 3	Location 4	Location 5
Availability of infrastructure	0.28	20	40	60	35	55
Size of the market	0.18	30	30	40	60	80
Industrial relations climate	0.15	80	30	50	60	50
Tax benefits and concessions	0.09	80	20	10	20	20
Availability of cheap labour	0.09	70	70	45	50	50
Nearness to port	0.20	20	40	90	50	60
Overall score for the locations		41.23	37.54	54.77	46.46	56.15
Ranking of the locations		4	5	2	3	1

Note: The overall scores computed in the table are more accurate as the relative importance of the factors are not rounded to two decimals, as shown in the sample computation for Location 3.

NOTES

EXAMPLE 2.2

A manufacturer of a certain industrial component is interested in locating a new facility in a target market and would like to know the most appropriate place in the target market to locate the proposed facility. The manufacturer feels that there are no location constraints in the target market (that is, any point in the target market is good enough). There are four supply points in the locality that will provide key inputs to the new facility—A, B, C, and D. A two-dimensional grid map of the target market in which we would like to locate a new facility, along with the distance coordinates of the four supply points, is available (see Figure 2.3). The annual supply from these four points to the proposed facility is 200, 450, 175 and 150 tonnes respectively. The situation is graphically shown in a two-dimensional plot in the figure. While the coordinates within parentheses show the distance from the origin of the target map to each of the supply points, the number that follows is the annual shipment (in tonnes) from these points to the proposed facility. Identify the most appropriate point in the grid map to locate the new facility.

Solution

Let us represent the given information using the following notations:

Number of existing demand (or supply) points in the grid map = n

Coordinates of Location i in the grid map = (x_i, y_i)

Quantum of shipment between existing demand (or supply) point i and proposed facility = W_i

Coordinates of the centre of gravity in the grid map = (X_c, Y_c)

In our example, the location of supply point B is represented as $(x_B, y_B) = (350, 400)$.

The coordinates of the centre of gravity are given by:

$$X_c = \frac{\sum_{i=1}^n (x_i) \times W_i}{\sum_{i=1}^n W_i} \text{ and } Y_c = \frac{\sum_{i=1}^n (y_i) \times W_i}{\sum_{i=1}^n W_i} \quad (2.1)$$

Using the information available in Figure 2.3 for Locations A to D and Eq. 2.1 one can compute the centre of gravity for the above example to be (366,376). The figure also has the plot of the centre of gravity. Therefore, the manufacturer might want to locate the new facility at this point.

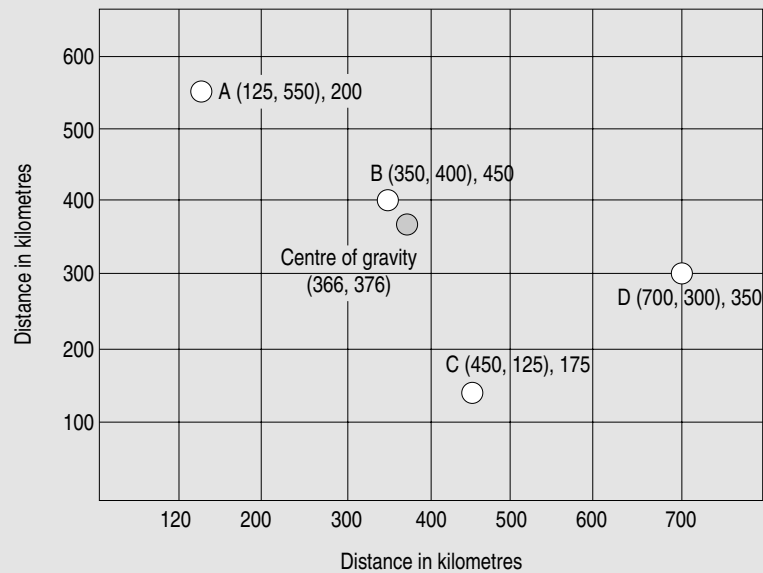


Figure 2.3 Grid map of the target market

NOTES

2.3.3 The Load–Distance Method

As opposed to the centre-of-gravity method, the load–distance method enables a location planner to evaluate two or more potential candidates for locating a proposed facility vis-à-vis the demand (or supply) points. The **load–distance method** provides an objective measure of the total load distance for each of the potential candidates. Choosing the location with best load distance among these will satisfy the objective of identifying an appropriate location for the proposed facility. Distance is measured using a Cartesian measure.

The **load–distance method** enables a location planner to evaluate two or more potential candidates for locating a proposed facility vis-à-vis the demand (or supply) points.

Let us use the following notations for the load–distance method:

Number of existing demand (or supply) points in the grip map

= n

Index used for existing demand (or supply) points = i

Coordinates of existing demand (or supply) points i in the grid map = (x_i, y_i)

Quantum of shipment between existing demand (or supply) point i and proposed facility = W_i

Number of candidates for the proposed facility = m

Index used for the candidates for the proposed facility = j

Coordinates of candidate j in the grip map = (X_j, Y_j)

NOTES

EXAMPLE 2.3

Consider Example 2.2. Assume that the manufacturer came to know that there are constraints in locating the new facility. Based on an initial survey of possible sites for the proposed facility, the manufacturer identified four candidates. Figure 2.4 has the location coordinates of the four candidates (numbered 1 to 4). What is the best location for the proposed new facility?

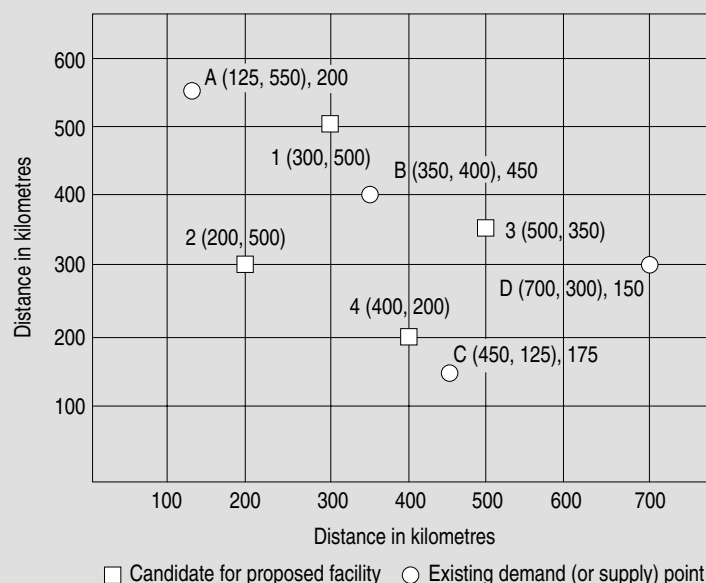


Figure 2.4 Location coordinates of the four candidates

Solution

Let us first summarize the coordinates of the existing supply points and the candidates for the proposed facility by tabulating them from the grid map for the target market. Let us also tabulate the quantum of shipment from these supply points to the proposed facility. Table 2.8 presents this information.

Using the information available in Table 2.8 and Eq. 2.2, one can compute the D_{ij} values between every pair of supply

TABLE 2.8 COORDINATES OF SUPPLY POINTS AND PROPOSED LOCATIONS

	Existing Supply Points			Candidates for Proposed Facility		
	x_i	y_i	W_i		X_j	Y_j
A	125	550	200	1	300	500
B	350	400	450	2	200	500
C	450	125	175	3	500	350
D	700	300	150	4	400	200

point and candidate for locating the new facility. The computation for A-1 is as follows:

$$\begin{aligned}
 D_{A1} &= \sqrt{(x_A - X_1)^2 + (y_A - Y_1)^2} \\
 &= \sqrt{(125 - 300)^2 + (550 - 500)^2} \\
 &= \sqrt{(175)^2 + (50)^2} \\
 &= 182.00
 \end{aligned}$$

Computing for all the pairs in this manner, one can obtain a matrix of D_{ij} , as shown in Table 2.9.

Using Eq. 2.3, one can obtain the LD_j values and select the candidate with the least value of LD . The computation of LD_j is available in Table 2.10.

The computation of LD_1 is as follows:

$$LD_1 = 182 \times 200 + 111.8 \times 450 + 403.89 \times 175 + 447.21 \times 150 = 2,24,474.41$$

As we see from the table, Candidate 1 has the least LD_j value and therefore, is the most appropriate place to locate the proposed new facility. It is interesting to note that Candidates 1 and 3 are very close to the centre of gravity that we computed in Example 2.2 and therefore, they have the minimum LD_j compared to the other two sites.

TABLE 2.9 D_{ij} VALUES

	1	2	3	4
A	182.00	90.14	425.00	445.11
B	111.80	180.28	158.11	206.16
C	403.89	450.69	230.49	90.14
D	447.21	538.52	206.16	316.23

TABLE 2.10 LD_j VALUES

1	2	3	4
2,24,474.41	2,58,801.57	2,27,410.05	2,45,000.8

The distance measure for the Cartesian coordinates between an existing demand (or supply) point i and a candidate j for the proposed facility (D_{ij}) is given by:

$$D_{ij} = \sqrt{(x_i - X_j)^2 + (y_i - Y_j)^2} \quad (2.2)$$

NOTES

NOTES

The load–distance for a candidate j for the proposed facility LD_j is nothing but the product of the distance between the candidate and all existing demand (or supply) points. It is computed as follows:

$$LD_j = \sum_{i=1}^n D_{ij} \times W_i \quad (2.3)$$

2.3.4 The Transportation Model

So far, we have confined ourselves to the issue of choosing one location for the proposed new facility out of several alternatives. However, in reality, there are several situations that prompt an organization to select multiple locations instead of just one. While the choice of these multiple locations could be on the basis of the factors described above, the multiple location problem introduces another dimension to location design. The location planner must also decide which of the demand points will be served by each of these locations and to what extent. Locating distribution centres for nationwide distribution of products is a typical example belonging to this category of decision making. Choosing multiple locations invariably requires that the combined set of locations will ensure minimum cost of transportation pertaining to flow to products in the distribution chain. Therefore, the problem is one of managing network flows to satisfy a set of demand points using a combination of supply points. The transportation model is ideally suited for solving this combinatorial optimization problem.

In the case of multiple plants and multiple demand points, the location problem can be solved using the standard transportation model.

The transportation model is an interesting variation of the basic linear programming (LP) model, which could be used to optimally identify a subset of supply points from a potential list that can satisfy the demand at various demand points. There is a unit transportation cost of shipping material from every supply point to every demand point, and by an optimal choice of supply points, the total cost of transportation is minimized. In the process of doing this, it is ensured that none of the capacity constraints in the supply point is violated. Similarly, it is also ensured that at every demand point the requirement is fully satisfied. Figure 2.5 shows the typical representation of a transportation problem.

In this example, there are four factories that could supply to five markets. The last column in the table shows the capacity available in each factory. For instance, Factory C has 700 units of capacity whereas Factory D has just 200 units of capacity. The last row shows the demand for the product in each of the five markets. The demand in Market 1 is 200 units and the demand in Market 5 is 400 units. Since the markets and the factories are physically separated,

	Market 1	Market 2	Market 3	Market 4	Market 5	Supply
Factory A	100	70	50	30	40	600
Factory B	30	90	40	120	50	400
Factory C	80	20	70	40	30	700
Factory D	20	40	90	90	80	200
Demand	200	450	300	550	400	1900

Figure 2.5 Typical representation of a transportation problem

one could expect different costs of transportation between each pair of markets and factories. This information is available in the cells of the table. The numbers in the cell (upper right hand side) denote the cost of transporting one unit from a factory to a market. In our example, it costs ₹100 to transport one unit from Factory A to Market 1.

An important decision in this problem is to decide which of the factories will serve the markets and how many units should be transported from a factory to a market. The transportation model provides the optimal combination of a factory with a particular market and also identifies how many units need to be shipped. Location problems in the case of multiple plants and multiple demand points follow exactly the same pattern; therefore, one can use the standard transportation model to solve the multi-location problem too. The transportation model is discussed in any standard and elementary level textbook on operations research. However, the application of the transportation model in a location problem is illustrated using an example.

EXAMPLE 2.4

A manufacturer of washing machines is in the process of locating regional warehouses in four geographical locations in south India to serve the markets. The markets are geographically and organizationally split into five segments. Based on the forecasting estimates by the marketing department, it has been found that the average monthly demand for the washing machine is 2000, 1500, 1200, 2800, and 2500 in each of the market segments. Based on this forecast and other costs including the fixed and variable costs of setting up warehouses, it has been decided to build four warehouses with a capacity to handle monthly requirements to the extent of 2900, 2300, 3700 and 1100 units, respectively. Due to the

NOTES

NOTES

geographical spread of the warehouses and the markets, the transportation cost per unit is different between these pairs of warehouses and market segments. Figure 2.6 presents the cost of transporting one unit. Identify the most cost-effective way of serving the markets from these warehouses.

	Market 1	Market 2	Market 3	Market 4	Market 5
Warehouse A	100	70	50	30	40
Warehouse B	30	95	40	125	50
Warehouse C	75	20	65	40	30
Warehouse D	20	40	95	85	80

Figure 2.6 The cost of transporting one unit

Solution

Constructing the Transportation Table

Let us first construct the initial transportation table by adding the available demand and supply information. Figure 2.7 has the initial table. In this case, we find that the demand and supply are exactly balanced. If the supply and demand are unbalanced, the first step is to make the transportation problem balanced by adding dummy supply (or demand) points. If the demand is greater than the supply, then we need to add a dummy supply point with zero cost of transportation. Similarly, if the supply is more than the demand, we need to add a dummy demand point with zero cost of transportation.

There are several heuristic methods to solve the transportation problem. Optimal methods are also available. Typically, the heuristic methods are used as the starting solution for the optimal method. One heuristic, Vogel's approximation method (VAM) is known to provide close-to-optimal solutions to the transportation problem.

Solution Using VAM

VAM is implemented with the following three simple steps:

- ⇒ *Step 1:* For each row and column, compute the difference between the minimum cost cell and the second smallest cell.
- ⇒ *Step 2:* Choose the row or column with the maximum difference. Assign as much as possible to the minimum cost cell in

the chosen row or column (in case of zero assign ϵ —denoting epsilon, an insignificantly small quantity—to the cell). Decrement the supply and demand values corresponding to the cell by the amount assigned. Delete either the row or the column—whichever has become zero, but not both.

- ⇒ *Step 3:* If there is only one row or column left, allocate to each of the remaining cells in the row or column and stop. Otherwise, return to Step 1.

The first iteration of VAM is shown in Figure 2.8

	Market 1	Market 2	Market 3	Market 4	Market 5	Supply
Warehouse A	100	70	50	30	40	2900
Warehouse B	30	95	40	125	50	2300
Warehouse C	75	20	65	40	30	3700
Warehouse D	20	40	95	85	80	1100
Demand	2000	1500	1200	2800	2500	10,000

Figure 2.8 The initial transportation table

Computing the Difference Between the Minimum Cost Cell and the Next Higher Cost Cell

The maximum value of the differences as computed in Figure 2.8 is 20. Since it occurs at two places, we break ties arbitrarily and choose Market 2. We shall assign the maximum to the least cost cell in the column. Using this logic, we assign 1500 to Warehouse C in that column and adjust the demand and supply values. Since the entire requirement of Market 2 is met with, we strike off the column. We also note that after this allocation,

	Market 1	Market 2	Market 3	Market 4	Market 5	Supply
Warehouse A	100	70	50	30	40	2900
Warehouse B	30	95	40	125	50	2300
Warehouse C	75	20	65	40	30	3700
Warehouse D	20	40	95	85	80	1100
Demand	2000	1500	1200	2800	2500	10,000

Figure 2.9 The final allocation table

NOTES

NOTES

the balance capacity available with Warehouse C is 2200. After making these adjustments, we return to Step 1 of the algorithm and recompute the differences and proceed with the allocation in the same way as mentioned above.

The final allocation table obtained in this manner is shown in Figure 2.9.

The final solution and the cost of the plan is as shown in Table 2.11.

TABLE 2.11 THE FINAL SOLUTION AND THE COST OF THE PLAN

Shipping From	Received at	Quantity	Unit Cost (₹)	Cost of Transportation (₹)
Warehouse A	Market 4	2800	30	84,000
Warehouse A	Market 5	100	40	4,000
Warehouse B	Market 1	900	30	27,000
Warehouse B	Market 3	1200	40	48,000
Warehouse B	Market 5	200	50	10,000
Warehouse C	Market 2	1500	20	30,000
Warehouse C	Market 5	2200	30	66,000
Warehouse D	Market 1	1100	20	22,000
Total cost of transportation				291,000

2.4 OTHER ISSUES IN LOCATION PLANNING

Recent trends in international markets point to a shift towards fewer facilities that could serve markets worldwide. For example, the HP DeskJet printer manufacturing facilities are available only at two places, Vancouver in Washington State in the United States, and Singapore. Several other appliance manufacturers are gravitating towards fewer and fewer distribution centres. These developments point to the fact that there are two aspects that could affect the location planning problem significantly. One is the availability of good transportation infrastructure and the other is the use of the Internet and IT infrastructure. Today's operations managers must know how these two tools affect location design.

The availability of good transportation infrastructure enables firms to invest in fewer and larger facilities and provide a good level of service to the customers.

The availability of good transportation infrastructure enables firms to invest in fewer and larger facilities and provide a good level of service to customers. Infrastructure includes availability of an all-weather high-speed road network, good railway infrastructure

that is capable of transporting bulk cargo, availability of ports that have faster turnaround times and good connectivity to the ports. The recent attempts by the Central Government pertaining to the Golden Quadrilateral Projects seek to address this issue. Once the project is complete, one can expect significant changes in the approach to location planning.

In the era of the Internet and IT, providing useful interfaces for customers may offset some of the shortcomings of location decisions. In recent years, the advent of the Internet has resulted in significant reductions in the cost of communication and coordination among trading partners. This has great relevance for the location planning problem that operations managers face. Due to the reduced cost of communication and coordination, it is possible to have fewer locations and still provide better customer service. Developing home pages that provide for high levels of customer interfacing and interaction may enable an organization to directly deal with the customers and provide good service even without a physical facility within close proximity of the customer.

Another issue pertaining to location is the supply management practice of leading Japanese manufacturers such as Toyota. Japanese manufacturing management techniques lay an emphasis on location planning in the overall context of the just-in-time manufacturing philosophy. In Japan, it is often the practice to locate a good number of suppliers in the vicinity (20–40km radius) of the manufacturer. Locating suppliers who are close by provides several advantages to the firm. It enables them carry almost zero inventory. Typically, a milk-van-type routine is followed, in which a truck goes round and collects hourly or half-day requirements of material from each of these suppliers and delivers them at the factory premises. Several Indian automobile manufacturers have similar practices with respect to locating their suppliers. The Ambattur and Guindy industrial estates in Chennai cater to the requirements of all major automobile manufacturers in Chennai. Similarly, Maruti has several suppliers located in the Gurgaon–Faridabad belt on the outskirts of Delhi. Locating suppliers close by helps, especially when the quality of the infrastructure is not good. Further, it helps minimize disruptions arising out of *bandhs*, strikes and natural calamities.

Finally, location issues in a service-intensive industry are different from those in a manufacturing industry. Since service quality is largely determined by the responsiveness of the service delivery system, locating service outlets as close to the demand point as possible is an important requirement in a service system. Therefore, nearness to the market will be more important than the cost of the site, as opposed to a manufacturing system, where the cost of the site would take precedence over location.

NOTES

2.5 SUMMARY

- Location issues have become more prominent in recent years on account of the globalization of markets. Multinational corporations have more opportunities to identify candidate locations for their manufacturing facilities.
- Factor cost advantages and expanding markets in developing countries have made these nations attractive for locating new facilities.
- Simple qualitative methods are useful for quickly screening an initial set of candidates and narrowing down the choice to one or two.
- The **load-distance and centre-of-gravity methods** help evaluate the suitability of candidate solutions from

a perspective of distance and the quantum of items to be transported between a location and the demand points.

- The *transportation method* helps in optimally identifying a set of k locations out of n candidate solutions.
- The availability of good transport infrastructure and recent developments in Internet technology suggest that it is possible to have fewer locations and still provide better customer service.
- Location decisions in service systems must address the requirement of speed of responsiveness. Therefore, locating service outlets as close as possible to the demand points may be highly desirable in service systems.

2.6 FORMULA REVIEW QUESTIONS

1. The coordinates of the centre of gravity

$$X_c = \frac{\sum_{i=1}^n (x_i) * W_i}{\sum_{i=1}^n W_i} \quad \text{and} \quad Y_c = \frac{\sum_{i=1}^n (y_i) * W_i}{\sum_{i=1}^n W_i}$$

2. Distance between demand point i and the proposed facility j $D_{ij} = \sqrt{(x_i - X_j)^2 + (y_i - Y_j)^2}$

3. The load-distance for a candidate j for the proposed facility $LD_j = \sum_{i=1}^n D_{ij} * W_i$

2.7 REVIEW QUESTIONS

1. How important are location decisions in operations management? What are the consequences of a bad location decision?
2. What factors drive the globalization of operations? Do these have any bearing on the location decision that an operations manager makes?
3. Will the choice of factors for selecting an appropriate location vary with the geographical spread of the potential candidates for location? Why?
4. Identify three important factors that a location planner may consider with respect to each of the following:
 - (a) A super-specialty intensive care unit
 - (b) A thermal power plant operating with coal as its fuel
 - (c) A state-of-the-art design centre for automobile manufacturing

- (d) A multi-cuisine restaurant
- (e) An agro-based handicrafts manufacturing unit

5. What are the pros and cons of using the location factor analysis method for location planning? Do you have any recommendations on how to use this method for location planning?
6. A manager intending to use the location planning model for analysis is not sure about which of the two models, the centre-of-gravity method and the load-distance method, should be used for his analysis. Prepare a one-page note that will help him identify an appropriate model for his consideration.
7. What are the key differences between using a transportation model and a load-distance method for solving a location problem?

2.8 PROBLEMS

1. A firm is considering three alternative locations for setting up its new factory. Each of these locations provides some advantages and some limitations. Therefore, it is necessary that some method of assessing the attractiveness of each site be arrived at. Based on a survey of its top management, the company has identified seven factors that will determine the appropriateness of a site for setting up the new factory. These factors and the score out of 100 for each of them are shown in Table 2.12.

TABLE 2.12 THE FACTORS AND THE SCORE OUT OF 100

Sl. No.	Factor for Consideration	Score (Out of 100)
1	Nearness to port	80
2	Existence of supplier infrastructure	70
3	Availability of skilled labour	90
4	Government policies and local taxes	50
5	Projected cost of operations	60
6	Quality of road infrastructure	70
7	Availability of educational infrastructure	40

The company collected some data pertaining to each site that it is considering, and, based on some criteria, was able to arrive at how each site rates against a score of 100 against each factor. In some cases, actual data was estimated on the basis of some projections. Table 2.13 has details pertaining to these:

TABLE 2.13 THE RATINGS FOR EACH SITE AGAINST EACH FACTOR

Sl. No.	Factor for Consideration	Site 1	Site 2	Site 3	Basis
1	Nearness to port	70 km	130 km	95 km	Actual data
2	Existence of supplier infrastructure	60	80	85	Score out of 100
3	Availability of skilled labour	50	70	85	Score out of 100
4	Government policies and local taxes	70	45	60	Score out of 100
5	Projected cost of operations	₹20,000,000	₹18,000,000	₹17,000,000	Actual data
6	Quality of road infrastructure	80	90	70	Score out of 100
7	Availability of educational infrastructure	60	80	80	Score out of 100

Identify the most appropriate site for locating the new factory on the basis of the above information.

(Hint: Suitably convert the actual data into scores out of 100 and then analyse)

2. A chain of stores trading in a host of retail goods for urban customers is planning to set up operations in one of the metropolitan cities in the country. The exact location for the new outlet is to be decided. In order to do that, the company conducted a market survey and found five pockets of consumption in the city. The distance coordinates of these five pockets from a reference point in the grid map pertaining to the city are as follows: Zone 1 (15, 22), Zone 2 (10, 40), Zone 3 (35, 15), Zone 4 (50, 5), and Zone 5 (40, 35). The annual demand for the goods (in tonnes) in the five zones is 200, 130, 80, 170, and 120, respectively.
 - (a) What is the centre of gravity for setting up the operation in the city?
 - (b) If the cost of transporting the goods is ₹300 per tonne, what is the cost implication of locating the stores at the centre of gravity?
3. The chain store officials found that locating the operations in the centre of gravity point, as obtained in the above problem, would cost them a lot. The operational costs of renting and maintaining the site will be ₹2,500,000 per annum. In view of this, they are now considering other options. They have identified three other candidate locations whose cost of operations will be less than the centre of gravity location by 10 per cent, 25 per cent and 18 per cent, respectively. The locational coordinates for these three are (20, 40), (40, 25), and (35, 45), respectively. What should the chain store do in this case? Does it make sense to set up operations in any of these three alternatives locations?
4. A manufacturer of certain household goods caters primarily to the south Indian market. At present, the manufacturer has located six stock points in the major south Indian cities of Hyderabad, Mysore, Chennai, Kochi, Pune, and Vizag to cater to the demand. There are four

warehouses from which the components are distributed. In order to save on the cost of operations, the warehouses are typically located away from these cities in different locations. The unit cost of transporting the items from each of these warehouses to the cities (₹/tonne), the capacity of the warehouses, and the demand at the city stock points per unit time (in tonnes) are all summarized in Figure 2.10. Identify the least cost schedule for deciding which warehouse will send the items to the cities.

5. Suppose, in the above problem, the company has decided to shut down its Kochi operations and to serve the requirements of Kochi from Mysore. Will the decision change? What will the cost impact of this decision be?

6. Consider Problem 4. Suppose the company has received a proposal for setting up another warehouse (Warehouse E) with a capacity of 900 units. The location of the warehouse is such that the unit costs of transporting to the six cities are as follows: Hyderabad–45, Mysore–75, Chennai–30, Kochi–65, Pune–90 and Vizag–80. The company would like to know how the allocation will change if the warehouse is made operational and the likely cost impact on the operation of the system.

(Hint: Its introduction of will result in an unbalanced transportation problem. Convert it into a balanced transportation problem and solve.)

	Hyderabad	Mysore	Madras	Kochi	Pune	Vizag	Supply
Warehouse A	60	70	10	30	90	90	1900
Warehouse B	70	25	40	55	80	100	1800
Warehouse C	50	80	45	75	85	30	3300
Warehouse D	80	40	50	45	90	60	1300
Demand	1750	750	2000	1800	1200	800	8300

Figure 2.10 Demand–supply and cost data for the problem

2.9 NET-WISE EXERCISES

- The BMW Group is the only manufacturer of automobiles which concentrates entirely on premium standards and outstanding quality for all its brands, and across all relevant segments. Click on the 'Production' icon on the top right side of the webpage (http://www.bmwgroup.com/e/0_0_www_bmwgroup_com/home/home.html), then you can click on 'Production worldwide' on the slide menu. To view the facilities and location click on 'Manufacturing facilities'. In addition, find a location in the map by clicking on the 'Place markers' in the map to know what is happening in each of these locations.
 - BMW being a global company, do you think they might have had a bearing on the location of their facilities?
 - Why do you think BMW setup an assembly plant in Chennai rather than a manufacturing plant?
- Blue Star is one of India's largest central air-conditioning and commercial refrigeration companies. It has 4 state-of-the-art manufacturing facilities. To view these facilities, go to the webpage (<http://www.bluestarindia.com>) and click on **Manufacturing** on the top menu bar. You can then click on **Manufacturing Facilities**, and choose the facilities you want to view.

After going through the Web site, answer the following questions:

 - Location issues play a very important role in setting up a manufacturing plant. Can you analyse why the plants have been set up at these places on account of the availability of electricity, water, transport facilities, etc?
 - Do you think that the spread of potential customers plays an important role in selecting the location?

2.10 CASE STUDY

INDCOSERVE—AN INDUSTRIAL COOPERATIVE OF TEA FACTORIES IN THE NILGIRIS

The Tamil Nadu Small Tea Growers Industrial Cooperative Tea Factories Federation Limited, usually called Indcoserve, was established in the year 1965 and is functioning under the administrative control of the Department of Industries and Commerce, Government of Tamil Nadu. With the establishment of the first Industrial Cooperative Tea Factory at Kundah in 1962, the small growers in the Nilgiris region in Tamil Nadu were able to realize a better price for their green leaf. In the four decades of its existence, Indcoserve has enabled its 15 member industrial cooperative tea factories to command 17 per cent of the total production of tea in the Nilgiris District by offering its expertise in technical, managerial, and commercial competence.

The 15 factories are functioning in the Nilgiris district with an aggregate installed capacity to produce over 27.3 million kg of tea per annum. Nearly 20,000 small growers supply green leaf to these factories that are geographically spread out in the region. Table 2.14 presents some basic details on these factories. The table has data for 17 factories. However, two factories—Kotagiri and Kilkotagiri—have discontinued their operations. At the outset, it is evident from the table that the 15 factories vary considerably on several parameters. For instance, Salisbury has the highest value in all the parameters. Nearly 20,000 small growers supply green leaf to these factories that are geographically spread out in the region.

Indcoserve factories have a serious problem of price realization. Except for the period 1997–1999, the price realized is much lower. The cost of production has gone up to 300 per cent of the

1989–1990 levels. This disparity between price and cost trends is highly undesirable and may threaten the viability of Indcoserve. Therefore, the management would like to assess the performance of these factories in order to make certain choices with respect to location of the factories and their future plans.

Data on the Performance of the 15 Factories

Data is available from all the factories for the period April 2004 to March 2006 (24 months). The data pertain to input material (green leaf) in terms of the quantity procured and the cost, production of tea in the factories, various elements that make up the production cost, data pertaining to cost of production excluding the green leaf cost, and price fetched by the tea in the market.

Green Leaf Purchased

The total green leaf procured during 2004–2005 and 2005–2006 were respectively 55.302 million kg and 48.027 million kg. The combined installed capacity for production of tea in all the 15 factories is 27.3 million kg. As per the existing norm, 4 kg of green leaf is required to produce 1 kg of tea. Therefore, the amount of green leaf procured is much lower than the productive capacity of the plants. The data pertaining to the quantity of green leaf (GL) purchased is available in Table 2.15.

The quantity procured across factories fluctuates wildly in every period. During all the 29 months, Salisbury procured the highest quantity of leaves (at an average of 0.824 million kg per month

TABLE 2.14 BASIC DATA ON THE 15 FACTORIES OF INDCOSERVE

Number	Factory Name	Installed Capacity (in Million Kg)	Number of People Employed	Acreage Covered
1	Kundah	2.81	65	2294
2	Kotagiri		Operations discontinued	
3	Karumbalam	1.00	34	1441
4	Kilkotagiri		Operations discontinued	
5	Mercunad	1.56	77	2268
6	Mahalinga	1.40	57	2641
7	Manjoor	2.81	87	2124
8	Ithalar	2.81	71	2362
9	Pandalur	2.01	98	2934
10	Kaikatty	1.25	57	2343
11	Kattabettu	1.56	45	1705
12	Salisbury	2.81	108	2931

(Continued)

TABLE 2.14 (Continued)

Number	Factory Name	Installed Capacity (in Million Kg)	Number of People Employed	Acreage Covered
13	Frontier	2.01	72	2872
14	Kinnakorai	1.25	30	1303
15	Bikkatty	1.56	49	1151
16	Ebbanad	1.09	18	1685
17	Bitherkad	1.40	52	1794

as opposed to the overall average of 0.309 million kg per month). For nine out of 15 factories, the maximum-to-minimum ratio was in excess of 300%). Such marked variations between Salisbury and other factories cannot be explained in any manner other than the acceptance that there were significantly varying procurement practices among the 15 factories. All the 15 factories exhibited significant price differences for the green leaf that they procured. These differences existed in each factory throughout the 29-month period of observation. Salisbury not only paid higher rates, but also procured the maximum quantity during the observed period.

Production of Tea

The data on production-related details are available in Table 2.16. Based on the data available, it was found that Salisbury was the highest producer in 28 out of 29 months of data that was analysed at an average production of 0.207 million kg per month (as opposed to an overall average of 0.08 million kg per month). Ebbanadu (0.027 million kg per month), Karumbalam (0.032 million kg per month) and Kinnakorai (0.035 million kg per month) are the lowest producers of tea during the observed period.

The level of technology used in tea manufacturing continues to be low in Indcoserve factories. Therefore, a useful measure for comparison is the labour productivity. Labour productivity is measured using “the number of kg of tea produced per person employed in the factory”. The relevant data is available in Table 2.16. It is evident from the table that productivity differences across the factories are very significant and that Salisbury scores higher than all others in the group. Salisbury employs the maximum number of peoples among the 15 factories.

All factories seem to have very similar, albeit high cost of production (excluding the cost of GL) close to the average of ₹18.68 per kg (Table 2.17). The only exceptions to this are Salisbury (₹12.93 per kg), Pandalur (₹14.97 per kg) and Frontier (₹15.27 per kg). A comparison of the maximum and minimum values observed during the 29 months of observation also shows that these factories had better cost control (evidenced by a narrower range).

Components of Cost of Production

The overall scenario points to significant differences among the factories with respect to the cost of production. In order to gain some more understanding of this aspect, data pertaining to one calendar year (2005–2006) on the various components of the cost of production is available in Table 2.17.

Packing material, and excise duty are very similar in all the factories. Perhaps it does not make sense to look at these aspects for cost cutting. Salisbury shows an interesting trend in the cost components of tea production. While it has the highest input cost, it has the lowest cost of other factors and overall has the lowest cost per kg of tea.

Average Sale Price Fetched

Tea is primarily sold through auctions in India. In the case of Indcoserve factories, they have three options to sell the tea; through their own auction centre (Teaserve), through other auction centres (such as Coonoor, Coimbatore, and Cochin in South India), and direct sales. However, a vast majority of the Indcoserve produce is sold only through Teaserve. The data pertaining to the price fetched through auctions (Teaserve) have been tabulated in Table 2.18. As in the case of other items of data, we observe fluctuations in the price fetched by Tea from the 15 factories. Salisbury continues to be a good performer in this dimension also. Salisbury tea fetches on an average a much higher price compared to other factories.

QUESTIONS FOR DISCUSSION

1. Develop a set of measures for assessing the performance of Indcoserve factories. Which are the best performing factories among them?
2. Advise Indcoserve on the location choices of the factories. Do you recommend the closure of any of these factories?
3. Based on your recommendations for location of the factories also list out the changes in operations that Indcoserve needs to do for the revised plan.

TABLE 2.15 GREEN LEAF PURCHASED IN THE 15 FACTORIES DURING APRIL 2004–MARCH 2006

Particulars	Kundah	Karumbalam	Mercunad	Mahalinga	Manjoor	Ithalar	Pandalur	Kaikkatt	Katrabettu	Salisbury	Frontier	Kinnakorai	Bikkatty	Ebbanad	Bitherkad
Green leaf purchased (in lakh kg)															
Average	4.06	1.16	2.96	2.84	2.71	3.87	4.88	2.09	2.03	8.24	3.61	1.31	2.13	1.32	2.75
Maximum	6.97	2.37	4.67	4.95	5.69	5.85	6.67	3.74	3.25	10.37	6.16	2.44	3.76	2.87	4.09
Minimum	2.04	0.32	1.56	1.37	1.58	1.79	3.57	1.05	1.13	5.67	2.47	0.70	1.02	0.30	1.69
Max/Min (%)	341%	733%	300%	361%	361%	326%	187%	355%	289%	183%	250%	347%	369%	958%	242%
Max/Avg (%)	172%	204%	158%	174%	210%	151%	137%	179%	161%	126%	171%	186%	176%	219%	148%
Green leaf rate paid (₹/kg)															
Average	6.72	6.87	6.64	6.69	6.71	6.72	6.81	6.74	6.79	7.29	6.42	6.55	6.65	6.50	6.70
Maximum	9.17	9.17	9.17	9.17	9.17	9.17	8.83	9.17	8.48	9.88	8.58	9.00	9.17	8.65	8.58
Minimum	5.00	4.92	4.92	5.00	5.00	5.28	5.08	4.93	5.15	5.61	4.85	4.95	4.68	5.27	5.67
(Max–Min)	4.17	4.25	4.25	4.17	4.17	3.88	3.75	4.23	3.33	4.27	3.73	4.05	4.48	3.38	2.92
Acreage covered	2294	1441	2268	2641	2124	2362	2934	2343	1705	2931	2872	1303	1151	1685	1794
Average kg of green leaf procured per acre	176.79	80.78	130.69	107.71	127.59	163.74	166.26	89.25	118.87	281.27	125.64	100.61	185.12	78.04	153.52

TABLE 2.16 TEA (PRODUCTION) IN THE 15 FACTORIES DURING APRIL 2004–MARCH 2006

Particulars	Kundah	Karumbalam	Mercunad	Mahalinga	Manjoor	Ithalar	Pandalur	Kaikatty	Kattabettu	Salisbury	Frontier	Kinnakorai	Bikkatty	Ebbanad	Bitherkad
Tea produced (in lakh kg)															
Average	1.07	0.32	0.81	0.76	0.73	1.01	1.19	0.56	0.54	2.07	0.90	0.35	0.66	0.27	0.68
Maximum	1.58	0.56	1.07	1.08	1.24	1.35	1.47	0.77	0.72	2.40	1.25	0.53	0.96	0.37	0.91
Minimum	1.02	0.24	0.78	0.79	0.87	0.81	1.15	0.58	0.56	1.79	1.04	0.34	0.64	0.32	0.63
Max/Min (%)	155%	231%	137%	137%	143%	167%	127%	134%	128%	134%	121%	155%	151%	118%	145%
Max/Avg (%)	147%	178%	133%	142%	169%	133%	123%	139%	134%	116%	139%	150%	145%	140%	133%
Labour productivity of the 15 factories															
Total employed	65	34	77	57	87	71	98	57	45	108	72	30	49	18	52
Avg. production per employee (kg)	1651	929	1046	1340	843	1427	1216	981	1194	1916	1250	1172	1351	1482	1314
Capacity utilization of the 15 factories															
Production of tea (2005–2006)	1.10	0.43	0.76	0.72	0.80	1.00	1.28	0.48	0.57	0.90	0.90	0.33	0.66	0.20	0.70
Capacity utilization	39%	43%	49%	51%	28%	36%	64%	38%	37%	32%	45%	26%	42%	18%	50%

*Installed capacity and production of tea are in million-kilograms per year.

TABLE 2.17 PRODUCTION COST EXCLUDING GREEN LEAF COST IN THE 15 FACTORIES DURING APRIL 2005–MARCH 2006

Particulars	Kundah	Karumbalam	Mercunad	Mahalinga	Manjoor	Ithalar	Pandalur	Kaikatty	Kattabettu	Salisbury	Frontier	Kinnakorai	Bikkatty	Ebbanad	Bitherkad
Cost of production (2005–2006) (₹ per kg)															
Total RM expenses	18.20	18.56	17.65	17.28	17.76	18.50	23.70	20.18	23.85	29.20	20.70	18.15	16.84	18.56	21.05
Variable costs															
Fuel charges	6.06	5.78	6.79	6.89	7.69	6.50	6.80	6.61	7.73	5.66	6.51	7.88	6.39	7.83	7.14
Other variable costs															
Wages	1.82	2.36	2.87	2.67	2.23	2.29	1.87	2.59	2.29	1.46	2.04	2.48	2.03	1.76	1.56
Packing material	0.89	1.02	0.90	0.73	1.01	0.84	0.78	1.10	0.93	0.87	0.84	0.85	0.79	0.87	0.93
Transport	0.42	0.10	0.29	0.33	0.18	0.21	0.21	0.27	0.12	0.08	0.09	0.35	0.08	0.32	0.07
Total other variable costs	3.13	3.48	4.06	3.73	3.42	3.34	2.86	3.96	3.34	2.41	2.97	3.68	2.90	2.95	2.56
Total variable costs	9.19	9.26	10.85	10.62	11.11	9.84	9.66	10.57	11.07	8.07	9.48	11.56	9.29	10.78	9.70
Overheads															
Factory overheads	2.37	1.56	2.04	2.57	2.11	2.25	2.24	3.71	2.90	2.23	1.99	2.47	2.27	7.97	3.79
General overheads	2.94	2.36	3.50	2.04	2.26	2.61	2.14	4.82	2.68	1.14	3.00	4.87	2.64	5.77	2.82
Total overheads	5.31	3.92	5.54	4.61	4.37	4.86	4.38	8.53	5.58	3.37	4.99	7.34	4.91	13.74	6.61
Other expenses															
Excise duty	0.31	0.30	0.30	0.30	0.30	0.32	0.33	0.31	0.30	0.30	0.31	0.30	0.31	0.30	0.31
Total	33.01	32.04	34.34	32.81	33.54	33.52	38.07	39.59	40.80	40.94	35.48	37.35	31.35	43.38	37.67
Cost excluding G.L.	15.75	15.38	18.04	16.52	16.90	15.99	15.83	21.35	18.36	13.14	16.43	19.75	15.02	26.52	18.46

* Installed capacity and production of tea are in million-kilograms per year.

TABLE 2.18 AVERAGE SALE PRICE IN THE 15 FACTORIES DURING APRIL 2004–MARCH 2006

Particulars	Kundah	Karumbalam	Mercunad	Mahalinga	Manjoor	Ithalar	Pandalur	Kaikkattay	Kattabettu	Salisbury	Frontier	Kinnakorai	Bikkattay	Ebbanad	Bitherkad
Sale average fetched (₹/kg)															
2004–05															
Average	41.18	42.45	38.12	38.89	39.88	40.11	45.22	39.53	42.63	46.68	41.47	38.56	40.38	34.76	44.15
Maximum	45.42	46.32	41.47	44.01	44.48	45.14	51.14	43.28	54.84	53.98	46.91	43.46	44.94	40.56	50.70
Minimum	33.51	38.60	33.18	32.66	34.13	34.32	40.07	34.13	33.49	40.90	36.41	32.16	34.93	29.40	38.29
(Max–Min)	11.91	7.72	8.29	11.35	10.35	10.82	11.07	9.15	21.35	13.08	10.50	11.30	10.01	11.16	12.41
2005–06															
Average	33.76	33.80	31.68	32.16	32.92	33.93	39.20	33.74	39.39	41.93	34.21	23.77	32.92	32.47	38.34
Maximum	48.11	41.73	45.90	44.61	49.60	47.18	47.95	45.18	47.99	48.79	44.36	43.02	45.58	43.13	51.20
Minimum	24.97	26.30	22.72	24.38	23.26	26.86	35.30	25.45	30.86	38.14	26.45	0.00	24.41	23.23	31.48
(Max–Min)	23.14	15.43	23.18	20.23	26.34	20.32	12.65	19.73	17.13	10.65	17.91	43.02	21.17	19.90	19.72

2.11 MULTIPLE-CHOICE QUESTIONS

- 2-1. What are impacts of globalization on location decisions of organizations?
A. Economic reforms, impact on location choice for organizations, multi-location (globalized) operations
B. Expanding markets, manpower recruitments, expanding financial market
C. Quality business management, location planning, Govt. laws and regulations
D. Worldwide competition, Quality of output, capitalist countries investment
- 2-2. What are the factors affecting location decisions?
A. Only the government regulatory and policy issues
B. Market related issues, cost-related issues, regulatory and policy issues, culture, climate, quality of life
C. Behavioral pattern and standard of living
D. All the above
- 2-3. The decision taken by an organization to set up their sites based on the geographical aspects of the location is termed as:
A. Location decision
B. Geographical decision
C. Infrastructural decision
D. None of the above
- 2-4. What are the two aspects that could affect the location planning problem significantly?
A. Govt. planning and rules, cost of communication
B. Availability of good transportation infrastructure, usage of internet and IT infrastructure
C. Technical know-how availability, manufacturing philosophy
D. Suppliers of raw material, customers availability
- 2-5. What pertain to the choice of appropriate geographical sites for locating the various manufacturing and service facilities of an organization?
A. Location decisions
B. Production decisions
C. Manpower decisions
D. Financial decisions
- 2-6. What are the effectiveness of regulatory issues?
A. TNGST **B.** GST, ST
C. IT **D.** VAT, FDI
- 2-7. The three steps required for location planning:
A. Identify the set of factors that could influence the location decision, establish the relative importance of these measures for the location decision, develop a methodology to assess the impact of the factors
B. Identify the set of factors that could influence the production, establish the relative importance of these measures for the production decision, develop a methodology to assess the value
C. Manpower identity, geographical identity, raw material identity
D. Competitors identity, warehouse identity, selling point identity
- 2-8. In the case of multiple plants and multiple demand points, the location problem can be solved using which of the below standard?
A. Land model
B. Transportation model
C. Labour market
D. Market survey
- 2-9. Which are the two tools that affect location design?
A. Availability of good transportation infrastructure, use of Internet and IT
B. Availability of raw materials, use of low scale labour
C. Availability of low cost factor, use of low maintenance
D. Availability of good demand, use of prompt supply
- 2-10. Location decision could be one of the two:
A. One facility-multiple candidates, and multiple facilities-multiple candidates
B. Two facilities-multiple candidates, and multiple facilities-one candidate
C. Three facilities-multiple candidates, and multiple facilities-two candidates
D. Four facilities-multiple candidates, and multiple facilities-three candidates

Answer Keys: 2-1. A 2-2. B 2-3. A 2-4. B 2-5. A 2-6. D 2-7. A 2-8. B 2-9. A 2-10. A

3

CHAPTER

PLANT LAYOUT: CONCEPT AND TYPES

3.1	Determinants of Process Characteristics in Operations
3.1.1	Volume
3.1.2	Variety
3.1.3	Flow
3.2	Types of Processes and Operations Systems
3.2.1	Continuous Flow Systems
3.2.2	Intermittent Flow
3.2.3	Jumbled Flow Systems
3.3	The Process–Product Matrix
3.4	Layout Planning
3.5	Types of Layouts
3.5.1	Process Layout
3.5.2	Product Layout
3.5.3	Group Technology Layout
3.5.4	Fixed Position Layout
3.6	Summary
3.7	Formula Review
3.8	Review Questions
3.9	Problems
3.10	Net-Wise Exercises
3.11	Mini Project
3.12	Case Study
3.13	Multiple-choice Questions

PROCESS DESIGN AT MILLTEC MACHINERY

Milltec Machinery Private Limited is a Bangalore-based manufacturer of equipment for rice and flour mills. During the short period for which they have been in operation, they achieved significant growth on account of the product range they have been able to bring to this sector of the industry. They manufacture a range of machines under four broad product categories: recleaning, hulling, polishing, and grading. Within each category, they have three or four sub-product categories and a range of sizes and operating characteristics, resulting in over 200 different products. Machine-tool manufacturing typically involves a certain amount of customization.

Designing appropriate manufacturing processes, selection of the right machine tools and other manufacturing resources is critical to improved performance and operational control at Milltec. Milltec employs the latest technology in manufacturing. It believes in producing machinery of international standards at affordable prices. Laser cutting is used in all sheet-metal operations to ensure precise assembly. Milltec uses the CNC press brake for uniform bending of sheet-metal parts. CNC machining centres are utilized to produce machined components. Milltec uses MiG welding stations for the welding operations.

Once the type and number of machines required for the manufacturing process is identified, the next important step is to design an appropriate layout of these resources on the shop floor. At Milltec, manufacturing is organized on a functional basis. The fabrication shop is spread over 750 m². The assembly and paint shops are spread over 2,000 m² and the machine shop over 500 m². Milltec has a dedicated engineering department to cater to the erection and commissioning of projects. It also has an R&D department.

Over the years, Milltec has realized the importance of addressing manufacturing process design and layout issues. Through a process of continuous improvement, Milltec employees and management have realized that several benefits arise out of designing an appropriate layout and workplace organization. These benefits include better production planning and control, less work-in-process inventory, reduced cost of production, better employee morale, and greater visibility of problems at the workplace.

Source: Company sources, <http://www.milltecmachinery.com>

NOTES

NOTES

Designing of processes is an important and early step in operations management. The designing of manufacturing processes essentially consists of certain choices we make with respect to the flow of parts in a manufacturing system. For instance, we need to decide on the types of machines to use, the number of machines of each type, and the manner of their placement on a shop floor. As a prerequisite to this, a detailed analysis of how each component will be manufactured is required. This would mean determining the number of steps involved in manufacturing, the machines used, and the time spent in each of these steps. It is more commonly referred to as process planning. Process planning is technical and is mainly a production engineering activity. We do not focus on this aspect of process design in this chapter.

Once the technical elements of process planning are finalized, at a higher level, the **design of processes** indicates how the manufacturing resources are organized in the operations system so that flow patterns are optimized and appropriate operations management tools can be used to effectively control the overall operations. Process design is defined by the nature of the activities that we pursue in an operating system. If the manufacturing system caters to a wide range of products, it demands a certain type of process design, as opposed to one that mass-produces just one or two variations of a product. By choosing an appropriate process, it is possible to streamline the product flow and deploy appropriate operations management practices commensurate to the process design.

The **design of processes** essentially consists of various choices that we make with respect to the flow of parts in a manufacturing system.

3.1 DETERMINANTS OF PROCESS CHARACTERISTICS IN OPERATIONS

Before we understand the various process choices available to an operations manager for configuring a manufacturing system, let us understand the factors that influence the choice of alternative processes. Generally, three important aspects have a significant influence on the process: volume, variety, and flow.

3.1.1 Volume

Volume indicates the average quantity of products produced in a manufacturing system. In reality, organizations differ on this from one end of the spectrum to the other. Turnkey project management organizations such as Larsen and Toubro (L&T) and Bharat Heavy Electricals Limited (BHEL) will typically have a production

Three important aspects have a significant influence on the process: *volume*, *variety*, and *flow*.

volume of just one. At the other extreme, organizations catering to the consumer non-durable and FMCG sectors have high volumes of production necessitating continuous production. Examples include the manufacture of spark plugs, electric bulbs, soaps, standard electrical fittings, and common household utilities. Many chemical process industries also belong to this category. Between these two extremes, we have mid-volume manufacturers who offer several versions of products. There are numerous examples of mid-volume manufacturing in consumer durables, white goods, and several industrial products. Hindustan Motors, for instance, manufactures several types of earth-moving equipment that belong to the mid-volume category.

3.1.2 Variety

Variety refers to the number of alternative products and variants of each product produced in a manufacturing system. Let us consider a watch manufacturer such as Titan Industries Limited. Titan offers more than 40,000 varieties of watches. These varieties stem not only from alternative product lines but also from variants in each product line. Similarly, consider a car manufacturer such as Tata Motors. The Tata Indica is available in several models. Basically, there are the petrol and diesel versions. However, within the petrol and the diesel versions, there are variations such as the GLS, GLX, VX, VXi, AX, and so on. While one version will have power steering, the second will have power steering and power windows, the third may have automatic transmission, and so on. A choice of colours adds another dimension of variety to passenger car manufacturing.

An increase in the variety of product offerings is likely to introduce variety in manufacturing processes. More varieties may mean alternative production resources, materials, skill of workers, and an increase in the number of stages of production. It may call for better operations management practices. Planning and scheduling may become more complex on account of these added choices in the manufacturing system.

One sees a similar phenomenon in service systems too. Assume that a travel agency was earlier engaged only in selling tickets for bus journeys from Delhi to Chandigarh. In the course of time, if the agency added booking of bus tickets to multiple destinations, rail tickets, air tickets for domestic travel, and air tickets for international travel, one can imagine how many alternative choices of processes, skill sets and resources are required by the travel agency. Operations management in the case of additional services is far more involved and complex.

3.1.3 Flow

Irrespective of the nature of activities involved, all manufacturing systems require certain input material with which the processes begin. The material undergoes a conversion process from the raw material stage to the finished goods stage. *Flow* indicates the

NOTES

NOTES

nature and intensity of this phenomenon. Knowledge of flow provides vital clues to the operations manager about production planning and control issues that need to be addressed. It also provides useful information about the complexities of operations management in an operations system.

Volume, variety, and flow are linked to one another. In general, volume and variety will have an inverse relationship. When the volume of production is very high, the firm is likely to be engaged in the manufacture of fewer varieties of products. On the contrary, if the firm caters to a wide range of products and services, then the production volume of each of these variations is likely to be very low. Figure 3.1 identifies typical examples belonging to three categories: (i) high volume–low variety, (ii) mid volume–mid variety, and (iii) high variety–low volume. Furthermore, volume and variety influence the flow patterns in a manufacturing system. Therefore, process design choices available to an operations manager could be understood by analysing alternative flow patterns. Using this approach, we shall understand how volume, variety, and flow influence the process design in an operations system.

3.2 TYPES OF PROCESSES AND OPERATIONS SYSTEMS

Process characteristics are largely determined by the flow of products in a manufacturing system. Therefore, it is useful to first identify the nature of flow in operations and then identify the process choices available to the operations manager for each category of flow. Generically, three types of flows can be identified in manufacturing systems: continuous, intermittent, and jumbled.

Three types of flows exist in manufacturing systems: continuous, intermittent, and jumbled.

3.2.1 Continuous Flow Systems

A continuous flow system is characterized by a streamlined flow of products in the operating system. Typically, in such systems, the conversion process begins with the input of raw material at one end. It progresses through the system in an orderly fashion to

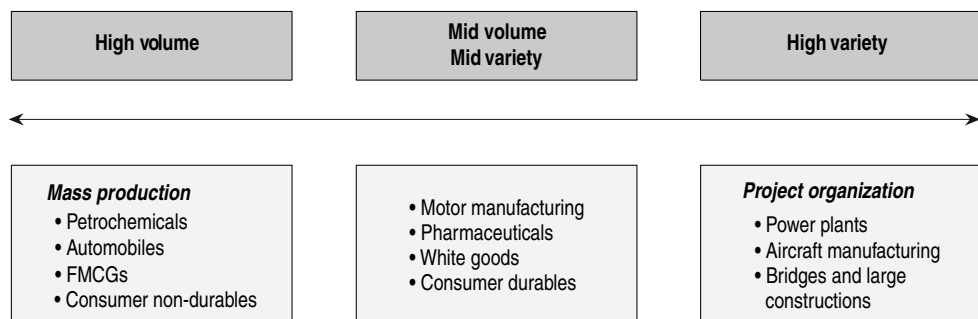


Figure 3.1 Relationship between volume and variety in operations systems

finally become finished goods. The production process is sequential and the required resources are organized in stages. Continuous flow is largely a result of technological constraints and a high volume of production. In the manufacture of petrochemicals, steel, pharmaceuticals, cement, and glass, the reason for continuous flow is technological constraint. For example, in a glass manufacturing unit, once the manufacturing process begins, it cannot be stopped until finished glass components come out of the system.

On the other hand, in a discrete manufacturing industry, the high volume of production of very few varieties results in a streamlined flow. Consider the case of Mico's manufacturing of spark plugs. Due to fewer variations and the large volume of production, the production system is configured in such a manner that an orderly flow is possible. Similarly, in a fast-food outlet offering very few variations for breakfast during the peak hours in a busy locality, the system will have a highly streamlined flow. Despite the similarity of streamlined flow in process industries and mass production systems, there are important differences with respect to operations management practices between the two. Therefore, it is useful to discuss these issues separately for these two categories.

Process Industries

Process industries have peculiar characteristics compared to discrete manufacturing industries. Process industries have a closed manufacturing set-up and finished goods are typically obtained at the end of the manufacturing process. In some cases, it is possible to derive products during the intermediate stages of the process in the form of by-products. An example of a process industry is paper manufacturing. Figure 3.2 illustrates the paper-manufacturing process in simple terms.

An analysis of the paper-manufacturing process will reveal several features of a process industry. As shown in the figure, there are three stages in the paper-manufacturing process: the preparatory stage, pulp making and papermaking. During the preparatory stage,

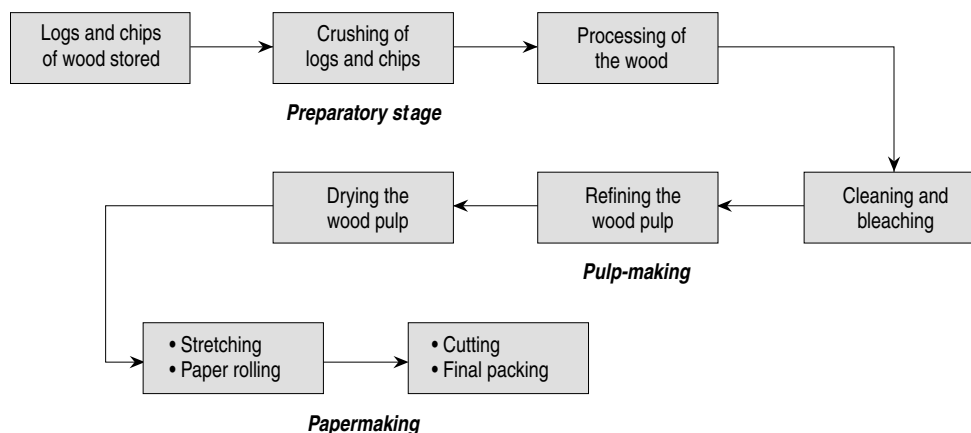


Figure 3.2 The paper-manufacturing process

N O T E S

NOTES

logs of wood and chips of various sizes are first fed into the system. The logs of wood and the chips are broken into smaller and uniform pieces in a crusher. After the pieces are cut into uniform pieces, they are “cooked” in a huge chamber where some chemicals are also added. During this process of cooking, the pieces are digested and the fibres are laid in a uniform pattern. This is done through a chemical process. After this process is complete, it is ready for making wood pulp. The pulp-making process is nothing but further refining of the wood fibre and cleaning and bleaching of the material. Finally, excess moisture is removed to form the pulp. At this stage, the pulp is almost devoid of moisture and is in a near-solid state. Therefore, it is possible to press it into the required thickness and remove further moisture. This is the last stage of the process known as papermaking. The dried and processed paper can be rolled into reels and cut to required lengths and shapes.

The papermaking example illustrates certain distinctive features of process industries:

- ⇒ Once the logs (raw materials) are fed into the system, it is not possible to stop the system until the dried wood pulp reaches the paper-rolling and cutting (final) stage.
- ⇒ Since the process is continuous, there should be a balance of capacity between all the stages in the manufacturing process to maintain an even flow of the material from the raw-material stage to the finished-goods stage. Otherwise, there will be a capacity mismatch, which may result in gross underutilization of resources.
- ⇒ The productivity of the system is directly related to the flow rate (or throughput) of the product.
- ⇒ Typically, process industries require huge capital investments in the manufacturing system as incremental addition at a later stage is not possible. Due to this, high productivity implies lower production costs and vice versa.
- ⇒ An important operations management task in process industries is making continuous process improvements and capacity de-bottlenecking to maximize the flow rate in the system.
- ⇒ As the flow rate directly determines productivity, it also implies that failure of any intermediate stage in the system will have an adverse effect on the cost as the entire system will have to be stopped (see Ideas at Work 3.2 for an illustration of this).

The need to keep the production facilities continuously running in a process industry directly translates into a need to be good in maintenance management. Therefore, process-industry firms invest more time in developing good maintenance practices.

Moreover, the amount of spare-parts inventory that they carry will be a significant portion of the overall inventory in the system. Ideas at Work 3.2 illustrates how disruptions in the production process, though unavoidable, can raise the costs incurred by

VIDEO INSIGHTS 3.1

A continuous or a batch manufacturing system typically used by process industries and mid-volume manufacturing organizations faces a different set of issues when it comes to various operations management practices. In order to understand this, it is valuable to make a factory visit to one such manufacturing system. Manufacturing of sugar and apparels is representative of such industries. To know more about how these are being manufactured, find the video links (Video Insights) in the Instructor Resources or Student Resources under section **Downloadable Resources** (<http://www.pearsoned.co.in/BMahadevan/>) subsections **Bonus Material**.

N O T E S

process-industry firms and increase their losses. Due to the continuous and closed-loop nature of the entire manufacturing process, process industries tend to have very little work-in-process inventory.

Although not evident from the paper-manufacturing example, process industries benefit vastly from the vertical integration of their operations as well as from the addition of several secondary conversion facilities. This is because, at certain intermediate stages of the processing, by-products are released from the manufacturing system. Having the facilities to process them into useful products will lower the overall cost of the production system and increase revenue generation opportunities. The other alternative will be to transport the by-products to another point of demand at an extra cost. One of the best examples of vertical integration can be seen in the development of various facilities by Reliance Industries Limited. Figure 3.3 provides a good illustration of the backward integration concept for process industries.

Mass production

The mass production system in the discrete manufacturing industry is another example of continuous and streamlined flow in the manufacturing system. In a mass production system, the volume of production is very high and the number of variations in the final product is low. Therefore, it is possible to organize the entire process by dedicating the required manufacturing resources for each product variant and arranging the resources one after the other, as per the manufacturing sequence. Such a structure is typically known as a product-line structure. Due to this arrangement, a streamlined flow is possible. Automobile and two-wheeler manufacture, the manufacture of electrical components (such as switches), and that of healthcare products (such as disposable syringes)—all are typical examples of high-volume production. Other examples include the manufacturing of several consumer non-durables.

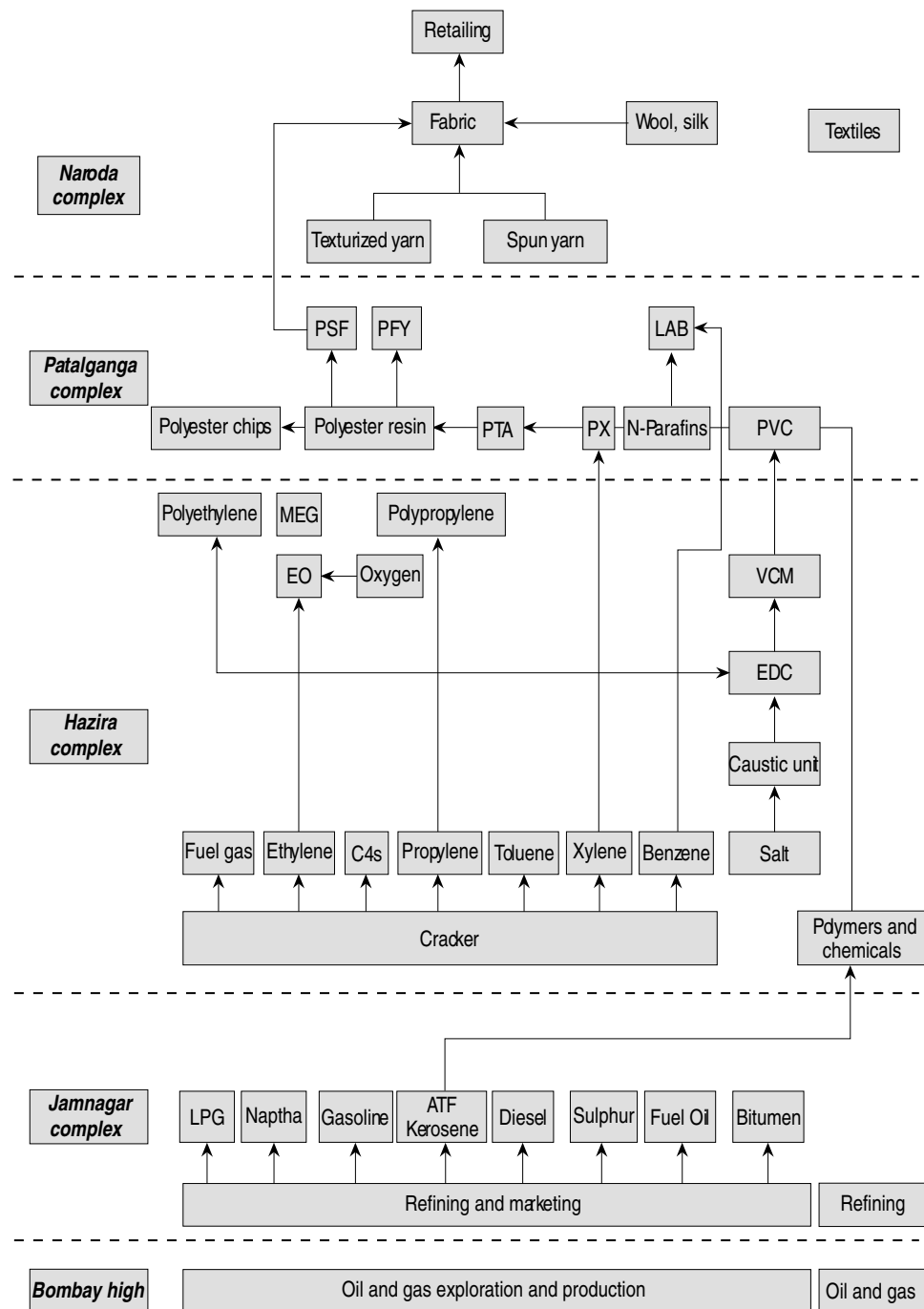


Figure 3.3 Backward integration in process industries: the case of Reliance Industries
 Source: Based on the author's research

Process design for streamlined flow can be visualized at two levels in a mass production system. At the overall level, each product will flow across departments in a streamlined fashion. Furthermore, within each department, there will be an orderly flow of components and materials. In operations management terminology, the manner in which the manufacturing resources

are arranged for mass production is referred to as a *flow shop*. Figure 3.4 depicts the department-level design as well as the typical flow-shop structure within each department.

As in the case of process industries, balancing of capacity across various stages of the manufacturing system is required to maximize throughput and productivity. Moreover, line stoppages due to breakdown and asynchronous production practices will also be detrimental to the working of the system. Therefore, better maintenance management practices and appropriate design of product-based layouts are important operations management practices to be followed in a mass-production system.

3.2.2 Intermittent Flow

All manufacturing firms do not have the benefit of high volume and low variety like a mass production system. As customer preferences increase, the need for creating more variety also increases and firms respond to this requirement by expanding the product line. Moreover, as competition increases, organizations try to retain their market share by introducing new products. An illustration of this is the passenger car segment in the country. The passenger car segment was delicensed in 1993. Until then, there were very few car manufacturers. They also offered fewer variations. However, once the competition increased and customer expectations went up, product variety also went up.

Intermittent flow systems are characterized by mid-volume, mid-variety products/services.

When variety increases, the volume of production for each variation will be less compared to a mass production system. Therefore, dedication of manufacturing resources for each

NOTES

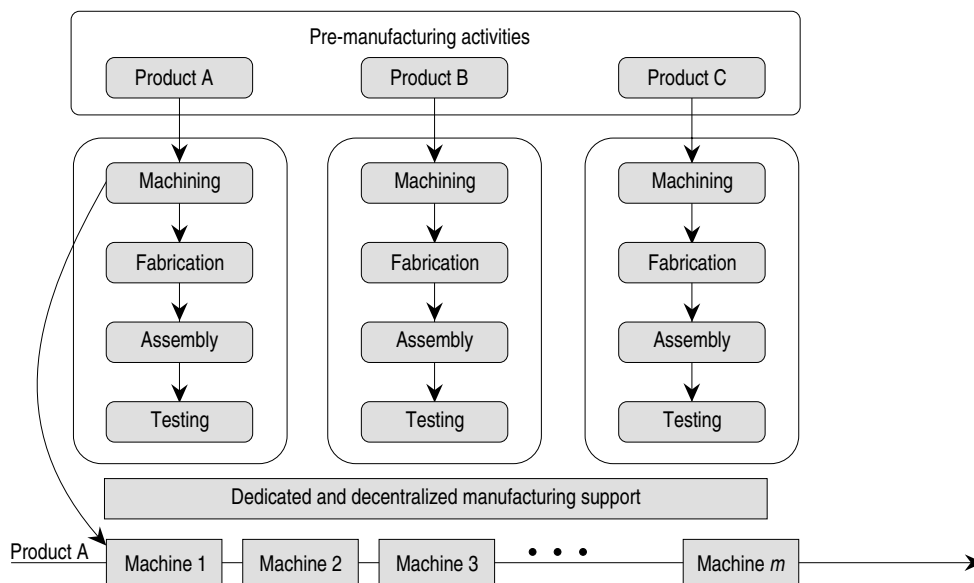


Figure 3.4 Process design and flow structure for a mass production system

NOTES

THE POLYESTER FILAMENT YARN PLANT AT RELIANCE INDUSTRIES LIMITED

Let us take the example of the polyester filament yarn plant of Reliance Industries Limited at Patalganga to understand how disruptions are a costly, though unavoidable, part of process-industry firms.

In simple terms, the manufacture of polyester filament yarn is done in three stages. In the first stage, the pre-polymerization activities include feeding in the required chemicals and catalysts for further processing. The second stage is continuous polymerization. During this stage, the polymer is created in molten form by a continuous polymerization unit and sent for spinning. In the third stage, the molten polymer enters into the spinning machine through packs. Packs are small cylindrical assemblies packed with steel powder that act as filters. When the molten polymer passes through the pack, the impurities contained in it are removed. The polymer finally enters the spinning machine, where it is finally wound on bobbins. There are metering pumps to monitor the flow of the polymer as it comes out of the continuous polymerization unit. There are 64 metering pumps that monitor the flow of polymer into each spinning machine. A single inverter controls all the 64 metering pumps.

Disruption of production for certain changeover requirements is unavoidable. Changeovers are necessitated by the need for changing the packs as packs have a finite life. Since packs act as filters, over a period of time, impurities accumulate in the pack and this increases the pressure. When the pressure reaches a certain threshold, the metering pumps will not be able to function properly. At this stage, the production needs to be disrupted for a pack change. A pack changeover involves the following steps: First, the metering pumps are stopped and the throughput of the continuous polymerization unit is reduced. After the existing packs are removed, the pre-fabricated ready-to-use packs are pre-heated to 280°C–300°C and then inserted. The metering pumps are then started and the continuous polymerization unit's throughput is restored to its original value over a period of time.

There are many losses due to this disruption. The continuous polymerization unit works continuously as stopping it will jeopardize the entire production unit. During the time the system is put on hold, the entire molten polymer produced will be wasted. There will also be losses due to quality until the unit stabilizes. Normally, when there is a throughput fluctuation, the viscosity of the polymer is altered and this results in waste and

downgraded quality of finished products. Similarly, after every disruption, the polymer takes about two and a half hours to change from bright to semi-dull. The output during this entire period is also wasted. Other costs include manpower costs and the opportunity cost of lost production.

The example clearly shows how important is it for process-industry firms to avoid breakdowns, unnecessary stoppages and changeovers. Therefore, investment in good maintenance practices and higher levels of spare-parts inventory are characteristic features of process industries.

Source: Based on the author's research.

variation may not be a feasible option. Manufacturing resources will be shared by a group of products. Since each product may have different processing requirements and sequences of operations, the flow will become complicated. Therefore, the manufacturing process is characterized as mid-volume, mid-variety systems and process design for this type of system must address issues arising out of intermittent flow.

Discrete and process industries have alternative ways by which they can address the flow complexities arising out of a mid-volume, mid-variety scenario. In the case of process industries, one way to minimize this flow complexity is to operate the system in batches. In one batch, one set of variations is manufactured and in the next batch, another set of variations is manufactured. In between these two batches, the necessary set-up and changeover of resources are made to facilitate smooth production and maximize productivity. Such an arrangement results in intermittent flow systems.

Reliance Industries Limited manufactures more than 30 grades of polyethylene in their Hazira plant. These include several variations

NOTES

VIDEO INSIGHTS 3.2

Automobile manufacturing has always been in the forefront of bringing new innovations in manufacturing technology, processes, and design of manufacturing systems. Therefore, one can develop a good understanding of the design of manufacturing systems by closely observing manufacturing of passenger cars. Manufacturing of Honda Amaze at the Tapakura plant in India and Mercedes Benz C class cars in an overseas plant demonstrates these concepts well. To know more about these, find the video links (Video Insights) in the Instructor Resources or Student Resources under section **Downloadable Resources** (<http://www.pearsoned.co.in/BMahadevan/>) subsections **Bonus Material**.

NOTES

used in industrial, medical, and food-grade applications. While dedicated processing facilities are available for some variations on account of their huge volume of production, several other grades need to share the same set of manufacturing facilities for production. There are variations in the manufacturing processes across these products. For example, food-grade applications require certain precautions to be taken. The nature of additives to the process is also different. Therefore, after one variation is manufactured, the plant is stopped and the required changes are made to the process before the next variation is produced. Similarly, in the manufacture of drugs and various formulations in a pharmaceutical industry, the batch-processing method is applied.

In the case of discrete manufacturing systems, the entire manufacturing set-up can be split into units in which similar processing requirements can be combined and manufacturing resources arranged to cater to these requirements. One batch of components may be produced in one of these units and another batch in yet another unit. In each batch, components of different product variations will be combined since they share the same processing requirement.

Figure 3.5 shows an illustration of the process design for intermittent flow in discrete manufacturing systems. As shown in the figure, the manufacturing resources are organized on the basis of similarity of manufacturing operations among several variations. Components of various product variants flow in an intermittent fashion from pre-manufacturing activities up to the final assembly and testing stage, taking alternative routes and timings to complete the process.

By now, it is evident that operations management issues in intermittent flow systems are very different from those of continuous flow systems. The notion of flow balancing is not an issue in the case of intermittent flow systems. However, capacity planning for catering to the overall requirements of the various varieties is important. Capacity estimation is difficult in an intermittent flow system compared to a continuous flow system. The major issue pertaining to the overall working of an intermittent flow system is the issue of

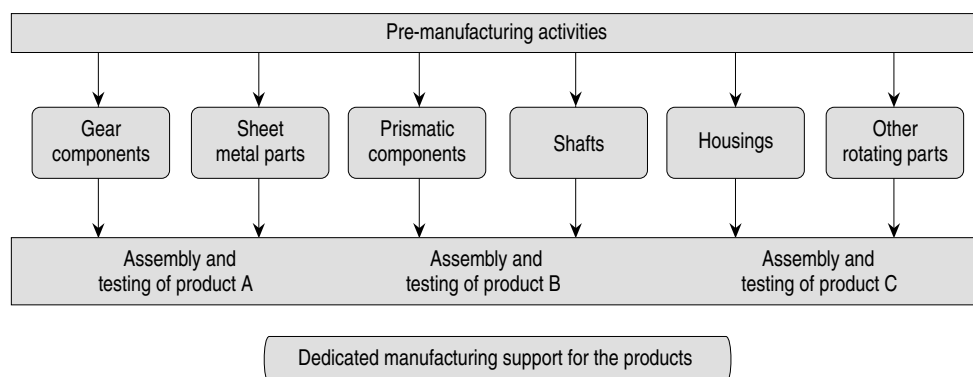


Figure 3.5 Process design for intermittent flow in a discrete manufacturing system

changeover from one variety to another. Production planning and scheduling must critically analyse the impact of alternative changeovers and develop an optimal changeover sequence while planning for production.

Designing an appropriate structure and manufacturing layout is critical to the working of batch manufacturing systems. Inappropriate choices in structure often lead to complicated material and information flows. This results in complex production planning and control problems. The tractability of the products and related information becomes difficult and may call for more paperwork, and control mechanisms to resolve them. Typically, the level of supervision and coordination will increase and organizations may resort to progress chasing and expediting of late jobs. All these will finally result in long lead times in manufacture, poor delivery reliability, excess and unwanted inventory, high overheads, and high overall cost of manufacturing.

Capacity estimation is a complex task in an intermittent flow system.

3.2.3 Jumbled Flow Systems

The third category of flow is jumbled flow, which occurs on account of non-standard and complex flow patterns characteristic of certain manufacturing systems. The flow pattern is non-standard and complex because there are unique process designs for each and every customer order. Moreover, customer orders are typically for one-off items and organizations cannot benefit from any batching and repetitive manufacturing practices, unlike continuous flow and intermittent flow systems. This adds to the complexity of the flow. All these characteristics demand different operations management practices, unlike the previous two categories of flows. In reality, two types of manufacturing systems have jumbled flow. These include project organizations and customized manufacturing systems.

Project organizations

Consider a turnkey project executor such as BHEL or L&T. The projects that they undertake are typically large-scale, involve high levels of customization and have long lead times (in the order of a few years). When the product or service offered is of a very large scale, it introduces problems that are not encountered in normal operating systems. In such a case, multiple entities are involved (architect, engineers, government regulatory bodies, construction workers, local communities and society and so on). It also involves multiple stages of the process (design of buildings, foundations,

Customer orders are typically for one-off items and organizations cannot benefit from any batching and repetitive manufacturing practices, unlike continuous flow and intermittent flow systems.

NOTES

PROCESS REDESIGN FOR IMPROVING FLOW

NOTES

An appropriate choice of processes can have a significant impact on reducing the complexity of managing operations in a manufacturing system. On the contrary, using an inappropriate process design will introduce avoidable complexities arising out of bad process flow leading to poor visibility, excessive efforts in production planning and control, and high investments in the process inventory. Several manufacturing organizations have realized the importance of addressing this aspect.

A case in point is the changes proposed by a manufacturer of earth-moving equipment having a large manufacturing set-up. The manufacturing set-up in the organization consists of a plate shop, fabrication shop, machine shop, heat treatment shop, gear shop, final drive assembly, and main assembly. The product profile and the product range were characterized by large variety and low volume, resembling one end of the spectrum of a mid-volume, mid-variety scenario. A certain amount of customization was inevitable as the product was manufactured, which resulted in more variety at the end product level.

The performance of the gear shop was crucial to improving the delivery, cost, and quality performance of the products manufactured in the company. A variety of spur, helical, and worm gears along with geared components are manufactured in the shop, which caters to final drive and swing machinery, the engine, and the transmission of various kinds of earth-moving equipment.

The components manufactured in the gear shop involve a large variety (over 625 at the time of the study) ranging in size from 50 mm diameter to 1,000 mm diameter. Moreover, many components involved intricate shapes and sizes. The shop had 107 machine tools which were arranged in three bays. A majority of the lathes were placed at the rear of the second and third bays. Similarly, gear-cutting machines were located on the left side of the second bay and grinding machines at the rear of the first bay. Given the complexity of operations involved in gear manufacturing and their large variety, the existing set-up posed serious problems in managing operations.

Based on the principles of group technology, a new process design for the gear shop was arrived at. To begin with, eight groups were formed on the basis of manufacturing similarity. For example, all components belonging to the group “large simple gears” would have the key “hob-shave” operation. Similarly, all components belonging to the “ring gear” group had “shape-grind” as the key operation. The eight groups were acceptable from a process-flow sequence point of view and would greatly reduce inter-bay movement. Based on further analysis, and taking additional constraints into consideration, seven groups were finally formed. With the addition of just four

more machines, the groups were made independent. In order to change over to the new layout, only 38 machine tools required relocation out of the 107 available in the shop.

The proposal also recommended changes in the manufacturing support functions. For instance, in the revised layout, inspection responsibility was decentralized and left to the groups. The required resources and floor space were identified. Major changes were proposed in the organization structure. The new set-up was to have seven heads of cells, who would be responsible to their internal customers. All “vital” manufacturing support functions, such as quality, tooling, and process plans, were made available to each of these heads; hence, accountability was likely to be high.

The major advantage of the proposed design was the drastic reduction in the distance travelled. The distance travelled by the components within the shop would be reduced from 68.5 km to 35.0 km, a reduction of more than 52 per cent. Such a reduction would result in drastic savings in material-handling costs, lead time and WIP inventory. Further improvements in material handling due to closer location of successive machines were also envisaged. Moreover, improved process flow and better traceability and control were the other benefits of the exercise.

Source: Adapted from B. Mahadevan, *The New Manufacturing Architecture* (New Delhi: Tata McGraw Hill, 1999), pp. 98–102.

NOTES

superstructures, electrical systems, heating and air-conditioning systems, fabrication activities, acoustics, and so on). The flow not only becomes jumbled but also complex and managing such a large set of activities requires different methods of planning and control.

Due to a high degree of customization, the degree of uncertainty in all the above cases is likely to be significant. Planning tools that incorporate uncertainty are essential in managing such situations. Second, the number of entities involved in accomplishing the various tasks is numerous. Furthermore, these entities have several complex interactions amongst themselves. The performance standards of one entity and deviations from the specifications at one part of the system have a significant impact on the others.

Virtually no process design is possible in such a situation. However, operations management principles are required to manage situations arising out of the jumbled flow inherent in project organizations. Scheduling of various activities and control of these is an essential requirement and appropriate tools are required. Similarly, handling uncertainty and decision making in the context of uncertainty is also an important requirement.

NOTES

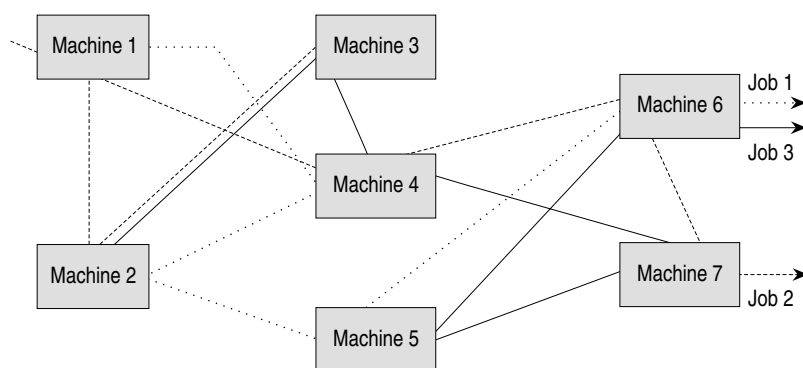


Figure 3.6 Process flow in job shops

Customized manufacturing systems

Another example of jumbled flow is a highly customized manufacturing system such as a tool room operation. While the production volume may not be as low as one unit as in a project organization, it could nevertheless be minimal. On the other hand, the number of customer orders could be very large, resulting in large variety. Since each customer order could potentially demand unique process requirements, the resulting flow in the system becomes highly jumbled. There are several examples of customized manufacturing available in practice. These include PCB fabricators, a variety of sheet metal fabricators, tool room operators and printing and publishing. In operations management, such shops are referred to as *job shops*. In job shops, machines of the same type are grouped together. This is inevitable because no single customer order can justify dedication of resources, on account of its low production volume. Figure 3.6 is an illustration of the process flow in a job shop.

Good scheduling practices could minimize the capacity-related problems in jumbled flow systems.

The operational complexity arising out of such a jumbled flow is high. The most significant issue is one of capacity management. Considerable time is lost due to the repeated set-up of processes. Since each customer order requires different levels of capacity, the estimation of capacity that is already committed and that which is remaining is very hard to make. Good scheduling practices could minimize such capacity-related problems. The other significant issue is managing operations in the system. Due to jumbled flow, criss-crossing of jobs in the system results in poor visibility. Problems are often hidden and a build-up of work in the process inventory takes place. Therefore, good process inventory management practices are essential. Cost accounting and estimation systems are crucial for job shops as there is a constant need to quote for specific customer orders.

Just-in-time (JIT) systems seek to create a streamlined flow of components and products in a manufacturing system.

Table 3.1 summarizes the salient features of various process flow alternatives and the key operations management issues relevant to the above discussions. It is evident from the table that continuous flow systems are relatively easier to plan and control. Capacity planning and scheduling are far simpler than the other two categories. At the other extreme, jumbled flow systems pose complex challenges to operations management. Therefore, in the past 20 years, manufacturing organizations the world over are making concerted efforts to streamline their flow and look for alternative methods of converting job shops. Similarly, firms are also investigating methods of implementing **just-in-time (JIT) systems** as they seek to create a streamlined flow of components and products in a manufacturing system.

3.3 THE PROCESS–PRODUCT MATRIX

So far we have discussed various aspects of the alternative process choices that an operations manager possesses and the relationship between process choices and flow patterns in a manufacturing system. The culmination of all these ideas is the process–product matrix. A process–product matrix depicts all these relationships in a compact form, as shown in Figure 3.7.

One dimension of the matrix represents product characteristics and the other process characteristics. Product characteristics essentially indicate the degree of customization and the volume of production. The left end of the spectrum represents low volume products which are less standardized, one of a kind, usually engineered to order. The right end of the spectrum represents high volume products that are highly standardized, mass produced, and positioned as commodity products. Volume and variety are negatively correlated

NOTES

TABLE 3.1 SALIENT FEATURES OF VARIOUS FLOW PATTERNS IN MANUFACTURING SYSTEMS

Flow Characteristics	Continuous	Intermittent	Jumbled
Product characteristics	High volume, very low variety	Mid volume, mid variety	Very high variety, low volume
Examples of production systems	Process industry, mass production systems in discrete manufacturing	Batch production in process and discrete manufacturing	Project organizations, tool rooms, general purpose fabricators
Issues of importance	Flow balancing, maintenance, capacity utilization and debottlenecking backward integration	Manufacturing system and layout design, changeover management, capacity planning and estimation	Capacity estimation, scheduling, production control, cost estimation
Operations management tools and techniques	Line balancing, maintenance management, process optimization planning and product layout design, flow shop scheduling, pull type scheduling, single piece flow design	Forecasting, capacity planning and estimation, optimized production optimization sequencing, group technology layout design, materials management	Project management and scheduling, capacity planning and scheduling, functional layout design, job order costing, work-in-process management

NOTES

capturing the value trade-off. On the other hand, process characteristics indicate the complexity and divergence in the process. The top end of the spectrum represents processes which are highly complex, with process flows / routing that is unique to a customer order. The bottom-end of the spectrum represents processes that are less complex, highly streamlined process flow/routing across customer orders. When organizations produce standardized products in high volumes, the production process flow will be highly streamlined. Similarly, when organizations produce one-of-a-kind customized products, the production process will be jumbled. As shown in the figure, these two situations represent the two extremes. As seen earlier, each of these situations involves unique operational management issue that need to be addressed. This is addressed by selecting an appropriate operations system in the diagonal of the product-process matrix. If the selected choice is below the diagonal, then the organization will incur out-of-pocket costs that will unnecessarily increase the cost-structure of producing the product and lower its profitability. If the selected configuration is above the diagonal, then organization has to watch out for the opportunity-cost of lost price-premiums thereby lowering its profitability. Clearly both of these are non-desirable to the organization. Hence, organizations have to apply their mind in determining the appropriate process choice that is aligned with the characteristics of the product; and that is observed in the diagonal of the above matrix.

3.4 LAYOUT PLANNING

Imagine yourself visiting a multi-specialty hospital for a master health check-up. What if the radiology department was located in the second floor, the general physicians were sitting in the ground

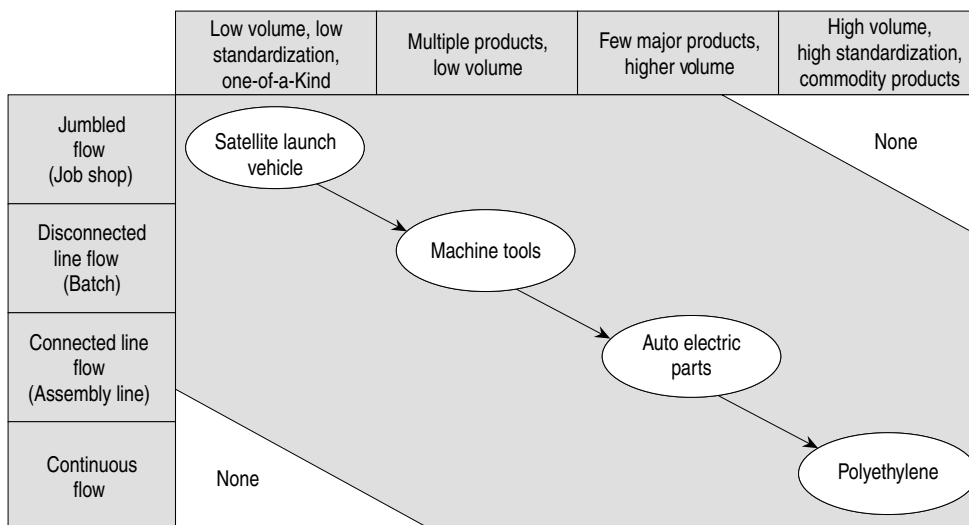


Figure 3.7 The process-product matrix

Source: Adapted from Hayes, R. H. and S. C. Wheelwright, "Link Manufacturing Process and Product Life Cycles," *Harvard Business Review* 57(1), (1979): 133–140. Reproduced with permission.

floor at the rear side, ECG and treadmill test facilities were located in the fourth floor and so on. Finally, imagine that you need to walk out of the main complex and go 50 metres away to another building to have your breakfast after you give your fasting food samples and return to the main complex to continue with the process. Such instances are not uncommon. How many times have you felt that you were made to walk too much in a hospital when you went for a health check-up or when you went to a financial institution asking for a loan sanction or to a government office to pay some utilities bill or make some inquiries? What is the core problem in these examples? How many times you have noticed that components travel excessively in a factory before they complete their processing requirements? In simple terms, these examples suggest that, with a better arrangement of resources, it is possible to improve operations and provide better service to the customers.

Layout planning in manufacturing and service organizations involves the physical arrangement of various resources available in the system to improve the performance of the operating system, thereby providing better customer service. Typically, in the case of a manufacturing organization, there may be over 200 machine tools of various kinds that need to be located in a machine shop. Similarly, in the case of a service organization such as a hospital or a hotel, there are various resources to be physically located. We can identify the best possible locations for various resources in an organization through a good layout planning exercise. Layout planning provides a set of tools and techniques that help an operations manager decide where to locate resources and also to assess the impact of the alternative choices that he/she may have for locating the resources.

Layout planning in manufacturing and service organizations involves the physical arrangement of various resources available in the system to improve the performance of the operating system, thereby providing better customer service.

A good layout design will ensure that a vast majority of jobs in a manufacturing system will have to travel shorter distances before completing their processing requirements. Similarly, in the case of service organizations, customers will have to walk shorter distances and spend less time in the system to complete their service requirements. This ensures that the costs and the lead time of the processes come down. On the other hand, a bad layout design will result in longer distances being covered before a process is completed. This creates several problems in organizations and key performance measures suffer. The most significant and visible effect is the time taken to complete the process. Longer distances would mean that more time is needed to complete the process. In the case of manufacturing organizations, more material-handling leads to higher material-handling costs. Eventually, in both service and manufacturing systems, this leads to poor quality.

NOTES

NOTES

The relationship between “volume– variety–flow” provides crucial inputs to the layout problem.

We have already discussed alternative configurations of manufacturing systems on the basis of volume–variety–flow. Revisiting this issue will provide the required inputs to the layout problem.

3.5 TYPES OF LAYOUTS

Over the years, operations management researchers and practitioners have evolved four major types of layouts. These include product layout, process layout, group technology layout and fixed position layout. The salient aspects of these layouts are described here.

As the flow becomes more cumbersome, the type of layout may significantly influence the ability of the operations manager to effectively plan and control operations on the shop floor.

3.5.1 Process Layout

A process or a functional layout is an arrangement of resources on the basis of the process characteristics of the resources. Consider a machine shop consisting of lathes (L), grinders (G), milling machines (M), and drilling machines (D). A sample process layout for this shop is shown in Figure 3.8. In the example, components belonging to Product A first visit a lathe, then a drilling machine, a milling machine, and finally a grinding machine. The sequence of visits is a function of the process plan and is available in a route card. The major implication of this design is that when the number of components manufactured is large, there will be enormous criss-crossing in the shop, as components need to visit machines in multiple combinations. This increases material handling and poses challenges for production control.

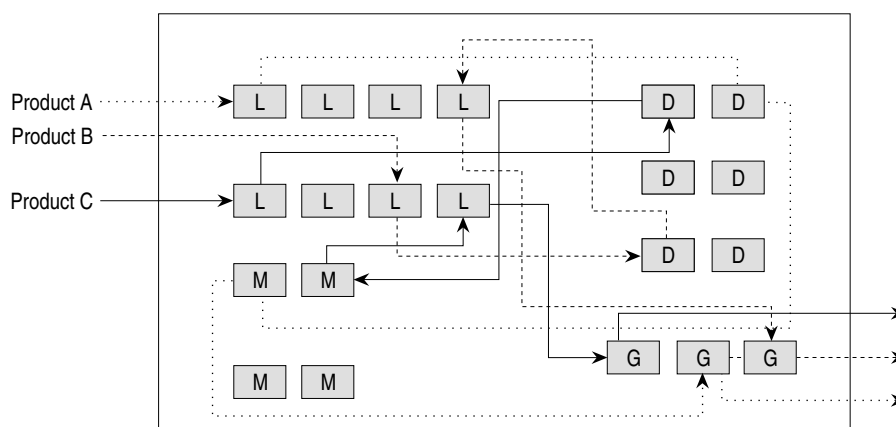


Figure 3.8 Example of a process layout

Each department in a **process layout** is typically organized into functional groups. Thus, all the lathes will be organized into a lathe department. Similarly, there will be a drilling department, a milling department, and so on. In the fabrication area a similar arrangement would be a welding department, a fitting department, a shearing department, and so on. All manufacturing support areas are also arranged on a functional basis. Examples include a maintenance department, quality control department, procurement, stores, and production planning.

A process layout is an arrangement of resources on the basis of the process characteristics of the resources available.

3.5.2 Product Layout

A product layout is an alternative design for the arrangement of resources. In this case, the order in which the resources are placed exactly follows the process sequence dictated by a product. For the above routing information the product layout will be as shown in Figure 3.9. In a product layout, the required set of resources for every product is dedicated. Due to this, the resources are arranged in the order of the machining requirements resulting in smooth component flow in the shop. Since each product will have its own set of resources, material handling is simpler and it is possible to invest in fixed-path material-handling systems to speed up material transfer between successive workstations. The production control issues are much simpler in a product type of layout compared to the process layout.

In a product layout, the resources are placed to follow exactly the process sequence dictated by a product.

Very often the final assembly in several manufacturing plants follows a product layout. The assembly workstations are designed in such a manner that at each workstation a part of the job is completed. The feeder stations are linked to the assembly workstations to ensure material availability. As the product moves through the assembly, the process is completed. Testing, final inspection,

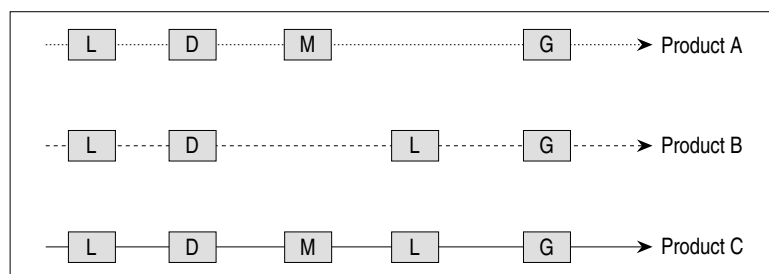


Figure 3.9 Example of a product layout

NOTES

NOTES

and even packing could be part of this layout, so that at the end of the line it is ready for dispatch to the market. If you visit the Chinchwad plant of Tata Motors at Pune, you will notice that there are nearly 180 workstations involved in the final assembly of the Tata Indica. As the chassis moves through these stations, the car is assembled in a progressive fashion, and at the end the car is ready for dispatch. In product layouts, the design of the layout is far more important as it results in a certain degree of operational control. Once the design of individual workstations is properly made, many problems are solved.

The notion of product and process layouts applies not only to manufacturing settings but also in a service setting. A classic example is the banking system. For several years, the layout in the banking operation was process based with the resources arranged on the basis of their function they do. Thus, we have savings bank clerks, current account clerks, cashiers, tellers and so on. However, in recent times, the layout is more product or customer based. In this arrangement, manpower and resources are organized on the basis of customer segments that need to be catered to. Figure 3.10 shows alternative layout choices between Bank A and Bank B.

In the case of Bank A, customers arrive at the waiting area and a member of the customer relationship staff attends to all the requirements of the customer. In contrast, in the case of Bank B, the staff is organized on the basis of functions. Depending on the nature of work involved, customers will visit different sections in the bank and get their services done.

Both process layouts and product layouts have certain advantages and disadvantages. Therefore, one needs to choose an

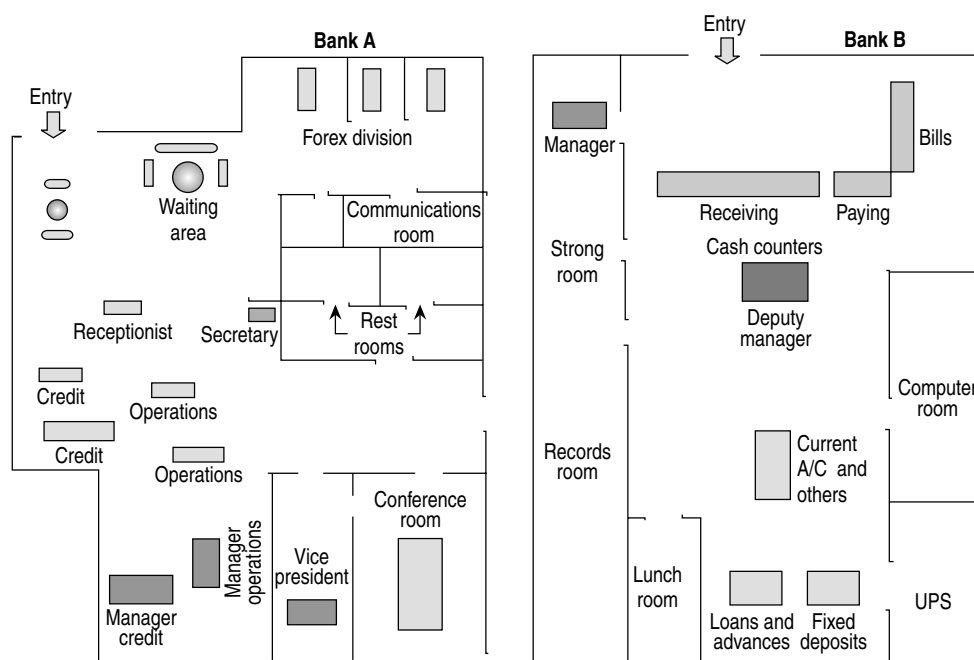


Figure 3.10 Product and process layouts: an example from banking

TABLE 3.2 ADVANTAGES AND DISADVANTAGES OF PRODUCT AND PROCESS LAYOUT

	Process Layout	Product Layout
	Sharing of specialized and costly equipments	Standardized product/ process routing
Advantages	More flexibility	Operational control is simpler
	Less vulnerable to breakdowns	High output rate is possible
	Large inventory build-up	Low tolerance for breakdowns
Disadvantages	Operational control difficult	Duplication of equipment leading to high cost
	Excess material handling	Lower flexibility due to dedication of resources

NOTES

appropriate layout depending on the specific circumstances. Table 3.2 lists some of the salient advantages and disadvantages of both these types of layouts. A product layout simplifies production planning and control problems. It also enables smooth production and high production rate. However, since a high level of dedication of resources is required it may increase the cost of operation and may be an unfeasible option in most cases. Only in the case of high-volume manufacturing can an organization justify the dedication of several resources.

The advantages of process layout include greater flexibility and better utilization of resources. However these are more than offset by several drawbacks. Organizations using process layouts often experience problems, which include the following:

- ⇒ Enormous criss-crossing of the jobs, resulting in poor visibility of flow of jobs. This makes progress chasing a difficult task.
- ⇒ Heavy load on material handling, leading to bottleneck situations in the material-handling system.
- ⇒ Difficulty in production planning, scheduling, and control of the jobs.
- ⇒ Very high throughput times, which in turn increases the investments in stocks, stock holding costs, damages, and material obsolescence.
- ⇒ Poor delegation of work and poor accountability.
- ⇒ Low motivation and low job satisfaction for the workforce.

Moreover, process layouts promote functional silos in organizations, which are considered to be undesirable today.

On account of these issues, another type of layout that combines the benefits of both process and product layouts is being widely adopted by organizations today.

NOTES

3.5.3 Group Technology Layout

There has been an increasing trend towards more variety, thereby pushing volumes down. The industrial fans and blowers division of ABB was manufacturing nearly 725 models. Titan Industries increased the number of its watch models from about 850 in 1993 to over 1,200 in 1996, an average of more than 100 new models every year. (Currently, the number of models that they offer has increased by orders of magnitude and is in excess of 40,000). Reliance Industries Limited, India's largest private sector company operating in petrochemicals, was planning to introduce several new grades of high-flow high-density polyethylene (HDPE) and low-density polyethylene (LDPE) with a view to entering five new markets of polyethylene in the year 1995–1996. With this, the number of variations that they would handle at the polyethylene plant was likely to be over 40. It is estimated that more than 70 per cent of manufacturing industries will have a mid-volume, mid-variety scenario. Therefore appropriate layout planning methods for mid-volume, mid-variety manufacturing is required.

Group technology (GT) is a philosophy that seeks to exploit the commonality in manufacturing and uses this as the basis for grouping components and resources. The implementation of GT is often known as *cellular manufacturing*.

Group technology (GT) layout provides an alternative method for configuring resources in organizations that have mid-volume, mid-variety product portfolios.

GT is a philosophy that seeks to exploit commonality in manufacturing and uses this as the basis for grouping components and resources. The implementation of GT is often known as *cellular manufacturing*. In cellular manufacturing, the available components are grouped into part families. An appropriate measure for manufacturing similarity is used to identify part families. Corresponding to each part family, machine groups are identified and the layout is formed accordingly. An example of a group technology layout for the process layout is shown in Figure 3.11.

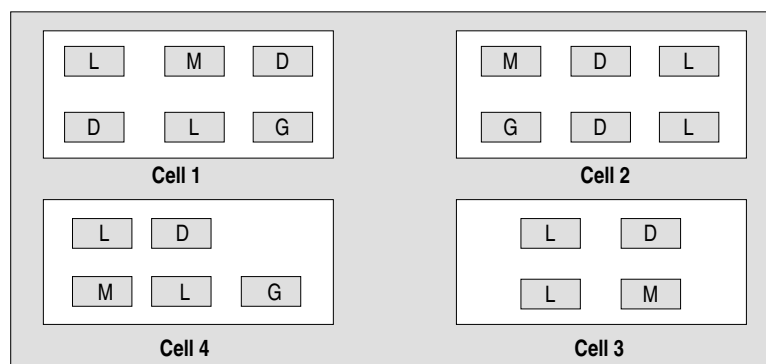


Figure 3.11 Example of a group technology layout

The benefits of GT are many. Once the part families and the machine groups are identified, the layout ensures that each cell has only a certain number of components to be processed. In essence, it is akin to breaking a monolithic structure into smaller, more manageable, and independent units of production. Because of this, production planning and control become much simpler. The components seldom travel outside their respective cells for processing. Therefore, material handling becomes easier and traceability improves. Moreover, employees will be able to relate better to their workplace and make concerted improvements in the process. The new structure will also help to implement several other operations management practices such as small group improvement activities, kaizen, and just-in-time manufacturing practices.

Fixed position layouts are typically employed in large-project type organizations.

3.5.4 Fixed Position Layout

There are several situations in which the product manufactured is very bulky, difficult to move and is often made in quantities of one. In such situations, the layout design ought to be very different. One cannot employ any of the layouts described so far. Typical examples include building very large machine tools and equipment, ship and aircraft building, and preventive and breakdown maintenance of such large systems. Since such equipment is very large and bulky, it dictates several choices with respect to the layout. The orientation of the equipment will dictate the placement of specific resources required for the process. Layout planning in such cases is often a question of a good workplace organization. By proper workplace organization it is possible to exercise greater control on the process and remove unwanted processes and delays from the system.

For reasons mentioned above, fixed position layouts are typically employed in large-project type organizations. Some examples include the nuclear engineering division of BHEL at Tiruchirapalli, the final assembly hanger of the Advanced Light Helicopter manufactured at Hindustan Aeronautics Limited's (HAL) helicopter division at Bangalore, and the Indian Space Research Organisation (ISRO) launch site at Sriharikota. The emphasis in fixed-positioning layouts is not so much on optimum positioning of resources required for the process. On the other hand, it should enable managers gain better control of material flow (see Ideas at Work 3.4 for an example).

NOTES

WORKPLACE ORGANIZATION AT THERMAX

NOTES

Thermax is a manufacturer of large and complex systems for heat exchange and environmental control applications. Their vapour absorption heat pump (chiller) shop is an example of a fixed position layout. Chillers are huge pieces of equipment weighing several tonnes. They are manufactured using a fixed position layout. The manufacturing process begins by first locating the lower shell of the chiller on the shop floor and progressively adding the components. The upper shell is finally mounted and the equipment is tested before packing and dispatch.

Previously, they were assembling as many as 40 chillers. The only consideration to start assembly was space availability to place the lower shell. As a result, these chillers were at various stages of production and were cluttering the shop. Production planning and control was very complex. It was difficult to understand why inventory was piling up and lead times were increasing. There were several reasons for chillers not being assembled. These included non-availability of material, labour, and other resources.

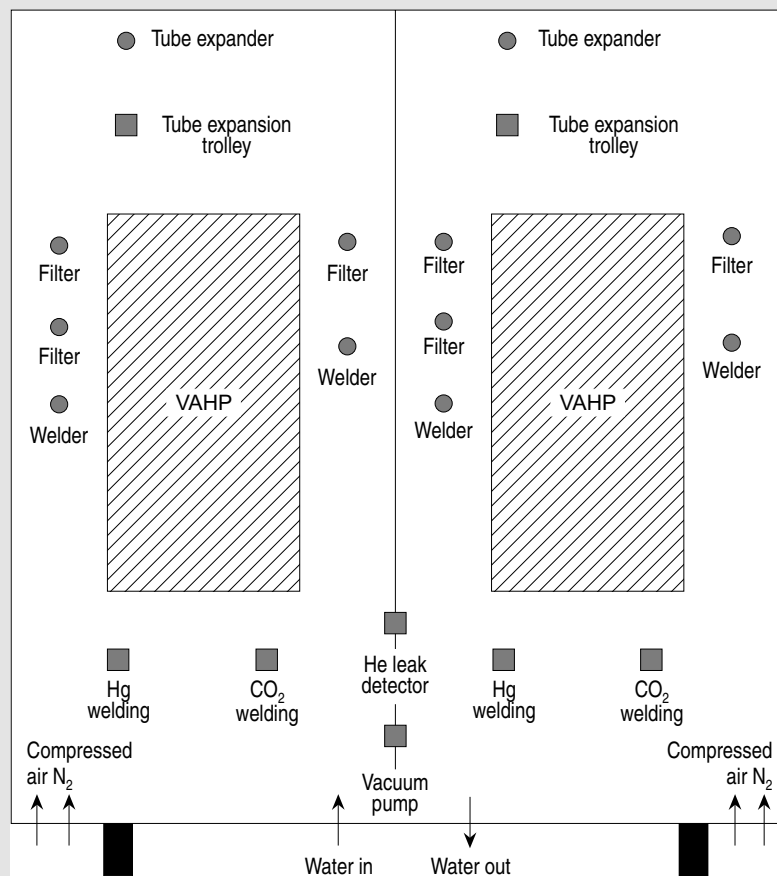


Figure 3.12 Details regarding the orientation of the workers and other requirements

An alternative layout was designed to address this problem. In the revised scheme of things, the workplace was clearly demarcated with finer details of how the lower shell of the chiller should be placed and how other resources were to be deployed for the production process. Even the minutest details regarding the orientation of the workers and other requirements such as air, carbon dioxide, and welding machines were worked out (see Figure 3.12). Eighteen such provisions were made. This ensured that not more than 18 chillers could be assembled in the shop at a time. Figure 3.12 has the revised layout.

The exercise had several benefits. There was a reduction in the work in progress from 40 to a maximum of 18. According to the manager of the shop, in the revised layout, problems became more visible. From his cabin, he could identify which of the cells were working and which were not. If he walked past the cell, he could know the exact nature of the problem in the non-working cells. The new layout made production control a visible and easy task.

Source: S. K. Wale, “Improving Manufacturing Effectiveness Through Implementation of Stall Built System in Chiller Shop at Thermax Ltd, Pune” (Bangalore: Unpublished report submitted at IIM, 1996)

NOTES

3.6 SUMMARY

- *Volume, variety and flow* exert significant influence on process design in organizations.
- Process industries and mass production systems generally have a streamlined flow of products.
- Mid-volume and mid-variety manufacturing systems have intermittent flow. Capacity estimation is difficult in such systems compared to continuous flow systems.
- Project organizations and customized manufacturing systems have jumbled flow. Capacity estimation and scheduling of jobs are quite difficult. Therefore, operations management complexity is high in jumbled flow systems.
- A *process-product matrix* depicts the relationship between process flow characteristics and volume of production in a manufacturing organization.
- Product layouts are useful for high volume-low variety situations. At the other extreme, fixed positions and project layouts are useful for high-variety situations.
- Product layout and process layouts are used in the discrete manufacturing industry. They have several advantages and disadvantages.
- Mid-volume and mid-variety manufacturing systems can benefit from a group technology (GT) layout.
- Several software packages are available for designing process layouts. CORELAP, ALDEP, COFAD, and CRAFT are among the popular packages in this group.
- Product layout design seeks to identify the minimum number of resources required to meet a targeted production rate and the tasks to be assigned to each of these resources, using a technique called line balancing.
- GT layouts are designed with the objective of subdividing a universe of machines and components into sub-groups such that each sub-group consists of part families and machine groups.
- A number of methods are available for designing GT layouts. These include production flow analysis (PFA), rank order clustering (ROC), mathematical programming and clustering methods.
- New manufacturing technologies such as *flexible manufacturing systems (FMS)* have the potential to simplify flow complexities in mid-volume and mid-variety manufacturing organizations due to increased flexibility.
- No. of stages, variety and flow characteristics determine the complexity of operations management. By a careful design of the process, some of the complexities can be minimized.

3.7 FORMULA REVIEW

- The total cost of the process layout, $TC = \sum_{i=1}^n \sum_{j=1}^n F_{ij} D_{ij} C_{ij}$
- Actual (desired) cycle time =
$$\frac{\text{Available time}}{\text{Actual(desired) production}}$$
- Minimum number of work stations required =
$$\frac{\text{Sum of all task times}}{\text{Cycletime}}$$
- Average resource utilization =
$$\frac{\text{Sum of all task times}}{\text{Number of workstations} \times \text{cycle time}}$$

3.8 REVIEW QUESTIONS

1. What are the key determinants of process characteristics in operations? Give specific examples to show how they influence process characteristics.
2. Is there any relationship between volume, variety and flow with respect to process design?
3. How is the continuous flow system different from the intermittent flow system? What are the key implications for operations managers?
4. Give three examples for each of the following and identify two important issues of interest to an operations manager:
 - (a) A mass production system
 - (b) A continuous flow system
 - (c) A flow shop
 - (d) An intermittent flow system
 - (e) Job shop
 - (f) A process industry
 - (g) A batch production system
 - (h) Project organization
5. As an operations manager, which areas would you focus on if you were to manage a jumbled flow process?
6. How does the choice of the layout affect the operational performance of an organization?
7. What is the relationship between the volume, variety and flow characteristics of an operating firm and its layout?
8. What are the different types of layouts? How should an organization decide on which layout to choose?
9. Is a group technology layout any different from a product layout? Give reasons for your answer.

10. Identify an appropriate layout for each of the following situations. Justify your choice in a sentence or two:
 - (a) A manufacturer of garments for Van Heusen
 - (b) A multi-cuisine restaurant in a posh residential area in Mumbai
 - (c) The overhaul of helicopters
 - (d) A fabricator of custom-made PCBs for a large number of electronic applications
 - (e) An eye hospital
 - (f) A motor manufacturer manufacturing 4 product groups for worldwide markets
 - (g) A manufacturer of large turbines for power sector applications
11. Suppose you are given three alternative designs for the layout of a shop floor in a manufacturing organization. How will you decide which of the three is the most appropriate?
12. Critically compare quantitative and qualitative approaches to designing process layouts.
13. How does a construction method of layout design differ from the improvement method? Which method is preferable and why?
14. What do you understand from the phrase “cellular manufacturing system”?
15. What are the reasons for the growing application of group technology concept for layout design?
16. What do you understand by the term FMS? What are the characteristic features of an FMS?
17. How does an FMS help in improving the process design in the case of mid-volume, mid-variety manufacturing?
18. What are the various types of automated material-handling devices available for use in a computer integrated manufacturing system?
19. What do you understand by the term complexity of operations management? Explain.

3.9 PROBLEMS

1. A mass producer of an industrial component currently has a daily production rate of 1,200 pieces. The company works for 16 hours a day. The production process currently employs five stations. There are 12 tasks involved in the manufacturing process and together they need 190 seconds.
 - (a) What is the actual cycle time for the component?
 - (b) Compute the average utilization of resources employed to manufacture the component.
 - (c) If the company wants to work with 85% utilization, by how much they should increase the daily production to achieve this?
 - (d) Suppose, there is a 10% increase in the demand. Do they need to make any changes in the layout?
2. A company procures a few components from the market and assembles them into a standard product and supplies it to its customers. There is enough demand for the product. If the company employs four stations for the assembly, it has 92% average utilization of the resources. The sum of the task time for the process is 180 seconds.
 - (a) What is the cycle time for the process?
 - (b) What is their daily production rate if they work in an 8-hour shift?
 - (c) If they add one more station in the assembly process what would be the new cycle time if the utilization drops to 79%?
 - (d) By how much can they increase the production on account of the reduced cycle time?
3. Following data are available in the design of a product-based layout in which a certain number of work stations are deployed. The sum of all task times is 200 seconds. The cycle time is 50 seconds and the average resource utilization is 80%. If one more work station is added, the utilization drops to 74.07% and the new cycle time is 45 seconds.
 - (a) How many work stations are used in the product layout design?
 - (b) What is the daily production rate if a day consists of 450 minutes?
 - (c) What is the maximum production possible in the current design?
4. A manufacturer of spark plugs for the mass market would like to design the final assembly shop and requires certain data for the process. The factory works for two shifts and the total available time is 15 hours per day.
 - (a) There are five tasks involved in the final assembly and the task times (in seconds) are 3.5, 4.5, 2.0, 3.0 and 4.0. What are the minimum and the maximum outputs possible from the factory?
 - (b) The daily production needs to be 4,000 plugs. What should be the cycle time of the operations at the assembly shop?
 - (c) If the firm wants to use three workstations, what will be the best output possible? By how much will the output increase if it adds one more workstation?
5. A fast-food restaurant that operates in Connaught Place, New Delhi, specializes in offering food during peak hours. The restaurant operates only for two hours in the morning and promises faster turnaround for customer orders. The restaurant needs to design the workstations at the front end of the kitchen to assemble all customer orders. The back end of the kitchen provides the required food items. A study of the tasks, their dependencies and times taken

on an average to assemble reveals the information shown in Table 3.3:

- If, during a typical peak hour operation, the restaurant is able to take 144 orders, what is the maximum permissible cycle time for the assembly workstations?
- If the restaurant desires to set a target of the above cycle time, what is the minimum number of workstations they should have at the assembly?
- Design the assembly with the minimum number of workstations. Can it serve 180 customers during the peak hour? If not, arrive at a feasible design.
- For the final design that you have made, compute the average utilization of workstations.
- The restaurant management is aware that the bottleneck is in terms of their capacity to serve. If they could increase the capacity, they would earn additional revenue. The marginal revenue from an order will be ₹100 and the marginal daily cost of adding a workstation is ₹2,700. What should the management do?

TABLE 3.3 DEPENDENCIES AND TIME TAKEN TO ASSEMBLE/SERVE FOOD

Task ID	Description	Task Time (Seconds)	Predecessor
A	Set up the serve tray	20	None
B	Arrange plates, cups, tissues	20	A
C	Place the main item in the plate	35	B
D	Fill in the side dishes	35	B
E	Place a filled in water cup	20	D
F	Arrange cutlery items in the plate	20	C, E
G	Verify the order	15	F
H	Prepare a bill for the order	35	G

3.10 NET-WISE EXERCISES

- Caterpillar is the largest maker of construction and mining equipment. To know more about the products offered by them, visit [http://india.cat.com/cda/layout?m = 81920&x = 7](http://india.cat.com/cda/layout?m=81920&x=7) and click on **Products**. Navigate through their product lines and offerings.

Haldia Petrochemicals is the one of the largest petrochemicals complexes in India. Go to [http:// www.haldiapetrochemicals.com/index.jsp](http://www.haldiapetrochemicals.com/index.jsp) and click on **Products** on the left side of the screen. Visit various sub-links in this section. Also click on the link **About Us**. There is a subsection on manufacturing processes. Visit other parts of the Web site.

Take a look at the photos of the plant loaded onto the main page.

Based on your understanding of the two organizations, compare and contrast the product flow and process characteristics.

- Identify three distinct features of Caterpillar and relate them to the design of the manufacturing process for the organization.
- Why would you classify Haldia Petrochemicals as a process industry? Identify three characteristics that you find in Haldia Petrochemicals that are peculiar to a process industry.

3.11 MINI PROJECT

- There are several manufacturing and service organizations in every part of the country. Select out of these three organizations that belong to any of the following categories: a process industry; a mass production system; a service set-up; a mid-volume, mid-variety manufacturer.

Visit the selected manufacturing and service organizations, make a formal plant tour of the organization and collect data on various aspects of operations management. After you have recorded all the key observations based on the visit, prepare a list of items on which you would like to collect additional information and obtain them. Also

interview the plant manager, the process planning team, and the line supervisor. Note down their key observations about the plant's functioning. Based on this work, prepare a final report that contains the following information:

- The process flow diagram of the plant
- The nature of the production system
- The key challenges for operations management
- Status of current operations management practices in the organization
- Specific areas in which improvement potential exist for the organization

2. There are alternative approaches to layout design. The use of software packages for this purpose has increased manifold in recent years. Prepare a project report to answer the following question pertaining to the use of software for layout design:

- (a) Scan the Internet and prepare a list of available software solutions for layout design. Briefly explain the salient features of the software.
- (b) Develop a classification scheme to list the software that you have identified. Can you recommend a specific set of application domains of the software on the basis of this classification scheme?
- (c) Select one software package from each of your classification schemes and critically compare them. Identify the specific advantages and disadvantages of each package.

3. There are several companies that have recently made significant changes in their layout design. Prominent in the list

are automobile and auto-components manufacturing firms. Identify one such company in your locality and perform the following studies and submit a report of your findings:

- (a) What were the major drivers of the layout change decision in the organization?
- (b) Enumerate the salient aspects of the alternative layout design process (methodology used to arrive at the new design, tools and techniques used, scope of the project, organizational aspects of driving the change process etc.)
- (c) Understand the implementation aspects of the layout design process (methodology of implementation, time taken, chronology of events leading to completion of the process etc.)
- (d) What were the main benefits that accrued to the organization as a result of this layout design change?
- (e) How did the layout change affect various people in the system including the production workers, supervisors, managers, customers, and suppliers?

3.12 CASE STUDY

Aerospace India Limited

Aerospace India Limited (AIL) has been serving the needs of the Indian defence sector for the past 20 years. The company has four divisions that manufacture various components for the aerospace sector. Unit 1 was established in 1970 to manufacture components for helicopters. The entire machining of about 2,230 components required for various versions of helicopters is done at the machine shop at Unit 1.

The Machine Shop at Ail

The machine shop at AIL is responsible for manufacturing all components required for the helicopters. It consists of more than 100 machine tools, which includes some high-precision machine tools such as an NC jig boring and a CNC gear grinding, besides many conventional lathes, turrets, milling machines, and drilling machines. The machine shop occupies an area of about 6,440 m².

The facilities in the machine shop are arranged in a functional fashion. These include a gear unit comprising facilities for gear manufacturing ranging from 50 mm to 600 mm diameter, a lathe unit for rotating parts, a milling unit, a turret unit, a drilling unit, a line spar milling, a grinding unit, CNC machines, a metrology unit, a tool room, an inspection unit, and production tool stores.

Manufacturing Requirements for New Product

With the progressively dwindling production of the existing range of helicopters, AIL was entrusted with the task of designing and manufacturing components in Unit 1 for a newer helicopter design by its customers. Irrespective of the many variations in the new product, most of the subsystems will have a commonality of about 85 per cent from the perspective of manufacturing components at the machine shop. Therefore, the manufacturing requirements of the new product are not much different from that of the existing ones. The

existing machining capacity at the machine shop is adequate to handle this additional load. The process plans and time estimates have already been made for the components to be machined at the machine shop in Unit 1.

Typically, the production of machined components for the new product involves the following production planning and control activities: The scheduling department issues route cards and job tickets based on the production target. The progress department collects these and draws the required material from the stores. Based on the scheduling prepared by the machine shop manager, the components are loaded on to different work centres. Components are inspected after the completion of every operation. Stage inspection is mandatory in production. After inspection, the components are loaded to the next operation. Component loading, operation completion, and the inspection cycle are repeated until all operations are completed. All components are subjected to 100 per cent inspection. However, if the components fall under the life-critical category, the inspection records are maintained for lifetime to trace the history in cases of accidents and failures pertaining to the component.

In the existing set-up, the machine shop is headed by a senior manager who is assisted by one manager and one deputy manager. Currently, even though the area of responsibility is clearly defined, interdependency is high due to the functional layout. It warrants frequent meetings between managers. One of the problems with the existing organization structure is the total lack of product focus. Many components that require turning or milling queue up in the respective sections. The supervisors and the section in-charge normally go by convenience and rush requirements. The senior manager is the only person responsible for the products and their sub-assemblies. The rest are more concerned with turning, milling, etc.

Implications of Maintaining the Status Quo

Since the manufacturing requirements for the new product are very similar to that for the existing one, one option to organize manufacture of new components is to maintain the status quo. This appears to be a soft option. It impinges on no one's working style and requirements. However, one of the senior managers was sceptical of this arrangement. "You see, with the existing set-up we will not be able to manage two different projects. It basically deprives us of product focus." This is so for several reasons, some of which are as follows:

1. A functional layout increases the cycle time. Since machine utilization is stressed over the completion of product families in a functional layout, components that do not deserve immediate priority are loaded to avoid idling, which causes bottlenecks for priority items. This leads to an increase in the WIP inventory.
2. Since the facilities are located far apart, the material-handling distance is large. Also, due to the number of operations, the frequency of material handling is high.
3. Given the complexity of scheduling for more than 2,000 components, functional layout often has to rely on the follow-up group, whose job is to "chase" the components.
4. In the absence of product focus, despite more than 80 per cent completion of loaded components, quite a few assemblies cannot be started, as the required items will be still under progress.
5. Both supervisors and operators develop a tendency to reduce set-up time by combining batches whenever possible. This results in increased batch quantity, even though it was not meant to be.
6. Manufacturing errors are not detected at an early stage. Due to an imbalance in the load, the inspection department often authorizes supervisors to carry out inspection, who do a superficial job.
7. Operators make frequent trips to production tool stores, the standard tools crib, the blueprint booth and the inspection department for various reasons. This increases non-value-added activities and lead time considerably.
8. With an increase in the number of batches and WIP, component traceability is lost. It requires some time to locate the batch required, as identification would be very difficult in the absence of route cards which will be either with the operator or with the inspection department.
9. Due to a focus on performance at the work-centre and section level, it appears that the responsibility of completing the component entirely as per the given schedule, cost, and quality is nobody's except that of the top management.
10. Due to lack of knowledge about other machining operations, workers generally do not appreciate another work area. This also creates a mindset against multi-functional skills.
11. Team spirit is absent, since the functional set-up *minimizes* chances of group interaction.

Design of the Manufacturing Process

Efforts were made to investigate alternate ways of organizing manufacturing for production of machined components for the new product. Initially, a meeting was convened to discuss the various possibilities. Mr Prakash, one of the managers from Unit 1, made a presentation based on his analysis of the problem and proposed two new alternatives. He suggested two alternative approaches to redesigning the machine shop for manufacturing machined components required for the new product. In the first approach, known as the assembly-module approach (AM), the entire machining is organized on the basis of modules of sub-assemblies. In the other approach, known as the machining similarity-based approach (MS), the grouping of components is based on the machining requirements.

Assembly-Module-Based Redesign

The manufacture of components required for the new product can be classified under 14 modules. Each module forms a sub-assembly. From the bill of materials, the components required for each sub-assembly have been segregated. There are about 731 components (Table 3.4 has the details). For example, 133 components pertain to the flight control system. Similarly, 25 components are required for the tail gearbox sub-assembly. For each component, the following information was collected:

TABLE 3.4 SUB-ASSEMBLIES AND MACHINING REQUIREMENTS FOR THE NEW PRODUCT

Sub-assembly	Number of Components
Flight control system (FCS)	133
Avionics and electrical system	25
Hydraulic system	12
Structures	59
Landing gear	29
Main rotor	76
Tail rotor	21
Power plant	74
Main gear box (MGB)	145
Tail gear box (TGB)	25
Intermediate gear box (IGB)	26
Auxiliary gear box (AGB)	25
Tail drive shaft (TDS)	24
Others	57
Gear components from existing project	41
Total	772

Note: *41 components from the existing project have been added to avoid omission of these during capacity calculations. These components require special purpose machines to be assigned for producing the new product.

part number, department number, work-centre number, set-up time, unit processing time, and the quantity required. Using this information, the work-centre-wise capacity requirements have been calculated for each module.

Using a minimum utilization value of 30 per cent, it was found that only seven modules could justify dedicating capacity. Hence, the other modules were combined with these to finally form seven modules. Some modules have more than one cell assigned to them. In addition, a separate cell designated as a common cell was formed with equipment that is both costly and has low utilization at the individual module level. Finally, this has resulted in 13 cells. Table 3.5 has the details. In the final

design, wherever the required machines were not available, recommendations have been made to procure them.

Machining Similarity-Based Redesign

In the second approach, the computation of the capacity requirements work-centre-wise, and the number of machines required for each cell are all done in the same way as described above. But there is a fundamental difference in the approach to forming cells. The 772 parts are grouped together based on their similarity in machining requirements. This would mean, for example, that a group of components requiring turning operations might have been grouped together. Similarly, a

TABLE 3.5 DETAILS OF MODULES AND CELLS AS PER THE AM APPROACH FOR REDESIGN

Sl. No.	Module	Cell Number	Number of Machines	Number of Componets	Number of Exceptions [#]	Number of Machines to be Procured
1	Flight control system (FCS)					
1(a)	FCS—I	1	7	93	3	1
1(b)	FCS—II	2	6	40	18	2
2	Structures					
2(a)	Avionics and electrical system, structures	3	5	84	20	1
2(b)	Hydraulic system, landing gear	4	4	41	18	1
3	Rotor group					
3(a)	Main and tail rotor	5	7	65	40	—
3(b)	Main and tail rotor	6	4	32	13	—
4	Power plant accessories	7	8	74	37	1
5	Main gear box (MGB)					
5(a)	MGB—I	8	5	78	32	1
5(b)	MGB—II	9	3	67	11	2
6	Gears (TGB, IGB, AGB)	10	15	76	44	1
7	Non-MGB					
7(a)	Others, gear components for existing project	11	7	86	30	3
7(b)	Others, tail drive shaft	12	3	36	9	—
8	Common cell	13	8	*	*	
	Total		82	772	275	13

Note: *Eight machines have been separately located as Cell 13. These machines were highly valued one-off machines and it was not possible to justify either locating them in a particular cell or procuring additional machines. About 79 components from four modules undergo processing in the cell.

[#]Out of the 275 exceptions, only 92 required movement across the modules. In other cases, it involved inter-cellular movement between cells belonging to the same module.

group of components going through a series of milling and finishing operations from many sub-assemblies might have been put together.

Production flow analysis was initially employed to identify cells. Forty cells were initially formed. However, the demand on many machines in these cells was far less. Moreover, the number of cells was too many. Using the same machine utilization cut-off value of 30 per cent, the 40 cells were merged and finally 8 cells were formed. Table 3.6 presents the summary details for this design. Finally, Mr Prakash presented a comparison of the two approaches based on several criteria (see Table 3.7 for details).

The task before the executives at AIL was to identify an appropriate configuration for the machine shop at Unit 1.

Questions for Discussion

1. Should AIL continue with the current manufacturing set-up? Articulate three reasons in support of your argument.
2. Classify the two alternatives provided by Mr Prakash using your knowledge of process and technology choices for manufacturing organizations.
3. What are the pros and cons of the two alternative designs that Mr Prakash has proposed for changing the manufacturing shop configuration?

TABLE 3.6 DETAILS OF CELLS AS PER THE MS APPROACH FOR REDESIGN

Cell number	Number of Machines	Number of Components	Number of Exceptions	Number of Machines to be Procured
1	10	112	29	2
2	6	80	4	1
3	7	127	6	2
4	5	47	—	1
5	10	104	25	1
6	11	122	16	1
7	7	89	39	1
8	22	91	9	—
Total	76	772	128	9

TABLE 3.7 A COMPARISON OF THE TWO APPROACHES

Si. No.	Criterion	AM	MS	Remarks
1.	Implementation cost related			
	(a) Number of cells	13	13	More supervisors in AM
	(b) Number of machines to be procured	8	9	More investment in AM
2.	Cell-size variation			
	(a) Components	32–93	47–127	Larger variations in MS
	(b) Machines	3–15	5–22	Larger variations in MS
	(c) Utilization of machines			
	Maximum	84	92	Capacity more balanced in MS
	Minimum	30	42	
	Average	67	84	Better utilization in MS
3.	No. of exceptions	275	128	More exceptions in AM
4.	Product focus	Well focused	Blurred	Simplified production planning in AM
5.	Production control	Satisfactory	Better	More exceptions in AM makes control difficult
6.	Quality control			
	(a) Requirement of inspection equipment and tools	Less	More	
	(b) Technical documents	Unique	Mixed	AM is better because of product focus
	(c) Ability to direct improvement efforts	More	Less	Each group in AM can identify distinctly to an end product/ sub-assembly
7.	Liaison with design	Simple	Complex	Lack of sub-assembly focus in MS makes it difficult
8.	Document control	Easy	Difficult	Standards vary between the modules. Hence it is easy to distribute documents related to these in AM
9.	Skill improvement/training	Focus on modules	Focus on machining processes	In AM, skill improvement with the end product as focus is possible.

Note: AM = assembly-module approach; MS = machining similarity-based redesign

3.13 MULTIPLE-CHOICE QUESTIONS

- 3-1. What are the three important aspects that have a significant influence on the process?
A. Volume, Variety and Flow
B. Knowledge, Material and Choices
C. Product, Workers and Skill
D. Quantity, Processing and Category
- 3-2. The third category of flow is:
A. Jumbled flow system
B. Alternative flow
C. Mass production
D. Raw material
- 3-3. What are the three important aspects that has a significant influence on the process?
A. Turn-around time, Capital, and Investment
B. Volume, Variety and Flow
C. Volume, Variety and its Quality
D. None of the above
- 3-4. Physical arrangement of various resources available in the system to improve the performance of the operating system, thereby providing better customer service is termed as:
A. Manpower recruitment
B. Layout planning
C. IT solutions
D. Service department
- 3-5. What are the different types of “Manufacturing/Plan” Layout?
A. Budget Layout, Financial Layout
B. Building Layout, Emergency Exit Layout, Organization Layout
C. Process Layout, Product Layout, Group Technology Layout, Fixed Position Layout
D. Production Layout, Sales Layout
- 3-6. What is the most important task in a process based industries?
A. Adhere to the safety protocol
B. To make continuous improvement and capacity de-bottlenecking
C. To achieve the outcome within the stipulated time limit
D. To achieve incremental addition in the productivity
- 3-7. The design of a process layout is:
A. One-step process
B. Two-step process
C. Three-step process
D. Four-step process
- 3-8. Fixed layouts are used in?
A. Small industries
B. Only applies to new venture for a period of one year
C. Medium scale industries
D. Large project-based industries
- 3-9. Layouts are designed with the objective of subdividing a universe of machines and components into sub-groups such that each group of components forms a part of the family and is endowed with a corresponding sub- group of machines known as:
A. GT layout
B. Machine group
C. Diagonalized structure
D. Production flow
- 3-10. What promises significant reduction in manufacturing lead time, better space utilization, and cost reduction due to better utilization of resources and manpower?
A. Multiple workers multiple machine layout
B. Multiple workers one machine layouts
C. One worker one machine layouts
D. One worker multiple machine layouts

Answer Keys: 3-1. A 3-2. A 3-3. B 3-4. B 3-5. C 3-6. B 3-7. C 3-8. D 3-9. B 3-10. D

4

C H A P T E R

PLANT LAYOUT : DESIGN

- 4.1 Performance Measures for Layout Design
- 4.2 Design of Process Layouts
 - 4.2.1 The Qualitative Approach to Layout Design
 - 4.2.2 The Quantitative Approach to Layout Design
 - 4.2.3 Software Packages for Layout Design
- 4.3 Design of Product Layouts
- 4.4 Design of Group Technology Layouts
- 4.5 One Worker–Multiple Machine Layouts
- 4.6 Technology Issues in Process Design
 - 4.6.1 Flexible Manufacturing Systems
 - 4.6.2 Automated Material-handling Systems
- 4.7 Summary
- 4.8 Formula Review
- 4.9 Review Questions
- 4.10 Problems
- 4.11 Net-Wise Exercises
- 4.12 Mini Project
- 4.13 Case Study
- 4.14 Multiple-choice Questions

PROCESS DESIGN AT MILLTEC MACHINERY

NOTES

Milltec Machinery Private Limited is a Bangalore-based manufacturer of equipment for rice and flour mills. During the short period for which they have been in operation, they achieved significant growth on account of the product range they have been able to bring to this sector of the industry. They manufacture a range of machines under four broad product categories: re-cleaning, hulling, polishing, and grading. Within each category, they have three or four sub-product categories and a range of sizes and operating characteristics, resulting in over 200 different products. Machine-tool manufacturing typically involves a certain amount of customization.

Designing appropriate manufacturing processes, selection of the right machine tools and other manufacturing resources is critical to improved performance and operational control at Milltec. Milltec employs the latest technology in manufacturing. It believes in producing machinery of international standards at affordable prices. Laser cutting is used in all sheet-metal operations to ensure precise assembly. Milltec uses the CNC press brake for uniform bending of sheet-metal parts. CNC machining centres are utilized to produce machined components. Milltec uses MiG welding stations for the welding operations.

Once the type and number of machines required for the manufacturing process is identified, the next important step is to design an appropriate layout of these resources on the shop floor. At Milltec, manufacturing is organized on a functional basis. The fabrication shop is spread over 750 m². The assembly and paint shops are spread over 2,000 m² and the machine shop over 500 m². Milltec has a dedicated engineering department to cater to the erection and commissioning of projects. It also has an R&D department.

Over the years, Milltec has realized the importance of addressing manufacturing process design and layout issues. Through a process of continuous improvement, Milltec employees and management have realized that several benefits arise out of designing an appropriate layout and workplace organization. These benefits include better production planning and control, less work-in-process inventory, reduced cost of production, better employee morale, and greater visibility of problems at the workplace.

Source: Company sources, <http://www.milltecmachinery.com>

4.1 PERFORMANCE MEASURES FOR LAYOUT DESIGN

The layout design should result in greater efficiency of the manufacturing system. This is the simple goal of any layout planning exercise. However, in order to assess how good the layout is, it will be useful to employ a few measures. Such measures help a layout designer not only to choose between alternative layout choices that he/she may have but also to set some targets for improvement in the future. Table 4.1 has a sample set of measures and the basis for measurement.

The most fundamental measure for layout design is the distance travelled by the components pertaining to a customer order in the system.

The most fundamental measure for layout design is the distance travelled by the jobs on the shop floor. Often organizations use a scale model of the existing layout and strings and pins to trace the path of the jobs to measure the total distance travelled. A good layout design must ensure that unnecessary movements and excessive material handling are avoided. Table 4.2 shows the impact of an existing layout in a manufacturer of earth-moving equipment with four product lines. Production and assembly of 1,080 components of Product A on average resulted in a total travel of 375.7 km. Products C and D were added much later in the factory. The average distance per item for all the four product lines confirmed that the company had not made any major layout changes in the last thirty years, despite adding new product lines. As products travel long distances, the lead time goes up, the overhead costs increase, and the investment in inventory also goes up.

TABLE 4.1 PERFORMANCE MEASURES FOR THE LAYOUT PROBLEM

Performance Measure	Basis for Measurement
Distance travelled by jobs in the shop floor	Kg-Metres of job movement for each product
Space utilization index	Minimum space required to actual space utilized
Material-handling costs	Rupees per month
Lead time of the processes	Hours per average product
Investment in work in progress	Rupees per month
Interdepartmental moves	Number and quantum of interdepartmental moves
Utilization of the resources	Per cent to total capacity
Ease of production control	Number of job cards and control documents generated; size of the progress chasing staff
Number of ownership changes	Number of times the job changes hands

NOTES

NOTES

TABLE 4.2 EFFECT OF LAYOUT ON DIFFERENT PRODUCT LINES

Product Line	Total Distance Travelled (In Metres)	Number of Items Manufactured*	Average Distance Per Item
Product A	375,655	1,080	347.83
Product B	415,125	757	548.38
Product C	288,710	301	959.17
Product D	297,110	405	734.60

Note: *The total distance travelled includes only those of the items manufactured on the shop floor. The number of items that were finally assembled into the final product includes many bought-out items in addition to these.

In addition to measures to assess the impact on lead time and inventory, another useful measure is the ease of production control. A better layout will enable an organization to have greater visibility of operations and ease of control. Finally, recent ideas pertaining to business process re-engineering suggest that if the jobs change fewer hands, then there are greater advantages from pinpointing responsibilities on groups of people. Jobs with fewer interdepartmental moves and ownership changes have a greater chance of getting completed faster. Therefore, these measures may also be useful.

4.2 DESIGN OF PROCESS LAYOUTS

The design of a process layout involves a three-step process. The first step is to identify the number of departments required and the space requirements for each department. The second step is to estimate the magnitude of flow across these departments. The last step is to use the above information to arrive at the final layout of the departments so that the cost of material handling is minimized. These steps are explained here:

⇒ *Step I—Identify the number of departments required:* Since process layouts are made on the basis of functional similarity of the resources, this is a fairly straightforward decision. For example, in a machine shop, all the lathes are grouped together to form a lathe shop, all the grinders are grouped to form the grinder shop, and so on. There will also be separate departments for receiving, final inspection, and shipping. Once these units are identified, it is possible to also estimate the sizes of each of these departments. The number of resources to be placed in the shop dictates the size. The distance between the blocks in the shop floor in which the departments are to be located could also be measured. This information will be of use in the layout design exercise. On the basis of these estimates, the required data for the layout problem could be gathered. Table 4.3 has the distance in metres between six blocks (A to F) for locating six departments on a hypothetical shop floor.

TABLE 4.3 DISTANCE BETWEEN SIX BLOCKS

	A	B	C	D	E	F
A		20	30	40	30	60
B			50	70	50	20
C				30	70	40
D					50	20
E						20
F						

NOTES

⇒ *Step II—Estimate the magnitude of flow:* The second step in the design process is to estimate the flow of material between departments and the cost of moving one unit across departments. In a process-focused layout, jobs visit various departments before the processing is complete. Further, the production requirements of each job in each resource will vary. Therefore, consolidating all this information will indicate not only the load on each resource but also the quantum of interdepartmental flow. Interdepartmental flow is the sum of flows between two departments irrespective of the direction of the flow. For instance, an interdepartmental flow of 100 units between Departments 1 and 2 may include 65 units of flow from Department 1 to Department 2 and 35 units of flow from Department 2 to Department 1. At the end of this consolidation exercise, one can arrive at a matrix of interdepartmental flows as indicated in Table 4.4 for a sample of six departments.

⇒ *Step III—Arrive at the final layout of the departments:* Once the basic data is available, appropriate methods could be used to solve the layout problem. Qualitative and quantitative approaches are available to arrive at the final layout of the departments. These are discussed here.

4.2.1 The Qualitative Approach to Layout Design

In this approach, some qualitative measures are used to decide which department is to be located next to another department. One simple method is to use the interdepartmental flow as the basis on which closeness between one department and another could be

TABLE 4.4 MATRIX OF INTERDEPARTMENTAL FLOWS

	1	2	3	4	5	6
1		100	30	40	0	30
2			90	0	150	20
3				0	70	40
4					50	20
5						70
6						

NOTES

established. However, often it is difficult to exactly quantify why two departments need to be close to each other or should not be located side-by-side. For example, even when the interdepartmental flow of jobs between a heat treatment shop and quality control section may be high, it is better not to locate them side-by-side. This is because the quality control section may use high precision gadgets for testing and calibration, and locating the heat treatment shop alongside may call for costly measures for isolation of particle impurities, pollution control, and cooling of the high temperature ambience. Similarly, it may not be desirable to locate a high tension testing facility, in an electrical goods manufacturing industry, in the midst of other shops for safety considerations.

The qualitative measure links some criteria to the closeness required between a pair of departments.

Based on such criteria, it is possible to arrive at qualitative measures for closeness requirements of a department vis-à-vis other departments. The qualitative measures link some criteria to the closeness required between a pair of departments. Figure 4.1 shows one of the popular methods used for qualitative ranking of closeness and an example of ranking for locating the six departments.

There are software packages available for the qualitative design of process layouts. These include ALDEP and CORELAP.

Using the closeness ratings between departments, layouts can be constructed. In the above example, Departments 1 and 3 are to be located close to each other. Similarly, Departments 1 and 6 should also be close to each other. Using this information, an initial layout can be constructed. Accommodating all the requirements may often turn out to be infeasible. However, alternative combinations may ensure better locations that improve the appropriateness of the layout. Therefore, one method is to begin with an initial layout that is feasible and progressively check the appropriateness of the new layout obtained. This may be done by

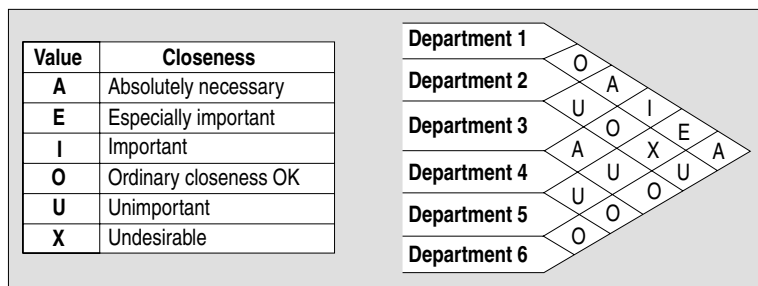


FIGURE 4.1 Qualitative ranking for designing process layouts

selectively exchanging parts of the layout plan through swapping of a subset of locations among each other. A pair-wise exchange could be a systematic approach to enumerate all the possible layouts that one can get from an initial layout.

Smaller layouts can be analysed using visual methods and by trial and error. However, if the number of departments involved is many, then the use of software packages may be inevitable for layout design. Software packages like ALDEP and CORELAP are used for the qualitative design of process layouts.

4.2.2 The Quantitative Approach to Layout Design

The quantitative method in making layout designs uses certain measures for assessing the impact of the layout and seeks to arrive at the best method for locating the departments. Let us assume that we are interested in arriving at an appropriate layout for six departments (1 to 6). The six departments are to be located in six locations (A to F). Since the basic purpose of a good layout design is to minimize excessive travel of jobs over long distances, let us use the material-handling cost as the basis for designing an appropriate layout.

Let,

C_{ij} denote the cost per unit of transporting a unit distance from Department i to Department j

F_{ij} denote the interdepartmental flow between Department i and Department j

D_{ij} denote the distance between Department i and Department j

n denote the number of departments to be laid out

The total cost of the plan is given by:
$$TC = \sum_{i=1}^n \sum_{j=1}^n F_{ij} D_{ij} C_{ij} \quad (4.1)$$

The interdepartmental flows, the distance between the blocks, and cost information are gathered during a layout planning exercise, as indicated in the previous section. Therefore, once the decision regarding which department will be located in which block is made, the total cost of the plan could be computed using Eq. 4.1. The use of the quantitative method requires some logic for arriving at the best layout. The above can be modelled as a mathematical programming problem with the objective function of minimizing the total cost of the plan. The mathematical programming problem is one of assigning the department to the blocks, subject to the constraint that one block can hold only one department. Further, the number of blocks assigned should be equal to the number of departments.

A popular heuristic for the assignment problem forms the basis for the computerized procedure known as **computerized relative allocation of facilities (CRAFT)**. CRAFT utilizes the improvement method for obtaining the best layout. In CRAFT, an initial

NOTES

NOTES

Computerized relative allocation of facilities (CRAFT) is a computerized procedure that utilizes the improvement method for obtaining the best layout.

feasible layout is formed and a series of improvement opportunities explored through a pair-wise exchange of departments. If there are n departments, a pair-wise comparison involves $n(n - 1)/2$ evaluations. After all these evaluations, the best possible pair-wise exchange is identified. If further reduction in the cost is possible by making this change, then the location of the two departments is changed. For example, in a six-department example, let us assume that the cost reduction by pair-wise exchange of departments is as shown in Table 4.5 for four pairs (we shall further assume that all other pair-wise exchanges result in an increase in material-handling costs).

Therefore, the location of Departments 3 and 5 will be exchanged. If the total cost of material handling at the beginning of this iteration was ₹20,000 the new material-handling cost will be ₹19,650. The procedure continues in this manner until no more improvement is possible.

4.2.3 Software Packages for Layout Design

Solving layout problems using manual methods is rather difficult when the number of departments is more than 10. In most realistic situations, such numbers are common. Therefore, the use of computerized methods is inevitable. In response to these requirements, several software packages have been developed.

There are three underlying approaches to the use of computer packages for designing layouts. One is the *construction method*. In this method, the solution begins with an empty layout solution and one department is added at a time. When all the departments are added to the shop floor, the final solution is obtained. Automated layout design programme (ALDEP), Computerized relationship layout planning (CORELAP), and plant layout evaluation techniques (PLANET) are three known software packages that utilize construction logic. In general, these software packages may result in departments having unrealistic shapes. For this reason, such methods are more useful in providing layout designers with some alternative ideas and initial solutions rather than a final solution.

TABLE 4.5 COST REDUCTION BY PAIR-WISE EXCHANGE OF DEPARTMENTS

Department Pair (i, j)	Cost Reduction
(1,4)	250
(3,5)	350
(2,4)	150
(5,6)	110

There are three underlying approaches to the use of computer packages for designing layouts: *the construction method*, *the improvement method*, and *performance evaluation modelling*.

On the other hand, in the *improvement method*, the starting point for a solution is a feasible layout. The feasible layout may not be the best layout. Therefore, the solution is improved progressively. When no more improvement is possible, the procedure stops and the available solution is the best solution. There are two known packages available in this category. These are computerized relative allocation of facilities (CRAFT) and computerized facilities design (COFAD). COFAD is a modification of CRAFT that incorporates the choice of a material-handling system in the layout design process.

The third approach to layout design is to use recent advances in object-oriented modelling techniques and build *performance evaluation models*. In this approach, software is used to build a model with 3-D objects of the layout that the designer has in mind, and to assess the impact of the design using several parameters. These parameters could include, for example, equipment and material-handling utilization, congestion in the material-handling network, build-up of inventory in the shop floor, and manufacturing lead time. Often, after building these models, a simulation analysis of the layout could be done to assess the parameters listed above. Since the cost of developing and analyzing these models has been progressively coming down, recent approaches to designing layouts are directed towards arriving at a configuration through performance evaluation.

4.3 DESIGN OF PRODUCT LAYOUTS

There are several examples of mass production systems in operation today. A two-wheeler manufacturer such as Bajaj Auto Limited sold nearly 3,422,400 motorcycles in 2013–2014, which translates into a monthly sale of 285,200. It is expected that the daily production of vehicles is nearly 11,400. Various sub-assemblies in the Bajaj plant need to be configured to match the production rate. Similarly, the final assembly stations also need to have the required number of resources at each station to meet the targeted demand. In such a scenario, much of the control and scheduling boils down to appropriately arriving at a balanced flow of components on the shop floor. A product layout design essentially caters to this objective by estimating the exact number and the sequence of resources required in the manufacturing system for a targeted production level. Essentially, the layout design seeks to identify the minimum number of resources required to meet a targeted production rate and the order in which these resources are to be arranged. In the process it seeks to establish a balance among the resources so that production is smooth. Therefore, the

NOTES

NOTES

technique employed for this purpose is known as the line-balancing technique.

Let us consider a mass production system with multiple tasks such as the manufacture of computers. If you visit the final assembly shop of the plant, you will invariably find a line layout. At one end of the assembly components will be fed into the system in multiple stations and the computer will be progressively assembled. Initially the mother board will be assembled with the required components. As it proceeds, the hard disk, optical drives, fans, power systems, serial, parallel and USB ports, and SMPS will be added to the box before it is finally secured with the cover plate. If you visit other mass producers like a car manufacturer such as Hyundai or an industrial valve manufacturer such as Audco India Limited you will notice a similar structure of manufacturing and assembly process. In all these cases, you will notice multiple tasks and precedence relationships among them.

Line balancing is a method by which the tasks are optimally combined without violating precedence constraints and a certain number of workstations are designed to complete the tasks.

Each task i requires a finite time, denoted by t_i and may have some precedence relationships with the other tasks. Further, the tasks require certain skills and resources such as machines and tools. **Line balancing** is a method by which the tasks are optimally combined without violating precedence constraints and a certain number of workstations are designed to complete the tasks. If there are three workstations, A, B, and C, in which 7 tasks are performed in a manufacturing system, then the workstation times are nothing but the summation of the task times assigned to each workstation. Let the workstation times be denoted as W_A , W_B and W_C . Clearly, a balanced design is one in which the workstation times do not vary widely. In such a situation, the resources will be uniformly utilized and the flow of material will be even. Further, there will be a good rhythm in operations. The maximum of the three workstation times determines the interval between the production of two successive components. If $W_A = 5$ seconds, $W_B = 3$ seconds and $W_C = 7$ seconds, one component will come out of the system every 7 seconds. This measure is known as the *cycle time*.

Cycle time could be considered as the reciprocal of production rate. If in a period of 20,000 seconds a shop produced 10,000 pieces of a component, then the production rate is $\frac{1}{2}$ per second. Conversely, the cycle time is 2 seconds. Cycle time could be actual or desired. If we compute on the basis of actual production, it represents the actual cycle time. On the other hand, if we compute the cycle time on the basis of what we desire the production to be, then it is desired cycle time. Maintaining the desired cycle time requires better management and work practices. The expression for cycle time is as follows:

$$\text{Actual (desired) cycle time} = \frac{\text{Available time}}{\text{Actual (desired) production}} \quad (4.2)$$

The problem of designing a balanced set of workstations suffers from the classical trade-off issue. If we combine more tasks into fewer workstations, we may require fewer workstations but the cycle time will be high, leading to reduced production rate. At the other extreme, if the tasks are kept separate and as many workstations are designed, we may increase the production rate beyond what is required at the risk of deploying more resources and workers with poor utilization. Therefore, solving the line-balancing problem calls for counterbalancing these opposing costs and striking the right trade-off between increased production and better utilization of resources.

Cycle time could be considered as the reciprocal of production rate.

The production rate, cycle time and the number of workstations are interrelated. Moreover, the number of workstations has a bearing on the average resource utilization. Using these measures it is possible to design an appropriate number of workstations. These relationships are indicated in the following expressions:

$$\text{Minimum no. of workstations required} = \frac{\text{Sum of all task times}}{\text{Cycle time}} \quad (4.3)$$

$$\text{Average resource utilization} = \frac{\text{Sum of all task times}}{\text{Number of workstations} \times \text{Cycle time}} \quad (4.4)$$

EXAMPLE 4.1

A factory working in two shifts each of 8 hours produces 24,000 electric bulbs using a set of workstations. Using this information, compute the actual cycle time of the plant operation. There are 8 tasks required to manufacture the bulb. The sum of all task times is equal to 12 seconds. How many workstations are required to maintain this level of production assuming that combining of tasks into those workstations is a feasible alternative?

Solution

Available time = $2 \times 8 \times 60 \times 60 = 57,600$ seconds

Actual production = 24,000 electric bulbs.

Therefore, using Eq. 4.2 we compute the cycle time for each bulbs as

$$\frac{57,600}{24,000} = 2.4 \text{ seconds.}$$

This means that the factory is producing 1 bulb every 2.4 seconds.

$$\text{No. of workstations required} = \frac{12}{2.4} = 5$$

Therefore, the tasks are to be split among the five stations such that each workstation will have 2.4 seconds as the sum of its task times.

NOTES

NOTES

EXAMPLE 4.2

A computer manufacturer needs to design assembly stations in the factory where the cabinet housing the hard disk, mother board and other accessories is to be made. The factory currently works for one shift of 8 hours. The tasks and their durations are given in Table 4.6. Figure 4.2 shows the precedence relationship among the tasks.

- If the cycle time is 80 seconds, what will the daily production of cabinets be?
- If the desired production rate is 320 cabinets per day, what is the maximum permissible cycle time?
- What is the maximum and minimum number of workstations required to maintain this daily production rate?
- Design an assembly set-up with 5 workstations and 6 workstations. What are the key inferences of this exercise?

TABLE 4.6 THE TASKS AND THEIR DURATION

Task	Description	Duration (Seconds)
A	Assemble and position the base unit	70
B	Install hard disk	80
C	Install mother board	40
D	Insert ports	20
E	Install speaker	40
F	Connect relevant modules to mother board and disk	30
G	Install controller	50
H	Visually inspect and close with a cover plate	50

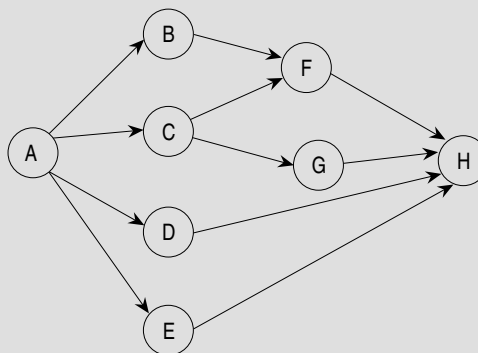


FIGURE 4.2 Precedence relationship among the tasks

Solution

(a) Total available time per day = $8 \times 60 \times 60 = 28,800$ seconds
 If the cycle time is 80 seconds, then the daily production of cabinets can be obtained using Eq. 4.2. Daily production rate is given by:

$$\begin{aligned} \text{Daily production rate} &= \frac{\text{Total available time}}{\text{Cycle time}} \\ &= \frac{28,800}{80} = 360 \end{aligned}$$

- (b) Since the desired production rate is only 320 cabinets, one can obtain the maximum permissible cycle time for the assembly stations.

$$\begin{aligned}\text{Maximum cycle time} &= \frac{\text{Total available time}}{\text{Desired production rate}} \\ &= \frac{28,800}{320} = 90 \text{ secon}\end{aligned}$$

The computations show that while a cycle time of 80 seconds may yield a much higher production rate, the workstations can be designed for a cycle time of up to 90 seconds without falling short of the desired daily production. Once the cycle time crosses 90 seconds, the manufacturer may not be able to produce 320 cabinets every day.

We know that the maximum number of workstations required to maintain the desired daily production rate is equal to the number of tasks that are to be performed. This is because we can simply assign one task to one workstation. In our example, the maximum number of workstations is eight, as there are eight tasks.

In our example, the sum of all task times is 380 seconds. Therefore, combining all the tasks into a single workstation will result in a cycle time of 380 seconds. This is not permissible. Therefore, the minimum number of workstations is dictated by the maximum cycle time permissible.

- (c) The minimum number of workstations is given by Eq. 4.3 as

$$\frac{380}{90} = 4.22 \approx 5$$

As long as we have five workstations, our cycle time may not exceed 90 seconds and we will be able to meet the production rate.

- (d) Design of assembly shop with five workstations (Table 4.7 presents the details)

We assign tasks to the five workstations on the basis of the following two criteria:

- (i) The workstation times should not exceed the maximum permissible cycle time of 90 seconds
- (ii) The precedence relationships among the tasks need to be honoured

The average utilization of the resources can be computed as follows (Eq. 4.4):

$$\begin{aligned}\text{Average utilization} &= \frac{\text{Sum of all task times}}{\text{No. of workstations} \times \text{Cycle time}} \\ &= \frac{380}{5 \times 90} = 84.4\%\end{aligned}$$

- (e) Design of assembly shop with six workstations (Table 4.8 presents the details):

We assign tasks to the six workstations using the same set of criteria as before.

$$\begin{aligned}\text{Average utilization} &= \frac{\text{Sum of all task times}}{\text{No. of workstations} \times \text{Cycle time}} \\ &= \frac{380}{6 \times 90} = 79.2\%\end{aligned}$$

NOTES

TABLE 4.7 DESIGN OF THE ASSEMBLY SET-UP WITH 5 WORKSTATIONS

	Workstation 1	Workstation 2	Workstation 3	Workstation 4	Workstation 5
Tasks assigned	A,D	B	C,G	E,F	H
Workstation times	90	80	90	70	50
Cycle time	90	90	90	90	90
Workstation idle time	0	10	0	20	40
Workstation utilization	100%	89%	100%	78%	56%

TABLE 4.8 DESIGN OF THE ASSEMBLY SET-UP WITH 6 WORKSTATIONS

	Workstation 1	Workstation 2	Workstation 3	Workstation 4	Workstation 5	Workstation 6
Tasks assigned	A	B	C,D	E,F	G	H
Workstation times	70	80	60	70	50	50
Cycle time	80	80	80	80	80	80
Workstation idle time	10	0	20	10	30	30
Workstation utilization	87.5%	100%	75%	87.5%	62.5%	62.5%

NOTES**Key Inferences**

The above computations clearly show the trade-off issues in line balancing. A design with five workstations will meet the current production rate with an average utilization of 84.4 per cent. However, when we increase the number of workstations by one, the utilization comes down by 4.8 per cent, but the production rate could increase by 11.1 per cent (since the daily production rate will increase from 320 to 360 on account of cycle time coming down to 80 seconds from 90 seconds). There are cost and strategic implications of these two alternatives and management could take an appropriate decision at the time of finalizing the design.

4.4 DESIGN OF GROUP TECHNOLOGY LAYOUTS

GT layouts are designed with the objective of subdividing a universe of machines and components into sub-groups such that each sub-group of components forms a part family and is endowed with a corresponding sub-group of machines known as a machine group. GT layout design is done with the systematic analysis of a machine-component incident matrix.

GT layout design is done with a systematic analysis of a machine-component incident matrix.

Consider an example of 20 components requiring processing in 10 machines. Let us denote the components using numbers and the machines using alphabets. A machine-component incident matrix (MCIM) indicates the relationship between the components and machines using a binary representation. Figure 4.3 has the MCIM for our hypothetical example.

		Components																			
Machines		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
	A	1			1			1													
	B			1	1	1			1												
	C		1	1		1			1												
	D	1			1			1													
	E													1	1	1		1	1		1
	F	1			1			1													
	G						1			1		1	1				1			1	
	H						1			1		1	1				1			1	
	I													1	1	1		1	1		1
	J						1			1		1	1				1			1	

FIGURE 4.3 MCIM for our hypothetical example

		Components																				
Machines		2	3	5	8	10	1	4	7	20	18	17	15	14	13	6	9	11	12	16	19	
	B	1	1	1	1	1																
	C	1	1	1	1	1																
	D						1	1	1													
	A						1	1	1													
	F						1	1	1													
	E									1	1	1	1	1	1							
	I									1	1	1	1	1	1							
	G																1	1	1	1	1	1
	H																1	1	1	1	1	1
	J																1	1	1	1	1	1

FIGURE 4.4 Rearranged rows and columns

In a GT layout design, the MCIM is analysed to detect possible sub-groups. If the sub-groups are non-overlapping, then the resulting design has independent resources and components. Each sub-group is also referred to as a *cell*. Therefore, the resultant structure obtained by re-configuring a manufacturing system with a GT layout is known as a *cellular manufacturing system*. Methods of identifying cells from the basic MCIM involve using a matrix manipulation or other methods to rearrange the rows and columns to obtain sub-groups. For our example, the cells are clearly identified by rearranging the rows and columns as shown in Figure 4.4.

The resultant structure obtained by reconfiguring a manufacturing system with GT layout is known as a *cellular manufacturing system*.

NOTES

NOTES

TABLE 4.9 THE FOUR CELLS WITH PART FAMILIES AND MACHINE GROUPS

	Part Families	Machine Groups
Cell 1	2, 3, 5, 8, 10	B, C
Cell 2	1, 4, 7	A, D, F
Cell 3	13, 14, 15, 17, 18, 20	E, I
Cell 4	6, 9, 11, 12, 16, 19	G, H, J

This structure of the matrix is known as a block diagonalized structure. The advantage of a block diagonalized structure is that it enables us to directly read the cells on visual inspection. In this case, four cells are formed with part families and machine groups as shown in Table 4.9.

A vast number of methods are available for identifying an appropriate set of sub-groups. The earliest and most popular among these is the *production flow analysis (PFA)* proposed by Burbidge. Later, the use of clustering techniques, matrix manipulation methods, graph theory, and mathematical programming were proposed by a number of researchers. The **rank order clustering (ROC) method** uses simple matrix manipulation methods to identify the cells in an MCIM. The ROC method rearranges the rows and columns of the MCIM in an iterative manner that will ultimately, and in a finite number of steps, result in a matrix form in which both the rows and columns are arranged in order of decreasing value when read as a binary word.

The earliest and most popular method for design of cellular manufacturing systems is the *production flow analysis (PFA)* proposed by Burbidge.

One worker–multiple machine layouts promise significant reduction in manufacturing lead time, better space utilization, and cost reduction due to better utilization of resources and manpower.

4.5 ONE WORKER–MULTIPLE MACHINE LAYOUTS

Earlier, machines were usually operated manually. Therefore, every machine needed to be attended by one worker. However, today, manufacturing organizations use semi-automatic machines and fully automatic machines. By semi-automatic machines, we mean those with automatic processing capability, but requiring a human interface for the loading and unloading of components into the machine. On the other hand, in a fully automatic machine, the machine has automatic processing capability and is linked with an automated material handling and material transfer mechanism. In this case, as soon as processing in one job is complete the material-handling system will shunt the completed job out of the machine into a material storage system or a conveyor

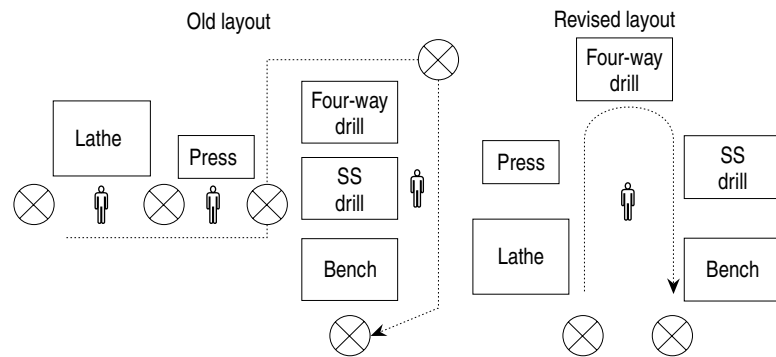


FIGURE 4.5 One worker-multiple machine layout

Source: N. Ravichandran, "A Journey Towards Manufacturing Excellence," CII Quality Summit 2000: 61-115.

(or some form of material-handling system) and will load the next job into the process.

There are numerous applications of semi-automatic machines in manufacturing systems. In the case of semi-automatic machines, an operator's task is to merely load the parts on to the machine and unload parts from the machines after the processing is over. Therefore, it is possible for one operator to manage more than one machine. The one worker-multiple machine opportunity adds new dimensions to the layout problem. It is possible to achieve effective utilization of floor space during the layout design. Moreover, material-handling tasks can be simplified and an optimal load-unload sequence that maximizes the output from the system may dictate the actual layout of the machines. On the other hand, an improperly designed layout, in the case of a one worker-multiple machine scenario, may result in worker fatigue, loss of productivity and deterioration of quality.

Figure 4.5 is an illustration of the nature of transformation and benefits that organizations obtain by employing one worker-multiple machine layouts. This example is one of several such cells formed in the manufacturing facility of Lucas TVS at Padi in Chennai. As is evident from the figure, the layout promises significant reduction in manufacturing lead time, better space utilization, cost reduction due to better utilization of resources and manpower, improvement in quality due to reduced material handling and, finally, a reduction in the distance travelled by the jobs before the completion of processing.

4.6 TECHNOLOGY ISSUES IN PROCESS DESIGN

Batch manufacturing has always had inherent limitations on account of the mid-volume, mid-variety nature of manufacturing. Work-in-process levels are generally high and machine utilization tends to be low. Jobs spend a high proportion of time waiting for a machine to be set up, waiting to be moved or waiting for other jobs on the machine to be completed. Batch production often requires

NOTES

NOTES

an army of progress chasers to keep jobs flowing through the manufacturing facilities. In batch type manufacturing, some studies conducted revealed that only 5 per cent of the total time spent in the shop contributed to actual metal cutting operation. The rest accounted for wastage of capacity, unnecessary waiting of jobs and so on. One way to improve productivity is to use technology to obtain a better process.

Advancements in the field of manufacturing technology have had an impact on process design. Versatile machines that are capable of handling the work of several traditional machines have provided some relief in simplifying process flow. These machines also provide considerable flexibility to address some of the issues pertaining to mid-volume, mid-variety manufacturing systems. Of the several developments, flexible manufacturing systems (FMS) have been most prominent. The development of manufacturing technology has been accompanied by developments in material-handling technology, which have also contributed to process improvements. We shall look at these developments in this section.

4.6.1 Flexible Manufacturing Systems

An **FMS** is a manufacturing system that usually consists of numerical control (NC) machines connected by an automated material-handling system. It is operated through central computer control and is capable of simultaneously processing a family of parts with low to medium demand, different process cycles and operation sequences. From the definition of an FMS, one can characterize its features as follows:

An **FMS** is a manufacturing system consisting usually of numerical control (NC) machines connected by an automated material-handling system. It is operated through central computer control.

- ⇒ It is an attempt to solve the production problem of mid-volume and mid-variety parts for which neither high production-rate transfer lines nor highly flexible standalone NC machines are suitable.
- ⇒ It is designed to process, simultaneously, several types of parts in the given mix.
- ⇒ It is equipped with sophisticated flexible machine tools that are capable of processing a sequence of different parts with negligible tool change over time.
- ⇒ Parts are transferred from machine to machine by a computer-controlled *material-handling system (MHS)*.
- ⇒ It consists of three subsystems: the machining system, the material-handling system and the control system.

A schematic representation of the structure of an FMS is given in Figure 4.6. Machine tools are the main machines used in

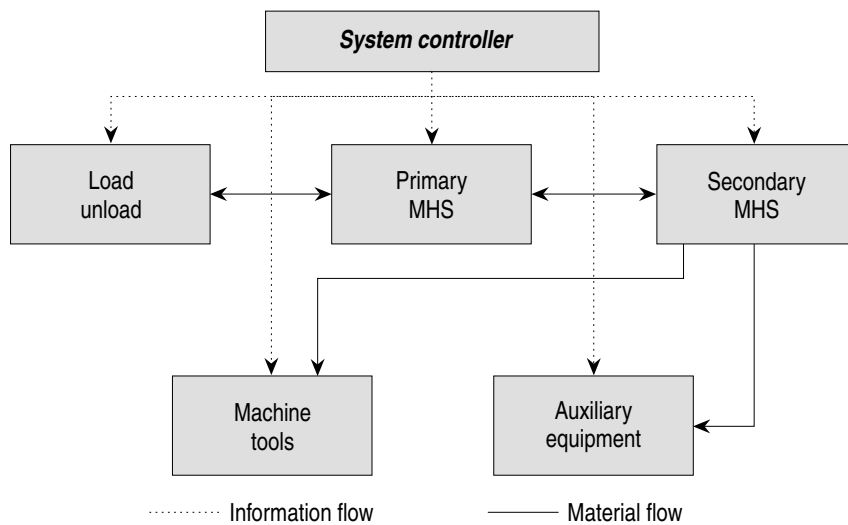


FIGURE 4.6 Structure of an FMS

manufacturing while auxiliary equipment are those used for other purposes, such as measuring quality using coordinate measuring machines. The primary MHS transports the work pieces from one location to another in an FMS. On the other hand, the secondary MHS transports work pieces from the machine table to the storage point in front of the machine tool.

Two general classes of FMS technology have been identified. The most recognized form is the multi-machine tool type, sometimes called the classical FMS. This is an automated production system for mid-volume, mid-variety production, consisting of machine tools tied together by an MHS. The second type of FMS is the standalone unit, usually consisting of a single machining centre or CNC lathe. This is sometimes referred to as a *flexible manufacturing cell (FMC)*. Such FMCs are equipped with automatic carousels or rotary tables for loading a small variety of work pieces. In some cases, an integral robot arm is employed for loading and unloading chores. Figure 4.7 shows typical machines used in flexible manufacturing systems.

FMS technology results in the reduction of direct and indirect labour force. With the level of automation that is employed in such

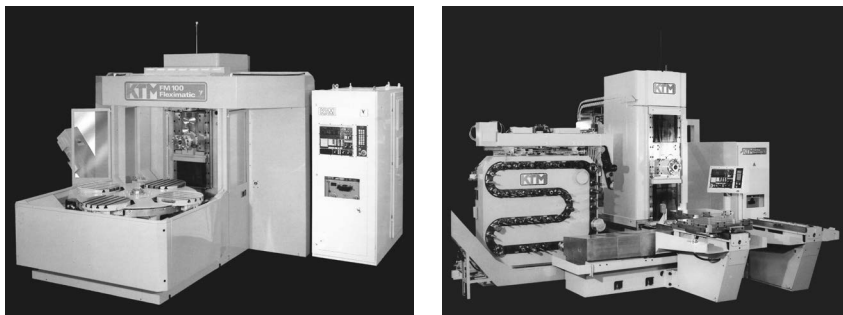


FIGURE 4.7 Typical machines used in FMS

NOTES

NOTES

systems, it is possible for a worker to tend to a group of machines in the system. The following is the role of human labour in the FMS:

- ⇒ Loading and unloading, tool set-up, tool replacement, work piece set-up off line
- ⇒ Maintenance of the system, multi-task monitoring (3 to 8 machines)
- ⇒ Supervision of the overall system, taking decisions using the information supplied by the computer system etc.

The technologically advanced features of the FMS, in part, simplify process design and complexities in flow in an intermittent flow system in batch manufacturing by offering several flexibilities, some of which are as follows:

- ⇒ *Machine flexibility*: This is related to the ease of making changes required to produce a given set of part types. The measurement of these changes includes, for example, the time taken to change tools in a tool magazine to produce a different part family, the time taken to assemble new fixtures, and so on. Technological progress, proper operation assignment and the technological capability of bringing part and tool together, along with other such factors, will improve this flexibility.
- ⇒ *Process flexibility or mix flexibility*: This is the ability to produce a given set of part types, each possibly using different materials in several ways. This, in essence, relates to the mix of jobs, which the system can process simultaneously. Process flexibility increases as the machine set-up cost decreases, and by having multipurpose adaptable CNC machines.
- ⇒ *Product flexibility*: It is the ability to produce a new set of products economically and quickly. Product flexibility can be measured from the time required to switch from one part mix to another.
- ⇒ *Routing flexibility*: This is the ability to handle breakdowns and to continue processing the given set of part types. This ability exists if either a part type can be processed via several routes or each operation can be performed by more than one machine.
- ⇒ *Volume flexibility*: This is a measure of the ability to operate an FMS profitably at different production volumes. This flexibility can be measured by how small the volumes can be for all part types with the system still being run profitably.
- ⇒ *Expansion flexibility*: It is the capability to build a system and expand it as the need arises, easily and in a modular fashion.

4.6.2 Automated Material-handling Systems

When manufacturing is automated, the material-handling system (MHS) must also be automated. Otherwise, it will not be possible

to ensure full utilization and smooth flow of work. Besides, an automated process cannot be totally controlled unless there is an MHS that is an integral part of the factory control system. An integrated MHS is made up of three basic elements, that is, an automated storage system, an automated transport system and an automated material transfer system.

An integrated MHS is made up of three basic elements:

- ⇒ an automated storage system,
- ⇒ an automated transport system, and
- ⇒ an automated material transfer system.

Automated storage system

An automated storage system is used for large-scale bulk storage as well as for small inline buffer storage. The *automated storage and retrieval system (AS/RS)* is the technology that takes the maximum advantage of the cubic space while inherently offering security and inventory control. With warehouse feeding by conveyor, an AS/RS can be used for storage operations ranging from maintenance supplies to raw materials for production, packaging and work in process. Automated storage and retrieval systems are in the forefront of warehouse automation.

As inventories have reduced, massive AS/RS operations are no longer warranted. *Carousels* have become sources of parts for assembly operations and the buffer storage of sub-assemblies during processing. They find increasing application as buffer storage units in flexible manufacturing systems. Two varieties of carousels are available—the horizontal type and the vertical type.

The *horizontal carousel* consists of a continuously looping conveyor belt, which carries segments of storage shelves or carriers. The *vertical carousel* is a cube-shaped storage structure containing two endless conveyor chain or cable loops that travel over the top and bottom sprockets. The loops are separated by supports, shelves, or carriers between them. Loads of various types and configurations can be placed on the carriers.

Automated transport and material transfer systems

Automated transport systems are used to move parts and products from the storage systems to the production operations while automated material transfer systems are used to load and unload materials at the workstations. It is customary to designate automated transport systems as primary MHS and automated transfer systems as secondary material-handling systems.

The *automated guided vehicle (AGV)* system forms one of the most exciting and dynamic tools in material handling today. While technological developments gave AGVs more flexibility and capability, it is market acceptance that has given AGVs a variety of applications to allow them to become the accepted standard material-handling method that they are today. AGVs are

N O T E S

NOTES

battery powered vehicles that can move and transfer materials by following prescribed paths around the manufacturing floor. They are neither physically tied to the production line nor driven by an operator like a forklift truck. Three basic techniques are used to guide AGVs:

- ⇒ Wires buried in the floor are electrically detected by the vehicle.
- ⇒ Lines painted on the floor are detected by a photoelectric sensor on the vehicle.
- ⇒ A computer controlled guidance system with wireless communication links to the AGVs.

AGVs can also communicate with factory computer control systems, either by making a physical link at each stopping point, or by infrared or digital radio transmission. This provides a means to track and control the flow of materials and alters the instructions to the guided vehicle. The central computer provides overall control functions such as dispatching, routing, traffic control, and collision avoidance. The recent advancements in AGVs include SGVs (self-guided vehicles) and free-ranging vehicles. AGVs usually complement an automated production line consisting of conveyors or transfer systems by providing the flexibility of complex and reprogrammable movement around the manufacturing process.

Assembly-line AGV applications are being introduced in manufacturing systems. These guided vehicles carry major sub-assemblies such as motors or transmissions to which parts are added in a serial assembly process. The vehicle proceeds into an assembly area and stops at an assembly workstation. When the process is completed, it enters the next assembly area, and this process is repeated several times. When the assembly process is complete, the finished assembly, say an engine block or a chassis, is unloaded from the vehicle, which is then sent to the start area for the assembly process. One such application was found at the Tata Motors plant in Jamshedpur. It uses an AGV system for rear axle assembly. The assembly line consists of 18 stations and the AGVs progressively move through these stations.

Conveyors exist in a wide variety of forms such as overhead, monorail, carry-and-free, power-and-free, underfloor drag-chain, floor-slot, gravity-feed, belt conveyors, and plastic link chains. All these systems are readily available in a wide variety of sizes, load-carrying capacities and speed capabilities. The intelligence for a conveyor system is provided by placing a large number of sensors around the conveyor and connecting them. It is this sensor and gating or routing technology that often complicates the implementation of a sophisticated conveyor system. These have a considerable impact on the overall cost of the system.

The use of a robot for materials handling can be beneficial for a number of reasons. The industrial application of robots is wide-ranging, particularly in assembly operations. In addition, robots

are predominantly used as secondary material-handling systems in FMS. They typically perform most of the loading and unloading chores at each machining centre or buffer location. In an effort to realize most of the advantages of robots and conveyors, the gantry loading system has been developed. Various types of gantry loaders are available.

FMS designers, over the past few years, have developed a rail-guided transport system, which is effectively a compromise between a conveyor system and an AGV. It is a vehicle similar in appearance to the AGV, but is constrained to run on rails. If an FMS can be arranged in a straight line, a rail-guided transport mechanism could well be an attractive means of load transfer between the various locations on an assembly line. Towline systems consist of carts that are powered either by an in-floor chain or by an overhead chain. The control systems available today make it feasible to programme tow carts and send them from one workstation to another. Towline systems are generally most advantageous when utilized in a relatively fixed path and in a consistent stations situation.

NOTES

4.7 SUMMARY

- *Volume, variety and flow* exert significant influence on process design in organizations.
- Process industries and mass production systems generally have a streamlined flow of products.
- Mid-volume and mid-variety manufacturing systems have intermittent flow. Capacity estimation is difficult in such systems compared to continuous flow systems.
- Project organizations and customized manufacturing systems have jumbled flow. Capacity estimation and scheduling of jobs are quite difficult. Therefore, operations management complexity is high in jumbled flow systems.
- A *process-product matrix* depicts the relationship between process flow characteristics and volume of production in a manufacturing organization.
- Product layouts are useful for high volume-low variety situations. At the other extreme, fixed positions and project layouts are useful for high-variety situations.
- Product layout and process layouts are used in the discrete manufacturing industry. They have several advantages and disadvantages.
- Mid-volume and mid-variety manufacturing systems can benefit from a group technology (GT) layout.
- Several software packages are available for designing process layouts. CORELAP, ALDEP, COFAD, and CRAFT are among the popular packages in this group.
- Product layout design seeks to identify the minimum number of resources required to meet a targeted production rate and the tasks to be assigned to each of these resources, using a technique called line balancing.
- GT layouts are designed with the objective of subdividing a universe of machines and components into sub-groups such that each sub-group consists of part families and machine groups.
- A number of methods are available for designing GT layouts. These include production flow analysis (PFA), rank order clustering (ROC), mathematical programming and clustering methods.
- New manufacturing technologies such as *flexible manufacturing systems (FMS)* have the potential to simplify flow complexities in mid-volume and mid-variety manufacturing organizations due to increased flexibility.
- No. of stages, variety and flow characteristics determine the complexity of operations management. By a careful design of the process, some of the complexities can be minimized.

4.8 FORMULA REVIEW

- The total cost of the process layout, $TC = \sum_{i=1}^n \sum_{j=1}^n F_{ij} D_{ij} C_{ij}$
- Actual (desired) cycle time = $\frac{\text{Available time}}{\text{Actual(desired) production}}$

- Minimum number of work stations required = $\frac{\text{Sum of all task times}}{\text{Cycle time}}$
- Average resource utilization = $\frac{\text{Sum of all task times}}{\text{Number of workstations} \times \text{cycle time}}$

4.9 REVIEW QUESTIONS

- What are the key determinants of process characteristics in operations? Give specific examples to show how they influence process characteristics.
- Is there any relationship between volume, variety and flow with respect to process design?
- How is the continuous flow system different from the intermittent flow system? What are the key implications for operations managers?
- Give three examples for each of the following and identify two important issues of interest to an operations manager:
 - A mass production system
 - A continuous flow system
 - A flow shop
 - An intermittent flow system
 - Job shop
 - A process industry
 - A batch production system
 - Project organization
- As an operations manager, which areas would you focus on if you were to manage a jumbled flow process?
- How does the choice of the layout affect the operational performance of an organization?
- What is the relationship between the volume, variety and flow characteristics of an operating firm and its layout?
- What are the different types of layouts? How should an organization decide on which layout to choose?
- Is a group technology layout any different from a product layout? Give reasons for your answer.
- Identify an appropriate layout for each of the following situations. Justify your choice in a sentence or two:
 - A manufacturer of garments for Van Heusen
 - A multi-cuisine restaurant in a posh residential area in Mumbai
 - The overhaul of helicopters
 - A fabricator of custom-made PCBs for a large number of electronic applications
 - An eye hospital
 - A motor manufacturer manufacturing 4 product groups for worldwide markets
 - A manufacturer of large turbines for power sector applications
- Suppose you are given three alternative designs for the layout of a shop floor in a manufacturing organization. How will you decide which of the three is the most appropriate?
- Critically compare quantitative and qualitative approaches to designing process layouts.
- How does a construction method of layout design differ from the improvement method? Which method is preferable and why?
- What do you understand from the phrase “cellular manufacturing system”?
- What are the reasons for the growing application of group technology concept for layout design?
- What do you understand by the term FMS? What are the characteristic features of an FMS?
- How does an FMS help in improving the process design in the case of mid-volume, mid-variety manufacturing?
- What are the various types of automated material-handling devices available for use in a computer integrated manufacturing system?
- What do you understand by the term complexity of operations management? Explain.

4.10 PROBLEMS

- A mass producer of an industrial component currently has a daily production rate of 1,200 pieces. The company works for 16 hours a day. The production process currently employs five stations. There are 12 tasks involved in the manufacturing process and together they need 190 seconds.
 - What is the actual cycle time for the component?
 - Compute the average utilization of resources employed to manufacture the component.
 - If the company wants to work with 85% utilization, by how much they should increase the daily production to achieve this?
 - Suppose, there is a 10% increase in the demand. Do they need to make any changes in the layout?
- A company procures a few components from the market and assembles them into a standard product and supplies it to its customers. There is enough demand for the product. If the company employs four stations for the assembly, it has 92% average utilization of the resources. The sum of the task time for the process is 180 seconds.
 - What is the cycle time for the process?
 - What is their daily production rate if they work in an 8-hour shift?
 - If they add one more station in the assembly process what would be the new cycle time if the utilization drops to 79%?
 - By how much can they increase the production on account of the reduced cycle time?
- Following data are available in the design of a product-based layout in which a certain number of work stations are deployed. The sum of all task times is 200 seconds. The cycle time is 50 seconds and the average resource utilization is 80%. If one more work station is added, the utilization drops to 74.07% and the new cycle time is 45 seconds.
 - How many work stations are used in the product layout design?
 - What is the daily production rate if a day consists of 450 minutes?
 - What is the maximum production possible in the current design?
- A manufacturer of spark plugs for the mass market would like to design the final assembly shop and requires certain data for the process. The factory works for two shifts and the total available time is 15 hours per day.
 - There are five tasks involved in the final assembly and the task times (in seconds) are 3.5, 4.5, 2.0, 3.0 and 4.0. What are the minimum and the maximum outputs possible from the factory?
 - The daily production needs to be 4,000 plugs. What should be the cycle time of the operations at the assembly shop?
 - If the firm wants to use three workstations, what will be the best output possible? By how much will the output increase if it adds one more workstation?
- A fast-food restaurant that operates in Connaught Place, New Delhi, specializes in offering food during peak hours. The restaurant operates only for two hours in the morning and promises faster turnaround for customer orders. The restaurant needs to design the workstations at the front end of the kitchen to assemble all customer orders. The back end of the kitchen provides the required food items. A study of the tasks, their dependencies and times taken on an average to assemble reveals the information shown in Table 4.10:
 - If, during a typical peak hour operation, the restaurant is able to take 144 orders, what is the maximum permissible cycle time for the assembly workstations?
 - If the restaurant desires to set a target of the above cycle time, what is the minimum number of workstations they should have at the assembly?
 - Design the assembly with the minimum number of workstations. Can it serve 180 customers during the peak hour? If not, arrive at a feasible design.
 - For the final design that you have made, compute the average utilization of workstations.
 - The restaurant management is aware that the bottleneck is in terms of their capacity to serve. If they could increase the capacity, they would earn additional revenue. The marginal revenue from an order will be ₹100 and the marginal daily cost of adding a workstation is ₹2,700. What should the management do?

TABLE 4.10 DEPENDENCIES AND TIME TAKEN TO ASSEMBLE/SERVE FOOD

Task ID	Description	Task Time (Seconds)	Predecessor
A	Set up the serve tray	20	None
B	Arrange plates, cups, tissues	20	A
C	Place the main item in the plate	35	B
D	Fill in the side dishes	35	B
E	Place a filled in water cup	20	D
F	Arrange cutlery items in the plate	20	C, E
G	Verify the order	15	F
H	Prepare a bill for the order	35	G

4.11 NET-WISE EXERCISES

1. Caterpillar is the largest maker of construction and mining equipment. To know more about the products offered by them, visit [http://india.cat.com/cda/layout?m = 81920&x = 7](http://india.cat.com/cda/layout?m=81920&x=7) and click on **Products**. Navigate through their product lines and offerings.

Haldia Petrochemicals is the one of the largest petrochemicals complexes in India. Go to [http:// www.haldia-petrochemicals.com/index.jsp](http://www.haldia-petrochemicals.com/index.jsp) and click on **Products** on the left side of the screen. Visit various sub-links in this section. Also click on the link **About Us**. There is a sub-section on manufacturing processes. Visit other parts of

the Web site. Take a look at the photos of the plant loaded onto the main page.

Based on your understanding of the two organizations, compare and contrast the product flow and process characteristics.

- (a) Identify three distinct features of Caterpillar and relate them to the design of the manufacturing process for the organization.
- (b) Why would you classify Haldia Petrochemicals as a process industry? Identify three characteristics that you find in Haldia Petrochemicals that are peculiar to a process industry.

4.12 MINI PROJECT

1. There are several manufacturing and service organizations in every part of the country. Select out of these three organizations that belong to any of the following categories: a process industry; a mass production system; a service set-up; a mid-volume, mid-variety manufacturer.

Visit the selected manufacturing and service organizations, make a formal plant tour of the organization and collect data on various aspects of operations management. After you have recorded all the key observations based on the visit, prepare a list of items on which you would like to collect additional information and obtain them. Also interview the plant manager, the process planning team, and the line supervisor. Note down their key observations about the plant's functioning. Based on this work, prepare a final report that contains the following information:

- (a) The process flow diagram of the plant
 - (b) The nature of the production system
 - (c) The key challenges for operations management
 - (d) Status of current operations management practices in the organization
 - (e) Specific areas in which improvement potential exist for the organization
2. There are alternative approaches to layout design. The use of software packages for this purpose has increased manifold in recent years. Prepare a project report to answer the following question pertaining to the use of software for layout design:
 - (a) Scan the Internet and prepare a list of available software solutions for layout design. Briefly explain the salient features of the software.

- (b) Develop a classification scheme to list the software that you have identified. Can you recommend a specific set of application domains of the software on the basis of this classification scheme?

- (c) Select one software package from each of your classification schemes and critically compare them. Identify the specific advantages and disadvantages of each package.

3. There are several companies that have recently made significant changes in their layout design. Prominent in the list are automobile and auto-components manufacturing firms. Identify one such company in your locality and perform the following studies and submit a report of your findings:

- (a) What were the major drivers of the layout change decision in the organization?
- (b) Enumerate the salient aspects of the alternative layout design process (methodology used to arrive at the new design, tools and techniques used, scope of the project, organizational aspects of driving the change process etc.)
- (c) Understand the implementation aspects of the layout design process (methodology of implementation, time taken, chronology of events leading to completion of the process etc.)
- (d) What were the main benefits that accrued to the organization as a result of this layout design change?
- (e) How did the layout change affect various people in the system including the production workers, supervisors, managers, customers, and suppliers?

4.13 CASE STUDY

Aerospace India Limited

Aerospace India Limited (AIL) has been serving the needs of the Indian defence sector for the past 20 years. The company has four divisions that manufacture various components for the aerospace sector. Unit 1 was established in 1970 to manufacture components for helicopters. The entire machining of about 2,230 components required for various versions of helicopters is done at the machine shop at Unit 1.

THE MACHINE SHOP AT AIL

The machine shop at AIL is responsible for manufacturing all components required for the helicopters. It consists of more than 100 machine tools, which includes some high-precision machine tools such as an NC jig boring and a CNC gear grinding, besides many conventional lathes, turrets, milling machines, and drilling machines. The machine shop occupies an area of about 6,440 m².

The facilities in the machine shop are arranged in a functional fashion. These include a gear unit comprising facilities for gear manufacturing ranging from 50 mm to 600 mm diameter, a lathe unit for rotating parts, a milling unit, a turret unit, a drilling unit, a line spar milling, a grinding unit, CNC machines, a metrology unit, a tool room, an inspection unit, and production tool stores.

MANUFACTURING REQUIREMENTS FOR NEW PRODUCT

With the progressively dwindling production of the existing range of helicopters, AIL was entrusted with the task of designing and manufacturing components in Unit 1 for a newer helicopter design by its customers. Irrespective of the many variations in the new product, most of the subsystems will have a commonality of about 85 per cent from the perspective of manufacturing components at the machine shop. Therefore, the manufacturing requirements of the new product are not much different from that of the existing ones. The existing machining capacity at the machine shop is adequate to handle this additional load. The process plans and time estimates have already been made for the components to be machined at the machine shop in Unit 1.

Typically, the production of machined components for the new product involves the following production planning and control activities: The scheduling department issues route cards and job tickets based on the production target. The progress department collects these and draws the required material from the stores. Based on the scheduling prepared by the machine shop manager, the components are loaded on to different work centres. Components are inspected after the completion of every operation. Stage inspection is mandatory in production. After inspection, the components are loaded to the next operation. Component loading, operation completion, and the inspection cycle are repeated until all operations are completed. All components are subjected to 100 per cent inspection. However, if the components fall under the life-critical category, the inspection records are maintained for lifetime

to trace the history in cases of accidents and failures pertaining to the component.

In the existing set-up, the machine shop is headed by a senior manager who is assisted by one manager and one deputy manager. Currently, even though the area of responsibility is clearly defined, interdependency is high due to the functional layout. It warrants frequent meetings between managers. One of the problems with the existing organization structure is the total lack of product focus. Many components that require turning or milling queue up in the respective sections. The supervisors and the section in-charge normally go by convenience and rush requirements. The senior manager is the only person responsible for the products and their sub-assemblies. The rest are more concerned with turning, milling, etc.

IMPLICATIONS OF MAINTAINING THE STATUS QUO

Since the manufacturing requirements for the new product are very similar to that for the existing one, one option to organize manufacture of new components is to maintain the status quo. This appears to be a soft option. It impinges on no one's working style and requirements. However, one of the senior managers was sceptical of this arrangement. "You see, with the existing set-up we will not be able to manage two different projects. It basically deprives us of product focus." This is so for several reasons, some of which are as follows:

1. A functional layout increases the cycle time. Since machine utilization is stressed over the completion of product families in a functional layout, components that do not deserve immediate priority are loaded to avoid idling, which causes bottlenecks for priority items. This leads to an increase in the WIP inventory.
2. Since the facilities are located far apart, the material-handling distance is large. Also, due to the number of operations, the frequency of material handling is high.
3. Given the complexity of scheduling for more than 2,000 components, functional layout often has to rely on the follow-up group, whose job is to "chase" the components.
4. In the absence of product focus, despite more than 80 per cent completion of loaded components, quite a few assemblies cannot be started, as the required items will be still under progress.
5. Both supervisors and operators develop a tendency to reduce set-up time by combining batches whenever possible. This results in increased batch quantity, even though it was not meant to be.
6. Manufacturing errors are not detected at an early stage. Due to an imbalance in the load, the inspection department often authorizes supervisors to carry out inspection, who do a superficial job.
7. Operators make frequent trips to production tool stores, the standard tools crib, the blueprint booth and the

inspection department for various reasons. This increases non-value-added activities and lead time considerably.

8. With an increase in the number of batches and WIP, component traceability is lost. It requires some time to locate the batch required, as identification would be very difficult in the absence of route cards which will be either with the operator or with the inspection department.
9. Due to a focus on performance at the work-centre and section level, it appears that the responsibility of completing the component entirely as per the given schedule, cost, and quality is nobody's except that of the top management.
10. Due to lack of knowledge about other machining operations, workers generally do not appreciate another work area. This also creates a mindset against multi-functional skills.
11. Team spirit is absent, since the functional set-up *minimizes* chances of group interaction.

DESIGN OF THE MANUFACTURING PROCESS

Efforts were made to investigate alternate ways of organizing manufacturing for production of machined components for the new product. Initially, a meeting was convened to discuss the various possibilities. Mr Prakash, one of the managers from Unit 1, made a presentation based on his analysis of the problem and proposed two new alternatives. He suggested two alternative approaches to redesigning the machine shop for manufacturing machined components required for the new product. In the first approach, known as the assembly-module approach (AM), the entire machining is organized on the basis of modules of sub-assemblies. In the other approach, known as the machining similarity-based approach (MS), the grouping of components is based on the machining requirements.

ASSEMBLY-MODULE-BASED REDESIGN

The manufacture of components required for the new product can be classified under 14 modules. Each module forms a sub-assembly. From the bill of materials, the components required for each sub-assembly have been segregated. There are about 731 components (Table 4.11 has the details). For example, 133 components pertain to the flight control system. Similarly, 25 components are required for the tail gearbox sub-assembly. For each component, the following information was collected: part number, department number, work-centre number, set-up time, unit processing time, and the quantity required. Using this information, the work-centre-wise capacity requirements have been calculated for each module.

Using a minimum utilization value of 30 per cent, it was found that only seven modules could justify dedicating capacity. Hence, the other modules were combined with these to finally form seven modules. Some modules have more than one cell assigned to them. In addition, a separate cell designated as a common cell was formed with equipment that is both costly and has low utilization at the individual module

TABLE 4.11 SUB-ASSEMBLIES AND MACHINING REQUIREMENTS FOR THE NEW PRODUCT

Sub-assembly	Number of Components
Flight control system (FCS)	133
Avionics and electrical system	25
Hydraulic system	12
Structures	59
Landing gear	29
Main rotor	76
Tail rotor	21
Power plant	74
Main gear box (MGB)	145
Tail gear box (TGB)	25
Intermediate gear box (IGB)	26
Auxiliary gear box (AGB)	25
Tail drive shaft (TDS)	24
Others	57
Gear components from existing project	41
Total	772

Note: *41 components from the existing project have been added to avoid omission of these during capacity calculations. These components require special purpose machines to be assigned for producing the new product.

level. Finally, this has resulted in 13 cells. Table 4.12 has the details. In the final design, wherever the required machines were not available, recommendations have been made to procure them.

MACHINING SIMILARITY-BASED REDESIGN

In the second approach, the computation of the capacity requirements work-centre-wise, and the number of machines required for each cell are all done in the same way as described above. But there is a fundamental difference in the approach to forming cells. The 772 parts are grouped together based on their similarity in machining requirements. This would mean, for example, that a group of components requiring turning operations might have been grouped together. Similarly, a group of components going through a series of milling and finishing operations from many sub-assemblies might have been put together.

Production flow analysis was initially employed to identify cells. Forty cells were initially formed. However, the demand on many machines in these cells was far less. Moreover, the

TABLE 4.12 DETAILS OF MODULES AND CELLS AS PER THE AM APPROACH FOR REDESIGN

Sl. No.	Module	Cell Number	Number of Machines	Number of Componets	Number of Exceptions [#]	Number of Machines to be Procured
1	Flight control system (FCS)					
1(a)	FCS—I	1	7	93	3	1
1(b)	FCS—II	2	6	40	18	2
2	Structures					
2(a)	Avionics and electrical system, structures	3	5	84	20	1
2(b)	Hydraulic system, landing gear	4	4	41	18	1
3	Rotor group					
3(a)	Main and tail rotor	5	7	65	40	—
3(b)	Main and tail rotor	6	4	32	13	—
4	Power plant accessories	7	8	74	37	1
5	Main gear box (MGB)					
5(a)	MGB—I	8	5	78	32	1
5(b)	MGB—II	9	3	67	11	2
6	Gears (TGB, IGB, AGB)	10	15	76	44	1
7	Non-MGB					
7(a)	Others, gear components for existing project	11	7	86	30	3
7(b)	Others, tail drive shaft	12	3	36	9	—
8	Common cell	13	8	*	*	
	Total		82	772	275	13

Note: *Eight machines have been separately located as Cell 13. These machines were highly valued one-off machines and it was not possible to justify either locating them in a particular cell or procuring additional machines. About 79 components from four modules undergo processing in the cell.

[#]Out of the 275 exceptions, only 92 required movement across the modules. In other cases, it involved inter-cellular movement between cells belonging to the same module.

number of cells was too many. Using the same machine utilization cut-off value of 30 per cent, the 40 cells were merged and finally 8 cells were formed. Table 4.13 presents the summary details for this design. Finally, Mr Prakash presented a

comparison of the two approaches based on several criteria (see Table 4.14 for details).

The task before the executives at AIL was to identify an appropriate configuration for the machine shop at Unit 1.

TABLE 4.13 DETAILS OF CELLS AS PER THE MS APPROACH FOR REDESIGN

Cell number	Number of Machines	Number of Components	Number of Exceptions	Number of Machines to be Procured
1	10	112	29	2
2	6	80	4	1
3	7	127	6	2
4	5	47	—	1
5	10	104	25	1
6	11	122	16	1
7	7	89	39	1
8	22	91	9	—
Total	76	772	128	9

TABLE 4.14 A COMPARISON OF THE TWO APPROACHES

Si. No.	Criterion	AM	MS	Remarks
1.	Implementation cost related			
	(a) Number of cells	13	13	More supervisors in AM
	(b) Number of machines to be procured	8	9	More investment in AM
2.	Cell-size variation			
	(a) Components	32–93	47–127	Larger variations in MS
	(b) Machines	3–15	5–22	Larger variations in MS
	(c) Utilization of machines			
	Maximum	84	92	Capacity more balanced in MS
	Minimum	30	42	
	Average	67	84	Better utilization in MS
3.	No. of exceptions	275	128	More exceptions in AM
4.	Product focus	Well focused	Blurred	Simplified production planning in AM
5.	Production control	Satisfactory	Better	More exceptions in AM makes control difficult
6.	Quality control			
	(a) Requirement of inspection equipment and tools	Less	More	
	(b) Technical documents	Unique	Mixed	AM is better because of product focus
	(c) Ability to direct improvement efforts	More	Less	Each group in AM can identify distinctly to an end product/ sub-assembly
7.	Liaison with design	Simple	Complex	Lack of sub-assembly focus in MS makes it difficult
8.	Document control	Easy	Difficult	Standards vary between the modules. Hence it is easy to distribute documents related to these in AM
9.	Skill improvement/training	Focus on modules	Focus on machining processes	In AM, skill improvement with the end product as focus is possible.

Note: AM = assembly-module approach; MS = machining similarity-based redesign

QUESTIONS FOR DISCUSSION

1. Should AIL continue with the current manufacturing set-up? Articulate three reasons in support of your argument.
2. Classify the two alternatives provided by Mr Prakash using your knowledge of process and technology choices for manufacturing organizations.
3. What are the pros and cons of the two alternative designs that Mr Prakash has proposed for changing the manufacturing shop configuration?

4.14 MULTIPLE-CHOICE QUESTIONS

- 4-1. What are the three important aspects that have a significant influence on the process?
A. Volume, Variety and Flow
B. Knowledge, Material and Choices
C. Product, Workers and Skill
D. Quantity, Processing and Category
- 4-2. The third category of flow is:
A. Jumbled flow system
B. Alternative flow
C. Mass production
D. Raw material
- 4-3. What are the three important aspects that has a significant influence on the process?
A. Turn-around time, Capital, and Investment
B. Volume, Variety and Flow
C. Volume, Variety and its Quality
D. None of the above
- 4-4. Physical arrangement of various resources available in the system to improve the performance of the operating system, thereby providing better customer service is termed as:
A. Manpower recruitment
B. Layout planning
C. IT solutions
D. Service department
- 4-5. What are the different types of “Manufacturing/Plan” Layout?
A. Budget Layout, Financial Layout
B. Building Layout, Emergency Exit Layout, Organization Layout
C. Process Layout, Product Layout, Group Technology Layout, Fixed Position Layout
D. Production Layout, Sales Layout
- 4-6. What is the most important task in a process based industries?
A. Adhere to the safety protocol
B. To make continuous improvement and capacity de-bottlenecking
C. To achieve the outcome within the stipulated time limit
D. To achieve incremental addition in the productivity
- 4-7. The design of a process layout is:
A. One-step process
B. Two-step process
C. Three-step process
D. Four-step process
- 4-8. Fixed layouts are used in?
A. Small industries
B. Only applies to new venture for a period of one year
C. Medium scale industries
D. Large project-based industries
- 4-9. Layouts are designed with the objective of subdividing a universe of machines and components into sub-groups such that each group of components forms a part of the family and is endowed with a corresponding sub- group of machines known as:
A. GT layout
B. Machine group
C. Diagonalized structure
D. Production flow
- 4-10. What promises significant reduction in manufacturing lead time, better space utilization, and cost reduction due to better utilization of resources and manpower?
A. Multiple workers multiple machine layout
B. Multiple workers one machine layouts
C. One worker one machine layouts
D. One worker multiple machine layouts

Answer Keys: 4-1. A 4-2. A 4-3. B 4-4. B 4-5. C 4-6. B 4-7. C 4-8. D 4-9. B 4-10. D

5

CHAPTER

PRODUCT AND SERVICE DESIGN

5.1	New Product Development
5.1.1	Generating New Products
5.1.2	Product Development System
5.2	Manufacturability and Value Engineering
5.3	Quality Function Deployment (QFD)
5.3.1	Organizing for Product Development
5.4	Process–Chain–Network (PCN) Analysis
5.4.1	Adding Service Efficiency
5.4.2	Documents for Services
5.5	Summary
5.6	Formula Review
5.7	Review Questions
5.8	Problems
5.9	Net-Wise Exercises
5.10	Mini Project
5.11	Case Study
5.12	Multiple-choice Questions

PROCESS DESIGN AT MILLTEC MACHINERY

NOTES

Milltec Machinery Private Limited is a Bangalore-based manufacturer of equipment for rice and flour mills. During the short period for which they have been in operation, they achieved significant growth on account of the product range they have been able to bring to this sector of the industry. They manufacture a range of machines under four broad product categories: re-cleaning, hulling, polishing, and grading. Within each category, they have three or four sub-product categories and a range of sizes and operating characteristics, resulting in over 200 different products. Machine-tool manufacturing typically involves a certain amount of customization.

Designing appropriate manufacturing processes, selection of the right machine tools and other manufacturing resources is critical to improved performance and operational control at Milltec. Milltec employs the latest technology in manufacturing. It believes in producing machinery of international standards at affordable prices. Laser cutting is used in all sheet-metal operations to ensure precise assembly. Milltec uses the CNC press brake for uniform bending of sheet-metal parts. CNC machining centres are utilized to produce machined components. Milltec uses MiG welding stations for the welding operations.

Once the type and number of machines required for the manufacturing process is identified, the next important step is to design an appropriate layout of these resources on the shop floor. At Milltec, manufacturing is organized on a functional basis. The fabrication shop is spread over 750 m². The assembly and paint shops are spread over 2,000 m² and the machine shop over 500 m². Milltec has a dedicated engineering department to cater to the erection and commissioning of projects. It also has an R&D department.

Over the years, Milltec has realized the importance of addressing manufacturing process design and layout issues. Through a process of continuous improvement, Milltec employees and management have realized that several benefits arise out of designing an appropriate layout and workplace organization. These benefits include better production planning and control, less work-in-process inventory, reduced cost of production, better employee morale, and greater visibility of problems at the workplace.

Source: Company sources, <http://www.milltecmachinery.com>

5.1 NEW PRODUCT DEVELOPMENT

5.1.1 Generating New Products

Because products die; because products must be weeded out and replaced; because firms generate most of their revenue and profit from new products—product selection, definition, and design take place on a continuing basis. Consider recent product changes: DVDs to video streaming, coffee shops to Starbucks lifestyle coffee, traveling circuses to Cirque du Soleil, landlines to cell phones, cell phone to smart phones, and an Internet of digital information to an Internet of “things” that connects you and your smart phone to your home, car, and doctor. And the list goes on. Knowing how to successfully find and develop new products is a requirement.

Student Tip

Societies reward those who supply new products that reflect their needs.

Aggressive new product development requires that organizations build structures internally that have open communication with customers, innovative product development cultures, aggressive R&D, strong leadership, formal incentives, and training. Only then can a firm profitably and energetically focus on specific opportunities such as the following:

1. *Understanding the customer* is the premier issue in new-product development. Many commercially important products are initially thought of and even prototyped by users rather than producers. Such products tend to be developed by “lead users”—companies, organizations, or individuals that are well ahead of market trends and have needs that go far beyond those of average users. The operations manager must be “tuned in” to the market and particularly these innovative lead users.
2. *Economic change* brings increasing levels of affluence in the long run but economic cycles and price changes in the short run. In the long run, for instance, more and more people can afford automobiles, but in the short run, a recession may weaken the demand for automobiles.
3. *Sociological and demographic change* may appear in such factors as decreasing family size. This trend alters the size preference for homes, apartments, and automobiles.
4. *Technological change* makes possible everything from smart phones to iPads to artificial hearts.
5. *Political and legal change* brings about new trade agreements, tariffs, and government requirements.
6. Other changes may be brought about through *market practice*, *professional standards*, *suppliers*, and *distributors*.

Operations managers must be aware of these dynamics and be able to anticipate changes in product opportunities, the products themselves, product volume, and product mix.

NOTES

NOTES

5.1.2 Product Development System

An effective product strategy links product decisions with other business functions, such as R&D, engineering, marketing, and finance. A firm requires cash for product development, an understanding of the marketplace, and the necessary human talents. The product development system may well determine not only product success but also the firm's future. Figure 5.1 shows the stages of product development. In this system, product options go through a series of steps, each having its own screening and evaluation criteria, but providing a continuing flow of information to prior steps.

Optimum product development depends not only on support from other parts of the firm but also on the successful integration of all 10 of the OM decisions, from product design to maintenance. Identifying products that appear likely to capture market share, be cost-effective, and be profitable but are, in fact, very difficult to produce may lead to failure rather than success.

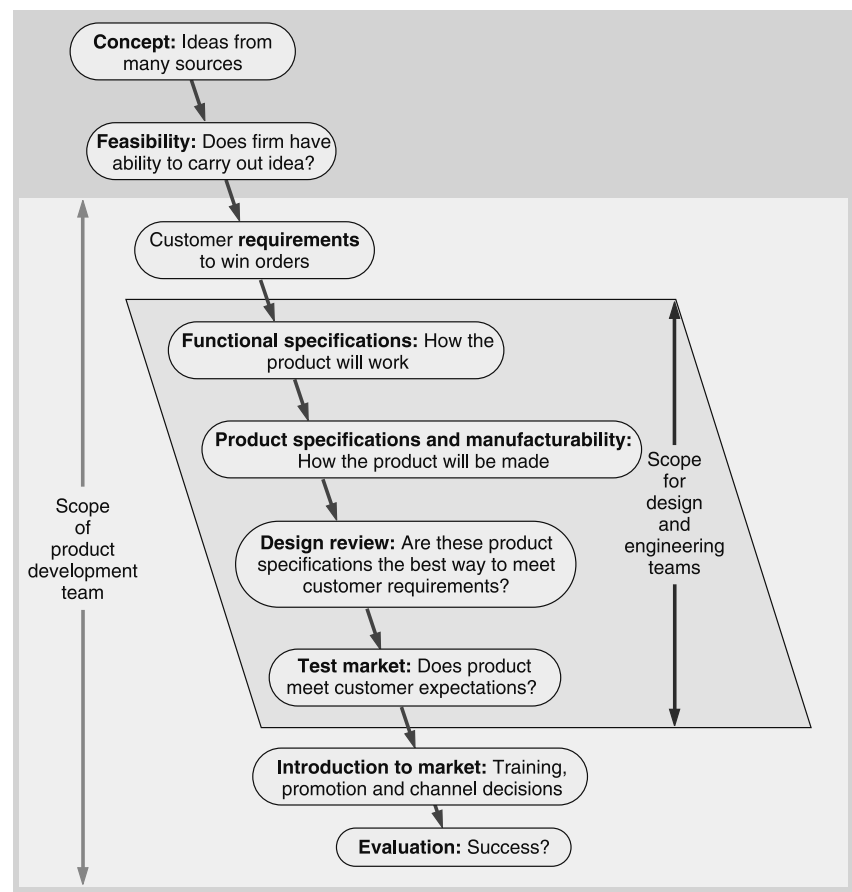


FIGURE 5.1 Product Development Stages

Product concepts are developed from a variety of sources, both external and internal to the firm. Concepts that survive the product idea stage progress through various stages, with nearly constant review, feedback, and evaluation in a highly participative environment to minimize failure.

5.2 MANUFACTURABILITY AND VALUE ENGINEERING

Manufacturability and value engineering activities are concerned with improvement of design and specifications at the research, development, design, and preproduction stages of product development. In addition to immediate, obvious cost reduction, design for manufacturability and value engineering may produce other benefits. These include:

Manufacturability and value engineering

Activities that help improve a product's design, production, maintainability, and use.

1. Reduced complexity of the product.
2. Reduction of environmental impact.
3. Additional standardization of components.
4. Improvement of functional aspects of the product.
5. Improved job design and job safety.
6. Improved maintainability (serviceability) of the product.
7. Robust design.

Manufacturability and value engineering activities may be the best cost-avoidance technique available to operations management. They yield value improvement by focusing on achieving the functional specifications necessary to meet customer requirements in an optimal way. Value engineering programs typically reduce costs between 15% and 70% without reducing quality, with every dollar spent yielding \$10 to \$25 in savings. The cost reduction achieved for a specific bracket via value engineering is shown in Figure 5.2.

5.3 QUALITY FUNCTION DEPLOYMENT (QFD)

Quality function deployment (QFD) refers to both (1) determining what will satisfy the customer and (2) translating those customer desires into the target design. The idea is to capture a rich understanding of customer wants and to identify alternative process solutions. This information is then integrated into the evolving product design. QFD is used early in the design process to help determine *what will satisfy the customer* and *where to deploy quality efforts*.

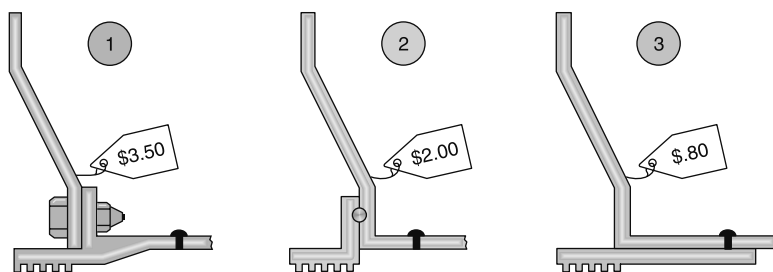


FIGURE 5.2 Cost Reduction of a Bracket via Value Engineering

NOTES

N O T E S

Quality function deployment (QFD)

A process for determining customer requirements (customer “wants”) and translating them into the attributes (the “hows”) that each functional area can understand and act on.

One of the tools of QFD is the **house of quality**, a graphic technique for defining the relationship between customer desires and product (or service). Only by defining this relationship in a rigorous way can managers design products and processes with features desired by customers. Defining this relationship is the first step in building a world-class production system. To build the house of quality, we perform seven basic steps:

House of quality

A part of the quality function deployment process that utilizes a planning matrix to relate customer “wants” to “how” the firm is going to meet those “wants.”

1. Identify customer *wants*. (What do customers want in this product?)
2. Identify *how* the good/service will satisfy customer wants. (Identify specific product characteristics, features, or attributes and show how they will satisfy customer *wants*.)
3. Relate customer *wants* to product *hows*. (Build a matrix, as in Example 1, that shows this relationship.)
4. Identify relationships between the firm’s *hows*. (How do our *hows* tie together? For instance, in the following example, there is a high relationship between low electricity requirements and auto focus, auto exposure, and number of pixels because they all require electricity. This relationship is shown in the “roof” of the house in Example 1.)
5. Develop importance ratings. (Using the *customer’s* importance ratings and weights for the relationships shown in the matrix, compute *our* importance ratings, as in Example 1.)
6. Evaluate competing products. (How well do competing products meet customer wants? Such an evaluation, as shown in the two columns on the right of the figure in Example 1, would be based on market research.)
7. Determine the desirable technical attributes, your performance, and the competitor’s performance against these attributes. (This is done at the bottom of the figure in Example 1.)

The following series of overlays for Example 1 show how to construct a house of quality.

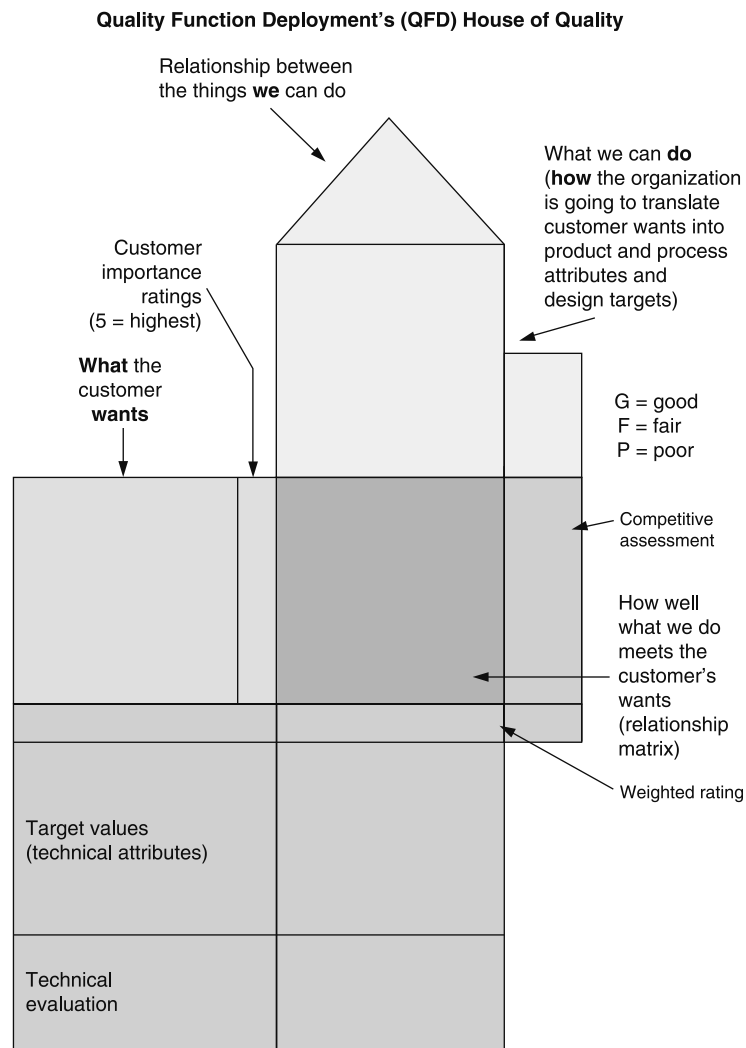
Another use of quality function deployment (QFD) is to show how the quality effort will be *deployed*. As Figure 5.3 shows, *design characteristics* of House 1 become the inputs to House 2, which are satisfied by *specific components* of the product. Similarly, the concept is carried to House 3, where the specific components

Example 1

CONSTRUCTING A HOUSE OF QUALITY

Great Cameras, Inc., wants a methodology that strengthens its ability to meet customer desires with its new digital camera.

APPROACH ► Use QFD's house of quality.



SOLUTION ► Build the house of quality for Great Cameras, Inc. We do so here using Overlays 1, 2, 3, and 4.

INSIGHT ► QFD provides an analytical tool that structures design features and technical issues, as well as providing importance rankings and competitor comparison.

LEARNING EXERCISE ► If the market research for another country indicates that “lightweight” has the most important customer ranking (5), and reliability a 3, what is the new total importance ranking for low electricity requirements, aluminum components, and ergonomic design? [Answer: 18, 15, 27, respectively.]

RELATED PROBLEMS ► 5.3, 5.4, 5.5, 5.6, 5.7

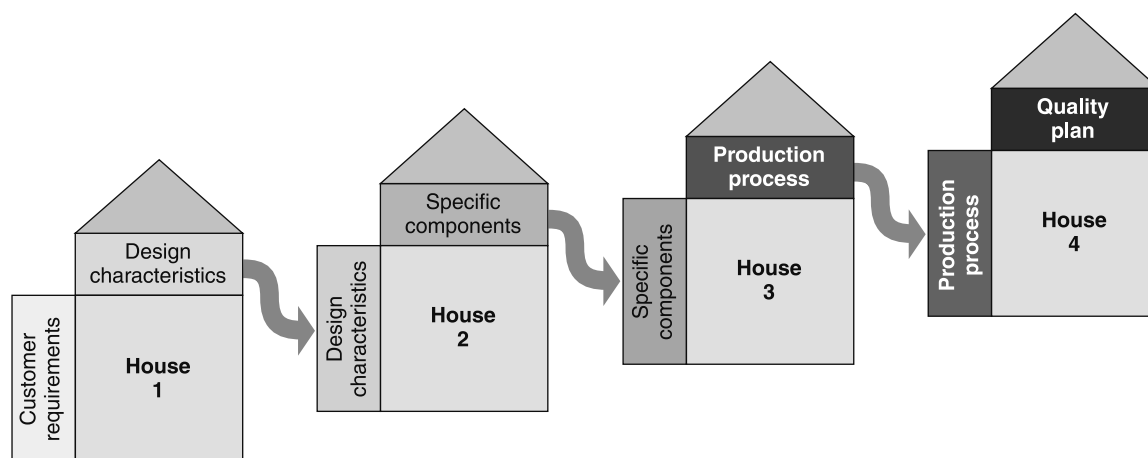


FIGURE 5.3 House of Quality Sequence Indicates How to Deploy Resources to Achieve Customer Requirements

NOTES

are to be satisfied through particular *production processes*. Once those production processes are defined, they become requirements of House 4 to be satisfied by a *quality plan* that will ensure conformance of those processes. The quality plan is a set of specific tolerances, procedures, methods, and sampling techniques that will ensure that the production process meets the customer requirements.

The QFD effort is devoted to meeting customer requirements. The *sequence* of houses is a very effective way of identifying, communicating, and deploying production resources. In this way we produce quality products, meet customer requirements, and win orders.

5.3.1 Organizing for Product Development

Let's look at four approaches to organizing for product development. *First*, the traditional U.S. approach to product development is an organization with distinct departments: a research and development department to do the necessary research; an engineering department to design the product; a manufacturing engineering department to design a product that can be produced; and a production department that produces the product. The distinct advantage of this approach is that fixed duties and responsibilities exist. The distinct disadvantage is lack of forward thinking: How will downstream departments in the process deal with the concepts, ideas, and designs presented to them, and ultimately what will the customer think of the product?

A *second* and popular approach is to assign a product manager to "champion" the product through the product development system and related organizations. However, a *third*, and perhaps the best, product development approach used in the U.S. seems to be the use of teams. Such teams are known variously as *product development teams*, *design for manufacturability teams*, and *value engineering teams*.

The Japanese use a *fourth* approach. They bypass the team issue by not subdividing organizations into research and development, engineering, production, and so forth. Consistent with the Japanese style of group effort and teamwork, these activities are all in one organization. Japanese culture and management style are more collegial and the organization less structured than in most Western countries. Therefore, the Japanese find it unnecessary to have “teams” provide the necessary communication and coordination. However, the typical Western style, and the conventional wisdom, is to use teams.

Product development teams are charged with the responsibility of moving from market requirements for a product to achieving a product success (refer to Figure 5.1). Such teams often include representatives from marketing, manufacturing, purchasing, quality assurance, and field service personnel. Many teams also include representatives from vendors. Regardless of the formal nature of the product development effort, research suggests that success is more likely in an open, highly participative environment where those with potential contributions are allowed to make them. The objective of a product development team is to make the good or service a success. This includes marketability, manufacturability, and serviceability.

Product development teams

Teams charged with moving from market requirements for a product to achieving product success.

Concurrent engineering implies speedier product development through simultaneous performance of the various stages of product development (as we saw earlier in Figure 5.1). Often the concept is expanded to include all elements of a product’s life cycle, from customer requirements to disposal and recycling. Concurrent engineering is facilitated by teams representing all affected areas (known as *cross-functional* teams).

Concurrent engineering

Simultaneous performance of the various stages of product development.

5.4 PROCESS-CHAIN-NETWORK (PCN) ANALYSIS

Process-chain-network (PCN) analysis, developed by Professor Scott Sampson, focuses on the ways in which processes can be designed to optimize interaction between firms and their customers.² A **process chain** is a sequence of steps that accomplishes an activity, such as building a home, completing a tax return, or

Process-chain-network (PCN) analysis

Analysis that focuses on the ways in which processes can be designed to optimize interaction between firms and their customers.

NOTES

NOTES

preparing a sandwich. A process participant can be a manufacturer, a service provider, or a customer. A network is a set of participants.

Process chain

A sequence of steps that accomplishes an identifiable purpose (of providing value to process participants).

Each participant has a *process domain* that includes the set of activities over which it has control. The domain and interactions between two participants for sandwich preparation are shown in the PCN diagram (Figure 5.4). The activities are organized into three *process regions* for each participant:

1. The *direct interaction* region includes process steps that involve interaction between participants. For example, a sandwich buyer directly interacts with employees of a sandwich store (e.g., Subway, in the middle of Figure 5.4).
2. The *surrogate (substitute) interaction* region includes process steps in which one participant is acting on another participant's resources, such as their information, materials, or technologies. This occurs when the sandwich *supplier* is making sandwiches in the restaurant kitchen (left side of Figure 5.4) or, alternately, when the *customer* has access to buffet ingredients and assembles the sandwich himself (right side of the figure). Under surrogate interaction, *direct* interaction is limited.
3. The *independent processing* region includes steps in which the sandwich supplier and/or the sandwich customer is acting on resources where each has maximum control. Most make-to-stock production fits in this region (left side of Figure 5.4; think of the firm that assembles all those prepackaged sandwiches

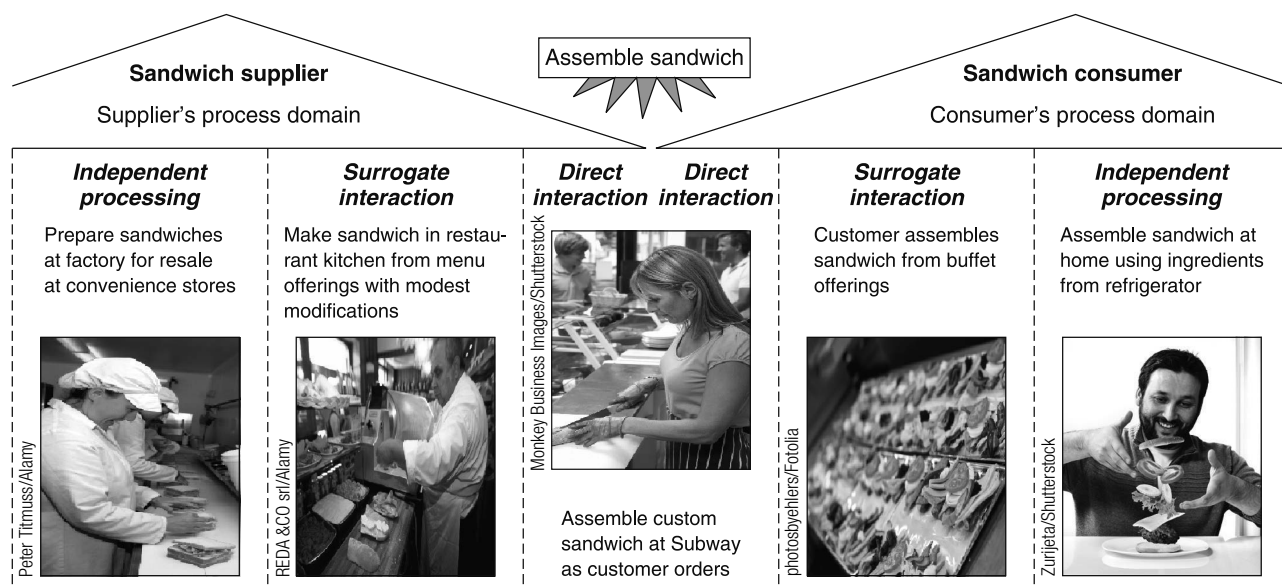


FIGURE 5.4 Customer Interaction Is a Strategic Choice

available in vending machines and convenience stores). Similarly, those sandwiches built at home occur to the right, in the customer's independent processing domain.

All three process regions have similar operating issues—quality control, facility location and layout, job design, inventory, and so on—but the appropriate way of handling the issues differs across regions. Service operations exist only within the area of *direct* and *surrogate interaction*.

From the operations manager's perspective, the valuable aspect of PCN analysis is insight to aid in positioning and designing processes that can achieve strategic objectives. A firm's operations are strategic in that they can define what type of business the firm is in and what value proposition it desires to provide to customers. For example, a firm may assume a low-cost strategy, operating on the left of Figure 5.4 as a manufacturer of premade sandwiches. Other firms (e.g., Subway) adopt a differentiation strategy with high customer interaction. Each of the process regions depicts a unique operational strategy.

Firms wanting to achieve high economies of scale or more control in their operations should probably position toward the independent processing region of their process domain. Firms intending to provide a value offering that focuses on customization should be positioned more toward the consumer's process domain. PCN analysis can be applied in a wide variety of business settings.

5.4.1 Adding Service Efficiency

Service productivity is notoriously low, in part because of customer involvement in the *design* or *delivery* of the service, or both. This complicates the product design challenge. We will now discuss a number of ways to increase service efficiency and, among these, several ways to limit this interaction.

Limit the Options Because customers may participate in the *design* of the service (e.g., for a funeral or a hairstyle), design specifications may take the form of everything from a menu (in a restaurant), to a list of options (for a funeral), to a verbal description (a hairstyle). However, by providing a list of options (in the case of the funeral) or a series of photographs (in the case of the hairstyle), ambiguity may be reduced. An early resolution of the product's definition can aid efficiency as well as aid in meeting customer expectations.

Delay Customization Design the product so that *customization is delayed* as late in the process as possible. This is the way a hair salon operates. Although shampoo and condition are done in a standard way with lower-cost labor, the color and styling (customizing) are done last. It is also the way most restaurants operate: How would you like that cooked? Which dressing would you prefer with your salad?

NOTES

Example 2

SERVICE DOCUMENTATION FOR PRODUCTION

First Bank Corp. wants to ensure effective delivery of service to its drive-up customers.

APPROACH ► Develop a “production” document for the tellers at the drive-up window that provides the information necessary to do an effective job.

SOLUTION ►

Documentation for Tellers at Drive-up Windows

Customers who use the drive-up teller windows rather than walk-in lobbies require a different customer relations technique. The distance and machinery between the teller and the customer raises communication barriers. Guidelines to ensure good customer relations at the drive-up window are:

- ◆ Be especially discreet when talking to the customer through the microphone.
- ◆ Provide written instructions for customers who must fill out forms you provide.
- ◆ Mark lines to be completed or attach a note with instructions.
- ◆ Always say “please” and “thank you” when speaking through the microphone.
- ◆ Establish eye contact with the customer if the distance allows it.
- ◆ If a transaction requires that the customer park the car and come into the lobby, apologize for the inconvenience.

Source: Adapted with permission from *Teller Operations* (Chicago, IL: The Institute of Financial Education, 1999): 32.

INSIGHT ► By providing documentation in the form of a script/guideline for tellers, the likelihood of effective communication and a good product/service is improved.

LEARNING EXERCISE ► Modify the guidelines above to show how they would be different for a drive-through restaurant. [Answer: Written instructions, marking lines to be completed, or coming into the store are seldom necessary, but techniques for making change and proper transfer of the order should be included.]

RELATED PROBLEM ► 5.10

NOTES

Modularization *Modularize* the service so that customization takes the form of changing modules. This strategy allows for “custom” services to be designed as standard modular entities. Just as modular design allows you to buy a high-fidelity sound system with just the features you want, modular flexibility also lets you buy meals, clothes, and insurance on a mix-and-match (modular) basis. Investments (portfolios of stocks and bonds) and education (college curricula) are examples of how the modular approach can be used to customize a service.

Automation Divide the service into small parts, and identify those parts that lend themselves to automation. For instance, by isolating check-cashing activity via ATM, banks have been very effective at designing a product that both increases customer service and reduces costs. Similarly, airlines have moved to ticketless service via kiosks. A technique such as kiosks reduces both costs and lines at airports—thereby increasing customer satisfaction—and providing a win-win “product” design.

Moment of Truth High customer interaction means that in the service industry there is a *moment of truth* when the relationship between the provider and the customer is crucial. At that moment, the customer's satisfaction with the service is defined. The moment of truth is the moment that exemplifies, enhances, or detracts from the customer's expectations. That moment may be as simple as a smile from a Starbucks barista or having the checkout clerk focus on you rather than talking over his shoulder to the clerk at the next counter. Moments of truth can occur when you order at McDonald's, get a haircut, or register for college courses. The operations manager's task is to identify moments of truth and design operations that meet or exceed the customer's expectations.

5.4.2 Documents for Services

Because of the high customer interaction of most services, the documents for moving the product to production often take the form of explicit *job instructions* or *script*. For instance, regardless of how good a bank's products may be in terms of checking, savings, trusts, loans, mortgages, and so forth, if the interaction between participants is not done well, the product may be poorly received. Example 2 shows the kind of documentation a bank may use to move a product (drive-up window banking) to "production." Similarly, a telemarketing service has the product design communicated to production personnel in the form of a *telephone script*, while a *manuscript* is used for books, and a *storyboard* is used for movie and TV production.

NOTES

5.5 SUMMARY

- Effective product strategy requires selecting, designing, and defining a product and then transitioning that product to production. Only when this strategy is carried out effectively can the production function contribute its maximum to the organization. The operations manager must build a product development system that has the ability to conceive, design, and produce products that will yield a competitive advantage for the firm. As products move through their life cycle (introduction, growth, maturity, and decline), the options that the operations manager should pursue change. Both manufactured and

service products have a variety of techniques available to aid in performing this activity efficiently.

- Written specifications, bills of material, and engineering drawings aid in defining products. Similarly, assembly drawings, assembly charts, route sheets, and work orders are often used to assist in the actual production of the product. Once a product is in production, value analysis is appropriate to ensure maximum product value. Engineering change notices and configuration management provide product documentation.

5.6 FORMULA REVIEW

- The total cost of the process layout, $TC = \sum_{i=1}^n \sum_{j=1}^n F_{ij} D_{ij} C_{ij}$
- Actual (desired) cycle time = $\frac{\text{Available time}}{\text{Actual(desired) production}}$

- Minimum number of work stations required = $\frac{\text{Sum of all task times}}{\text{Cycle time}}$
- Average resource utilization = $\frac{\text{Sum of all task times}}{\text{Number of workstations} \times \text{cycle time}}$

5.7 REVIEW QUESTIONS

- What are the key determinants of process characteristics in operations? Give specific examples to show how they influence process characteristics.
- Is there any relationship between volume, variety and flow with respect to process design?
- How is the continuous flow system different from the intermittent flow system? What are the key implications for operations managers?
- Give three examples for each of the following and identify two important issues of interest to an operations manager:
 - A mass production system
 - A continuous flow system
 - A flow shop
 - An intermittent flow system
 - Job shop
 - A process industry
 - A batch production system
 - Project organization
- As an operations manager, which areas would you focus on if you were to manage a jumbled flow process?
- How does the choice of the layout affect the operational performance of an organization?
- What is the relationship between the volume, variety and flow characteristics of an operating firm and its layout?
- What are the different types of layouts? How should an organization decide on which layout to choose?
- Is a group technology layout any different from a product layout? Give reasons for your answer.
- Identify an appropriate layout for each of the following situations. Justify your choice in a sentence or two:
 - A manufacturer of garments for Van Heusen
 - A multi-cuisine restaurant in a posh residential area in Mumbai
 - The overhaul of helicopters
 - A fabricator of custom-made PCBs for a large number of electronic applications
 - An eye hospital
 - A motor manufacturer manufacturing 4 product groups for worldwide markets
 - A manufacturer of large turbines for power sector applications
- Suppose you are given three alternative designs for the layout of a shop floor in a manufacturing organization. How will you decide which of the three is the most appropriate?
- Critically compare quantitative and qualitative approaches to designing process layouts.
- How does a construction method of layout design differ from the improvement method? Which method is preferable and why?
- What do you understand from the phrase “cellular manufacturing system”?
- What are the reasons for the growing application of group technology concept for layout design?
- What do you understand by the term FMS? What are the characteristic features of an FMS?
- How does an FMS help in improving the process design in the case of mid-volume, mid-variety manufacturing?
- What are the various types of automated material-handling devices available for use in a computer integrated manufacturing system?
- What do you understand by the term complexity of operations management? Explain.

5.8 PROBLEMS

- A mass producer of an industrial component currently has a daily production rate of 1,200 pieces. The company works for 16 hours a day. The production process currently employs five stations. There are 12 tasks involved in the manufacturing process and together they need 190 seconds.
 - What is the actual cycle time for the component?
 - Compute the average utilization of resources employed to manufacture the component.
 - If the company wants to work with 85% utilization, by how much they should increase the daily production to achieve this?
 - Suppose, there is a 10% increase in the demand. Do they need to make any changes in the layout?
- A company procures a few components from the market and assembles them into a standard product and supplies it to its customers. There is enough demand for the product. If the company employs four stations for the assembly, it has 92% average utilization of the resources. The sum of the task time for the process is 180 seconds.
 - What is the cycle time for the process?
 - What is their daily production rate if they work in an 8-hour shift?
 - If they add one more station in the assembly process what would be the new cycle time if the utilization drops to 79%?
 - By how much can they increase the production on account of the reduced cycle time?
- Following data are available in the design of a product-based layout in which a certain number of work stations are deployed. The sum of all task times is 200 seconds. The cycle time is 50 seconds and the average resource

utilization is 80%. If one more work station is added, the utilization drops to 74.07% and the new cycle time is 45 seconds.

- (a) How many work stations are used in the product layout design?
 - (b) What is the daily production rate if a day consists of 450 minutes?
 - (c) What is the maximum production possible in the current design?
4. A manufacturer of spark plugs for the mass market would like to design the final assembly shop and requires certain data for the process. The factory works for two shifts and the total available time is 15 hours per day.
- (a) There are five tasks involved in the final assembly and the task times (in seconds) are 3.5, 4.5, 2.0, 3.0 and 4.0. What are the minimum and the maximum outputs possible from the factory?
 - (b) The daily production needs to be 4,000 plugs. What should be the cycle time of the operations at the assembly shop?
 - (c) If the firm wants to use three workstations, what will be the best output possible? By how much will the output increase if it adds one more workstation?
5. A fast-food restaurant that operates in Connaught Place, New Delhi, specializes in offering food during peak hours. The restaurant operates only for two hours in the morning and promises faster turnaround for customer orders. The restaurant needs to design the workstations at the front end of the kitchen to assemble all customer orders. The back end of the kitchen provides the required food items. A study of the tasks, their dependencies and times taken on an average to assemble reveals the information shown in Table 5.1:
- (a) If, during a typical peak hour operation, the restaurant is able to take 144 orders, what is the maximum permissible cycle time for the assembly workstations?
 - (b) If the restaurant desires to set a target of the above cycle time, what is the minimum number of workstations they should have at the assembly?

TABLE 5.1 DEPENDENCIES AND TIME TAKEN TO ASSEMBLE/SERVE FOOD

Task ID	Description	Task Time (Seconds)	Predecessor
A	Set up the serve tray	20	None
B	Arrange plates, cups, tissues	20	A
C	Place the main item in the plate	35	B
D	Fill in the side dishes	35	B
E	Place a filled in water cup	20	D
F	Arrange cutlery items in the plate	20	C, E
G	Verify the order	15	F
H	Prepare a bill for the order	35	G

- (c) Design the assembly with the minimum number of workstations. Can it serve 180 customers during the peak hour? If not, arrive at a feasible design.
- (d) For the final design that you have made, compute the average utilization of workstations.
- (e) The restaurant management is aware that the bottleneck is in terms of their capacity to serve. If they could increase the capacity, they would earn additional revenue. The marginal revenue from an order will be ₹100 and the marginal daily cost of adding a workstation is ₹2,700. What should the management do?

5.9 NET-WISE EXERCISES

1. Caterpillar is the largest maker of construction and mining equipment. To know more about the products offered by them, visit <http://india.cat.com/cda/layout?m=81920&x=7> and click on **Products**. Navigate through their product lines and offerings.

Haldia Petrochemicals is the one of the largest petrochemicals complexes in India. Go to <http://www.haldiapetrochemicals.com/index.jsp> and click on **Products** on the left side of the screen. Visit various sub-links in this section. Also click on the link **About Us**. There is a subsection on manufacturing processes. Visit other parts of the Web site.

Take a look at the photos of the plant loaded onto the main page.

Based on your understanding of the two organizations, compare and contrast the product flow and process characteristics.

- (a) Identify three distinct features of Caterpillar and relate them to the design of the manufacturing process for the organization.
- (b) Why would you classify Haldia Petrochemicals as a process industry? Identify three characteristics that you find in Haldia Petrochemicals that are peculiar to a process industry.

5.10 MINI PROJECT

1. There are several manufacturing and service organizations in every part of the country. Select out of these three organizations that belong to any of the following categories: a process industry; a mass production system; a service set-up; a mid-volume, mid-variety manufacturer.

Visit the selected manufacturing and service organizations, make a formal plant tour of the organization and collect data on various aspects of operations management. After you have recorded all the key observations based on the visit, prepare a list of items on which you would like to collect additional information and obtain them. Also interview the plant manager, the process planning team, and the line supervisor. Note down their key observations about the plant's functioning. Based on this work, prepare a final report that contains the following information:

- (a) The process flow diagram of the plant
 - (b) The nature of the production system
 - (c) The key challenges for operations management
 - (d) Status of current operations management practices in the organization
 - (e) Specific areas in which improvement potential exist for the organization
2. There are alternative approaches to layout design. The use of software packages for this purpose has increased manifold in recent years. Prepare a project report to answer the following question pertaining to the use of software for layout design:
 - (a) Scan the Internet and prepare a list of available software solutions for layout design. Briefly explain the salient features of the software.
 - (b) Develop a classification scheme to list the software that you have identified. Can you recommend a specific set of application domains of the software on the basis of this classification scheme?
 - (c) Select one software package from each of your classification schemes and critically compare them. Identify the specific advantages and disadvantages of each package.
 3. There are several companies that have recently made significant changes in their layout design. Prominent in the list are automobile and auto-components manufacturing firms. Identify one such company in your locality and perform the following studies and submit a report of your findings:
 - (a) What were the major drivers of the layout change decision in the organization?
 - (b) Enumerate the salient aspects of the alternative layout design process (methodology used to arrive at the new design, tools and techniques used, scope of the project, organizational aspects of driving the change process etc.)
 - (c) Understand the implementation aspects of the layout design process (methodology of implementation, time taken, chronology of events leading to completion of the process etc.)
 - (d) What were the main benefits that accrued to the organization as a result of this layout design change?
 - (e) How did the layout change affect various people in the system including the production workers, supervisors, managers, customers, and suppliers?

5.11 CASE STUDY

De Mar's Product Strategy

De Mar, a plumbing, heating, and air-conditioning company located in Fresno, California, has a simple but powerful product strategy: Solve the customer's problem no matter what, solve the problem when the customer needs it solved, and make sure the customer feels good when you leave. De Mar offers guaranteed, same-day service for customers requiring it. The company provides 24-hour-a-day, 7-day-a-week service at no extra charge for customers whose air conditioning dies on a hot summer Sunday or whose toilet overflows at 2:30 A.M. As assistant service coordinator Janie Walter puts it: "We will be there to fix your A/C on the fourth of July, and it's not a penny extra. When our competitors won't get out of bed, we'll be there!"

De Mar guarantees the price of a job to the penny before the work begins. Whereas most competitors guarantee their work for 30 days, De Mar guarantees all parts and labor for one year. The company assesses no travel charge because "it's not fair to charge customers for driving out." Owner Larry Harmon says: "We are in an industry that doesn't have the best reputation. If we start making money our main goal, we are in trouble. So I stress customer satisfaction; money is the by-product."

De Mar uses selective hiring, ongoing training and education, performance measures, and compensation that incorporate customer satisfaction, strong teamwork, peer pressure, empowerment, and aggressive promotion to

implement its strategy. Says credit manager Anne Semrick: “The person who wants a nine-to-five job needs to go somewhere else.”

De Mar is a premium pricer. Yet customers respond because De Mar delivers value—that is, benefits for costs. In 8 years, annual sales increased from about \$200,000 to more than \$3.3 million.

QUESTIONS FOR DISCUSSION

1. What is De Mar's product? Identify the tangible parts of this product and its service components.
2. How should other areas of De Mar (marketing, finance, personnel) support its product strategy?
3. Even though De Mar's product is primarily a service product, how should each of the 10 strategic OM decisions in the text be managed to ensure that the product is successful?

Source: Reprinted with the permission of The Free Press, from *On Great Service: A Framework for Action* by Leonard L. Berry.

Micromax Dominating the Indian Market

Micromax is an Indian consumer electronics company headquartered in Gurgaon, Haryana. The company was established as an IT software company operating in the embedded devices domain, it later entered the mobile handset business. By 2010, it was one of the largest domestic companies making handsets in the low-cost feature phone segment in India. As of Q3 2014, Micromax was the tenth largest smartphone vendor in the world. It started selling mobile telephones in 2008, focusing on low pricing to compete with international brands. Micromax's co-founder Rahul Sharma once saw a public call office being powered by a truck battery because of frequent power cuts in its locale. It prompted him to launch a feature telephone with an extended battery life. Micromax launched Xli, its first telephone with a month-long battery back-up. In 2014, Micromax's sales exceeded those of Samsung to become the mobile telephone manufacturer shipping the most telephones in one quarter in India.

Some of the reasons for their success is providing best value for money to the consumers and they managed to create a compelling supply chain. Company sourced their handsets from China, ensuring low manufacturing cost. It tied up with world-class manufacturers such as FoxConn, who had experience in manufacturing for Apple and Samsung. This ensured that the quality of the handsets is maintained at the global standard. Secondly, they adopted Android very early. They realized the potential of the Android OS and switched completely to it. They were able to deliver amazing features at low prices and there by attract youth and early android adopters. Thirdly, they have made their distribution channel very efficient in such a way that whereas a big international brand typically requires 18 months to launch a product, Micromax can do that in a month or two. It has achieved this by performing a thorough analysis of the needs of their customers and identification of their markets. Initially they used to source

pre-built handsets from China, but now they source just the components and assemble them in India, customizing them to the needs. Micromax is basically on a path to glory. It has been challenging the global companies such as Nokia, Motorola, Samsung and others ever since its inception. Micromax has achieved this huge success primarily by identification of the right market, thoroughly understanding the needs of those markets and catering to them with world-class products.

In order to be competitive in future, Micromax announced a strategic partnership with digital payments company TranServ and global payments technology major Visa to offer the next generation of payments solutions in India in April 2016. In future Micromax expects as much as 80% of its smartphone revenue to come from 4G-enabled devices by 2017 and expects the shift to 4G to be much faster than the move to 3G.

Adapted from https://en.wikipedia.org/wiki/Micromax_Informatics, as accessed on February 2, 2017 at 3.57pm

<http://cmuscm.blogspot.in/2014/01/micromax-indias-largest-rural-mobile.html>, as accessed on February 2, 2017 at 3.58pm

Muntazir Abbas, 'Rural India poised for a mobile revolution as customers upgrade to smartphones', <http://economictimes.indiatimes.com/tech/internet/rural-india-poised-for-a-mobile-revolution-as-customers-upgrade-to-smartphones/article-show/52158748.cms>, as accessed on February 2, 2017 at 3.58pm

QUESTIONS FOR DISCUSSION

1. Discuss the factors responsible for the success of micromax.
2. What product innovation has micromax done in order to compete with established companies in India?
3. What should micromax do to stay competitive in future?

5.12 MULTIPLE-CHOICE QUESTIONS

- 5-1.** A product's life cycle is divided into four stages, including:
A. introduction.
B. growth.
C. maturity.
D. all of the above.
- 5-2.** Product development systems include:
A. bills of material.
B. routing charts.
C. functional specifications.
D. product-by-values analysis.
E. configuration management.
- 5-3.** A house of quality is:
A. a matrix relating customer "wants" to the firm's "hows."
B. a schematic showing how a product is put together.
C. a list of the operations necessary to produce a component.
D. an instruction to make a given quantity of a particular item.
E. a set of detailed instructions about how to perform a task.
- 5-4.** Time-based competition focuses on:
A. moving new products to market more quickly.
B. reducing the life cycle of a product.
C. linking QFD to PLM.
D. design database availability.
E. value engineering.
- 5-5.** Products are defined by:
A. value analysis.
B. value engineering.
C. routing sheets.
D. assembly charts.
E. engineering drawings.
- 5-6.** A route sheet:
A. lists the operations necessary to produce a component.
B. is an instruction to make a given quantity of a particular item.
C. is a schematic showing how a product is assembled.
D. is a document showing the flow of product components.
E. all of the above.
- 5-7.** The three process regions in a process-chain-network diagram are:
A. manufacture, supplier, customer
B. direct and surrogate, customer, provider
C. independent, dependent, customer interaction
D. direct interaction, surrogate interaction, independent processing
- 5-8.** Decision trees use:
A. probabilities.
B. payoffs.
C. logic.
D. options.
E. all of the above.

Answer Keys: 5-1. D 5-2. C 5-3. A 5-4. A 5-5. E 5-6. A 5-7. D 5-8. E

6

CHAPTER

PRODUCTIVITY AND PRODUCTION

6.1	Planning Hierarchies in Operations
6.2	Aggregate Operations Planning
6.3	The Need for Aggregate Operations Planning
6.4	Alternatives for Managing Demand
6.4.1	Reservation of Capacity
6.4.2	Influencing Demand
6.5	Alternatives for Managing Supply
6.5.1	Inventory-based Alternatives
6.5.2	Capacity Adjustment Alternatives
6.6	Basic Strategies for Aggregate Operations Planning
6.6.1	Level Strategy
6.6.2	Chase Strategy
6.6.3	Mixed Strategy
6.7	Aggregate Operations Planning Methods
6.7.1	Heuristic Methods for AOP
6.7.1	Optimal Methods for AOP
6.8	Operations Research Tools for Operations Planning
6.8.1	Aggregate Operations Planning Using the Transportation Model
6.8.2	The Linear Programming Model for AOP
6.8.3	The Linear Decision Rule (LDR)
6.9	Master Operations Scheduling
6.10	Summary
6.11	Review Questions
6.12	Problems
6.13	Case Study
6.14	Multiple-choice Questions

OPERATIONS PLANNING AT AMAZON TO IMPROVE SERVICE EFFECTIVENESS

NOTES

Amazon.com, started in 1995, is the largest internet retailer that offers new, used, and refurbished items in a number of categories, including books, music, food, apparel, kitchenware, and consumer electronics. Amazon's Customer Service Operations (CSO) provides service to customers through certain features on the company web site. These features allow customers to perform various activities, including tracking orders and shipments, reviewing estimated delivery dates, and cancelling unshipped items. Customers place orders and follow up on orders on the company web site. Customers who cannot resolve issues using the features on web site can either call the company's 800 number or send e-mail messages to customer service.

The e-mail messages and voice calls are annually estimated to be in millions with the peak just before and after Christmas. Operations planning to handle this requirement invariably amount to capacity planning of its contact centers. They must make decisions about hiring and training at internally managed centers and about the volume of voice calls and e-mail messages to allocate to external service providers.

The handling time for customer requests manifesting as voice calls and e-mail exchanges depends on attributes such as product type, customer type, and purchase type. Amazon uses these attributes to categorize customer requests to reflect the skill sets needed to resolve different issues. It has eight Planning Groups (PGs) dedicated to processing the requests. The internally managed contact centers are assigned to specific PGs and trained to handle both voice and e-mail contacts.

Operation planning begins with forecasting demand by product line, which is then converted into a forecast of orders. This would eventually determine the weekly forecasts of e-mails and voice contacts for the eight categories over the planning horizon. After the development of these contact forecasts, the capacity-planning team assesses the contact-handling capacity of each PG for each week of the planning horizon. If the capacity is less than forecast, resources are then hired externally.

Each week, the capacity-planning team in CSO employs a solution approach developed using an optimization model to address this operations planning problem. When the planning horizon is 52 weeks, the optimization model consists of approximately 134,000 constraints and almost 16,000 variables, where about 1,000 of these are both binary and integer.

With the optimization model the capacity-planning team is able to examine a number of scenarios and consider uncertainty by performing sensitivity analysis on the inputs to the planning process. After analyzing the output, they inform the HR team

the number of new CSRs Amazon will need to hire. They are also able to estimate the transfers needed into and out of the PGs, and future quantum of customer request volumes for the external service providers.

Source: Adapted from Kebblis, M.F. and Chen, M. (2006). 'Improving Customer Service Operations at Amazon.com', *Interfaces*, 36(5), pp 433-445.

NOTES

Operating systems in a business play the vital role of providing a continuous flow of goods and services to customers. For instance, a bicycle manufacturer may be producing at a daily rate of 10,000 bicycles in its factory. Similarly, an airline operating between the metropolitan cities of Mumbai and Delhi may be offering 2,000 passenger seats daily. However, these operating systems are not like a water tap, where, with a turn of the knob, the flow alters. Changing the daily production rate at the bicycle manufacturer to 15,000 or the seating capacity in the case of the airline to 2,500 are not simple and instantaneous decisions.

Operating systems are characterized by complex relationships between people, resources, market, and materials, and often involve lead time, affecting several of the processes. For example, if the order placed with suppliers is increased by 20 per cent, it requires considerable time at the suppliers' end to adjust the capacity to match the supply. Therefore, prior planning and strategies for dealing with real-time changes play a very important role in managing operating systems and ensuring a continuous flow of goods and services to the market. We address these issues in this chapter and several others in the book. In this chapter, we discuss planning methodologies that address capacity requirements in the medium term (typically a one-year period).

6.1 PLANNING HIERARCHIES IN OPERATIONS

It is important to understand that planning the operations in a manufacturing or a service system happens at different levels and at different time horizons. On the other hand, control of operations always happens in a short-term horizon (such as weekly, daily, or even on a shift-by shift-basis). Figure 6.1 graphically depicts the various steps involved in planning the operations in a hierarchical fashion.

We saw in the previous chapter how demand for products/services is estimated using medium-level forecasting techniques. Based on this input, a firm needs to begin the planning exercise. At the highest level, business planning is done typically for a period of one year. For example, organizations following the financial year between April and March perform a detailed planning exercise between November and February and arrive at the plan for the next year. The business plan is strategic in nature and addresses

NOTES

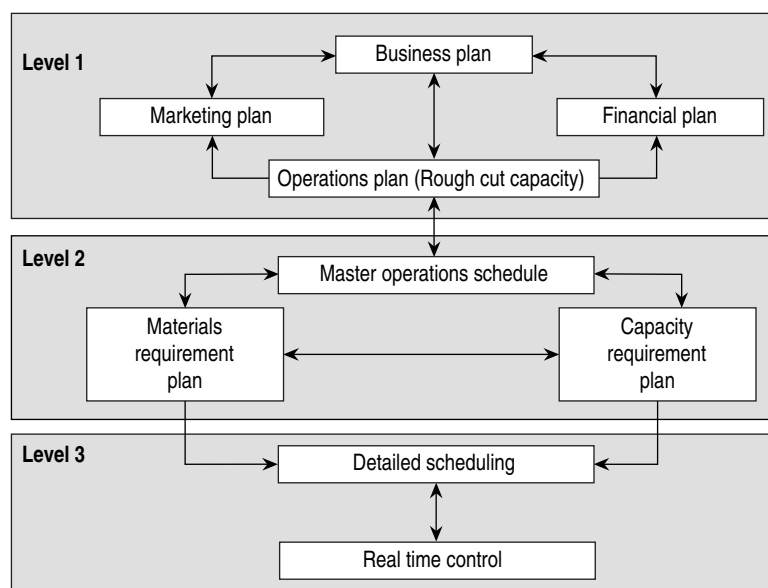


FIGURE 6.1 A hierarchical approach to planning

the key level of resources to be committed to meet a targeted market segment by answering the following questions:

- What is the likely demand for the next financial year?
- Should we meet the projected demand entirely or a portion of the projected demand?
- What are the implications of this decision on the overall competitive scenario and the firm's standing in the market?
- How is this likely to affect the operating system and planning in other functional areas of the business, such as marketing and finance?
- What resources should we commit to meet the chosen demand during the planning horizon?

These questions enable the top management to link the strategic intent to various operational choices to be made during the planning horizon. Moreover, they provide some clarity on the level of resources to be committed and the investments and policy decisions to be taken to execute the operational plans.

Once the level of resources to be committed is arrived at, rough-cut capacity planning needs to be done. During this stage, planning is done to adjust the demand and the available capacity on a period-by-period basis and ensure that the available capacity could match the demand. At this level of planning, information pertaining to products and resources is aggregated. For example, if a manufacturing organization is offering nine variants of a product, they are all aggregated into one "equivalent" model for the purpose of capacity calculations.

Consider for example the case of a fan manufacturer such as Crompton Greaves Limited (CGL). Suppose CGL manufactures

three variations of fans: standard, deluxe, and premium. Assume that manufacturing a unit of standard fan requires 1 hour of machine capacity. Let us also assume that the deluxe model requires 1.5 hours and the premium one 2 hours. Then, this information could be used to convert the demand for all variants into a single measure of capacity. For example if the demand for standard is 100, deluxe is 200, and premium is 75, then the aggregated requirement in terms of machine-hours is $100 \times 1.00 + 200 \times 1.5 + 75 \times 2 = 450$ hours.

Similarly, if a service provider offers three levels of service—silver, gold, and platinum—that may place varying demands on the resources, they will be aggregated into one service version on some basis for the purpose of planning. Therefore, planning done at this level is only rough-cut and is usually done for a period of 9–12 months.

At the second level, detailed planning is done to ensure that capacity, material, and other resources are available to meet the projected demand on a period-by-period basis. The capacity, material, and resource requirements are linked to specific variations in the products to be manufactured and services to be offered during the planning horizon. In the earlier example of the service firm, assume that the projected demand for the three types of services is 300, 200, and 50, respectively. If they need six different labour skills and four different kinds of equipment in varying proportions, the planning exercise is done to ensure adequate availability of these labour skills and equipment. Clearly, such a detailed capacity planning is done at a time much closer to the planning horizon. Therefore, detailed planning is done typically for a period of 1–3 months duration.

The final level of detailed planning is done almost at the time the plan needs to be executed. During this exercise, decisions concerning the use of specific resources to manufacture goods as per plan are taken. For example, after deciding to manufacture 1,500 bicycles of a particular variety during a month, the exact scheduling will be done. This means that the set of machines to be used for manufacturing and the time when the “job” needs to be launched into the system in order to meet the targeted completion time are decided. This level of planning is usually done during a period of 1–4 weeks.

After the plans are worked out to this level of detail, the actual execution of the plan is done using an operational control system. Basically, the plans are monitored and resources are committed as per plan to complete the requirements. However, in real time, the plans get disturbed due to resource unavailability, breakdowns, absenteeism, quality problems, and so on. Any problems arising in real time are addressed to ensure the targets are met with. It may call for a certain degree of re-planning.

An earlier planning exercise invariably results in a certain level of re-planning at a later stage.

NOTES

NOTES

The hierarchy in planning offers several advantages to the planner. Splitting the decision-making framework into various stages enables the planner to reduce the complexity of the planning process. Moreover, as more information is available, it can be incorporated into the planning exercise. When doing the first-level planning, the planner tends to make several assumptions about markets, products, and the competitive scenario for the next 9–12 months. However, at the third level of planning, the planner gets recent information and uses it to validate and modify several assumptions. On this basis, the planner may perform a re-planning exercise for the next 1–3 months. Therefore, it is important to realize that an earlier planning exercise invariably results in a certain level of re-planning at a later stage.

6.2 AGGREGATE OPERATIONS PLANNING

As the name suggests, it is a planning exercise done for operations using data at an aggregate level. Aggregate operations planning serves the critical role of translating the business plans and strategic intent to operational decisions. Using an **aggregate operations planning (AOP)** exercise, firms arrive at the quantity and timing of resources to be committed to ensure continuous flow of goods and services to customers. Usually, the decisions involve the amount of resources (productive capacity and labour hours) to be committed, the rate at which goods and services need to be produced during a period, and the inventory to be carried forward from one period to the next.

An **aggregate operations planning (AOP)** decision deals with the amount of resources (productive capacity and labour hours) to be committed, the rate at which goods and services need to be produced during a period, and the inventory to be carried forward from one period to the next.

For example, at the end of an aggregate operations planning exercise, a garments manufacturer may arrive at the following plan:

- ⇒ Produce 9,000 metres of cloth everyday during the period January–March, increase it to 11,000 metres during April–August, and change the operations rate to 10,000 metres during September–December.
- ⇒ Carry 10 per cent of monthly production as inventory during the first nine months of production.
- ⇒ Work on a one-shift basis throughout the year with 20 per cent overtime during July–October.

As we can see in this example, three critical decisions are made: the rate of production, the amount of inventory to carry, and the amount of resource (in terms of working hours) to be committed on a period-by-period basis. We shall see later that there are several options available for the planner and each has its own implications on cost and availability.

TABLE 6.1 AGGREGATE UNITS FOR OPERATIONS PLANNING: AN EXAMPLE

Product	Aggregate Unit of Capacity
Phenyl acetic acid	Metric tons
Data-entry systems	Numbers
Mini-computer	Value (ex-factory) in ₹
Printed circuit board	Square metres
Alloy iron castings	Metric tons
Cement	Metric tons
Multispeciality clinic	Patient-bed days
Bank	No. of accounts
Insurance Firm	No. of policies

The entire planning exercise is done on the basis of some aggregate unit. There is no single basis on which the demand data is aggregated. The only requirement is the need to establish equivalences between variations. For example, suppose that one standard model of a bicycle takes 30 machine-hours to manufacture, a deluxe model takes 60 machine-hours, and a sports model takes 90 machine-hours. It is obvious that manufacturing one deluxe model is equivalent to manufacturing two standard models and manufacturing one sports model is equivalent to manufacturing three standard models from a resource-consumption perspective. Then a monthly demand of 1,000 bicycles of the basic model, 500 of the deluxe model, and 250 of the sports model can be aggregated as 2,750 basic models on the basis of machine-hours.

Usually, the aggregated unit turns out to be some measure of the capacity. For example, a steel processing mill may make several variations of steel but may prefer to do the AOP on the basis of number of heats that can be extracted from a furnace. A textile manufacturer may manufacture several varieties of cloth but may plan on the basis of metres of cloth production. Similarly, a PCB manufacturer may fabricate thousands of customised PCBs but may plan on the basis of square metres of board that it can process. Therefore, AOP can be viewed as a capacity planning exercise over the medium term of 9–12 months. Table 6.1 is an extract of the aggregate measures of capacity in manufacturing and service organizations.

6.3 THE NEED FOR AGGREGATE OPERATIONS PLANNING

There are several reasons for every organization, whether engaged in product manufacturing or service delivery, to plan operations at an aggregate level. The following ones merit some attention:

⇒ *Demand fluctuations:* Organizations hardly experience stable or even demand. As we saw in the previous chapter, several

NOTES

NOTES

sectors of the manufacturing and service industry experience a significant upswing in demand during certain periods. The demand for garments in India is high between August and October due to the festive season. Prior planning is required to meeting surges in demand.

- ⇒ *Capacity fluctuations:* While demand fluctuations occur on account of seasonality, there are fluctuations in capacity too. The capacity available in the month of February will be 10 per cent lower than that in the month of May on account of fewer calendar (and working) days. Moreover, scheduled and unscheduled plant shutdowns have a significant impact on capacity availability.
- ⇒ *Difficulty level in altering operation rates:* Operational systems are complex. Varying the rate of operation from one level to another requires some amount of prior planning and coordination with related systems on the supplier and distributor sides. Thus, an organization manufacturing diesel engines at the rate of 4,000 engines per day cannot unilaterally change it to 5,000 engines. It needs to ensure that the required material, capacity, and other resources are available for the incremental plan. Similarly, it needs to ensure that the distributors have the required capacity to handle the additional load on them. Similar is the case of a restaurant offering multi-cuisine menu.
- ⇒ *Benefits of multi-period planning:* It is known that planning just for a particular period with no consideration of potential events in the near future amounts to a knee-jerk reaction than an attempt to reach optimal and cost-saving decisions. For example, while planning the production for a month, an organization is better off by taking into consideration the likely scenario in the next few months. If the estimates indicate a rising demand in the next few months, it is better to produce at a slightly higher, albeit even pace and accumulate inventory to handle the additional demand in the future than to react every month. Such month-by-month reactions to the market are not only expensive but also not feasible for the reasons discussed earlier.

It is evident that for several reasons, supply and demand may not match exactly on a period-by-period basis in an organization. For instance, during the festive season, a manufacturer of appliances, such as refrigerators, may experience higher demand than average since many employees might have received bonuses, special festival allowances, and advances. If the manufacturer has sufficient capacity to meet this demand, then there is no problem. On the other hand, if the capacity is less, then the organization has to either forego the unfulfilled demand or try to influence the demand or the supply.

Therefore, aggregate operations planning is done in an organization to match the demand with the supply on a period-by-period

Aggregate operations planning is done in an organization to match the demand with the supply on a period-by-period basis in a cost-effective manner.

NOTES

basis in a cost-effective manner. Matching the demand with the supply can be done in any one of the following ways:

- (a) For the supply capacity that an organization has during a period, ways by which the demand could be suitably modified are identified so that both are matched.
- (b) For the targeted demand to be fulfilled in every period, ways by which the supply is altered are identified so that both are matched.
- (c) During every period, both the demand and the supply are adjusted in such a way that both are matched.

Figure 6.2 provides a simple framework for the AOP exercise. The forecasting module in an organization merely indicates the likely demand for its product or service during a period. It does not throw light on the availability of sufficient capacity to meet the demand or the manner in which an organization needs to address issues pertaining to supply–demand mismatch. All these fall under the purview of the aggregate planning exercise.

The aggregate operations planning exercise begins by exploring alternative ways of modifying demand and arriving at the period-by-period net demand during the planning horizon. The second step is to further explore the possibilities of modifying the supply so that at the end of the exercise the demand and the supply are matched on a period-by-period basis. On the basis of this, various decisions on the level of inventory to be carried during every period, the rate at which goods and services are produced and the number of people to be engaged are arrived at. If required, additional arrangements for outsourcing and augmenting capacity are also made. In the final analysis, all decisions with respect to use of capacity are taken through an aggregate operations planning exercise.

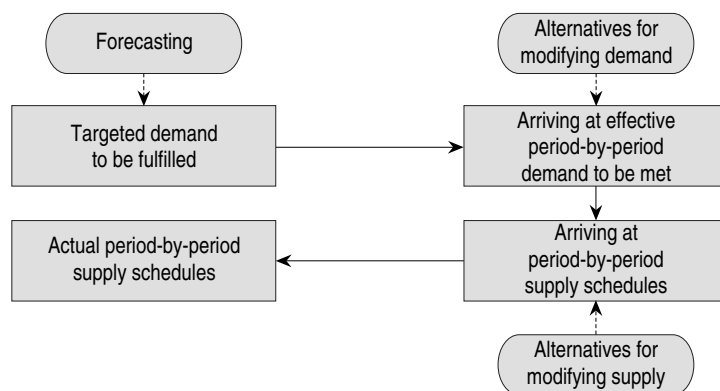


FIGURE 6.2 An aggregate operations planning framework

NOTES

6.4 ALTERNATIVES FOR MANAGING DEMAND

When you want to visit your doctor, don't you take an appointment? Do you make use of differential tariffs and make long distance calls (STD/ISD) to your friends or family members late at night? Why do you think hotel rates are high in Darjeeling during the peak season? It is often the practice in organizations to manage demand using some strategies. These strategies basically aim at shifting the demand from one period to another without losing them for want of capacity. Normally, capacity is far in excess of demand during non-peak hours and vice versa during peak hours. Therefore, any attempt to even out the demand by shifting the demand from peak hours to non-peak hours is a desirable step. The aggregate operations planning exercise provides a few alternatives to achieve this goal.

6.4.1 Reservation of Capacity

One method used to manage demand is to moderate and shift the excess demand out of a busy period to any future period without completely losing it. Consider, for instance, a multi-speciality healthcare unit in which patients arrive for consultation. If there is no reservation of capacity (in this case, the doctor's time for consultation), some patients may go unattended and the hospital might lose the opportunity of servicing the demand. Therefore, an appointment system helps to capture the demand by scheduling the service on another day when there is sufficient capacity. Capacity reservation is a very popular method in service systems.

6.4.2 Influencing Demand

Another method to manage demand is to influence the demand during a particular period through some mechanisms and induce customers to voluntarily shift the demand count structures, and limited period product promotions. Consider the differential tariffs promoted by telecommunication companies in the country. The demand for bandwidth tends to be low during late night and early morning (10 p.m. to 9 a.m.). Therefore, to induce customers to use the bandwidth during this time, special rates (quarter-charges, for example) are applied. There are several examples similar to this, which induce customers to shift the demand from peak to non-peak periods. This includes special off-season discounts for several seasonal products such as summer resorts, raincoats, winter wear, and weekend tariff structures for airline tickets. Similarly, several years ago, white goods manufacturers such as Videocon and BPL used to offer a "wait-buy-save" scheme. In this scheme, customers willing to wait for a period of about five weeks would get a refrigerator with a substantial discount on the product price and it would also be home delivered according to the promised schedule. It is clear that strategies for influencing demand stem

largely from a variety of initiatives in the marketing function in an organization. Therefore, close coordination and planning by operations planning and marketing functions is important to implement this set of strategies.

Although both these alternatives appear to be very similar there are some differences. In the case of reservation of capacity, there is an explicit approach to book capacity in the future. On the other hand, promotional schemes merely encourage the customers to use the capacity. There is still an element of uncertainty in the extent to which capacity is likely to be used during the promotional (non-peak) period. Therefore, an accurate capacity planning exercise could be done only in the case of the former. This explains why the appointment system is a useful tool for managing capacity in service systems efficiently.

6.5 ALTERNATIVES FOR MANAGING SUPPLY

Organizations have several options to modify supply to meet the demand during a period. In all these cases, the plan seeks to increase or decrease the available capacity to match the supply to the demand during the planning horizon. Supply can be modified through one of the alternatives: (i) inventory-based alternatives, (ii) capacity adjustment alternatives, or (iii) capacity augmentation alternatives.

In the case of inventory-related strategies, changes are made in the inventory levels to match the demand with supply whereas in the case of capacity related strategies, changes in capacity availability cause corresponding changes in the supply.

Supply can be modified through one of the following alternatives: (i) inventory-based alternatives, (ii) capacity adjustment alternatives, or (iii) capacity augmentation alternatives.

6.5.1 Inventory-based Alternatives

Matching the demand with supply could be done in four ways using inventory. One is to *build inventory* during periods of lean demand and consume them during periods of high demand. For instance, consider a textile manufacturer such as Arvind Mills. Suppose the forecast demand for denim during the months of April–June is 35,000 metres per month and during July–September is 55,000 meters per month. Let us also assume that the production capacity is 45,000 metres per month. Then, the firm could maintain the production at a level of 45,000 metres during April–September and carry inventory from the first three months to meet the demand during the next three months. Similarly, in a fast-food restaurant, a pre-processed inventory of certain food items is built in the preceding period in order to handle the high demand during peak periods.

Another option is to **back order** the current periods demand in a future period. **Back ordering** is a method of pushing an order to

NOTES

NOTES

Back ordering is a method of pushing an order to a future period on account of insufficient inventory and/or capacity to supply during the current period.

a future period on account of insufficient inventory and/or capacity to supply during the current period. For example, in a watch-manufacturing firm, if the demand for watches during the month of October is 40,000 watches and if the firm has capacity to manufacture only 30,000 watches and has on-hand inventory of 5,000 watches, then the balance 5,000 watches is scheduled in the month of November, although the demand is for the month of October.

Another possibility is the *stock-out*. By stock-out, we mean a deliberate strategy of leaving a portion of the demand unfulfilled. In the above example, it is also possible that the watch manufacturer can go to its customers and express its inability to supply 5,000 watches.

The fourth variation is to *backlog* the orders. In this method, instead of promising the customer a due date and later incurring a back order or a stock out, the organization could specify a waiting time for supply. Thus, during periods of high demand, the backlog of orders could be large.

It is clear from the above that inventory-related alternatives ensure that an organization will continue to produce goods and services at a uniform rate. The demand–supply mismatch is absorbed through alternative plans, depending on the inventory availability. However, organizations incur several types of costs while implementing these alternatives. In the case of building inventory, carrying costs are incurred. Moreover, the organization could also incur costs of obsolescence. On the other hand, the back order/backlog/stock-out alternatives impose direct costs such as cost of shortage and other indirect costs pertaining to loss of customer goodwill. These costs are real, difficult to quantify and in some cases, significant too.

6.5.2 Capacity Adjustment Alternatives

Another possibility available to organizations is to adjust the available capacity to meet the demand during the planning horizon. Two factors affect capacity in any organization: the number of working hours and the number of people employed. Therefore, by varying these parameters one can adjust the capacity to match the demand. The variations include the following:

Hiring/lay-off of workers

In several service systems and in certain areas of manufacturing systems, it is possible to increase/decrease capacity and operations rate by varying the number of workers. For instance, in a construction project, employing more people on the site could hasten the completion of the project. This is true of several project-oriented situations such as maintenance, overhaul, new-product

development, software development, and consulting. Similarly, in a service system, it is customary to hire temporary workers to augment capacity during peak hours. Therefore, hiring additional workers during periods of high demand and laying-off excess workers during periods of lean demand could help an organization match supply with demand. The cost of this includes cost of hiring, initial training, lost productivity due to learning effects, and severance costs during lay-off. Moreover, only semi-skilled and less-skilled workers can be frequently hired and laid off. Furthermore, frequent hiring and laying off causes low morale and motivational problems among the existing workers. These considerations limit the nature of jobs and the extent to which these options could be used.

Varying shifts

The actual capacity utilized could be different from the installed capacity in most organizations. For example, a CNC machine could theoretically work for 24 hours a day. If it can produce components at the rate of 1,000 pieces per hour, then the installed capacity for the machine is 24,000 components per day. However, in reality, the machine may be working for only one shift of eight hours. Therefore the realised capacity is only 8,000 pieces per day. If there is an increased demand for the component, then it is possible to add another shift of eight hours to increase the production to 16,000 components per day. Similarly, during periods of low demand, the number of shifts could be reduced.

Adjusting capacity by varying shifts is a feasible option only in the case of discrete manufacturing industries, as process industries work in a continuous fashion. Shift changes introduce additional costs in an organization. Several fixed and semi-fixed components of the overheads remain unchanged when the number of shifts is reduced. Moreover, the productivity levels decrease as the number of shifts increases. These costs need to be appropriately captured while analysing the suitability of this option. Service systems can also employ a similar strategy to match the supply to demand.

Varying working hours

Consider the example of submission of income-tax returns by individual tax payers. The last date for submission is 31 July. In order to meet the peak demand, it is possible to extend the working hours from the normal practice of 9 a.m. to 5 p.m. to perhaps 8.00 a.m. to 8.00 p.m. in the last week of July. The Income Tax Department and several other service systems employ such a practice to match the supply to demand.

From the above example it is clear that the other alternative for organizations to adjust the capacity is to vary the number of working hours instead of varying the number of shifts. This is normally achieved through overtime/undertime adjustments. In the overtime (OT) alternative, employees are allowed to work for

N O T E S

NOTES

additional time over and above the normal shift timing. Thus, a person working in a shift for eight hours will be permitted to work for another three hours on an overtime basis. Industrial regulations stipulate certain conditions for overtime working. This includes a certain premium to be paid to the employee and limits the number of hours permissible under overtime. Moreover, the productivity of the employee suffers on account of fatigue due to long working hours.

On the other hand, during periods of lean demand, organizations may require their employees to work less than the normal time, which is referred to as undertime (UT). The recessionary trends and slowing market conditions in the world over have forced many organizations to resort to undertime. Several global software firms required their employees to take a compulsory two-week annual vacation during December 2008–January 2009. Another example of undertime is retaining bench staff in several service organizations. During UT, there is loss of productivity and employees continue to get their normal pay. Therefore, additional costs of the OT plan must include the OT premium, and those of the UT plan must include the opportunity costs of idle time and lost productivity.

Capacity augmentation alternatives

Another option for an organization to address the supply–demand mismatch is to augment its capacity. Capacity can be augmented by subcontracting (outsourcing) a part of the requirement during a period. Consider an organization such as Indian Telephone Industries (ITI). During a month, if the demand for PCB board fabrication is in excess of the available capacity, ITI can subcontract the work to an outside vendor for the excess requirement. In this case, the materials, process plans and drawings, production, quality, and inspection specifications are supplied to the chosen vendor with a due date for supplying the completed PCBs. Developing good sources for subcontracting and ensuring that they conform to quality and delivery norms is a critical process. This results in additional costs related to identifying, setting up the relationship, and transacting with an outside vendor for subcontracting.

Other options for augmenting capacity include re-scheduling maintenance plans, launching de-bottlenecking projects and adding new capacity. In these cases, the annualized costs of these operations constitute the cost of the option. If the demand projections for the future indicate a growing trend, then capacity augmentation could be considered on a permanent basis. Otherwise, temporary capacity augmentation should be considered. In the case of service organizations similar examples can be found. When there was a huge pile up of IT returns to be processed, the Income Tax department outsourced the process to private players.

An organization can augment its capacity by subcontracting (outsourcing), rescheduling maintenance, de-bottlenecking projects, and adding new capacity.

TABLE 6.2 ALTERNATIVES FOR AGGREGATE OPERATIONS PLANNING

Alternatives	Description of the Alternative	Costs
For managing demand	Reservation of capacity	Planning and scheduling costs
	Influencing demand	Marketing-oriented costs
For managing supply	Inventory-based alternatives	
	(a) Build inventory	Inventory-holding costs
	(b) Backlog/backorder/shortage	Shortage/loss of good will costs
	Capacity adjustment alternatives	
	(a) Overtime/undertime	OT premium, lost productivity
	(b) Variable number of shifts	Shift-change costs
	(c) Hire/lay-off workers	Training/hiring costs, morale issues
	Capacity augmentation alternatives	
	(a) Subcontract/outsource	Transaction costs for subcontract
	(b) De-bottleneck	Annualized de-bottlenecking cost
	(c) Add new capacity	Annualized cost of new capacity

Table 6.2 lists all the available alternatives and the costs associated with each of these alternatives.

6.6 BASIC STRATEGIES FOR AGGREGATE OPERATIONS PLANNING

So far, we have identified several alternatives for bridging the supply–demand gap during the planning horizon. However, we are still uncertain about the specific combination of these that should be used, and the implications of these in practice. There are two generic approaches to AOP that can use a combination of these alternatives: *level strategy* and *chase strategy*. These two approaches represent alternative modes of thinking towards employing the available alternatives for AOP. Table 6.3 summarizes the key differences between these two strategies.

6.6.1 Level Strategy

In the **level strategy**, the emphasis is on not disturbing the existing operations at all. This implies that the system would employ a constant workforce and/or maintain constant working hours. In this strategy, inventory plays the vital role of linking one period with the other. Therefore, firms often employ inventory-related alternatives to address the supply–demand mismatch. During periods of lean demand, anticipation inventory is built and during periods of high demand, the anticipation inventory is consumed

In the **level strategy**, the emphasis is on not disturbing the existing operations at all.

NOTES

TABLE 6.3 TWO GENERIC STRATEGIES FOR AOP

Strategy	AOP Alternatives	Key Features
Level strategy	Inventory-based alternatives	
	(a) Build inventory (b) Backlog/backorder/shortage	Inventory as the critical link between the periods; made-to-stock environments; products with low risks of obsolescence
Chase strategy	Capacity adjustment alternatives	
	(a) Overtime/undertime (b) Variable number of shifts (c) Hire/lay-off workers	No inventory carried from one period to another; made-to-order and project environments; several service systems
	Capacity augmentation alternatives	
	(a) Subcontract/outsource (b) De-bottleneck	

NOTES

and other alternatives such as backordering/shortage are made use of to match supply with demand. Clearly, inventory-related alternatives are useful only when the risks of carrying inventory are low. Therefore, several sectors of industry operating in a made-to-stock environment and products with low technological obsolescence are suitable candidates for using this strategy.

In a constant workforce strategy, an organization having a certain number of workers may not hire new workers or lay off excess workers in response to changes in demand. They will utilize other means of addressing the supply–demand mismatch. This is especially true of highly skilled employees. In such cases, the high costs of hiring, training, and laying off may be responsible for several organizations resorting to a constant workforce strategy. A case in point is the recent experiences of firms operating in the information technology (IT) sector in India. As the global recession continues to affect several economies there could be a drop in the demand for IT services in the country. Firms such as Infosys and Wipro will resort to undertime strategy rather than laying off software engineers. At most, they may temporarily halt fresh recruitment of employees.

The other possibility is to maintain constant working hours. A constant operations rate is often achieved by maintaining constant working hours. For instance, a firm operating on a single-shift basis will continue to work in the same fashion. A constant operations rate is an appropriate method for achieving a level strategy in a highly automated and mechanized set-up. In such an organization, the operational rate is a function of the number of working hours. Process industries and highly mechanized batch-processing units fall under this category of such industries—they include two-wheeler and passenger-car manufacturing and petrochemical and pharmaceutical manufacturing.

In several organizations, level strategy is obtained only by maintaining constant working hours and a constant number of workers. These firms may employ a variety of automated and semi-automated set-ups for offering products and services.

6.6.2 Chase Strategy

At the other end of the spectrum is the **chase strategy**. In this method of AOP, very little or no inventory is carried from one period to another. Rather, the supply–demand mismatch is addressed during each period by employing a variety of capacity-related alternatives. For example, during periods of high demand, additional workers are hired, the number of working hours is increased, workers are permitted to do overtime, and more capacity is obtained by outsourcing the unmet demand. Similarly, during periods of low demand, some workers are laid off, others are permitted to go on undertime, and the number of working hours is reduced by reducing the number of shifts, and, in extreme situations, even the duration of the shift. Clearly, these strategies are more appropriate when it is not possible to stock inventory. Several service systems and made-to-order project type of organizations fall under this category. As we can see from our discussion of AOP alternatives, capacity augmentation and capacity adjustment alternatives are suitable for a chase strategy. However, for reasons already described (and due to the costs of these alternatives), organizations may prefer to exploit capacity adjustment alternatives before employing capacity augmentation alternatives.

In the **chase strategy**, the supply–demand mismatch is addressed during each period by employing a variety of capacity-related alternatives.

6.6.3 Mixed Strategy

The chase and the level strategies represent two pure forms of aggregate operations planning. In reality, one can expect organizations to utilize a combination of the available alternatives to devise a strategy for aggregate operations planning. **Mixed strategy** is therefore defined as one that employs a combination of the available alternatives for AOP. The following examples illustrate this concept:

- ⇒ *Mixed strategy 1*: Hire a certain number of workers at the beginning of the planning horizon, maintain a constant operations rate, and utilize inventory-based alternatives to match supply with demand.
- ⇒ *Mixed strategy 2*: Do not hire or lay off workers, adjust the operations rate by varying the number of shifts, use inventory and subcontracting to match supply with demand.

Mixed strategies employ a combination of the available alternatives for APP.

NOTES

NOTES

⇒ *Mixed strategy 3*: Do not hire or lay off workers, use subcontracting during periods of high demand and build inventory during periods of low demand.

In all these examples, we see that the strategy seeks to maintain level operations during periods of low demand by building inventory and chase the demand by additional alternatives during periods of high demand. Thus, the focus shifts between level and chase. The cost structure of the alternatives will determine the total cost of each of these strategies.

In reality, there can be numerous combinations of the AOP alternatives. Each combination could form one mixed strategy with a certain cost of the plan. Therefore, arriving at the best AOP strategy involves enumerating the pure as well as the numerous mixed strategies and choosing the one with the least cost. Aggregate planning methods seek to fulfil this requirement.

EXAMPLE 6.1

A manufacturer of electrical switchgears is in the process of preparing the aggregate production plan for the next year. Let us assume that a good measure of capacity is the number of working hours available per month. Table 6.4 presents details pertaining to the forecast demand for the “equivalent” model of switchgears and the number of working days available during the planning horizon.

The following relevant details are also available:

1. The manufacturer currently works on a single-shift basis and employs 125 workers.
2. One unit of switchgear requires 100 hours of production time.
3. It is expected that at the beginning of the planning horizon, there will be a finished-goods inventory of 200 switchgears.
4. Inventory carrying costs are ₹1,000 per switchgear per month and unit shortage/backlogging costs are 200 per cent of unit carrying cost.

Devise a level production strategy with constant workforce and constant working hours and compute the cost of the plan.

Solution

Computing the Period-by-period Demand-supply Mismatch

The first step in AOP is to compute the demand–supply mismatch during the planning horizon. Using the information available this is computed and presented in Figure 6.3. Since the firm works for eight hours a day employing 125 people, the capacity available is 1,000 hours per day. Similarly, since each switchgear requires 100 hours, we can compute the number of

TABLE 6.4 FORECAST DEMAND FOR MODELS OF SWITCH GEARS AND THE NUMBER OF WORKING DAYS

Month	Demand (in Units)	Number of Working Days
April	250	23
May	220	22
June	300	21
July	290	24
August	260	22
September	180	22
October	200	19
November	220	23
December	250	21
January	200	23
February	240	20
March	270	24

hours required during a period. Figure 6.3 shows the relevant calculations.

Level Production Strategy

In this strategy, we use inventory-related alternatives only. During periods of low demand we accumulate inventory and during periods of high demand we consume the inventory built during lean periods. During each period we have an opening inventory.

	A	B	C	D	E	F	G	H	I	L
2										
3										
5		(A)	(B)	(C) = (A)*(B)	(D)	(E)	(F)	(G) = (D)*(E)*(F)	(H) = (G)-(C)	
6	Month	Demand (in units)	Hours required per unit of production	Demand (hours)	No. of Working Days	Working Hours per Day	No. of Workers	Capacity Available (hours)	Supply - Demand (hours)	
8	April	250	100	25,000	23	8	125	23,000	(2,000)	
9	May	220	100	22,000	22	8	125	22,000	-	
10	June	300	100	30,000	21	8	125	21,000	(9,000)	
11	July	290	100	29,000	24	8	125	24,000	(5,000)	
12	August	260	100	26,000	22	8	125	22,000	(4,000)	
13	September	180	100	18,000	22	8	125	22,000	4,000	
14	October	200	100	20,000	19	8	125	19,000	(1,000)	
15	November	220	100	22,000	23	8	125	23,000	1,000	
16	December	250	100	25,000	21	8	125	21,000	(4,000)	
17	January	200	100	20,000	23	8	125	23,000	3,000	
18	February	240	100	24,000	20	8	125	20,000	(4,000)	
19	March	270	100	27,000	24	8	125	24,000	(3,000)	

FIGURE 6.3 Computation of demand-supply mismatch period by period

	A	B	C	D	E	F	G	H	I	L
18										
19										
20	Level Production Strategy									
21		(A)	(B)	(C) = (B)-(A)	(D) = (C)/100	(E)	(F) = (D)*(E)	(G) = (E+F)/2		
22	Month	Demand (hours)	Capacity Available (hours)	Supply - Demand (hours)	Supply - Demand (units)	Opening Inventory	Closing Inventory	Average Inventory	Cost of Inventory	
23										
24	April	25,000	23,000	(2,000)	(20)	200	180	190	190,000	
25	May	22,000	22,000	-	-	180	180	180	180,000	
26	June	30,000	21,000	(9,000)	(90)	180	90	135	135,000	
27	July	29,000	24,000	(5,000)	(50)	90	40	65	65,000	
28	August	26,000	22,000	(4,000)	(40)	40	-	20	20,000	
29	September	18,000	22,000	4,000	40	-	40	20	20,000	
30	October	20,000	19,000	(1,000)	(10)	40	30	35	35,000	
31	November	22,000	23,000	1,000	10	30	40	35	35,000	
32	December	25,000	21,000	(4,000)	(40)	40	-	20	20,000	
33	January	20,000	23,000	3,000	30	-	30	15	15,000	
34	February	24,000	20,000	(4,000)	(40)	30	(10)	15	35,000	
35	March	27,000	24,000	(3,000)	(30)	(10)	(40)	-	80,000	
36										
37	Total cost of the plan								830,000	
38										
39										

FIGURE 6.4 Total cost of the level production strategy

After computing the excess/shortfall of inventory, we compute the closing inventory, which becomes the opening inventory for the next period. We compute the cost of the plan by computing the costs of holding inventory and backloging/shortages. Figure 6.4 shows the relevant calculations and results.

We see from the computations that the opening inventory of 200 units has enabled the switchgear manufacturer to avoid

shortages/backlogging until January. However, during February and March shortage/backlogging costs are incurred. The total cost of the plan is ₹830,000.

The costs related to backloging/shortage are hard to quantify in practice. Therefore, enough care should be exercised in interpreting the usefulness of this strategy.

EXAMPLE 6.2

Consider Example 6.1. Assume the switchgear manufacturer has no opening stock of inventory and chooses to devise a chase production strategy. The following additional information is available:

1. Overtime costs are ₹40 per hour and undertime costs are ₹20 per hour.
2. Temporary workers can be hired on a monthly basis. The relevant cost of this option including firing them at the end of the month amounts to ₹7,500 per worker.

	A	C	F	H	J	K	L	M
19								
20								
21	Chase Production Strategy Using OT/UT Alternatives							
22		(A)	(B)	(C)=(B)-(A)	(D)	(E)	(F)=(D)*40+(E)*20	
23	Month	Demand (hours)	Capacity Available (hours)	Supply - Demand (hours)	Overtime (hours)	Undertime (hours)	OT/UT cost	
24								
25	April	25,000	23,000	(2,000)	2,000	0	80,000	
26	May	22,000	22,000	-	-	0	-	
27	June	30,000	21,000	(9,000)	9,000	0	360,000	
28	July	29,000	24,000	(5,000)	5,000	0	200,000	
29	August	26,000	22,000	(4,000)	4,000	0	160,000	
30	September	18,000	22,000	4,000	-	4,000	80,000	
31	October	20,000	19,000	(1,000)	1,000	0	40,000	
32	November	22,000	23,000	1,000	-	1,000	20,000	
33	December	25,000	21,000	(4,000)	4,000	0	160,000	
34	January	20,000	23,000	3,000	-	3,000	60,000	
35	February	24,000	20,000	(4,000)	4,000	0	160,000	
36	March	27,000	24,000	(3,000)	3,000	0	120,000	
37								
38	Total cost of the plan						1,440,000	
39								
40								

FIGURE 6.5 Total cost of the chase strategy using the OT/UT option

Evaluate the following options for chase strategy and offer your suggestions to the switchgear manufacturer.

- Utilizing overtime and undertime alternatives
- Using hiring and laying-off alternatives for capacity adjustment

Solution

(a) Chase strategy using OT/UT

In this strategy, the demand–supply mismatch is addressed through use of OT during periods of high demand and UT during periods of low demand. For example, consider the computations in Example 6.1. During April, the demand for capacity is 25,000 hours and the available capacity is only 23,000 hours. Therefore, 2,000 hours of OT will be used at the rate of ₹40 per hour. Similarly, during the month of September, the demand is only 18,000 hours whereas the supply is 22,000 hours. Therefore, 4,000 hours of UT will be allowed during September at the rate of ₹20 per hour.

Since chase strategy does not result in carrying over inventory, there will be no inventory carrying costs. By computing

the extent of OT/UT required and the costs associated with this in every period one can arrive at the cost of the plan. Figure 6.5 shows all the relevant calculations. The total cost of the plan is ₹1,440,000.

(b) Chase strategy using hire-lay-off

In the hire–lay-off alternative we compute the number of workers to be hired/laid off by examining the demand–supply mismatch. When the demand greater than the supply, we resort to hiring. For the given example, consider the month of April. The demand is in excess of supply by 2,000 hours. If we hire one worker, he/she will contribute 184 hours as there are 23 working days and each day we have eight working hours. Therefore, the number of workers to be hired during April is $2000/184 = 10.86 \cong 11$ workers. In the case of a lean period, we merely incur UT costs as we do not fire the existing permanent workers. For example, in the month of January we incur UT cost of ₹60,000 ($3,000 \times 20$).

Since the number of workers to be hired/laid off can only be integers, we round off the number to be hired to the next higher integer. In this process we also incur some idle time by

Month	Demand (Hrs.)	Capacity available (Hrs.)	Supply - Demand (Hrs.)	No. of workers hired	Under time (Hrs)	Hiring/ Laying off & UT costs
April	25,000	23,000	(2,000)	11	24	82,980
May	22,000	22,000	-	-	-	-
June	30,000	21,000	(9,000)	54	72	406,440
July	29,000	24,000	(5,000)	27	184	206,180
August	26,000	22,000	(4,000)	23	48	173,460
September	18,000	22,000	4,000	-	4,000	80,000
October	20,000	19,000	(1,000)	7	64	53,780
November	22,000	23,000	1,000	-	1,000	20,000
December	25,000	21,000	(4,000)	24	32	180,640
January	20,000	23,000	3,000	-	3,000	60,000
February	24,000	20,000	(4,000)	25	-	187,500
March	27,000	24,000	(3,000)	16	72	121,440
Total cost of the plan						1,572,420

FIGURE 6.6 Total cost of the chase strategy using the hire/lay-off option

retaining a little more capacity than needed. The cost of the plan takes all these into consideration. Figure 6.6 shows all the relevant details for the computation. The total cost of this plan is ₹1,572,420.

Clearly, the cost of the hire–lay-off plan is higher than the cost of the OT/UT plan. Moreover, there are other costs that are hard to quantify in the former case. For instance, frequent hiring/lay-off of workers may not be feasible in several sectors

of industry involving skilled sets of workers. The training costs and lost productivity during the learning phase would have been grossly under-estimated. Moreover, such practices could result in low morale/motivation among the workers, leading to productivity and quality problems. These costs are difficult to quantify. Therefore, it is a good practice to deploy the hire–lay-off plan only as a last resort.

EXAMPLE 6.3

Consider Example 6.2. Assume that on the basis of these computations, the switchgear manufacturer has come up with a plan that employs the following alternatives:

- Hire 25 more workers at the beginning of April.
- Lay off the 25 workers at the end of August.
- Maintain constant working hours (1 shift of 8 hours) throughout the year.

- Absorb the demand–supply mismatch by building inventory during periods of lean demand and by resorting to OT during periods of excess demand.

Assume the cost of hiring to be ₹7,500 per worker and the cost of firing to be ₹5,000 per worker. Evaluate the cost of this plan and compare it with the earlier alternatives. What are the key inferences?

Mixed Strategy — OT, Inventory, Hire, Lay-off										
Month	Opening Inventory	Demand (hours)	Effective Demand	Capacity Available (hours)	Overtime (hours)	Closing Inventory	Average Inventory	Cost of Inventory		
April	0	25,000	25,000	27,600	-	26	13	13,000		
May	26	22,000	19,400	26,400	-	70	48	48,000		
June	70	30,000	23,000	25,200	-	22	46	46,000		
July	22	29,000	26,800	28,800	-	20	21	21,000		
August	20	26,000	24,000	26,400	-	24	22	22,000		
September	24	18,000	15,600	22,000	-	64	44	44,000		
October	64	20,000	13,600	19,000	-	54	59	59,000		
November	54	22,000	16,600	23,000	-	64	59	59,000		
December	64	25,000	18,600	21,000	-	24	44	44,000		
January	24	20,000	17,600	23,000	-	54	39	39,000		
February	54	24,000	18,600	20,000	-	14	34	34,000		
March	14	27,000	25,600	24,000	1,600	-	7	7,000		
Cost of hiring (25 workers at INR 7,500 per worker)								187,500		
Cost of lay-off (25 workers at INR 5,000 per worker)								125,000		
Cost of overtime (at INR 40 per hour)								64,000		
Cost of inventory								436,000		
Total cost of the plan								812,500		

FIGURE 6.7 Total cost of the mixed strategy

Solution**The Impact of the Hiring-lay-off Decision**

25 workers are hired at the beginning of April and laid off at the end of August. This would mean that during the months of April to August the number of workers available at the switch-gear manufacturing facility is 150. The capacity available (in hours) during this time is computed on this basis. For example, during May the capacity available is $(22 \text{ days} \times 8 \text{ hours per day} \times 150) = 26,400$ hours. The capacity available during the months of September to March is the same as computed in Example 6.1, as the number of workers remains at 125. Figure 6.7 shows these capacity calculations (see Column 5).

The decision affects the cost of the plan due to hiring charges at the rate of ₹7,500 per worker for 25 workers in the beginning of April and laying-off charges at the rate of ₹5,000 per worker for 25 workers by the end of August.

Absorbing the Supply-demand Mismatch through Inventory and OT

We first begin with the opening inventory and adjust the demand to get the net demand. For example, during the month of May, the opening inventory is 26 units. This means 2,600 hours of demand is already satisfied with this opening inventory (since each unit requires 100 hours). Therefore, the net demand is the period demand minus what is already available as inventory. If the net demand is more than the supply for the month, we resort to OT for the excess requirement. On the other hand, if the net demand is less than the available capacity, we maintain the production rate and build additional inventory. The calculations are shown in Figure 6.7.

We find from Figure 6.7 that except during the month of March, there is an inventory build-up in all other periods. Only during the month of March are 1,600 hours of OT employed at the rate of ₹40 per hour. Figure 6.7 summarizes the total cost of this strategy. The total cost of the plan is ₹812,500.

Key Inferences

We first summarize the cost of each of the strategies that we analysed.

Cost of level strategy: ₹830,000

Cost of chase strategy (OT/UT): ₹1,440,000

Cost of chase strategy (hiring/lay-off): ₹1,572,420

Cost of mixed strategy: ₹812,500

It has been found in several situations that the cost of chase strategy is higher than other strategies. This is primarily due to the costs of disturbing the operating system, both at the manufacturer and at the supply-chain-members' level. Crash purchasing of material, uneven utilisation of resources, and payment of premium for making various resources available at the last minute are some of the reasons for high cost of chase strategy.

The examples we have discussed demonstrate the fact that levelling and chasing are two pure strategies; in reality, therefore, organizations should use a mixed strategy by utilizing several combinations of the available alternatives for AOP. It is interesting to note that for the given cost structure, a mixed strategy has resulted in the lowest cost. Therefore, it is worthwhile to explore all the available combinations of AOP alternatives and select the one with the least cost. Clearly, it requires robust models to accomplish this task.

EXAMPLE 6.4

A manufacturer of industrial motors produces four varieties of motors. The Type 4 motor requires a larger amount of customization than the other three. The production planning department has estimated the number of hours required for manufacturing one unit of motor of each variety. Type 1 requires 125 hours; Type 2, 175 hours; Type 3, 200 hours; and Type 4, 350 hours. The forecasts for the four varieties are given in Table 6.5.

TABLE 6.5 FORECASTS FOR THE FOUR VARIETIES

Month	Forecast of Motor Demand (units)			
	Type 1	Type 2	Type 3	Type 4
April	100	40	60	10
May	90	30	30	5

June	80	50	50	10
July	130	40	70	20
August	70	30	30	15
September	90	50	50	5
October	100	40	50	10
November	110	30	40	15
December	60	60	80	5
January	110	20	30	10
February	120	40	40	5
March	140	50	70	10

The overtime premium is estimated to be ₹20 per hour while undertime costs are 50 per cent of the OT premium. The cost of carrying inventory is ₹15 per hour per month and the

backordering/shortage costs are 200 per cent of inventory carrying costs. The company can subcontract capacity from a reputed vendor and the additional costs related to subcontracting is ₹25 per hour of capacity.

Evaluate the following alternatives for AOP:

- Prepare a level strategy. Use inventory/backlogging to absorb the demand–supply mismatch.
- Use constant production rate for the first six months, change the production rate to another level and have a constant production rate for the next six months. Use inventory/backlogging to absorb demand–supply mismatch.
- Maintain a production level of 31,000 hours per month and use OT/UT alternatives to chase the demand.
- Maintain a production level of 31,000 hours per month. Build inventory during periods of low demand and use

subcontracting when the demand exceeds available capacity.

Solution

Aggregation of Product Demand

Since there are four products, we use the number of hours as the basis for aggregation and convert the demand in units of each variety to demand in hours. Figure 6.8 shows the details. Let us consider the month of October. Since the demand for the number of Type 1 motors is 100, and since it requires 125 hours for each unit, the demand for Type 1 in the month of October is 12,500 hours. Similarly, the demand for Type 2, Type 3 and Type 4 are 7,000 hours, 10,000 hours, and 3,500 hours, respectively. Therefore, the aggregate demand during October is 33,000 hours. All the values in Figure 6.8 are calculated in a similar fashion.

Chapter 14 v1 [Compatibility Mode] - Microsoft Excel										
F21										
	A	B	C	D	E	F	H	J	K	L
1										
2	Hours/unit	125	175	200	350					
3		Forecast of motor demand (units)				Forecast of motor demand (hours)				
4	Month	Type 1	Type 2	Type 3	Type 4	Type 1	Type 2	Type 3	Type 4	Total
5										
6	April	100	40	60	10	12,500	7,000	12,000	3,500	35,000
7	May	90	30	30	5	11,250	5,250	6,000	1,750	24,250
8	June	80	50	50	10	10,000	8,750	10,000	3,500	32,250
9	July	130	40	70	20	16,250	7,000	14,000	7,000	44,250
10	August	70	30	30	15	8,750	5,250	6,000	5,250	25,250
11	September	90	50	50	5	11,250	8,750	10,000	1,750	31,750
12	October	100	40	50	10	12,500	7,000	10,000	3,500	33,000
13	November	110	30	40	15	13,750	5,250	8,000	5,250	32,250
14	December	60	60	80	5	7,500	10,500	16,000	1,750	35,750
15	January	110	20	30	10	13,750	3,500	6,000	3,500	26,750
16	February	120	40	40	5	15,000	7,000	8,000	1,750	31,750
17	March	140	50	70	10	17,500	8,750	14,000	3,500	43,750
18										
19	Total	1200	480	600	120	150,000	84,000	120,000	42,000	396,000
20	Average	100	40	50	10	12,500	7,000	10,000	3,500	33,000
21										
22										
23										
24										
25										

FIGURE 6.8 Aggregate units of the forecasted demand

	A	F	H	J	K	L	M	N
24								
25								
26	Plan (a): Level Strategy with a Monthly Production of 33,000 Hours							
27	Month	Demand (hours)	Production Rate (hours)	Opening Inventory (hours)	Production - Demand (hours)	Closing Inventory (hours)	Cost of Inventory/Backorder	
28	April	35,000	33,000	-	(2,000)	(2,000)	60,000	
29	May	24,250	33,000	(2,000)	8,750	6,750	50,625	
30	June	32,250	33,000	6,750	750	7,500	106,875	
31	July	44,250	33,000	7,500	(11,250)	(3,750)	168,750	
32	August	25,250	33,000	(3,750)	7,750	4,000	30,000	
33	September	31,750	33,000	4,000	1,250	5,250	69,375	
34	October	33,000	33,000	5,250	-	5,250	78,750	
35	November	32,250	33,000	5,250	750	6,000	84,375	
36	December	35,750	33,000	6,000	(2,750)	3,250	69,375	
37	January	26,750	33,000	3,250	6,250	9,500	95,625	
38	February	31,750	33,000	9,500	1,250	10,750	151,875	
39	March	43,750	33,000	10,750	(10,750)	-	80,625	
40								
41	Average (12 months)	33,000			Total cost of the plan		1,046,250	
42								
43								

FIGURE 6.9 Computations for Plan A

Plan A: Level strategy

A level strategy requires a constant production rate. From Figure 6.8, we find that the average requirement during the planning horizon to be 33,000 hours. Therefore, every month, the production will be 33,000 hours of motors. We then compute the demand–supply mismatch during each period (as in Example 6.1) and use inventory-related strategy. In the month of April, the demand is in excess of supply by 2,000 hours. Therefore, there will be a backordering cost of ₹60,000 ($2,000 \times 30$). There will be no opening and finishing inventory and therefore no inventory carrying costs. Similarly, there will be no inventory carrying costs in the month of July. However, in the month of June, there will be an inventory carrying cost of ₹106,875 ($15 \times (6750 + 7500)/2$). Figure 6.9 shows all the relevant computations. The total cost of the plan is ₹1,046,250.

Plan B: Two levels for constant production rate

Plan B differs from Plan A in the manner the production rate is fixed. In the case of Plan A, there is a single level at which the production rate is set. On the other hand, in Plan

B, the production level is initially set at the average of the total requirement in the first six months and then changed to another level, which is set at the average of the total requirement of the next six months. The other calculations pertaining to the demand–supply mismatch, inventory holding, and backordering costs follow this as explained under Plan A.

Figure 6.10 presents the detailed computation. The total cost of the plan is ₹787,500.

By merely fine-tuning the production level with an alternative strategy, we were able to bring down the cost by nearly 25 per cent.

Plan C: Chase strategy with OT/UT

The chase strategy results in no carryover of inventory from one period to another. If the demand is in excess of supply, OT is employed (as in the case of July) and OT cost is incurred. On the other hand, if the demand is less than the supply, UT is employed (as in the case of August) and UT cost is incurred. The demand–supply mismatch computation and the computation pertaining to OT/UT costs are similar

	A	F	H	J	K	L	M	N
24								
25								
26	Plan (a): Level Strategy with a Monthly Production of 33,000 Hours							
27	Month	Demand (hours)	Production Rate (hours)	Opening Inventory (hours)	Production - Demand (hours)	Closing Inventory (hours)	Cost of Inventory/Backorder	
28	April	35,000	33,000	-	(2,000)	(2,000)	60,000	
29	May	24,250	33,000	(2,000)	8,750	6,750	50,625	
30	June	32,250	33,000	6,750	750	7,500	106,875	
31	July	44,250	33,000	7,500	(11,250)	(3,750)	168,750	
32	August	25,250	33,000	(3,750)	7,750	4,000	30,000	
33	September	31,750	33,000	4,000	1,250	5,250	69,375	
34	October	33,000	33,000	5,250	-	5,250	78,750	
35	November	32,250	33,000	5,250	750	6,000	84,375	
36	December	35,750	33,000	6,000	(2,750)	3,250	69,375	
37	January	26,750	33,000	3,250	6,250	9,500	95,625	
38	February	31,750	33,000	9,500	1,250	10,750	151,875	
39	March	43,750	33,000	10,750	(10,750)	-	80,625	
40								
41	Average (12 months)	33,000			Total cost of the plan		1,046,250	
42								
43								

FIGURE 6.10 Computations for Plan B

to those in the above examples. Figure 6.11 shows the complete computations for this plan. The total cost of this plan is ₹982,500.

Expectedly, chase strategy is expensive. Due to knee-jerk reactions on a month-by-month basis, some opportunities for cost reduction are ignored. For example, we notice that there is 4,250 hours of UT in the month of January. By utilizing this, we could have avoided 750 hours of OT in February and 3,500 hours of OT in March. This would have resulted in a cost savings of ₹11,250.

Plan D: Mixed strategy

In the fourth plan, knee-jerks reaction are avoided to an extent by carrying inventory from a period of low demand to a future period of high demand. However, the cost of the plan is likely to be higher on account of the high cost of subcontracting. Also, the cost of carrying inventory is more than the cost of UT.

Figure 6.12 shows all the relevant computations for the plan, which confirms this.

Key inferences

Example 6.4 points to a few important guidelines for aggregate operations planning.

- First, the chase strategy could prove to be more expensive than level strategy in situations involving items that are not made-to-order and vice versa. This is because some benefits of carrying inventory for a limited period and maintaining constant production rate are missed in chase strategy.
- It may be worthwhile to explore fewer production-level changes in a planning horizon even while implementing level strategy.
- Finally, mixed strategies provide a fertile ground for identifying the most appropriate AOP for an organization.

	A	F	H	J	K	L	M	N
72								
73								
74	Plan (c): Chase Strategy with OT/UT							
75	Month	Demand (hours)	Production Rate (hours)	Opening Inventory (hours)	Production - Demand (hours)	Closing Inventory (hours)	Cost of OT/UT	
76	April	35,000	31,000	-	(4,000)	-	80,000	
77	May	24,250	31,000	-	6,750	-	67,500	
78	June	32,250	31,000	-	(1,250)	-	25,000	
79	July	44,250	31,000	-	(13,250)	-	265,000	
80	August	25,250	31,000	-	5,750	-	57,500	
81	September	31,750	31,000	-	(750)	-	15,000	
82	October	33,000	31,000	-	(2,000)	-	40,000	
83	November	32,250	31,000	-	(1,250)	-	25,000	
84	December	35,750	31,000	-	(4,750)	-	95,000	
85	January	26,750	31,000	-	4,250	-	42,500	
86	February	31,750	31,000	-	(750)	-	15,000	
87	March	43,750	31,000	-	(12,750)	-	255,000	
88								
89					Total cost of the plan		982,500	
90								
91								

FIGURE 6.11 Computations for Plan C

	A	F	H	J	K	L	M	N
98								
99								
100	Plan (d): Mixed Strategy (Constant Production, Inventory, Sub-contracting)							
101		(A)	(B)	(C)	(D) = B+C-A			
102	Month	Demand (hours)	Production Rate (hours)	Opening Inventory (hours)	Net Capacity (hours)	Closing Inventory (hours)	Cost of Mixed Strategy	
103	April	35,000	31,000	-	(4,000)	-	100,000	
104	May	24,250	31,000	-	6,750	6,750	50,625	
105	June	32,250	31,000	6,750	5,500	5,500	91,875	
106	July	44,250	31,000	5,500	(7,750)	-	235,000	
107	August	25,250	31,000	-	5,750	5,750	43,125	
108	September	31,750	31,000	5,750	5,000	5,000	80,625	
109	October	33,000	31,000	5,000	3,000	3,000	60,000	
110	November	32,250	31,000	3,000	1,750	1,750	35,625	
111	December	35,750	31,000	1,750	(3,000)	-	88,125	
112	January	26,750	31,000	-	4,250	4,250	31,875	
113	February	31,750	31,000	4,250	3,500	3,500	58,125	
114	March	43,750	31,000	3,500	(9,250)	-	257,500	
115								
116					Total cost of the plan		1,132,500	
117								
118								

FIGURE 6.12 Computations for Plan D

NOTES

6.7 AGGREGATE OPERATIONS PLANNING METHODS

It is evident from these four examples that organizations need to devise a methodology to generate several alternatives for AOP and evaluate these alternatives before selecting the most appropriate one. In the absence of this, organizations may not be sure if the chosen strategy is the best. Moreover, they may not also know how much they could have improved by enumerating other strategies. Aggregate operations planning methods fulfil this objective. There are several methods available to the planner to evaluate alternative strategies before selecting one. Broadly, the available methods could be classified into heuristic methods and optimal methods.

6.7.1 Heuristic Methods for AOP

In heuristic methods, alternative strategies are generated on some basis and the alternatives are evaluated. The best among them is selected for implementation. Some of the known heuristic methods for AOP are the trial-and-error and search methods.

The trial-and-error method

In this method, a few options are considered. Each option employs a certain combination of AOP alternatives that we have already discussed. The total cost of each of these options is computed and the best one is chosen. Examples 6.1–6.4 illustrate this method for AOP. The trial-and-error method is simple to implement, particularly with the help of Excel-based spreadsheets. Since the method involves generating a limited set of alternatives, the usefulness of the method critically depends on the choice of the alternatives available for consideration. Managers with considerable experience in an organization and a good understanding of the cost structures of the AOP alternatives may be in a position to generate good candidates for evaluation.

Search methods

Search methods are useful in ensuring that “promising” candidates are not missed out in the enumeration process. In all the search methods, certain parameters are chosen and alternatives are generated by systematically varying the parameter values. As the parameter values are modified, newer alternatives are considered and the quality of the solution is evaluated. The search usually ends on the basis of some stopping criterion. The stopping criterion usually depends on the rate of improvement of the solution and/or the number of iterations done. At the end of the search, the best solution is identified.

A generalized search procedure is the random search process. In this process, sampling values for the parameters under consideration randomly creates alternative AOP plans. For example, suppose the parameters for consideration are desired production rate,

number of employees, number of working hours and the inventory at the beginning of the planning horizon. Using a random number generator, a candidate solution for AOP could be sampled. As each plan is evaluated, the existing solution is updated. The procedure stops when the quality of the solution does not improve significantly and the best solution at the time of terminating the procedure is identified to be the solution to the AOP problem. Figure 6.13 illustrates the generalized search procedure for AOP.

Heuristic methods offer some advantages. First is the simplicity with which the method can be implemented. In several cases, one can set up the heuristic in a spreadsheet and quickly analyse the alternatives. It usually does not demand high levels of mathematical sophistication on the part of the user. However, the methods suffer from some drawbacks. Heuristic methods generally do not enumerate all the possible candidates in the solution space. Rather, they enable candidates with some knowledge of the problem to eliminate some from consideration. Therefore, the ability of the user to correctly identify the candidate solutions is critical to the usefulness of the method.

The other limitation of the trial-and-error method is that there is no guarantee of obtaining the best solution. Furthermore, users may not even know how far the best solution obtained from the method is from the optimal solution to the problem. Therefore, before using heuristic methods, users need to make a realistic assessment of these issues. With such a perspective, the use of heuristic methods could provide some promising solutions to the AOP problem.

6.7.2 Optimal Methods for AOP

Another method to develop the AOP is to build mathematical programming models of the problem and use the available solution

NOTES

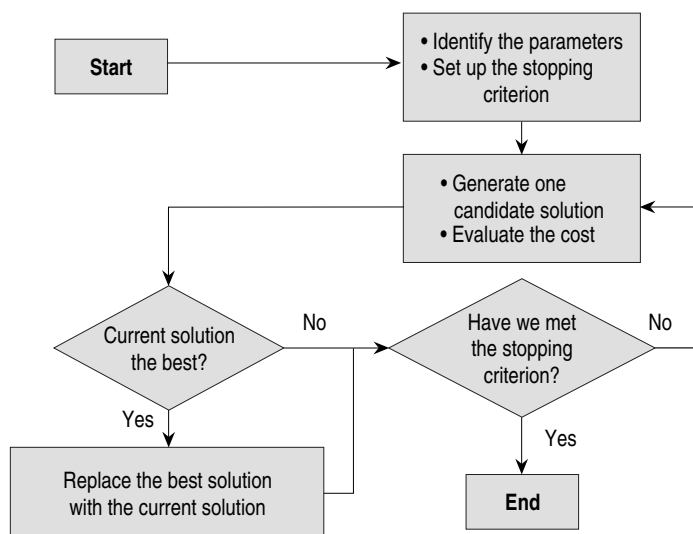


FIGURE 6.13 Search procedure for AOP

NOTES

procedures to solve the problem. Typical examples include the use of the linear programming (LP) and the transportation and dynamic programming (DP) models for AOP. We discuss some of the optimal methods for AOP in the Section 6.8.

Unlike heuristic methods, optimal methods systematically enumerate the entire solutions space and the optimal solution is obtained. While optimal methods guarantee the best solution, they may pose limitations arising out of the size of the problem that could be solved, the complexity of the solution procedure, and the nature of assumptions made in utilizing a model for solution.

Optimal methods include the use of linear programming (LP) and the transportation and dynamic programming (DP) models.

6.8 OPERATIONS RESEARCH TOOLS FOR OPERATIONS PLANNING

Operations research (OR) was developed during the Second World War to systematically model and analyse several business problems. The optimum utilization of resources, maximization of profit, and minimization of cost were some of the pressing requirements of various government and private sector institutions at that time. Operations research tools provided them with a basis for meeting these requirements while taking several decisions. The application of OR tools in several areas of business happened in rapid succession in the post-World War scenario. Operations management problems are extensively addressed by OR researchers. In this section we will see how well known OR tools such as linear programming (LP) and transportation can be applied to operations planning problems.

6.8.1 Aggregate Operations Planning Using the Transportation Model

The transportation model is a special case of linear programming that deals with the issue of shipping commodities from multiple sources to multiple destinations. The objective is to determine a shipping schedule that minimizes the total shipping cost while satisfying supply and demand constraints. Table 6.6 shows the transportation table, which contains all the relevant information for solving the problem. C_{11} is the cost of transporting one unit of the item from Supply A to Demand Point 1, and so on. S_A is the supply constraint of Supply Point A. Similarly, D_1 is the demand at Demand Point 1. The basis of the transportation model is the existence of multiple demand points and multiple supply points and the development of an optimum minimum cost method to satisfy the demand using the supply points. This structure is amenable for solving several other problems, like inventory control, operations planning, scheduling and personnel assignment.

TABLE 6.6 GENERALIZED REPRESENTATION OF A TRANSPORTATION TABLE

	Demand 1	Demand 2	Demand 3	Demand 4	Demand 5	Supply Constraints
Supply A	Ca1	Ca2	Ca3	Ca4	Ca5	Sa
Supply B	Cb1	Cb2	Cb3	Cb4	Cb5	Sb
Supply C	Cc1	Cc2	Cc3	Cc4	Cc5	Sc
Supply D	Cd1	Cd2	Cd3	Cd4	Cd5	Sd
Demand Requirements	D1	D2	D3	D4	D5	

Recall our discussion on aggregate operations planning. The central issue in AOP is to balance the supply with the demand on a period-by-period basis. This structure of AOP lends itself to straightforward application of the transportation model for solving AOP problems. Let us consider the various supply options for satisfying each period demand. These options are nothing but the various AOP alternatives that we discussed in the chapter. For instance, for a period i the supply options that an organization may consider are regular time production (RP), overtime production (OP) and subcontracting (SC). Each of these alternatives has differential cost and limits of production. Using this information, one can construct a transportation table. Example 6.5 illustrates how one can model the AOP using transportation.

The central issue in aggregate production planning is to balance the supply with the demand on a period-by-period basis. This structure of APP lends itself to a straightforward application of the transportation model for solving APP problems.

6.8.2 The Linear Programming Model for AOP

The transportation model we just discussed cannot include AOP alternatives such as hiring and lay-off of workers. Furthermore, a linear programming (LP) model is a more generalized version of the transportation model and therefore can accommodate all AOP alternatives. In the LP model for average production planning, an inventory balance equation and the regular operations rate balance equation links each period with the succeeding period. During every period, there will be an increase in the inventory level due to regular operations and subcontracting. On the other hand, the inventory level decreases due to demand for the product. Similarly, the regular operations rate increases due to the additional hiring of

NOTES

USE OF SPREADSHEETS FOR AGGREGATE PRODUCTION PLANNING

The complexity of production planning increases with variability in demand. The problem would have been much simpler if all the customers were to confirm their order well in advance. As discussed in the chapter, to avoid shortage, a company can either build inventory in anticipation or have additional capacity. As the numerical examples demonstrate, the best plan is a combination, allowing some backlog to accumulate, building anticipation inventory and providing variable capacity.

Use of spreadsheet software such as Microsoft Excel makes it easier to study the various alternatives that an organization may have in order to obtain the best plan. Furthermore, the Solver, an add-on utility, can help in building useful linear programming models and exploring optimal solutions. Figure 6.14 shows one such application of Excel for aggregate production planning.

Since variations in demand and cost parameters are very common, spreadsheet modelling of the aggregate production planning problem will allow the user to perform “what if” sensitivity analysis on the base case. The figure shows the effect of introducing hiring–lay-off costs in the aggregate production planning problem. It is evident from the figure that in the base case with a constant workforce of five employees, the best plan was to allow back orders in the first three months, whereas in the case of introduction of hiring–lay-off costs, the back orders were minimal. Instead, the workforce varied from three to seven.

The discussed example demonstrates how spreadsheets could be used to solve aggregate production planning problems. With the easy availability of computers and increase in the computing power, modern-day operations management will invariably have several such tools that help the operations manager make quick and sensible decisions to solve the problems that he/she faces.

Source: Based on Crandall, R.E., “Production Planning in a Variable Demand Environment,” *Production and Inventory Management Journal* 39, no. 4 (1998): 34–41.

Microsoft Excel - ideas at work 14.2.xls

File Edit View Insert Format Tools Data Window Help

90% Calibri 11 B I U

T10

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1																	
2																	
3																	
4																	
5																	
6																	
7																	
8																	
9																	
10																	
11																	
12																	
13																	
14																	
15																	
16																	
17																	
18																	
19																	
20																	
21																	
22																	
23																	

Aggregate Production Planning Using Excel

Cost parameters (all values in INR)

Opening inventory (units)	0	Ending inventory (units)	0	Per worker hiring cost	4,000
No. of employees (beginning)	5	No. of employees (ending)	5	Per worker layoff cost	5,000
Average no. of working days	20			Normal wages per hour	100
Average working hours per day	8			OT premium per hour (%)	50%
Employee hours per month per employee	160			Inventory carrying cost/unit/m	200
Labour hours required per unit	20			Backorder cost/unit/month	500

Base Case: Constant Workforce

Planning Horizon (Month)

	1	2	3	4	5	6	7	8	9	10	11	12	Total
Demand	65	65	35	30	30	15	30	20	30	60	80	75	535
Production	45	45	40	40	40	40	40	40	40	40	45	45	500
Inventory (Backorder)	-20	-40	-35	-25	-15	10	20	40	50	30	-5	-35	
No. of employees	5	5	5	5	5	5	5	5	5	5	5	5	
OT hours	100	100	0	0	0	0	0	0	0	0	100	100	400
Total costs	#####	#####	97,500	92,500	87,500	82,000	84,000	88,000	90,000	86,000	97,500	#####	1,137,500

(a) Base case: constant workforce

Microsoft Excel - ideas at work 14.2.xls

File Edit View Insert Format Tools Data Window Help

90% Calibri 11 B I U

U22

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1																	
2																	
3																	
4																	
5																	
6																	
7																	
8																	
9																	
10																	
11																	
12																	
13																	
14																	
15																	
16																	
17																	
18																	
19																	
20																	
21																	
22																	
23																	
24																	

Aggregate Production Planning Using Excel

Cost parameters (all values in INR)

Opening inventory (units)	0	Ending inventory (units)	0	Per worker hiring cost	4,000
No. of employees (beginning)	5	No. of employees (ending)	5	Per worker layoff cost	5,000
Average no. of working days	20			Normal wages per hour	100
Average working hours per day	8			OT premium per hour (%)	50%
Employee hours per month per employee	160			Inventory carrying cost/unit/m	200
Labour hours required per unit	20			Backorder cost/unit/month	500

Case 2: Variable Workforce through Hiring and Layoff

Planning Horizon (Month)

	1	2	3	4	5	6	7	8	9	10	11	12	
Demand	65	65	35	30	30	15	30	20	30	60	80	75	535
Production	56	56	32	32	24	24	24	40	56	56	56	40	496
Inventory (Backorder)	-9	-18	-21	-19	-25	-16	-22	-2	24	20	-4	-39	
No. of employees	7	7	4	4	3	3	3	5	7	7	7	5	
Hiring/Layoff costs	8000	0	15000	0	5000	0	0	8000	8000	0	0	10000	
OT hours	0	0	0	0	0	0	0	0	0	0	0	0	0
Total costs	124,500	121,000	89,500	73,500	65,500	56,000	59,000	89,000	124,800	116,000	114,000	109,500	1,142,300

(b) Variable workforce through hiring and lay-off

FIGURE 6.14 Application of Microsoft Excel to aggregate production planning

EXAMPLE 6.5

A manufacturer needs to perform AOP and has the following information about the problem:

- Demand information:** The demand forecast for the next four quarters is 100,000, 50,000, 70,000, and 60,000 units respectively. It is the company's policy to plan for an additional inventory of 10 per cent of next period's forecast. Moreover, planning needs to take into consideration that 25,000 units of inventory will be available at the beginning of Period 1 and 40,000 units must be made available at the end of Period 4.
- Supply options and costs:** The manufacturer can produce 80,000 units through regular production and add another 20 per cent through overtime. Adequate capacities are available with the supply base and it is estimated that up to 100,000 additional units can be obtained through subcontracting. Regular production costs ₹80 per unit while overtime production costs ₹120 per unit. The cost per unit if subcontracted is ₹105 per unit. It is also possible to carry over inventory from one period to another at a cost of ₹2 per unit per period.

TABLE 6.7 THE TRANSPORTATION TABLE

		Demand				Supply Constraints
		Period 1	Period 2	Period 3	Period 4	
Supply	Initial Inventory	0	2	4	6	25,000
	Period 1 RP	80	82	84	86	80,000
	Period 1 OP	120	122	124	126	16,000
	Period 1 SC	105	107	109	111	100,000
	Period 2 RP		80	82	84	80,000
	Period 2 OP		120	122	124	16,000
	Period 2 SC		105	107	109	100,000
	Period 3 RP			80	82	65,000
	Period 3 OP			120	122	13,000
	Period 3 SC			105	107	100,000
	Period 4 RP				80	80,000
	Period 4 OP				120	16,000
	Period 4 SC				105	100,000
	Demand Requirement	105,000	57,000	76,000	110,000	

Use the transportation method to model the problem.

Solution

Requirements for Each Period

Period 1: $100,000 + 10\% \times 50,000 = 105,000$ units

Period 2: $50,000 + 10\% \times 70,000 = 57,000$ units

Period 3: $70,000 + 10\% \times 60,000 = 76,000$ units

Period 4: $60,000 + 10\% \times 100,000 = 70,000 +$ ending inventory of 40,000 = 110,000 units

(Since no information about the next period forecast is available, to simplify the process we use the current year's Period 1 forecast as a substitute)

Cost of AOP Alternatives

Regular time: ₹80

Overtime: ₹120

Subcontracting: ₹105

Inventory can be carried at the rate of ₹2 per unit per period. For example, a regular time production during Period 1 costs ₹80 per unit in that period, ₹82 per unit in Period 2, ₹84 per unit in Period 3, and ₹86 per unit in Period 4 on account of carrying inventory. Similar costs are incurred in the case of the other alternatives.

Based on the above information and the supply constraints of the AOP alternatives, the transportation table can be constructed as shown in Table 6.7.

Some portions of the table have been excluded from consideration. This is because there are no feasible AOP alternatives available in those cells. For example, the items obtained from Period 2 AOP alternatives (RP, OP, and SC) cannot be used to meet the demand for Period 1. By introducing backorders as another AOP alternative one can make these cells feasible. One can solve Table 6.7 using the standard solution procedures available.

workers and decreases due to laying off workers. Using this logic, it is possible to construct a set of equations pertaining to every period in a planning horizon. There are also other constraints such as minimum quantity for subcontracting capacity and restrictions on the extent to which overtime can be utilized in a system. The LP model formulation using such information is discussed here.

Notations for the model

The notations for the LP model are as follows:

Cost parameters:

C_r = Per unit cost of regular production

C_o = Per unit cost of overtime production

C_s = Per unit cost of subcontracted units

C_h = Per unit cost related to hiring of workers

C_l = Per unit cost related to laying off workers

C_i = Per unit costs related to inventory

Decision variables for the time period t :

R_t = Number of units produced in regular time

O_t = Number of units produced using overtime

S_t = Number of units obtained through subcontracting

H_t = Number of additional units obtained through hiring of workers

L_t = Number of units reduced through laying off workers

I_t = Inventory during the period

Other parameters:

D_t = Projected demand during the period

K = Minimum amount to be subcontracted

α = Maximum allowable overtime as a proportion of regular production

NOTES

Objective function:

$$\text{Min } TC_{APP} = \sum_{t=1}^N (C_r R_t + C_o O_t + C_s S_t + C_h H_t + C_l L_t + C_i I_t) \quad (6.1)$$

Subject to the constraints:

Amount produced in regular time during a period is:

$$R_t = R_{t-1} + H_t - L_t \quad \forall t = 1, 2, 3, \dots, N \quad (6.2)$$

The inventory balance equation is:

$$I_t = I_{t-1} + (R_t + O_t + S_t) - D_t \quad \forall t = 1, 2, 3, \dots, N \quad (6.3)$$

The overtime constraint is:

$$O_t \leq \alpha R_t \quad \forall t = 1, 2, 3, \dots, N \quad (6.4)$$

The subcontracting constraint is:

$$S_t \geq K \quad \forall t = 1, 2, 3, \dots, N \quad (6.5)$$

$$R_t, O_t, H_t, L_t, S_t, I_t \geq 0 \quad \forall t = 1, 2, 3, \dots, N \quad (6.6)$$

In this model, the total cost of aggregate planning is minimized subject to the meeting of the constraints. Equation 6.1 shows all the cost components associated with APP. Equations 6.2 and 6.3 are the production rate balance and the inventory balance. Equations 6.4 and 6.5 pertain to the constraints for the subcontracting and overtime alternatives of APP. Equation 6.6 is the non-negativity constraint. For a planning horizon of one year split into 12 months, the above formulation will result in 48 constraint equations and 72 variables. As this is a linear programming model, the most important assumption that we are making is that all costs are linear in the decision variables. In certain circumstances, this assumption may not be appropriate.

6.8.3 The Linear Decision Rule (LDR)

An interesting approach to the AOP is the linear decision rule proposed by Holt, Modigliani, Muth, and Simon. In the LP formulation, the assumption was that all costs are linear. However, in reality, this could be a restrictive assumption. In the case of LDR, they assume that all the relevant costs, including inventory costs and the costs of changing production level and number of workers, are represented by quadratic functions. The decision variables in LDR are the production rate (P_t), workforce (W_t) and the inventory (I_t) during every period. Four types of costs are captured—the cost of regular production (C_{RP}), cost of hiring/lay-off (C_{HL}), cost of overtime (C_{OT}) and cost of inventory (C_{IN}). The formulation is as follows:

$$\text{Min } TC_{\text{LDR}} = \sum_{t=1}^N C_{RP} + C_{HL} + C_{OT} + C_{IN}$$

Where the four components of the costs are modelled as:

$$C_{RP} = (c_1 W_t)$$

$$C_{HL} = c_2 (W_t - W_{t-1})^2$$

$$C_{OT} = c_3 (P_t - c_4 W_t)^2 + c_5 P_t - c_6 W_t$$

$$C_{IN} = c_7 (I_t - c_8 - c_9 D_t)^2$$

Subject to the constraint:

$$I_t = I_{t-1} + P_t - D_t \quad \forall t = 1, 2, 3, \dots, N$$

In the above formulation, there are 9 different cost coefficients. Depending on the specific cost structure and the nature of operations of a manufacturing set-up, these coefficients are to be arrived at. As mentioned before, D_t is the expected demand for the period. The most appealing feature of LDR is that the optimal policy for P_t and W_t has a simple form involving a linear equation. However, the computational advantage of this simple policy structure is less significant today than at the time this procedure was proposed. This is primarily due to improvements in availability of computing power and software packages for solving LP and NLP problems today. However, LDR is useful for the insights it offers with respect to solving AOP problems using mathematical modelling methods.

6.9 MASTER OPERATIONS SCHEDULING

Master operations scheduling (MOS) represents the critical linkage between planning and execution of operations. It is the next stage in the operations planning process in any organization after aggregate operations planning is done. Aggregate operations planning is a rough-cut capacity planning exercise on the basis of forecast quantities of products and services. MOS makes use of actual customer orders for the purpose of capacity planning and resource allocation to specific customer orders. While aggregate operations planning ensures that adequate capacity is available in a period-by-period basis, organizations need to relate the capacity needs of specific varieties of products and services that they offer to the overall capacity available. MOS is the process by which disaggregation of varieties is done. Using this information, additional planning is carried out to assign the required capacity to each variety. At this stage, planning for the material required for operations during each period is also done.

Master operations scheduling is required for a variety of reasons:

- ⇒ The most important reason is that at the stage of aggregate operations planning, the forecast demand is normally taken

NOTES

NOTES

- for the purpose of estimating the capacity required. However, when actual orders are received, this information provides a better input and another opportunity for the planning exercise.
- ⇒ Moreover, the AOP exercise is to ensure availability of capacity in broad terms. However, as we approach the planning horizon, it is important to relate material and capacity availability to specific varieties of products and services that an organization plans to produce during a planning horizon. Therefore, MOS uses actual and the most recent information while revisiting the planning problem and ensures specific material and capacity availability in a particular time period.
 - ⇒ As customer orders get amended and/or cancelled, the information has a bearing on capacity availability. Therefore, as order inflow happens in real time, the marketing department may want to know if there is scope for accepting newer order inquiries. The marketing department may also want to know if delivery commitments can be made to customers while accepting orders. The MOS module serves the important purpose of computing capacity available to promise.

The critical linkages between MOS and other planning processes in an organization are shown in Figure 6.15. As shown in this figure, MOS takes inputs from the order inflow system and AOP to perform the disaggregation exercise. Based on the disaggregation, capacity planning and materials planning are performed to ensure their adequate availability. If there is some infeasibility in either capacity or material availability, the plans are altered to arrive at feasible production plans. Therefore, MOS determines *what needs to be ultimately produced, not what is demanded*.

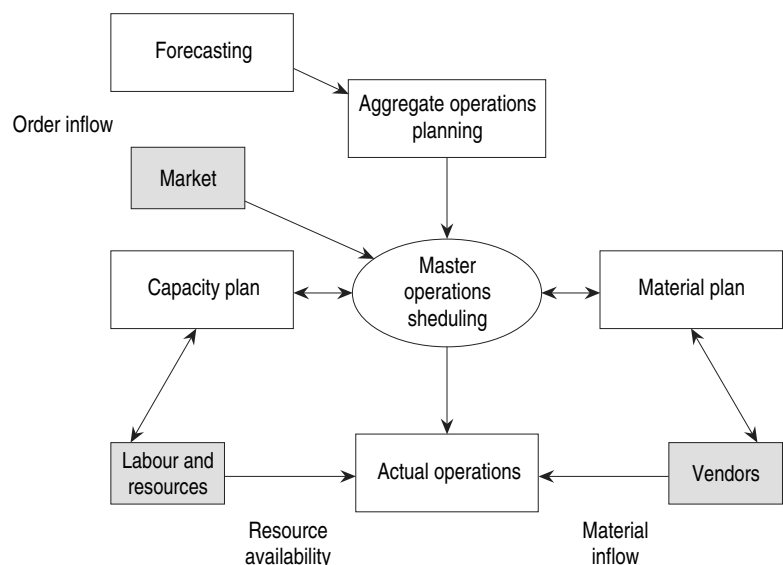


FIGURE 6.15 Master operations scheduling: a planning tool

Master production scheduling (MOS) is the process by which disaggregation of varieties is done.

There are two stages involved in MOS. The first step is to update the projected demand based on earlier forecast and current market information. The second step involves disaggregation of information and relating it to specific material and capacity requirements. After the disaggregation process, it is possible to accurately estimate the amount of material and capacity required for planned units of production of each variety. If either capacity or material is not available, that information is made use of in MOS and plans are re-worked. This is an iterative process and the actual plan is finally arrived at. The two steps are explained in some detail here.

MOS determines what needs to be ultimately produced and not what is demanded.

Step 1: Incorporating current market information into the operations plan

MOS exercise is normally done several times during the planning horizon. Therefore, at each instance it is possible that additional information is available to the planner. Let us assume that the AOP exercise was done during September–October for the next financial year beginning April. At that point in time the demand for June, July and August would have been based on some forecasting. If the MOS exercise was carried out in the month of April it is possible that some customers would have already placed firm orders. Therefore, while planning is done at that time it is important that firm orders are also taken into consideration. If the forecast quantities are higher than the firm orders it is a normal practice to work on the basis of forecast quantities. On the other hand if the firm orders are more than the forecast quantity, then it is desirable to incorporate this information for the purpose of planning. Moreover, between one MOS exercise and the other order cancellations, order amendments and new orders are received. Therefore, the MOS exercise must include all this information at the time of planning. Table 6.8 presents an example that illustrates this.

Step 2: Disaggregation of information

You may recall that at the time of AOP an aggregate unit was chosen and different varieties of a product or service were converted to an “equivalent” product/service. Therefore, it is imperative that the process be reversed at the time of detailed planning. The MOS exercise performs this process known as disaggregation. Let us revisit the example in Section 6.1, where we described three different service levels offered by a service provider. Assume that one unit of the “silver” category of service requires 20 man-hours, one unit of the “gold” category requires 40 man-hours and one unit of the “platinum” category requires 70 man-hours. Table 6.8 provides

NOTES

NOTES

TABLE 6.8 DISAGGREGATION PROCESS IN MOS

Capacity Planned Using AOP		18,000
Capacity required/unit	Silver	20
	Gold	40
	Platinum	70

Type of Service	Demand Status	Planning Horizon		
		Month One	Month Two	Month Three
Silver	Forecast	100	120	140
	Firm order	120	90	30
Gold	Forecast	200	240	180
	Firm order	180	200	60
Platinum	Forecast	80	100	90
	Firm order	50	110	20
MOS Quantity		Shaded area represents MOS quantity.		
Capacity required (for MOS quantity)		16,000	19,700	16,300

information on forecast and firm orders status for the next three months for each category of service. Using this information, one can disaggregate the total requirement and relate specifically to each category of service.

Aggregate operations planning and master operations scheduling are the first two steps in every planning and control activity in an organization. These form the critical inputs for further planning and control of operations, as we will see in the next chapter.

6.10 SUMMARY

- *Operations planning* serves to translate business plans into operational decisions. The decisions include the amount of resources (productive capacity and labour hours) to commit, the rate at which to produce, and the inventory to be carried forward from one period to the next.
- AOP is done to match the demand and the available capacity on a period-by-period basis using a set of alternatives available to an organization to modify the demand and/or the supply.
- Alternatives for modifying demand include the reservation of capacity and methods of influencing (changing) the demand during a period.
- Alternatives for modifying the supply include inventory variations, capacity adjustment, and capacity augmentation.
- The AOP exercise employs two generic strategies: *chase* and *level production*. A chase strategy is often found to be expensive and hard to implement in organizations.
- In reality, a *mixed strategy* using a combination of alternatives is employed in an AOP exercise. It uses a variety of alternatives for modifying supply.
- The structure of a transportation model lends itself to studying the AOP problem. Linear programming can also be used to model the AOP problem.
- MOS involves disaggregation of information and ensuring the required capacity and material are available as per the plan.

6.11 REVIEW QUESTIONS

- Why do you need aggregate units while planning for operations?
- Give three examples each from the manufacturing and service sectors of industry for aggregation of products and services.
- Assume that you are a garment manufacturer. Your current production rate is 20,000 pieces per day. If you want to increase it to 30,000 per day, what problems do you envisage in making this change?
- Identify a suitable aggregate unit for each of the following situations:
 - A business school offering four different long-duration academic programmes and three different short-duration programmes
 - A manufacturer of colour televisions of 30 different varieties
 - A service provider offering inter-city transport facility using six different classes of vehicles
 - A multi-specialty hospital
 - A financial supermarket such as ICICI Bank
 - A fast-food restaurant offering North Indian and South Indian dishes
 - A law firm offering a variety of legal services
 - A manufacturer of custom-made PCBs
 - A BPO service provider offering back-office services to a variety of domestic and MNC clients
- What do you mean by hierarchical operations planning? Are there any benefits in this exercise? Use an example to demonstrate hierarchical operations planning.
- How does aggregate operations planning impact each of the following functional areas in an organization?
 - Marketing
 - Finance
 - Strategy
 - Materials and procurement
- What considerations will you bear in mind while selecting either demand-oriented alternatives or supply-oriented alternatives during an AOP exercise?
- Which AOP alternatives are more suited to a service organization? Why?
- What are the key differences between level strategy and chase strategy in AOP formulation?
- Identify suitable AOP alternatives for an organization manufacturing made-to-order items. Will your recommendation change if the items are made-to-stock?
- "Mixed strategies are always superior to pure strategies in an AOP exercise." Comment on this statement.
- What are the pros and cons of using optimal methods for solving the AOP problem?
- How is the master production scheduling exercise different from an AOP exercise?
- How are materials planning and capacity planning related to MOS?

6.12 PROBLEMS

- A manufacturer of ceiling fans offers three different versions of the product to its customers: basic (three blades), improved (four blades) and deluxe (enhanced performance features). Table 6.9 has information on the machine-hours required for each of these variations and the number of fans to be manufactured in the next quarter. Assume 25 working days per month.

TABLE 6.9 DATA FOR PROBLEM 1

Fan Type	April	May	June	M/c hours per unit
Basic	140	160	160	9.50
Improved	110	150	150	12.00
Deluxe	120	120	140	18.50

- Choose an appropriate aggregate unit for planning the production.
- For the chosen aggregate unit, estimate the capacity required.

- If the manufacturer operates two shifts each of eight hours, how many machines are required for meeting the demand during the next quarter?
- A manufacturing organization employs 220 people in the factory. The manufacturing is largely manual; therefore, man-hours are a good measure of the capacity of the system. The factory currently works for a single shift of eight hours. The number of working days for the next six months is as follows: 24, 22, 25, 19, 21, and 23. The company uses this information to plan its capacities. The demand for its product during the next six months is such that with the current level of employment, the company will have 80 per cent capacity utilization. During the next six months, the number of working days remains the same but the demand is likely to go up by 35 per cent. Under these circumstances, analyse the situation and answer the following questions:
 - What is the total capacity available in the factory as of now?
 - The company is considering three options to meet the impending demand for capacity:

- (i) Introduce two hours of overtime for the next six months
- (ii) Introduce one hour of overtime for the next six months
- (iii) Introduce two hours of overtime for the first three months and two hours of idle time thereafter

Evaluate these options from the capacity perspective and suggest a suitable option for the company.

3. Suppose, in Problem 2, the following are the costs pertaining to overtime, inventory and shortage and idle time:
- (a) Overtime premium: ₹4 per hour
 - (b) Idle time costs: ₹2 per hour
 - (c) Cost of holding inventory: ₹10 per hour
 - (d) Cost of shortages is 200 per cent of cost of holding inventory.

Use this additional information and suggest an appropriate option to the management.

4. Hot Spot is a fast-food restaurant located in a prominent locality in the city of Bangalore. Hot Spot offers three different menus for breakfast: standard, extended, and custom-made. The standard offering takes nearly 20 minutes of one worker and the extended requires 30 minutes. On the other hand, custom-made requires three workers to spend 25 minutes each. Hot Spot is interested in identifying the right number of workers required in its restaurant. Using the data given in Table 6.10, estimate the aggregate capacity required at Hot Spot during the next four weeks.

TABLE 6.10 DATA FOR PROBLEM 4

Type of Service Offered	Demand During the Next 4 Weeks			
	1	2	3	4
Standard	2,000	2,200	2,800	2,500
Extended	1,700	1,600	1,300	1,000
Custom-made	500	600	200	400

5. Assume that Hot Spot currently employs twenty nine workers in its restaurant. If Hot Spot works for six days a week, 10 hours a day compute the following:
- (a) What is the capacity available at Hot Spot in its current level of working?
 - (b) Is the capacity adequate to meet the demand? If not, how much additional capacity do they need?
 - (c) What alternatives would you recommend to Hot Spot to solve their capacity problem?
 - (d) If Hot Spot is ready to recruit new workers, how many should they recruit?
6. A manufacturing firm is performing an AOP exercise for its product. One unit of product requires 100 hours of the capacity. An extract of the aggregate planning exercise for the firm is given in the table. Table 6.11 has details on the number of working days, number of people employed and

TABLE 6.11 DATA FOR PROBLEM 6

Month	Demand (in Units)	Demand (Hours)	No. of Working Days	Working Hours Per Day	No. of Workers	Capacity Available (Hours)	Supply-Demand (Hours)
April	160	16,000	23	8	90		
May	150	15,000	22	8	90		
June	140	14,000	21	8	90		
July	160	16,000	24	8	90		
August	120	12,000	22	8	90		
September	150	15,000	22	8	90		
October	160	16,000	19	8	90		
November	180	18,000	23	8	90		
December	160	16,000	21	8	90		
January	180	18,000	23	8	90		
February	190	19,000	20	8	90		
March	150	15,000	24	8	90		
Total							
Average							

working hours per day. Complete the table and answer the following questions:

- What is the capacity required for the planning horizon? Does the company have adequate capacity to meet the demand?
- Will the company fall short of capacity if you analyse on a period-by-period basis? By how much does it fall short?
- If the cost of shortage is ₹5 per hour of capacity falling short, what is the cost that the company may incur on account of shortages?
- If the cost of having excess capacity is ₹1 per hour, what will the cost of excess capacity be?
- If the company chose to adopt this plan, what is the total cost of the plan? Identify the AOP alternatives and the AOP strategy that the company has deployed in this plan.

- A manufacturer of garments is interested in making an aggregate production plan for its range of products manufactured. From the perspective of capacity usage, the range of products could be divided into three: basic garments, office wear and trendy garments. A basic garment required on an average 2 hours. The ratio of the time required for the office wear and trendy garment to the basic garment is 2 and 4, respectively. The company can source 20 per cent additional capacity through overtime. The overtime premium to be paid for the labour is 50 per cent. Material cost is 125 per cent of labour cost. Currently the company is operating at a capacity of 20,000 hours. The labour charges are paid at the rate of ₹40 per hour.

The company can also subcontract the work to nearby units for a premium of 20 per cent. However, every time a subcontracting arrangement is made, it involves a fixed cost of ₹3,000 also. Short term funds for working capital at 24 per cent per annum is available for the company without any restrictions. The company cannot afford to run into shortages as it is a competitive industry. The forecasted demand for the next six periods is as shown in Table 6.12.

TABLE 6.12 DEMAND FORECAST FOR THE NEXT 6 PERIODS

	Basic	Office wear	Trendy
Period 1	5,000	2,000	1,500
Period 2	4,500	1,800	1,900
Period 3	5,500	2,200	2,000
Period 4	5,200	2,100	1,750
Period 5	5,500	1,600	1,800
Period 6	5,000	1,900	1,500

- Set up a transportation table for the given problem.
- What is the least cost plan for the company?
- Suppose the premium for subcontracting is only 12 per cent, will the plan change?

- A major manufacturer of vacuum cleaners is preparing the AOP for the four quarters of the next year. They are currently operating at a production level of 24,000 vacuum cleaners per quarter. The number of employees working now is 480. The current capacity represents the regular time production. The company can additionally produce a maximum of 2,000 vacuum cleaners per quarter through overtime. Subcontracting capacity to the extent of 750 vacuum cleaners per quarter is also available. The cost of production (including materials) through regular time, overtime, and subcontracting are ₹7,300, ₹8,000 and ₹10,000 respectively. Unused regular time capacity costs ₹800 per vacuum cleaner. The company plans to start the planning horizon with an opening inventory of 2,700 vacuum cleaners and wants to end with an inventory of 2,500. No back orders are allowed and the cost of carrying the vacuum cleaners is ₹1,000 per quarter. It costs ₹9,000 to hire a worker and ₹6,000 to lay off. The demand for the planning horizon (next four quarters) is 6,700, 28,000, 49,000, and 30,000 respectively.

The company is considering two options with respect to the workforce, as shown in Table 6.13.

TABLE 6.13 THE TWO PLANS UNDER CONSIDERATION

Quarter	1	2	3	4
Plan 1	480	520	620	480
Plan 2	570	570	570	570

- Advise a suitable course of action for the company.
- The company wants to evaluate the policy of having opening and closing inventories. Help them with some computations that provide some implications of carrying inventory.

- Channapatna near Bangalore is well known for specialized wooden toys. A toy manufacturer in Channapatna has determined the demand (in units) to serve the forecasted demand for the next year. This data is shown in Table 6.14.

TABLE 6.14 THE DEMAND FOR THE TOYS (IN UNITS)

Month	1	2	3	4	5	6
Demand	500	800	1,000	1,400	2,000	1,800
Month	7	8	9	10	11	12
Demand	1,350	1,100	900	2,400	2,900	1,000

Each worker is paid a monthly salary of ₹8,000 and can produce 10 toys per month. The owner does not resort to overtime, instead, he prefers to hire and fire workers depending on his requirement. He can hire and train a new employee for ₹8,000 and lay off for ₹2,000. In this way, he enables the local community to acquire some skills and

gainfully employ themselves over time. The cost of holding inventory is ₹125 per month. At present there are 130 employees in the manufacturing unit.

(a) Prepare a level production strategy for the toy manufacturer.

(b) Can you also develop a chase strategy for the toy manufacturer?

(c) Identify two more options for the toy manufacturer.

(d) Prepare a comparative analysis of the plans and suggest a suitable course of action for the toy manufacturer.

6.13 CASE STUDY

New Age Electronics Limited

Bala Athreya, the director (operations) of a popular brand of electronic voltage stabilizer, was on the verge of finalizing the aggregate production plan for his manufacturing plant when he received a letter from his marketing director about a new marketing and promotional campaign that he plans to launch to boost the sales of voltage stabilizers. What is the implication of this to him? Should he re-plan the campaign, or should he convince the marketing director about the need to take another look at his plan?

INTRODUCTION

New Age Electronics Limited (NAEL) is a Hyderabad-based manufacturer that specializes in the manufacture of electronic voltage stabilizers. Electronic voltage stabilizers are increasingly popular on account of better performance, longer working life, light weight, and low cost. NAEL was promoted by Bala Athreya, an electronics engineer with about 15 years' experience prior to starting NAEL with two other friends. NAEL manufactures three variations of voltage stabilizers, primarily differing on account of the type of application they are put to and the rating. These three can be designated as VS-TV, VS-RF and VS-IN. One unit of VS-TV requires two hours of labour. For the other two models VS-RF and VS-IN the labour hours required are 2.5 hours and 3.25 hours, respectively. Due to their early entry into the market, NAEL has been enjoying a good market share of the electronic voltage stabilizers.

PRODUCTION OF VOLTAGE STABILIZERS

The factory is currently working on a single shift with a capacity equivalent to 60,000 hours of labour. Over the last six months, the order inflow has been on the rise at NAEL, and Bala had to schedule overtime since the demand was in excess of 60,000 hours. The factory can increase the production up to 80,000 hours through overtime although it means paying more by way of overtime (OT) premium. By resorting to a second shift, NAEL can produce anywhere between 60,000 hours and 120,000 hours. A second shift has certain benefits. Instead of paying an OT premium of ₹15 per hour, it is enough if NAEL pays ₹3 per hour as shift allowance. After carefully analysing the demand trend in the past six months, Bala recently took a decision to operate the factory

on a two-shift basis and made the change two months back. However, by changing over to two shifts from one, NAEL incurred a one-time shift change cost of ₹50,000. Henceforth, Bala plans to operate only on a two-shift basis for the coming year as the demand is likely to go up.

COSTS PERTAINING TO PRODUCTION AND INVENTORY

NAEL incurs several types of costs, just as any manufacturer would. The first set of costs relate to inventory. As it manufactures an electrical product, NAEL does not wish to carry too much inventory of finished goods. There are frequent changes in the components that NAEL uses in its products. Therefore the cost of carrying inventory is ₹18 per hour of average inventory held. Shortage costs are double the average inventory carrying costs at NAEL. The second element of cost is the cost of changing the production level. Changing the production level (either up or down) requires NAEL to make several adjustments with its suppliers, the internal departments. It also calls for re-planning and manpower scheduling adjustments to suit the revised production plan. The costing department recently did an elaborate exercise and estimated the cost of change in production level to be ₹12 per hour.

PRODUCTION PLANNING

NAEL is currently in the process of developing an aggregate production plan for the next financial year. Detailed estimates of the demand were made and based on this a strategic planning exercise at the corporate level was done to arrive at the potential demand for the next year. During the strategic planning exercise the marketing department highlighted the increased competition as more players are bringing similar products to the market. According to some market information, a number of new players were planning to launch major marketing campaigns with a view to capture some share of the market. According to NAEL top management, these new entrants would not make significant inroads into their market share. The new competitors were most likely to take away the market from traditional voltage stabilizer manufacturers. After detailed discussions, the final projections for the next year were finalized. Table 6.15 has the demand projection for the next year.

TABLE 6.15 DEMAND PROJECTION FOR THE NEXT YEAR

Month	VS-TV	VS-RF	VS-IN
1	12,150	12,750	6,100
2	9,050	18,700	5,600
3	22,900	15,000	11,000
4	6,750	8,900	4,100
5	10,750	11,550	4,200
6	10,250	17,500	6,100
7	26,550	19,200	9,200
8	6,950	11,200	6,800
9	11,250	12,400	5,700
10	9,950	15,500	7,500
11	28,750	20,000	9,100
12	7,050	11,000	7,200

Bala and his team analysed the demand pattern for the next year and came up with certain observations. For instance, the demand for the stabilizers had pronounced seasonality and peaked during Months 3, 7, and 11. Further, being an electrical product, the cost of carrying inventory was high, therefore the amount of inventory build-up they could do to meet the peak demand was a question that they needed to answer. After some discussions, they came with two possible production plans:

- To maintain a constant production level of 80,000 hours except during peak periods when it would be increased to 100,000 hours
- To maintain the production at the level of the monthly demand, but to increase it to the peak production level of

120,000 hours during the peak months in order to meet the demand

They were not clear about which one to choose as there are different implications for each of these choices. One of the major constraints for NAEL is the ability of their supply chain partners to respond to the changes in production plan. Up to 5 per cent increase or decrease is generally accepted without any discussion. Mere e-mail communication will do. Another 5 per cent change (both increase and decrease) requires that NAEL discusses this with them, have a mutual agreement, and give them at least 45 days notice. Any larger magnitude of change in excess of 10 per cent is generally resisted. It also requires a much longer time frame to respond.

LATER DEVELOPMENT

While Bala and his team were working on finalizing the production plan for the next year, the marketing director informed them that they would like to launch a promotional campaign to counteract the new entrants. The specifics of the plans are as follows. The marketing plan would cost ₹2 million and the budget will be spent during Months 2 to 5. Due to this the increase in sales (across all the three variations) expected would be Month 3: 10 per cent; Month 4: 15 per cent, Month 5: 18 per cent, Month 6: 12 per cent and Month 7: 5 per cent.

QUESTIONS FOR DISCUSSION

- What is your advice to Bala Athreya with respect to production planning for the next year? Which of the two plans would you recommend? Why?
- What are the practical difficulties that Bala and his team will face while implementing the plan? How should they handle them?
- What are the implications of the marketing plans at NAEL? Bala needs to attend a meeting to discuss this issue. Prepare a brief for the meeting.

6.14 MULTIPLE-CHOICE QUESTIONS

- 6-1.** The hierarchy in planning, offers several advantages to the:
A. Supplier
B. Planner
C. Customer
D. Manufacturer
- 6-2.** Using which exercise, firms arrive at the quantity and timing of resources to be committed to ensure continuous flow of goods and services to customers:
A. Organization planning
B. Energy planning
C. Aggregate operations planning
D. Inventory planning
- 6-3.** What are needs for aggregate operations planning?
A. Demand fluctuations, Capacity fluctuations, Difficulty level in altering operation rates, Benefits of multi-period planning
B. Market fluctuations, Production fluctuations, Difficulty level in altering selling rates, Benefits of single period planning
C. Weather fluctuations, Sales fluctuations, Difficulty level in altering demand rates, Benefits of two years planning
D. Govt. policy fluctuations, Procurement fluctuations, Difficulty level in altering supply rates, Benefits of three years planning
- 6-4.** What are alternatives for managing demand?
A. Managing Demand, Reservation of Utility
B. Reservation of Capacity, Influencing Demand
C. Managing Production, Reservation of Supply
D. Managing Sales, influencing Schedule
- 6-5.** Supply can be modified through which one of the alternatives:
A. Inventory-based alternatives, capacity adjustment alternatives, capacity augmentation alternatives
B. Manpower-based alternatives, organization-based alternatives, inventory augmentation alternatives
C. Manufacturing-based alternatives, Operation adjustment alternatives, costing augmentation alternatives
D. Procurement-based alternatives, Supply adjustment alternatives, Supply augmentation alternatives
- 6-6.** In which the emphasis is on not disturbing the existing operations at all?
A. Chase strategy
B. Mixed strategy
C. Level strategy
D. Inventory strategy
- 6-7.** In which of the below methods we use linear programming (LP) and the transportation and dynamic programming (DP) models?
A. The trial-and- error methods
B. Search methods
C. Optimal methods
D. Start methods
- 6-8.** What model is a special case of linear programming that deals with the issue of shipping commodities from multiple sources to multiple destinations?
A. The Linear Programming Model for AOP
B. Aggregate Operations Planning Using the Transportation Model
C. The Linear Decision Rule (LDR)
D. The supply and Demand Rule
- 6-9.** What represents the critical linkage between planning and execution of operation?
A. Cost/estimation scheduling
B. Product/service scheduling
C. Finance/accounts scheduling
D. Master Operations scheduling
- 6-10.** An earlier planning exercise invariably results in a certain level of:
A. Re-planning at a later stage
B. Re-planning at a beginning stage
C. Re-planning at a middle stage
D. Re-planning at an End stage

Answer Keys: 6-1. B 6-2. C 6-3. A 6-4. B 6-5. A 6-6. C 6-7. C 6-8. B 6-9. D 6-10. A

7

CHAPTER

MANUFACTURING ECONOMICS: FOUNDATION CONCEPTS

7.1	Dependent Demand Attributes
7.2	Planning A Framework: The Basic Building Blocks
7.2.1	Multiple Levels in Products
7.2.2	Product Structure
7.2.3	The Bill of Materials (BOM)
7.2.4	Time Phasing the Requirement
7.2.5	Determining the Lot Size
7.2.6	Incorporating Lead-time Information
7.2.7	Establishing the Planning Premises
7.3	Summary
7.4	Review Questions
7.5	Problems
7.6	Net-Wise Exercises
7.7	Mini Projects
7.8	Multiple-choice Questions

ERP IMPLEMENTATION AT HPCL

NOTES

Hindustan Petroleum Corporation Limited (HPCL) is a Fortune 500 company with an annual turnover of over ₹914.48 million for the financial year ended March 2007. The company has successfully implemented JD Edwards EnterpriseOne ERP spanning about 400 locations and 4,000 employees across the nation.

The ERP implementation programme was started in 2000. The availability of sound technical infrastructure was necessary for implementing such a large initiative. This meant building a centralized data centre where the servers and applications could be hosted as well as having efficient connectivity for all locations spread across India's geography. Around 400 locations were completed over a period of two and a half years. As each location was selected for inclusion, the related infrastructure (like WAN, lease lines, VPN and VSATs) was built simultaneously depending on the availability.

As HPCL enjoys a national presence, it was important to determine the path for the ERP rollout. Several business processes were analysed for determining the points affecting the process look. The project was kicked off by a 25-member in-house team comprising representatives from all functional areas. After the completion of the first phase of the project, which included system configuration, more employees were inducted into the team and its size expanded to 100 people. In addition to this, a team of 100 consultants was also working on it.

The locations due for the "Go-Live" stage in a particular month were given top priority and a "work backwards" system was followed to arrive at separate tasks and their deadlines for those locations. These tasks included the procurement of hardware, applications for communication links, and testing and liaising with local people at the location.

Post-implementation, the company noticed substantial improvement in efficiency. For example, before the implementation, the annual financial accounts closure would take approximately a month while the monthly closing would take 15 to 20 days. In 2007–08, the monthly and quarterly closing was completed in about five days while the annual closing took a mere 10 to 12 days.

The second major benefit of the centralized ERP implementation was that it compelled the company to look at various other services that could be offered to customers with the help of IT. It was also realized that they could improve transparency by making more information available to their customers, vendors, and transporters on a real-time basis. With the help of the Web, they can log in and check the status of loads, orders, and payments.

The implementation gave the company an opportunity to relook at all its business processes from the beginning. Several processes in manufacturing, planning, procurement, and employee benefits that were earlier done manually were automated. The ERP enabled the generation of various day-end MIS reports on a regular basis. This helped to revamp the way in which reporting was done to the top management.

HPCL's ERP implementation can potentially provide several clues to other large organizations wishing to derive benefits from ERP/IT adoption.

Source: Based on Abhishek Raval, "HPCL Refines Business Process with ERP Implementation," accessed at <http://biztech2.in.com/india/casestudies/enterprise-solutions/hpcl-refines-business-processes-with-erp-implementation/28091/0>.

NOTES

Material requirements planning (MRP) is a computerized planning system developed during the 1960s to address the vexing problem of high inventories in manufacturing organizations. Before the introduction of MRP systems, organizations were using the same methodology for controlling all types of inventories. However, MRP systems drew the attention of managers to the fact that the planning of requirements for production, such as raw material and work in progress, require a different approach from that used for managing finished goods. MRP systems exploit certain unique characteristics of the production items. They utilize information on lead time, inventory status, and master production schedule to ensure production items are available at the time of requirement. In this chapter, we will try to understand the logic behind MRP systems. We will also see how the idea applied to controlling materials could be extended to other resources required in any operations system. The culmination of this interesting extension of the MRP logic are the modern-day enterprise resources planning (ERP) systems that serve as the backbone of the IT-enabled strategic planning and control in any operations system. Therefore, these planning methodologies can be broadly defined as resources planning.

Material requirements planning (MRP) is a computerized information system that aids the planning of materials in manufacturing organizations.

Before we discuss the issues in resources planning, it will be useful for us to understand the key differences between the two types of inventories that exist in an operations system. The first is called **operating inventory**. This includes all the

Operating inventory includes all the resources (broadly of material and capacity) that are available for the operating system to consume in the production process.

NOTES

resources (broadly of material and capacity) that are available for the operating system to consume in the production process. Consider an automobile manufacturer such as Hyundai. The inventory of tyres available in the stores is meant for satisfying the production requirements. If Hyundai has 4,000 tyres suitable for its Santros model at the beginning of the shift, it can assemble 1,000 Santros during the shift (we do not count the extra tyre provided as standby in this computation). Similarly, if it takes 1.5 hours to paint an assembled car at the paint shop, the capacity of paint shop required for 1,000 Santros is 1,500 hours. There is no statistical estimation involved in arriving at the number of Santros that Hyundai can assemble on that day or the capacity required at the paint shop.

On the other hand, operations systems carry another type of inventory known as *distribution inventory*. Distribution inventories are meant for market consumption. In our example, finished Santros waiting for dispatch belong to the distribution inventory. If the distribution inventory of Santros is 1,000, we cannot come to the conclusion that 1,000 Santros will indeed be sold. It depends on the demand in the open market and may be subject to statistical fluctuations. Unlike the operating inventory, it requires alternative estimation techniques to approximately guess the number of Santros likely to be demanded on that day.

One can give similar examples in the service system as well. Consider a hotel in the Andheri area of Mumbai. The number of rooms to be made available and the number of people to be served lunch and dinner are examples of distribution inventories. On the other hand, the amount of raw materials made available for use in the kitchen and the number of waiters and housekeeping staff deployed in the hotel on that day are examples of operating inventories.

Look at the converse of the problem we just discussed. How do we decide how much inventory to carry? In the case of operating inventories, it is a matter of using arithmetic to arrive at this decision, whereas in the case of distribution inventories, the decision is a little more complex. In other words, while it is difficult to decide how many Santros to assemble on a particular day, it is very easy to compute the exact number of tyres required on that day once we have decided on the assembly schedule. In this chapter, we focus our attention on operating inventories and developing a methodology to help managers plan for these inventories. Planning for intermediate components and raw material/bought-out items using this methodology is known as materials requirement planning. However, one can extend this idea of planning to capacity and other resources in an operating system. Therefore, we broadly define this methodology as resources planning. We will discuss the planning issues pertaining to distribution inventories in the next chapter.

7.1 DEPENDENT DEMAND ATTRIBUTES

Why should a manager address planning issues associated with operating inventories differently from those associated with distribution inventories? In order to answer this question, we need to know more about the demand attributes of these inventories. As we have already discussed, operating inventories have dependent demand. Distribution inventories, on the other hand, have independent demand. All other differences stem from this attribute of demand.

The foremost difference stems from the nature of the demand. There is no uncertainty in a dependent demand item. Moreover, the demand is lumpy, manifesting at specific points in time in response to a requirement in the system. Consider the domestic fan manufacturer, Crompton Greaves Limited. If Crompton Greaves decides to produce 20,000 fans in May 2014, the number of blades that it requires is 60,000 (since each assembled fan is fitted with three blades). Going one step further, if each blade consumes two hours of machine time, 350 g of a grade of steel, and 50 ml of paint, then it requires 120,000 machine-hours, 21,000 kg of steel, and 3,000 litres of paint if it were to manufacture the 60,000 blades during the month. In this example, we notice that a parent-child relationship between inventories causes the dependency. The same cannot be said in the case of independent demand items.

Due to certainty, the goal for a planning manager is to make the dependent demand items available to exactly match the requirement. It is feasible to attain a 100 per cent service level in this case. On the other hand, due to uncertainty, independent demand items cannot be made available at a 100 per cent service level. The extra inventory required increases substantially as we approach 100 per cent service level in this case. Therefore, managers plan for a certain targeted service level in the case of independent demand items.

Since dependent demand items exhibit some structure and have a causal relationship with other items in the system, it is possible to estimate the demand by appropriate planning methodologies. Moreover, as our example demonstrated, it is possible to accurately estimate how much to order. These issues are far more complex in the case of independent demand items. However, in the case of dependent demand items, the timing of the order is very crucial. This is because the dependency relationship among the items requires that items arrive exactly at the required time. Otherwise, shortage and schedule slippages could cascade through the system and create severe schedule disruptions. Table 7.1 lists the key differences between dependent demand and independent demand items.

Since dependent demand items exhibit some structure and have a causal relationship with other items in the system, it is possible to estimate the demand through simple planning methodologies.

NOTES

NOTES

TABLE 7.1 KEY DIFFERENCES BETWEEN DEPENDENT DEMAND AND INDEPENDENT DEMAND ITEMS

Attribute	Dependent Demand	Independent Demand
Nature of demand	No uncertainty; dependent; parent–child relationships cause dependency	Considerable uncertainty, independent
Goal	Make resources availability meet requirements exactly	Make availability meet estimated demand for a targeted service level
Service level	100% service level a necessity, feasible to achieve	100% service level not feasible
Demand occurrence	Often lumpy	Often continuous
Estimation of demand	By production planning	By forecasting
How much to order?	Known with certainty	Estimate based on past consumption
When to order?	Very critical, can be estimated	Cannot be answered directly

7.2 PLANNING A FRAMEWORK: THE BASIC BUILDING BLOCKS

Before we unfold the planning methodology for dependent demand items, let us look at the following aspects of the planning framework: (a) multiple levels of dependency, (b) the product structure—the bill of materials (BOM), (c) time-phasing the requirement, (d) determining the lot size, (e) incorporating lead-time information, and (f) establishing the planning premises.

7.2.1 Multiple Levels in Products

Clearly, a parent–child relationship in dependent demand items causes dependency. In reality, multiple levels of relationships exist in a product. While computing the requirement for an item, it is important that we proceed level by level. Otherwise, the amount of inventory that we make available will be different from what is required. Let us study the example of a basic telephone instrument to illustrate this idea. As shown in Figure 7.1, the telephone instrument consists of a base unit and a handset. The handset has a speaker and a microphone and is covered by a pair of cover plates firmly held together using a pair of screws. There is a wire that connects the handset to the base unit. In this simple example, let us consider the parent–child relationship.

This iterative process of computing all requirements at a particular level and then moving down one level is known as an **explosion** in resources planning literature.

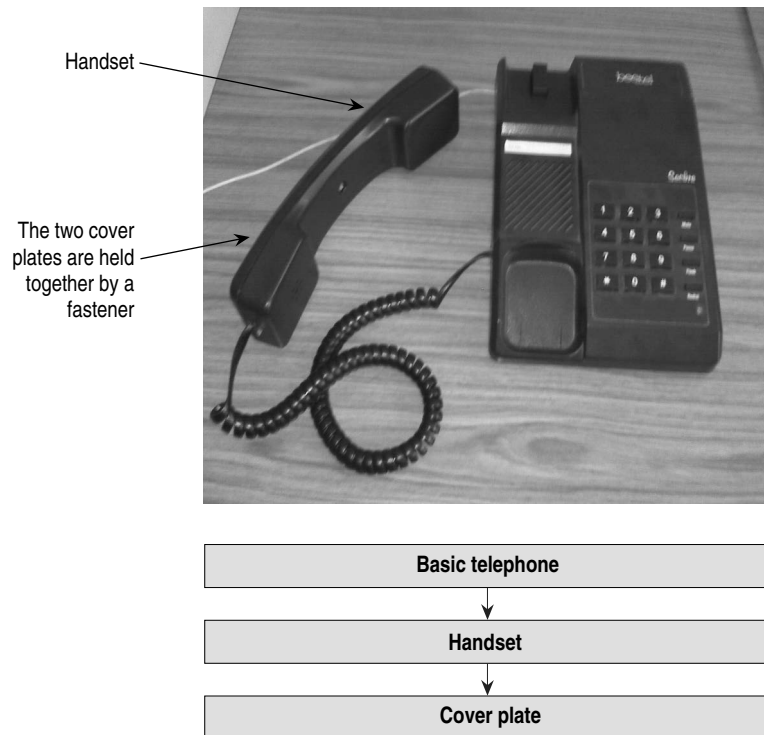


FIGURE 7.1 A telephone instrument

Let us assume that we have in stock an inventory of 30 telephones, 27 handsets and 16 cover plates. Suppose that we want to plan for the production of 100 telephones during the next week. How much inventory do we need in each of these components? One method is to merely subtract the on-hand inventory from the requirement at each level to arrive at this number (see Method A in Figure 7.2). Another method is to proceed level by level in the computation. In this method, after computing the requirement at one level, we use the parent-child relationship to compute the requirement at the next level (see Method B in the Figure 7.2).

As we see in Figure 7.2, both the methods result in different numbers. However, the second calculation truly depicts the

Method A		
Basic Telephone		
Required		100
On-hand inventory		30
Planned quantity		70
Handset		
Required		100
On-hand inventory		27
Planned quantity		73
Cover Plate		
Required		200
(Each handset requires two cover plates)		
On-hand inventory		16
Planned quantity Manufacturing		184

Method B		
Basic Telephone		
Required	CMR	100
On-hand inventory		30
Planned quantity		70
Handset		
Required		70
On-hand inventory		27
Planned quantity		43
Cover Plate		
Required		86
(Each handset requires two cover plates)		
On-hand inventory		16
Planned quantity		70

FIGURE 7.2 Methods for calculating inventory requirement

NOTES

At the next level, we can identify the sub-assemblies that make up each assembly. The base unit assembly consists of the operating unit (consisting of the basic circuitry and processor), the panel board (consisting of the buttons to dial the numbers), the control panel (consisting of controls for volume, pulse/tone settings etc.), a pair of cover plates, four lugs (to support the instrument on any surface), and four screws to cover the entire unit using the base plates form the next level. One can similarly identify all the components and proceed downwards until we reach the basic raw material/bought-out components. The resulting graphical structure is the product structure. Figure 7.4 shows the product structure for the basic telephone unit. When common items appear in more than one level, they must be represented at the lowest level of their appearance. This is known as low level coding. In our example, the item “screw” appears at Level 2 and Level 4, but we have represented them at Level 4. Low level coding does not alter the logic behind planning for the requirement but merely serves to improve the efficiency of processing.

The product structure graphically depicts the dependency relationships among various items that make up the final product.

The product structure graphically depicts the dependency relationships among various items that make up the final product. Every level in the product structure has a parent relationship with those below it. In Figure 7.4, the numbers in parentheses show the number of items at a particular level needed to assemble one unit of its parent. Thus, 12 buttons and 12 screws are required to make one button assembly. Similarly, one speaker, one microphone, two cover plates, and two screws make up one handset.

Since knowledge of the dependency is fundamental to planning operating inventories, the product structure is very important. However, in several real-life examples, it is not possible to represent the dependency information in the form of a product structure. This is because the number of components that make

NOTES

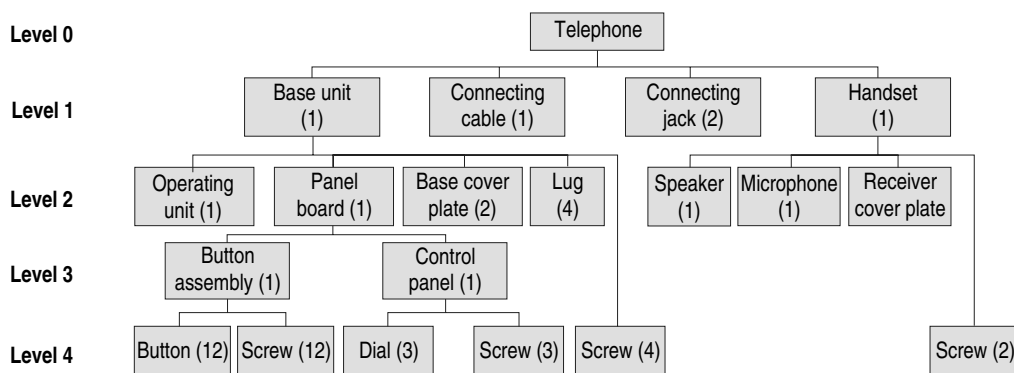


FIGURE 7.4 The product structure of a telephone

EXAMPLE 7.1

Consider a product, code-named A, manufactured by a company. Product A is made of three Sub-assemblies B, C, and D. One unit of B, four units of C, and two units of D are required for assembling one unit of Product A. Sub-assembly B is made of two units of E and four units of F. Sub-assembly C is made of two units of E and four units of D, whereas Sub-assembly D is made of one unit of G and three units of H.

- Develop a product structure for Product A.
- How many units of D are required to manufacture 10 units of A?
- Will the answers change if there is already an inventory of 10 units each of C and D?

Solution

- From a description of the product and its sub-assemblies, we note that Item D occurs at two different levels. Therefore, we employ low-level coding for Item D. The product structure is shown in Figure 7.5. The product has three levels.
- If we assume that no inventory of sub-assemblies or the end product exists in the system, then the number of D's required is computed as follows:

Number of C's required for assembling 10 units of A: 40
 Number of D's required for assembling 40 units of C: 160
 Number of D's required for assembling 10 units of A: 20
 Therefore, the total number of D's required for manufacturing A = 180

- Since an inventory of the intermediate sub-assemblies exists, it is important to compute the number of D's required using a level-by-level approach.

Number of C's required for assembling 10 units of A: 40
 Number of C's already available: 10
 Net requirement of C's: 30
 Number of D's required for assembling 30 units of C: 120
 Number of D's required for assembling 10 units of A: 20
 Total requirements of D's: 140
 Number of D's already available: 10
 Net requirement of D's for assembling 10 units of A: 130

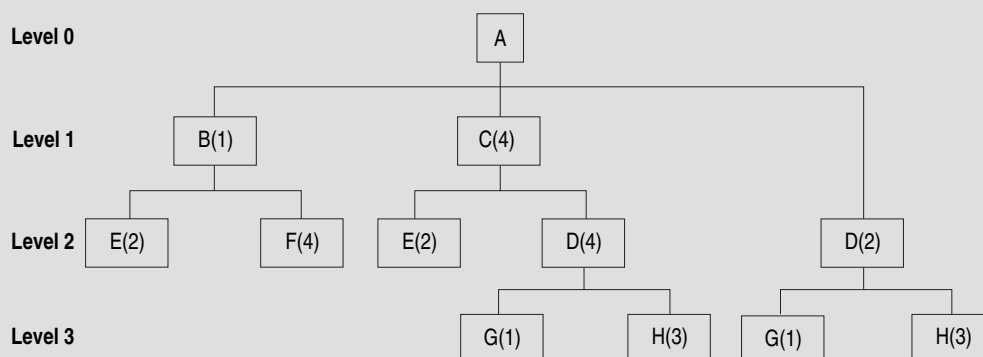


FIGURE 7.5 The product structure

up a final product could be numerous and the number of levels involved could also be many. Instead, one can represent this information using a standard data structure. Such a structure is known as the *bill of materials (BOM)*.

7.2.3 The Bill of Materials (BOM)

A **bill of materials** is a list of all the parts, ingredients, or materials required to assemble or put together one unit of a product. A BOM essentially consists of the complete list of each part in the product

The **bill of materials** is a list of all the parts, ingredients, or materials required to assemble or put together one unit of a product.

NOTES

structure, the components that are directly used in the part, and the quantity of each component needed to make one unit of that particular part. The data set also includes a short description and the unit of measure for each part. Clearly, a BOM is an alternative representation of a product structure. It provides an efficient methodology to represent complex product structures having multiple levels and numerous items. Codes are used to denote the level at which the item occurs in the product structure, and the number the parent requires to assemble one unit.

A variety of formats are available for BOMs. The simplest format is the *single-level BOM*. It consists of a list of all components that are directly used in a parent item. An *indented BOM* is a form of multi-level BOM. It exhibits the final product as Level 0 and all its components as Level 1. The level numbers increase as you proceed down the tree structure. If an item is used in more than one parent within a given product structure, it appears more than once, under every sub-assembly in which it is used. A third variation is the *modular BOM*. Modular BOMs are very useful to represent product structures with several varieties.

In the telephone example, let us assume that four different colours (grey, blue, black, and beige) are available. Further, let us assume that three different control panels (simple, deluxe, and elegant) and four different memory settings (20-call memory, 50-call memory, 100-call memory and 250-call memory) are available. This means that we can make 48 ($4 \times 3 \times 4$) unique offerings to the customer. Therefore, we need 48 different BOMs to represent each variant. One way to solve this problem is to have a modular BOM. By picking up one variation for each of the three attributes, it is possible to construct one unique BOM for the purpose of planning. In several other cases such as automobiles and IT hardware, the number of unique offerings could be as high as 250,000 considering the numerous options available for each variation. In such situations, it is prudent to use a modular BOM. Table 7.2(a), (b), and (c) illustrate the alternative BOMs for our telephone example.

The content of the Bill of Materials (BOM) differs in an organization based on the user department and use purpose as well. For the same telephone example, the content of the BOM used by the engineering/design function, may differ from that used by the manufacturing, or sales, or service functions. Accordingly, they are referred to as engineering BOM (EBOM), manufacturing BOM (MBOM) or production BOM (PBOM), sales BOM, and service BOM (SBOM) in many organizations. In our telephone example, the engineering BOM may involve unique components / sub-assemblies that require different technical specifications or engineering drawings. These drawings are usually produced

TABLE 7.2(A) A SINGLE-LEVEL BOM FOR A TELEPHONE INSTRUMENT

Item Code	Item Description	UOM	Quantity per Product
1000	Basic telephone	Each	1
1010	Base unit assembly	Each	1
1020	Handset assembly	Each	1
1030	Connecting cable	Metre	1
1040	Connecting jack	Each	2
1050	Speaker	Each	1
1060	Microphone	Each	1
1070	Receiver cover plate	Each	2
1080	Panel board assembly	Each	1
1090	Operating unit	Each	1
1100	Lug	Each	4
1110	Base cover plate	Each	2
1120	Button assembly	Each	1
1130	Control panel assembly	Each	1
1140	Buttons	Each	12
1150	Control dials	Each	3
1160	Screw	Each	21

TABLE 7.2(B) AN INDENTED BOM FOR A TELEPHONE INSTRUMENT

Level	Item Code	Item Description	UOM	Quantity per Product
0 ----	1000	Basic telephone	Each	1
- 1 ---	1010	Base unit assembly	Each	1
-- 2 --	1080	Panel board assembly	Each	1
--- 3 -	1120	Button assembly	Each	1
---- 4	1140	Buttons	Each	12
---- 4	1160	Screw	Each	12
--- 3 -	1130	Control panel assembly	Each	1
---- 4	1150	Control dials	Each	3
---- 4	1160	Screw	Each	3
-- 2 --	1090	Operating unit	Each	1
-- 2 --	1100	Lug	Each	4
-- 2 --	1110	Base cover plate	Each	2
-- 2 --	1160	Screw	Each	4
- 1 ---	1020	Handset assembly	Each	1
-- 2 --	1050	Speaker	Each	1
-- 2 --	1060	Microphone	Each	1
-- 2 --	1070	Receiver cover plate	Each	2
-- 2 --	1160	Screw	Each	2
- 1 ---	1030	Connecting cable	Metre	1
- 1 ---	1040	Connecting jack	Each	2

TABLE 7.2(C) A MODULAR BOM FOR A TELEPHONE INSTRUMENT

Item Code	Item Description	UOM	Quantity per Product
1000	Basic telephone	Each	1
1010	Base unit assembly	Each	1
1020	Handset assembly	Each	1
1030	Connecting cable	Metre	1
1040	Connecting jack	Each	2
1050	Speaker	Each	1
1060	Microphone	Each	1
107X	Receiver cover plate		
1071	Receiver cover plate: Grey	Each	2
1072	Receiver cover plate: Blue	Each	2
1073	Receiver cover plate: Black	Each	2
1074	Receiver cover plate: Beige	Each	2
1080	Panel board assembly	Each	1
109X	Operating unit		
1091	Operating unit: 20 calls		
1092	Operating unit: 50 calls		
1093	Operating unit: 100 calls		
1094	Operating unit: 200 calls		
1100	Lug	Each	4
111X	Base cover plate		
1111	Base cover plate: Grey	Each	2
1112	Base cover plate: Blue	Each	2
1113	Base cover plate: Black	Each	2
1114	Base cover plate: Beige	Each	2
1120	Button Assembly	Each	1
113X	Control panel assembly		
1131	Control panel assembly: Simple	Each	1
1132	Control panel assembly: Deluxe	Each	1
1133	Control panel assembly: Elegant	Each	1
1140	Buttons	Each	12
1150	Control dials	Each	3
1160	Screw	Each	21

NOTES

in CAD software and the master drawings are stored on a PLM system. For the same product, the manufacturing BOM may involve unique components / sub-assemblies sourced from different plants or vendors. These BOM components are uniquely

NOTES

identifiable in the ERP system and referred by the procurement team for sourcing. Similarly, for the same product, the sales BOM and the service BOM may be different as appropriate for the concerned teams. These usage-based BOMs can be represented in any of the formats discussed earlier viz. single-level BOM, indented BOM, or modular BOM.

7.2.4 Time Phasing the Requirement

The computing of the requirement of items is based on simple arithmetic. Let us consider a two-month period as the planning horizon. Let us use the following notations:

Gross requirement for an item during the period: A

Inventory of the item available for the period: B

Net requirement of an item during the period: C

We know that $C = A - B$. If C is positive, we need to schedule an order for the item so that it arrives at the beginning of the period. The “how much” question is answered but not “when”. To understand this, let us return to the telephone example and look at the issue more closely. Dependent demand items exhibit another important property—the lumpy nature of demand. In our telephone example, let us consider the requirement of base cover plates. Based on the planning cycle of the parent, the demand for base cover plates may suddenly crop up during a particular week. For instance, if it was decided to produce 100 units of base unit assembly during Week 3, 60 units during week 5 and 75 units during Week 8, then the gross requirements of base cover plates are 200, 120, and 150, respectively. In between these periods, there is no need for base cover plates. Let us also suppose that the on-hand inventory during the period is 350, computed on the basis of what is physically available and what is likely to be supplied during the two-month period for which we do the planning.

Using these equations, we find that:

B: On hand = 350

A: Gross requirement = 470 (200 + 120 + 150)

C: Net requirement = 120

Therefore, we may conclude that we need to place an order for 120 base cover plates to meet the planned requirements. While this computation is correct, it may not reveal “when” to place the order and if there are any problems that the organization may face by planning in this manner. Let us look at the same data in a different fashion in Table 7.3:

Table 7.3 reveals important information to the planner arising out of the lumpy nature of demand that is characteristic of dependent demand items. First, we find that the on-hand inventory, due to arrive

TABLE 7.3 TIME PHASED ESTIMATION OF NET REQUIREMENT

	Period (Week)								
	0	1	2	3	4*	5	6	7	8
On Hand**	50				300				
Gross Requirements				200		120			150
Net Requirement				150					120

Notes: *Increase in on-hand inventory by 300 at the end of Week 4 (beginning of Week 5) is due to an order placed earlier;

**On-hand data pertains to the inventory at the end of the period.

at the beginning of Week 4 (on account of an earlier order) is “out of phase” with the requirement. Although the order can cover the requirement for the first five weeks, it is wrongly scheduled. Second, it is also clear from the computation that our order of 120 needs to arrive at the manufacturing system only at the beginning of Week 8. There is no need to order much in advance and keep the base cover plates idle in the stores.

This illustration highlights another important principle associated with resources planning. “Time phasing” of the information is a critical step in the process. In this case, we used a time bucket of “weeks”. The planner needs to choose an appropriate time bucket for the system for which planning is done and analyse the inventory-related information. Time bucketing has significant implications for the computational efficiency and the accuracy of information. In the example being discussed, by aggregating all the data to a two-month bucket, we first obtained gross information by using just four data points (on hand, on order, gross requirement, and net requirement). Later, we obtained better information by using a weekly bucket involving 25 data points (one data point for on-hand inventory at the end of time 0 and 24 data points for the eight weeks, as can be seen in Table 7.3). This is a managerial trade-off and needs to be resolved before using the system for planning purposes.

It is also important to realize that the accuracy and integrity of the inventory data is crucial for obtaining good schedules for the components. Organizations incorporate periodic checking routines (called cycle counting) to ensure that the integrity of the data is established. Physical inventory is verified with the inventory data and corrective measures are taken if there are any discrepancies.

It is evident that computing the net requirements and appropriately time phasing the requirement is the first step in the planning process for dependent demand items. A sample illustration for the operating unit of the telephone example is given in Table 7.4.

NOTES

TABLE 7.4 A SAMPLE ILLUSTRATION FOR THE OPERATING UNIT

Item: Operating Unit	Period (Week)								
Computing the net requirements	0	1	2	3	4	5	6	7	8
Gross requirements				200		120		130	160
On hand*	50	50	50						
Net requirement				150		120		130	160

Note: *On-hand data pertains to the inventory at the end of the period

NOTES

7.2.5 Determining the Lot Size

Lot sizing is the process of determining the size of the order quantities for each component in a product. The lot-sizing decision also affects the lead time required for the manufacture/purchase of components. While deciding on an appropriate lot size, the planner must keep in mind the influence of two sets of costs. Large lot sizes will require fewer set-ups (and associated set-up costs) but may result in carrying a huge inventory for a longer time. On the other hand, smaller lot sizes require several set-ups, thereby increasing the set-up costs. Balancing these two costs is central to the lot-sizing decision. Let us look at three lot-sizing rules commonly considered by planners.

Lot sizing is the process of determining the size of the order quantities for each component in a product.

Lot-for-lot (LFL)

Lot-for-lot is perhaps the simplest of the lot-sizing rules. Under this rule, the lot size is equal to the net requirement during every period in the planning horizon. If, for a component, the net requirements for the next 4 weeks are computed to be 120, 0, 200, and 50, then the order sizes scheduled to arrive during these four weeks are 120, 0, 200, and 50, respectively. Table 7.5 shows a sample illustration of the rule for the operating unit of the phone.

TABLE 7.5 THE LFL RULE FOR THE OPERATING UNIT OF A TELEPHONE

Item: Operating Unit	Period (Week)								
Lot size rule: Lot for lot	0	1	2	3	4	5	6	7	8
Gross requirements				200		120		130	160
On hand*	50	50	50						
Net requirement				150		120		130	160
Lot size (planned receipt)				150		120		130	160

Note: *On-hand data pertains to the inventory at the end of the period.

The LFL rule is appropriate under the following conditions:

- (a) The cost of carrying inventory from one period to the next is very high compared to the cost of set-up.
- (b) The demand for the item is sparse and highly discontinuous (several data points in the planning horizon have zero demand).

One may find that in situations with complex product configurations and several varieties in the end product, these assumptions hold as we approach the final assembly stage.

Fixed order quantity (FOQ)

The other alternative is to use a fixed order quantity. In this rule, irrespective of the nature of the demand, orders are always placed for a fixed order quantity. After computing the net requirements for the planning period, orders are scheduled such that they arrive at the first point of demand. The next order is scheduled to arrive at the time when the first order is insufficient to meet the net requirements for the period. The procedure continues in this fashion until the requirements for the entire period are covered by the orders. Table 7.6 presents a sample illustration for the FOQ lot-sizing rule for the operating unit of the telephone. In this case, the first point of demand for operating units occurs during Week 3. Therefore, an order quantity of 300 is scheduled to arrive at the beginning of the week. The quantity that arrives during Week 3 satisfies the net requirements until Week 6. Therefore, the next order of 300 is scheduled to arrive at the beginning of Week 7.

Alternative methods could be used for computing the fixed order quantity. One method is to estimate the economic order quantity for the item on the basis of set-up/ordering costs and carrying cost. In the case of components manufactured “in-house”, there are preferred run lengths or economic run lengths. Similarly, for items bought from outside, minimum order quantity restrictions imposed by the suppliers, other considerations such as truck load capacity, and the prevailing quantity discounts structure will influence the choice of order quantity.

N O T E S

TABLE 7.6 THE FOQ RULE FOR THE OPERATING UNIT OF A TELEPHONE

Item: Operating Unit	Period (Week)								
Lot size rule: Fixed order quantity	0	1	2	3	4	5	6	7	8
Gross requirements				200		120		130	160
On hand*	50	50	50						
Net requirement				150		120		130	160
Lot size (planned receipt)				300				300	

Note: *On-hand data pertains to the inventory at the end of the period

NOTES

FOQ provides an alternative perspective to the “carrying cost–ordering cost” trade-off. When the unit value of the item is low and there is a continuous and more or less stable demand for the item, it is appropriate to use this rule. Items that are at the lower ends of a product structure share a greater degree of commonality among numerous end-product variations. Therefore, the demand is likely to be continuous, justifying the choice of FOQ rule. Moreover, they tend to have a low unit cost. Hence, the cost of carrying any leftover inventory from one period to the next is likely to be very low. In reality, depending on the net requirements, the order placed will be in integral numbers of FOQ (i.e., $k \times \text{FOQ}$, where k is an integer).

Periodic order quantity (POQ)

In the periodic order quantity method, an order is placed such that it covers the requirements of P periods. It does not mean that we need to place an order during every P period. It merely suggests that if we plan an order at a particular time, the quantity should cover the net requirements of P successive periods. The choice of P could be made in alternative ways. One way is to use the review cycle. If an organization reviews decisions every two or three weeks, it could be an appropriate time to also plan the orders. The other method is to use the economic order quantity (Q^*) and the average demand during the period to arrive at P . The expression is given by:

$$\text{Number of periods } (P) = \frac{\text{Economic order quantity } (Q^*)}{\text{Average demand during the planning period}} \quad (7.1)$$

Let us return to the example of operating units. The total net requirement for the 8-week period is 560 ($150 + 120 + 130 + 160 = 560$). Let us suppose that the economic order quantity is 210. Therefore, using Eq. 7.1, we obtain P to be 3 periods. Table 7.7 shows the implementation of the POQ rule for operating units with $P = 3$. An order is scheduled to arrive at the beginning of Week 3 and the quantity ordered is sufficient to cover the net requirements during Weeks 3 to 5. Similarly, another order is scheduled to arrive at the beginning of Week 7.

TABLE 7.7 IMPLEMENTATION OF THE POQ RULE FOR OPERATING UNITS WITH $P = 3$

Item: Operating Unit	Period (Week)								
Lot size rule: Periodic order quantity	0	1	2	3	4	5	6	7	8
Gross requirements				200		120		130	160
On hand*	50	50	50						
Net requirements				150		120		130	160
Lot size (planned receipt)				270				290	

Note: *On-hand data pertains to the inventory at the end of the period

7.2.6 Incorporating Lead-time Information

We observed earlier that timing of the orders in dependent demand items is very crucial. Errors in scheduling will have a cascading effect right through the product structure. One important aspect of timing is to incorporate lead-time information in the planning exercise. In our telephone example, let us consider the following situation:

During Week 8, we want to assemble 200 sets of telephones. A look at the product structure indicates that in order to accomplish this task, we need 200 base unit assemblies, 200 handset assemblies, 400 connecting jacks and 200 metres of connecting cable. Let us suppose that it takes one week to assemble 200 base units and we have no inventory in stock. This implies that we should launch a work order to the assembly shop at the beginning of Week 7 to assemble the required number of base units. Since assembling base units take one week, by beginning the assembly in Week 7 itself, they will be able to deliver 200 base units exactly at the beginning of Week 8.

At the next level, let us assume that the operating units are not available in stock and it takes two weeks to manufacture the operating units. Therefore, in order to make 200 operating units available at the assembly shop at the beginning of Week 7, we need to send a work order to the production shop at the beginning of Week 5 to begin the manufacturing process. The process is graphically portrayed in Figure 7.6.

This illustration (Figure 7.6) shows that lead-time information is required to correctly schedule the launch of work order for the components. At every level of planning, the component schedule gets offset to the extent of lead time. For a planned receipt of items during a period, we merely offset the time to the extent of lead time to arrive at the planned order release. If:

Planned receipt for the component during period t is PR_t ,

Lead time for the component is L , and

Planned order release for the component during $t - L$ is POR_{t-L} , then

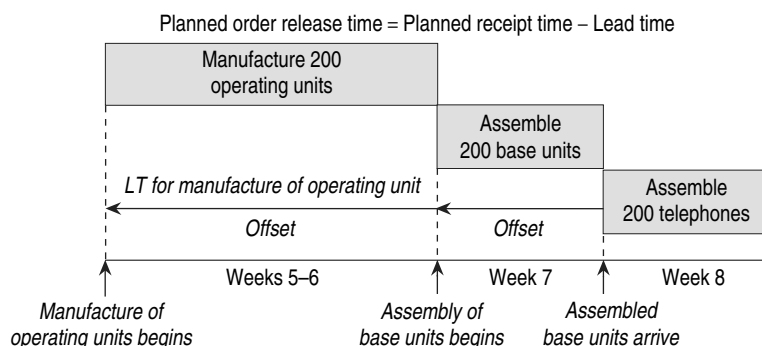


FIGURE 7.6 The impact of lead time in resources planning

NOTES

NOTES

$$POR_{t-L} = PR_t \quad (7.2)$$

Accurate computation of component lead times is therefore important in obtaining realistic schedules for the components. Lead time consists of set-up time, processing time, waiting time, and the move time between work stations involved in the manufacture of the component. While estimating the process time and the set-up time is relatively easy, estimating the waiting time is difficult. It calls for detailed data collection and analysis. Alternatively, if the item is bought from outside, ordering, supplier lead time for delivery, inspection and issue determine the lead time for the component.

Let us return to our example of planning the requirements for the operating unit. If we assume that the lot-sizing rule used is LFL and the lead time for the manufacture of operating units is two weeks, we can then arrive at the planned receipt and the planned order release using this additional information. Figure 7.7 shows the calculations.

7.2.7 Establishing the Planning Premises

In the examples discussed earlier, we have not addressed the key issue, that is, the basis on which the component level planning needs to be done. We address that issue now. Planning for dependent demand items begins with a simple goal, that is, make availability meet the requirements. Therefore, the first step in the planning process is to know exactly what the requirement is. In organizations, a master production schedule (MPS) is drawn for this purpose. A master production schedule disaggregates the information contained in an aggregate plan, links it to specific varieties of the product being manufactured, and specifies the exact timing of the requirement.

Let us assume that our telephone manufacturer makes four varieties of phones. At the time of planning, we need to know the exact number of telephones of each variety that should be produced in each time period and the associated BOM. An MPS serves this purpose and “fixes” the requirements of each end product. An illustrative example is shown in Table 7.8.

In this example, the aggregate plan of 1,200 instruments for the first month and 1,400 in the second month is split into the specific

Item: Operating unit; LT = 2 weeks		Period (week)							
Lot size rule: Lot for lot	0	1	2	3	4	5	6	7	8
Gross requirements				200		120		130	160
On hand*	50	50	50						
Net requirements				150		120		130	160
Planned receipts				150		120		130	160
Planned order releases		150		120		130	160		

FIGURE 7.7 Computation for planned receipt and planned order release

Note: *On-hand data pertains to the inventory at the end of the period

TABLE 7.8 MPS FOR A TELEPHONE MANUFACTURER

Master Production Schedule	Period (Week)							
Item	1	2	3	4	5	6	7	8
Aggregate production plan								
Total Production				1,200				1,400
Phone– basic	80	190	110		100	100	100	100
Phone– deluxe	0	50	0	50	0	50	0	50
Phone- home office	75	120	40	20	70	60	0	0
Phone- enterprise	90	125	125	125	125	245	200	200

varieties of phones to be manufactured during each week. Eighty units of the basic telephone is the gross requirement for the first week, 190 units for the second week, and so on. Using this information and the on-hand inventory, the net requirements for the basic telephones can be computed for all the weeks. Further, using the lot-sizing rule appropriate, lots are identified for the planned order receipt and using the lead-time data, the orders can be offset to schedule planned order releases.

Once the planned order release of the end product is arrived at in this fashion, it is possible to use the dependency information available in a product structure or BOM to estimate the gross requirements of the lower level items. However, some lower level items may have “independent” demand as well, since they are consumed directly in the market as spares. The MPS also identifies the spares requirement of lower level components. Therefore, for the lower level items, gross requirements may arise in two ways; one as a dependent demand of its parent and the other as spares. These are to be added up while planning for the item.

Let us assume that based on the on-hand inventory of basic telephones, base unit assemblies and operating units as well as the scheduled receipts of these during the planning period, the planned order releases for base unit assemblies are computed to be 140 for Week 3, 80 for Week 5, 130 for Week 7, and 120 for Week 8. Let us further assume that the MPS has projected the demand for operating units as spares to be 60 units in Week 3, 40 units in Week 5, and 40 units in Week 8. In general, we can compute the gross requirements of components for a period t at any level using the following notations:

NOTES

MPS for telephone manufacturer		Period (week)							
		1	2	3	4	5	6	7	8
Operating unit				60		40			40

Item: Base unit assembly		Period (week)							
		1	2	3	4	5	6	7	8
Planned order releases				140		80		130	120

Item: operating unit		Period (week)							
		1	2	3	4	5	6	7	8
Gross requirements				200		120		130	160

FIGURE 7.8 Computation of gross requirements

NOTES

The planned order release of the parent = POR_t

The quantity required to assemble one unit of the parent = $BOMQ_t$

Independent demand of the component as spares = ID_t

Gross requirements of a component = GR_t

$$GR_t = (POR_t \times BOMQ_t) + ID_t \quad (7.3)$$

In our telephone example, each base unit assembly requires one operating unit. Therefore one can use the given information and Equation 7.3 to compute the gross requirements of operating unit for the planning period. Figure 7.8 explains the computation of gross requirements in the case of the operating unit of the phone.

7.3 SUMMARY

- Dependent demand items exhibit certain characteristics such as lumpy demand and parent-offspring relationships that can be exploited while planning for materials in organizations.
- Bill of materials (BOM) and product structures depict the dependency relationships among various components in a manufacturing set-up.
- Various lot-sizing rules can be utilized to batch requirements and schedule orders. The lot-sizing rules strike an appropriate trade-off between carrying cost and ordering cost.
- MRP logic involves a four-step logical process: net-lot-offset-explode. Through an iterative process, the material

requirements of all components in an organization during a planning horizon are arrived.

- Regeneration and net change are the two methods used for updating the MRP schedules.
- Capacity requirements planning (CRP) uses logic similar to that of MRP.
- Furthermore, the logic of MRP can be extended to several other spheres of an operations system. MRP II systems provide this functionality. Modern-day ERP systems are further extensions of MRP logic to other areas of business such as marketing, finance, and personnel.

7.4 REVIEW QUESTIONS

1. What do you mean by dependent demand and independent demand? Is knowledge of this useful to an organization? If so, in what way is this information useful?
2. Consider each of the following operating systems and identify three examples of dependent and independent demand items.
 - (a) A hospital providing coronary health care services
 - (b) A university offering post-graduate level courses
 - (c) A manufacturer of ceiling fans
 - (d) A transportation service provider for industrial products
 - (e) The corporation office of the city of Bangalore
 - (f) A computer facilities maintenance provider
 - (g) A BPO offering call centre services to clients
3. What are the key attributes of a dependent demand item? Explain with one example each from manufacturing and service organizations.
4. Why should an organization plan the requirements for a dependent demand item differently from that of an item with independent demand?
5. Suppose that an organization uses normal inventory control methods to plan for its dependent demand items. What will the impact of this be on material planning? Under what conditions will the planning result in satisfactory results?
6. What is meant by a bill of materials (BOM)? Why do organizations need BOMs?
7. Are BOMs and product structures the same? Explain with an example.
8. An organization engaged in the manufacture of electrical appliances is not clear if they need to use a modular or a single-level BOM. Prepare an advisory note clearly specifying the conditions under which each of these alternatives is appropriate.
9. Suppose that an organization prepared an MRP on the basis of incorrect data on lead time of the components. What will the impact of this be on the actual operations?
10. What is meant by the “explosion” of a BOM? Explain with a simple example
11. Identify the appropriate lot-sizing rule for each of the following components. Justify your choice of the rule.
 - (a) Steel plates for manufacturing automobile doors
 - (b) Eye surgeon time (in hours) in an eye hospital
 - (c) Bed linen for a five-star hotel
 - (d) Packing cartons for a computer manufacturer
 - (e) Sliced vegetables for use in 12 vegetarian dishes in a fast food joint
 - (f) Aircraft engines for Hindustan Aeronautics Limited's advanced light helicopter
12. What is a time bucket? Why do you need it in an MRP exercise?
13. A firm performing an MRP exercise has been using a time bucket of “months” so far. Beginning next month they have chosen to change the time bucket to “weeks”. What are the implications of this change to the organization?
14. What do you mean by safety stock and safety lead time in MRP? When should organizations use these in MRP?
15. What are the key implications to a manager in using the regenerative and net change updation of MRP schedules?
16. An organization was performing the MRP re-run schedule every week. It decided to change the time between two such re-runs to 10 days. What are the implications of this change for the organization?
17. Data integrity is a major issue for many organizations while using an MRP system. Comment on this statement.
18. How is MRP II different from MRP? What are the additional advantages that an organization will obtain by using an MRP II system?
19. “Organizations can obtain realistic plans by properly integrating MPS, MRP, and CRP through an iterative process.” Comment on this statement.

7.5 PROBLEMS

1. Omega, a manufacturer of office supplies, produces a desktop calendar unit using a simple process and the following components. The desktop calendar unit is basically assembled using a base plate, a sliding unit and a calendar assembly. The sliding unit is made up of a plastic tray and a set of 30 cards to store telephone and address information. The sliding unit is fixed to the base plate using a pair of 3 mm screws. The calendar assembly is made up of two U-shaped hooks and a calendar (consisting of about 400 pages containing the daily calendar details). The calendar assembly is fixed to the base plate by clamping the U-shaped hooks using a pair of 3 mm screws each.

Each base plate is made of plastic sheets and requires 150 square inches of plastic sheet. Similarly, the sliding unit is also made of the same plastic sheet and requires 30 square inches of the plastic sheet. Each U-shaped hook is made from 7 inches of 4mm steel rod. The calendar is bought directly from a supplier. Finally, the product is packed in a specially made carton box and sealed using 12 inches of tape. There are also four bar-coded labels stuck to each carton box after it is sealed. This makes it a packaged calendar ready for shipping.

- Develop a product structure for the desk calendar unit.
- Using the given information, prepare a bill of material for the desktop calendar unit.

2. In Problem 1, assume that Omega offers the following alternatives:

- Two different varieties [one is the basic version and the other has provision for keeping pens/pencils (these affect only the base plate)]
- Five different colour choices in the desktop calendar unit (these affect the base plate and the sliding unit)
- Three different configurations of the calendar itself (one page per day, half a page per day, and month planner sheet at the beginning of every month in addition to one page per day)

- How many different varieties of desktop calendar units can the organization offer to the customer?
- Develop a BOM for the product range offered by Omega.

3. A gearbox manufacturer has 20 gearboxes in stock. Each gearbox has four gears. There are 200 gears already in stock. The gears are made from gear blanks. The stock of gear blanks in the stores is 100. Each gear blank requires 30 kilograms of alloy steel. The stores have 7,000 kilograms of alloy steel. Compute the requirement of components for manufacturing 570 gearboxes in the next month.

4. Consider the product structure given in Figure 7.9 pertaining to a product manufactured by Oriental Housings & Seals. The numbers in parentheses in the figure indicates the number of units of the item required to assemble one unit of its parent. Use the information available in the product structure to answer the following questions:

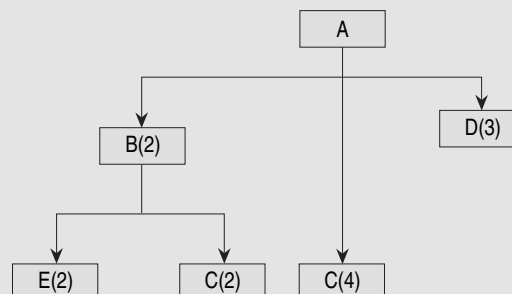


FIGURE 7.9 Product structure for Oriental Housings & Seals product

- How many units of C are required to manufacture one unit of Product A? Did you make any assumption in computing this value?
 - How much inventory of each component is required for satisfying a demand of 100 units of the final product if: (i) there is no stock of finished goods (ii) there is a stock of 30 units of the final product?
 - Consider Problem (a). Will the computations change if there is some stock of Item B? Why and by how much? (Hint: Assume that you have x units of Item B to proceed with the analysis)
5. In Problem 4, the lead times for manufacture/assembly of the components are as follows:
- | | |
|-------------|---------|
| Product A | 2 weeks |
| Component B | 1 week |
| Component C | 2 weeks |
| Component D | 2 weeks |
| Component E | 1 week |
- How early can Oriental deliver an order of 100 units of the product to the customer if (i) there is no stock of Item B (ii) there is a stock of 220 units of Item B?
 - Will the results change if there is a stock of Product A?
6. Given in Table 7.9 is a partially completed MRP working for Component X. Using the information provided, complete the table.

TABLE 7.9 INCOMPLETE TABLE FOR PROBLEM 6

Component X		Lot Size: 2 Periods						
BOM Quantity: 1		Lead Time: 1						
		0	1	2	3	4	5	6
Gross requirement		200	150	100	200	120	200	
On-hand inventory	300							
Net requirement								
Planned receipts								
Planned order releases								

TABLE 7.10 DATA ABOUT COMPONENT XX (PROBLEM 7)

Component XX	Lot Size: ?						
BOM Quantity: 1	Lead Time: 1						
	0	1	2	3	4	5	6
Gross requirement		200	350	400	200	450	200
On-hand inventory	600						
Net requirement							
Planned receipts							
Planned order releases							

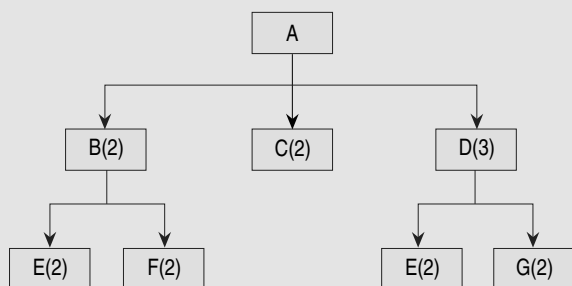
7. Consider Component XX, for which the MRP exercise needs to be done. The relevant information pertaining to the component has been extracted from the company records and reproduced in Table 7.10. Currently, no decision has been made about the lot-sizing rule. Company estimates suggest that the cost of carrying inventory is ₹10 per unit per period and the cost of scheduling an order is ₹500 per order. The economic order quantity is 150 units.

Use this information to analyse the following five lot-size rules and suggest an appropriate lot-sizing rule for the component.

- Lot for lot
- POQ = 3 periods
- POQ based on the economic order quantity
- FOQ = 500
- FOQ = 1,000

8. Consider the product structure of a manufacturing firm, as shown in Figure 7.10. The company is interested in performing an MRP exercise for the next eight weeks to cover the requirements indicated in the MPS. The MPS requirements for the next eight weeks are 20, 80, 100, 200, 100, 0, 120, and 150.

The relevant information on lead time, inventory status and lot sizing are available in Table 7.11. Develop the MRP schedule for the final product as well as for all the components and answer the following questions:

**FIGURE 7.10** Product structure for Problem 8**TABLE 7.11** LEAD TIME, INVENTORY STATUS, AND LOT-SIZING RULES FOR PROBLEM 8

Item	Lead Time (Weeks)	Inventory Status	Lot-sizing Rule
A	1	100	Lot for lot
B	2	200	Lot for lot
C	1	400	2 periods
D	2	1,000	3 periods
E	2	2,000	2 periods
F	2	1,200	500
G	1	2,000	1,000

- Do we need to schedule order releases for Component B? If so, when and how much?
- How long will the on-hand inventory of Item D satisfy the requirement?
- How many orders need to be placed for Component F?
- If the cost of scheduling an order is ₹2,000, what is the total cost of ordering for the MRP plan?

9. Figure 7.11 shows the product structure for a manufacturing firm. In addition to the final product (Product A), Component B is sold as a spare in the market. The MPS for Product A and Component B and other relevant information for the problem are given in Tables 7.12 and 7.13.

Perform an MRP exercise to cover the requirements of the MPS and answer the following questions:

- What is the gross requirement for Component B during the six periods?
- How many orders do we have to place for Component C and in what quantities?
- Suppose the lead time for Component G is two weeks instead of one week, what will its impact be on the planning process?
- Suppose that the economic order quantity for Component C is 320 units. What will the appropriate POQ lot-sizing rule be for Component C?

10. Consider the following MRP (Table 7.14) for Component C, which is used in a product manufactured by an organization.

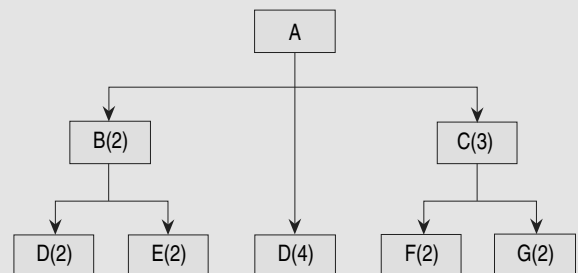
**FIGURE 7.11** Product structure for Problem 9

TABLE 7.12 MPS FOR THE NEXT SIX PERIODS

Period	1	2	3	4	5	6
Product A	100	80	220	140	70	160
Component B	40	30	70		60	50

TABLE 7.13 LEAD TIME, INVENTORY STATUS, AND LOT-SIZING RULE FOR PROBLEM 9

Item	Lead time (Weeks)	Inventory Status	Lot-sizing Rule
A	2	200	Lot for lot
B	2	900	Lot for lot
C	1	750	Lot for lot
D	2	2,500	2 periods
E	1	300	3 periods
F	2	1,000	500
G	1	600	1,000

Of late, the organization has been experiencing frequent shortages of this component. It has been estimated that on an average, the company loses ₹5,000 by way of lost

TABLE 7.14 MRP FOR COMPONENT C

Component C Lot Size: LFL

BOM Quantity: Lead Time: 1

	0	1	2	3	4	5	6
Gross requirement		100	250	160	0	200	70
On-hand inventory	400	300	50	0	0	0	0
Net requirement		0		110	0	200	70
Planned receipts		0		110	0	200	70
Planned order releases			110	0	200	70	0

business and goodwill due to these shortages in a six-week period. Therefore, it has been decided to include a safety stock of 100 units. Rework the planning details for this component by incorporating the safety stock requirement.

11. In Problem 10, assume that the organization decided to implement a safety lead time of one week instead of safety stock.
 - (a) Rework the planning details for this component by incorporating the safety lead time requirement.
 - (b) Compare the two results and list two significant observations that you have made.

7.6 NET-WISE EXERCISES

1. Visit the Web sites of the major ERP vendors (SAP, PeopleSoft, Oracle, Ramco, J.D. Edwards). Study the product offerings of all these vendors and prepare a detailed report to answer the following questions:
 - (a) What are the key components of the ERP system offered by the major vendors?
 - (b) How do these vendors differentiate their offerings?
 - (c) If an organization wishes to choose one of the four vendors, what factors will you suggest to the organization to enable appropriate selection?
 - (d) Are there industry-specific offerings of ERP by these vendors? If so, which sectors of the industry are covered and how well?

7.7 MINI PROJECTS

1. Take a simple household gadget or appliance such as a dining table set or a sofa set with a centre table. Carefully analyse the assembly of the appliance and identify the parts that make up the appliance. Identify the major assemblies and sub-assemblies of the appliance.
 - (a) Using this information prepare a complete BOM and a product structure for the appliance
 - (b) What are the various ways by which you think product varieties can be created for this appliance? Identify them and prepare an indented BOM for the entire set of product offerings.
 - (c) Assume an MPS and other relevant data for the components that you have identified for the appliance.

Can you set up a standard table for performing the MRP exercise for the appliance?

2. Set up the table created in Problem 1 in a Microsoft Excel worksheet (assume a planning horizon of 6 periods). Use the worksheet to accomplish the following tasks:
 - (a) Perform the MRP exercise for the appliance that you have chosen.
 - (b) Assume that the MPS is wrong. Quantities have been underestimated to the extent of 10 per cent during Periods 4 and 5 and overestimated to the extent of 5 per cent during Period 6. What is the likely impact of this on customer commitment?

7.8 MULTIPLE-CHOICE QUESTIONS

- 7-1.** Operating inventories are dependent on:
A. Supply
B. Demand
C. Production
D. Procurement
- 7-2.** What are the two types of inventories exist in an operating system?
A. Operating and distribution inventories
B. Product and operating inventories
C. Distribution and product inventories
D. None of the above
- 7-3.** A structured approach that develops schedules for launching orders for materials in any manufacturing system and ensures the availability of these at the right time and the right place is:
A. Capacity requirement planning
B. Inventory requirement planning
C. Material requirement planning
D. Production requirement planning
- 7-4.** Regeneration is one method to update:
A. Service requirements planning schedule
B. Inventory requirement planning schedule
C. Capacity requirement planning schedule
D. Material requirements planning schedule
- 7-5.** What is the technique that applies the MRP logic to address the capacity issues in an organization?
A. Capacity Requirements Planning
B. Material Requirements Planning
C. Distribution Requirement Planning
D. Energy Requirement Planning
- 7-6.** To plan distribution inventories in the downstream supply chain, consisting of depots and stocking points at the dealer level is termed as:
A. Capacity Requirements Planning
B. Material Requirements Planning
C. Energy Requirements Planning
D. Distribution Requirements Planning
- 7-7.** The iterative process of computing all requirements at a particular level and then moving down one level is known as:
A. Level by level analysis
B. Top to bottom approach
C. Scheduling
D. Explosion
- 7-8.** Which is that organization-wide planning system that utilizes some common software and an integrated database for planning and control purposes?
A. Capacity Requirements Planning
B. Material Requirements Planning
C. Energy Requirements Planning
D. Distribution Requirements Planning
- 7-9.** In which of the below, planning is based on resources?
A. Capacity Requirements Planning
B. Resources Planning in Services
C. Distribution Requirements Planning
D. Service Requirements Planning
- 7-10.** What are the two types of inventories exist in an operating systems?
A. Operating inventory and distribution inventory
B. Fixed inventory and fluctuation inventory
C. Physical inventory and computerized inventory
D. First in First out and Last in Last out

Answer Keys: 7-1. B 7-2. A 7-3. C 7-4. D 7-5. A 7-6. D 7-7. D 7-8. C 7-9. B 7-10. A

8

CHAPTER

MANUFACTURING ECONOMICS: PLANNING SYSTEMS

8.1	MRP Logic
8.2	Using the MRP System
8.2.1	Updating MRP Schedules
8.2.2	Safety Stock and Safety Lead Time
8.3	Capacity Requirements Planning (CRP)
8.4	Distribution Requirement Planning (DRP)
8.5	Manufacturing Resources Planning (MRP II)
8.6	Enterprise Resource Planning (ERP)
8.7	Resources Planning in Services
8.8	Summary
8.9	Review Questions
8.10	Problems
8.11	Net-Wise Exercises
8.12	Mini Projects
8.13	Multiple-choice Questions

ERP IMPLEMENTATION AT HPCL

NOTES

Hindustan Petroleum Corporation Limited (HPCL) is a Fortune 500 company with an annual turnover of over ₹914.48 million for the financial year ended March 2007. The company has successfully implemented JD Edwards EnterpriseOne ERP spanning about 400 locations and 4,000 employees across the nation.

The ERP implementation programme was started in 2000. The availability of sound technical infrastructure was necessary for implementing such a large initiative. This meant building a centralized data centre where the servers and applications could be hosted as well as having efficient connectivity for all locations spread across India's geography. Around 400 locations were completed over a period of two and a half years. As each location was selected for inclusion, the related infrastructure (like WAN, lease lines, VPN and VSATs) was built simultaneously depending on the availability.

As HPCL enjoys a national presence, it was important to determine the path for the ERP rollout. Several business processes were analysed for determining the points affecting the process look. The project was kicked off by a 25-member in-house team comprising representatives from all functional areas. After the completion of the first phase of the project, which included system configuration, more employees were inducted into the team and its size expanded to 100 people. In addition to this, a team of 100 consultants was also working on it.

The locations due for the "Go-Live" stage in a particular month were given top priority and a "work backwards" system was followed to arrive at separate tasks and their deadlines for those locations. These tasks included the procurement of hardware, applications for communication links, and testing and liaisoning with local people at the location.

Post-implementation, the company noticed substantial improvement in efficiency. For example, before the implementation, the annual financial accounts closure would take approximately a month while the monthly closing would take 15 to 20 days. In 2007-08, the monthly and quarterly closing was completed in about five days while the annual closing took a mere 10 to 12 days.

The second major benefit of the centralized ERP implementation was that it compelled the company to look at various other services that could be offered to customers with the help of IT. It was also realized that they could improve transparency by making more information available to their customers, vendors, and transporters on a real-time basis. With the help of the Web, they can log in and check the status of loads, orders, and payments.

The implementation gave the company an opportunity to relook at all its business processes from the beginning. Several processes in manufacturing, planning, procurement, and employee benefits that were earlier done manually were automated. The ERP enabled the generation of various day-end MIS reports on a regular basis. This helped to revamp the way in which reporting was done to the top management.

HPCL's ERP implementation can potentially provide several clues to other large organizations wishing to derive benefits from ERP/IT adoption.

Source: Based on Abhishek Raval, "HPCL Refines Business Process with ERP Implementation," accessed at <http://biztech2.in.com/india/casestudies/enterprise-solutions/hpcl-refines-business-processes-with-erp-implementation/28091/0>.

NOTES

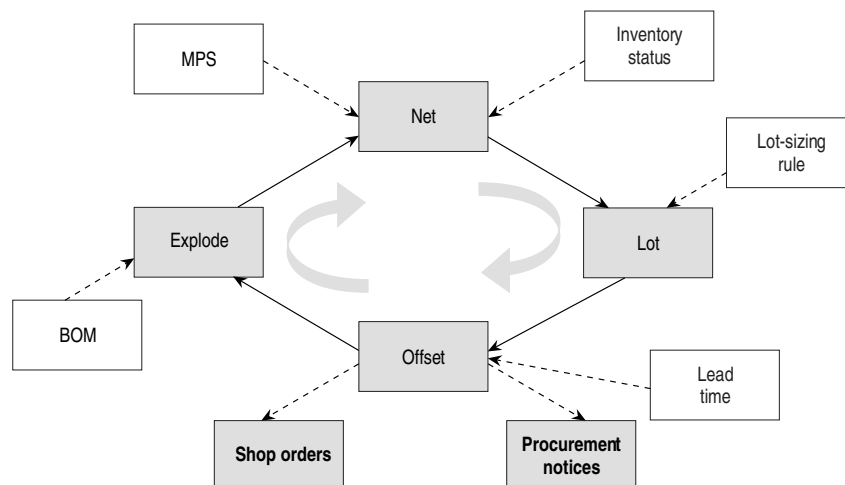
8.1 MRP LOGIC

Material requirements planning is a structured approach that develops schedules for launching orders for materials in any manufacturing system and ensures the availability of these at the right time and at the right place. It uses the basic building blocks of resources planning that we described in the previous section to develop these schedules. Figure 8.1 shows the core logic of the MRP process, its input, and output.

As shown in the figure, four key processes drive the MRP procedure. These processes occur in a cyclic fashion. The first process is the "net" process. The MPS for the end product provides information on the gross requirements for the end product. By utilizing the information available in the inventory records, the "net" process computes the net requirements for the end product. The second process is the "lot" process. Once the net requirements are computed, the lot-sizing rule is used to schedule planned receipts of the product. The third process is the "offset" process. Once the planned receipts are identified, lead-time information is used to offset and obtain the planned order releases for the product. The planned order releases are either work orders for a manufacturing shop to assemble as many components as per the schedule or a purchase order to obtain sub-assemblies from outside.

Once these three processes are completed, the requirements for the end products are estimated and orders are scheduled. Then the next step is to cascade the process down the product structure and repeat the procedure with all the components at the next level in the product structure. This process is the last in the cycle denoted as "explode". In order to perform the explosion process, BOM data is required. The planned order releases of a parent creates dependent demand for the offspring as specified in the BOM. This becomes the gross requirements for the offspring. The procedure continues iteratively, level by level, until the lowest level is reached and all component schedules are determined.

NOTES

**FIGURE 8.1** Material requirements planning core logic

Therefore, the key inputs for the MRP processes are MPS, BOM, inventory status, lead-time data, and lot-sizing rule. We have already discussed in detail in the earlier sections the role each one of them plays in accurately determining the quantity and timing of the material requirements. As we proceed through the lower level components, two types of outputs are generated from the MRP system. The first output is a work order. Work orders are generated for items that are manufactured in-house. In our telephone example, we would have generated work orders for base set and handset assemblies as they are likely to be done in-house. The second output is a procurement notice. Procurement notices are generated for items that are bought from outside and directly used in the assembly. It triggers the purchase ordering process in an organization. In our telephone example, it is possible that the manufacturer might directly source connecting cables and jacks instead of producing them in-house. In such a situation, the outcome of the MRP process will be a procurement notice.

8.2 USING THE MRP SYSTEM

Perhaps the most significant impact that a well-designed MRP system could provide to an organization is the reduction in inventory. MRP systems were first developed in the early 1960s and organizations that started using them reported dramatic reductions in their inventory. The reasons are obviously related to the logic of exploiting the peculiar characteristics of dependent demand items. Using traditional EOQ-based inventory control systems will often result in having inventory when not required. The other advantage of the MRP system is the increased visibility of items and their dependencies through a BOM representation of products being manufactured. Further, it could potentially inculcate a certain discipline in the planning process.

EXAMPLE 8.1

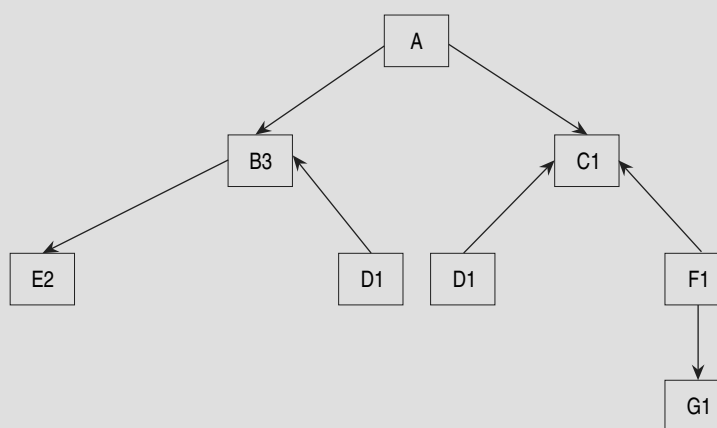
A manufacturing organization needs to plan the materials required for the next six weeks for the manufacture of its end-product (Product A), as per a master production schedule. In addition to the end product, there is an independent requirement of Component C, as it is sold as a spare in the market. The master production schedule for both the end product and the spares are given in Table 8.1. In order to assemble one unit of Product A, Components B to G are required. Figure 8.2 shows the product structure. In the figure, the number alongside each component denotes the number of each component required to assemble its immediate parent. An extract of the inventory status (Table 8.2) reveals the inventory on hand. There are no pending orders for delivery. Different lot-sizing rules are used for the components of Product A. Moreover, the components have different lead times. All this information is available in Table 8.2. Perform an MRP exercise to estimate the quantity and timing of the components required for the manufacture of Product A as per the MPS.

Solution

We perform the MRP exercise by using the four-step process given in Figure 7.8. We begin with Product A. Figure 8.3 shows the calculation of the net requirements. Since the lot-sizing rule is lot-for-lot, the net requirements and the planned receipts are the same. We offset the planned receipt by lead time to obtain the planned order releases. Figure 8.3 shows the calculations.

Now, we continue the process by exploding the product structure and moving to the next level. There are two components at this level: Components B and C. In our example, if an order needs to be released for assembling 100 units of Product A in Week 1, then we need 300 units of Component B and 100 units of Component C at the beginning of the week so that we can launch the work order. Therefore, the planned order releases of a parent determine the gross requirements for the offspring. We repeat the process once for B and then for C before we explode to the next level. The two tables in Figure 8.4 show the calculations for Components B and C.

While computing the gross requirements for C, we take into consideration both the dependent demand of C (as indicated in the product structure) as well as the independent demand (as indicated in the MPS in Table 8.1). Consider Period 2. The planned order releases for Product A during this period is 200. Therefore, there is a gross requirement of 200 units of C. There is also an independent demand of 50 units of C during Period 2. Therefore, the gross requirement for Component C is 250 ($200 + 50 = 250$). Similar computations have been made for all other periods also.

**FIGURE 8.2** The product structure**TABLE 8.2** INVENTORY STATUS, LEAD TIME, AND LOT-SIZING RULE

Component	On hand	Lead time	Lot size
A	150	1	LFL
B	1,000	2	LFL
C	300	1	LFL
D	750	2	3 periods
E	700	6	3 periods
F	200	1	400
G	500	3	500

TABLE 8.1 MPS FOR THE NEXT SIX PERIODS

	Period 1	Period 2	Period 3	Period 4	Period 5	Period 6
Product A	100	150	200	100	0	200
Component C		50	60			70

Component D is used by both B and C. Therefore, while arriving at the gross requirements of D, we take into consideration the planned order releases of both the parents. For example, during Period 1, the planned order releases of Components B and C are 200 and 50, respectively. Since they both require one component of D, the gross requirement for D during Period 1 is 250. Since the first instance of planned

receipt is Week 3, we add up the requirements of three weeks (Weeks 3 to 5) in order to implement the POQ policy and schedule a planned order release during the beginning of Week 1.

We perform similar computations for the remaining items in the product structure. Figure 8.5 presents the detailed calculations for Components D and E, while Figure 8.6 has the calculations for Components F and G.

	A	B	C	D	E	F	G	H	I	J	K
23											
24											
25											
26		Product A							Lot size	LFL	
27									Lead time	1	
28											
29				0	1	2	3	4	5	6	
30		Gross requirement			100	150	200	100		200	
31		On hand inventory		150	50	0	0	0	0	0	
32		Net requirement			0	100	200	100	0	200	
33		Planned receipts			0	100	200	100	0	200	
34		Planned order releases			100	200	100	0	200	0	
35											

FIGURE 8.3 Calculations for Product A

	A	B	C	D	E	F	G	H	I	J	K
47											
48		Component B							Lot size	LFL	
49		BOM quantity	3						Lead time	2	
50											
51				0	1	2	3	4	5	6	
52		Gross requirement			300	600	300	0	600	0	
53		On hand inventory		1,000	700	100	0	0	0	0	
54		Net requirement			0	0	200	0	600	0	
55		Planned receipts			0	0	200	0	600	0	
56		Planned order releases			200	0	600	0	0	0	
57											
58											
59											
60		Component C							Lot size	LFL	
61		BOM quantity	1						Lead time	1	
62											
63				0	1	2	3	4	5	6	
64		Gross requirement			100	250	160	0	200	70	
65		On hand inventory		300	200	0	0	0	0	0	
66		Net requirement			0	50	160	0	200	70	
67		Planned receipts			0	50	160	0	200	70	
68		Planned order releases			50	160	0	200	70	0	
69											

FIGURE 8.4 Calculations for components B and C

File Edit View Insert Format Tools Data Window Help										
D90 1700										
	A	B	C	D	E	F	G	H	I	J
68										
69										
70										
71										
72										
73		Component D							Lot size	POQ-3
74		BOM quantity		1 for B					Lead time	2
75			1 for C							
76			0	1	2	3	4	5	6	
77		Gross requirement		250	160	600	200	70	0	
78		On-hand inventory	750	500	340	0	0	0	0	
79		Net requirement		0	0	260	200	70	0	
80		Planned receipts				530				
81		Planned Order releases		530		0	0	0	0	
82										
83										
84										
85		Component E							Lot size	POQ-3
86		BOM quantity		2 for B					Lead time	6
87			0	1	2	3	4	5	6	
88		Gross requirement		400	0	1,200	0	0	0	
89		On-hand inventory	1,700	1,300	1,300	100	100	100	100	
90		Net requirement		0	0	0	0	0	0	
91		Planned receipts		0	0	0	0	0	0	
92		Planned order releases		0	0	0	0	0	0	
93										
94										

FIGURE 8.5 Detailed calculations for components D and E

File Edit View Insert Format Tools Data Window Help										
P120										
	A	B	C	D	E	F	G	H	I	J
99										
100										
101		Component F							Lot size	400
102		BOM quantity		1 for C					Lead time	1
103			0	1	2	3	4	5	6	
104		Gross requirement		50	160	0	200	70	0	
105		On hand inventory	200	150	0	0	0	0	0	
106		Net requirement		0	10	0	200	70	0	
107		Planned receipts		0	400	0	0	0	0	
108		Planned order releases		400	0	0	0	0	0	
109										
110										
111										
112		Component G							Lot size	500
113		BOM quantity		1 for F					Lead time	1
114			0	1	2	3	4	5	6	
115		Gross requirement		400	0	0	0	0	0	
116		On hand inventory	500	100	100	100	100	100	100	
117		Net requirement		0	0	0	0	0	0	
118		Planned receipts		0	0	0	0	0	0	
119		Planned order releases		0	0	0	0	0	0	
120										
121										

FIGURE 8.6 Calculations for components F and G

NOTES

Despite their simplicity and their initial success, MRP installations faced several problems after implementation.

Despite the simplicity and initial success, MRP installations faced several problems after implementation. In several cases, MRP systems suffer from three major problems:

- ⇒ The data integrity is low. The quality of the solution is only as good as the data used for the computation. If the lead-time data is wrong, organizations may either have too much inventory or frequent shortages. Similarly, if the inventory status is wrong it could jeopardize the entire computation.
- ⇒ Users did not have the discipline of updating the required databases as and when changes were taking place elsewhere in the organization. If the R&D department creates new designs and revisions in existing product design, this data needs to be incorporated in the BOM file. Failure to do so will mean introducing errors in the process, resulting in inappropriate planning.
- ⇒ There are uncertainties associated with several issues that lie outside the control of the people and the system (for instance, bad supply management resulting in many uncertainties in lead time and quantity delivered, and so on).

The net result of these problems is that the predictions made by MRP systems may often turn out to be less accurate and the system may have to be rerun often. This could also result in several production schedule changes and consequent delays in the downstream supply chain.

Moreover, there are other limitations in using the MRP system. The amount of computation involved in generating component-wise schedules for the planning horizon is large. Real-life examples require thousands of iterations that consume time. In some cases, it is not uncommon to have a single run of an MRP extending for about 12–16 hours. Although speed and availability of computing power keep increasing continuously, this issue still merits some attention and puts realistic limits to the frequency of generation of MRP schedules.

Therefore, an organization needs to incorporate certain aspects into the MRP planning framework to minimize problems arising out of these issues. Alternative methods are available to re-run an MRP system and they have implications on the accuracy, cost, and time pertaining to the exercise. However, there are methods available to handle some of the uncertainties in the system and thereby reduce the risk of shortage, but such alternatives have cost implications as well. We shall look at some of these alternatives in the following sections.

8.2.1 Updating MRP Schedules

In real-life situations, plans become obsolete over time due to several changes in the environment. For example, a customer might

have cancelled an order or amended the order quantity and delivery schedule. A supplier could have defaulted in the supply schedule. Similarly, there could have been some unexpected disruptions in the manufacturing and assembly schedules within the manufacturing system. In each of these cases, the MRP and the schedules for order releases and purchase become inaccurate and call for a certain amount of re-planning. Several such instances happen on a daily basis in any manufacturing system. Therefore, the critical issue that a manager needs to resolve while using the MRP system is the frequency with which the MRP schedules are re-run. Before addressing this issue let us first understand the methods available for updating schedules in MRP.

Regeneration is one method to update MRP schedules. In this method, the MRP system is run from scratch. Based on the changed information, one can start from Level 0 and run the MRP logic right up to the bottom level. The implication of this in real-life situations is that large computational efforts are required. It also amounts to 100 per cent replacement of the existing MRP information. The second method of updating the MRP schedule is known as *net change*. In this method, instead of running the entire MRP system, schedules of components pertaining to portions where changes have happened are updated. Let us consider a product structure with three levels. Suppose one of the components in Level 2 is likely to be made available two periods later, one can analyse this information and process only the relevant records to arrive at an alternative MRP schedule. Clearly, the net change method of updating MRP schedule modifies only a subset of data as opposed to regeneration. Therefore, it is likely to be computationally more efficient than the regeneration method. Moreover, it may be possible to run it in frequent intervals.

Regeneration is a method to update the MRP schedules where the MRP system is run from scratch.

The decision to use net change or regeneration depends on the magnitude of changes that occur in an organization. If the number of changes to be incorporated in the system tends to be large, then organizations are better off using the regenerative method for updating MRP schedules. One sees a similar kind of logic being applied to bringing out the next edition of the telephone directory in a large metropolitan city such as Mumbai. Every year, they may come up with a new telephone directory (equivalent to regenerative method). However, between two editions of the telephone directory, they bring out a small supplement announcing the changes (equivalent to net change). The cost of running an MRP system and the number of changes happening in the planning horizon influence the type of updating procedure employed and the frequency of updation.

NOTES

NOTES

8.2.2 Safety Stock and Safety Lead Time

Uncertainties in the system that are outside the control of an MRP system is a reality that organizations need to face and plan for. Generally, two types of uncertainties are prevalent: the quantity of components received, and the timing of receipt. First is the uncertainty with respect to supply quantity. Poor quality input material could result in quantity loss on account of rejections. Alternatively, the reliability of suppliers may also result in uncertainty in quantity. In the case of components manufactured in-house, there could be uncertainty in supply quantity due to changes in the batch quantity of upstream stages. Therefore, it may be desirable to plan for a safety stock to absorb these uncertainties. The inclusion of safety stock in MRP computation is fairly straightforward. At the time of “netting” the requirements, an order is scheduled to arrive when the on-hand quantity falls below zero. Instead, the order needs to be scheduled when the on-hand inventory falls below the safety stock. Figure 8.7 shows the MRP schedules for a component both without any safety stock and with a safety stock of 50.

In this figure, the first table shows the MRP schedule where no safety stocks are assumed. We see that the inventory on hand could satisfy the requirements up to Period 4 (since the total requirement up to this period is only 190). In order to satisfy the requirements for Periods 5 and 6 quantities are scheduled after taking the lead time into consideration. On the other hand, the second table shows that although sufficient inventory is available to meet the requirements

	A	B	C	D	E	F	G	H	I	J	K
4											
5											
6		Component XX			Safety stock		Nil		Lot size		LFL
7		BOM quantity	3						Lead time		2
8											
9				0	1	2	3	4	5	6	
10		Gross requirement			75	75	0	40	90	60	
11		On hand inventory		200	125	50	50	10	0	0	
12		Net requirement			0	0	0	0	80	60	
13		Planned receipts			0	0	0	0	80	60	
14		Planned order releases			0	0	80	60	0	0	
15											
16											
17		Component XX			Safety stock		50		Lot size		LFL
18		BOM quantity	3						Lead time		2
19											
20				0	1	2	3	4	5	6	
21		Gross requirement			75	75	0	40	90	60	
22		On hand inventory		200	125	50	50	10	0	0	
23		On hand inventory (net of safety stock)		150	75	0	0	0	0	0	
24		Net requirement			0	0	0	40	90	60	
25		Planned receipts			0	0	0	40	90	60	
26		Planned order releases			0	40	90	60	0	0	
27											

FIGURE 8.7 MRP schedules for a component without any safety stock and with a safety stock of 50

of Period 4, the on-hand inventory falls below the safety stock of 50. Therefore, 40 units are scheduled to arrive during the beginning of Period 4. It is clear from this example that inclusion of safety stock will result in carrying more inventory throughout the period. Moreover, it may significantly alter the ordering pattern. Therefore, managers need to exercise careful thought before fixing safety stock levels for the components in an MRP system.

Safety lead time while planning for the components is quite similar to safety stock. Safety lead time is incorporated in MRP systems by offsetting the planned receipts to the extent of the safety lead time. Let us assume that the planned receipt for a component during Period 5 is 1,200 units. If the safety lead time is one week, then by offsetting the planned receipts by one week and scheduling the receipt to Period 4, one can ensure that the uncertainties related to timing of delivery are largely addressed. Figure 8.8 illustrates the use of safety lead time in MRP. Compared to the first table in the figure, which has no safety lead time, the planned receipts and the planned order releases in the second table are further offset to the extent of the safety lead time of one week. Therefore, one can infer that incorporating safety lead time does not inflate the lead time—it merely shifts the planned order release schedule.

NOTES

8.3 CAPACITY REQUIREMENTS PLANNING (CRP)

Consider a hypothetical plan of Maruti Udyog Limited (MUL) to produce 20,000 Omni vans, 15,000 Altos, and 9,000 Zens during a

	A	B	C	D	E	F	G	H	I	J	K
31											
32											
33		Component XX			Safety lead time	Nil			Lot size	LFL	
34		BOM quantity	3						Lead time	2	
35											
36				0	1	2	3	4	5	6	
37		Gross requirement			75	75	0	40	90	60	
38		On hand inventory		200	125	50	50	10	0	0	
39		Net requirement			0	0	0	0	80	60	
40		Planned receipts			0	0	0	0	80	60	
41		Planned order releases			0	0	80	60	0	0	
42											
43											
44											
45		Component XX			Safety lead time	1 week			Lot size	LFL	
46		BOM quantity	3						Lead time	2	
47											
48				0	1	2	3	4	5	6	
49		Gross requirement			75	75	0	40	90	60	
50		On hand inventory		200	125	50	50	10	0	0	
51		Net requirement			0	0	0	0	80	60	
52		Planned receipts (before incorporating safety LT)			0	0	0		80	60	
53		Planned receipts (after incorporating safety LT)						80	60		
54		Planned order releases			0	80	60	0	0	0	
55											

FIGURE 8.8 The use of safety lead time in MRP

NOTES

month. Using MRP logic, MUL can schedule the arrival of materials, sub-assemblies, and components at various shops in the factory. MRP thus addresses the material aspect of the planning problem. However, in addition to material, we need capacity in the form of resources such as machines and skilled labour. What is the guarantee that there is sufficient capacity available in the factory to complete the tasks as per the MRP schedule? Clearly, MRP merely addresses “what needs to be produced” during each time period in a planning horizon.

However, MUL should also take into consideration “what can be produced” during each time period in a planning horizon and do the required planning to match these two. Therefore, capacity requirement planning (CRP) is necessary to ensure that what needs to be produced during a period can in fact be produced. **CRP** is a technique that applies logic similar to MRP to address the capacity issues in an organization. Similar to MRP, CRP develops schedules for planned releases of capacities to specific work orders as identified in an MRP schedule. The output of an MRP process becomes the basis for the CRP exercise.

Capacity requirement planning (CRP) is a technique that applies the MRP logic to address the capacity issues in an organization.

The notion of dependency applies very well to capacities also. Every manufacturing process generates a dependent demand for the resources involved in the conversion process. In the MUL example, if it takes three hours to complete the painting of one Alto and 20 minutes to complete the final inspection process using a specialised photo-sensitive gadget, then 45,000 hours of paint-shop capacity and 5,000 hours of the testing facility are required during the month. Just as the BOM provides information on materials required to complete one unit of the parent, a routing file or a process plan will provide information on the resources that are required to complete the processing of one unit of the parent. The basis for the MRP is an MPS. Similarly, CRP systems employ the detailed schedule generated by an MRP system as the basis for capacity planning. Just as MRP systems perform netting of the requirements using on-hand inventory data at the beginning of a planning horizon, a CRP system uses the capacity status as the starting point for the “netting” process. Figure 8.9 illustrates the underlying logic of CRP.

There are striking similarities between MRP and CRP and therefore organizations use logic similar to that of MRP for performing CRP (see Table 8.3 for a comparative picture of both). However, there are certain important issues that one needs to understand about MRP–CRP–MPS interfaces. In a simple hierarchical mode, MPS will drive MRP and MRP in turn will drive CRP. If both the schedules are feasible, then the plan is finalized. On the other hand, if there are mismatches between capacity and material schedules

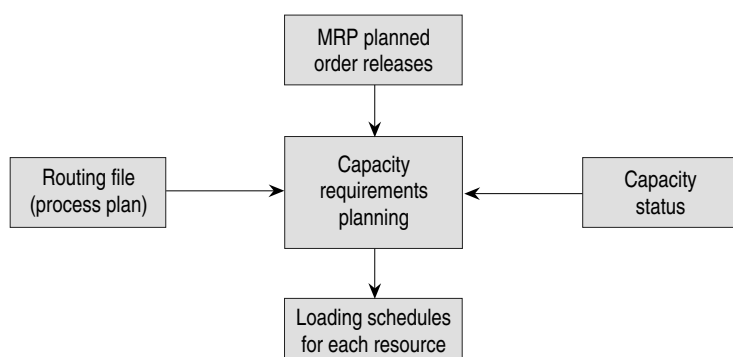


FIGURE 8.9 The underlying logic of CRP

TABLE 8.3 MRP AND CRP: A COMPARATIVE PICTURE

Criteria for Comparison	Material Requirements Planning (MRP)	Capacity Requirements Planning (CRP)
Input	Bill of material (product structure)	Routing information (process plan)
	MPS	MRP schedules
	Lead-time data	Lead-time data
	Inventory status	Capacity status
	Lot-sizing rules	
Output	Purchase orders	Capacity loading schedules
	Work orders	Capacity usage profiles

due to non-availability of capacity to meet the MRP schedule, then it calls for a few iterations of the process. The MRP schedules are first modified to obtain feasible MRP and CRP schedules. However, if this is not possible, then the MPS is modified until feasible schedules are obtained for both material and capacity. The process is presented in Figure 8.10.

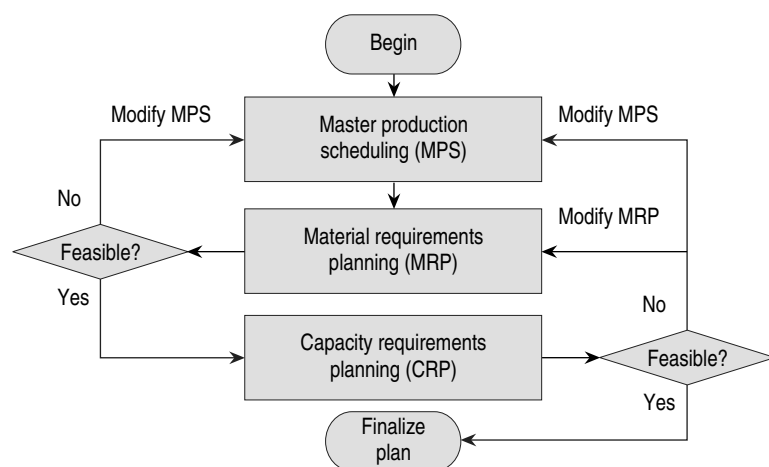


FIGURE 8.10 MPS-MRP-CRP: Iterative process of planning

NOTES

NOTES

8.4 DISTRIBUTION REQUIREMENT PLANNING (DRP)

The logic of MRP is sufficiently general to apply to several other areas of business. One extension of MRP is to plan distribution inventories in the downstream supply chain, consisting of depots and stocking points at the dealer level. Consider a downstream supply chain consisting of a factory warehouse, four distribution centres, and 20 dealers geographically spread out in the country. The demand at the dealers' level aggregates up in the chain and eventually reaches the factory warehouse. Transporting the products through the supply chain involves a certain lead time. Further, the products are transported in some preferred lot sizes. In general, ordering decisions happen based on a periodic review of stock and the emerging demand. This causes the demand at the next level in the supply chain to be lumpy. Therefore, it calls for time phasing of the planning process.

It is easy to notice that the logic of MRP could be readily applied to planning distribution requirements. Using a time phased data, organizations could schedule the planned receipt of products at various points in the supply chain and offset these by the required lead time to arrive at planned order releases for dispatch of material to the demand points. A dealer may review the inventory on hand and the likely demand during the planning period and arrive at the net quantity to order. Based on his lot-sizing considerations, he may lot the orders and determine the planned receipts. Further, by offsetting the order to the extent of the lead time required for receiving the order (due to transportation time and administrative processes) he may arrive at planned order releases for each order. This becomes the gross requirement for the distribution centres. The entire process could be repeated at the distribution centres and the planned order releases of the four distribution centres eventually influence the MPS at the factory.

A distribution requirement planning (DRP) exercise will help organizations and their supply chain partners to jointly plan and reduce investment in inventory in the supply chain. They will be in a position to respond to changes in the demand, such as sudden surges and drops, and have a cost-effective operation. They will also be able to offer a high level of service to their customers. However, unlike the MRP exercise, a DRP exercise relies on key information pertaining to planned order releases outside the domain of an organization. Retailers should be willing to share with dealer the planned order releases and their estimates of upcoming demand, for better planning. Similarly, dealers need to share similar information with the manufacturer. If the information exchange is not proper and if there is no data integrity, then the value of the entire exercise will be severely undermined. It will eventually result in inventory build-up or shortages, poor service, and increased costs of the operation in the supply chain.

8.5 MANUFACTURING RESOURCES PLANNING (MRP II)

The previous sections show the applicability of MRP logic to other domains of the business. Furthermore, the availability of computing power and software for storage and the manipulation of large chunks of data has increased ever since organizations began to use MRP systems. Therefore, it was logical that newer systems were developed to expand the application of MRP into other domains of business where dependency relationships exist. In the 1980s, organizations began to incorporate several modules in the MRP systems. This extended version is known as **manufacturing resources planning (MRP II)**.

Manufacturing resources planning (MRP II) is an extended version of MRP.

A typical MRP II system will consist of the following modules:

- ⇒ Business planning
- ⇒ Purchasing
- ⇒ Forecasting/demand management
- ⇒ Inventory control
- ⇒ Order entry and management
- ⇒ Shop-floor control
- ⇒ Master production scheduling (MPS)
- ⇒ Distribution requirements planning (DRP)
- ⇒ Material requirements planning (MRP)
- ⇒ Service requirements planning (SRP)
- ⇒ Capacity requirements planning (CRP)
- ⇒ Accounting

As we see from this list, MRP II covers all activities, from business planning to servicing the customer. In reality, the business planning exercise triggers dependency relationships for all resources in an organization. The forecasting/demand management module and the order entry system essentially interface with the outside world and bring recent information into the planning system. Based on these, production planning, MPS, and other requirements planning can be done. Since the outcome of these exercises is to procure items and services from outside and perform in-house activities as per plan, the relevant modules have also been included to close the gap. Essentially, the focus is on planning for all the resources that an operations system requires.

The advantage of MRP II lies in its ability to provide numerous feedback loops between different modules and minimize re-planning on a piece-meal basis. We saw in an earlier section that with the inclusion of CRP it was possible to close the gap between material and capacity. Similarly, adding DRP could address

NOTES

OPEN SOURCE ERP SOLUTION FOR VENKATESWARA HATCHERIES

NOTES

Venkateswara Hatcheries (VH) was established in 1971. By 2013, it was a ₹60 billion conglomerate and is arguably the largest fully-integrated poultry group in Asia. Uttara Foods and Feeds Limited (UFFL), a division of VH group is a leading manufacturer of quality poultry feed. Venky's Feed, and Venky's Chicken are among the forerunners in their respective markets. UFFL has production plants spread across eight locations in India.

Until 2012, UFFL was using an in-house IT solution for managing its operations. With the gradual expansion and resultant globalization of business, the existing IT solution was not enough. The limitations of the standalone IT solution became apparent when UFFL began seriously exploring the proposals for poultry seed farming in Bangladesh and Vietnam. Poor infrastructure in these countries prevented the company from laying its own corporate lease lines, and privacy concerns ruled out cloud or public domain as an option. Also, language and local tax conditions posed challenges as well.

The IT team at UFFL chose OpenBravo, an Open Source ERP vendor. The Open Source solution gave UFFL the opportunity to experiment with new ideas and build new capabilities. In collaboration with solution architects from OpenBravo, UFFL's IT team developed the new ERP solution. OpenBravo provided a localization feature, which allows it to operate in 64 different languages. UFFL could make use of this for their proposed operations in Bangladesh and Vietnam. UFFL went live with the new ERP in early 2011. UFFL has more than 250 users on the OpenBravo system.

OpenBravo's web service addressed UFFL's twin needs of protecting data and enabling mobility. The company has also developed a mobile application which its sales force can use to place purchase and transaction orders. The system has enabled UFFL to discover new capabilities. For instance, they have developed a new desktop application that allows employees working from remote areas with poor internet connectivity access an offline module to continue work uninterrupted. The web service functions can auto-synchronize any incremental work as soon as connectivity is established. The biggest benefit, however, has been the increase in transparency.

ERP implementation often calls for a good deal of change management as users will be required to modify their behaviours on many counts. One example in the case of UFFL was the reporting system. With the new solution, UFFL standardized its reporting structure. However, as users were very used to seeing reports in a certain format, the company launched a massive awareness and training exercise to educate users about the merits of the new system and turned their reluctance into approval.

Source: Adopted from <http://www.cio.in/case-study/venky-s-develops-erp-poultry-business-open-source>; Last accessed on 14 June 2014.

distribution-related issues. As more and more gaps are closed, it promotes a centralised approach to planning and promises to bring additional benefits arising out of integration.

8.6 ENTERPRISE RESOURCE PLANNING (ERP)

Enterprise resource planning is an organization-wide planning system that utilizes some common software and an integrated database for planning and control purposes. An ERP embeds all the organization processes into the software and creates a workflow mechanism such that different organizational players engaged in planning and control of a variety of activities can make use of the system. Viewed alternatively, an **ERP** system is a mammoth transaction engine that runs on some common software.

Enterprise resource planning is an organization-wide planning system that utilizes some common software and an integrated database for planning and control purposes.

Since activities in an organization typically have hundreds of processes comprising of thousands of activities, the software is split into modules representing different functional domains. Each module has a set of inputs, processes, and outputs. Moreover, each module is closely interconnected with several other modules (see Figure 8.11 for an illustration of this). Experts say that the power of ERP software lies in its ability to manage these interfaces well, thereby providing tighter integration. The following are the typical modules in ERP software:

- ⇒ Sales and distribution
- ⇒ Production planning
- ⇒ Logistics
- ⇒ Accounts payable/receivable, treasury
- ⇒ Operational (shop floor) control

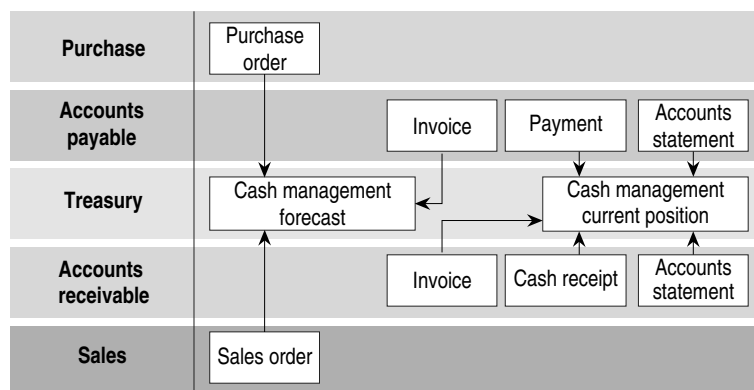


FIGURE 8.11 The role of ERP in interfacing functional areas of business

NOTES

NOTES

- ⇒ Purchasing
- ⇒ Finance and cost control
- ⇒ Human resources

In addition, other tools are also available for generating Web interfaces, coding and programme generation for specialised requirements, report generation routines, data import/export, and a library of best practices from which an organization could choose processes for implementation. A wide number of software options are available for ERP today. The most popular among them include SAP, Oracle, Ramco Systems, PeopleSoft, and JD Edwards. Ramco Systems provide ERP software using a cloud based architecture focusing on small and medium enterprises. Initially, ERP software were proprietary standards that were expensive and inflexible. However, all of them are now available as cloud based and customizable. They are also relatively less expensive and amenable to a greater degree of customization.

The heart of ERP is the organization-wide integration of several activities. This ranges from the integration of functions to markets, divisions, plants, products, and customers across the globe. The greatest benefit organizations get from implementing ERP is its ability to link various functional areas of business tightly through the software. Figure 8.11 is an illustration of the idea. Let us consider the issue of cash management activity pertaining to treasury. Purchase and sales order information directly affects the cash forecasting exercise. Similarly, the accounts payable and accounts receivable functions affect the actual cash management position. The ERP software will ensure that with the input new information in each of these modules, the treasury module will get automatically updated with this information.

Prior to the implementation of ERP, an organization re-engineers its processes. It also identifies the level of customization it needs on various functionalities in the software. Usually, a consulting firm assists the whole process and the software is implemented in a project mode.

Through the efficient use of a host of IT-enabled mechanisms, ERP promises to cut down cycle time, transaction costs, layers of decision making, and thereby improve responsiveness and flexibility. The other benefits include a greater sense of empowerment, better visibility of information, and improved customer service. Moreover, an ERP system could be the backbone of IT infrastructure for an organization with which several outward-facing IT applications can be built in the future. All these will eventually lead to improving the competitiveness of the organization at the marketplace.

8.7 RESOURCES PLANNING IN SERVICES

Consider a typical eye hospital, such as the Aravind Eye Hospital in Madurai, which conducts a large number of eye surgeries. A typical eye surgery requires certain resources. Let us look at four

key resources: eye surgeon, operation theatre (OT) and medical equipment, pre-operative care, and post-operative care. Let us assume that a typical eye surgery requires 1 hour each of the eye surgeon and the OT, 8 hours of pre-operative care, and 20 hours of post-operative care. If we need to plan to perform 1,000 operations in a 6-day week, one can compute the amount required of each of these resources. For example, the hospital requires 1,000 hours each of the eye surgeon and OT, the 8,000 hours of pre-operative care, and 20,000 hours of post-operative care. Furthermore, we also know that an item in this list could in turn depend on a host of other resources. Pre-operative care may mean the requirement of certain skills of medical professionals, diagnostic and curative equipment, medical consumables, medicines, support staff, and so on.

There are striking parallels between the eye hospital example and that of the telephone set, which was used to develop the logic of MRP. The end product in this case is a treated and discharged patient (similar to the telephone instrument in the manufacturing example). In the telephone examples, four sub-assemblies were required to complete the product. In this case, the product structure in the previous level will have post-operative care as a single item.

Therefore, as in the manufacturing situation, one can develop a bill of resources (BOR) in the case of service organizations and employ a similar planning exercise to schedule service deliveries and the associated resources and materials.

NOTES

SERVICES RESOURCES PLANNING(SRP): AN ERP SOLUTION FROM RAMCO

NOTES

Professional services industry is globally becoming more competitive forcing companies to move up the value chain, incorporate best business methods, predict and analyze near-term requirements, optimize costs, and identify new and stable revenue generation opportunities. Services industry faces multiple challenges ranging from the use of multiple standalone applications for the daily functioning of internal departments. This results in lack of control over project expenses and cash outflow.

Ramco SRP provides an end-to-end solution for customers of service industry by tightly integrating key functions such as HR and Finance with Project Management. Its core functionality is project based billing where projects can be Fixed Bid/ Milestone or Time and Material (T and M) in nature. SRP is also equipped with built-in adaptors to integrate with project management tools such as Microsoft Projects and Primavera. It further provides comprehensive project tracking with revenue recognition. Figure 8.12 provides the product guide for Ramco's SRP.

Ramco's SRP provides industry specific applications for the services sector. ITes/BPO industry is project-centric, making project management very essential for its success. SRP supports milestone-based, T and M based as well as capped T and M based projects. In addition, it also helps in comprehensive project tracking by integrating with MS Project and Primavera. SRP provides an integrated platform for process-centric consulting companies where timesheet entry, invoice generation, and billing are all inter-linked to one another. This reduces data redundancy and manual data entry into excel spreadsheets. Being a resource-centric industry, workforce management plays a key role in staffing companies. SRP application for staffing company includes processes and the flexibility and variations required for recruitment and selection, onboarding and resource deployment, and a transparent client billing process from Time/Attendance approval through payroll till invoicing.

SRP provides seamless real-time flow of information between multiple departments within the company such as HR, Finance, Project Control Office, and Resource Management Group. Employees, who form the core of services industry, can be managed very efficiently by optimizing their skills and effort across multiple projects or assignments simultaneously. On the whole, SRP reduces the time and effort spent in manual activities through process automation and also makes day-to-day transactions of the organization paperless by providing a single electronic source of information as a global data repository.

Source: Adopted from <http://www.ramco.com/hcm/why-ramco-hcm/services-resources-planning/>; Last accessed on 15 June 2014.

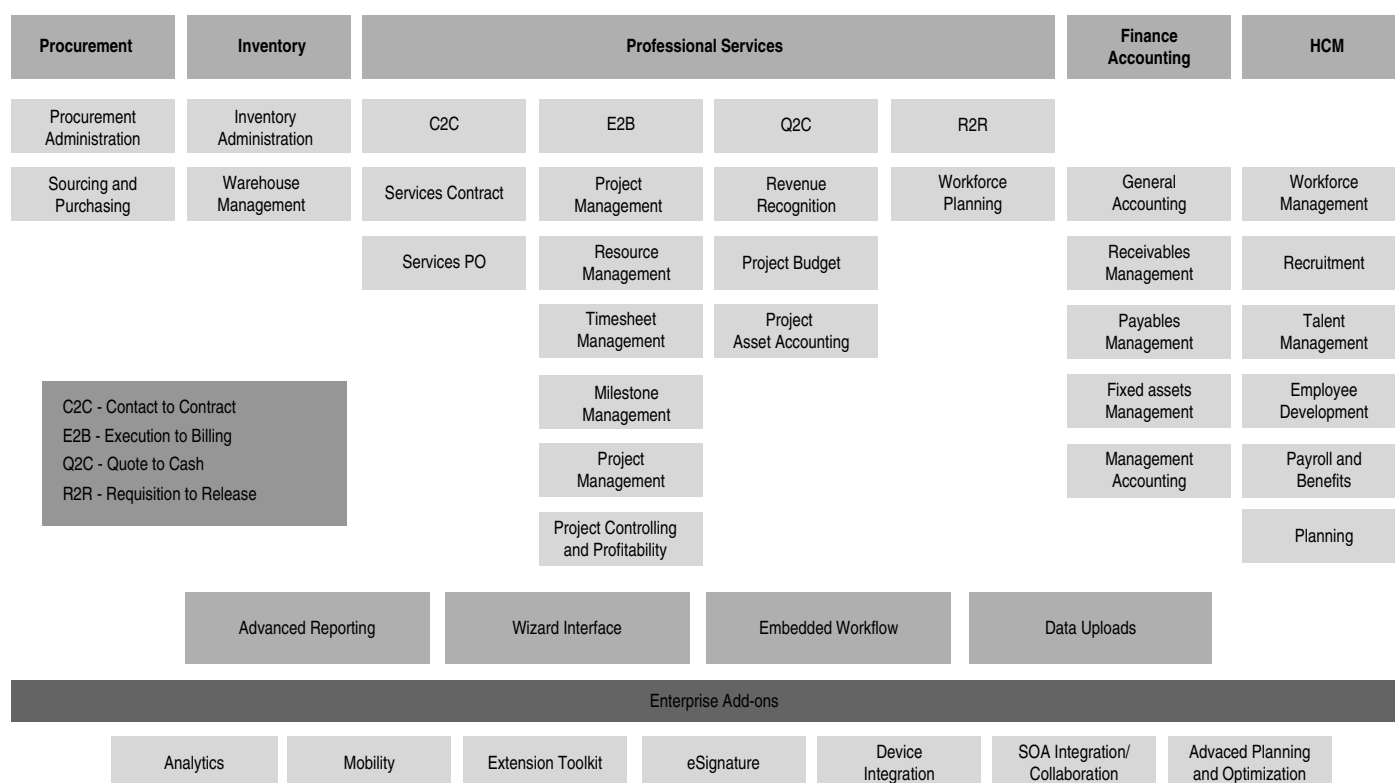


FIGURE 8.12 Product guide for Ramco services resources planning

8.8 SUMMARY

- Dependent demand items exhibit certain characteristics such as lumpy demand and parent-offspring relationships that can be exploited while planning for materials in organizations.
- Bill of materials (BOM) and product structures depict the dependency relationships among various components in a manufacturing set-up.
- Various lot-sizing rules can be utilized to batch requirements and schedule orders. The lot-sizing rules strike an appropriate trade-off between carrying cost and ordering cost.
- MRP logic involves a four-step logical process: net-lot-offset-explode. Through an iterative process, the material

requirements of all components in an organization during a planning horizon are arrived.

- Regeneration and net change are the two methods used for updating the MRP schedules.
- Capacity requirements planning (CRP) uses logic similar to that of MRP.
- Furthermore, the logic of MRP can be extended to several other spheres of an operations system. MRP II systems provide this functionality. Modern-day ERP systems are further extensions of MRP logic to other areas of business such as marketing, finance, and personnel.

8.9 REVIEW QUESTIONS

1. What do you mean by dependent demand and independent demand? Is knowledge of this useful to an organization? If so, in what way is this information useful?
2. Consider each of the following operating systems and identify three examples of dependent and independent demand items.
 - (a) A hospital providing coronary health care services
 - (b) A university offering post-graduate level courses
 - (c) A manufacturer of ceiling fans
 - (d) A transportation service provider for industrial products
 - (e) The corporation office of the city of Bangalore
 - (f) A computer facilities maintenance provider
 - (g) A BPO offering call centre services to clients
3. What are the key attributes of a dependent demand item? Explain with one example each from manufacturing and service organizations.
4. Why should an organization plan the requirements for a dependent demand item differently from that of an item with independent demand?
5. Suppose that an organization uses normal inventory control methods to plan for its dependent demand items. What will the impact of this be on material planning? Under what conditions will the planning result in satisfactory results?
6. What is meant by a bill of materials (BOM)? Why do organizations need BOMs?
7. Are BOMs and product structures the same? Explain with an example.
8. An organization engaged in the manufacture of electrical appliances is not clear if they need to use a modular or a single-level BOM. Prepare an advisory note clearly specifying the conditions under which each of these alternatives is appropriate.
9. Suppose that an organization prepared an MRP on the basis of incorrect data on lead time of the components. What will the impact of this be on the actual operations?
10. What is meant by the “explosion” of a BOM? Explain with a simple example
11. Identify the appropriate lot-sizing rule for each of the following components. Justify your choice of the rule.
 - (a) Steel plates for manufacturing automobile doors
 - (b) Eye surgeon time (in hours) in an eye hospital
 - (c) Bed linen for a five-star hotel
 - (d) Packing cartons for a computer manufacturer
 - (e) Sliced vegetables for use in 12 vegetarian dishes in a fast food joint
 - (f) Aircraft engines for Hindustan Aeronautics Limited's advanced light helicopter
12. What is a time bucket? Why do you need it in an MRP exercise?
13. A firm performing an MRP exercise has been using a time bucket of “months” so far. Beginning next month they have chosen to change the time bucket to “weeks”. What are the implications of this change to the organization?
14. What do you mean by safety stock and safety lead time in MRP? When should organizations use these in MRP?
15. What are the key implications to a manager in using the regenerative and net change updation of MRP schedules?
16. An organization was performing the MRP re-run schedule every week. It decided to change the time between two such re-runs to 10 days. What are the implications of this change for the organization?
17. Data integrity is a major issue for many organizations while using an MRP system. Comment on this statement.
18. How is MRP II different from MRP? What are the additional advantages that an organization will obtain by using an MRP II system?
19. “Organizations can obtain realistic plans by properly integrating MPS, MRP, and CRP through an iterative process.” Comment on this statement.

8.10 PROBLEMS

1. Omega, a manufacturer of office supplies, produces a desktop calendar unit using a simple process and the following components. The desktop calendar unit is basically assembled using a base plate, a sliding unit and a calendar assembly. The sliding unit is made up of a plastic tray and a set of 30 cards to store telephone and address information. The sliding unit is fixed to the base plate using a pair of 3mm screws. The calendar assembly is made up of two U-shaped hooks and a calendar (consisting of about 400 pages containing the daily calendar details). The calendar assembly is fixed to the base plate by clamping the U-shaped hooks using a pair of 3 mm screws each.

Each base plate is made of plastic sheets and requires 150 square inches of plastic sheet. Similarly, the sliding unit is also made of the same plastic sheet and requires 30 square inches of the plastic sheet. Each U-shaped hook is made from 7 inches of 4mm steel rod. The calendar is bought directly from a supplier. Finally, the product is packed in a specially made carton box and sealed using 12 inches of tape. There are also four bar-coded labels stuck to each carton box after it is sealed. This makes it a packaged calendar ready for shipping.

- (a) Develop a product structure for the desk calendar unit.
- (b) Using the given information, prepare a bill of material for the desktop calendar unit.

2. In Problem 1, assume that Omega offers the following alternatives:

- Two different varieties [one is the basic version and the other has provision for keeping pens/pencils (these affect only the base plate)]
 - Five different colour choices in the desktop calendar unit (these affect the base plate and the sliding unit)
 - Three different configurations of the calendar itself (one page per day, half a page per day, and month planner sheet at the beginning of every month in addition to one page per day)
- (a) How many different varieties of desktop calendar units can the organization offer to the customer?
- (b) Develop a BOM for the product range offered by Omega.

3. A gearbox manufacturer has 20 gearboxes in stock. Each gearbox has four gears. There are 200 gears already in stock. The gears are made from gear blanks. The stock of gear blanks in the stores is 100. Each gear blank requires 30 kilograms of alloy steel. The stores have 7,000 kilograms of alloy steel. Compute the requirement of components for manufacturing 570 gearboxes in the next month.

4. Consider the product structure given in Figure 8.13 pertaining to a product manufactured by Oriental Housings & Seals. The numbers in parentheses in the figure indicates the number of units of the item required to assemble one unit of its parent. Use the information available in the product structure to answer the following questions:

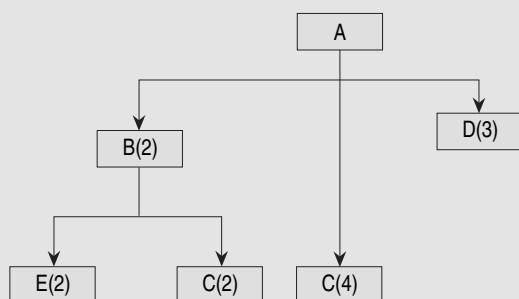


FIGURE 8.13 Product structure for Oriental Housings & Seals product

- (a) How many units of C are required to manufacture one unit of Product A? Did you make any assumption in computing this value?
- (b) How much inventory of each component is required for satisfying a demand of 100 units of the final product if: (i) there is no stock of finished goods (ii) there is a stock of 30 units of the final product?
- (c) Consider Problem (a). Will the computations change if there is some stock of Item B? Why and by how much? (Hint: Assume that you have x units of Item B to proceed with the analysis)
5. In Problem 4, the lead times for manufacture/assembly of the components are as follows:

Product A	2 weeks
Component B	1 week
Component C	2 weeks
Component D	2 weeks
Component E	1 week

- (a) How early can Oriental deliver an order of 100 units of the product to the customer if (i) there is no stock of Item B (ii) there is a stock of 220 units of Item B?
- (b) Will the results change if there is a stock of Product A?

6. Given in Table 8.4 is a partially completed MRP working for Component X. Using the information provided, complete the table.

TABLE 8.4 INCOMPLETE TABLE FOR PROBLEM 6

Component X	Lot Size: 2 Periods						
BOM Quantity: 1	Lead Time: 1						
	0	1	2	3	4	5	6
Gross requirement		200	150	100	200	120	200
On-hand inventory	300						
Net requirement							
Planned receipts							
Planned order releases							

7. Consider Component XX, for which the MRP exercise needs to be done. The relevant information pertaining to the component has been extracted from the company records and reproduced in Table 8.5. Currently, no decision has been made about the lot-sizing rule. Company estimates suggest that the cost of carrying inventory is ₹10 per unit per period and the cost of scheduling an order is ₹500 per order. The economic order quantity is 150 units.

TABLE 8.5 DATA ABOUT COMPONENT XX (PROBLEM 7)

Component XX	Lot Size: ?						
BOM Quantity: 1	Lead Time: 1						
	0	1	2	3	4	5	6
Gross requirement		200	350	400	200	450	200
On-hand inventory	600						
Net requirement							
Planned receipts							
Planned order releases							

Use this information to analyse the following five lot-size rules and suggest an appropriate lot-sizing rule for the component.

- Lot for lot
- POQ = 3 periods
- POQ based on the economic order quantity
- FOQ = 500
- FOQ = 1,000

8. Consider the product structure of a manufacturing firm, as shown in Figure 8.14. The company is interested in performing an MRP exercise for the next eight weeks to cover the requirements indicated in the MPS. The MPS requirements for the next eight weeks are 20, 80, 100, 200, 100, 0, 120, and 150.

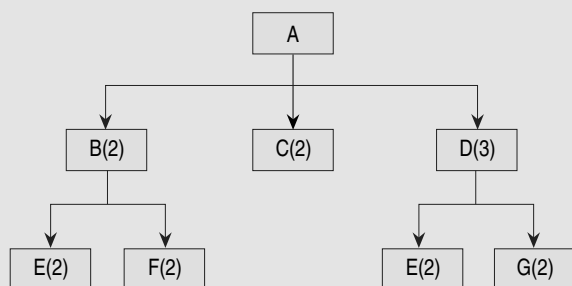


FIGURE 8.14 Product structure for Problem 8

The relevant information on lead time, inventory status and lot sizing are available in Table 8.6. Develop the MRP schedule for the final product as well as for all the components and answer the following questions:

TABLE 8.6 LEAD TIME, INVENTORY STATUS, AND LOT-SIZING RULES FOR PROBLEM 8

Item	Lead Time (Weeks)	Inventory Status	Lot-sizing Rule
A	1	100	Lot for lot
B	2	200	Lot for lot
C	1	400	2 periods
D	2	1,000	3 periods
E	2	2,000	2 periods
F	2	1,200	500
G	1	2,000	1,000

- Do we need to schedule order releases for Component B? If so, when and how much?
- How long will the on-hand inventory of Item D satisfy the requirement?
- How many orders need to be placed for Component F?
- If the cost of scheduling an order is ₹2,000, what is the total cost of ordering for the MRP plan?

9. Figure 8.15 shows the product structure for a manufacturing firm. In addition to the final product (Product A), Component B is sold as a spare in the market. The MPS for Product A and Component B and other relevant information for the problem are given in Tables 8.7 and 8.8.

Perform an MRP exercise to cover the requirements of the MPS and answer the following questions:

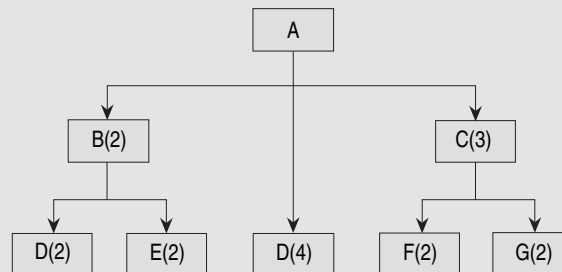


FIGURE 8.15 Product structure for Problem 9

TABLE 8.7 MPS FOR THE NEXT SIX PERIODS

Period	1	2	3	4	5	6
Product A	100	80	220	140	70	160
Component B	40	30	70		60	50

TABLE 8.8 LEAD TIME, INVENTORY STATUS, AND LOT-SIZING RULE FOR PROBLEM 9

Item	Lead time (Weeks)	Inventory Status	Lot-sizing Rule
A	2	200	Lot for lot
B	2	900	Lot for lot
C	1	750	Lot for lot
D	2	2,500	2 periods
E	1	300	3 periods
F	2	1,000	500
G	1	600	1,000

- What is the gross requirement for Component B during the six periods?
- How many orders do we have to place for Component C and in what quantities?
- Suppose the lead time for Component G is two weeks instead of one week, what will its impact be on the planning process?
- Suppose that the economic order quantity for Component C is 320 units. What will the appropriate POQ lot-sizing rule be for Component C?

10. Consider the following MRP (Table 8.9) for Component C, which is used in a product manufactured by an organization.

Of late, the organization has been experiencing frequent shortages of this component. It has been estimated that on an average, the company loses ₹5,000 by way of lost business and goodwill due to these shortages in a six-week period. Therefore, it has been decided to include a safety stock of 100 units. Rework the planning details for this component by incorporating the safety stock requirement.

11. In Problem 10, assume that the organization decided to implement a safety lead time of one week instead of safety stock.
- Rework the planning details for this component by incorporating the safety lead time requirement.
 - Compare the two results and list two significant observations that you have made.

TABLE 8.9 MRP FOR COMPONENT C

Component C	Lot Size: LFL						
BOM Quantity: 1	Lead Time: 1						
	0	1	2	3	4	5	6
Gross requirement		100	250	160	0	200	70
On-hand inventory	400	300	50	0	0	0	0
Net requirement		0		110	0	200	70
Planned receipts		0		110	0	200	70
Planned order releases			110	0	200	70	0

8.11 NET-WISE EXERCISES

- Visit the Web sites of the major ERP vendors (SAP, PeopleSoft, Oracle, Ramco, J.D. Edwards). Study the product offerings of all these vendors and prepare a detailed report to answer the following questions:
 - What are the key components of the ERP system offered by the major vendors?
 - How do these vendors differentiate their offerings?
- If an organization wishes to choose one of the four vendors, what factors will you suggest to the organization to enable appropriate selection?
- Are there industry-specific offerings of ERP by these vendors? If so, which sectors of the industry are covered and how well?

8.12 MINI PROJECTS

- Take a simple household gadget or appliance such as a dining table set or a sofa set with a centre table. Carefully analyse the assembly of the appliance and identify the parts that make up the appliance. Identify the major assemblies and sub-assemblies of the appliance.
 - Using this information prepare a complete BOM and a product structure for the appliance
 - What are the various ways by which you think product varieties can be created for this appliance? Identify them and prepare an indented BOM for the entire set of product offerings.
 - Assume an MPS and other relevant data for the components that you have identified for the appliance.
- Can you set up a standard table for performing the MRP exercise for the appliance?
- Set up the table created in Problem 1 in a Microsoft Excel worksheet (assume a planning horizon of 6 periods). Use the worksheet to accomplish the following tasks:
 - Perform the MRP exercise for the appliance that you have chosen.
 - Assume that the MPS is wrong. Quantities have been underestimated to the extent of 10 per cent during Periods 4 and 5 and overestimated to the extent of 5 per cent during Period 6. What is the likely impact of this on customer commitment?

8.13 MULTIPLE-CHOICE QUESTIONS

- 8-1.** Operating inventories are dependent on:
A. Supply
B. Demand
C. Production
D. Procurement
- 8-2.** What are the two types of inventories exist in an operating system?
A. Operating and distribution inventories
B. Product and operating inventories
C. Distribution and product inventories
D. None of the above
- 8-3.** A structured approach that develops schedules for launching orders for materials in any manufacturing system and ensures the availability of these at the right time and the right place is:
A. Capacity requirement planning
B. Inventory requirement planning
C. Material requirement planning
D. Production requirement planning
- 8-4.** Regeneration is one method to update:
A. Service requirements planning schedule
B. Inventory requirement planning schedule
C. Capacity requirement planning schedule
D. Material requirements planning schedule
- 8-5.** What is the technique that applies the MRP logic to address the capacity issues in an organization?
A. Capacity Requirements Planning
B. Material Requirements Planning
C. Distribution Requirement Planning
D. Energy Requirement Planning
- 8-6.** To plan distribution inventories in the downstream supply chain, consisting of depots and stocking points at the dealer level is termed as:
A. Capacity Requirements Planning
B. Material Requirements Planning
C. Energy Requirements Planning
D. Distribution Requirements Planning
- 8-7.** The iterative process of computing all requirements at a particular level and then moving down one level is known as:
A. Level by level analysis
B. Top to bottom approach
C. Scheduling
D. Explosion
- 8-8.** Which is that organization-wide planning system that utilizes some common software and an integrated database for planning and control purposes?
A. Capacity Requirements Planning
B. Material Requirements Planning
C. Energy Requirements Planning
D. Distribution Requirements Planning
- 8-9.** In which of the below, planning is based on resources?
A. Capacity Requirements Planning
B. Resources Planning in Services
C. Distribution Requirements Planning
D. Service Requirements Planning
- 8-10.** What are the two types of inventories exist in an operating systems?
A. Operating inventory and distribution inventory
B. Fixed inventory and fluctuation inventory
C. Physical inventory and computerized inventory
D. First in First out and Last in Last out

Answer Keys: 8-1. B 8-2. A 8-3. C 8-4. D 8-5. A 8-6. D 8-7. D 8-8. C 8-9. B 8-10. A

9

CHAPTER

INVENTORY MANAGEMENT: CONCEPTS

9.1	Understanding Inventory
9.1.1	Functions of Inventory
9.2	Inventory Planning for Independent Demand Items
9.2.1	Continuous Demand
9.2.2	Uncertainty of Demand
9.3	Types of Inventory
9.3.1	Seasonal Inventory
9.3.2	Decoupling Inventory
9.3.3	Cyclic Inventory
9.3.4	Pipeline Inventory
9.3.5	Safety Stock
9.4	Inventory Costs
9.4.1	Inventory-carrying Cost
9.4.2	Cost of Ordering
9.4.3	Cost of Shortages
9.5	Inventory Control for Deterministic Demand Items
9.5.1	Problems in the EOQ model
9.6	Handling Uncertainty in Demand
9.7	Summary
9.8	Formula Review
9.9	Review Questions
9.10	Problems
9.11	Mini Projects
9.12	Case Study
9.13	Multiple-choice Questions

INVENTORY MANAGEMENT PROVIDES COMPETITIVE ADVANTAGE AT AMAZON.COM

NOTES

When Jeff Bezos opened his revolutionary business in 1995, Amazon.com was intended to be a “virtual” retailer—no inventory, no warehouses, no over-head—just a bunch of computers taking orders for books and authorizing others to fill them. Things clearly didn’t work out that way. Now, Amazon stocks millions of items of inventory, amid hundreds of thousands of bins on shelves in over 150 warehouses around the world.

Additionally, Amazon’s software is so good that Amazon sells its order taking, processing, and billing expertise to others. It is estimated that 200 million items are now available via the Amazon Web site.

Bezos expects the customer experience at Amazon to be one that yields the lowest price, the fastest delivery, and an error-free order fulfillment process so no other contact with Amazon is necessary. Exchanges and returns are very expensive. Managing this massive inventory precisely is the key for Amazon to be the world-class leader in warehouse automation and management. The time to receive, process, and position the stock in storage and to then accurately “pull” and package an order requires a labor investment of less than 3 minutes. And 70% of these orders are multiproduct orders. This underlines the high benchmark that Amazon has achieved. This is world-class performance. When you place an order with Amazon.com, you are doing business with a company that obtains competitive advantage through inventory management. This Global Company Profile shows how Amazon does it.



1. You order three items, and a computer in Seattle takes charge. A computer assigns your order—a book, a game, and a digital camera—to one of Amazon’s massive U.S. distribution centers.
2. The “flow meister” at the distribution center receives your order. She determines which workers go where to fill your order.

Source: Marilyn Newton/Reno Gazette-Journal

3. Amazon’s current system doubles the picking speed of manual operators and drops the error rate to nearly zero.

Source: Bernhard Classen/Alamy





5. All three items converge in a chute and then inside a box. All the crates arrive at a central point where bar codes are matched with order numbers to determine who gets what. Your three items end up in a 3-foot- wide chute—one of several thousand—and are placed into a corrugated box with a new bar code that identifies your order. Picking is sequenced to reduce operator travel.
6. Any gifts you've chosen are wrapped by hand. Amazon trains an elite group of gift wrappers, each of whom processes 30 packages an hour.

Source: REX/Newscom



4. Your items are put into crates on moving belts. Each item goes into a large yellow crate that contains many customers' orders. When full, the crates ride a series of conveyor belts that wind more than 10 miles through the plant at a constant speed of 2.9 feet per second. The bar code on each item is scanned 15 times, by machines and by many of the 600 workers. The goal is to reduce errors to zero—returns are very expensive.

Source: Ben Cawthra/Sipa USA/Newscom



7. The box is packed, taped, weighed, and labeled before leaving the warehouse in a truck. A typical plant is designed to ship as many as 200,000 pieces a day. About 60% of orders are shipped via the U.S. Postal Service; nearly everything else goes through United Parcel Service.
8. Your order arrives at your doorstep. In 1 or 2 days, your order is delivered.

Source: Adrian Sherratt/Alamy

NOTES

The term **inventory** refers to any idle resource that can be put to some future use. Manufacturing and several service organizations have significantly invested in inventory. Often, investment in inventory has a direct bearing on the profitability of a firm. Experiences in the last twenty years suggest that the world-class performance of a firm hinges on the firm's ability to cut investment in inventory to very low levels. Automobile assembly plants and component manufacturers are known to carry very little inventory and yet provide good quality finished goods at the right time. In addition to cutting down cost, reduced inventory levels help an organization improve quality, planning systems, and supply chain coordination. They also reduce wastage and obsolescence. Hence, inventory planning and control continues to derive considerable attention of the management in organizations.

NOTES

The term **inventory** denotes any idle resource that could be put to some future use.

9.1 UNDERSTANDING INVENTORY

As Amazon.com well knows, inventory is one of the most expensive assets of many companies, representing as much as 50% of total invested capital. Operations managers around the globe have long recognized that good inventory management is crucial. On the one hand, a firm can reduce costs by reducing inventory. On the other hand, production may stop and customers become dissatisfied when an item is out of stock. *The objective of inventory management is to strike a balance between inventory investment and customer service.* You can never achieve a low-cost strategy without good inventory management.

All organizations have some type of inventory planning and control system. A bank has methods to control its inventory of cash. A hospital has methods to control blood supplies and pharmaceuticals. Government agencies, schools, and, of course, virtually every manufacturing and production organization are concerned with inventory planning and control.

In cases involving physical products, the organization must determine whether to produce goods or to purchase them. Once this decision has been made, the next step is to forecast demand, as discussed in Chapter 4. Then operations managers determine the inventory necessary to service that demand. In this chapter, we discuss the functions, types, and management of inventory. We then address two basic inventory issues: how much to order and when to order.

9.1.1 Functions of Inventory

Inventory can serve several functions that add flexibility to a firm's operations. The four functions of inventory are:

1. *To provide a selection of goods for anticipated customer demand and to separate the firm from fluctuations in that demand.* Such inventories are typical in retail establishments.
2. *To decouple various parts of the production process.* For example, if a firm's supplies fluctuate, extra inventory may be necessary to decouple the production process from suppliers.
3. *To take advantage of quantity discounts,* because purchases in larger quantities may reduce the cost of goods or their delivery.
4. *To hedge against inflation* and upward price changes.

9.2 INVENTORY PLANNING FOR INDEPENDENT DEMAND ITEMS

In manufacturing organizations, finished goods and spare parts typically belong to the category of independent demand items. While planning for a dependent demand item is done to meet

manufacturing requirements, in the case of independent demand items, it is done to meet customer requirements. Two attributes characterize and distinguish independent demand items: They are in continuous demand, and there is an element of considerable uncertainty in the demand for such items.

In manufacturing organizations, finished goods and spare parts typically belong to the category of independent demand items.

9.2.1 Continuous Demand

Independent demand items are in continuous demand. Consider the sale of consumer appliances of manufacturers such as Videocon or LG. The demand for 32" LCD colour television panels in a particular city will be continuous. When there is a continuous demand for an item, constant availability of items and periodic replenishment of stock are important elements of the planning process. Non-availability of items translates into lost sales, poor customer goodwill, and additional costs in servicing the promised deliveries.

9.2.2 Uncertainty of Demand

There is an element of considerable uncertainty of demand in the case of independent demand items. On the other hand, in the case of dependent demand items, the demand is always derived and hence known with certainty. For example, the number of tyres required per month for new vehicles is known accurately once the monthly production is finalized. However, the number of spark plugs required for sale as spare parts needs to be estimated on some basis. Therefore, inventory planning for independent demand items should include some cushion for handling uncertainties whenever they are significant.

Given these requirements, inventory planning of independent demand items must address the following two key questions:

Inventory planning of independent demand items must address the two key questions: *how much, and when?*

NOTES

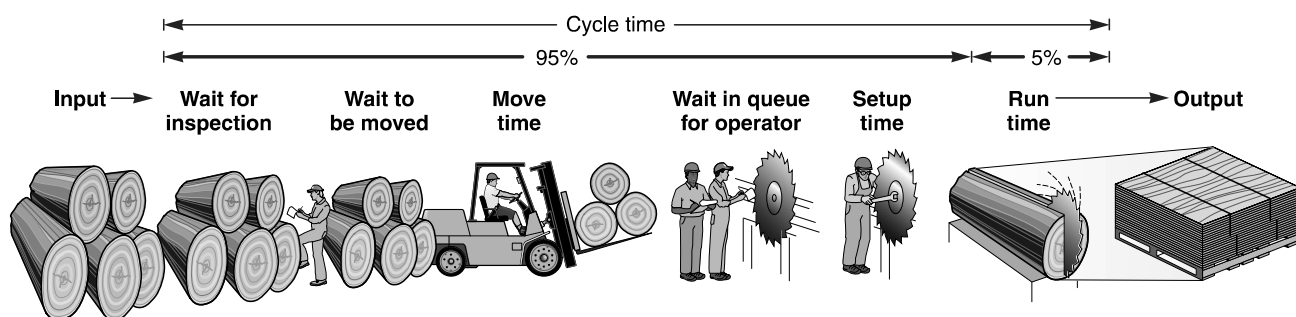


FIGURE 9.1 The Material Flow Cycle

Most of the time that work is in-process (95% of the cycle time) is not productive time.

NOTES

- (a) *How much?* Good inventory planning must devise means by which the planner can continuously replenish the stock as it depletes over time. The replenishment could be of a fixed quantity or variable. It could be based on the costs associated with inventory. It could also satisfy certain practical considerations such as truck capacity, quantity discounts, and minimum order quantity restrictions posed by the supplier.
- (b) *When?* The other issue is the timing of replenishment. Since demand is continuous and uncertain, the timing of replenishment is equally crucial. There are numerous ways of addressing the timing. It could be based on consumption patterns, the value of the item, managerial preferences for review, and other practical considerations.

9.3 TYPES OF INVENTORY

Before planning for inventory, it is important to know why organizations carry inventory and what factors influence the level of investment. Five types of inventory are normally found in most organizations.

9.3.1 Seasonal Inventory

Organizations carry inventory to meet fluctuations in demand arising out of seasonality. During festival periods, the demand for consumer durables may be high due to an increase in disposable income in the hands of consumers. In order to meet this surge in demand, there is inventory build-up during non-peak periods. Similarly, in a fast-food restaurant, a certain amount of inventory build-up happens during non-peak hours to handle the increase in demand during peak hours. In both these cases, inventory plays the crucial role of addressing short-term capacity issues in an organization.

Five types of inventory are normally found in organizations: *seasonal inventory*, *decoupling inventory*, *cyclic inventory*, *pipeline inventory*, and *safety stock*.

9.3.2 Decoupling Inventory

Manufacturing systems typically involve a series of production and assembly workstations. Raw material passes through these stages before it is converted into finished goods. Each stage behaves idiosyncratically on account of varying process times, downtimes, and resource availability. Therefore, the planning and control of such multi-stage production processes becomes very complex. One way to simplify the production planning and control problem is to decouple successive stages using inventory at some intermediate points. Each stage will have an input buffer and an output buffer. The output buffer of the preceding stage becomes the input buffer of the succeeding stage. Inventory decisions in this case

require analysis of workstation capacities, resource availability, and bottlenecks.

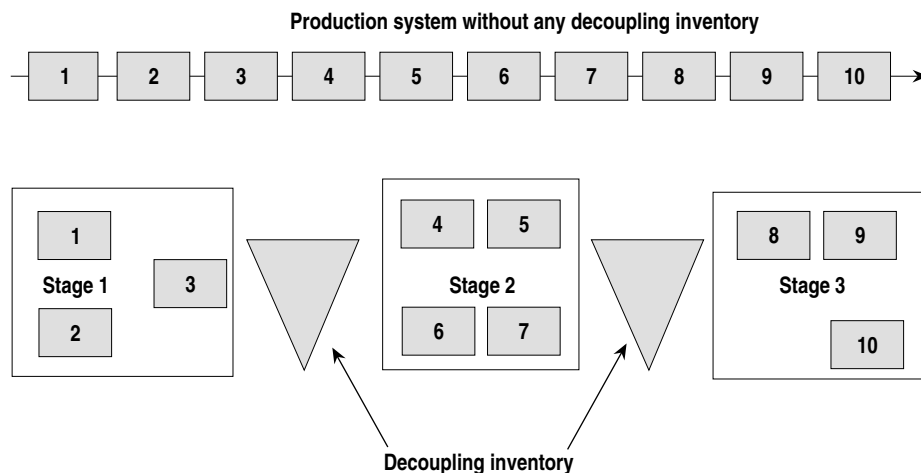
Figure 9.2 illustrates the use of decoupling inventory. In the first case, there is no decoupling inventory. In this case, all the ten stations of production are to be closely monitored and flow balance is to be ensured on a continuous basis. If there is a problem in one of the stations, it will affect all upstream and downstream stations. Further, the vagaries of demand uncertainty will directly affect the system's performance. Planning and control is challenging, especially when the number of stations is very large. On the other hand, the second case in Figure 9.2 has decomposed the production into three major stages. Each stage is linked to the other using decoupling inventory. This enables each stage to work with reasonable levels of independence. Further, the adverse impact of one stage will not immediately affect the others. Therefore, a certain amount of decentralized planning and control is possible.

9.3.3 Cyclic Inventory

It is customary for organizations to order inventory in repeated cycles and consume them over time. For example, a hospital may order disposable syringes in quantities of 10,000. If the average consumption rate is 500 per day, then it takes 20 days to deplete one order. On the twenty-first day, another order of 10,000 will arrive and it will be consumed over the next 20 days, and so on. Cyclic inventory goes through a saw-toothed pattern as shown in Figure 9.3. Each cycle begins with replenishment and ends with complete depletion of the inventory.

If Q is the order quantity per cycle,

$$\text{The average cyclic inventory} = \frac{Q+0}{2} = \frac{Q}{2} \quad (9.1)$$



NOTES

NOTES

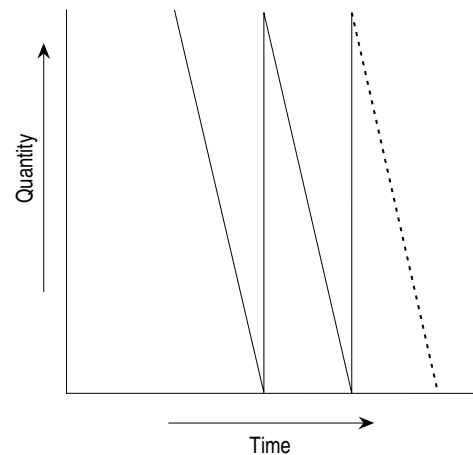


FIGURE 9.3 Cyclic inventory

9.3.4 Pipeline Inventory

Pipeline inventory pertains to the level of inventory that organizations carry in the long run due to non-zero lead time for order, transport, and receipt of material from the suppliers. Because of the geographical distances between the buyers and the suppliers and the host of business processes involved in ordering and receiving material, there is a time delay between order placement and order receipt. The inventory carried to take care of these delays is known as pipeline inventory.

Consider the example of the hospital. Suppose it takes three days to supply disposable syringes. Then the hospital should place an order at the end of Day 17 in order to replenish the stock to 10,000. This means that an order will be placed when the inventory level reaches 1,500 (3 days at the rate of 500 syringes per day). In general, if the lead time for supply is L , and the mean demand per unit time is μ , then the pipeline inventory is $L \times \mu$.

In the long run, the system will always have this inventory in order to take care of lead time for supply. The only way organizations can reduce this inventory is to cut down lead time for procurement, production, and distribution. Locating suppliers nearby, re-engineering the business processes related to procurement and distribution of materials, and using efficient alternatives for logistics are some of the ways in which an organization can reduce lead time and pipeline inventory.

9.3.5 Safety Stock

Organizations also have additional investment in inventory to buffer against uncertainties in demand and supply of raw material and components. We know from the elementary theory of probability distributions that when demand is stochastic, carrying average inventory will ensure that the demand is met only 50 per cent of the time (in the long run). However, in order to improve the availability

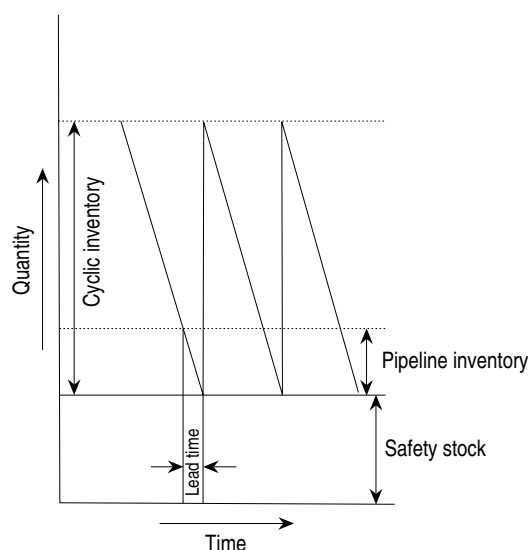


FIGURE 9.4 Cyclic inventory, pipeline inventory, and safety stock

to meet uncertain demand, an additional quantity known as safety stock is kept. Greater investment in safety stock leads to a lower probability of inventory going out of stock. Similarly, the higher the uncertainty, the greater is the need for safety stock. Safety stock only serves to prevent shortages in the short run. However, in the long run, the demand will tend toward an average value and the safety stock will not be consumed. Figure 9.4 represents an inventory system with safety stock and other types of inventory.

Reducing safety stock calls for reducing uncertainty in the system. Investment in superior planning and forecasting systems and good supplier development practices will help organizations reduce uncertainty in the system. In the case of components manufactured in-house, better production planning and improved maintenance of resources will help reduce uncertainty in the system.

Reducing safety stock calls for reducing uncertainty in the system.

Cyclic inventory, pipeline inventory, and safety stock are critically linked to both the “how much” and “when” questions in inventory planning. Lead time influences the “when” decision directly and determines the level of pipeline inventory in the system. Similarly, cyclic inventory is the outcome of the “how much” decision that an inventory planner makes. Safety stock influences both “how much” and “when” in an indirect sense. However, in order to understand how these decisions are made, one needs to understand various costs associated with inventory planning and control.

Cyclic inventory, pipeline inventory, and safety stock are critically linked to the “how much” and “when” questions in inventory planning.

NOTES

NOTES

9.4 INVENTORY COSTS

There are several costs associated with inventory planning and control. These costs could be classified under three broad categories:

- ⇒ The cost of carrying inventory. All costs related to maintaining inventory in organizations will be classified under this.
- ⇒ The cost associated with ordering material and replenishing it in cyclic intervals.
- ⇒ The cost arising out of shortages.

Inventory control models should take these into consideration and aim at minimizing the sum of all these costs. Sometimes, the unit cost of the item for which inventory planning is done is also a relevant cost for decision making. This is because when large quantities are ordered, there could be some discount in the unit cost of the item.

9.4.1 Inventory-carrying Cost

Organizations need to spend a considerable amount of money to carry inventory. The most significant component is the interest for the short-term borrowals for the working capital required for inventory investment. The second significant cost relates to the cost of stores and warehousing and the administrative costs related to maintaining inventory and accounting for it. The elements of storing and warehousing costs include the following:

- ⇒ Investment in store space and storage and retrieval systems
- ⇒ Software for maintaining the inventory status
- ⇒ Managerial and other administrative manpower to discharge various activities related to stores
- ⇒ Insurance costs
- ⇒ Cost of obsolescence, pilferage, damages, and wastage

In the case of high-tech electronics and short-shelf-life items, such as active chemical compounds, food items, and pharmaceutical formulations, obsolescence costs could be significant. Table 9.1 shows a sample computation of carrying cost for a company. The components of inventory-carrying costs exert considerable pressure on an organization to keep inventory levels low.

EXAMPLE 9.1

A manufacturer of transformers requires copper (both in plate and wire form) as a key ingredient. The average weekly requirement of copper is 200 tonnes. The lead time for the supply of copper is two weeks. If the manufacturer places monthly orders of copper, analyse the various types of inventory in the system.

Solution

Order quantity $Q = 1$ month requirement = 800 tonnes.

$$\text{Cyclic inventory in the system} = \frac{Q}{2} = \frac{800}{2} = 400 \text{ tonnes.}$$

$$\text{Lead time } L = 2 \text{ weeks}$$

$$\text{Average weekly demand } \mu = 200 \text{ tonnes}$$

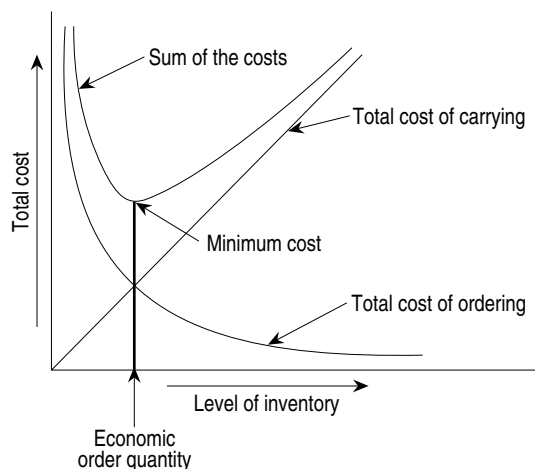
$$\text{Pipeline inventory} = L \times \mu = 200 \times 2 = 400 \text{ tonnes.}$$

TABLE 9.1 COMPUTATION OF CARRYING COST: AN ILLUSTRATION

Sl. No.	Item of Expenditure (Annual)	Amount (₹)
1	Stationary	75,000.00
2	Insurance premium	375,000.00
3	Establishment expenses and overheads	275,000.00
4	Salary of stores personnel	1,100,000.00
	Total expenditure	1,750,000.00
	Average value of the inventory in stores	35,000,000.00
	Warehousing cost (in % inventory value)	5.00%
a	Cost of warehousing	5.00%
b	Cost of capital (assumed)	15.00%
c	Obsolescence (estimated historically)	2.00%
d	Damages, spoilage, etc. (estimated historically)	1.00%
	Carrying cost (%)	23.00%

However, the ability of an organization to reduce inventory depends on the nature of business processes in place, the lead time for procurement, and the quality of planning, as pointed out in the earlier section.

Note that all the costs related to carrying inventory are directly related to the level of inventory. Graphically, it can be shown as a simple linear relationship (Figure 9.5). Let C_c denote the inventory-carrying cost per unit per unit time. Since the interest component is the predominant part of C_c , the usual practice is to represent C_c as a percentage of the unit cost of the item. Thus, if C_u is the unit cost of the item, then $C_c = iC_u$, where i is in percentage. For example, if the unit cost of an item is ₹5,000, the annual interest

**FIGURE 9.5** Behaviour of ordering and carrying costs of inventory**NOTES**

NOTES

charges are 12 per cent, and the other annual costs related to carrying inventory are 3 per cent, then the inventory-carrying cost is 15 per cent of the unit cost, that is, ₹750 per unit per year.

For an order quantity of Q , the average inventory carried by an organization is $Q/2$. Therefore,

$$\text{Cost associated with carrying inventory} = \left(\frac{Q}{2} \times C_c \right) \quad (9.2)$$

9.4.2 Cost of Ordering

Replenishment of cyclic inventory is achieved by ordering material with the suppliers. Organizations perform a series of tasks related to ordering material. These include search and identification of appropriate sources of supply, price negotiation, contracting and purchase-order generation, follow-up and receipt of material, and eventual stocking in the stores after necessary accounting and verification. We have discussed several aspects pertaining to this in some detail. All these tasks involve manpower, resources, and time that can be classified under cost of ordering. Since several of these costs are fixed in nature, ordering a larger quantity could reduce the total costs of ordering. A larger order quantity Q will require fewer orders to meet a known demand D , and vice versa. The relationship between Q and the total cost of ordering is graphically shown in Figure 9.5. Table 9.2 shows a sample computation of ordering cost for a company.

The number of orders to be placed to satisfy a demand of $D = D/Q$. If C_0 denotes the cost of ordering per order, then:

$$\text{Total ordering cost} = \left(\frac{D}{Q} \times C_0 \right) \quad (12.3)$$

Sometimes, components required for use in an organization are sourced from within. A division or a department of the organization may be specially manufacturing the required components for internal consumption. In such situations, the costs associated with setting up the required machinery will represent the cost of ordering. For example, let us assume that a component is produced in-house. In order to manufacture the component, two machines are to be set up and it takes 60 minutes and 120 minutes respectively in these machines. It also requires consumables and tools for the purpose of setting up the machine. Finally, direct labour time and supervisor time needs to be devoted for the activity. By estimating the cost of all these factors, we can arrive at the cost of set-up.

The cost of carrying and the cost of ordering are fundamentally two opposing cost structures in inventory planning. For instance, when the order quantity becomes smaller, the total cost of carrying inventory decreases. On the other hand, since a large number of orders are to be placed to satisfy the demand,

**TABLE 9.2 COMPUTATION OF ORDERING COST:
AN ILLUSTRATION**

Sl. No.	Item of Expenditure (Annual)	Amount (₹)
1	Stationary	80,000.00
2	Telephone	40,000.00
3	Other communication expenses	60,000.00
4	Salary of purchase department personnel	1,100,000.00
5	Inwards goods inspection section expenses	350,000.00
6	Other expenses and overheads	200,000.00
	Total expenditure	1,830,000.00
	No. of purchase orders generated	600.00
	Average cost of ordering	3,050.00

the total ordering costs will go up. Conversely, when the order quantity is increased, while the total cost of ordering will come down owing to fewer orders being placed, the total cost of carrying inventory will increase due to an increase in the average cyclic inventory in the system. Balancing these two opposing costs will be central to inventory planning and control in any organization. Figure 9.5 gives a graphical representation of the sum of these two costs.

9.4.3 Cost of Shortages

Despite careful planning, organizations are likely to run out of stock for several reasons. There could be a sudden surge in demand. Alternatively, the suppliers might not have delivered the material as per schedule, or a lot could have been rejected because of defective components. Such events disrupt production and have a cascading effect down the supply chain. Delivery schedules are missed, leading to customer dissatisfaction and loss of goodwill. It also introduces additional costs arising out of pushing the order back and rescheduling the production system to accommodate these changes. Rush purchases, uneven utilization of available resources, and lower capacity utilization further escalate the costs in the system. All these form part of the cost of shortage.

The cost of carrying and the cost of ordering are fundamentally two opposing cost structures in inventory planning.

The cost of shortage per unit is denoted by C_s . Since the effects of shortage are vastly intangible, it is difficult to accurately estimate C_s . In practice, managers tend to use other measures to incorporate the cost of shortage if estimation of C_s proves to be futile.

NOTES

NOTES

9.5 INVENTORY CONTROL FOR DETERMINISTIC DEMAND ITEMS

Let us consider a situation in which the demand for an item is continuous and is known with certainty. Since the demand is known, we exclude the possibility of having shortages. Better inventory control requires that we answer the “how much” and “when” questions by balancing the total costs of carrying inventory and ordering. For an order quantity of Q , we can compute the total cost of carrying and ordering from Eqs. 9.2 and 9.3.

Total cost of the plan = Total cost of carrying inventory + Total cost of ordering

$$TC(Q) = \left(\frac{Q}{2} \times C_c \right) + \left(\frac{D}{Q} \times C_0 \right) \quad (9.4)$$

When the total cost is minimum, we obtain the most economic order quantity (EOQ).

By taking the first derivative of Eq. 9.4 with respect to Q and equating it to zero, we can obtain the EOQ. Differentiating Eq. 9.4 with respect to Q we obtain:

$$\frac{dTC(Q)}{dQ} = \left(\frac{C_c}{2} \right) - \frac{C_0 D}{Q^2}$$

Note that the second derivative is positive and hence we obtain the minimum cost by equating the first derivative to zero. Equating the first derivative to zero and rearranging the terms, we obtain Eq. 9.5.

Denoting EOQ by Q^* , we obtain the expression of Q^* as:

$$Q^* = \sqrt{\frac{2C_0 D}{C_c}} \quad (9.5)$$

$$\text{Optimal number of orders} = \frac{D}{Q^*} \quad (9.6)$$

$$\text{Time between orders} = \frac{Q^*}{D} \quad (9.7)$$

Q^* answers the “how much” question directly. Every time the inventory depletes to zero, it is economical to place an order equal to Q^* . Similarly, time between orders answers the “when” question. Since the demand is continuous, the demand rate will determine “when” to place an order. For instance, if the annual demand for an item is 2,500 units and if there are 250 working days, the daily demand is 10. If Q^* is 300, then it is implied that an order needs to be placed every 30 days.

EXAMPLE 9.2

A two-wheeler component manufacturing unit uses large quantities of a component made of steel. Although these are production items, the demand is continuous and inventory planning could be done independent of the production plan. The annual demand for the component is 2,500 boxes. The company procures the item from a supplier at the rate of ₹750 per box. The company estimates the cost of carrying inventory to be 18 per cent per unit per annum and the cost of ordering as ₹1,080 per order. The company works for 250 days in a year. How should the company design an inventory control system for this item? What is the overall cost of the plan?

Solution

Annual demand for the item (D) = 2,500 boxes

Number of working days = 250

The average daily demand = $\frac{2500}{250} = 10$ boxes

Unit cost of the item (C_u) = ₹750.00 per box

Inventory-carrying cost = $iC_u = 0.18 \times 750.00 = ₹135.00$ per unit per year

Cost of ordering (C_0) = ₹1,080.00 per order

The “how much” decision:

Economic order quantity(Q^*)

$$= \sqrt{\frac{2C_0D}{C_c}} = \sqrt{\frac{2 \times 1080 \times 2500}{135}} = 200 \text{ Boxes}$$

Number of orders to be placed

$$= \frac{D}{Q^*} = \frac{2500}{200} = 12.5 \approx 13$$

The “when” decision:

Time between orders

$$= \frac{Q}{D} = \frac{200}{2500} = 0.08 \text{ year} = 0.08 \times 250 = 20 \text{ days.}$$

Total cost of the plan

$$\begin{aligned} &= TC(Q^*) = \left(\frac{Q^*}{2} \times C_c \right) + \left(\frac{D}{Q^*} \times C_0 \right) \\ &= \frac{200}{2} \times 135 + \frac{2500}{200} \times 1080 = ₹27,000 \end{aligned}$$

Hence, the manufacturer will place an order for 200 boxes of the component once in every 20 days and will incur a total cost of ₹27,000 for the plan. Any quantity above or below will increase the cost of the plan.

9.5.1 Problems in the EOQ model

We have made several assumptions while deriving Q^* . The salient among these include the following:

- The demand is known with certainty and is continuous over time.
- There is an instantaneous replenishment of items.
- The items are sourced from an external supplier.
- There are no restrictions on the quantity that we can order.
- There are no preferred order quantities for the items.
- No price discount is offered when the order size is large.

In practice, several of these assumptions do not hold. Despite this, the EOQ model can be applied with suitable modifications because it is robust. Because of the square root in the formula, changes in model parameters, such as demand and cost, have less impact on the solution. If demand is not known accurately, one can use representative figures for D and assess the impact of changes on the total cost of the plan (see Problem 4 at the end of the chapter). The assumption of instantaneous replacement can be easily taken care of by placing an order ahead. If the lead time is L , then by placing an order L periods ahead of depletion of inventory, one can ensure

NOTES

NOTES

The EOQ model is robust and can be applied in several real-life settings with some modifications.

availability of material. Similarly, if the items are manufactured in-house, it is possible to estimate the economic run length by a simple modification to Eq. 9.5.

Assumptions (d), (e), and (f) indicate the need for examining the suitability of utilizing order quantities different from Q^* . Large order sizes result in price discounts. There are other reasons for placing a larger order than what the EOQ suggests. Sometimes, it will be economical to transport a truckload of items and save on transportation cost. In other cases, the supplier may impose a minimum order quantity. In all these cases, the relevant cost for analysis includes cost elements other than carrying and ordering. An analysis of these relevant costs will help the inventory planner make an appropriate choice in order quantity. See Example 9.3 for an illustration of this.

9.6 HANDLING UNCERTAINTY IN DEMAND

With some modifications, the EOQ model can accommodate several practical considerations. However, when the demand is uncertain, instances of shortages occur while operating the system with EOQ. There are two ways of handling the issue of shortages. One

EXAMPLE 9.3

Consider Example 9.2. Assume that the carrying cost of the component remains ₹135 per unit per year and that the supplier is willing to offer a discount on the unit price as per the following structure:

Up to 399 boxes = No discount

400 – 799 boxes = 2 per cent discount

800 – 1,000 boxes = 3 per cent discount

What should the company do in this case?

Solution

The economic order quantity is 200 boxes and the unit price of the item is ₹750.

There are two other order quantities at which the unit price changes on account of discount.

At $Q_1 = 400$, unit price of the item = $0.98 \times 750 = ₹735.00$

At $Q_2 = 800$, unit price of the item = $0.97 \times 750 = ₹727.50$

Since there is a discount on the price as the order quantity is varied, the total cost comparison between alternatives can be made only after incorporating the purchase price.

Total cost for Q^*

$$\begin{aligned} TC(Q^*) &= \left(\frac{Q^*}{2} \times C_c \right) + \left(\frac{D}{Q^*} \times C_o \right) + D \times C_u \\ &= \frac{200}{2} \times 135 + \frac{2500}{200} \times 1080 + 2500 \times 750 \\ &= ₹1,902,000 \end{aligned}$$

Total cost for Q_1

$$\begin{aligned} TC(Q_1) &= \left(\frac{Q_1}{2} \times C_c \right) + \left(\frac{D}{Q_1} \times C_o \right) + D \times C_u \\ &= \frac{400}{2} \times 135 + \frac{2500}{400} \times 1080 + 2500 \times 735 \\ &= ₹1,871,250 \end{aligned}$$

Total cost for Q_2

$$\begin{aligned} TC(Q_2) &= \left(\frac{Q_2}{2} \times C_c \right) + \left(\frac{D}{Q_2} \times C_o \right) + D \times C_u \\ &= \frac{800}{2} \times 135 + \frac{2500}{800} \times 1080 + 2500 \times 727.5 \\ &= ₹1,875,125 \end{aligned}$$

Since $TC(Q_1)$ is the lowest among the alternatives, the firm can make use of the discount offered and reset the order quantity to 400 boxes.

is to incorporate shortage cost in the model. The other is to use the concept of service level. **Service level** is the desired probability of not running out of stock between the time an order is placed with a supplier and the order is received. For example, if an organization wants to operate with a 90 per cent service level, it means that in the long run, the organization is able to meet the demand on 90 per cent of all occasions.

Service level is the desired probability of not running out of stock between the time an order is placed with a supplier and the order is received.

Consider an item with uncertain demand. When the item is available in stock, there is no risk of shortage. However, when an order is placed, the demand can be met only with the available stock until the stock is replenished by an order. Hence, if an organization is protected from the risks of shortage during the supply lead time, then inventory planning is better. The concept of service levels seeks to achieve this objective. Let us understand the concept of service level through a simple example.

Consider an item with uncertain demand that has a lead time of one week. Based on observed consumption patterns one can estimate the weekly demand. Table 9.3 has illustrative data for the item. The last two columns show the cumulative frequency of demand exceeding the lower class. For example, the weekly demand exceeded 30 units during 98.25 per cent of the observed occasions and exceeded 180 units during 18.42 per cent of the occasions. The information in the table can also be plotted as a frequency ogave (see Figure 9.6).

From the data, one can compute the average demand during lead time to be 143 units. If the organization stocks 143 units at

NOTES

TABLE 9.3 THE SERVICE LEVEL CONCEPT: AN ILLUSTRATION

Demand During Lead Time	Frequency of Occurrence	Demand Exceeding Lower Class	
		Cumulative Frequency	Cumulative Percentage
0–30	2	114	100.00%
31–60	5	112	98.25%
61–90	11	107	93.86%
91–120	20	96	84.21%
121–150	25	76	66.67%
151–180	30	51	44.74%
181–210	13	21	18.42%
211–240	5	8	7.02%
241–270	2	3	2.63%
271–300	1	1	0.88%

NOTES

the time of placing the order with the supplier, then in the long run, it will be able to meet the demand during lead time only on 50 per cent of the occasions. In other words, the organization has a 50 per cent service level if it operates with this inventory level. Clearly, this will be an undesirable situation and managers would want to offer better service. From the data, we can compute the number of units to be stocked to offer 90 per cent and 95 per cent service levels to be 203 and 224 units, respectively. By stocking 60 more units (203–143), the organization can improve the service level from 50 per cent to 90 per cent. Similarly, by investing an additional 21 units, it can further increase the service level to 95 per cent. This additional stock is known as “safety stock” because it provides the needed safety to an organization to handle uncertain demand during lead time.

The concept of service levels permits managers to sidestep problems related to estimating shortage cost per unit. Using this concept, it is possible for managers to relate the amount of additional investment required and the improvement in service and strike appropriate trade-offs between additional investment and service. In the example we just discussed, a 5 per cent increase in service level from 90 per cent to 95 per cent entails an additional investment of 21 units. Depending on the unit cost of the item in question, the cost of carrying and the competitive scenario, managers can strike a better trade-off in their choice towards the appropriate level of service that needs to be offered.

The concept of service level permits managers to sidestep problems related to estimating shortage cost per unit.

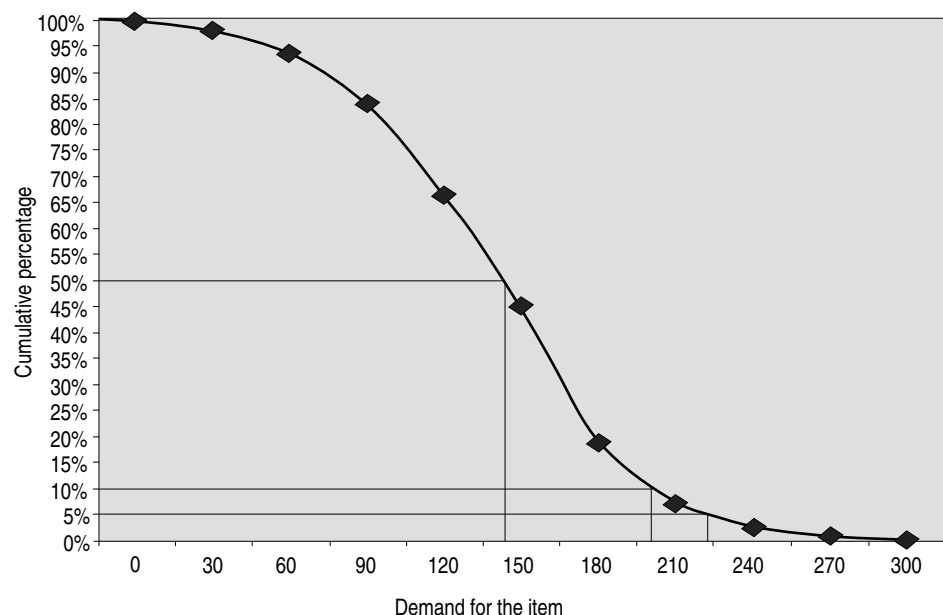


FIGURE 9.6 Frequency ogive of weekly demand

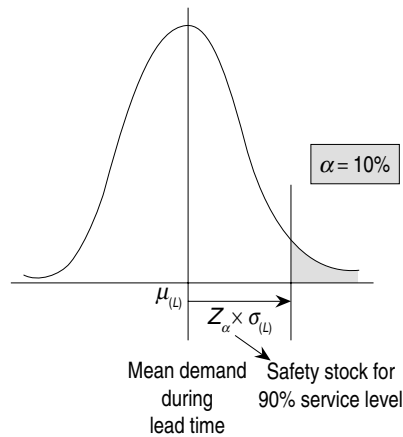


FIGURE 9.7 Illustration of service level and the safety stock concept

These computations can be generalized by using a few notations and replacing the empirical distribution for demand by a theoretical distribution such as normal distribution.

Let the demand during lead time follow a normal distribution with:

$\mu_{(L)}$, the mean demand during lead time and
 $\sigma_{(L)}$, the standard deviation of demand during lead time

Let $(1 - \alpha)$ denote the desired service level, where α denotes the probability of a stockout.

Z_α is the standard normal variate corresponding to an area of $(1 - \alpha)$ covered on the left side of the normal curve.

An expression for safety stock (SS) is given by $SS = Z_\alpha \times \sigma_{(L)}$. Figure 9.7 represents the principle of safety stock and service level.

Due to the introduction of safety stock, the total cost of the plan will change. The total cost for an order quantity equal to Q^* is given by:

Total cost of the plan = Cost of ordering + Cost of carrying cyclic inventory + Cost of carrying safety stock

$$TC(Q^*) = \left(\frac{D}{Q^*} \times C_0 \right) + \left(\frac{Q^*}{2} \times C_c \right) + (Z_\alpha \times \sigma_{(L)}) \times C_c \quad (9.8)$$

If the organization places an order when the stock on hand is equal to the sum of mean demand during lead time and safety stock, then it will be able to provide a service level of $(1 - \alpha)$ in the long run. Therefore, an expression for reorder point (ROP) for systems with uncertain demand is given by:

$$ROP = \mu_{(L)} + Z_\alpha \times \sigma_{(L)} \quad (9.9)$$

NOTES

9.7 SUMMARY

- Every organization carries five different types of inventory. This includes *cyclic stock*, *pipeline inventory*, *safety stock*, *decoupling inventory*, and *seasonal inventory*. The purpose of carrying these types and the manner to assess the level of investment required in each of these vary.
- *Inventory planning* is done in order to minimize the total cost of the plan. The costs include the unit cost of the item for which planning is done, the cost of carrying inventory, the cost of ordering, and the cost of shortages.
- The key decisions in any inventory planning scenario is to answer the “how much” and the “when” questions. These decisions are made by balancing various costs associated with inventory.
- The *EOQ model* is useful for inventory planning in the case of multi-period deterministic demand situations. The EOQ model is robust to model parameters and could be suitably modified to incorporate some real-life situations such as quantity discounts and non-zero lead time for supply.
- *Service level* is a useful concept for modelling inventory planning in the case of stochastic demand. Safety stocks can be built commensurate to the desired service level.
- A *fixed order quantity* (Q system) or continuous review system of inventory planning and control is useful for B-class and C-class items of inventory. A popular application of the continuous review system in organizations is the two-bin system.
- A *fixed order interval* or a periodic review system (P system) is useful for planning and control of high-value or A-class items. The P system is more responsive to changes in demand patterns than the Q system.
- Selective control of inventories is achieved through alternative classification methodologies. The ABC, VED, and XYZ classifications are often used by organizations.

9.8 FORMULA REVIEW

Total cost associated with carrying inventory = $\left(\frac{Q}{2} \times C_c\right)$

Total ordering cost = $\left(\frac{D}{Q} \times C_o\right)$

Total cost of inventory, $TC(Q) = \left(\frac{Q}{2} \times C_c\right) + \left(\frac{D}{Q} \times C_o\right)$

Economic Order Quantity (EOQ), $Q^* = \sqrt{\frac{2C_oD}{C_c}}$

Optimal number of orders = $\frac{D}{Q^*}$

Time between orders = $\frac{Q^*}{D}$

For the perpetual review inventory system

Re-order Point (ROP) = $\mu_{(L)} + Z_\alpha \times \sigma_{(L)}$

Total cost, $TC(Q^*) = \left(\frac{D}{Q^*} \times C_o\right) + \left(\frac{Q^*}{2} \times C_c\right) + (Z_\alpha \times \sigma_{(L)}) \times C_c$

For the periodic review inventory system

Order up to level, $S = \mu_{(L+R)} + Z_\alpha \times \sigma_{(L+R)}$

Optimal quantity (Q) to order in a single period inventory system is given as $P(d \leq Q) \leq \frac{C_{us}}{C_{us} + C_{os}}$

9.9 REVIEW QUESTIONS

- Which of the following constitutes independent demand items?
 - Maintenance spares for machine tools in a manufacturing firm
 - Carburettors for a petrol engine manufacturer
 - Fresh linen in a five-star hotel
 - Radiography services in a multi-specialty hospital
 - Number of tyres for an automobile manufacturer
 - Replacement clutch cables for two-wheelers
 - Halogen lamps for automotive headlights
 - Number of operations management text books in a business school
 - Number of bottles of mineral water in a restaurant
 - Number of lunches for the mid-day Chennai- Mumbai flight
- How is inventory planning for independent demand items different from that for dependant demand items?
- Do organizations need to carry inventory? Why?
- What is the relationship between inventory investment and profitability in an organization?
- Consider the inventory planning problem. Write down the relevant total cost equation in each of the following situations:
 - The organization can fetch a discount on the purchase price if large quantities are ordered
 - There is a benefit in transportation costs if ordered in full truckloads
 - The supplier is willing to set up elaborate systems for easy ordering, monitoring, and follow-up of orders if the order quantity is large

- (d) There is uncertainty in demand and the organization wishes to carry safety stock
- What are the difficulties in using shortage cost measures in inventory planning models?
 - Can you identify four alternatives that organizations use as a basis for fixing the order quantity?
 - On what basis would you recommend the periodic review system of inventory control?
 - Why should organizations adopt a selective system of inventory control? If you were asked to recommend a suitable classification scheme, how would you go about the task?
 - What is the relationship between the service level and safety stock?
 - There are N locations at which an organization faces demand for an item. The demand for the item is uncertain

and hence the organization needs to carry safety stock. If it were to make the choice between centralized storage of the item and localized storage of the item in each of the location, which alternative will result in lower investment in inventory? (**Hint:** Assume that all locations have the same mean and standard deviation for demand).

- What is the effect of the cost parameters on the economic order quantity? Perform a “what if” analysis using a spreadsheet by varying these cost parameters. What did you deduce from the results?
- When is it appropriate to use the ABC classification scheme and the FSN classification scheme?
- Suppose you develop a classification scheme for selective control of inventories using ABC and VED classification as two dimensions. Relate the discussions pertaining to service level to this two-dimensional classification scheme.

9.10 PROBLEMS

- Consider an item that is ordered once per month. The daily requirement is 200 and the lead time for supply is two days. There are 25 working days in a month. The cost of ordering is ₹300 per order and the cost of carrying is ₹150 per unit per year.
 - Draw a sketch showing the cyclic and pipeline inventory in the system
 - What is the reorder point for this item?
 - What will the cost of this plan be?
- An auto-component manufacturer requires a certain steel forging in large quantities. The annual requirement is 40,000 pieces, each costing ₹450. The ordering cost is ₹600 per order and the carrying cost is ₹100 per unit per year.
 - What is the optimal order quantity?
 - How frequently should the manufacturer place the order with the supplier?
 - Compute the total ordering cost and total carrying cost. Do you notice anything?
- Consider Problem 2. Perform a “what if” analysis for the following scenarios and compute the total cost of the plan for ordering an optimal order quantity:
 - The actual demand is –12 per cent, –8 per cent, –4 per cent, 0 per cent, 4 per cent, 8 per cent, and 12 per cent of the demand mentioned in the previous problem.
 - The actual ordering cost has a similar variation as that of the demand
 - The actual carrying cost has a similar variation as that of the demand

Plot the total cost in each of the three scenarios and make your observations.
- Excel Toys, a manufacturer of plastic toys for children, requires constant supply of high-density polyethylene

(HDPE) in the resin form. HDPE is available in the resin form and one tonne costs approximately ₹2,000. The manufacturer runs his factory for 250 days a year and the daily consumption rate for HDPE is four tonnes. The suppliers normally take a week to replenish an order. The ordering cost is ₹2,000 per order and the carrying cost is 20 per cent.

- Estimate the cyclic and pipeline inventory in the system at Excel Toys.
 - What is the total cost of the plan?
 - Suppose if the supplier insists on a minimum order quantity of 75 tonnes, what is your recommendation to Excel? Will the recommendation change if the minimum order quantity is 150 tonnes?
 - Suppose Excel launches some improvement initiatives in the company and brings down the ordering cost by 20 per cent. Recompute the economic order quantity.
 - If Excel spent a sum of ₹10,000 towards improvements in the system, when will the improvement efforts bring payback? (**Hint:** Estimate the savings in costs due to reduced ordering cost)
- Consider Example 9.2 in this chapter. If the company continues with the same plan even when there is an increase in the demand by 10 per cent, what will the impact be? What do you infer from the result?
 - A manufacturing organization has been consuming a certain item in large quantities and is currently procuring the item from Supplier A. The price offered by the supplier is ₹400 per piece. The ordering cost is ₹2,800 per order and the carrying cost is ₹350. The annual demand for the item is 10,000. The supplier is currently not offering any discount. However, another supplier, Supplier B, is willing to offer the following discount structure:

Up to an order size of 999 = No discount

For an order size of 1,000 –1,999 = 2 per cent discount in price

For an order size of 2,000 and above = 5 per cent discount in price

Switching over to this supplier means incurring an initial cost of ₹15,000. This cost is primarily to set-up new communication systems with the new supplier. What should the company do in the light of the new offer?

7. Siemens India Limited manufactures switchgears at their Kalwa plant. One of the key components, *Kombishraube* (contacts), is manufactured in-house. If the machine producing these contacts manufactures at the rate of 800 pieces per hour and the demand rate is 450 pieces per hour, what is the economic run length for this production process? How often should they set up the machine if the factory works for 250 days in a year on a single shift (of 8 hours) basis? The cost of setting up the machine is ₹3,000 per set-up and the cost of carrying the components is ₹400 per unit per year.

8. Consider an item with the following demand attributes:

Mean daily demand = 20; Standard deviation of daily demand = 8.

Compute the safety stock required for a continuous review system in each of the situations given in Table 9.4:

What do you infer from these calculations?

TABLE 9.4 SITUATIONS FOR PROBLEM 8

Lead Time	Service Level
1 week	90%, 95%, 99%
10 days	90%, 95%, 99%
2 weeks	90%, 95%, 99%
3 weeks	90%, 95%, 99%

9. For Problem 8, compute the order up to levels and safety stock for a periodic review system with a two-week review period.
10. The weekly requirement of furnace oil in a foundry has a mean of 3,000 litres and a standard deviation of 900 litres. A litre of furnace oil costs ₹400 and the carrying costs are estimated at 25 per cent. If the foundry uses a continuous review system of inventory control with a desired service level of 90 per cent, then

- (a) What will the reorder point and safety stock for furnace oil be if the lead time for supply is 1 week?
- (b) What will the service level be if the reorder point is changed to 3,100 litres?
- (c) If the reorder point is left unchanged when there is a 5 per cent increase in the mean demand, how will it affect the service level?
- (d) If the foundry estimates that the monetary value of the benefit that it may obtain is ₹3,000 when increasing the service level from 90 per cent to 95 per cent, what is the appropriate service level for the item?

11. The demand for springs in the coil form for an auto component manufacturer is continuous and uncertain. The manufacturer places orders for the springs with a local supplier who takes two weeks to deliver. The cost of the spring is ₹400. The weekly demand for the springs has a mean of 500 and a standard deviation of 100. The cost of ordering is ₹2,000 per order and the cost of carrying is 18 per cent. The supplier specifies a minimum order quantity of 2,000. Assume a service level of 90 per cent.

- (a) Is it feasible to operate an EOQ based system for this item?
- (b) Design periodic and continuous review systems of inventory control.
- (c) What is the total cost of the plan in each of these cases?

12. Oriental Healthcare is a multi-specialty hospital catering to a variety of illnesses connected to the heart and respiratory system. The demand for a class of medical consumable is generally random. Initially, Oriental followed a practice of ordering 200 boxes every two weeks. This practice was not found to be satisfactory. After some review, they resorted to ordering 300 boxes every two weeks. Even this practice was not found satisfactory. Recently, an examination of the stores records over a period of 10 weeks revealed the following weekly consumption pattern (Table 9.5):

The supplier of the item takes (on an average) two weeks to deliver once the order is placed. The item costs ₹450 per box. The ordering cost is ₹2,000 per order and the carrying cost is 20 per cent. What should Oriental do in the light of this information? Will they be better off with the new recommendation? Assume a service level of 90 per cent.

13. A pharmacy needs to decide on monthly procurement quantity of a critical yet a highly perishable drug that has a shelf-life of just one month. The unit price of drug is ₹7,000 and its cost is ₹6,000. As per the historical data, the monthly demand for the drug is random with a certain distribution. The details are given in Table 9.6. Since the drug is an emergency drug, the associated hospital offered to reimburse ₹3,000 for

TABLE 9.5 WEEKLY CONSUMPTION PATTERN

Week no.	Consumption (Units)
1	120
2	109
3	142
4	140
5	110
6	145
7	144
8	120
9	160
10	110

each unit of drug left unsold. At the same time, the store would be penalized ₹2,000 per unit by the hospital for any unmet demand.

TABLE 9.6 HISTORICAL DATA ON DEMAND DISTRIBUTION

Demand	22	24	26	28	30	32	34	36
Proba- bility	0.05	0.10	0.10	0.15	0.20	0.20	0.12	0.08

What should be the monthly order size the pharmacy must plan for?

14. In problem number 13 consider the following. The hospital is contemplating a change in the contractual terms with the pharmacy. Suppose the hospital is willing to make an offer of either withdrawing the reimbursement or penalty. What should the pharmacy owner do now?
15. A manufacturer of electric ovens has made an estimate (shown in Table 9.7) for annual consumption and unit cost of the components that are used for manufacture. Perform an ABC analysis and advise how the manufacturer should plan and control for inventory.

TABLE 9.7 ANNUAL CONSUMPTION AND UNIT COST OF COMPONENTS

Item No.	Item Name	Unit of Measurement	Annual Consumption	Unit Value (₹)
1	Copper wire/plate	tonnes	200	7,000.00
2	Steel plate	tonnes	400	5,000.00
3	Ceramic insulators	pieces	40,000	30.00
4	Brass fasteners	pieces	60,000	1.00
5	Steel fasteners	pieces	40,000	1.25
6	Ceramic bushes	pieces	20,000	10.00
7	Paint	litre	400	200.00
8	Electrical cable	metre	20,000	3.00
9	Power sockets	pieces	10,000	7.00
10	Plugs	pieces	10,000	8.00
11	Brass washers	pieces	80,000	0.50
12	Plastic components	pieces	60,000	1.25

9.11 MINI PROJECTS

1. Identify a sector of industry of your choice and select three companies operating in that sector. Analyse their annual report and identify portions of the annual report that have a direct impact on inventory-related issues.
 - (a) Prepare a report highlighting your understanding of the performance of the companies with respect to inventory planning and control. Can you identify some possible explanations for your observations pertaining to the inventory data?
 - (b) Among the three, which one is better in managing their inventory? How do these firms compare with the industry average?
 - (c) Can you make some comparisons with similar firms elsewhere (outside the country)?
 - (d) What are your key inferences from this exercise?
2. Study the reports produced by ICICI (Financial Performance of Companies) and the Centre for Monitoring Indian Economy on performance of corporate sector.
 - (a) Identify the inventory investment across different sectors of our industry. Do you see any significant trends in these movements?
 - (b) How has the performance been in the last five years? What factors do you think might have contributed to this?
 - (c) Which sectors have performed better and which have performed the worst? Can you identify some reasons for these?
3. Analyse the multiplier effect of inventory using the Du Pont equation for return on investment for a pair of close competitors. What are your key inferences from this exercise?

9.12 CASE STUDY

MML Limited

The Background

MML Limited was incorporated as a private limited company and later converted into a public limited company in 1988. Though the company started with the manufacture of soaps and perfumes, it has added a variety of products to diversify its product line. Today, it is a conglomerate with businesses in soaps, perfumes, plastics, petrochemicals, paints, industrial electronics, and agro-business. Their unit at Hosur manufactures electronic components. The inventory position has been deteriorating over the last few years. The year-end inventory position, as revealed by the annual reports of the conglomerate, showed an increase from ₹1,230 million as on 30 June 2004 to ₹2,490 million as on 31 March 2007.

The conglomerate has been operating in a highly centralized fashion. All major planning activities were done at the corporate office. However, many inputs for this decision making were made available by all the operating units. The headquarters sent the tentative monthly production plan two months prior to the commencement of production. The final production plan arrived 5 days prior to start of the production. It was not unusual for the corporate office to change the production plans while production was progressing. This was communicated through telephone, telex, fax, or mail. Once the production plan was made, the requirement of each item was also computed using the bill of materials. Table 9.8 presents the tentative and final plans, and the actual production of the Hosur unit for 2005–06 and Table 9.9 presents the consumption pattern of the “A-class” item.

The Existing Buying Process

The company has been following a centralized ordering system, which was operated from its headquarters. The advantages of this system according to the organization were:

1. Better control over inventory-related costs.

2. Since the corporate office compiled the net requirements of all operating units, the order quantities for the suppliers were large, sometimes inducing them to offer generous quantity discounts.
3. In case of sudden shortages of an item at a location, the corporate office could advise another location to transfer stocks because it knew the safety stocks at each location.

The corporate office advises the purchase departments at the units on the following:

1. What to order?
2. How much to order?
3. When to order?
4. On whom to place the order?
5. What should be the delivery date?
6. What will be the price?

On receipt of this, the purchase department at the operating units took the responsibility for the procurement of items on the basis of the advice and subsequently made the payment to suppliers.

For the corporate office to take decisions on these items, it needed the following:

1. Stock levels of each item under inventory management.
2. The production plan for the following month.
3. The maximum and the minimum stock levels to be maintained for each item.
4. The previous month's consumption.

The operating unit provided the information pertaining to Item 1. Items 3 and 4 were policy decisions made at the corporate office based on historical data and future plans and Item 2 was market-related information that was forecast.

TABLE 9.8 TENTATIVE PLAN, FINAL PLAN, AND ACTUAL PRODUCTION FOR SOME PRODUCTS

Month	Production	Product A	Product B	Product C	Product D	Product E	Product F
Apr 2005	Tentative	6,000	2,500	19,000	19,000	33,500	2,500
	Final	6,500	2,500	17,000	15,000	30,000	5,000
	Actual	7,370	2,427	14,145	13,249	24,206	2,370
May 2005	Tentative	6,500	2,500	21,000	NA	35,000	5,000
	Final	6,500	2,000	14,500	4,000	38,000	5,000
	Actual	5,438	2,078	16,897	10,177	35,899	4,840
Jun 2005	Tentative	6,500	2,500	21,000	NA	35,000	5,000
	Final	6,500	2,000	14,500	10,500	35,000	7,500
	Actual	5,882	2,123	13,145	14,671	35,936	9,230
Jul 2005	Tentative	6,000	2,500	13,500	14,000	35,000	5,000
	Final	5,000	2,500	24,500	12,000	35,000	5,000
	Actual	9,137	2,661	28,287	11,960	34,567	2,491
Aug 2005	Tentative	4,000	2,500	6,000	16,000	33,500	5,000
	Final	6,000	2,500	5,000	12,500	30,000	5,000
	Actual	9,170	2,688	7,883	13,631	31,984	2,340
Sep 2005	Tentative	6,000	3,000	9,500	15,000	35,000	5,000
	Final	8,000	4,000	8,000	12,000	36,000	9,000
	Actual	7,042	4,018	8,262	15,825	31,281	3,933
Oct 2005	Tentative	4,000	4,000	7,500	10,000	36,000	2,500
	Final	4,000	4,000	7,500	19,000	35,000	2,500
	Actual	3,060	4,320	13,246	18,111	36,204	872
Nov 2005	Tentative	3,000	4,000	10,000	12,000	31,500	5,000
	Final	3,500	3,500	14,500	10,000	38,000	5,000
	Actual	7,351	3,448	16,406	12,006	39,679	3,392
Dec 2005	Tentative	7,500	2,000	7,000	13,000	38,000	5,000
	Final	7,500	3,500	12,000	13,000	40,000	5,000
	Actual	8,938	2,214	12,828	14,320	37,762	8,330
Jan 2006	Tentative	5,000	2,500	6,500	20,000	37,000	5,000
	Final	NA	2,500	2,500	33,500	38,000	2,500
	Actual	1,634	2,983	4,149	25,376	34,133	5,040
Feb 2006	Tentative	7,500	2,500	11,000	19,500	30,000	2,500
	Final	4,500	2,500	4,000	22,945	36,000	NA
	Actual	3,504	2,660	4,032	23,790	35,978	2,301
Mar 2006	Tentative	7,000	3,000	6,000	20,000	32,500	5,000
	Final	4,000	3,000	6,000	27,000	36,000	5,000
	Actual	2,004	3,101	6,079	26,138	39,393	5,070

The Hosur Unit

The unit manufactures electronic components that form part of the control devices of a variety of equipment. Basically, there are minor variations in the basic configuration, which result in 13 different products as they roll out of the assembly. The variations are mainly due to the differences in the rating and the number of various electronic components used to build the product and the variations in the electronic circuit design.

The Purchase Department

A senior manager (purchase) assisted by a manager (purchase) headed the purchase department of the Hosur unit. There were four buyers attached to the department. The buyers were given responsibility to procure materials of specific groups of materials. There was one buyer for semiconductor components, another for raw materials and a third buyer for imported components. Another buyer was looking after consumables, office equipment, and capital equipment. Although imported components formed a small percentage of the total purchase, the procedure and modalities were different, necessitating an exclusive buyer to handle them. Two clerical staff undertook the responsibility for all the typing, data preparation and entry

into the computer, preparation of purchase orders, and other such support activities. The purchase department was provided with a desktop computer, which was connected to a local area network.

The senior manager was interacting with the corporate office concerning matters such as source development, supplier rating, and preparation of purchase budgets. In addition, he was liaising with other departments such as stores, quality control, finance, production planning, and design. This was an ongoing activity to help take many decisions on issues such as value engineering, make or buy, new material development, import substitution, etc.

Recently, there was an exercise carried out by the purchase personnel to collect data useful for making certain policy decisions. On analysing the past three years' records it was found that on an average, 450 purchase orders were sent out per year. An ABC analysis was carried out and the average lead time taken by the vendors for the supply of such items were computed. Table 9.9 presents the relevant data on the unit price, lead time, and consumption patterns of "A-class" items. It was also found that the currently the average investment in inventory is to the tune of ₹35 million.

TABLE 9.9 CONSUMPTION PATTERN FOR A-CLASS ITEMS

	Item 1	Item 2	Item 3	Item 4	Item 5	Item 6	Item 7	Item 8	Item 9	Item 10
Apr 2005	11,058	2,180	—	3,499	2,675	131	4,608	3,024	743	2,345
May 2005	12,899	2,854	—	5,931	5,484	287	4,582	1,768	537	2,867
Jun 2005	12,595	2,937	—	4,474	2,627	270	9,204	—	891	1,489
Jul 2005	15,532	3,071	28	5,595	2,445	311	9,727	4,563	522	3,187
Aug 2005	12,786	1,585	4,253	3,606	2,044	148	6,077	2,078	719	2,150
Sep 2005	13,021	1,390	12,683	3,505	2,071	232	8,016	677	834	2,722
Oct 2005	14,260	2,433	7,153	4,435	2,737	314	10,104	2,989	1,238	1,867
Nov 2005	14,864	3,233	11,160	5,016	2,567	268	10,220	1,754	947	2,439
Dec 2005	15,067	2,868	15,621	4,349	2,702	223	6,973	2,456	—	1,196
Jan 2006	12,914	2,970	19,131	3,404	3,590	306	10,795	2,278	459	2,380
Feb 2006	13,365	2,202	921	3,656	3,199	200	6,679	3,219	855	2,100
Mar 2006	15,338	2,234	2,705	3,812	3,051	276	10,504	2,345	1,167	1,939
Unit cost (₹)	98.00	395.00	154.00	216.00	271.00	3,180.80	95.80	160.00	340.00	112.50
Lead time (days)*	30	38	30	45	60	50	20	15	37	30

Notes:

*The lead time is an average figure based on six months' data.

- The consumption details are expressed in units of items. The units of measurement are different for the items. For example, the unit of measurement for Item 3 is boxes (each box contains 250 items) whereas the unit of measurement for Item 8 is kg.
- Item 6 is sold by all the suppliers in integral units of 250 only. This has been attributed to the fact that once the seal is broken it should be used within three days.
- Items 7 and 10 are supplied by the same supplier, who is the only available domestic source.

TABLE 9.10 SOME COST DETAILS EXTRACTED FROM THE STUDY GROUP REPORT AND APPORTIONMENT DETAILS

No	Item	Expenditure*	Purchase and Stores**	Marketing**	Production**	Design**	Finance**	General Administration and Others**
1	Stationery expenses	1,854,000	9	20	17	15	15	24
2	Telephone bills, telex, etc.	1,095,600	16	25	9	7	10	33
3	Taxi, auto (travel expenditure)	1,145,000	8	45	4	3	12	28
4	Postage, etc.	330,000	9	18	—	5	14	54
5	Courier expenses	380,000	10	20	3	—	5	62
6	Freight, demurrage	655,400	40	—	—	—	—	60
7	Insurance premium for stores	742,500	100	—	—	—	—	—
8	Handling costs: fork-lift truck maintenance (depreciation)	765,000	35	—	54	—	—	11
9	Batteries replacement	1,224,500	40	—	50	—	—	10
10	Maintenance of tractors and trailers	149,000	75	5	—	—	20	—
11	Air conditioning	645,000	10	2	7	1	3	77
12	Utilities	2,235,400	7	4	41	3	4	41

Notes: * All values are yearly expenditure in rupees; ** All numbers are expressed as percentages. For limited purposes many departments have been combined with general administration.

Inward Goods Stores and Inspection Section

The Hosur unit had an inward goods stores and inspection section that performed the task of receiving consignments of supplies, conducting inspections, and initiating the necessary follow-up action. The section employed two load/unload workers on a temporary basis each for a monthly wage of ₹1,200. As soon as a consignment arrived, a goods received note (GRN) was prepared after inspecting the materials for any damages. If there were damages, the insurance details were verified and the consignment sent to the claims section. The claims section liaised with either the supplier or the insurance company for claims.

Those consignments that were free of damage were set aside for marking codes and placing at the desired location. Simultaneously, the details were filled in the GRN. The GRN details were used to update the stock position and copies of it were sent to other departments such as finance and purchase.

The stores employed a clerk, an inspector and an assistant manager. In addition one forklift truck operator was employed.

Recent Developments

Recently, the company took a decision to decentralize most of the decision-making processes. Under this scheme, each unit was focused on certain areas of the business. The individual unit heads entered into an agreement with the corporate office regarding the target for sales turnover, profitability, and a few

other broad parameters. The unit took all operational decisions with the required autonomy.

A study group was constituted at the Hosur unit to analyse the various expenditures incurred at the plant level. The unit head initiated this as soon as decentralization was put into effect. Until recently, such statistics was not compiled and analysed at the unit level. The units would merely send a host of weekly and monthly reports to the corporate office. The corporate staff performed the task of analysing the data and informing the various unit heads regarding the variances from budgetary provisions. A portion of the cost data gathered so far by the study group is given in Table 9.10. Certain service items such as forklift truck maintenance, insurance premiums freight, and demurrage were allocated to specific departments using the basis developed for costing purposes. Table 9.10 also shows the apportionment details for the items.

QUESTIONS FOR DISCUSSION

1. What are the reasons for high levels of inventory at MML Ltd.? Can you offer suggestions for bringing down the level of inventory?
2. Based on the data provided, compute the cost of carrying inventory at the Hosur unit of MML Ltd.
3. Design an appropriate inventory control system for the A-class items identified by MML Ltd.

9.13 MULTIPLE-CHOICE QUESTIONS

- 9-1.** In manufacturing organizations, finished goods and spare parts typically belong to the category of:
A. Dependent demand
B. Independent demand
C. Continuous demand
D. Uncertainty of demand
- 9-2.** How many types of Inventory are normally found in organizations? What are they?
A. One type of inventory: Physical inventory
B. Two types of inventory: First in First out inventory, Last in First out inventory
C. Three types of inventory: Discriminant inventory, Transit inventory, Buffer inventory
D. Five types of inventory: Seasonal inventory, Decoupling inventory, Cyclic inventory, Pipeline inventory and safety stock
- 9-3.** What are the several costs associated with inventory planning and control?
A. Inventory-carrying Cost, Cost of Ordering, Cost of Shortages
B. Storing Cost, Opportunity Cost, Purchase Cost
C. Transportation Cost, Deterioration Cost, Obsolescence Cost
D. Stock Out Cost, Holding Cost, Shrinkage Cost
- 9-4.** When the total cost is minimum, we obtain the most economic?
A. Resource utilization
B. order quantity
C. And effective process flow
D. None of the above
- 9-5.** When the item is available in stock, there is:
A. No risk of shortage
B. Risk of maintaining the stock
C. Production line can be shut down
D. None of the above
- 9-6.** What are the Inventory Control Systems?
A. Perpetual, Periodic
B. Inventory Management, Technology
C. Continuous Review, Periodic Review
D. Barcode, Radio Frequency
- 9-7.** Selective control of Inventories are:
A. Re-ordering level, Maximum level and Economic Order Quantity
B. HML Analysis, SDE Analysis and Golf Analysis
C. SOS Analysis, FSN Analysis, XYZ Analysis
D. ABC Classification, XYZ Classification, VED Classification and Combination of more than one classification
- 9-8.** In a single-period demand the unfilled demand cannot be:
A. Pre-ordered to the next period
B. Back-ordered to the next period
C. Post-ordered to the next period
D. None of the above
- 9-9.** What is the problem in inventory planning and control?
A. Opportunities for better inventory control and management often exist outside of the materials management function in organization
B. Low Product Turnover Failure to track demand can lead to a lower than expected product turnover due to low demand
C. Excess Inventory
D. Failure to keep track of stock
- 9-10.** Despite their simplicity and initial success, which installation faces several problems after implementations:
A. MRP **B.** SAP
C. ERP **D.** DTP

Answer Keys: 9-1. B 9-2. D 9-3. A 9-4. B 9-5. A 9-6. C 9-7. D 9-8. B 9-9. A 9-10. A

10

CHAPTER

INVENTORY CLASSIFICATION AND PLANNING SYSTEMS

10.1	Introduction to Inventory Planning Systems
10.1.1	ABC Analysis
10.1.2	Record Accuracy
10.1.3	Cycle Counting
10.1.4	Control of Service Inventories
10.1.5	Inventory Models
10.1.6	Inventory Models for Independent Demand
10.2	Inventory Control Systems
10.2.1	The Continuous Review (Q) System
10.2.2	The Periodic Review (P) System
10.2.3	Issues in the P and Q Systems of Inventory Control
10.3	Selective Control of Inventory
10.3.1	ABC Classification
10.3.2	Other Classification Schemes for Selective Control
10.4	Inventory Planning for Single-Period Demand
10.5	Other Issues in Inventory Planning and Control
10.6	Summary
10.7	Formula Review
10.8	Review Questions
10.9	Problems
10.10	Mini Project
10.11	Case Study
10.12	Multiple-choice Questions

INVENTORY MANAGEMENT IN A CONSUMER PRODUCTS COMPANY

N O T E S

A consumer products company dealing in cosmetics and other personal-care products was exploring ways to reduce inventory levels across their outbound supply chain and improve inventory record accuracy at their storage points. The company had a supply chain network of three factories with bonded stock rooms (BSRs) attached for dispatch to the depots and 35 depots for servicing distributors. Goods moved from the factory to the BSRs. The BSRs dispatched stocks to one centralized depot. Other depots received stocks from this depot and sold them to distributors.

These depots were holding high levels of inventory of old/withdrawn stocks and damaged stocks for a long time (over three months). The total average inventory holding at the BSRs was 8.2 weeks of sales and at the depots was 6.5 weeks of sales. There were several reasons for high levels of inventory. Some of them are discussed here.

Sales and dispatch forecasts were not in line with actual sales. Furthermore, there was no process to periodically review and refine the annual forecasts utilizing market feedback. Stocking across all points in the distribution chain was driven by a push-oriented system that did not have any provision for factoring in market requirements. Actual safety stocks maintained at depots were significantly higher than the target safety stocks agreed on at the beginning of the operating year. No system was in place to monitor and correct this practice. There was also a high level of old/damaged/slow-moving stocks. Dead stock was allowed to accumulate in the system mainly because there was an absence of visibility into inventory details across stocking points. The process to monitor and act on dead stock was not adhered to and records of slow-moving/old/damaged stocks were not maintained methodically at the stocking points.

A study was conducted focusing on the inventory-related issues at the BSRs and depots. This included inventory holding as a proportion of sales, practices employed for tracking goods in the warehouse, and the proportion of fast- and slow-moving stocks to the total inventory. The study also looked at the inventory planning process pertaining to forecast accuracy, the process of reviewing and revising forecasts, the level of safety stock at each location, combined with the process to review and reset the same.

An IT solution was implemented for computing the forecast using consolidated orders, with factoring for promotions and seasonality. The IT solution also enabled the organization to calculate safety stock levels based on the number of weeks of sales target. Demand planning and forecasting were made a periodic

NOTES

activity using the IT solution to align forecasting with market orders and actual sales. The process of setting safety stocks at depots was made periodic and dynamic, based on updated sales data. Furthermore, norms were set to act on damaged/old and other dead stocks. Clear steps were laid down regarding the liquidation or destruction of these stocks. An accountability chain was set up in the organization to monitor and authorize activities in this regard, based on the visibility provided by the IT solution.

The overall benefit of the exercise was that the organization was able to ensure availability of fresh stocks in the market. This was achieved mainly by reducing inventory levels across the chain and also through better stock management at the depots. The company achieved a stock-level reduction from 8.2 weeks to 5.5 weeks at the BSRs and from 6.5 weeks to 4 weeks at the depots. Transparency of saleable and damaged stocks quantities across the supply chain resulted in more accurate demand planning, stock allocation, and production.

Source: K. Ravichandran and Debjyoti Paul, “Best Practices in Inventory Management,” <http://forumcentral.sify.com/athena/login/casestudyinventory.pdf>. Last accessed on 15 December 2008.

10.1 INTRODUCTION TO INVENTORY PLANNING SYSTEMS

Operations managers establish systems for managing inventory. In this section, we briefly examine two ingredients of such systems: (1) how inventory items can be classified (called *ABC analysis*) and (2) how accurate inventory records can be maintained. We will then look at inventory control in the service sector.

10.1.1 ABC Analysis

ABC analysis divides on-hand inventory into three classifications on the basis of annual dollar volume. ABC analysis is an inventory application of what is known as the *Pareto principle* (named after Vilfredo Pareto, a 19th-century Italian economist). The Pareto principle states that there are a “critical few and trivial many.” The idea is to establish inventory policies that focus resources on the *few critical* inventory parts and not the many trivial ones. It is not realistic to monitor inexpensive items with the same intensity as very expensive items.

ABC analysis

A method for dividing on-hand inventory into three classifications based on annual dollar volume

To determine annual dollar volume for ABC analysis, we measure the *annual demand* of each inventory item times the *cost per unit*. *Class A* items are those on which the annual dollar volume

is high. Although such items may represent only about 15% of the total inventory items, they represent 70% to 80% of the total dollar usage. *Class B* items are those inventory items of medium annual dollar volume.

These items may represent about 30% of inventory items and 15% to 25% of the total value. Those with low annual dollar volume are *Class C*, which may represent only 5% of the annual dollar volume but about 55% of the total inventory items.

Graphically, the inventory of many organizations would appear as presented in Figure 10.1.

Criteria other than annual dollar volume can determine item classification. For instance, high shortage or holding cost, anticipated engineering changes, delivery problems, or quality problems may dictate upgrading items to a higher classification. The advantage of dividing inventory items into classes allows policies and controls to be established for each class.

Policies that may be based on ABC analysis include the following:

1. Purchasing resources expended on supplier development should be much higher for individual A items than for C items.
2. A items, as opposed to B and C items, should have tighter physical inventory control; perhaps they belong in a more secure area, and perhaps the accuracy of inventory records for A items should be verified more frequently.
3. Forecasting A items may warrant more care than forecasting other items.

Better forecasting, physical control, supplier reliability, and an ultimate reduction in inventory can all result from classification systems such as ABC analysis.

10.1.2 Record Accuracy

Record accuracy is a prerequisite to inventory management, production scheduling, and, ultimately, sales. Accuracy can be maintained by either periodic or perpetual systems. *Periodic*

NOTES

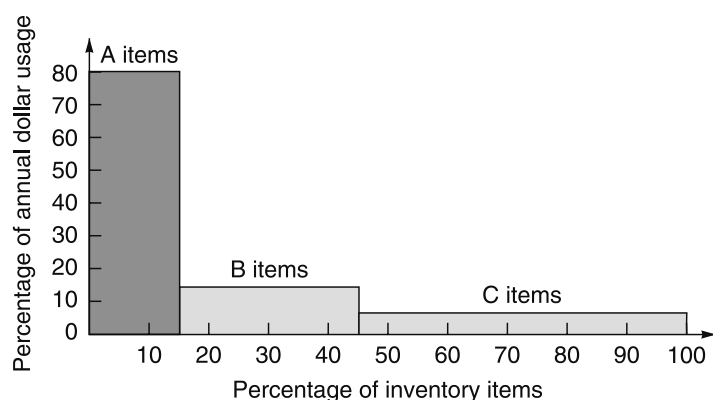


Figure 10.1 Graphic Representation of ABC Analysis

NOTES

systems require regular (periodic) checks of inventory to determine quantity on hand. Some small retailers and facilities with vendor-managed inventory (the vendor checks quantity on hand and resupplies as necessary) use these systems. However, the downside is lack of control between reviews and the necessity of carrying extra inventory to protect against shortages.

A variation of the periodic system is a *two-bin system*. In practice, a store manager sets up two containers (each with adequate inventory to cover demand during the time required to receive another order) and places an order when the first container is empty.

Alternatively, *perpetual inventory* tracks both receipts and subtractions from inventory on a continuing basis. Receipts are usually noted in the receiving department in some semiautomated way, such as via a bar-code reader, and disbursements are noted as items leave the stockroom or, in retailing establishments, at the point-of-sale (POS) cash register.

Regardless of the inventory system, record accuracy requires good incoming and outgoing record keeping as well as good security. Stockrooms will have limited access, good housekeeping, and storage areas that hold fixed amounts of inventory. In both manufacturing and retail facilities, bins, shelf space, and individual items must be stored and labelled accurately. Meaningful decisions about ordering, scheduling, and shipping, are made only when the firm knows what it has on hand.

10.1.3 Cycle Counting

Even though an organization may have made substantial efforts to record inventory accurately, these records must be verified through a continuing audit. Such audits are known as **cycle counting**. Historically, many firms performed annual physical inventories. This practice often meant shutting down the facility and having inexperienced people count parts and material. Inventory records should instead be verified via cycle counting. Cycle counting uses inventory classifications developed through ABC analysis. With cycle counting procedures, items are counted, records are verified, and inaccuracies are periodically documented. The cause of inaccuracies is then traced and appropriate remedial action taken to ensure integrity of the inventory system.

A items will be counted frequently, perhaps once a month;

B items will be counted less frequently, perhaps once a quarter; and

C items will be counted perhaps once every 6 months.



Omniceil

In this hospital, these vertically rotating storage carousels provide rapid access to hundreds of critical items and at the same time save floor space. This Omnicell inventory management carousel is also secure and has the added advantage of printing bar code labels.

Cycle counting

A continuing reconciliation of inventory with inventory records.

10.1.4 Control of Service Inventories

Although we may think of the service sector of our economy as not having inventory, that is seldom the case. Extensive inventory is held in wholesale and retail businesses, making inventory management crucial. In the food-service business, control of inventory is often the difference between success and failure. Moreover, inventory that is in transit or idle in a warehouse is lost value. Similarly, inventory damaged or stolen prior to sale is a loss. In retailing, inventory that is unaccounted for between receipt and time of sale is known as shrinkage. Shrinkage occurs from damage and theft as well as from sloppy paperwork. Inventory theft is also known as pilferage. Retail inventory loss of 1% of sales is considered good, with losses in many stores exceeding 3%. Because the impact on profitability is substantial, inventory accuracy and control are critical.

Applicable techniques include the following:

1. *Good personnel selection, training, and discipline:* These are never easy but very necessary in food-service, wholesale, and retail operations, where employees have access to directly consumable merchandise.
2. *Tight control of incoming shipments:* This task is being addressed by many firms through the use of Universal Product Code (or bar code) and radio frequency ID (RFID) systems that read every incoming shipment and automatically check tallies against purchase orders. When properly designed, these systems—where each stock keeping unit (SKU; pronounced “skew”) has its own identifier—can be very hard to defeat.
3. *Effective control of all goods leaving the facility:* This job is accomplished with bar codes, RFID tags, or magnetic strips on merchandise, and via direct observation. Direct observation can be personnel stationed at exits (as at Costco and Sam’s Club wholesale stores) and in potentially high-loss areas or can take the form of one-way mirrors and video surveillance.

Successful retail operations require very good store-level control with accurate inventory in its proper location. Major retailers lose 10% to 25% of overall profits due to poor or inaccurate inventory records.¹ (See the *OM in Action* box, “Retail’s Last 10 Yards.”)

Shrinkage

Retail inventory that is unaccounted for between receipt and sale.

Pilferage

A small amount of theft.

10.1.5 Inventory Models

We now examine a variety of inventory models and the costs associated with them.

NOTES



Robin Nelson/ZUMA Press, Inc./Alamy

Pharmaceutical distributor McKesson Corp., which is one of Arnold Palmer Hospital's main suppliers of surgical materials, makes heavy use of bar-code readers to automate inventory control. The device on the warehouse worker's arm combines a scanner, a computer, and a two-way radio to check orders. With rapid and accurate data, items are easily verified, improving inventory and shipment accuracy.

NOTES

A handheld reader can scan RFID tags, aiding control of both incoming and outgoing shipments.



Independent vs. Dependent Demand

Inventory control models assume that demand for an item is either independent of or dependent on the demand for other items. For example, the demand for refrigerators is *independent* of the demand for toaster ovens. However, the demand for toaster oven components is *dependent* on the requirements of toaster ovens.

10.1.6 Inventory Models for Independent Demand

In this section, we introduce three inventory models that address two important questions: *when to order* and *how much to order*. These *independent* demand models are:

1. Basic economic order quantity (EOQ) model
2. Production order quantity model
3. Quantity discount model

The Basic Economic Order Quantity (EOQ) Model

The **economic order quantity (EOQ)** model is one of the most commonly used inventory control techniques. This technique is relatively easy to use but is based on several assumptions:

Economic order quantity (EOQ) model

An inventory-control technique that minimizes the total of ordering and holding costs.

1. Demand for an item is known, reasonably constant, and independent of decisions for other items.
2. Lead time—that is, the time between placement and receipt of the order—is known and consistent.
3. Receipt of inventory is instantaneous and complete. In other words, the inventory from an order arrives in one batch at one time.
4. Quantity discounts are not possible.
5. The only variable costs are the cost of setting up or placing an order (setup or ordering cost) and the cost of holding or storing inventory over time (holding or carrying cost). These costs were discussed in the previous section.
6. Stockouts (shortages) can be completely avoided if orders are placed at the right time.

Production Order Quantity Model

In the previous inventory model, we assumed that the entire inventory order was received at one time. There are times, however, when the firm may receive its inventory over a period of time. Such cases require a different model, one that does not require the instantaneous-receipt assumption. This model is applicable under two situations: (1) when inventory continuously flows or builds up over a period of time after an order has been

Production order quantity model

An economic order quantity technique applied to production orders.

placed or (2) when units are produced and sold simultaneously. Under these circumstances, we take into account daily production (or inventory-flow) rate and daily demand rate.

Quantity Discount Models

Quantity discounts appear everywhere—you cannot go into a grocery store without seeing them on nearly every shelf. In fact, researchers have found that *most* companies either offer or receive quantity discounts for at least some of the products that they sell or purchase. A **quantity discount** is simply a reduced price (P) for an item when it is purchased in larger quantities.

Quantity discount

A reduced price for items purchased in large quantities.

NOTES**10.2 INVENTORY CONTROL SYSTEMS**

The previous sections provide organizations with the necessary building blocks to put inventory control systems in place. In practice, organizations employ some methods to manage and control inventory. Prominent among them include the use of continuous

EXAMPLE 10.1

The daily demand for an item is stochastic and follows the normal distribution with a mean of 100 and a standard deviation of 20. The supplier of the item takes two weeks to deliver the item from the date the order is placed. What will be the appropriate reorder point for 90 per cent and 95 per cent service level? The cost of ordering is ₹1,000 per order and the carrying cost is ₹250 per unit per year. There are 250 working days in a year. If the organization places orders in fixed quantities of 500, what will be the total cost of the plan?

Solution

Mean daily demand = 100

Number of working days in a year = 250

Total annual demand = 25,000

Lead time for supply (L) = 2 weeks = 14 days

Therefore, mean demand during LT, $\mu_{(L)} = 14 \times 100 = 1,400$

Standard deviation of daily demand = 20

Variance of daily demand = 400

Variance of the demand during LT = $14 \times 400 = 5,600$

Standard deviation of demand during LT, $\sigma_{(L)} = 74.83$

For 90 per cent service level, $\alpha = 10$ per cent and $(1 - \alpha) = 90$ per cent.

From standard normal tables (Z_α) = 1.28

Safety stock (SS) = $Z_\alpha \times \sigma_{(L)} = 1.28 \times 74.83 = 95.78 \approx 96$

Therefore, ROP = $\mu_{(L)} + Z_\alpha \times \sigma_{(L)} = 1,400 + 96 = 1,496$

Total cost of the plan,

$$\begin{aligned} TC(Q) &= \left(\frac{D}{Q^*} \times C_o \right) + \left(\frac{Q}{2} \times C_c \right) + \left((Z_\alpha \times \sigma_{(L)}) \times C_c \right) \\ &= \frac{25,000}{500} \times 1000 + \frac{500}{2} \times 250 + 96 \times 250 \\ &= ₹136,500 \end{aligned}$$

For 95 per cent service level, the corresponding values are:

$\alpha = 5$ per cent, $(1 - \alpha) = 95$ per cent and $Z_\alpha = 1.645$, SS = 123.09 ≈ 124 and ROP = $1,400 + 124 = 1,524$.

Total cost of the plan

$$= \frac{25,000}{500} \times 1000 + \frac{500}{2} \times 250 + 124 \times 250 = ₹143,500$$

review of inventory and periodic review of inventory. Let us understand the working of these two systems, and also how various decision parameters of these systems can be estimated using the building blocks provided in previous sections.

10.2.1 The Continuous Review (Q) System

Organizations widely use a continuous review system called a *two-bin* system. In operation, the available inventory is stocked in two bins, first in a smaller bin and the balance in a larger bin. As the material is consumed, the larger bin is emptied first. As soon as the larger bin is empty, an order is placed with a supplier for a predetermined quantity, Q , and until the material arrives in the stores, the smaller bin is consumed. During replenishment, the smaller bin is filled in first and the cycle continues. Can you guess what will be the capacity of the smaller bin? A little thought will help you appreciate that the capacity of the smaller bin is indeed the reorder point as per the calculations that we performed in the previous section.

Figure 10.2 shows the general working of a Q system. The inventory position in the system is continuously monitored to check if it has reached the reorder point. Once it reaches the reorder point, an order is placed for a fixed quantity of Q . Due to this design choice of ordering fixed quantities, Q systems are also known as fixed order quantity systems. After a lead time of L , the inventory arrives in the system and the physical inventory increases. In Figure 10.2, the continuous line indicates the on-hand inventory in the system. When there is a pending order, the physical inventory in the system is less but the total inventory in the system must take into consideration scheduled receipts in the future. This is shown by the dotted line of inventory movement. The two constant dotted lines in the bottom of the figure represent

The “when” decision in a Q system is answered by reorder point.

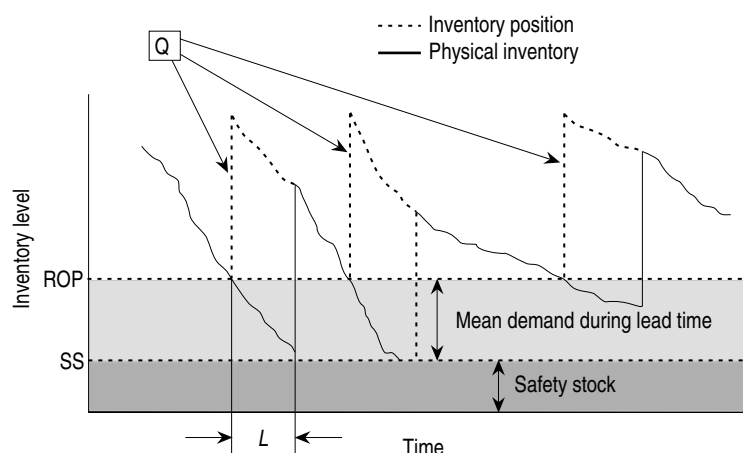


Figure 10.2 Continuous review (Q) system

the safety stock built into the system and the average demand during lead time (LT). The summation of these two quantities is the reorder point for the system.

The “when” decision is answered by ROP. We can employ the computations in the previous section to arrive at ROP. The “how much” decision is answered by our choice of Q . Several alternatives are available to fix Q .

- ⇒ One is to compute EOQ and fix Q at this value.
- ⇒ The other is to fix Q to be the minimum order quantity that the supplier insists, if EOQ value is lesser than this.
- ⇒ The third option is to have any other preferred quantity arising out of practical considerations such as full truckload requirement, quantity discounts, savings in transportation costs, economies of scale, or any other consideration.

The total cost of using a Q system can be computed using Eq. 10.1 for a chosen value of Q .

$$TC(Q^*) = \left(\frac{D}{Q^*} \times C_0 \right) + \left(\frac{Q^*}{2} \times C_c \right) + \left((Z_a \times \sigma_{(L)}) \times C_c \right) \quad (10.1)$$

10.2.2 The Periodic Review (P) System

An alternative model for inventory control, known as the periodic review system, operates differently from the Q system. In a periodic review system, the inventory level in the system is reviewed at fixed intervals of time. Therefore, these systems are also known as fixed order interval systems. At the time of review, an order is placed to replenish the inventory to a predetermined level, S . The parameter S , known as the order up to level, dictates the order quantity. If at any time R , a review is made and if the inventory position in the system is I_R , then order quantity $Q_R = S - I_R$. Figure 10.3 illustrates the working of the P system.

The two decisions “when” and “how much” are made in a different fashion compared to the Q system. The review period R determines when to order. There are several alternatives available for determining R . One way is to use the EOQ model, derive the time between orders, and use it for R . Managers tend to review the performance of departments on a periodic basis. R can be linked to such intervals. Alternatively, the MRP system used in some organizations has some time buckets and it is possible to fix R on this basis. In a petrochemical manufacturing unit in western India, the review period is linked to the value of items under inventory control. In the case of low-value items, the review is once in 60 days, and in the case of medium-value items, there is a monthly review. In the case of high-value items, there is a fortnightly review of inventory and ordering decisions are taken.

In a periodic review system, the inventory level in the system is reviewed at fixed intervals of time.

NOTES

NOTES

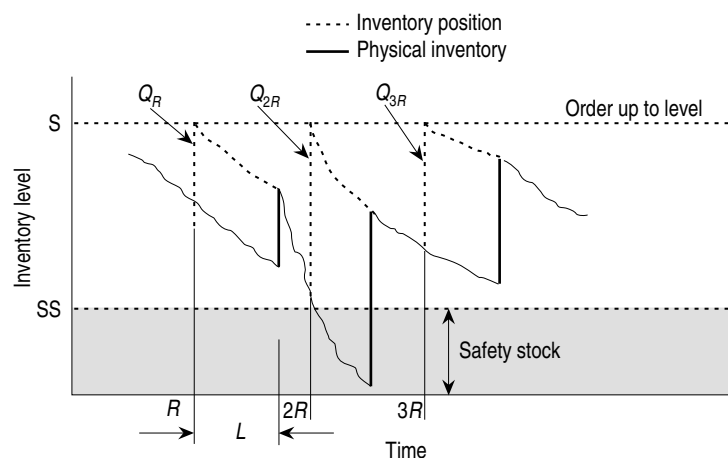


Figure 10.3 Periodic review (P) system

The order-up-to-level S determines how much to order in a P system.

The order up to level S determines how much to order. Hence, it is important to derive an appropriate expression for S . Let us consider an instant of time t , when a review is taken and an order is placed with the supplier. The organization gets the next opportunity to review the inventory R periods later. An order placed at that point in time will reach the organization only after L time periods, which is the lead time for supply. Hence, at every review, the organization needs to protect itself from the risks of shortages for a period of $(R + L)$ periods. Therefore, an expression for S can be derived by modifying Eq. 10.2 as follows:

$$ROP = \mu_{(L)} + Z_{\alpha} \times \sigma_{(L)} \quad (10.2)$$

Order up to level S = Mean demand during $(L + R)$ + Safety stock for the period $(L + R)$.

$$S = \mu_{(L+R)} + Z_{\alpha} \times \sigma_{(L+R)} \quad (10.3)$$

Where, $\mu_{(L+R)}$ = Mean demand during $(L + R)$ and $\sigma_{(L+R)}$ = Standard deviation of demand during $(L + R)$

10.2.3 Issues in the P and Q Systems of Inventory Control

The Q system is based on perpetual monitoring of the inventory level in the system. This calls for elaborate methods of keeping track of available inventory. To overcome this problem, organizations use the two-bin system. In recent years, several manufacturing firms have used a variety of visual control systems to manage the Q system. For instance, an automobile component manufacturer in South India stocks steel ingots in its stores with three colour-coded level indicators. When the stock level depletes from

the green zone to yellow zone, an order is placed. When the stock level depletes further down to the red zone, close follow-up and monitoring of supply is done to avoid shortage. In other words, the yellow zone corresponds to the reorder point in the Q system and the red zone to the safety stock.

The Q system is less responsive to changes in demand. When there is a decline in demand, the ROPs move to the right. Moreover, the system will continue to order Q even when there is a decline in demand, leading to a situation of keeping the inventory for a longer period. Therefore, ordering high-value components using the Q system during periods of falling demand could increase the risk of excessive stocking of inventory for longer periods and resultant obsolescence. Similarly, if there is an increase in demand, the ROPs shift to the left, necessitating frequent orders and increasing both the ordering cost and the risks of shortages. One way of addressing this problem is to order variable quantities depending on the demand.

The Q system is less responsive to changes in demand and poses greater difficulty in ordering multiple items from the same supplier.

Another issue in using the Q system is the difficulty of ordering multiple items from the same supplier. Consider three items—A, B, and C—purchased from a supplier using the Q system of inventory control. Since demand attributes can vary between the three, the three items can reach the reorder point on different days. This will call for multiple orders being placed with the same supplier within a short span of time. In addition to increasing the costs of ordering, the administrative mechanisms related to follow-up and receipt of material against each order is also increased. On the other hand, the organization could have benefited from placing a single order clubbing the requirements for Items A, B, and C. This would have resulted in significant reduction in ordering costs, monitoring and follow-up costs.

The P system is responsive to demand and enables ordering multiple items from the same supplier at the same time.

However, the P system overcomes several of these limitations of the Q system and offers many advantages over the Q system. This includes more responsiveness to demand, the ability to order multiple items from the same supplier at the same time, ease of planning and control, greater chances for linkages with MRP, and other planning systems in the organization. Table 10.1 compares the salient features of P and Q systems of inventory control. The safety stock investment in the P system, however, is more than that of the Q system because it requires protection from shortage for a longer period of time.

The safety stock investment in the P system is more than that of the Q system because it requires protection from shortage for a longer period of time.

NOTES

TABLE 10.1 COMPARISON OF THE Q AND P SYSTEMS OF INVENTORY CONTROL

Criterion	Continuous Review (Q) System	Periodic Review (P) System
How much to order	Fixed order quantity: Q	$S = \mu_{(L+R)} + Z \times \sigma_{(L+R)}$ $Q_R = S - I_R$
When to order	$ROP = \mu_{(L)} + Z_\alpha \times \sigma_{(L)}$	Every R periods
Safety stock	$SS = Z_\alpha \times \sigma_{(L)}$	$SS = Z_\alpha \times \sigma_{(L+R)}$
Salient aspects	Implemented using two-bin system	More safety stock More responsive to demand
	Suited for medium and low value items	Ease of implementation

EXAMPLE 10.2

A manufacturing organization is using a certain raw material, which is consumed in large quantities by various products. The raw material is procured from a local supplier. An extract of the relevant records from the stores indicate the following details about the component:

Mean of weekly demand: 200

Standard deviation of weekly demand: 40

Unit cost of the raw material: ₹300

Ordering cost: ₹460 per order

Carrying cost percentage: 20 per cent per annum

The purchase department has indicated that the lead time for procurement of this raw material is two weeks. Past experience with the supplier suggests that there is no uncertainty with respect to the lead time. The organization has been using EOQ for the purpose of scheduling orders. However, there is a general feeling that it is not working satisfactorily. It is not uncommon for the organization to experience stock-outs. Work out the parameters of the P and Q systems of inventory control.

Solution**EOQ Model**

Weekly demand = 200

Number of weeks per year = 52

Annual demand, $D = 200 \times 52 = 10,400$

Ordering cost, $C_0 = ₹460$ per order

Unit cost of the item = ₹300

Carrying cost, $C_c = ₹60$ per unit per year

Economic order quantity

$$= \sqrt{\frac{2C_0 D}{C_c}} = \sqrt{\frac{2 \times 460 \times 10400}{54}} = 399.33 \approx 400$$

$$\text{Time between orders} = \frac{400}{10400} = \frac{2}{52} = 2 \text{ weeks}$$

Since there is a two-week lead time, an order needs to be placed two weeks ahead of complete depletion of the inventory, that is, by the time the inventory position in the system reaches 400 units. The order quantity is 400. There is no safety stock to protect the system from shortages arising out of uncertain demand. The system can be improved either by using a Q system or a P system, as shown here.

Q System

Standard deviation of weekly demand = 40

Lead time, $L = 2$ weeks

Mean demand during L , $\mu_{(L)} = 2 \times 200 = 400$

Standard deviation of demand during L , $\sigma_{(L)} = \sqrt{2} \times 40 = 56.57$

For a service level of 95 per cent,

$SS = Z_\alpha \times \sigma_{(L)} = 1.645 \times 56.57 = 93.05 \approx 93$

$ROP = \mu_{(L)} + Z_\alpha \times \sigma_{(L)} = 400 + 93 = 493$

Using EOQ as the fixed order quantity, the Q system can be designed such that, as the inventory position in the system reaches 493, an order is placed for 400 units. In the long run this will ensure a service level of 95 per cent.

P System

Using the time between orders derived from the EOQ model as the basis for review period

Review period, $R = 2$ weeks

Mean demand during $(L + R)$, $\mu_{(L+R)} = 200 \times (2 + 2) = 800$

Standard deviation of demand during $(L + R)$, $\sigma_{(L+R)} = \sqrt{(2+2)} \times 40 = 80$

For a service level of 95 per cent,

$SS = Z_\alpha \times \sigma_{(L+R)} = 1.645 \times 80 = 131.6 \approx 132$

Order up to position, $S = \mu_{(L+R)} + \sigma_{(L+R)} = 800 + 132 = 932$

The P system can be designed such that the inventory position in the system is reviewed every two weeks and an order is placed to restore the inventory position back to 932 units. This will ensure a service level of 95 per cent.

10.3 SELECTIVE CONTROL OF INVENTORY

Managing inventory invariably amounts to handling a large number and variety of items. For instance, an automobile manufacturer such as Ashok Leyland may have an inventory of about 50,000 items in their stores. A vast majority of them will be consumed (albeit in varying rates) and will require a mechanism for monitoring inventory on-hand, establishing reorder points, order quantity, and the type of inventory control system that needs to be adopted. Clearly, establishing the same level of monitoring and control for all the items may not be practically feasible. Therefore, organizations devise suitable ways of categorizing the items and adopt mechanisms that have variable levels of control on the different categories of items.

Consider two items in inventory, a simple fastener that costs less than a rupee and a specialized electronic sub-assembly that costs several thousands of rupees. Managing the latter item with close monitoring and tighter control will benefit the organization more than exercising greater managerial attention and control on fasteners. If there were errors in judging the demand for the item, or if extra items were ordered, the impact of these factors will be minimal in the case of fasteners. Also, if there were shortages, the response time to procure fasteners will be very low. On the other hand, in the case of the electronics sub-assembly, the impact will be significant. It is therefore obvious that managers need alternative methods and levels of control while dealing with a multitude of items in the inventory. Selective control of inventory achieves this objective.

10.3.1 ABC Classification

It is clear from the arguments outlined earlier that the unit cost or total cost of items consumed could be one basis on which items could be classified. The ABC classification of inventories is based on the cost (or value) of items consumed. Very high-value items are “A-class items” and may require tighter control. Medium-value items are categorized as “B class” and low-value items as “C class”.

Let us consider an example of twenty items in inventory, as shown in Table 10.2. For the twenty items, the unit value and annual consumption are available. Using this, it is possible to compute the cumulative percentage of items and cumulative value of consumption (see the last two columns of Table 10.2) and rank them in the descending order of cumulative value. Figure 10.4 shows a graphical representation of this data.

As shown in Table 10.2, 5–10 per cent of the items typically contributes to 70–80 per cent of the value and constitutes A-class items. These items require greater control and close monitoring. By virtue of constituting a small percentage of the total in terms of volume, it will be practically feasible to exercise close monitoring and greater control. At the other extreme, about 60–70 per

NOTES

TABLE 10.2 SAMPLE DATA FOR ABC CLASSIFICATION

Item No.	Unit Value (₹)	Annual Consumption	Cumulative Value (₹)	Cumulative Number of Items (%)	Consumption Value (%)
1	30,000	80	2,400,000	5.00%	62.96%
2	450	1,200	540,000	10.00%	77.12%
3	590	400	236,000	15.00%	83.32%
4	25,000	9	225,000	20.00%	89.22%
5	600	200	120,000	25.00%	92.37%
6	4,500	15	67,500	30.00%	94.14%
7	400	100	40,000	35.00%	95.19%
8	30	1,000	30,000	40.00%	95.97%
9	145	200	29,000	45.00%	96.73%
10	2,300	12	27,600	50.00%	97.46%
11	9	1,500	13,500	55.00%	97.81%
12	11	1,000	11,000	60.00%	98.10%
13	2,000	5	10,000	65.00%	98.36%
14	4	4,000	16,000	70.00%	98.78%
15	120	120	14,400	75.00%	99.16%
16	20	500	10,000	80.00%	99.42%
17	10	1,000	10,000	85.00%	99.69%
18	80	100	8,000	90.00%	99.90%
19	25	100	2,500	95.00%	99.96%
20	1	1,500	1,500	100.00%	100.00%
Total			3,812,000		

NOTES

cent of items constitute about 5–10 per cent of the total value. These are the C-class items. These items require simple control mechanisms such as level-based reordering. The middle portion of about 10–25 per cent by value and about 20–30 per cent by number of items constitute B-class items. These items require a moderate level of control.

The periodic review and the continuous review systems of inventory control can be linked to the category of items. Since A-class items require closer control and better response to changes in the demand pattern, periodic review systems are more appropriate. In the case of “B” class items, continuous review systems are appropriate. “C” class items can have simple level-based rules for inventory control. C-class items are often available readily off the shelf and it is possible to obtain them by ordering over the phone. Hence, issuing blanket purchase orders for a year and following up with specific requests for supply against the purchase order is often practiced in the case of C-class items.

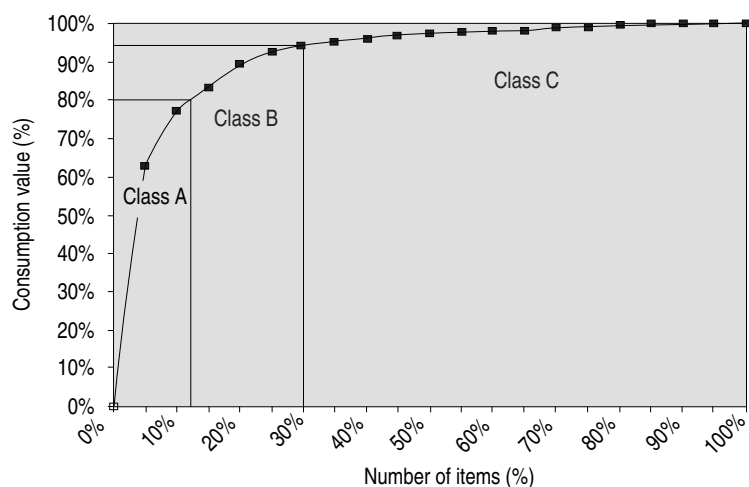


Figure 10.4 ABC classification of inventories

NOTES

10.3.2 Other Classification Schemes for Selective Control

Using the consumption value as the basis for categorizing inventory is just one method. In practice, organizations use a wide variety of measures for selective control. Each of these has relevance in specific contexts. Let us look at a few alternative categorization schemes:

1. *On the basis of unit cost of the item (XYZ classification):*
(a) high unit cost (X-class item); (b) medium unit cost (Y-class item); (c) low unit cost (Z-class item)

This classification is based only on the unit cost, whereas ABC classification, takes the consumption pattern also into account. A very high-value item often turns out to be specially made to order, complex, and may call for lengthy supply identification procedures. If the item is consumed in small quantities, it may be classified as a “B-class item” and may be denied adequate managerial attention.

XYZ classification is based only on the unit cost, whereas ABC classification takes into account the consumption pattern as well.

2. *On the basis of movement of inventory (FSN classification):*
(a) fast-moving, (b) slow-moving, (c) non-moving.

This method of classification is ex-post, as opposed to ABC classification, which is future oriented. Items that have not been moving for sometime incur carrying costs and may call for managerial attention for disposal decisions. On the other hand, fast-moving items can be controlled using available inventory control systems.

VED classification is relevant in the case of maintenance items.

NOTES

3. *On the basis of criticality of items (VED classification):* (a) vital, (b) essential, (c) desirable.

This classification is relevant in the case of maintenance items. In several cases a vital item for maintenance may not be very expensive. For instance, a wide variety of oil seals are used in hydraulic systems. If the oil seals are not available, the entire equipment becomes inoperative and has a ripple effect on the entire system. However, by virtue of having smaller unit value or lower consumption value, these items may escape the attention of the management without this classification scheme in place.

4. *On the basis of sources of supply:* (a) imported, (b) indigenous (national suppliers), (c) indigenous (local suppliers).

Imported items are usually of high value. Moreover, they have long lead times owing to several statutory and procedural complexities involved in the buying process. Therefore, they may require tighter control. On the other hand, items procured from the local suppliers are available “off the shelf”.

It is often possible to combine more than one classification scheme and make use of them to further sub-categorize the inventory and devise appropriate inventory control system for each of them.

It is often possible to combine more than one classification scheme and make use of them to further sub-categorize the inventory and devise an appropriate inventory control system for each of them. In practice, organizations have utilized a variety of these combinations to implement appropriate inventory control systems for the items.

10.4 INVENTORY PLANNING FOR SINGLE-PERIOD DEMAND

So far we have considered the problem of inventory planning in the case of recurring demand that requires continuous replenishment. However, there are a number of situations that require inventory planning for a single-period demand. In a single-period demand, the unfulfilled demand cannot be back-ordered to the next period because the demand ceases to exist after the period for which planning is done. In other cases, even though demand exists in the future, what is ordered for a period cannot be used for future periods due to the perishable nature of the item. Examples include the demand for morning newspapers, tickets for journeys, advertising space for a mega entertainment event, and expensive maintenance spares. This type of inventory models are referred to as news vendor model. Often, there is a high degree of uncertainty involved in estimating the demand for a single period.

In a single-period demand, the unfulfilled demand cannot be back-ordered to the next period.

THE INVENTORY CONTROL SYSTEM OF A PETROCHEMICAL MANUFACTURER

A few years ago, a well-known petrochemical manufacturer in the country devised a simple yet effective system of inventory control in one of its manufacturing locations. It adopted a two-dimensional classification scheme to classify its inventory. Based on the consumption value, the items were classified as A-, B-, or C-class items using the ABC classification. Similarly, based on the unit value, items were classified as X, Y, or Z using XYZ classification.

The management adopted a periodic review control system for the AX items. The review period was weekly, fortnightly, or monthly. The BY items were controlled using continuous review (reorder-level based) methods. The system parameters for the BY items were as follows:

Maximum inventory = Reorder level + Order quantity

Reorder level = Safety stock + Lead time consumption

Safety stock = 1 month consumption for indigenous items and 2 months for imported items

The CZ items had no specific control mechanism in place. It was customary to have one blanket purchase order per year and request for 3 or 4 deliveries in a year.

Source: Based on data gathered from the author's own research.

Planning for appropriate levels of inventory in such situations requires careful balancing of two opposing costs. The main driver of costs in this inventory planning is the fact that uncertain demand manifests only for a period. Therefore, carrying lesser inventory than demand directly results in lost opportunity to make profit. This represents the *cost of understocking*. In the same way, any excessive inventory cannot be consumed afterwards. At the most, the unused inventory may fetch some salvage value and the resulting loss on account of this could be termed as the *cost of overstocking*.

Let C_{os} = Cost of overstocking per unit

C_{us} = Cost of understocking per unit

Q = Optimal number of units to be stocked

d = Single-period demand

$P(d \leq Q)$ = The probability of the single-period demand being at most Q units

If $d > Q$, then we incur costs on account of cost of understocking. On the other hand, if $d < Q$, then we incur the cost of overstocking. At a very low value of Q , we tend to experience costs arising out of understocking and as we increase Q incrementally we will approach optimal Q . At very high values of Q , we will incur overstocking costs. By incremental analysis, we find that while taking a decision to stock Q units, we would like to ensure that:

NOTES

NOTES

The expected cost of overstocking $\leq$ The expected cost of understocking

$$P(d \leq Q) \times C_{os} \leq P(d > Q) \times C_{us}$$

$$P(d \leq Q) \times C_{os} \leq \{1 - P(d \leq Q)\} \times C_{us}$$

$$P(d \leq Q) \times (C_{os} + C_{us}) \leq C_{us}$$

$$P(d \leq Q) \leq \frac{C_{us}}{C_{us} + C_{os}} \quad (10.4)$$

Therefore, we choose that largest value of Q that satisfies Eq. 10.4 as the optimal value.

10.5 OTHER ISSUES IN INVENTORY PLANNING AND CONTROL

In practice, inventory control systems vary from theoretical formulations in many ways. Moreover, there is a lack of understanding of certain critical issues that influence inventory management. One serious drawback of inventory control systems in practice is that they tend to stand aloof from reality. Any inventory control system should have provisions for linking key parameters such

EXAMPLE 10.3

Navratri is a popular festival in India. In South India, beautifully painted dolls made of clay are bought by customers during this festival. After the festival, there is no demand for these dolls. A manufacturer of dolls needs to decide on the optimal stock of dolls that he needs to carry in his inventory to satisfy the demand during the festival time. The item fetches a sales value of ₹1,300 per box. The cost of production is ₹1,000 per box. After the festival is over, the items at best can be salvaged at a value of ₹800 per box. Table 10.3 presents the distribution of demand for the item during the festival time. What is the optimal quantity to stock?

TABLE 10.6 DISTRIBUTION OF DEMAND FOR EXAMPLE 10.3

Number of units demanded	Probability	Cumulative probability
0	0.05	0.05
100	0.15	0.20
200	0.20	0.40
300	0.25	0.65
400	0.20	0.85
500	0.15	1.00

Solution

Since each unfulfilled demand results in a foregone profit, the cost of understocking is the profit per box. Similarly, by salvaging each unsold box after the festival, the manufacturer loses an amount equal to the difference between the cost of manufacture and salvage value, which represents the cost of overstocking.

Selling price per box of the item: ₹1,300.00

Cost of production: ₹1,000.00

Cost of understocking, C_{us} : ₹300.00

Salvage value: ₹800.00

Cost of overstocking, C_{os} : ₹200.00

As per Eq. 10.4, the optimal quantity to stock is:

$$P(d \leq Q) \leq \frac{C_{us}}{C_{us} + C_{os}} = P(d \leq Q) \leq \frac{300}{200} \leq 0.60$$

On examination of the cumulative probability values in the last column of the demand table, we notice that Q lies between 200 and 300. We round up the value of Q to 300. Therefore, the manufacturer should plan for an inventory of 300 boxes for sale during the festival.

as reorder level and safety stock to actual changes in consumption pattern, measured in terms of mean and standard deviation of demand. In the absence of this linkage, the model parameters become outdated and result in either piling up of inventory or in frequent shortages.

With the installed base of ERP systems and other MIS systems for transactional data capture, it is possible to upgrade computerized inventory control systems from an ordinary MIS system generating a host of shortage reports and accurate inventory status records to one of a decision-support system. On a periodic basis, these computerized systems could analyse the consumption patterns of items, detect impending changes, and adjust the model parameters of the inventory control system.

Better management of inventory does not happen only because of good inventory planning tools. Organizations need to take several other measures to bring down inventory. A frequently encountered issue is part number proliferation. Over the years, an organization creates a vast number of part numbers and managing such a large variety of parts in inventory could be a herculean task. Efforts to reduce part numbers by variety reduction measures are an important aspect of better inventory management. Similarly, an organization having several manufacturing/service divisions must engage in inter-divisional joint planning efforts towards standardization of classification and coding systems and stock holding of expensive items such as capital spares. These efforts go a long way in better inventory control.

Opportunities for better inventory control and management often exist outside of the materials management function in organizations.

Opportunities for better inventory control and management often exists outside of the materials management function in organizations. For example, alternative methods of launching material onto the shop floor, such as kit launching, helps reduce inventory mismatch and non-moving inventory. In a kit launching system, items required for manufacture and assembly are always launched into the system in exact quantities of ship sets or assembly sets. This reduces the chances of ending up with excess and unwanted work in progress at the end of the planning period. Another example relates to performance measurement systems in organizations. Management control systems that emphasize on utilization-based measures for rewards and incentives tend to promote inventory build-up in the system. Better structures and systems also contribute positively to reducing inventory build up in the system.

An inventory planning manager should take all these into consideration while actively pursuing the models suggested in this chapter. Such an approach will ensure greater chances of reducing inventory in the system.

NOTES

10.6 SUMMARY

- Every organization carries five different types of inventory. This includes *cyclic stock*, *pipeline inventory*, *safety stock*, *decoupling inventory*, and *seasonal inventory*. The purpose of carrying these types and the manner to assess the level of investment required in each of these vary.
- *Inventory planning* is done in order to minimize the total cost of the plan. The costs include the unit cost of the item for which planning is done, the cost of carrying inventory, the cost of ordering, and the cost of shortages.
- The key decisions in any inventory planning scenario is to answer the “how much” and the “when” questions. These decisions are made by balancing various costs associated with inventory.
- The *EOQ model* is useful for inventory planning in the case of multi-period deterministic demand situations. The EOQ model is robust to model parameters and could be suitably

modified to incorporate some real-life situations such as quantity discounts and non-zero lead time for supply.

- *Service level* is a useful concept for modelling inventory planning in the case of stochastic demand. Safety stocks can be built commensurate to the desired service level.
- A *fixed order quantity* (Q system) or continuous review system of inventory planning and control is useful for B-class and C-class items of inventory. A popular application of the continuous review system in organizations is the two-bin system.
- A *fixed order interval* or a periodic review system (P system) is useful for planning and control of high-value or A-class items. The P system is more responsive to changes in demand patterns than the Q system.
- Selective control of inventories is achieved through alternative classification methodologies. The ABC, VED, and XYZ classifications are often used by organizations.

10.7 FORMULA REVIEW

Total cost associated with carrying inventory = $\left(\frac{Q}{2} \times C_c\right)$

Total ordering cost = $\left(\frac{D}{Q} \times C_o\right)$

Total cost of inventory, $TC(Q) = \left(\frac{Q}{2} \times C_c\right) + \left(\frac{D}{Q} \times C_o\right)$

Economic Order Quantity (EOQ), $Q^* = \sqrt{\frac{2C_oD}{C_c}}$

Optimal number of orders = $\frac{D}{Q^*}$

Time between orders = $\frac{Q^*}{D}$

For the perpetual review inventory system

Re-order Point (ROP) = $\mu_{(L)} + Z_\alpha \times \sigma_{(L)}$

Total cost, $TC(Q^*) = \left(\frac{D}{Q^*} \times C_o\right) + \left(\frac{Q^*}{2} \times C_c\right) + ((Z_\alpha \times \sigma_{(L)}) \times C_c)$

For the periodic review inventory system

Order up to level, $S = \mu_{(L+R)} + Z_\alpha \times \sigma_{(L+R)}$

Optimal quantity (Q) to order in a single period inventory

system is given as $P(d \leq Q) \leq \frac{C_{us}}{C_{us} + C_{os}}$

10.8 REVIEW QUESTIONS

- Which of the following constitutes independent demand items?
 - Maintenance spares for machine tools in a manufacturing firm
 - Carburettors for a petrol engine manufacturer
 - Fresh linen in a five-star hotel
 - Radiography services in a multi-specialty hospital
 - Number of tyres for an automobile manufacturer
 - Replacement clutch cables for two-wheelers
 - Halogen lamps for automotive headlights
 - Number of operations management text books in a business school

- Number of bottles of mineral water in a restaurant
 - Number of lunches for the mid-day Chennai- Mumbai flight
- How is inventory planning for independent demand items different from that for dependant demand items?
 - Do organizations need to carry inventory? Why?
 - What is the relationship between inventory investment and profitability in an organization?
 - Consider the inventory planning problem. Write down the relevant total cost equation in each of the following situations:

- (a) The organization can fetch a discount on the purchase price if large quantities are ordered
 - (b) There is a benefit in transportation costs if ordered in full truckloads
 - (c) The supplier is willing to set up elaborate systems for easy ordering, monitoring, and follow-up of orders if the order quantity is large
 - (d) There is uncertainty in demand and the organization wishes to carry safety stock
6. What are the difficulties in using shortage cost measures in inventory planning models?
 7. Can you identify four alternatives that organizations use as a basis for fixing the order quantity?
 8. On what basis would you recommend the periodic review system of inventory control?
 9. Why should organizations adopt a selective system of inventory control? If you were asked to recommend a suitable classification scheme, how would you go about the task?
 10. What is the relationship between the service level and safety stock?
 11. There are N locations at which an organization faces demand for an item. The demand for the item is uncertain and hence the organization needs to carry safety stock. If it were to make the choice between centralized storage of the item and localized storage of the item in each of the location, which alternative will result in lower investment in inventory? (**Hint:** Assume that all locations have the same mean and standard deviation for demand).
 12. What is the effect of the cost parameters on the economic order quantity? Perform a “what if” analysis using a spreadsheet by varying these cost parameters. What did you deduce from the results?
 13. When is it appropriate to use the ABC classification scheme and the FSN classification scheme?
 14. Suppose you develop a classification scheme for selective control of inventories using ABC and VED classification as two dimensions. Relate the discussions pertaining to service level to this two-dimensional classification scheme.

10.9 PROBLEMS

1. Consider an item that is ordered once per month. The daily requirement is 200 and the lead time for supply is two days. There are 25 working days in a month. The cost of ordering is ₹300 per order and the cost of carrying is ₹150 per unit per year.
 - (a) Draw a sketch showing the cyclic and pipeline inventory in the system
 - (b) What is the reorder point for this item?
 - (c) What will the cost of this plan be?
 2. An auto-component manufacturer requires a certain steel forging in large quantities. The annual requirement is 40,000 pieces, each costing ₹450. The ordering cost is ₹600 per order and the carrying cost is ₹100 per unit per year.
 - (a) What is the optimal order quantity?
 - (b) How frequently should the manufacturer place the order with the supplier?
 - (c) Compute the total ordering cost and total carrying cost. Do you notice anything?
 3. Consider Problem 2. Perform a “what if” analysis for the following scenarios and compute the total cost of the plan for ordering an optimal order quantity:
 - (a) The actual demand is –12 per cent, –8 per cent, –4 per cent, 0 per cent, 4 per cent, 8 per cent, and 12 per cent of the demand mentioned in the previous problem.
 - (b) The actual ordering cost has a similar variation as that of the demand
 - (c) The actual carrying cost has a similar variation as that of the demand
- Plot the total cost in each of the three scenarios and make your observations.
4. Excel Toys, a manufacturer of plastic toys for children, requires constant supply of high-density polyethylene (HDPE) in the resin form. HDPE is available in the resin form and one tonne costs approximately ₹2,000. The manufacturer runs his factory for 250 days a year and the daily consumption rate for HDPE is four tonnes. The suppliers normally take a week to replenish an order. The ordering cost is ₹2,000 per order and the carrying cost is 20 per cent.
 - (a) Estimate the cyclic and pipeline inventory in the system at Excel Toys.
 - (b) What is the total cost of the plan?
 - (c) Suppose if the supplier insists on a minimum order quantity of 75 tonnes, what is your recommendation to Excel? Will the recommendation change if the minimum order quantity is 150 tonnes?
 - (d) Suppose Excel launches some improvement initiatives in the company and brings down the ordering cost by 20 per cent. Recompute the economic order quantity.
 - (e) If Excel spent a sum of ₹10,000 towards improvements in the system, when will the improvement efforts bring payback? (**Hint:** Estimate the savings in costs due to reduced ordering cost)
 5. Consider Example 10.2 in this chapter. If the company continues with the same plan even when there is an increase in the demand by 10 per cent, what will the impact be? What do you infer from the result?

6. A manufacturing organization has been consuming a certain item in large quantities and is currently procuring the item from Supplier A. The price offered by the supplier is ₹400 per piece. The ordering cost is ₹2,800 per order and the carrying cost is ₹350. The annual demand for the item is 10,000. The supplier is currently not offering any discount. However, another supplier, Supplier B, is willing to offer the following discount structure:

Up to an order size of 999 = No discount

For an order size of 1,000–1,999 = 2 per cent discount in price

For an order size of 2,000 and above = 5 per cent discount in price

Switching over to this supplier means incurring an initial cost of ₹15,000. This cost is primarily to set-up new communication systems with the new supplier. What should the company do in the light of the new offer?

7. Siemens India Limited manufactures switchgears at their Kalwa plant. One of the key components, *Kombishraube* (contacts), is manufactured in-house. If the machine producing these contacts manufactures at the rate of 800 pieces per hour and the demand rate is 450 pieces per hour, what is the economic run length for this production process? How often should they set up the machine if the factory works for 250 days in a year on a single shift (of 8 hours) basis? The cost of setting up the machine is ₹3,000 per set-up and the cost of carrying the components is ₹400 per unit per year.

8. Consider an item with the following demand attributes:

Mean daily demand = 20; Standard deviation of daily demand = 8.

Compute the safety stock required for a continuous review system in each of the situations given in Table 10.4:

What do you infer from these calculations?

TABLE 10.4 SITUATIONS FOR PROBLEM 8

Lead Time	Service Level
1 week	90%, 95%, 99%
10 days	90%, 95%, 99%
2 weeks	90%, 95%, 99%
3 weeks	90%, 95%, 99%

9. For Problem 8, compute the order up to levels and safety stock for a periodic review system with a two-week review period.
10. The weekly requirement of furnace oil in a foundry has a mean of 3,000 litres and a standard deviation of 900 litres. A litre of furnace oil costs ₹400 and the carrying costs are estimated at 25 per cent. If the foundry uses a continuous review system of inventory control with a desired service level of 90 per cent, then

- (a) What will the reorder point and safety stock for furnace oil be if the lead time for supply is 1 week?
- (b) What will the service level be if the reorder point is changed to 3,100 litres?
- (c) If the reorder point is left unchanged when there is a 5 per cent increase in the mean demand, how will it affect the service level?
- (d) If the foundry estimates that the monetary value of the benefit that it may obtain is ₹3,000 when increasing the service level from 90 per cent to 95 per cent, what is the appropriate service level for the item?

11. The demand for springs in the coil form for an auto component manufacturer is continuous and uncertain. The manufacturer places orders for the springs with a local supplier who takes two weeks to deliver. The cost of the spring is ₹400. The weekly demand for the springs has a mean of 500 and a standard deviation of 100. The cost of ordering is ₹2,000 per order and the cost of carrying is 18 per cent. The supplier specifies a minimum order quantity of 2,000. Assume a service level of 90 per cent.

- (a) Is it feasible to operate an EOQ based system for this item?
- (b) Design periodic and continuous review systems of inventory control.
- (c) What is the total cost of the plan in each of these cases?

12. Oriental Healthcare is a multi-specialty hospital catering to a variety of illnesses connected to the heart and respiratory system. The demand for a class of medical consumable is generally random. Initially, Oriental followed a practice of ordering 200 boxes every two weeks. This practice was not found to be satisfactory. After some review, they resorted to ordering 300 boxes every two weeks. Even this practice was not found satisfactory. Recently, an examination of the stores records over a period of 10 weeks revealed the following weekly consumption pattern (Table 10.5):

TABLE 10.5 WEEKLY CONSUMPTION PATTERN

Week no.	Consumption (Units)
1	120
2	109
3	142
4	140
5	110
6	145
7	144
8	120
9	160
10	110

The supplier of the item takes (on an average) two weeks to deliver once the order is placed. The item costs ₹450 per box. The ordering cost is ₹2,000 per order and the carrying cost is 20 per cent. What should Oriental do in the light of this information? Will they be better off with the new recommendation? Assume a service level of 90 per cent.

13. A pharmacy needs to decide on monthly procurement quantity of a critical yet a highly perishable drug that has a shelf-life of just one month. The unit price of drug is ₹7,000 and its cost is ₹6,000. As per the historical data, the monthly demand for the drug is random with a certain distribution. The details are given in Table 10.6. Since the drug is an emergency drug, the associated hospital offered to reimburse ₹3,000 for each unit of drug left unsold. At the same time, the store would be penalized ₹2,000 per unit by the hospital for any unmet demand.

What should be the monthly order size the pharmacy must plan for?

TABLE 10.6 HISTORICAL DATA ON DEMAND DISTRIBUTION

Demand	22	24	26	28	30	32	34	36
Probability	0.05	0.10	0.10	0.15	0.20	0.20	0.12	0.08

14. In problem number 13 consider the following. The hospital is contemplating a change in the contractual terms with the pharmacy. Suppose the hospital is willing to make an offer of either withdrawing the reimbursement or penalty. What should the pharmacy owner do now?
15. A manufacturer of electric ovens has made an estimate (shown in Table 10.7) for annual consumption and unit cost of the components that are used for manufacture. Perform an ABC analysis and advise how the manufacturer should plan and control for inventory.

TABLE 10.7 ANNUAL CONSUMPTION AND UNIT COST OF COMPONENTS

Item No.	Item Name	Unit of Measurement	Annual Consumption	Unit Value (₹)
1	Copper wire/plate	tonnes	200	7,000.00
2	Steel plate	tonnes	400	5,000.00
3	Ceramic insulators	pieces	40,000	30.00
4	Brass fasteners	pieces	60,000	1.00
5	Steel fasteners	pieces	40,000	1.25
6	Ceramic bushes	pieces	20,000	10.00
7	Paint	litre	400	200.00
8	Electrical cable	metre	20,000	3.00
9	Power sockets	pieces	10,000	7.00
10	Plugs	pieces	10,000	8.00
11	Brass washers	pieces	80,000	0.50
12	Plastic components	pieces	60,000	1.25

10.10 MINI PROJECT

- Identify a sector of industry of your choice and select three companies operating in that sector. Analyse their annual report and identify portions of the annual report that have a direct impact on inventory-related issues.
 - Prepare a report highlighting your understanding of the performance of the companies with respect to inventory planning and control. Can you identify some possible explanations for your observations pertaining to the inventory data?
 - Among the three, which one is better in managing their inventory? How do these firms compare with the industry average?
 - Can you make some comparisons with similar firms elsewhere (outside the country)?
 - What are your key inferences from this exercise?
- Study the reports produced by ICICI (Financial Performance of Companies) and the Centre for Monitoring Indian Economy on performance of corporate sector.
 - Identify the inventory investment across different sectors of our industry. Do you see any significant trends in these movements?
 - How has the performance been in the last five years? What factors do you think might have contributed to this?
 - Which sectors have performed better and which have performed the worst? Can you identify some reasons for these?
- Analyse the multiplier effect of inventory using the Du Pont equation for return on investment for a pair of close competitors. What are your key inferences from this exercise?

10.11 CASE STUDY

MML Limited**The Background**

MML Limited was incorporated as a private limited company and later converted into a public limited company in 1988. Though the company started with the manufacture of soaps and perfumes, it has added a variety of products to diversify its product line. Today, it is a conglomerate with businesses in soaps, perfumes, plastics, petrochemicals, paints, industrial electronics, and agro-business. Their unit at Hosur manufactures electronic components. The inventory position has been deteriorating over the last few years. The year-end inventory position, as revealed by the annual reports of the conglomerate, showed an increase from ₹1,230 million as on 30 June 2004 to ₹2,490 million as on 31 March 2007.

The conglomerate has been operating in a highly centralized fashion. All major planning activities were done at the corporate office. However, many inputs for this decision making were made available by all the operating units. The headquarters sent the tentative monthly production plan two months prior to the commencement of production. The final production plan arrived 5 days prior to start of the production. It was not unusual for the corporate office to change the production plans while production was progressing. This was communicated through telephone, telex, fax, or mail. Once the production plan was made, the requirement of each item was also computed using the bill of materials. Table 10.8 presents the tentative and final plans, and the

TABLE 10.8 TENTATIVE PLAN, FINAL PLAN, AND ACTUAL PRODUCTION FOR SOME PRODUCTS

Month	Production	Product A	Product B	Product C	Product D	Product E	Product F
Apr 2005	Tentative	6,000	2,500	19,000	19,000	33,500	2,500
	Final	6,500	2,500	17,000	15,000	30,000	5,000
	Actual	7,370	2,427	14,145	13,249	24,206	2,370
May 2005	Tentative	6,500	2,500	21,000	NA	35,000	5,000
	Final	6,500	2,000	14,500	4,000	38,000	5,000
	Actual	5,438	2,078	16,897	10,177	35,899	4,840
Jun 2005	Tentative	6,500	2,500	21,000	NA	35,000	5,000
	Final	6,500	2,000	14,500	10,500	35,000	7,500
	Actual	5,882	2,123	13,145	14,671	35,936	9,230
Jul 2005	Tentative	6,000	2,500	13,500	14,000	35,000	5,000
	Final	5,000	2,500	24,500	12,000	35,000	5,000
	Actual	9,137	2,661	28,287	11,960	34,567	2,491
Aug 2005	Tentative	4,000	2,500	6,000	16,000	33,500	5,000
	Final	6,000	2,500	5,000	12,500	30,000	5,000
	Actual	9,170	2,688	7,883	13,631	31,984	2,340
Sep 2005	Tentative	6,000	3,000	9,500	15,000	35,000	5,000
	Final	8,000	4,000	8,000	12,000	36,000	9,000
	Actual	7,042	4,018	8,262	15,825	31,281	3,933
Oct 2005	Tentative	4,000	4,000	7,500	10,000	36,000	2,500
	Final	4,000	4,000	7,500	19,000	35,000	2,500
	Actual	3,060	4,320	13,246	18,111	36,204	872
Nov 2005	Tentative	3,000	4,000	10,000	12,000	31,500	5,000
	Final	3,500	3,500	14,500	10,000	38,000	5,000
	Actual	7,351	3,448	16,406	12,006	39,679	3,392
Dec 2005	Tentative	7,500	2,000	7,000	13,000	38,000	5,000

(Continued)

TABLE 10.8 (Continued)

Month	Production	Product A	Product B	Product C	Product D	Product E	Product F
	Final	7,500	3,500	12,000	13,000	40,000	5,000
	Actual	8,938	2,214	12,828	14,320	37,762	8,330
Jan 2006	Tentative	5,000	2,500	6,500	20,000	37,000	5,000
	Final	NA	2,500	2,500	33,500	38,000	2,500
	Actual	1,634	2,983	4,149	25,376	34,133	5,040
Feb 2006	Tentative	7,500	2,500	11,000	19,500	30,000	2,500
	Final	4,500	2,500	4,000	22,945	36,000	NA
	Actual	3,504	2,660	4,032	23,790	35,978	2,301
Mar 2006	Tentative	7,000	3,000	6,000	20,000	32,500	5,000
	Final	4,000	3,000	6,000	27,000	36,000	5,000
	Actual	2,004	3,101	6,079	26,138	39,393	5,070

TABLE 10.9 CONSUMPTION PATTERN FOR A-CLASS ITEMS

	Item 1	Item 2	Item 3	Item 4	Item 5	Item 6	Item 7	Item 8	Item 9	Item 10
Apr 2005	11,058	2,180	—	3,499	2,675	131	4,608	3,024	743	2,345
May 2005	12,899	2,854	—	5,931	5,484	287	4,582	1,768	537	2,867
Jun 2005	12,595	2,937	—	4,474	2,627	270	9,204	—	891	1,489
Jul 2005	15,532	3,071	28	5,595	2,445	311	9,727	4,563	522	3,187
Aug 2005	12,786	1,585	4,253	3,606	2,044	148	6,077	2,078	719	2,150
Sep 2005	13,021	1,390	12,683	3,505	2,071	232	8,016	677	834	2,722
Oct 2005	14,260	2,433	7,153	4,435	2,737	314	10,104	2,989	1,238	1,867
Nov 2005	14,864	3,233	11,160	5,016	2,567	268	10,220	1,754	947	2,439
Dec 2005	15,067	2,868	15,621	4,349	2,702	223	6,973	2,456	—	1,196
Jan 2006	12,914	2,970	19,131	3,404	3,590	306	10,795	2,278	459	2,380
Feb 2006	13,365	2,202	921	3,656	3,199	200	6,679	3,219	855	2,100
Mar 2006	15,338	2,234	2,705	3,812	3,051	276	10,504	2,345	1,167	1,939
Unit cost (₹)	98.00	395.00	154.00	216.00	271.00	3,180.80	95.80	160.00	340.00	112.50
Lead time (days)*	30	38	30	45	60	50	20	15	37	30

Notes:

*The lead time is an average figure based on six months' data.

- The consumption details are expressed in units of items. The units of measurement are different for the items. For example, the unit of measurement for Item 3 is boxes (each box contains 250 items) whereas the unit of measurement for Item 8 is kg.
- Item 6 is sold by all the suppliers in integral units of 250 only. This has been attributed to the fact that once the seal is broken it should be used within three days.
- Items 7 and 10 are supplied by the same supplier, who is the only available domestic source.

actual production of the Hosur unit for 2005–06 and Table 10.9 presents the consumption pattern of the “A-class” item.

The Existing Buying Process

The company has been following a centralized ordering system, which was operated from its headquarters. The advantages of this system according to the organization were:

1. Better control over inventory-related costs.
2. Since the corporate office compiled the net requirements of all operating units, the order quantities for the suppliers were large, sometimes inducing them to offer generous quantity discounts.
3. In case of sudden shortages of an item at a location, the corporate office could advise another location to transfer stocks because it knew the safety stocks at each location.

The corporate office advises the purchase departments at the units on the following:

1. What to order?
2. How much to order?
3. When to order?
4. On whom to place the order?
5. What should be the delivery date?
6. What will be the price?

On receipt of this, the purchase department at the operating units took the responsibility for the procurement of items on the basis of the advice and subsequently made the payment to suppliers.

For the corporate office to take decisions on these items, it needed the following:

1. Stock levels of each item under inventory management.
2. The production plan for the following month.
3. The maximum and the minimum stock levels to be maintained for each item.
4. The previous month's consumption.

The operating unit provided the information pertaining to Item 1. Items 3 and 4 were policy decisions made at the corporate office based on historical data and future plans and Item 2 was market-related information that was forecast.

The Hosur Unit

The unit manufactures electronic components that form part of the control devices of a variety of equipment. Basically, there are minor variations in the basic configuration, which result in 13 different products as they roll out of the assembly. The variations are mainly due to the differences in the rating and the number of various electronic components used to build the product and the variations in the electronic circuit design.

The Purchase Department

A senior manager (purchase) assisted by a manager (purchase) headed the purchase department of the Hosur unit. There were four buyers attached to the department. The buyers were given responsibility to procure materials of specific groups of materials.

There was one buyer for semiconductor components, another for raw materials and a third buyer for imported components. Another buyer was looking after consumables, office equipment, and capital equipment. Although imported components formed a small percentage of the total purchase, the procedure and modalities were different, necessitating an exclusive buyer to handle them. Two clerical staff undertook the responsibility for all the typing, data preparation and entry into the computer, preparation of purchase orders, and other such support activities. The purchase department was provided with a desktop computer, which was connected to a local area network.

The senior manager was interacting with the corporate office concerning matters such as source development, supplier rating, and preparation of purchase budgets. In addition, he was liaising with other departments such as stores, quality control, finance, production planning, and design. This was an ongoing activity to help take many decisions on issues such as value engineering, make or buy, new material development, import substitution, etc.

Recently, there was an exercise carried out by the purchase personnel to collect data useful for making certain policy decisions. On analysing the past three years' records it was found that on an average, 450 purchase orders were sent out per year. An ABC analysis was carried out and the average lead time taken by the vendors for the supply of such items were computed. Table 10.9 presents the relevant data on the unit price, lead time, and consumption patterns of “A-class” items. It was also found that the currently the average investment in inventory is to the tune of ₹35 million.

Inward Goods Stores and Inspection Section

The Hosur unit had an inward goods stores and inspection section that performed the task of receiving consignments of supplies, conducting inspections, and initiating the necessary follow-up action. The section employed two load/unload workers on a temporary basis each for a monthly wage of ₹1,200. As soon as a consignment arrived, a goods received note (GRN) was prepared after inspecting the materials for any damages. If there were damages, the insurance details were verified and the consignment sent to the claims section. The claims section liaised with either the supplier or the insurance company for claims.

Those consignments that were free of damage were set aside for marking codes and placing at the desired location. Simultaneously, the details were filled in the GRN. The GRN details were used to update the stock position and copies of it were sent to other departments such as finance and purchase.

The stores employed a clerk, an inspector and an assistant manager. In addition one forklift truck operator was employed.

Recent Developments

Recently, the company took a decision to decentralize most of the decision-making processes. Under this scheme, each unit was focused on certain areas of the business. The individual unit heads entered into an agreement with the corporate office regarding the target for sales turnover, profitability, and a few other broad parameters. The unit took all operational decisions with the required autonomy.

TABLE 10.10 SOME COST DETAILS EXTRACTED FROM THE STUDY GROUP REPORT AND APPORTIONMENT DETAILS

No	Item	Expenditure*	Purchase and Stores**	Marketing**	Production**	Design**	Finance**	General Administration and Others**
1	Stationery expenses	1,854,000	9	20	17	15	15	24
2	Telephone bills, telex, etc.	1,095,600	16	25	9	7	10	33
3	Taxi, auto (travel expenditure)	1,145,000	8	45	4	3	12	28
4	Postage, etc.	330,000	9	18	—	5	14	54
5	Courier expenses	380,000	10	20	3	—	5	62
6	Freight, demurrage	655,400	40	—	—	—	—	60
7	Insurance premium for stores	742,500	100	—	—	—	—	—
8	Handling costs: fork-lift truck maintenance (depreciation)	765,000	35	—	54	—	—	11
9	Batteries replacement	1,224,500	40	—	50	—	—	10
10	Maintenance of tractors and trailers	149,000	75	5	—	—	20	—
11	Air conditioning	645,000	10	2	7	1	3	77
12	Utilities	2,235,400	7	4	41	3	4	41

Notes: * All values are yearly expenditure in rupees; ** All numbers are expressed as percentages. For limited purposes many departments have been combined with general administration.

A study group was constituted at the Hosur unit to analyse the various expenditures incurred at the plant level. The unit head initiated this as soon as decentralization was put into effect. Until recently, such statistics was not compiled and analysed at the unit level. The units would merely send a host of weekly and monthly reports to the corporate office. The corporate staff performed the task of analysing the data and informing the various unit heads regarding the variances from budgetary provisions. A portion of the cost data gathered so far by the study group is given in Table 10.10. Certain service items such as forklift truck maintenance, insurance premiums freight, and demurrage were allocated to specific departments using the basis developed for costing purposes. Table 10.10 also shows the apportionment details for the items.

QUESTIONS FOR DISCUSSION

1. What are the reasons for high levels of inventory at MML Ltd.? Can you offer suggestions for bringing down the level of inventory?
2. Based on the data provided, compute the cost of carrying inventory at the Hosur unit of MML Ltd.
3. Design an appropriate inventory control system for the A-class items identified by MML Ltd.

10.12 MULTIPLE-CHOICE QUESTIONS

- 10-1.** In manufacturing organizations, finished goods and spare parts typically belong to the category of:
A. Dependent demand
B. Independent demand
C. Continuous demand
D. Uncertainty of demand
- 10-2.** How many types of Inventory are normally found in organizations? What are they?
A. One type of inventory: Physical inventory
B. Two types of inventory: First in First out inventory, Last in First out inventory
C. Three types of inventory: Discriminant inventory, Transit inventory, Buffer inventory
D. Five types of inventory: Seasonal inventory, Decoupling inventory, Cyclic inventory, Pipeline inventory and safety stock
- 10-3.** What are the several costs associated with inventory planning and control?
A. Inventory-carrying Cost, Cost of Ordering, Cost of Shortages
B. Storing Cost, Opportunity Cost, Purchase Cost
C. Transportation Cost, Deterioration Cost, Obsolescence Cost
D. Stock Out Cost, Holding Cost, Shrinkage Cost
- 10-4.** When the total cost is minimum, we obtain the most economic?
A. Resource utilization
B. order quantity
C. And effective process flow
D. None of the above
- 10-5.** When the item is available in stock, there is:
A. No risk of shortage
B. Risk of maintaining the stock
C. Production line can be shut down
D. None of the above
- 10-6.** What are the Inventory Control Systems?
A. Perpetual, Periodic
B. Inventory Management, Technology
C. Continuous Review, Periodic Review
D. Barcode, Radio Frequency
- 10-7.** Selective control of Inventories are:
A. Re-ordering level, Maximum level and Economic Order Quantity
B. HML Analysis, SDE Analysis and Golf Analysis
C. SOS Analysis, FSN Analysis, XYZ Analysis
D. ABC Classification, XYZ Classification, VED Classification and Combination of more than one classification
- 10-8.** In a single-period demand the unfilled demand cannot be:
A. Pre-ordered to the next period
B. Back-ordered to the next period
C. Post-ordered to the next period
D. None of the above
- 10-9.** What is the problem in inventory planning and control?
A. Opportunities for better inventory control and management often exist outside of the materials management function in organization
B. Low Product Turnover Failure to track demand can lead to a lower than expected product turnover due to low demand
C. Excess Inventory
D. Failure to keep track of stock
- 10-10.** Despite their simplicity and initial success, which installation faces several problems after implementations:
A. MRP **B.** SAP
C. ERP **D.** DTP

Answer Keys: 10-1. B 10-2. D 10-3. A 10-4. B 10-5. A 10-6. C 10-7. D 10-8. B 10-9. A 10-10. A

11

CHAPTER

TOTAL QUALITY MANAGEMENT

11.1	The Quality Revolution
11.2	Quality Gurus
11.2.1	William Edwards Deming
11.2.2	Joseph M. Juran
11.2.3	Philip B. Crosby
11.2.4	Karou Ishikawa
11.2.5	Shigeo Shingo
11.2.6	Genichi Taguchi
11.3	Definitions of Quality
11.3.1	Conformance to Specifications
11.3.2	Fulfilling Customer Needs
11.3.3	Fitness for Use
11.4	Total Quality Management
11.4.1	Commitment of Top Management
11.4.2	Employee Involvement
11.4.3	Addressing Training Requirements
11.5	Quality Management Tools
11.5.1	Histograms
11.5.2	Pareto Diagrams
11.5.3	Cause and Effect (Fishbone) Diagrams
11.5.4	CEDAC
11.5.5	Poka Yoke
11.5.6	Quality Function Deployment
11.5.7	Quality Costing

11.6	Quality Certifications and Awards
11.6.1	Quality Awards
11.6.2	Quality Certifications
11.7	Design of Quality Assurance Systems
11.8	Summary
11.9	Review Questions
11.10	Problems
11.11	Net-Wise Exercises
11.12	Mini Projects
11.13	Case Study
11.14	Multiple-choice Questions

POKA YOKE FOR DEFECT PREVENTION

Brakes India Limited (BIL) is a well-known manufacturer of automotive braking systems. BIL's commitment to quality is evident from the fact that they won the Deming Award in 2003, as well as the other quality certifications that they have obtained. One of the methods they use for defect prevention and attaining parts per million (ppm) quality levels is poka yoke.

In their Sholingur plant, it was observed that the defect percentage in a particular process was higher than what was normally expected. The operators collected weekly data and used a simple system to classify the defects into various categories. Every week, an analysis of the data was done to identify high-defect areas. Wrong loading of the component for the second operation was the source of the problem (see Figure 11.1). After careful observation of the process, it was found that the existing operating procedure involved loading the component onto a chuck and using a hammer to gently hit the component to ensure that the chuck had firmly gripped the component. In the process, the worker was occasionally damaging one edge of the component due to fatigue.

Once this problem was identified, the process planning department developed a hydraulic mechanism that would perform the hammering. In the revised process, workers had to merely press a button which would activate the hydraulic mechanism to securely locate the component by hammering. By implementing this revised procedure, they eliminated defects arising out of the wrong loading of components.

Implementing this improvement brought down the defect levels significantly. Later, the group went on to identify more areas for defect prevention to bring the quality to the desired ppm levels.

Source: K. Sundar, "Poka Yoke and Continuous Improvement at Brakes India, Sholingur" (MPT Project Report. IIM Bangalore, 1995).

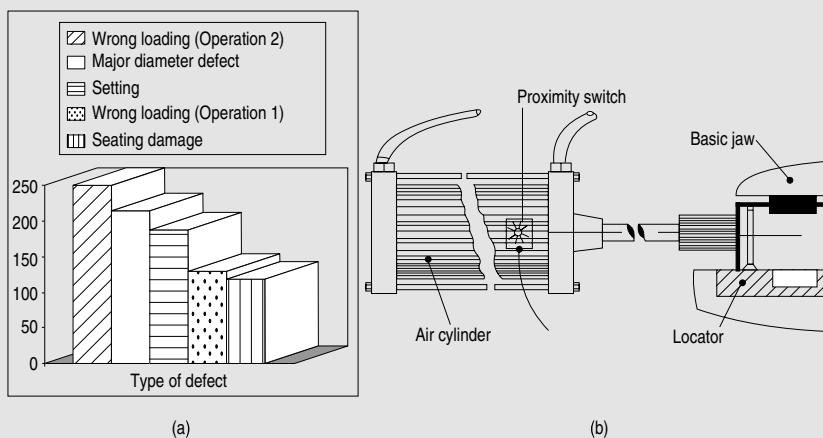


FIGURE 11.1 Wrong loading of the component for the second operation

NOTES

NOTES

Modern-day quality management is characterized by the use of several simple yet powerful tools such as those used by Brakes India Limited. It also requires that the employees participate in the improvement process and take initiatives that help the organization achieve superior quality in the long run. In this chapter, we shall look at various aspects of quality management. Globally, several organizations have adopted new quality management practices to compete effectively in the market.

The resounding success of several manufacturing and service firms in recent times has invariably been linked to excellent practices pertaining to quality management. If you consider the auto-component manufacturers in India, many of them have won the Deming Award for quality, the largest number outside Japan. Similarly, India has the largest number of CMM Level 5-certified software companies in the world. With such international recognition in quality, these two sectors of our industry were able to compete globally. In the past, firms such as Harley Davidson faced critical trade-offs while they made an effort to improve quality. For instance, providing a high-quality product would have meant higher cost, or compromising on flexibility or delivery commitments to the customer. However, successful firms like Kawasaki demonstrated that they need not make any trade-offs with cost or other performance attributes such as delivery and flexibility to offer better quality. On the other hand, they showed that it was possible to make significant improvements in quality and at the same time make the products or services available at a lower cost than that offered by the competition. Much of the success enjoyed by Japanese manufacturers during the 1970s and the 1980s is attributed to this phenomenon.

The resounding success of several manufacturing and service firms in recent times was invariably linked to excellent quality management practices.

In this chapter, we shall see how traditional thinking on quality management has given way for an alternative known as total quality management (TQM). We will also see various components of a good TQM programme and the enabling mechanisms required for implementing a TQM programme.

11.1 THE QUALITY REVOLUTION

The success of Japanese manufacturing firms in the 1970s and the 1980s is largely attributed to a quality revolution sweeping across these firms. Japanese manufacturing firms penetrated established markets in the West by the sheer performance gaps that they created through the quality of the products that they brought to the market. Based on a study of air-conditioner manufacturers in both Japan and the United States during the early 1980s, David Garvin concluded that the best air-conditioner manufacturer in Japan brought the product to the market at a quality level 1000 times

better than the worst manufacturer in the United States. He further showed that the extra cost of making good-quality Japanese goods was much lower than the cost of fixing defective products made by the American manufacturers.

The quality revolution was possible because of new thinking on what constituted good quality and methods. Several new tools were utilized to assess the performance of an organization with respect to quality. Over time, several other tools helped organizations make sustained improvements in quality. At the same time, the top management realized the need for addressing certain mindset issues and providing a leadership role. The roles of middle managers and supervisors underwent a change from control to facilitation of the process of building quality into products and services. Workers were responsible for not only output but also quality.

Such improvements and wide-ranging changes did not happen all of a sudden. A rich repository of knowledge developed by several influential thinkers facilitated this process. Alternative definitions were coined for quality and people in organizations were trained in the new scheme of things. All these eventually led to the quality revolution.

11.2 QUALITY GURUS

The progress made in quality and productivity by organizations is mainly due to the teachings of quality gurus. Between 1950 and 1980, several people suggested alternative approaches to raise quality standards. Therefore, without knowing the salient aspects of these teachings, we may not be able to understand the current best practices in quality management. We shall focus on the teachings of some of the well-known quality gurus.

11.2.1 William Edwards Deming

William Edwards Deming is considered to be the father of the Japanese quality management systems. The Japanese owe much of their success in manufacturing to the teachings of Deming when he was in Japan from 1949–1958. In recognition of his contributions, the highest honour for excellence in quality management, the Deming Prize, is named after him. The Emperor of Japan hands over the Deming Prize to the winners of this award. Several TVS Group companies from India are Deming award winners.

Deming's basic premise regarding quality management was that it is possible to provide good-quality products and services to customers at a much lower price and firms can operate profitably in the long run. He believed, however, that only good firms know this and the others were ignorant of this simple fact. In order to have a systematic approach to quality management, Deming proposed a four-step process, **plan-do-check-act**, popularly known as the **PDCA cycle**. He believed that every successful

NOTES

NOTES

Deming proposed a four-step process, **plan-do-check-act**, popularly known as the **PDCA cycle**.

quality improvement and management programme would invariably follow the plan-do-check-act cycle of events. He proposed a 14-point agenda for quality improvement.

11.2.2 Joseph M. Juran

Juran visited Japan immediately after Deming and imparted to Japanese business executives the key principles behind quality management. Juran's approach to quality management complements that of Deming's. He believed that the quality problems faced by most companies are due the constraints imposed by the top management rather than by those at the operational level. Juran proposed a simple definition of quality—"fitness for use". This definition suggests that quality be viewed from both external and internal perspectives. The interdependency of these functions emphasizes the need for company-wide quality management. Juran also insisted that the senior management play an active role in the quality management process.

Juran's approach to quality management focuses on three important aspects known as **Juran's trilogy**: quality planning, quality control, and quality improvement.

Juran emphasized the need for investing in superior quality systems, including a sound quality planning infrastructure. His approach to quality management focuses on three important aspects, known as **Juran's trilogy**:

- ⇒ *Quality planning*: The process of preparing to meet quality goals
- ⇒ *Quality control*: The process of making quality goals during operations
- ⇒ *Quality improvement*: The process of breaking through to unprecedented levels of performance

11.2.3 Philip B. Crosby

Philip B. Crosby was the corporate vice-president for quality at International Telephone and Telegraph (ITT) in Japan for 14 years, after working his way up from the position of a line inspector. The essence of Crosby's quality philosophy is embodied in what he calls the "absolutes of quality management". The absolutes are:

- ⇒ Absolute I: Definition of quality is conformance to standards
- ⇒ Absolute II: The system of quality is prevention
- ⇒ Absolute III: The performance standard is zero defects
- ⇒ Absolute IV: Measurement of quality is the price of non-conformance
- ⇒ Absolute V: There is no such thing as a quality problem

Crosby introduced the notion of zero-defect performance and hence popularized the phrase “do it right the first time”.

The five absolutes of quality provide a refreshingly different view on the issue of quality management. Crosby believed that the performance standard should be zero defects and hence popularized the phrase “do it right the first time”. “Zero defects” implies that errors should not be expected or accepted as inevitable. Through a system of quality costing, Crosby provided a practical application of his absolute that measurement of quality is the price of non-conformance. Quality cost data are useful to draw the top management’s attention to the problem, to select opportunities for corrective action, and to track quality improvement over time. We shall see in a later section of this chapter how the concept of quality costing can help organizations improve quality management practices.

11.2.4 Karou Ishikawa

Karou Ishikawa was the most prominent figure in Japanese quality management until his death in 1989. Dr Ishikawa influenced the development of a participative, bottom-up approach to quality management. He promoted greater involvement of all employees, from the top management to the front-line staff, reducing reliance on quality professionals and quality departments. He advocated collecting and analysing factual data using simple visual tools, statistical techniques and teamwork as the foundation for implementing quality. Amongst his several contributions, the fishbone diagram (also known as the cause–effect diagram) is widely used in several manufacturing and service organizations worldwide for continuous quality improvement. In a later section of this chapter, we shall discuss several quality control tools.

Ishikawa promoted the concept of company-wide quality through use of quality control (QC) circles. He also proposed the fishbone diagram to analyse quality problems at the workplace.

In a nutshell, Dr Ishikawa promoted the concept of company-wide quality. To help implement the philosophy of participation and to put the quality control tools to gainful exploitation in the hands of everyone in the organization, he developed quality control (QC) circles.

11.2.5 Shigeo Shingo

Shigeo Shingo’s primary contribution to quality management is to suggest an approach for complete elimination of errors. He developed a systematic approach to understand why errors creep into any operation. Further, he devised a method to completely eliminate such errors over a period of time. In this manner, Shingo provided

NOTES

NOTES

an operational angle to achieving zero defects, as proposed by Crosby. This method is called **poka yoke** or mistake-proofing. The idea is to handle errors as they occur. Since identifying the errors is vital to the success of poka yoke, it must be carried out rigorously and thoroughly. Once an error (or a defect) is found, a detailed analysis is done to understand what caused the error. Based on the analysis, alternative methods are identified to eliminate the cause once and for all.

Shigeo Shingo proposed a method called **poka yoke** or mistake-proofing for eliminating defects once and for all.

11.2.6 Genichi Taguchi

Taguchi's focus was on achieving quality by reducing variations in processes. His approach is based on the basic premise that variations in processes eventually manifest as variations in the quality performance of the product or service. Therefore he advocated the need to keep the variations in the processes under control. A different method of measuring quality, the *loss function*, is central to Taguchi's approach. The loss function measures quality. A schematic representation of the loss function is given in Figure 11.2.

Taguchi proposed a three-stage approach to the design process: **systems design, parameter design, and tolerance design.**

As shown in the figure, the loss function establishes a financial measure of user dissatisfaction with a product's performance as it deviates from a target value. The loss function captures the extent to which a process goes off target. Minimizing the loss calls for controlled experimentation in the process and setting up of various process parameters. Taguchi proposed detailed procedures for the design of experiments that must precede setting various process parameters.

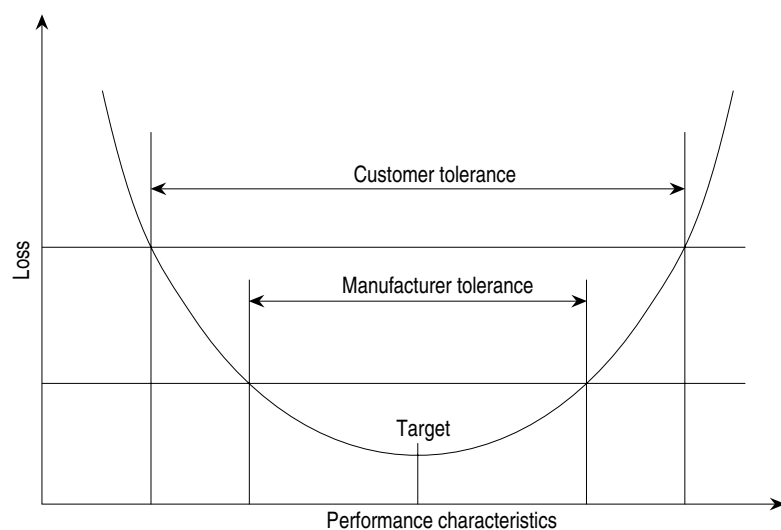


FIGURE 11.2 A schematic representation of Taguchi's loss function

11.3 DEFINITIONS OF QUALITY

The quality gurus provided alternative approaches to build quality into the products and services that a company offers. The critical differences in their approaches stem from their definition of quality. For instance, Deming's emphasis on process improvements and use of statistical process control charts (SPC) clearly point to a definition of quality that is linked to specifications. On the other hand, Taguchi's loss function or Juran's notion of "fitness for use" extends the definition of quality to a larger domain. Therefore, it is important to understand the various definitions of quality and the implications of these alternative definitions.

11.3.1 Conformance to Specifications

The easiest definition of quality is conformance to specifications. This definition focuses on the manufacturing process and motivates the quality manager to develop mechanisms for ensuring that specifications are met with. Every component of a manufactured product has specifications. As long as the specifications are met, it is assumed that quality objectives are met with. Such a definition is very crucial for a manufacturing firm as it enables the management to develop unambiguous instructions for the worker on the shop floor to maintain quality. Just as we have specifications for products, specifications for services can also be developed. For instance, in a hotel, one could set a specification that check-in of arriving guests should be done in 3 minutes $\pm$ 30 seconds. Or, in a bank, the time for servicing a customer at the teller's counter could be specified and the process monitored and controlled to meet this specification.

11.3.2 Fulfilling Customer Needs

In the previous definition of quality, we made an important assumption that the specifications are right. Consider a two-wheeler such as the Kinetic Nova. Let us assume that the specifications include an overall weight of 115 kg, a height of 42 inches, and so on. If the manufacturing process is able to ensure that the scooters manufactured meet these specifications, have they met the quality objectives? The answer depends on whether they met the customer needs: If the customers have felt a strong need for a lighter scooter (less than 95 kg), with moderate height (less than 40 inches) then perhaps the scooter is not of good quality. Therefore, this definition of quality focuses not merely on the manufacturing process but also on the process of arriving at the specifications themselves. It emphasizes the need to have a system of collecting information regarding customers' perceptions about the product and preferences before the specifications are arrived at.

An alternative method to define quality is to understand the various dimensions of quality. Garvin proposed that quality has eight dimensions.

NOTES

NOTES

11.3.3 Fitness for Use

Quality can also be defined as “fitness for use”. Such a definition of quality demands that an organization understands the preferred methods of use by prospective customers over an extended period of time. In order to develop this understanding, an organization needs to focus on customer relationships, the design, and the manufacturing processes. A variation of this definition is minimizing the loss to society. This definition of quality emphasizes a life-cycle approach to building quality. Ensuring quality in this manner also demands that considerable efforts are made at the concept and design stages of a product.

An alternative method to define quality is to understand its various dimensions. Garvin proposed eight dimensions for quality:

- ⇒ *Performance*: The customer expects a certain level of performance from a product. It relates to the operational characteristics of the product. An industrial cooling application, for instance, requires that the fan be operational for a certain duration, without stopping at a particular temperature.
- ⇒ *Features*: In addition to meeting the basic performance requirements of the product, features provide additional attributes that enhance the quality.
- ⇒ *Reliability*: The expectation that the product will perform satisfactorily for a period of time is known as reliability. For instance, if a two-wheeler performs without any major repair or maintenance for five years, it may indicate a certain level of reliability.
- ⇒ *Conformance*: Meeting the specifications and standards of design.
- ⇒ *Durability*: How long the product lasts before it requires a replacement. Clearly, durability and reliability are related.
- ⇒ *Serviceability*: The ease with which the product can be serviced.
- ⇒ *Aesthetics*: Customers also value the aesthetics of the product. One can relate this idea to the various feel-good factors that a customer evaluates before making a choice of, say, competing brands of passenger cars.
- ⇒ *Safety*: Safety aspects denote the assurance to the customer that there are no hazards in using the product.
- ⇒ *Other perceptions*: Customers may also have a host of subjective perceptions such as brand name, image, impact of advertising, etc. in his/her assessment of quality.

Irrespective of the nature of definitions, there are some important implications for quality managers:

- ⇒ *Quality is both qualitative and quantitative*: Attributes of good quality are 100 per cent quantitative on the shop floor. A process engineer cannot ensure quality if the specifications of the product and the processes are ambiguous. On the contrary, on the

ultimate customer side, attributes of good quality are 100 per cent qualitative. Customers usually recognize good quality, not by a set of attributes, but by experience and intangible issues. Thus, a customer using similar products from two competitors will find it much easier to rate one to be of better quality than the other. However, if you ask the customer to articulate clearly the attributes that made him/her conclude why one product was better than the other, he/she may have difficulty. Therefore, a good quality management system should enable the manager to understand the qualitative attributes that influence a customer, and have a method of translating these into unambiguous quantifiable parameters for design and manufacturing. The critical success factor lies in handling this challenge.

- ⇒ *Changing definitions*: The definition of quality appears to be constantly changing over time. While it reflects the need to keep up with changing ground realities, the implication for organizations is to constantly invest in new capabilities and skills.
- ⇒ *Quality is a moving target*: The operational aspects of the zero-defects measure of quality clearly demonstrate that quality is a moving target. For instance in their journey towards zero-defect status, organizations may change their measure of quality initially from percentage of defects to parts per million, and later to parts per billion. In such a scenario, how are organizations going to keep up their employees' enthusiasm in meeting the targeted quality standards? The implications for organizations are that there are mindset issues to be addressed in achieving a good quality management programme.

11.4 TOTAL QUALITY MANAGEMENT

The culmination of the discussions in the previous sections is the notion of total quality management (TQM). Modern quality management systems are built on the concept of TQM. **Total quality management** can be defined as an organization-wide effort to develop the systems, tools, techniques, skills, and mindset required to establish a quality assurance system that is responsive to emerging market needs. Such an effort will provide distinctive advantages to a firm and will enable it to compete effectively in the market. A good TQM programme consists of the following important elements:

- (a) Role of the top management
- (b) Employee involvement for continuous improvement
- (c) Addressing the training requirements of employees
- (d) Tools and techniques for quality assurance and continuous improvement

Total quality management (TQM) can be defined as an organization-wide effort to develop the systems, tools, techniques, skills, and mindset required to establish a quality assurance system that is responsive to emerging market needs.

NOTES

NOTES

11.4.1 Commitment of Top Management

Total quality management calls for radical changes in the thinking among employees. First, the term “total” connotes “everyone”, “everywhere” and “every time”. Implementing this simple slogan is not an easy task. Involving everyone means that right from the CEO down to the lowest level employee in an organization, all must play their role in quality assurance. This would require that all employees have role clarity with respect to quality matters. In the traditional method, production workers were not expected to worry about quality. Their job was to engage in jobs allotted to them and take care of their productivity bonuses and earnings. The quality control department would assess quality and initiate the required actions, should there be a quality problem. In the TQM philosophy, the responsibility for quality shifts to the production workers.

Similarly, in the past, middle managers and top managers played the role of controllers. They directed improvement efforts and established tight controls and supervision to ensure that quality goals were met. On the contrary, in TQM, the focus shifts to one of facilitating the production workers in their efforts towards producing quality goods and services. This calls for a certain shifting of process ownership to the workers and their supervisors. The success of the TQM programme depends on how well this transition is made.

The notion of “every time” puts a different pressure on the traditional methods of operations. There is a greater degree of intangibility and uncertainty with respect to outcomes in the long term. On the other hand, short-term goals and benefits are like low-hanging fruits. They are not only easily reachable but also tangible. Managers are averse to working for long-term benefits and tend to compromise on them in order to post impressive performances in meeting short-term goals. If, for some reason, there is a quality problem in the production line, the manufacturing managers will choose to ignore it and continue to produce. When quality control rejects the components, they could be reworked later. Therefore, the top management has to play an important role in ensuring that “every time” is followed in letter and spirit. This may call for leadership and guidance in minimizing conflicts between short-term and the long-term goals.

Nothing is taken seriously when employees in organizations come to know that the changes do not have the top management’s involvement and commitment. This is a significant aspect of the management of change. The top management also needs to play a vital role in communication. They should not miss even a single opportunity that comes their way to demonstrate their support and

The term “total” in total quality management connotes “everyone”, “everywhere”, and “every time”.

QUALITY POLICY IN SOME ORGANIZATIONS

GE Healthcare

Quality and continuous improvement are part of our culture, it is embedded in our processes, our recruitment, and our training. We take responsibility for the projects we execute, for maintaining the operation of the products we deliver, and for the traceability required by the authorities.

Hero Honda

Excellence in quality is the core value of Hero Honda's philosophy. We are committed at all levels to achieve high quality in whatever we do, particularly in our products and services that will meet and exceed customer's growing aspirations through:

- ⇒ Innovation in products, processes, and services.
- ⇒ Continuous improvement in our total quality management systems.
- ⇒ Teamwork and responsibility.

Sasken

Achieve excellence through commitment and innovation surpassing the expectations of our customers. In this endeavor, we shall

- ⇒ Set up and implement world-class systems and processes
- ⇒ Sensitize and raise our quality consciousness
- ⇒ Continuously strive for improvements to make quality a way of life
- ⇒ Meet all applicable environmental, health, and safety standards and demonstrate corporate citizenship on an ongoing basis

Sources: http://www.gehealthcare.com/euen/hospital_solutions/quality_certification.html; http://www.herohonda.com/co_policy.htm#qp; and <http://www.sasken.com/aboutus/quality.html>, accessed on 25 December 2008.

NOTES

involvement in the TQM implementation process. Moreover, issues pertaining to “everyone” and “every time” require clear directions and a leadership role from the CEO of an organization. Only then can the TQM process get a fair chance at success.

A well-defined quality policy serves to signal the seriousness of an organization and the nature of its commitment to various stakeholders in the system. Therefore, the first step in launching a TQM programme pertains to stating a quality policy. Many Indian organizations have recently set forth their quality policies and have communicated it to all their stakeholders. It is not surprising that the top management takes a major responsibility in this process and thereby steers the TQM programme. They help establish an unambiguous quality policy and communicate it to all employees in the organization.

NOTES

11.4.2 Employee Involvement

Shifting the responsibility of quality management to the workers must be accompanied by the necessary organizational changes enabling the employee to take on the responsibility. The important change is process ownership by the employee. **Process ownership** is the transformation of the role of employees from the function of merely accomplishing the assigned tasks related to a process to one of thinking through the various aspects of the process and taking on a conscious role in the overall management of the process.

Consider two alternative roles for employees. In the first case, the employee will be merely focusing on producing a component using a machine. Maximizing production (and thereby the productivity bonus) may be of interest to the employee. The employee may not be motivated to maintain cleanliness in the working area or ensure good maintenance of the machine and other resources. Further piling up of inventory and problems related to the quality of the component are not his concern. There may be separate specialists to attend these problems.

In the second case, in addition to the production, the employee will have the ultimate responsibility of maintaining all the aspects of the process. He will be responsible and accountable for all local issues pertaining to the process. This manner of relating to the entire process builds a sense of ownership in the employee and makes him/her get involved more intimately in the process. Employees eventually tend to take a greater interest in the overall improvement of the process, leading to better quality of the product. Therefore, employee involvement is an important aspect of TQM.

Employee involvement invariably requires employee empowerment. Employees must be allowed to take decisions pertaining to stopping the line when quality problems surface and they need mechanisms to work in small groups to make continuous improvements in the process. Each employee must also be able to improve the housekeeping of the resources in his/her work area and improve the upkeep and maintenance of machines and other resources. These require several organizational and procedural changes in the organization.

The other aspect of empowerment pertains to equipping employees with the required skills and tools for managing quality. Figure 11.3 shows the normal steps involved in quality improvement through a series of projects done in small groups. In order to be problem solvers, employees need problem-solving tools. They need to be trained in the required quality control tools. Small group improvement projects are the basic building blocks of a

Process ownership is the transformation of the role of the employees from a mere function of accomplishing the assigned tasks to one of taking a conscious role in the overall management of the process.

TQM programme for improving the quality over a period of time. This calls for a set of skills oriented towards working in a team; in other words, interpersonal skills to work with fellow employees. Moreover, employees need better communication skills to deal with one another in a group and with their subordinates and supervisors. Therefore, addressing these training needs is an important component of the TQM implementation process.

NOTES

11.4.3 Addressing Training Requirements

Successful TQM programmes demand substantial investment in communication. The top management is first educated regarding the need, the required changes and the potential benefits. The process then continues, until the lowest level in the organizational hierarchy is reached. The top management should be aware of the impending changes and the benefits. Only out of such an understanding can there be involvement and support in resolving conflicts that arise during the implementation stage. Middle-level managers often form part of the team of change agents. Moreover, changes may result in some of them giving up their traditional role of control over their subordinates. They need adequate exposure and clarity in the required changes, failing which they become insecure and try to stonewall the efforts. The workforce, the union, and the supervisors as well as the staff and managers belonging to the manufacturing-support and non-manufacturing areas should be appraised of the nature of changes and the potential benefits.

Communicating the need for alternative procedures, new methods of empowerment, and the benefits that are likely to accrue to the organization is just one aspect of training. The other aspect relates to educating all the employees in using the new quality control tools and techniques to be adopted for quality management. At the operational level, several new quality control tools are available. Employees need to be trained in using these tools. Supervisors require training in problem-solving approaches and working in small group improvement teams. Supervisors and the middle managers require several new skills

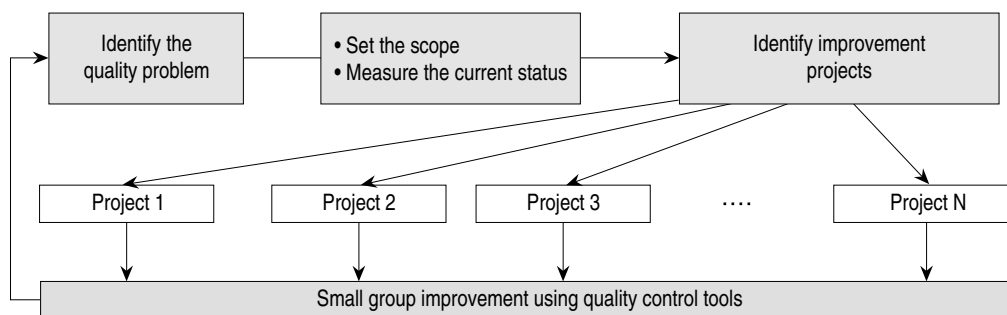


FIGURE 11.3 Quality improvement through small group improvement projects

NOTES

for setting up a good quality management system at work. These include the following:

- ⇒ Identifying problem areas
- ⇒ Setting targets for improvement
- ⇒ Participating in the improvement process
- ⇒ Facilitating the process owners and monitoring and guiding them over time

11.5 QUALITY MANAGEMENT TOOLS

The greatest strength of a quality management system such as TQM is the emphasis laid on using simple yet powerful tools for tracking various aspects related to quality management. Over the years, several QC tools have been developed. These tools are widely applied in practice too. The QC tools fall under five broad categories. Table 11.1 has a description of these categories and the tools available in each category.

Let us take a look at the categories of tools discussed in the table:

- ⇒ *Tools for highlighting quality problems:* These tools bring impending quality problems in a process to the attention of the operational personnel and alert him/her to the issue. The most

Quality management tools available tools fall under five broad categories:

- ⇒ Tools for highlighting quality problems
- ⇒ Tools for identifying specific improvement opportunities
- ⇒ Tools for analysing problems and their root causes
- ⇒ Tools for operational planning
- ⇒ Tools for strategic planning

TABLE 11.1 TOOLS AND TECHNIQUES FOR TQM

Purpose of the Tool	Quality Control Tools	Management Tools
Highlighting problems	⇒ Control charts	
Identifying specific improvement opportunities	⇒ Histograms ⇒ Check sheets ⇒ Pareto diagrams ⇒ Scatter diagrams ⇒ Graphs	
Analysing problems and their root causes	⇒ Cause and effect (Fishbone) diagram ⇒ CEDAC	⇒ Affinity diagram ⇒ Relationship diagram
Operational planning for building quality into products/services		⇒ Tree diagram ⇒ Matrix diagram ⇒ Matrix data analysis ⇒ Process decision programme chart (PDPC) ⇒ Arrow diagram ⇒ Poka yoke
Strategic planning		⇒ Quality function deployment (QFD) ⇒ Quality costing

commonly used tool is the process control chart. By carefully selecting a set of important process parameters, it is possible to set up a continuous monitoring system using statistical principles to identify when the process is likely to go out of control. An out-of-control situation points to deteriorating quality in the process. It triggers closer observation and further analysis to nip the problem in the bud.

- ⇒ *Tools for identifying specific improvement opportunities:* Once an impending quality problem is brought to the attention of the concerned supervisor, manager and the employees involved in the work area, additional tools are required to zero in on the exact problem. This could be done by collecting some additional data, systematically plotting the data and analysing it. Several tools are available to perform this task. The most popular among them include check sheets, histograms, and pareto diagrams. Using these tools, it is possible to locate exactly where the problem lies.
- ⇒ *Tools for analysing problems and their root causes:* The next step in quality assurance is to conduct a systematic analysis of the problems and their root causes. Only when the root causes are known can solutions be found for the elimination of the problem. The most popular tool used in this category is the cause and effect diagram (also known as fishbone diagram) and a variation of it known as the cause and effect diagram with action card (CEDAC).

These categories are tools for operational control. Once a quality problem manifests, these tools help in recognizing it and identifying the root causes. Once the causes and remedies are identified, it is possible to rectify the problem. However, a good quality management system also requires tools for planning purposes. Some of the improvement methods discovered while using the quality control tools can be permanently incorporated into operations in the future. Therefore, the use of quality tools that help prior planning minimizes the occurrence of quality problems during operations. Two categories of planning tools are advocated in TQM.

- ⇒ *Tools for operational planning:* Operational planning tools pertain to prior planning done to minimize quality problems in operations. For instance, if some mistake-proof methods are incorporated in the system, it results in complete elimination of the problem.
- ⇒ *Tools for strategic planning:* The strategic tools help an organization link the quality management initiatives to the requirements in the market place and provide overall direction for operational planning and improvement activities. Two popular tools in this category include the quality function deployment (QFD) and quality costing.

We shall now at some of these tools in greater detail.

NOTES

NOTES

11.5.1 Histograms

The **histogram** is a simple method of graphically representing the frequency distribution of multiple attributes of interest. Based on a simple count of the number of occurrences of each attribute, a histogram can be constructed. In a manufacturer of earth moving equipment a study revealed that poor quality resulted in various snags in the assembly plant. Due to a large number of adjustment snags, the throughput time was much higher. The data and the accompanying graph in Figure 11.4 shows the various causes for adjustment snags.

The value of a histogram lies in its ability to graphically portray the various causes pertaining to the problem as well as the magnitude of these causes. Clearly, employees would like to focus on major problem areas and initiate further data collection to understand why problems occur in those areas. For instance, in this example, problems due to reworking are very significant. Therefore, employees would like to know more about the causes of such problems.

A **histogram** is a simple method of graphically representing the frequency distribution of multiple attributes of interest.

11.5.2 Pareto Diagrams

The process of data collection helps to understand where the problem lies and what the nature of the problem is. Using this information, it is easy to identify which aspects need improvement. The difficulty, however, is deciding where to start the improvement process. A **pareto diagram** is a method by which clear-cut priorities are established for directing the improvement efforts. Moreover, it helps organize the basic data in a systematic manner such that the improvement team detects the significant aspects of improvement and initiates necessary corrective actions.

Let us look at the histogram shown in Figure 11.4 in an alternative fashion by plotting the data in decreasing order of occurrence.

A **pareto diagram** is a method by which clear-cut priorities are established for directing the improvement efforts.

Causes for adjustment snags	Number of occurrences
Leakage	250
Missing components	240
Fouling	50
Reworks	260
Poor routing	50
Loose fitting	150

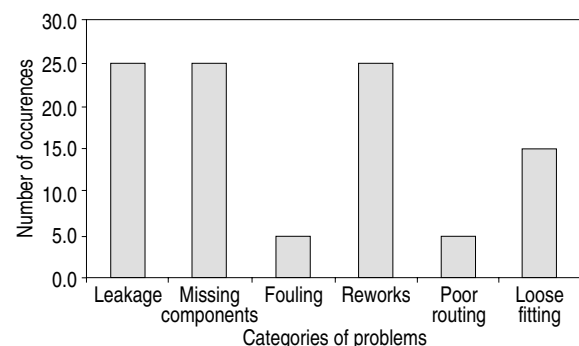


FIGURE 11.4 Histogram for the causes of adjustment snags

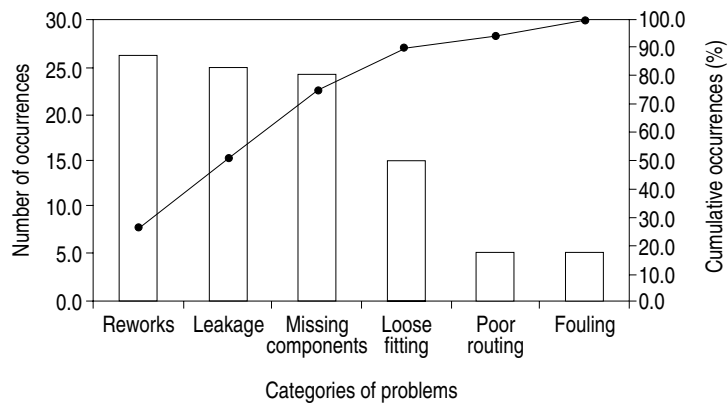


FIGURE 11.5 A pareto diagram for the causes of adjustment snags

The revised histogram is presented in Figure 11.5. The revised representation of the histograms clearly establishes priorities for improvement. In the case of adjustment, snags, reworks, leakages and missing components are important elements that require immediate attention.

11.5.3 Cause and Effect (Fishbone) Diagrams

The earlier tools help to decide where to focus the improvement efforts for reducing defects. However, they do not point to the root causes of the problem. The **cause and effect diagram**, or the **fishbone diagram**, is a generic methodology developed to trace problems to their root causes. Since a total elimination of the problem requires understanding the root causes of the problem, this technique is vital for quality improvement.

At a more general level, a cause and effect diagram has a structure similar to that shown in Figure 11.6. The technique is based on the assumption that good or bad quality is on account of various causes pertaining to the process. Typically, the causes are due to the choice of material, work method, and equipment used as well as the impact of labour practices. A fishbone diagram helps the quality improvement team to analyse problems in a structured manner and identify the root causes.

The cause and effect diagram is a very useful tool for problem solving in the small group improvement efforts that an organization wants to pursue. The tool enables members of the group to engage in a group discussion. Based on the discussion on each component of the fishbone diagram, it is possible to identify a set of potential causes for the problem. Organizations typically use the right side of the fishbone diagram to set the target for improvement. For example, in the case of Figure 11.6, we may like to set a target of 100 parts per million defects as the objective for the exercise and direct improvement efforts towards this.

The **cause and effect diagram**, or the **fishbone diagram**, is a generic methodology developed to trace the problems to their root causes.

NOTES

NOTES

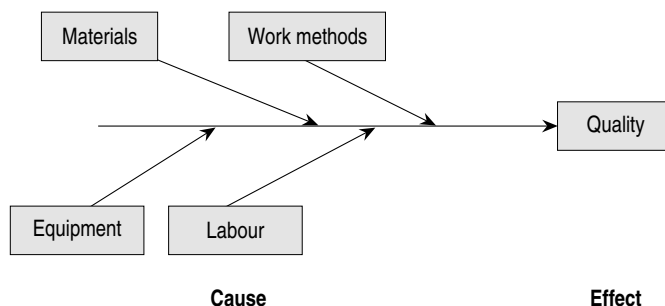


FIGURE 11.6 A generic representation of a cause-and-effect diagram

11.5.4 CEDAC

A variation of the cause and effect diagram is CEDAC. In this, the “AC” stands for “action cards”. In the cause and effect diagram, there is a fishbone structure and the members of the small group improvement activity need to discuss on the problem, causes and solutions. On the other hand, in the case of CEDAC, this entire process is left open to all employees of the organization by adding two sets of cards to the fishbone. One set of cards is known as the *problem cards* and another set is known as the *solution cards*. By placing a visual board with the fishbone structure and problem and solution cards in two separate bins alongside the board, CEDAC enables any employee to contribute to problem solving. Normally, the problem and solution cards are colour-coded for easy identification. Figure 11.7 shows one such CEDAC system that was implemented as part of a study in an automobile braking system manufacturing plant in India.

A variation of the cause and effect diagram is CEDAC. In this, “AC” stands for “action cards”.



FIGURE 11.7 An example of CEDAC implementation

The implementation works as follows. Any employee who thinks that he/she knows what the problem or solution to the problem could be will pick up an appropriate card. He/she will then write his/her suggestions and affix it at the appropriate leg of the fishbone. Figure 11.7 is a snapshot of the status of the board after a week of its installation. Even in such a short period of time, nearly a dozen problem cards and another dozen solution cards have already been affixed to the board. It is clear from this example that the CEDAC system establishes an atmosphere of continuous improvement. It enables all the employees in an organization to make full use of their accumulated knowledge and experience. Furthermore, it helps change the organizational culture by empowering people to fully participate in the continuous improvement process.

11.5.5 Poka Yoke

Poka yoke is a Japanese term for “mistake-proofing of operations”. Shingo proposed this method while he improved the Toyota Production System. The basic principle behind this method is that several defects that creep into an operation are usually avoidable. Errors and defects have a cause and effect relationship. If inadvertent errors in any process are not recognized and methods are not established to eliminate them, they will eventually manifest as defects. Only by a careful scrutiny of the process is it possible to identify the root causes of the defects. The root causes could then be completely eliminated by redesigning the operations and incorporating methods by which the process will not allow the errors to happen in the future.

Poka yoke is best explained with an actual example. Consider a component manufactured by an automobile component manufacturer that requires a through hole to be drilled using a hand-operated drilling machine. If the worker operated the hand drill imperfectly, it is likely that he/she would not have traversed the drilling tool till the very end. Consequently, a through hole is not drilled. It may appear as though such errors are unlikely to creep into a manufacturing system. However, in reality, when a large number of pieces are produced, workers tend to commit such mistakes out of sheer mental and physical fatigue. In the automobile component manufacturing firm, the defect was found at a much later stage, during assembly.

When the worker at the assembly section wanted to insert another component he found that there was no hole (since the component is not drilled right through). Discovering this at this stage meant tearing down the assembly to remove the component and replacing it with a good-quality component. This resulted in considerable loss in productivity and added to the cost. One way to avoid this simple quality problem is to undertake mistake-proofing in the drilling process so that imperfect drilling of a component is completely avoided. Figure 11.8 shows the new arrangement.

NOTES

NOTES

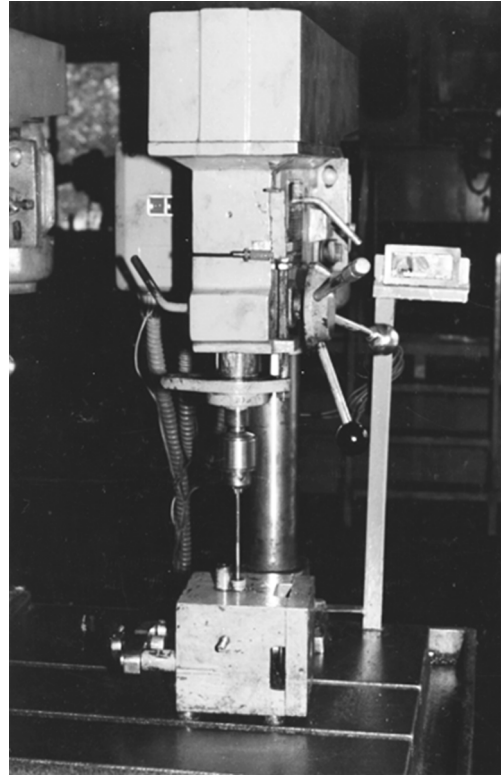


FIGURE 11.8 Defect prevention through operations planning

In the new arrangement, there is a pair of sensors controlling the hand drilling process. While the sensor at the top activates the forward movement of the handle, the sensor at the bottom activates the return path for the handle. Once the worker begins the forward movement of the spindle, there is no way he can retract the spindle, except by the activation of the lower sensor. In this process, it is ensured that a through hole is drilled. By incorporating these features in the process through prior planning, quality is built into the product.

11.5.6 Quality Function Deployment

Quality function deployment (QFD) is a tool developed to address these requirements. It is a technique introduced in Japan by Yoji Akao in 1966 and used extensively by Toyota. According to Akao, QFD “is a method for developing a design quality aimed at satisfying the consumer and then translating the consumer’s demand into design targets and major quality assurance points to be used throughout the production phase.” The technique aims to capture what the customer needs, and link it to operations to ensure that it is achieved.

Based on a set of structured information gathering, QFD helps an organization to capture the qualitative attributes of the quality as perceived by the customers and translate them into meaningful quantitative measures.

MATRIX DIAGRAM FOR ENHANCING THE COMPETITIVENESS OF A PRODUCT

The matrix diagram is one of the quality management tools used in operational planning and for building quality into products and services. In a typical matrix diagram method, a two-dimensional matrix is constructed to analyse the quality problem and identify areas that require further improvement. Once the two dimensions of the matrix are identified, the relevant data is collected and represented in the matrix diagram. The visual method helps the management to easily identify areas that require improvement and to focus their efforts.

Several years ago, Hindustan Motors, a manufacturer of earth-moving equipment, undertook an improvement project to enhance the competitiveness of one of its products, the Hindustan 1035N dumper. The first step in this process was to identify key areas that it needed to improve vis-à-vis its competitors. Once Hindustan Motors was clear about the areas that required improvement, it was easier to focus improvement efforts through a small group “project-by-project” approach. The use of the matrix method came in handy for this purpose. To construct the matrix diagram, the two dimensions were first identified. On one dimension, the product attributes valued by the market were categorized as less important, order-qualifying, and order-winning. On the other dimension, the performance of 1035N Dumper vis-à-vis the competing products was classified as worst, same and better. Ten attributes were identified and for each attribute a 9-point scale was developed to rate the 1035N dumper along both these dimensions. Using this scheme, a matrix diagram was constructed (see Figure 11.9 for details).

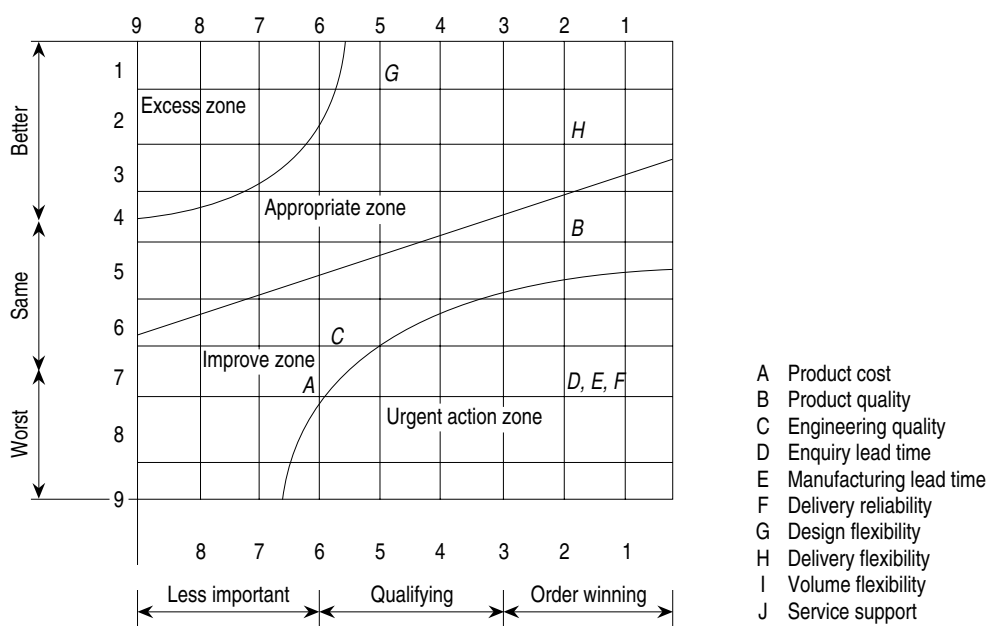


FIGURE 11.9 A matrix diagram to identify improvement opportunities

NOTES

Based on this analysis, it was found that four attributes required immediate action. These included enquiry lead time, manufacturing lead time, delivery reliability and volume flexibility. Further analysis clearly showed that in order to enhance the competitiveness of the product, throughput time reduction was very important. Based on this analysis, some throughput-time reduction exercises were taken up.

Source: N. Ravichandran, "A Framework for Creating Competitive Advantage Through Throughput Time Reduction" (MPT Report, Indian Institute of Management, Bangalore, 1996).

Based on a set of structured information-gathering processes, QFD helps an organization to capture the qualitative attributes of the quality as perceived by customers, and translate them into meaningful quantitative measures. Once this transition is made, their functional aspects are deployed at various levels and control systems are established to monitor the extent to which these are effectively deployed.

QFD utilizes a set of four "house of quality" to translate the customer requirements/preferences into design features. It is a conceptual map of inter-functional planning and communication that helps rationalize design priorities from the preferences of diverse stakeholders. Each house of quality captures certain information and uses a structured methodology to translate customer needs to actionable ideas and quantifiable attributes pertaining to the product/service. Figure 11.10 is a broad representation of the four houses. As is clear from the figure, during each stage of the process, specific details are culled out of the basic understanding of customer needs. In the first stage, design attributes are arrived at. During the second stage, the design attributes are linked to actionable items. The actions identified in the second stage are the basis for the third stage in aiming at specific decisions to be implemented. Finally, in the last stage, specific details pertaining to each process parameter are arrived at. The four houses are sequentially linked to one another. Therefore, the outcome of

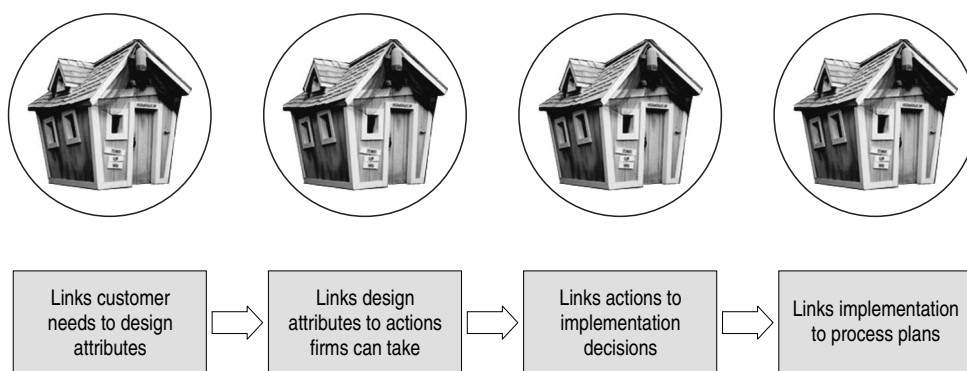


FIGURE 11.10 The four houses of quality

each preceding stage becomes the input for the next stage. Each stage is known as a “house of quality” simply because it is constructed in the shape of a house.

Since the first stage of house of quality translates customer attributes into design attributes, it sometimes needs to address conflicts between various customer attributes. For example, sometimes a customer may want a lightweight product, but also require strength and sturdiness to handle failures. If these two features are not conflicting, then it would have been pretty straightforward to translate into an actionable design attribute. Sometimes, it may be possible for the designer to select materials that are lightweight and sturdy to handle crashes. Otherwise, house of quality provides a structured approach to resolve this conflict by examining the relative importance of various customer attributes. It also considers the customer’s willingness to pay to resolve these trade-offs. A similar approach is followed in each of these houses to address conflicting requirements to eventually translate the customer attribute into production process parameters.

11.5.7 Quality Costing

A useful tool for quality management is quality costing. **Quality costing** is a technique by which all the costs related to the delivering of a certain level of quality are captured and appropriately classified for the purpose of evaluating the quality management efforts in an organization. By translating the quality performance into monetary terms, quality costing captures the attention of the top management. Furthermore, it enables firms to prioritize various spending opportunities in the area of quality improvement. Juran advocated the notion of quality costing and Crosby encouraged firms to put a good quality costing system in place. According to Crosby, firms that begin measuring quality costs will first realize that nearly 25 per cent of their sales turnover is spent on quality management.

Quality costing is a technique by which all the costs related to the delivering of quality are captured and classified for evaluation of the quality management efforts in an organization.

Quality costs in any organization fall under four categories (Figure 11.11). *Prevention costs* include all costs incurred to prevent, reduce, or eliminate the occurrence of defects. Expenses incurred towards training of employees in better methods, vendor development and small group improvement projects are some of the examples in the category. On the other hand, *appraisal costs* pertain to all expenses incurred in ascertaining the quality of a product, component or a service. The third category of costs relates to *costs of internal failures*. An appraisal activity may reveal defects and may call for scrapping components altogether or necessitate rework in several situations. This results in

NOTES

NOTES

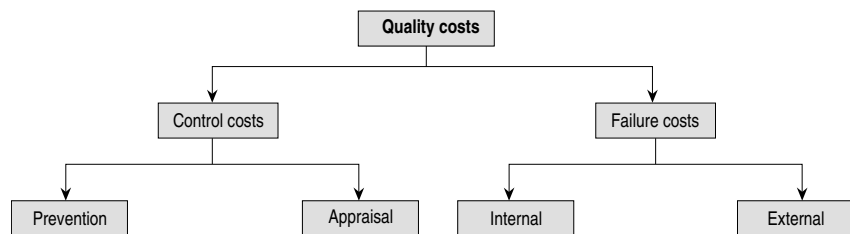


FIGURE 11.11 Classification of quality costs

lost productivity and added costs for rectification. All these costs are classified as internal failure costs. The final category is the *external failure costs*. All costs related to rectifying the defects after the product is shipped out of a manufacturer's premises will be classified as external failure costs. The costs include warranty, cost of product returns and replacement, on-field maintenance, and any compensation for the damages incurred. It also includes the cost of lost customer goodwill.

Quality costing has several useful purposes. First, by translating quality parameters into monetary terms, it promotes quality as an important business parameter and helps capture the attention of the top management of an organization. Quality costs can be used for motivational purposes with regard to improvement at all levels in the organization. It is often found that many organizations are unclear about how to improve. In such cases, the use of quality costs could very well provide some direction. This is so as it enables an organization to set targets and budgets pertaining to quality improvement.

11.6 QUALITY CERTIFICATIONS AND AWARDS

Benchmarking exercises are an important component of good quality management. In the absence of these, an organization may not know how well it performs in quality management vis-à-vis its competitors. Amongst the several alternatives for benchmarking, one useful method is to make use of third-party quality rating services to assess the status of an organization's quality management practices. Use of third-party rating systems offers several advantages to an organization. It provides a neutral, unbiased, and fair method of evaluating its quality management practices. With such a rating, an organization may find it much easier to convince its prospective customers about its performance in quality management. This is especially true if the organization needs to tap overseas markets. Moreover, quality certifications and awards provide additional reputation effects to an organization and help the organization increase its market share and profitability in the long run.

The second advantage is that a third party rating provides a uniform nomenclature and helps understand the status of quality management cutting across geographical and sectoral domains. Finally, a third party will be better equipped to constantly invest

in newer and better methods of measurement and it would be wasteful for every organization to individually invest in skill and expertise building in these areas. Therefore, quality certification and awards have become not only popular but also important for every organization to compete in a globalized market.

11.6.1 Quality Awards

Different countries have instituted quality awards to motivate business organizations towards achieving excellent performance. Table 11.2 has a list of awards and awarding agencies. The Deming Prize was one of the earliest awards instituted for excellence in quality management. Although only a few companies outside Japan have bagged the award, a majority of them are from India (Table 11.3). Although these awards differ from one another in the manner in which winners are selected, all of them have a system of identifying key components and a scoring

NOTES

TABLE 11.2 A LIST OF IMPORTANT QUALITY CERTIFICATIONS AND AWARDS

Name of the Award/ Certification	Awarding Agency	Applicable to Whom	Remarks
<i>Some well-known quality awards</i>			
The Deming Prize	Union of Japanese Scientists and Engineers (JUSE)	No geographical restrictions for applicants	More than one may be selected for the prize
The Malcom Baldrige National Quality Award	National Institute of Standards and Technology (NIST), U.S.	Open only to corporations based in the United States	Only one winner is selected for each category
European Quality Award	European Foundation for Quality Management (EFQM)	Open only to European companies	Only one winner selected for each category
The CII-EXIM Business Excellence Award	Confederation of Indian Industries (CII)	Any company in India can apply	Only one winner is selected for each category
The Rajiv Gandhi National Quality Award	Bureau of Indian Standards (BIS)	Indian companies can apply	Four categories (large-scale manufacturing, small-scale manufacturing, service sector, best of all)
<i>Some well-known quality certifications</i>			
ISO 9000 series, ISO 14000 series (Environmental issues), QS 9000 series (Automotive sector), TL 9000 series (Telecommunication Sector)	International Organization for Standardization	Any company can apply for the certification	Rating agencies assess and recommend certification. Certificates valid for three years
OHSAS 18001	An association of national standards bodies	Companies can apply to the respective national standards bodies	Deals with occupational health and safety management

TABLE 11.3 LIST OF DEMING PRIZE WINNERS (INDIAN COMPANIES)

Year	Name of the Company
1998	Sundaram-Clayton Limited, Brakes Division
2001	Sundaram Brake Linings Limited (India)
2002	TVS Motor Company Limited (India)
2003	Brakes India Limited, Foundry Division (India)
	Mahindra and Mahindra Limited, Farm Equipment Sector (India)
	Rane Brake Linings Limited (India)
	Sona Koyo Steering Systems Limited (India)
2004	SRF Limited - Industrial Synthetics Business Lucas-TVS
	Indo-Gulf Fertilizers Limited
2005	Krishna Maruti Limited, Seat Division (India)
	Rane Engine Valves Limited (India)
	Rane TRW Steering Systems Limited,
	Steering Gear Division (India)
2007	Asahi India Glass Limited, Auto Glass Division (India)
	Rane (Madras) Limited (India)
	Reliance Industries Limited, Hazira Manufacturing Division
2008	Tata Steels Limited
2010	National Engineering Industries Limited
2011	Sandan Vikas Limited (India)
2012	SRF Limited, Chemicals Business
	Mahindra and Mahindra Limited, Farm Equipment Sector, Swaraj Division
2013	RSB Transmissions (India) Limited, Auto Division (Jamshedpur Unit 1, Pune, Pant Nagar Plants)

NOTES

In India, two quality awards are given; the Rajiv Gandhi National Quality award and the CII-EXIM Business Excellence Award.

system to rate the performance of each applicant against each parameter. Visits to applicant organizations and scrutinizing an extensive number of documents submitted in support of each component of the award form part of the process of selecting winners.

There two quality awards in India: the Rajiv Gandhi National Quality Award given by the Bureau of Indian Standards, and an award for business excellence instituted by the Confederation of Indian Industries along with the Export-Import (EXIM) Bank

ISO 9000 is a generic set of standards that enable organizations to build a quality management and assurance systems.

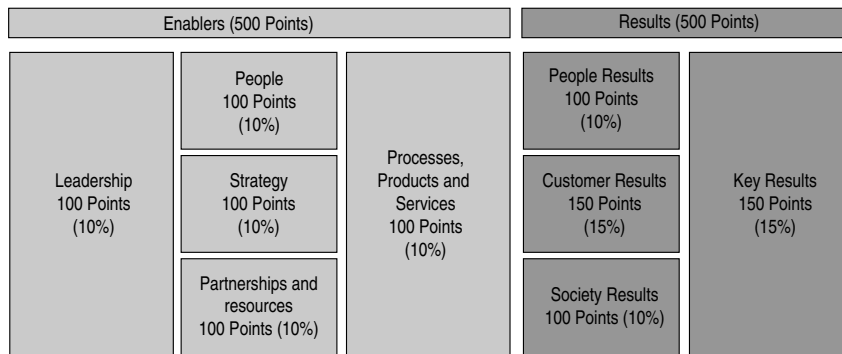


FIGURE 11.12 The CII-EXIM business excellence award model

Source: Adapted from <http://www.cii.in/uploads/SMB899.pdf>

of India. Figure 11.12 shows the model utilized for the CII-EXIM Business Excellence Award. The model closely follows the one proposed by the European Foundation for Quality Management (EFQM). Out of a total 1000 points, 500 points each are awarded for enablers and results.

11.6.2 Quality Certifications

Every country has a national standards body that is mandated by the government to establish standards and certify quality. The Bureau of Indian Standards is the Indian body for quality certification. However, over the years there has been a considerable convergence of alternative standards towards the International Organization for Standardization (ISO). ISO 9000 is a generic set of standards that enables organizations to build quality management and assurance systems. Individual sectors of industry have added more components to the basic ISO 9000 system to obtain their own standards for quality management. For example, automobile majors developed QS 9000. Similarly, the players in the telecommunications industry developed TL 9000. Growing concerns about environmental issues and occupational safety have necessitated the development of newer standards such as ISO 14000 and OHSAS 18000.

The software sector has adopted a unique quality certification programme developed by the Software Engineering Institute (SEI) at the Carnegie Mellon University, known as the **Capability Maturity Model (CMM)**. CMM is a model used by many organizations to identify the best practices useful in helping them increase the maturity of their processes. In 2000, CMM was upgraded to Capability Maturity Model Integration (CMMI). According to SEI, the goals in developing CMMs include addressing software

The software sector has adopted a unique quality certification programme known as the **Capability Maturity Model (CMM)**, used by organizations to identify the best practices useful in helping them increase the maturity of their processes.

NOTES

SRF'S TQM JOURNEY TOWARDS THE DEMING PRIZE

NOTES

SRF's journey towards the Deming Prize started many years ago. SRF had a joint venture with a Japanese company called Denso—a motor component manufacturer—in the early 1980s. In several interactions with them, the top management realized that the real catalyst behind the Japanese revolution in quality was total quality management. SRF started experimenting at that point of time with QC circles. However, it was a dismal failure because there was no preparation, no groundwork done for actually introducing QC circles—it was just something the management tried to push down and impose on its employees. SRF gave up on this effort after trying it for a few years.

The motivation to return to the quality movement seriously happened in the early 1990s. SRF sent eight or ten of their top executives to Japan to attend a programme organized by JUSE, the Union of Japanese Scientists and Engineers, which gives away the Deming Awards in Japan. When they returned, they resolved that the only way to make a change in their organization was to anchor it to TQM. In 1995, SRF engaged Professor Kume—one of the doyens of the TQM movement in Japan, an individual Deming Prize winner and a former chairman of the Deming examination committee—to help with the TQM initiative.

SRF faced credibility issues. Restarting something that has failed once is like going in on a losing wicket. There were many doubters, even at the higher levels of management, who believed it was not going to work, and that, being a Japanese technique, TQM was culturally unsuited to India. In 1994, SRF went through major industrial strife, and for almost four and a half months, the factory was not working optimally. The workers were not ready to listen to the management. However, there was a possibility that this would prove a catalyst to enthuse the workers, because the TQM movement was ultimately all about involvement, from the top right down to the lowest levels. Therefore, while SRF was determined to go ahead with TQM movement, the management did not push anything with the workers.

SRF first started with a massive communication drive explaining how it would be in the interest of the workers to return to normal working conditions, and how imbibing the TQM methodology and the concepts of TQM would provide them opportunities for growth. Simultaneously, an intensive education and training programme was started for top-level and then for middle-level executives on the philosophy of TQM, the training required, and the kind of changes that they would have to make in their mindsets.

When SRF started the plant in Chennai in 1974–75, 60 per cent to 70 per cent of the workers were illiterate. The TQM programme helped them to become literate so that they could start participating in the new culture and gave them immense satisfaction. They

can understand what a histogram is, what a pareto diagram is, what the seven tools of QC are, and they actively participate in QC circles and put forward ideas through the kaizen movement.

The TQM journey facilitated SRF to put in place better workplace management systems. SRF implemented Daily Management, a rigorous system of ten elements, cutting across both manufacturing and non-manufacturing. This was deployed in all the units with a diagnosis score on a 100-point scale (see Figure 11.13 for an illustration). These are increasing levels of difficulty. At each level of capability of the organization, it helped the SRF executives learn more and understand the entire aspect of management better, as relevant to TQM. In later years they devised a system of “management diagnosis”, which included many elements that did not exist in the earlier framework, like leadership, strategy and technology development. This is a much more robust system of measuring the capability of the organization, and the management sets targets around these.

The SRF journey offers several interesting lessons for launching a successful TQM movement. One cannot expect to transform a company in two or three years. Organizations are likely to face a situation where people will start losing interest. Perseverance is very important, not just at the top, but at all levels; and communication is very important too. Organizations must not be looking merely for an end result; they must be looking for the journey. The moot question to keep asking all the time in a TQM journey is: are we continuously improving?

Source: B. Mahadevan, “Journey to the Deming: Implications for Corporate Transformation and Competitiveness—In Conversation with Arun Bharat Ram,” *IIMB Management Review*, Vol. 17, no. 2 (2005): 59–70.

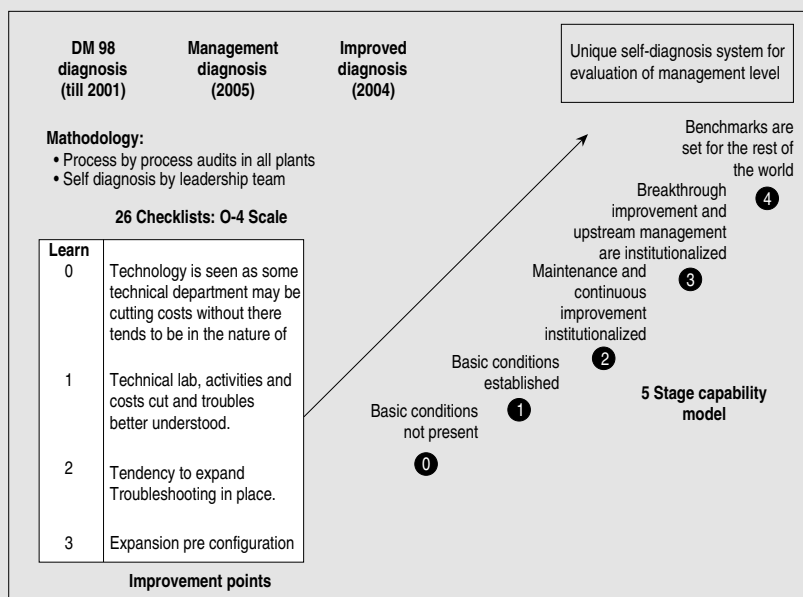


FIGURE 11.13 An illustration of SRF's daily management system

NOTES

NOTES

engineering and other disciplines that have an effect on software development and maintenance and providing integrated process improvement reference models. Moreover, the models will also enable efficient improvement across disciplines relevant to software development and maintenance. Currently, six variations of CMM are available. These include the following:

- ⇒ Capability Maturity Model Integration (CMMI)
- ⇒ People Capability Maturity Model (P-CMM)
- ⇒ Software Acquisition Capability Maturity Model (SA-CMM)
- ⇒ Capability Maturity Model for Software (SW-CMM)
- ⇒ Systems Engineering Capability Maturity Model (SE-CMM)
- ⇒ Integrated Product Development Capability Maturity Model (IPD-CMM)

11.7 DESIGN OF QUALITY ASSURANCE SYSTEMS

A quality assurance system is one that has all the components that promote total quality management in an organization. Based on the discussions in this chapter, we can clearly identify the key elements of a robust quality assurance system. Figure 11.14 is a graphical representation of such a system, the components of which are briefly here:

- ⇒ *Top management commitment*: One of the important lessons of TQM is that the fundamental building block of a quality assurance system is the top management's commitment to quality.
- ⇒ *Employee involvement*: The other pre-requisite for a robust quality assurance system is to develop a good atmosphere for employee involvement. Appropriate choice of quality management tools could further promote group working and inculcate a sense of ownership of the process.
- ⇒ *Understanding customer wants and translating them into meaningful measures*: Without a good understanding of what the customer wants and what the market values, it is futile to establish a good quality management system. It is equally important that we have methods by which this is translated into meaningful measures at the workplace. Without this, all efforts directed towards quality will fetch products and services that customers are not interested in. Therefore, an important element of a good quality assurance system is to resolve this issue. Tools such as QFD are very useful for this purpose.
- ⇒ *Identifying quality problems*: A good quality assurance system requires a methodology by which quality problems are brought to the attention of the employees and managers as quickly as possible. Control charts and other such visual control techniques enable an organization to achieve this requirement of a quality assurance system.

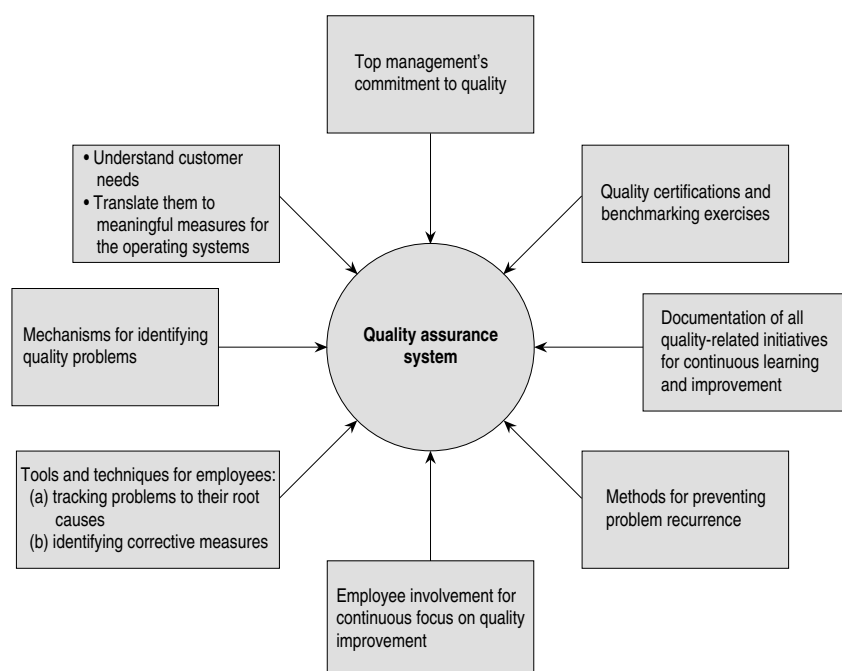


FIGURE 11.14 Elements of a quality assurance system

NOTES

- ⇒ *Providing tools and techniques for problem solving:* Lasting solutions to quality problems are found only when the problems are traced to their root causes and corrective measures are initiated. In the absence of this, organizations constantly face quality problems and incur high costs in rectifying them. Therefore, a quality assurance system will provide alternative tools and techniques, such as the fishbone diagram and CEDAC, to track the problems to their root causes. It will also promote small group improvement projects for continuous improvement.
- ⇒ *Mistake-proofing of operations:* Higher levels of performance in quality management are attained when an organization rises to the level of avoiding quality problems by careful redesign of its operations and standard operating procedures. Tools such as poka yoke are valuable for such initiatives.
- ⇒ *Documentation for continuous learning and improvement:* Organizations tend to benefit greatly from learning curve effects. Revisions and changes in operating procedures that lead to better practices need to be properly documented. Such practices build a strong knowledge base of quality management and help face a variety of situations in the future. Quality certification systems such as ISO 9000 equip an organization to perform this task very well.
- ⇒ *Benchmarking:* The final aspect of a good quality assurance system lies in its ability to constantly point to new areas of

NOTES

improvement that an organization needs to make to remain competitive in the market. In addition to a firm-initiated internal benchmarking exercise, rating by an independent third party helps this process a great deal. Therefore, every organization should endeavour to work towards obtaining quality awards and certifications. Therein lies the ultimate test of the strength of its quality assurance system.

11.8 SUMMARY

- The resounding success of Japanese manufacturing firms has invariably been linked to excellent practices in quality management in the last 30 years.
- Much of the progress that organizations made in quality and productivity management is attributed to the teachings of quality gurus such as Deming, Juran, Crosby, Taguchi, Shingo and Ishikawa.
- A good quality management system must enable a manager to understand the qualitative attributes that influence a customer and have a method of translating these into unambiguous quantifiable parameters for design and manufacturing.
- A total quality management programme consists of four components: top management commitment, employee involvement in continuous improvement initiatives, training and skill development, and investment in robust systems.
- Several tools and techniques are available that help in highlighting problems, identifying specific improvement opportunities, analysing problems and their root causes, and operational and strategic planning for building quality into products and services.
- Alternative certification procedures and award mechanisms are available to recognize excellent quality management systems in organizations.

11.9 REVIEW QUESTIONS

1. What is the relevance of quality in today's competitive scenario? Was it any different during the 1970s and the 1980s?
2. Enumerate the key benefits that an organization derives by improving the quality of its products and services.
3. Compare the teachings of Deming and Juran and make a critical report of your key inferences from them. Were they advocating the same idea?
4. Identify three ways by which an organization can benefit from the teachings of the following gurus:
 - (a) Philip Crosby
 - (b) Karou Ishikawa
 - (c) Genechi Taguchi
5. For each of the products/services mentioned here, define quality from the customers' perspective:
 - (a) A laptop computer
 - (b) A weekend holiday in a beach resort in Goa
 - (c) A motorbike
 - (d) Legal services obtained from a lawyer
 - (e) A day-care centre (crèche)
 - (f) An electric bulb
6. For each of the products/services in Review Question 5, define quality from the perspective of the firm providing the products/services. Are these definitions the same or different? Explain your observations.
7. What are the challenges that an organization faces when it attempts to define quality? Is it possible to overcome these and if so how?
8. What is poka yoke? Can you identify one situation each in your classroom, hostel, and college administrative office that can potentially benefit from the application of this concept?
9. What do you mean by the term total quality management? Is there anything special about "total" in TQM?
10. If an organization practices employee involvement, how is it likely to be benefited?
11. How can the top management of an organization influence the quality management process?
12. What is the use of the QC tools in quality management?
13. What are the elements of a quality costing system? How does an organization benefit from a quality costing system?
14. In an organization if the prevention and appraisal costs are very low, what, according to you, will the magnitude of the other components of the quality costs be? Explain your answer with supporting arguments.
15. What is a quality house? What is the use of a quality house?
16. An organization does not know if it needs to go for ISO 9000 certification or ISO 14000 certification. They are also not aware why they should go for a certification. Prepare a one-page report to resolve their confusion.
17. How does an organization benefit by working towards obtaining a quality award? Explain with reference to the CII-EXIM Business Excellence Award.

11.10 PROBLEMS

1. A hospital buys medical consumables from several suppliers. The bad quality of consumables sometimes affects the hospital operations critically. Therefore, the hospital follows a quality control routine. Based on a weekly analysis of the quality of cotton and dressing material used in large quantities in the surgery section of the hospital, the following causes of defects were found out (see Table 11.4).

- (a) Draw a histogram of the problem for further analysis.
- (b) Draw a pareto diagram of the data in Table 11.4. Based on the pareto diagram, what course of action would you recommend?

2. A bank operating in a prominent residential area in the metropolitan city of New Delhi is facing tough competition from several of its competitors. Improving the quality of services has become a necessary condition for the branch to survive. One of the major problems is the excessive delay that customers experience in the bank. After a brainstorming exercise, a group of bank employees came up with the possible causes for poor quality of the service. The following is the list:

TABLE 11.4 CAUSES OF DEFECTS

Sl. No.	Description	Number of Occurrences
1	Soiled and dirty	23
2	Cuts and discontinuities	12
3	Improper specifications	19
4	Weight loss	31
5	Defective packing	21

- (a) The number of employees are too few.
- (b) The use of technology was minimal.
- (c) Work distribution among employees is lopsided.
- (d) Employees have not been trained properly in using the system.
- (e) There is a space constraint in the bank with the result that the layout of the service counters is inconvenient.
- (f) There are procedural delays in the bank.
- (g) The existing performance measures do not motivate employees to provide faster service.
- (h) Employees have varied patterns of service delivery, leading to uneven service time.
- (i) Nobody in the bank is aware of how much time a customer waits.
- (j) Customers are not realistic in their expectations.
- (k) The number of accounts handled is far in excess of capacity.
- (l) There is too much variety in the services offered by the branch.
- (m) There is too much absenteeism among employees.

Represent the problem using a fishbone diagram. What future course of action should the group of employees take with respect to the problem?

3. The quality of the food services in a students' hostel mess is a cause of great concern among the students. Use the fishbone diagram and show how one should go about studying this problem.
4. A manufacturing company is keen to understand the notion of quality costing and its relevance to quality management. Based on one month's work by a task force, the following data have been collected from the accounts of the company, pertaining to the last three years. Table 11.5 has various items of cost (in thousand ₹):
 - (a) Develop a quality costing system on the basis of the given data.
 - (b) What is the cost of quality as a percentage of turnover if the total turnover for the three years were ₹200 million, ₹240 million and ₹300 million during 2002, 2003 and 2004, respectively?
 - (c) Plot the various components of the quality cost. Do you see any trend in the behaviour of the components of quality cost over time? Explain your key observations of this exercise.

TABLE 11.5 THE ITEMS OF COST

Number	Description of Item of Cost	2002	2003	2004
1	Warranty costs	12,532	11,540	12,320
2	Cost of rework	9,348	18,540	17,439
3	Training costs	234	567	893
4	Supplier development costs	1,238	2,345	3,459
5	On-site repair of products	8,345	7,934	7,839
6	Cost of testing	4,567	5,672	5,789
7	Loss from product returns	4,562	4,234	2,389
8	Inspection costs	7,812	12,127	17,283
9	Calibration of testing devices	1,678	1,482	2,093
10	Scrap	2,568	4,298	6,238

11.11 NET-WISE EXERCISES

1. To get an overview of the Malcolm Baldrige Quality Award program, visit the URL: <http://www.nist.gov/baldrige>. Click on 'About Us' and read about the history of Malcom Baldrige National Quality Awards for background information about the award. Click on the 'FAQs'. There are several side bars in 'FAQs' menu. Go through them to know more about the award.

Next, visit the URL

<http://http://cii-iq.in/cii/sectors/BE/CII-Exim%20Bank%20Award%20for%20Excellence.html>. There are details on various aspects of the award in several links in this page. Go through these links.

- (a) What do you think is the basic purpose of the Malcolm Baldrige Award and the CII-EXIM Bank Award? How do they achieve this objective?
- (b) Compare and contrast the two awards. Define three criteria for comparison and use them for answering this question.

11.12 MINI PROJECTS

1. Form a small group improvement team of six students. Visit the local branch of the bank or a hospital and get their approval to study the quality problems that they face for possible improvements and prepare a final report based on the entire exercise. The project should involve the following important steps:

- (i) Develop an appropriate definition for quality pertaining to the problem taken up for the study in consultation with the employees in the organization.
- (ii) Identify the metrics that needs improvement and set realistic targets for the metrics.
- (iii) Collect data and use the available QC tools to analyse the problems, their root causes, and potential improvements.
 - (a) Estimate the projected improvements and the changes required in the system to improve the quality
 - (b) Discuss your results with the organization and understand the feasibility of your recommendations. Make further refinements of the proposed recommendations
 - (c) Write a detailed report based on your experiences in the project

2. Consider the following three quality awards; the CII-EXIM Business Excellence Award, the Deming Award, and Malcom Baldrige Award. Make a detailed study of these awards by visiting their respective websites and collecting information from secondary data such as published literature and reports. Based on this study, prepare a report to answer the following questions:

- (a) What are the salient elements of each of these awards? Is there an underlying philosophy that governs selection of successful companies in each of these awards?
- (b) Select one award-winning firm for each of these categories of awards and study them. Briefly document the experience of one firm each that has bagged these awards.
- (c) What is the nature of the efforts required from a firm to get these awards? Is there a time frame involved? How do firms that qualify for these awards benefit from the exercise?
- (d) What is the status of firms that obtain these awards? How do they perform after they obtain these awards? Are there any significant disadvantages or problems these firms go through in the process of obtaining the award?

3. Quality certification is an important aspect in quality management. Prepare a detailed report on the basis of your own study to answer the following questions:

- (a) What are the emerging standards in the area of quality management? Why are they important?
- (b) What are the industry specific initiatives in the area of quality certification? How are they different from generic standards?
- (c) What benefits accrue to a firm that obtains a quality certification? Are there any prior experiences that confirm these findings?

11.13 CASE STUDY

Bangalore Garment Exports Limited

Bangalore Garment Exports Limited (BGEL) is a garments manufacturer in Bangalore primarily engaged in the manufacture of gents shirts against specific client orders. The garment sector is a competitive one and competing merely on price is an unsustainable strategy in the long run. The company has been in operation for the last 12 years and has earned a reputation for their delivery

reliability and cost. However, they have not been able to establish a robust quality system. While they take care of the customer requirements through stringent inspection standards, they end up with a much higher proportion of rejects (seconds) than the industry average in the process. BGEL has made some initiatives in the recent past to address the issue of quality.

Operating Unit

BGEL has both manual and mechanized operations for shirt-making. The current capacity is for a daily production of 6500 shirts. While cutting and basic stitching are done using high-speed machines, there are portions of shirt-making that involves skilled labour in addition to the machines. In all, there are 65 machines and of there are 12 cutting machines, the rest being stitching machines. BGEL employs over 400 people; of these, 290 are engaged in production. The production workers are functionally organized into four sections; laying and cutting of cloth, body stitching, making slacks and attaching it to the body of the shirt, and finally, attaching the collar to the shirt. After this, it goes to the QC department for final checking and QC clearance. After the QC clearance, it is sent for packaging and dispatch. The production flows in this sequence, and each section employs a number of production workers in order to meet the daily demand.

There are separate sections for routine, preventive, and breakdown maintenance of the machines; quality control; and production planning. All these sections, as well as the production sections, have supervisors who are responsible for output and performance of the units. At the beginning of each shift, the production supervisor in the section assigns jobs to each worker based on the daily requirements. The production workers receive incentive pay on a "piece rate" basis, once they exceed the normal production in a shift.

Quality Management Issues

One of the major problems at BGEL has been the high proportion of rejection of the final product. Last year, the QC section initiated a system of recording daily rejects using a display board. The display board also recorded some significant events pertaining to the day along with the rejects data to help the management take corrective actions. The snapshot of the data on daily rejects for a period of ten days is given in Table 11.6.

Management Actions on Quality Management

The display board was primarily meant to highlight to the management the quality issues at BGEL. Every day, the management reviews the status of the board and "instructs" the supervisors on the nature of the actions to be taken to bring down the number of rejects. The supervisors in turn modify the work practices and/or the instructions to the workers working under them and monitor compliance of the requirements. The management also engaged an expert in textile technology to educate the supervisors on modern methods in textile manufacturing and the trends in usage of new machinery. On the other hand, the maintenance workers were sent to one-week training programmes on better maintenance practices. The management also appointed an expert on quality assurance so that better methods would be put in place to improve the quality. The display board was one of the suggestions of the expert to improve quality.

According to the QC section, the skill levels of the employees and absence of a quality monitoring system are the two major reasons for poor quality. The other issues are high levels of absenteeism and a high turnover of the production workers. Despite several efforts by the management in the past year, there has not been much improvement in the quality standards at BGEL. This is a major cause of concern for the MD of BGEL.

QUESTIONS FOR DISCUSSION

1. Comment on the current status of the quality management system at BGEL. Why have they not been able to make improvements in quality?
2. What are your observations based on the display board? Can you identify three areas that BGEL need to address in order to improve their quality standards?

Suggest a suitable quality management system and a plan of action for BGEL for the next one year.

TABLE 11.6 A SNAPSHOT OF THE DISPLAY BOARD AT BGEL

	Daily Production	Daily Rejects	Remarks on Significant Events
Day 1	5957	298	Stoppage of machines, shortage of material to work on, high absenteeism
Day 2	6043	423	Schedule and specification changes not properly communicated, skill inadequacies
Day 3	5450	600	Machine failure, work reassignment to handle high absenteeism
Day 4	5878	353	Machine failure, specifications not met
Day 5	6012	541	Coordination problems among production sections, worker skill issues
Day 6	6123	490	Coordination problems among production sections, specifications not met
Day 7	5875	764	Skills mismatch, defective and improper instructions given to workers, specifications not met
Day 8	5367	590	Coordination delays, rework due to wrong assembly of pieces
Day 9	5590	839	Wrong specs used, malfunctioning of machines, missing instructions
Day 10	5670	510	Defects due to work reassignments, Too many rush orders squeezing capacity

11.14 MULTIPLE-CHOICE QUESTIONS

- 11-1.** The quality revolution is possible because of:
A. New Thinking
B. Re-engineering
C. Radical Thinking
D. Discovering from the past
- 11-2.** Who is the father of the Japanese quality management systems?
A. Joseph M. Juran
B. William Edwards Deming
C. Philip B. Crosby
D. Karou Ishikawa
- 11-3.** Definitions of quality and the implications of alternative definitions are:
A. Specified Performance Range, Design Standard, Delivery Satisfaction
B. Improvement in Product Quality, Standard Grade, Character Nature
C. Value Level, Meeting or Exceeding Customer Expectations, Structured Methodology
D. Conformance to Specifications, Fulfilling Customer Needs, Fitness for Use
- 11-4.** A good Total Quality Management (TQM) programme consist of the following important elements:
A. Management behavior, Recognition, Training
B. High standard of work, Permanent climate to meet demand, Services to satisfy customers
C. Role of the top management, Employees involvement, Addressing the training requirement of employees and Tools/Techniques for quality assurance and continuous improvement
D. Team work, Leadership, Communication
- 11-5.** Under how many categories Quality management tools available tools will fall and what are they?
A. One category: Quality Control
B. Two categories: Organizational structure, Check Sheet to collect data
C. Three categories: Cause and Effect/Problems and solutions, Information
D. Five categories: Tools for highlighting quality problems, Tools for identifying specific Improvement opportunities, Tools for analyzing problems and their root causes, Tools for operational planning, Tools for strategic planning
- 11-6.** Which award is a generic set of standards that enable organizations to build a quality management and assurance systems?
A. QS 9000
B. ISO 9000
C. ISO 14000
D. OHSAS 18000
- 11-7.** What are the elements of a quality assurance systems?
A. Process of verification, Confirmation of design, Confirmation of development
B. Confirmation of production, Customer satisfaction, Performance specification
C. Product testing, Failure testing, statistical control
D. Top management commitment, Employee involvement, Understanding customer wants and translating them into meaningful measures, Identifying quality problems, Providing tools and techniques for problem solving
- 11-8.** The resounding success of several manufacturing and service firms in recent times was invariably linked to:
A. Excellent quality management practices
B. Excellent financial management practices
C. Excellent manpower management practices
D. Excellent Logistic management practices
- 11-9.** An alternative method to define quality is to:
A. Understand the various dimensions of quality
B. Understand the various dimensions of quantity
C. Understand the various dimensions of supply
D. Understand the various dimensions of demand
- 11-10.** What can be defined as an organization-wide effort to develop the systems, tools, techniques, skills, and mind-set required to establish a quality assurance system?
A. Total Quality Management
B. Total Logistic Management
C. Total Production Management
D. Total Quantity Management

Answer Keys: 11-1. A 11-2. B 11-3. D 11-4. C 11-5. D 11-6. B 11-7. D 11-8. A 11-9. A 11-10. A

12

CHAPTER

THEORY OF CONSTRAINTS

12.1	The Need for Scheduling
12.1.1	Scheduling: Alternative Terms
12.1.2	Planning-related Terms
12.1.3	Technological Constraints-related Terms
12.1.4	Administration-related Terms
12.1.5	The Loading of Machines
12.2	The Scheduling Context
12.2.1	Scheduling Rules
12.2.2	Performance Criteria
12.3	Scheduling of Flow Shops
12.3.1	Johnson's Rule
12.4	Scheduling of Job Shops
12.5	Operational Control Issues in Mass Production Systems
12.5.1	Machine Redeployment
12.5.2	Altering Operator Allocations
12.5.3	Adjusting Material Feed Rates
12.6	Operations Planning and Control Based on the Theory of Constraints
12.6.1	Measures of Performance
12.6.2	The Analogy of Marching Soldiers
12.6.3	Synchronous Manufacturing
12.6.4	Constraints in a Manufacturing System
12.6.5	The Drum-Buffer-Rope (DBR) Methodology
12.7	Summary
12.8	Formula Review
12.9	Review Questions
12.10	Problems
12.11	Mini Projects
12.12	Multiple-choice Questions

SCHEDULING THE ORDER PROCESSING OPERATION AT BLOCKBUSTER DISTRIBUTION CENTER

NOTES

Blockbuster Incorporated, a chain of VHS, DVD, Blu-ray, and video game rental stores, has developed a highly specialized distribution network. The company maintains a single distribution center in which it receives products from suppliers, and processes and packs them for shipping to stores across United States. Because of a dynamic changes in product mix, Blockbuster's weekly aggregate demand can experience a peak-to-average ratio of 4:1 or more. Forecasting this change in volume is difficult because each new product is truly a new release with no history. To accommodate the supply chain requirements, flexible methods and processes are adopted in order to handle the high level of volatility and to maintain sufficient robustness to virtually eliminate late deliveries. At the distribution center, processing and packing are scheduled through multiple processing departments that compete for use of shared merge conveyors and shared sortation systems. Blockbuster's general processing and packing goal is on-time delivery of products to stores while controlling costs.

An order refers to a request to pick or process individual pieces of a product (e.g., 1,000 pieces of a particular DVD title) for a retail store. A job is a collection of orders for a set of stores. Once a job has been processed, the completed pieces of products are placed in cartons that proceed through a system of conveyors, merge, and sortation points to a consolidation area, where the contents from the individual cartons are combined with other same-store cartons. When the consolidation has taken place, cartons are shipped to retail stores using a third-party logistics network. Figure 12.1 schematically represents the consolidation process.

Typically, about 400 jobs must be processed in a week. Cartons produced in these departments must then compete for capacity in up to nine subsequent shared resources (merge and sortation points). The operating schedule consists of two 12-hour shifts per day, six days per week. Approximately 10.5 production hours are typically available per employee per shift. The penalty for late shipments is very high in this industry. Therefore scheduling the orders through the distribution network and shipping them on time are very important.

A feasible schedule would mean an assignment of each job to a department such that the capacity at each department and at each merge and sortation point is respected. Due to high volatile demand, the previous scheduling model often generated an unrealistic plan that required excessive resources. A new scheduling model was developed that can create a realistic schedule, thus eliminating the need to maintain significant additional staffing. The scheduling model was introduced for testing in October 2006 and was implemented in January 2007.

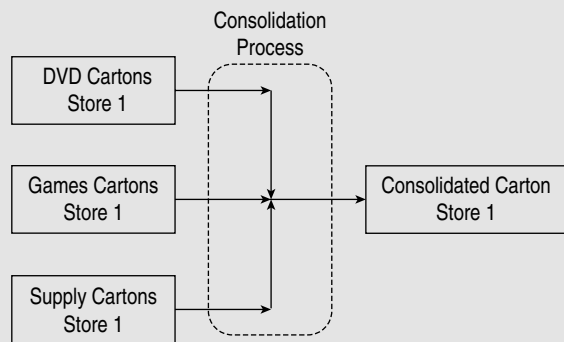


FIGURE 12.1 The consolidation process at Blockbuster distribution center

The cost savings resulting from implementing the scheduling solution were mainly on account of three reasons. Initially, it is cost savings due to reduced staffing. The total number of employee shifts was 31,445. In comparison, using the new scheduling approach, the actual staffing in 2007 was 23,229 employee shifts. Second, the capacity utilization improved because of the balanced workload solutions generated by the scheduling model. Prior to implementing the scheduling model, Blockbuster had to maintain capacity for wide fluctuations in peak volumes. The use of the model has effectively smoothed the fluctuations, resulting in improved capacity utilization. Third, Blockbuster has experienced savings because more cartons can flow through the entire process. This significantly increased the volume of cartons, thus reducing transportation costs.

Source: Adopted from Chung, C. Dawande, M., Rajamani, D. and Sriskandarajah, C. (2011), 'A Short-Range Scheduling Model for Blockbuster's Order-Processing Operation', *Interfaces*, 41(5):466–484.

NOTES

12.1 THE NEED FOR SCHEDULING

Operations management requires planning and control at various levels and time horizons. So far, we have discussed operations management issues in the long term and the medium term. Configuration of an operations system at the right location and choice of appropriate process and product designs are issues associated with the long term. On the other hand, aggregate planning and material and capacity requirements planning are issues associated with the medium term. In the same manner, there is a host of issues that one needs to address in short-term planning. In this chapter, we focus on planning and control issues associated with the short term. In the short term, there is more emphasis on operational control than on planning. This, however, does not imply that there is no planning involved. In fact, planning does half the job of control.

NOTES

“Short term” denotes a time horizon close to real time. Typically, in a majority of manufacturing and service organizations, this would imply a horizon ranging from a day to at most a week or two. In some special cases, “short term” may mean a few hours or a shift, as in the case of a power transmission firm or an event management firm. One question that comes to our mind is: When we have exercised so much care in the previous stages of planning, why do we still need to continue with the planning exercise? Before we proceed with the concepts and techniques of operations scheduling, it is best to clarify this issue. Planning for the short term becomes inevitable for three important reasons:

1. As we approach real time, several kinds of additional information become available to an organization. Use of this new information makes the plans far more robust and representative of reality. For instance, some customer orders could be cancelled, some new orders booked, and the terms of the existing orders could be revised. Obviously, using this information makes operations more accurate than choosing to simply ignore them.
2. The occurrence of random events is inevitable in business. These include the sudden breakdown of a machine, absenteeism of skilled labour, delays in the supply of key raw material, and a sudden revision of job priorities. It is not possible to include these details accurately while planning for the medium term. Broad assumptions and approximations are made about these at that time of planning. Therefore, we need to revisit these assumptions and fine-tune our planning and decision-making methodology.
3. In the short term, we need to focus on micro-resources, a single machine, a set of workers, and so on. Such a focus is neither possible nor warranted in medium or long-term planning.

In the short term, several questions need to be answered. How can jobs be assigned to various work centres? Within each work centre, how can the jobs be rank-ordered? How can other resources such as skilled workers and special gadgets be assigned to the operating system? How do we react to a breakdown in the system? How can the performance of the operating system be measured? All these questions are equally applicable to both manufacturing and service systems. In this chapter, we address many of these questions and develop methodologies to address these questions using a defined framework. We refer to the collective framework for addressing these issues as **scheduling of operations**.

Operations scheduling has a direct linkage with MRP systems in an organization. Recall that the function of an MRP exercise is to generate feasible production orders and procurement notices. *Production orders* are nothing but specific sets of instructions to

Operations scheduling uses a defined framework to address issues associated with the use of available resources and the delivery of products and services as promised to the customers.

utilize a set of resources for the manufacture and/or assembly of the components under question during a particular period of time. Consider a sub-assembly, X. If the MRP schedule indicates that during Week 16 a designated shop, S1, should manufacture 200 units of X, then it is clear that during Week 16, the planner needs to ensure that the job is assigned to Shop S1 and provide capacity for manufacturing 200 units of X. During the same week, there could be several jobs assigned to Shop S1. Therefore, the shop planners also need to prioritize all such jobs in the shop with respect to each machine they visit, and ensure completion of the job without delay. This example indicates that the output of an MRP exercise becomes the input for job scheduling.

12.1.1 Scheduling: Alternative Terms

In practice and in literature, several terms are used in scheduling. These terms convey important concepts related to scheduling. Yet, those unfamiliar with the subject are often confused by these terms and experience considerable difficulty in understanding their significance. We will, therefore, define these terms here and use them consistently in the rest of the chapter.

12.1.2 Planning-related Terms

Planners often use a pair of terms—loading and scheduling. **Loading** is defined as a planning methodology that is used to assign an adequate number of jobs to the resources in an operating system during a planning horizon (of, say, a week). Loading is done to ensure that the resources in an operating system are utilized in the best possible way. Moreover, loading also ensures that the workload among competing resources is not lopsided, and hence does not create an uneven strain on the operating system. In a dental hospital, let us assume that there are three dental specialists visiting the outpatient department for four days. On each day, let us assume that they spend three hours advising or treating patients. The loading function will ensure that the specialists are adequately engaged and that no specialist is grossly over or underutilized. (We make the important assumption here that the patients do not make the choice of the consultant. If the patient has a choice, then the loading methodology will suggest shifting appointments from a particular day to another day or that the patient consult another specialist.)

Loading is a planning methodology that is used to assign an adequate number of jobs to the resources in an operating system during a planning horizon.

Scheduling is defined as the process of rank-ordering the jobs in front of each resource with a view to optimize some chosen performance measure. If there are n jobs waiting in front of a machine, using scheduling, the planner could rank-order them for

NOTES

NOTES

Scheduling is the process of rank-ordering the jobs in front of each resource to maximize some chosen performance measure.

the purpose of processing. Rank-ordering could be done to minimize the waiting time of all the jobs in the queue. In the dental hospital example, if it is decided to allow 10 patients for consulting during a three-hour schedule, some rules are utilized to schedule the ten patients one after the other. In service systems, the rule is often first-cum-first served. But in manufacturing systems, several other possibilities exist.

12.1.3 Technological Constraints-related Terms

Two more terms are used in this field: routing and sequencing. These terms pertain to the technological constraints of the jobs to be scheduled in the operating system. **Routing** is defined as the order in which the resources available in a shop are used by the job for processing. The order is a function of the technological constraint associated with the job. The information is normally available in a routing file or a route card. In a manufacturing system, the routing denotes the order in which operations are performed on a component. For example, a spindle production may involve cutting the required length using a shearing operation, rough machining using a turning operation, heat treatment for stress relief, grinding for finishing the operation, and, finally, chromium plating—in that order. Similarly, routing information is utilized in a service system also. A person aspiring to obtain a U.S. visa from the U.S. Consulate needs to go through three or four stages of the process—submission of papers and drafts, initial verification of documents and basic clearance, interview by the visa officer, and final clearance for the visa. There could be rejections as a candidate proceeds through the system. This information is crucial for resource deployment and operations scheduling at the U.S. Consulate.

Routing is the order in which the resources are used by jobs for processing.

Sequencing is the ordering of the operations of the jobs in the operating system. It combines the routing and scheduling information to provide a basis for the planners to ensure that jobs are processed in the shop without violating technological constraints.

Sequencing is the ordering of the operations of the jobs in the operating system.

12.1.4 Administration-related Terms

In organizations, dispatching and expediting are terms frequently associated with the scheduling of operations. **Dispatching** is defined as the administrative process of authorizing the processing

Dispatching is the administrative process of authorizing processing of jobs.

of jobs by resources in the operating system, as identified by the scheduling system. The authorization could be in the form of a shop order or information communicated electronically to the respective work centre. Dispatching physically signals the assignment of jobs to work centres and places, and the onus for reviewing the progress is on the controllers. *Expediting*, on the other hand, is the administrative process of reviewing the progress of the job in the operating system and ensuring that it travels through various stages of the process as indicated in the routing, without harming the performance measure. If there are deviations and delays, it calls for pushing the job forward through the system.

12.1.5 The Loading of Machines

When there are multiple copies of the same machine type, the first step in scheduling is to assign the jobs to the resources. As we have seen already, this is achieved by loading. Let us look at a simple illustration of this idea. Suppose three milling machines are available on a shop floor and each of them provide a weekly capacity of 45 hours (we assume that each day, 7.5 hours are available and the factory works for 6 days a week). During an upcoming week, let us say that 30 jobs require processing in the machine and a total of 110 hours of machine capacity is required. While the required capacity is available, it still requires some planning so that the jobs are assigned to these machines such that none of the machines are either overloaded or under-loaded. This is achieved using loading. For this purpose, one can use the assignment method, a variation of the linear programming model.

The assignment method is useful when one job is assigned to one machine and there are as many machines available for assignment as there are jobs. If there are fewer jobs than machines, one can still use the assignment method by adding some fictitious jobs and solving the standard assignment problem. On the other hand, if there are fewer machines than jobs, other methods are available for loading jobs on machines. In this case, several jobs are assigned to each machine as long as sufficient capacity is available for processing and other constraints are taken care of.

12.2 THE SCHEDULING CONTEXT

The nature and the complexity of the scheduling problem are dependent on the context in which we solve the problem. Moreover, past research suggests that even the type of performance measure

The nature and the complexity of the scheduling problem are dependent on the context in which we solve the problem.

NOTES

EXAMPLE 12.1

A manufacturer of dies for the automotive component industry utilizes a job shop structure. During a planning horizon, the manufacturer needs to first assign five jobs to the five machines that are in one section of the factory. Table 12.1 provides details on the processing time of the jobs in each of these machines if assigned. The differences in the processing time are attributed to differences in the ratings of these machines. Use the assignment method to solve the loading problem.

Solution

We use the standard assignment method for solving the problem. We first subtract each row by the minimum number in the row. The resultant table is as shown in Table 12.2.

After that, we subtract each column by the minimum number in the column. There is a change only in Column 2 as other

columns already have a zero. The table after column subtraction is as shown in Table 12.3.

Since the minimum number of lines required to cover all the zeros is equal to the number of rows, we can use Table 12.3 to perform the final assignment. The assignments are as follows:

- ⇒ Job 1 assigned to Machine 2;
- ⇒ Job 2 assigned to Machine 1;
- ⇒ Job 3 assigned to Machine 3;
- ⇒ Job 4 assigned to Machine 5;
- ⇒ Job 5 assigned to Machine 4.

This assignment results in a minimum total processing time of 80 time units.

TABLE 12.1 PROCESSING TIMES FOR THE JOBS

	Machine 1	Machine 2	Machine 3	Machine 4	Machine 5
Job 1	15	17	16	22	18
Job 2	16	19	20	19	16
Job 3	19	20	17	19	20
Job 4	17	23	22	18	14
Job 5	20	19	24	16	17

TABLE 12.2 RESULTANT MATRIX AFTER THE ROW OPERATION

	Machine 1	Machine 2	Machine 3	Machine 4	Machine 5
Job 1	0	2	1	7	3
Job 2	0	3	4	3	0
Job 3	2	3	0	2	3
Job 4	3	9	8	4	0
Job 5	4	3	8	0	1

TABLE 12.3 RESULTANT MATRIX AFTER THE COLUMN OPERATION

	Machine 1	Machine 2	Machine 3	Machine 4	Machine 5
Job 1	0	0	1	7	3
Job 2	0	1	4	3	0
Job 3	2	1	0	2	3
Job 4	3	7	8	4	0
Job 5	4	1	8	0	1

that we use for scheduling jobs can influence the ease with which we could obtain the solution. Therefore, a good introduction to the parameters of the problem is essential before we analyse the solution techniques available for scheduling. The following parameters adequately describe the scheduling problem that one tries to solve:

⇒ *Number of machines*: The number of machines, or more generally, the number of resources for which the planner proposes to perform the loading and scheduling function is the first parameter for the problem. Let us denote the number of machines by m . The simplest case is $m = 1$. In this case, the jobs that require processing in that machine need to be simply rank-ordered on some basis. However, as m becomes larger, the problem size increases as numerous combinations are introduced. Moreover, if multiple machines are available for the jobs as parallel resources for processing, we need to develop methods by which one out of those multiple machines is chosen for assigning the job.

As the number of machines and jobs increases, the problem size also increases as numerous combinations are introduced.

⇒ *Number of jobs*: The number of jobs is the second parameter for the scheduling problem. We denote the number of jobs by n . If the number of jobs increases, it introduces more complexity into the scheduling problem. As multiple jobs compete for the same set of resources, additional methods are required to assign the jobs to the resources.

⇒ *Shop configuration*: “Shop configuration” refers to the manner in which m machines are organized on the shop floor and the flow pattern of the jobs utilizing these machines. Two alternative configurations—flow shop and job shop—are available for configuring machines in a manufacturing system. Since these configurations differ in terms of the deployment of resources, the scheduling methodology also differs. Moreover, we shall also see that the complexity of the problem varies between these two configurations.

The simplest configuration is **flow shop**. In a flow shop, the resources are organized one after the other in the order the jobs are processed. In a flow shop, jobs start their processing at Machine 1. After completing processing, they visit Machines 2, 3, and so on before they exit the system from Machine m . Figure 12.2 graphically portrays a pure flow shop. A pure flow shop is one in which all the jobs visit all the machines in the same order (beginning at Machine 1 and ending at Machine m). However, in a mixed flow shop, some jobs are allowed to skip machines in between.

The simplest shop configuration is **flow shop**. In a flow shop, the resources are organized one after the other in the order the jobs are processed.

NOTES

NOTES

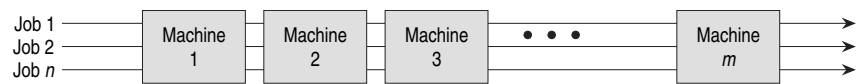


FIGURE 12.2 A pure flow shop

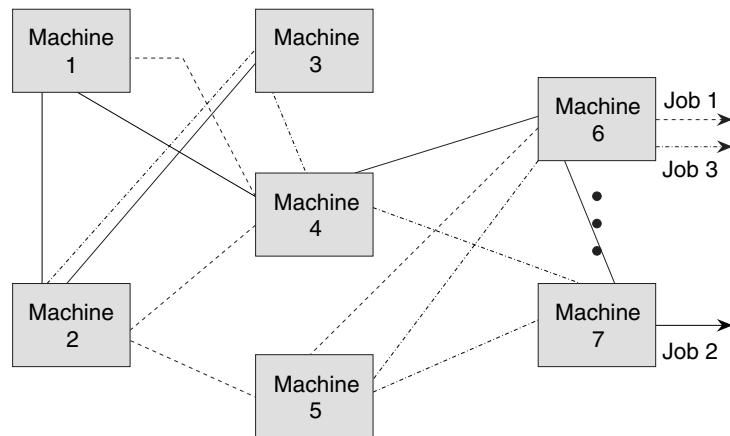


FIGURE 12.3 Illustration of a job shop with 7 machines and 3 jobs

Since in a pure flow shop all jobs follow the same order of visiting the machines, the scheduling function is essentially reduced to one of ordering the jobs in front of the first machine. Once the jobs are ordered in front of the first machine, it follows the same order in the rest of the machines. Since there are n jobs, there are $n!$ ways in which one can draw up alternative schedules in the shop. It is easy to note that scheduling complexity increases in the case of a mixed flow shop.

The second configuration is the **job shop**. In a job shop, machines are not organized in any processing order. Rather, similar types of resources are grouped together. Every job follows a unique order in which it visits the machines for processing. Moreover, there is no requirement for all jobs to visit a particular machine for their first operation or a particular machine for their last operation (as in the case of a flow shop). Figure 12.3 presents a simple illustration of a job shop consisting of seven machines. In this example, the route that Job 1 follows for processing is 1 – 4 – 2 – 5 – 6. On the other hand, Job 2 follows the sequence 3 – 2 – 1 – 4 – 6 – 7 and Job 3 follows 2 – 3 – 4 – 7 – 5 – 6.

In a **job shop**, machines are not organized in any processing order. Rather, similar types of resources are grouped together.

Since each job has its own unique route, the set of jobs visiting each machine is not the same. Therefore, the ordering of jobs in front of each machine is to be separately determined. There are $n!$ ways of rank-ordering jobs in front of a machine. Since there are m machines in the shop, the number of alternative schedules that

SCHEDULING OF PHYSICIAN IN THE EMERGENCY DEPARTMENT OF A HOSPITAL

NOTES

Physicians at a branch of the emergency department at Cincinnati Children's Hospital Medical Center complained that their schedules were too erratic because of the multitude of operating requirements, regulatory constraints, physician preferences, and holiday requests. They wanted their work schedules to be more predictable by using repeating patterns. They also wanted more consideration given to both their working preferences and their need to fulfill personal life choices.

Crew scheduling in services is always more complex than scheduling manufacturing resources in a factory. For instance, in this case, the physicians had several considerations that needed to be honoured to build a workable schedule. Some of them are as follows: physicians never wanted to work in more than two consecutive weekends. They would prefer to be assigned, at the most four shifts in any given week. Furthermore, they would not like to have more than two consecutive overnight shifts assigned to them during Monday–Thursday.

One of the methods adopted to address crew scheduling is cyclic scheduling as it has certain advantages. For instance, the physicians might know their work schedules in advance so that they could organize their lives during their time-off better. On the downside, it is impossible to build a cyclic schedule that considers diverse requests for time-off and the assigned legal holidays, because the uniqueness of each period conflicts with the repetition of a predetermined cycle. A critical aspect of cyclic scheduling is the choice of cycle length. Short cycles provide more repeatability and can be more easily remembered by the workers. However, they offer fewer possibilities for the inclusion of all the required attributes, in particular for work-load and work-pattern necessities.

Based on several considerations, an eight-week cycle was finally selected. The physicians felt that this cycle length was neither too short nor too long, and would permit a balanced distribution of weekend shifts. A solution was developed using integer programming to build cyclic schedules that can be repeated throughout the year. These schedules are flexible enough to handle incorporating holidays, work assignments, and vacation requests *ex post*.

Three months after the cyclic schedule was implemented, the physicians were asked about their satisfaction with the cyclic schedule. The physicians stated that the new scheduling method provided them with more predictability over their workweek, which helped them plan their activities outside of work. Moreover, having a template for the entire year gives the physicians the ability to plan their vacation requests. They also found the new work patterns addressing their work-life balance better.

Source: Adopted from Ferrand, Y., Magazine, M., Rao, U.S. and Glass, T.F. (2011), 'Building Cyclic Schedules for Emergency Department Physicians', *Interfaces*, 41(6):521–533.

NOTES

one can draw for a job shop is given by $(n!)^m$. Clearly, the number of alternatives is far too high for one to complete by enumerating these alternatives and arriving at the best schedule.

12.2.1 Scheduling Rules

A critical parameter for the scheduling problem is the basis on which competing jobs queueing in front of a machine are rank-ordered. Alternative rules can be used for establishing priorities. These rules are known as scheduling rules. The following are the most frequently used rules:

- ⇒ *Shortest processing time (SPT)*: This rule simply chooses the job with the least processing time from the competing list and schedules it ahead of the others. For example, let us consider four jobs waiting in front of a machine for processing. If the processing times (in minutes) are 12, 9, 22, and 11, then the order in which the jobs will be scheduled using the SPT rule is Job 2–Job 4–Job 1–Job 3.
- ⇒ *Longest processing time (LPT)*: This rule is the reverse of the SPT rule. The job with the longest processing time is scheduled ahead of other competing jobs. For our example, the order in which the four jobs are scheduled is exactly in reverse (Job 3–Job 1–Job 4–Job 2).
- ⇒ *Earliest due date (EDD)*: It is possible to establish priorities on the basis of the due date for the jobs. The logic here is that a job that is due tomorrow needs to be scheduled ahead of another that is due the day after even though the second job may require lesser processing time compared to the first. Therefore, jobs can be rank-ordered on the basis of due dates. If the chronological order of the due dates of the four jobs is Job 4–Job 3–Job 2–Job 1, then EDD rule will rank the jobs in that order.
- ⇒ *Critical ratio (CR)*: The SPT and LPT rules take only the processing time information into consideration. On the other hand, EDD considers only the due dates of the jobs. In reality, it will be desirable to consider both while scheduling. A job with a long processing time due in the near future requires greater attention than a job with very little processing time but due at the same time. The critical ratio estimates the criticality of the job by computing this information using a simple ratio:

$$\text{Critical ratio (CR)} = \frac{\text{Remaining time}}{\text{Remaining work}} = \frac{\text{Due date} - \text{Current date}}{\text{Remaining processing time}} \quad (12.1)$$

A smaller CR value indicates that the job is more critical. Particularly, if $CR < 1$, it denotes that the available time is not sufficient and the job is already running late. On the contrary, a larger CR value ($CR > 1$) indicates that some slack is available for the job.

A smaller CR value indicates that the job is more critical.

- ⇒ *First-come-first-served (FCFS)*: This rule schedules the jobs simply in the order of job arrival. There will be no consideration of the processing time or any other information such as the due date of the jobs. This is akin to the working of Indian Railways' computerized passenger reservation system in various cities.
- ⇒ *Random order (RAN)*: It is also possible to assign priorities on a random basis. Suppose we get four random numbers from a random number generator, as follows: 563, 125, 490, and 923. Then one can use this set of numbers for rank-ordering the four jobs (on the basis of ascending order of the three digit numbers). The resulting order will be Job 2–Job 3–Job 1–Job 4.

Table 12.4 illustrates these five rules for the four-job example. In reality, it is possible to construct alternative scheduling rules on the basis of the specifics of the problem that one is trying to solve.

12.2.2 Performance Criteria

The scheduling problem needs to be solved for the purpose of meeting certain characteristics in the operating system. Therefore, the performance criterion is an important parameter for the scheduling problem. Three types of performance criteria are normally considered in operations scheduling. These include *completion-based*, *due date-based* and *inventory-/cost-based* measures.

Three types of performance criteria are normally considered in operations scheduling: completion-based, due date-based, and inventory-/cost-based measures.

Completion-based measures

In completion-based measures, an assessment of the time taken to complete the operations is made and used for the purpose of evaluating the goodness of the chosen scheduling rule. Two measures are frequently used: flow time and make span. **Flow time** is the elapsed time between releasing a job into the shop and the time

Flow time is the elapsed time between releasing a job into the shop and the time of completion of the job.

NOTES

TABLE 12.4 ILLUSTRATION OF THE SCHEDULING RULES (CURRENT TIME = 0)

Job No.	Processing Time (Minutes)	Order of Arrival	Due by	CR	Random Number	Rank Ordering of Jobs Based on					
						SPT Rule	LPT Rule	EDD	CR	FCFS	RAN
1	12	1	23	1.92	0.233	3	2	2	3	1	1
2	2	2	24	2.67	0.857	1	4	3	4	2	3
3	9	3	30	1.36	0.518	4	1	4	1	3	2

NOTES

of completion of the job. Flow time equals the sum of processing time of all operations of the job and other non-productive time, including waiting and moving in the shop at various work centres. Consider the following notations for Job i :

Release time of the job: R_i

Completion time of the job: C_i

Flow time of the job: $F_i = (R_i - C_i)$ (12.2)

Alternative variations of these measures can be computed. For example, maximum flow time, mean flow time, maximum completion time, and mean completion time can also be computed for the set of jobs under consideration. However, the maximum completion time is denoted by another measure known as make span.

Make span is the time taken to complete all the jobs released into the shop for processing. Even when all the jobs are released at the same time, jobs come out of the shop at different times due to time differences in processing. Therefore, make span is the completion time of the last job in the shop.

Make span is the time taken to complete all the jobs released into the shop for processing.

Make span (maximum completion time):

$$C_{\max} = \max_i \{C_i\} \quad (12.3)$$

Flow time, also known as *throughput time*, is an appropriate measure for jobs that are continuously launched on the shop floor. It provides a good measure of the length of the time the job resides on the shop floor. This is directly related to the work-in-progress inventory likely to be accumulating in the shop. On the other hand, if the shop launches batches of components and requires setting up the entire system for another batch of components, then make span will be a more appropriate measure.

Due date-based measures

These measures evaluate the efficacy of the scheduling rule with reference to the due date of the jobs being scheduled. The completion date is compared with the due date and some assessment of the rule is made. *Lateness* is the difference between completion time and due date. If the due date for Job i is denoted as D_i , then

$$\text{Lateness of the job, } L_i = (C_i - D_i) \quad (12.4)$$

If a job is completed ahead of its due date, then there will be negative **lateness** (earliness) as per Equation 12.4. For the set of jobs being scheduled, one can compute additional measures such as mean lateness and maximum lateness.

Lateness is the difference between completion time and the due date.

There is an alternative measure used in scheduling literature known as *tardiness*. If a job is completed ahead of time, instead of computing a negative value for L_i , if we take zero, then the resulting measure is known as tardiness. In other words, tardiness captures information of only those jobs that are late for the purpose of evaluation of the scheduling rule. Therefore,

$$\text{Tardiness of the job, } T_i = \max(0, L_i) \quad (12.5)$$

As in the case of other measures, one can compute the mean, maximum, and minimum values of tardiness for the set of jobs scheduled using a particular scheduling rule. The third measure is the number of tardy jobs, denoted by n_T .

Inventory-/Utilization-based measures

Inventory- or utilization-based measures are based on the actual inventory build-up in the shop or the utilization of the shop resources. The measures include the mean number of jobs waiting, the mean number of finished jobs, the mean number of unfinished jobs, and the mean utilization of machines (or other resources) in the shop. It is also possible to convert these numbers into monetary terms by appropriate cost-based measures. Although these measures are very meaningful, computing these numbers is neither straightforward nor easy. These measures are typically used while the scheduling problem is studied using a simulation model.

It is possible to obtain optimum schedules for certain combinations of shop configurations, scheduling rules, and performance criteria.

Past research in the area of scheduling has established that it is possible to draw optimum schedules for certain combinations of shop configurations, scheduling rules, and performance criteria. Therefore, choice of these parameters could significantly influence the complexity of the problem and the feasibility of obtaining optimum schedules.

12.3 SCHEDULING OF FLOW SHOPS

It is not an easy task to arrive at the optimum schedule for a manufacturing shop. Numerous combinations are possible for scheduling jobs in a shop. In a flow shop, the number of combinations grows non-linearly (on account of the $n!$ factor). Even for a small problem involving 10 jobs, the number of alternatives to be enumerated is 3,628,800. Therefore, it will not only be difficult but also futile to resort to complete enumeration of these alternatives. However, we have noted earlier that a combination of the shop

Development of context-specific scheduling rules is a better investment of managerial time to solve the scheduling problem in organizations.

N O T E S

EXAMPLE 12.2

A manufacturing organization is in the process of assessing the usefulness of SPT and EDD rules for the purpose of scheduling four jobs on a machine. Table 12.5 presents information on the processing time and due dates for the jobs. Assume that all the jobs are available for scheduling at the beginning of the planning horizon so that the release time is zero. Further, due dates in the table denote the time at which the component is due at its next stage of manufacturing.

Compute the relevant performance measures and comment.

TABLE 12.5 PROCESSING TIME AND DUE DATES FOR THE JOBS

Job Number	Process Time	Due by
1	4	6
2	7	9
3	2	19
4	8	17

Solution

Job sequencing using SPT rule: 3–1–2–4.

Job sequencing using EDD rule: 1–2–4–3.

In order to understand when the jobs get completed, we need to estimate the beginning and end times for all jobs. In the case of SPT, Job 3 is first scheduled in the machine. At the end of 2 time units, Job 1 is scheduled and it takes 4 time units. Therefore, at the end of 6 time units, Job 2 is scheduled, and finally, at the end of 13 time units, Job 4 is scheduled. Job 4 completes its processing by 21 time units, which is the make span. Using this information we can compute all the measures. The workings for both SPT and EDD are given in Tables 12.6 and 12.7. It may be noted that all the jobs are indeed available for processing at time zero. Therefore, the release time for all the jobs is zero. Due to the scheduling rule, there are differences with respect to when they undergo the processing.

Some Observations:

Both the scheduling rules result in the same make span. This is because there is only one machine involved and all the jobs undergo processing in this machine. However, the rules differ in their performance. EDD rule performs better in due date-based measures even at the expense of increasing the number of tardy jobs by one.

TABLE 12.6 PERFORMANCE OF THE SPT RULE

Scheduling Rule: SPT					
Processing Order	Release Time (R_i)	Completion Time (C_i)	Flow Time (F_i)	Lateness	Tardiness
2	0	6	6	0	0
3	0	13	13	4	4
1	0	2	2	– 17	0
4	0	21	21	4	4
Mean		10.50	10.50	– 2.25	2.00
Maximum		21.00	21.00	4.00	4.00
Minimum		2.00	2.00	– 17.00	0.00
Number of tardy jobs = 2; Make span = 21					

TABLE 12.7 PERFORMANCE OF THE EDD RULE

Scheduling Rule: EDD					
Processing Order	Release Time (R_i)	Completion Time (C_i)	Flow Time (F_i)	Lateness	Tardiness
1	0	4	4	– 2	0
2	0	11	11	2	2
4	0	21	21	2	2
3	0	19	19	2	2
Mean		13.75	13.75	1.00	1.50
Maximum		21.00	21.00	2.00	2.00
Minimum		4.00	4.00	– 2.00	0.00
Number of tardy jobs = 3; Make span = 21					

configuration, the scheduling rule, and the performance measure could lend itself to identifying the optimum combination without much computational effort. We shall concentrate on one such algorithm pertaining to the flow shop here.

12.3.1 Johnson's Rule

Let us consider a flow shop with two machines and n number of jobs to be scheduled. Let us also assume that we are interested in minimizing the make span. Johnson proposed a simple algorithm that will provide the optimum schedule for the shop. The algorithm is as follows:

Step 1: Let t_{1i} denote the processing time of Job i in Machine 1 and t_{2i} denote the processing time in Machine 2.

Step 2: Identify the job with the least processing time in the list. If there are ties, break the tie arbitrarily.

- (a) If the least processing time is for Machine 1, place the job at the front of the sequence immediately after any jobs already scheduled.
- (b) If the least processing time is for Machine 2, place the job at the back of the sequence immediately before any jobs already scheduled.
- (c) Remove job i from the list.

Step 3: If there are no more jobs to be scheduled, go to Step 4. Otherwise, go back to Step 1.

Step 4: The resulting sequence of jobs is the best schedule to minimize the make span of the jobs.

Johnson's rule is quite simple and it applies to a restrictive set of assumptions about the number of machines and the shop configuration. However, there have been some extensions of Johnson's rule that apply to a wider set of situations involving flow shops. The first variation is Jackson's rule that generalized Johnson's rule to a situation involving two machines with jobs skipping one of the two machines. Thus, the flow shop becomes a job shop with four sets of jobs possible. One set containing jobs visiting only Machine 1, the second set visiting only Machine 2, the third set visiting Machine 1 and then Machine 2, and the fourth set visiting Machine 2 first and then Machine 1. The second extension of the Johnson's rule is the CDS method. This method extends Johnson's rule to a multi-machine situation and derives schedules that are near-optimal.

There are several realistic assumptions that one needs to take into consideration while scheduling the jobs that makes the problem mathematically intractable. For example, there are situations of no job waiting during certain intermediate stages of processing as in the case of a heat treatment operation with a pre-heating schedule. There are other situations that warrant minimum waiting before being scheduled for the next operation, as in the

NOTES

EXAMPLE 12.3

Five jobs are to be scheduled in two machines in a manufacturing shop. All the five jobs undergo processing in both the machines (flow shop). Table 12.8 provides information on the processing time in both the machines.

- Identify the best sequence using Johnson's rule.
- Develop a Gantt chart for the schedule and compute the flow times and make span for the jobs for this sequence.

TABLE 12.8 PROCESSING TIME

Processing Time		
Job no.	Machine 1	Machine 2
1	4	7
2	6	3
3	2	3
4	7	7
5	8	6

Solution**Johnson's Rule:**

The least processing time is for Job 3. Since it is for Machine 1, we schedule the job at the front end of the sequence:

Job 3 – – – –

Job 3 is removed from the list. The next least processing time is for Job 2. Since the processing time is for Machine 2, we schedule the job at the end of the sequence:

Job 3 – – – Job 2

The next job with least processing time in Machine 1 is Job 1. Therefore, we schedule it immediately after Job 3 at the front:

Job 3 Job 1 – – Job 2

The fourth job with least processing time in Machine 2 is Job 5. Therefore, we schedule the job immediately before Job 2 at the back of the sequence:

Job 3 Job 1 – Job 5 Job 2

Finally, we schedule Job 4 in the remaining slot. Since no more jobs are available, we end the procedure with the following optimum schedule for the problem:

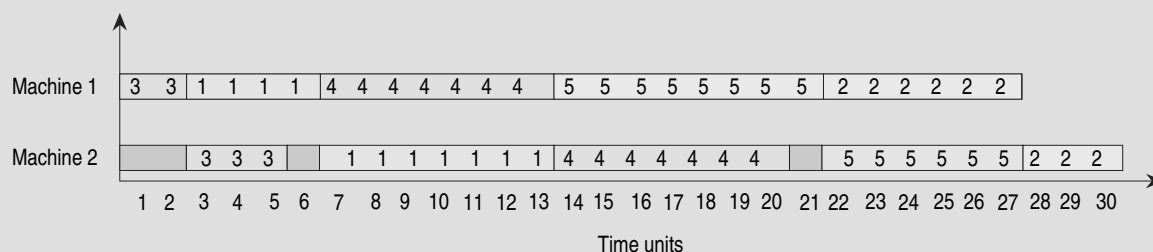
Job 3 Job 1 Job 4 Job 5 Job 2

Gantt Charts for the Schedule:

Graphical representations of the information on the start and end times of resource usage by various jobs are often used in practice. The Gantt chart is a simple graphical tool using which this can be accomplished. Gantt charts improve the understanding of the problem and aid in visual control of the problem. Moreover, they make communication of the schedules and monitoring of the progress much easier. We will demonstrate the use of Gantt charts using the schedule developed in this example.

In this example, the scheduling sequence has already been identified. Therefore, one can graphically plot the start and end times of the jobs in both machines. Job 3 takes two minutes in Machine 1, during which time Machine 2 is idle. The job requires three minutes in Machine 2. Therefore, Machine 2 will be processing Job 3 during minutes 3–5. One can proceed in this fashion to arrive at all the start and end times for the jobs. The resulting Gantt chart is shown in Figure 12.4.

The graphical representation of the schedule helps operating personnel in easy identification and control. The make span for the five jobs is 30 time units. Moreover, one can directly identify the flow time for each job. The flow time for the jobs is Job 1: 13, Job 2: 30, Job 3: 5, Job 4: 20, and Job 5: 27. The Gantt chart also shows the idle time for machines and waiting time for the jobs (if any). In this example, Machine 2 is idle for 4 time units in a cycle of 30 time units. No jobs are waiting for the machines in this case.

**FIGURE 12.4** The resulting Gantt chart

case of a painting application. Further complications can arise on account of sequence-dependent set-up times between two jobs.

As more and more such constraints are introduced into the problem, obtaining optimum schedules evades the planner. The realistic alternative is one of obtaining good enough and workable schedules. Production planners benefit in this process by gainful exploitation of the available computing. Development of context-specific scheduling rules and heuristic procedures will serve the management better and save time in solving the scheduling problem in organizations.

12.4 SCHEDULING OF JOB SHOPS

As we have already noticed, scheduling is much more complicated in the case of a job shop. Unlike the flow shop, developing optimal methods is difficult and computationally complex. Complete enumeration is also not possible as the number of schedules $[(n!)^m]$ grows exponentially with n and m . Therefore, the use of appropriate scheduling rules and performance criteria are the only viable options for the planner. Use of graphical methods such as Gantt charts is very useful for job-shop scheduling. However, unlike the case of flow shops, the performance of various scheduling rules cannot be easily predicted in a job shop. Therefore, one can resort to simulation modelling of the system and evaluate alternative scheduling rules.

The use of graphical methods such as Gantt charts is very useful for job-shop scheduling.

12.5 OPERATIONAL CONTROL ISSUES IN MASS PRODUCTION SYSTEMS

So far, we have concentrated on flow shops and job shops and issues pertaining to scheduling and operational control in these systems. However, there are several examples of mass production systems in operation today. For example, a two-wheeler manufacturer such as Bajaj Industries produces over 9,000 two wheelers everyday. Various sub-assemblies in the Bajaj plant need to be configured to match the production rate. Similarly, the final assembly stations should also have the required number of resources at each station to meet the targeted demand. In such a scenario, much of control and scheduling boils down to appropriately arriving at a balanced flow of components on the shop floor. There are two ways of achieving this objective:

- (a) Estimate the number of resources required at each station of the manufacturing system for a targeted production level. Conversely, if there are a certain number of resources available, estimate the production level and adjust the plans accordingly.

NOTES

EXAMPLE 12.4

Consider a job shop with five machines and four jobs. Each job has a processing sequence in which it visits three out of the five machines. Each job also has a due date for completion. The relevant information is given in Table 12.9.

- Use SPT and EED rules to obtain the schedules for the jobs.
- Compute the make span, mean lateness and number of tardy jobs.

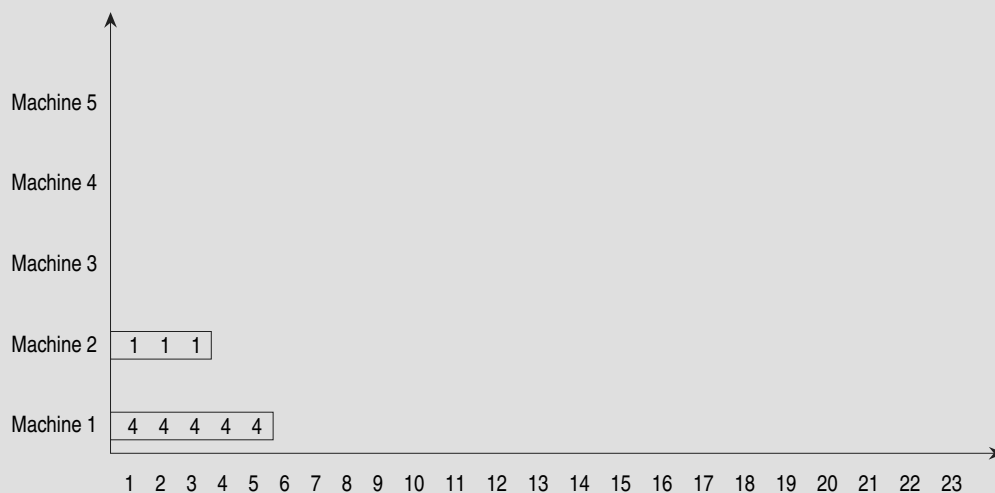
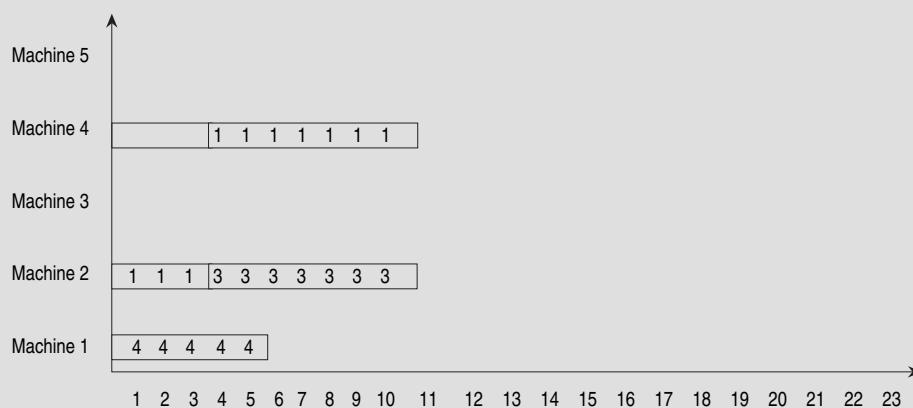
Solution

We use a Gantt chart to develop the schedule for the shop.

TABLE 12.9 DETAILS FOR EACH JOB

Jobs	Processing Time (Machine Visited in the Sequence)*			Due by
	1	2	3	
Job 1	3 (2)	7 (4)	3 (5)	10
Job 2	6 (1)	3 (2)	7 (5)	12
Job 3	7 (2)	3 (4)	4 (3)	9
Job 4	5 (1)	4 (3)	5 (4)	14

Note: *The numbers in the parenthesis indicates the machine in which the job gets processed

**FIGURE 12.5** The partial Gantt chart for the SPT rule**FIGURE 12.6** The Gantt chart with the additional schedules**SPT Rule:**

Jobs 2 and 4 require processing in Machine 1. However, since the processing time of Job 4 is less than for Job 1 it is scheduled first in the machine using SPT. Similarly, Job 1 is first

scheduled in Machine 2. The partial Gantt chart is shown in Figure 12.5.

At the end of time unit 3, Job 1 exits Machine 2 and it is time to take the next scheduling decision. Job 1 will be scheduled in Machine 4 as there is no other job competing for that machine.

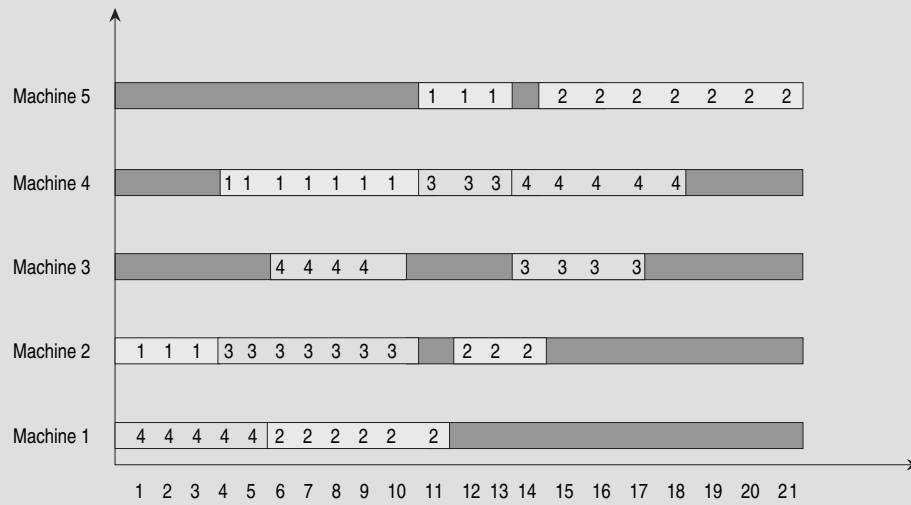


FIGURE 12.7 The final Gantt chart for the SPT rule

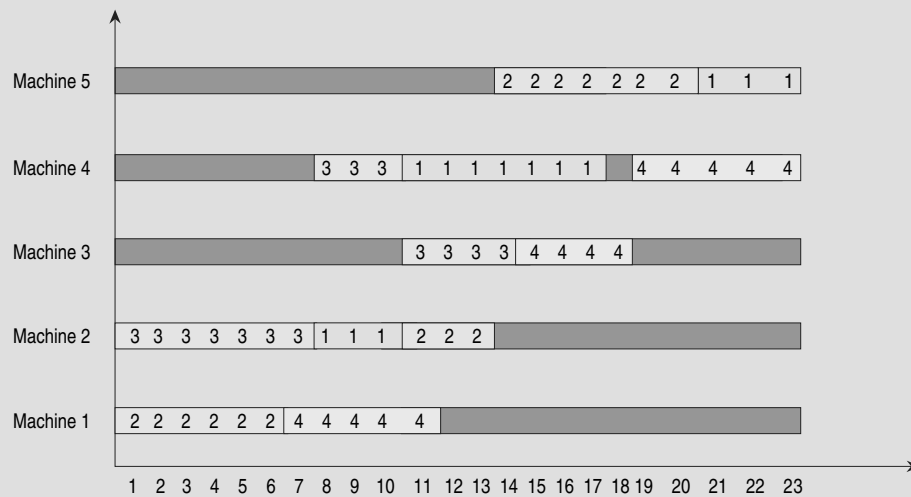


FIGURE 12.8 The Gantt chart for the EDD schedule

Similarly Job 3 will be scheduled in Machine 2. The Gantt chart in Figure 12.6 has these additional schedules included.

One can proceed in this fashion to arrive at the final chart for the SPT rule, as shown in Figure 12.7. It may be noted that the SPT rule is invoked only when more than one job competes for a resource. The make span for the problem is 21 units as evident from the Gantt chart.

The EDD rule schedule is drawn using a similar logic. When there are competing jobs for a resource, the EDD criterion is used. The Gantt chart for the EDD schedule is given in Figure 12.8.

In this case, the make span has increased by two time units to 23. Table 12.10 presents a summary of the performance measures for SPT and EDD rules. Interestingly, the EDD rule has fared badly on completion time, make span and lateness. This example clearly demonstrates the complexity of job shops.

Predicting the performance of the rules is not as easy as in the case of a flow shop. In both the cases, all the jobs were tardy.

TABLE 12.10 PERFORMANCE MEASURES FOR SPT AND EDD RULES: A SUMMARY

Job Number	SPT			EDD	
	Due	C_i	Lateness	C_i	Lateness
1	10	13	3	23	13
2	12	21	9	20	8
3	9	17	8	14	5
4	14	18	4	23	9
Average		17.25	6.00	20.00	8.75

NOTES

- (b) Given a certain availability of resources at various stages of the production system and a targeted daily production, estimate the rate at which the entire system should work and accordingly modify the work practices and resource deployment to meet the target.

The first method lays emphasis on designing operational control systems for mass production. On the other hand, the second method focuses on the ability to craft alternative operational control strategies to meet the changing targets on a period-to-period basis. The former is a traditional application of the concept and is often referred to as the *line-balancing technique*, while the latter is the modern Japanese approach to manage mass production systems. A detailed description of line balancing is available in the book.

The line-balancing technique does not solve all the operational control problems in a mass production system. It provides a basis for the initial design and deployment of resources. However, mass production systems face constant revisions in the production rate. These changes occur due to changes in the market demand and revisions in component off-take by customers. Consider an auto-component manufacturing firm, such as the Chennai-based Lucas TVS that supplies alternators to Maruti Udyog Limited (MUL). Suppose the daily requirement of MUL is 480 alternators during the first week of a month. If Lucas works for two shifts of eight hours each, then the hourly production of alternators is $480/16 = 30$. Furthermore, the cycle time for alternator assembly is $60/30 = 2$ minutes. This means that the production system responsible for manufacturing alternators for MUL should work at a uniform pace with a cycle time of two minutes everywhere. The final assembly will have to be tuned to two minutes, and so will all the feeder shops to the final assembly.

Suppose that there has been a revision in the daily requirement by the third week of the month. Due to increased production at their plant, let us assume that MUL wants a daily supply of 600 alternators. To cope up with the increase in the demand, Lucas will have to work with a new cycle time 20 per cent less than the earlier measure (that is, 1.6 minutes). The final assembly as well as all the feeder units will have to be tuned to this new requirement. This measure therefore provides a rhythm for the overall functioning of the shop. In mass production systems, this measure is known as TAKT time. Operational control issues in mass production systems begin with the computation of TAKT time. Based on this, several choices are made to ensure that the actual cycle time is as per the computation of TAKT. These choices can be grouped under the following categories: machine redeployment, altering operator allocations, and adjusting material feed rates.

Operational control issues in mass production systems begin with the computation of TAKT time.

12.5.1 Machine Redeployment

Mass production systems lay emphasis on the flexibility of the varying deployment of resources to alternative product groups. Therefore, one method of responding to changing TAKT time is to alter the assignment of machines (or manufacturing cells) to specific product groups. By incremental addition/deletion of machines, it is possible to adjust the production rate and cycle time.

12.5.2 Altering Operator Allocations

Operators in a mass production system are often expected to be multi-skilled. Therefore, one way of adjusting the cycle time is to alter the worker allocations. If a worker was attending to three machines earlier, then by adding additional workers to the manufacturing cell, the cycle time can be reduced, and vice versa. Such practices are common in mass production systems. Table 12.11 presents an illustrative example of this. In this example, for a required daily output of 400 units, the output required per shift is 200 units. Since the net available production time is 420 minutes, the TAKT time comes to 126 seconds. When the daily output is increased from 400 to 450, it required adjustment of the TAKT time from 126 seconds to 112 seconds in order to continue working on a two-shift basis. However, the number of workers has increased from 28 to 32 (2 additional workers per shift). Typically, employees will be deployed from another product line to this one to meet the new daily requirement. On the other hand when the daily output falls to 371, two workers need to be taken out of the production line (one per shift) and redeployed elsewhere.

12.5.3 Adjusting Material Feed Rates

Mass production systems have a streamlined material flow, often achieved by installing a conveyor system. The speed at which the

TABLE 12.11 AN ILLUSTRATION OF WORKER DEPLOYMENT FOR ADJUSTED TAKT TIME

Required output per day	400	450	371
Number of shifts per day	2	2	2
Required output per shift	200	225	185.5
Net available production time (minutes)	420	420	420
TAKT time (seconds)	126	112	136
Work content (seconds)	1,764	1,764	1,764
Number of operators required per shift	14	16	13
Total number of operators required	28	32	26

NOTES

NOTES

conveyor moves between stations is a function of the cycle time. For instance, if two stations are five metres apart and the TAKT time is 30 seconds, then the conveyor needs to move between stations at a speed of 1 metre every 6 seconds (10 metres per minute). On the other hand, if the TAKT time changes to 20 seconds, the conveyor speed should be adjusted to 1 metre every 4 seconds (15 metres per minute). Furthermore, the tasks to be completed at every workstation needs to be completed within 20 seconds.

12.6 OPERATIONS PLANNING AND CONTROL BASED ON THE THEORY OF CONSTRAINTS

In this chapter, we have discussed various techniques for scheduling operations in a job shop and a flow shop. However, despite the use of these, there is a common problem of high WIP inventory in several manufacturing organizations. Moreover, manufacturing organizations suffer from the “end-of-the-period” syndrome. Typically, as organizations approach the end of a reporting period (the year-end is a classic example, but there are monthly and quarterly reporting cycles also), workflow gets considerably complex and inventory piles up beyond normal limits. Several reasons are attributed to this and significant among them are the following:

- (a) Organizations do not use appropriate measures for performance.
- (b) Operations scheduling does not take into consideration some ground realities in the shop floor.

Measures based on standard costing, such as variances at the department or even work-centre level, promote local optimum at the expense of global optima. Therefore, it is possible to create mountains of inventory, yet products may not flow evenly out of the manufacturing system. Similarly, scheduling on the basis of balancing capacity at individual work-centre level may not ensure a smooth flow of material through the manufacturing system.

12.6.1 Measures of Performance

The **theory of constraints** developed by Eliyahu Goldratt addresses these concerns and proposes alternative methodologies for operations scheduling. Goldratt also proposed three new measures of performance for manufacturing firms.

- ⇒ *Operational measures*: The operational measures for a company are the throughput, inventory, and operational expenses. These measures could be used to make all decisions pertaining to the use of manufacturing resources.

According to the **theory of constraints**, it is possible to increase the throughput even while the operating expenses and inventory are being reduced.

- ⇒ *Throughput*: Throughput is the quantity of money generated by a firm through sales over a period of time. The standard definition of throughput is total output, but this definition considers only sales and not finished goods (FG). According to Goldratt, this is because finished goods do not generate revenue for the firm until they are sold.
- ⇒ *Inventory*: The quantity of money invested in materials that a firm intends to sell is defined as inventory. Contrary to the traditional logic (where inventory valuation includes value-added components to estimate WIP and FG inventory), here, it is a simple measure of the value of the material only.
- ⇒ *Operating expenses*: Operating expenses include all expenses incurred by a firm to convert inventory into throughput over a period of time.

According to the theory of constraints, it is possible to increase the throughput even while the operating expenses and inventory are being reduced. This is identified as the single most important goal of managing manufacturing organizations. The principles governing the theory of constraints are best explained using the analogy of marching soldiers.

12.6.2 The Analogy of Marching Soldiers

The problems associated with scheduling and operational control of manufacturing systems can be best explained using the analogy of a column of marching soldiers. Assume that a batch of 10 soldiers begin the march from a reference point. Initially, all the 10 soldiers will be one behind the other with an equal gap. However, as they begin to march, certain differences will be noticed. If we observe the marching soldiers, we may find that the distance between the first and the last soldier is more than what it was at the starting position. This is because as the soldiers engage in marching, differences in the speeds and capacities of each soldier begin to manifest. If the third soldier is slower than the first two, then the gap between the third and the other two soldiers ahead of him will increase with time. If we observe the marching soldiers after they have covered, say, 5 km, the gap will increase even further.

Scheduling and operational control problems in manufacturing organizations share considerable similarity with this analogy of marching soldiers. The ground to be covered in the marching-soldiers example is the same as the raw material in the case of manufacturing organizations, and the ground already covered corresponds to finished goods. On the other hand, the mounting gap between the first and the last soldier is equivalent to the work-in-process inventory in a manufacturing organization (see Figure 12.9 and Table 12.12 for a graphical representation of this).

As in the case of the soldiers, resources in manufacturing organizations differ from one another in their ability to process components. Some machines may have more than the required

NOTES

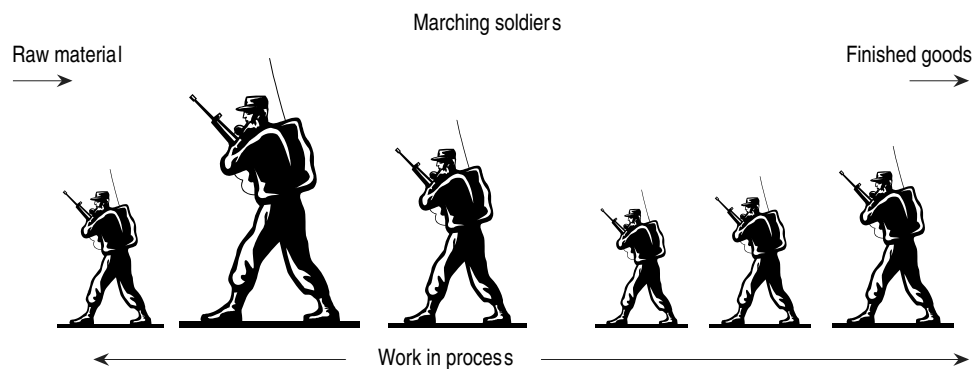


FIGURE 12.9 Marching soldiers: an analogy for planning and control in manufacturing systems

NOTES

capacity to process in a planning horizon. Some other resources may have less than the total capacity required for the planning horizon. Statistical fluctuations and dependent events are characteristic of resources in a manufacturing organization. All these generate similar effects in a manufacturing system as that of increasing the gap between the soldiers. However, we know that increasing WIP inventory and manufacturing lead time is directly related to the increase in the gap between the soldiers as they march. Therefore, operational control is primarily a function of making sure that the gap between the soldiers does not increase with time.

The *theory of constraints* is a systematic body of knowledge that recognizes these effects in a manufacturing system and uses specific methods to improve the performance of the system under these conditions. **Synchronous manufacturing** is a specific application of the theory of constraints to scheduling and operational control of the manufacturing systems.

Synchronous manufacturing is a specific application of the theory of constraints to scheduling and operational control of manufacturing systems.

TABLE 12.12 A COMPARISON OF MARCHING SOLDIERS AND A PRODUCTION SYSTEM

Resource	Marching Soldiers	Production System
Processing	Ground to be covered	Raw material to be processed
WIP	Gap between the first and the last soldier	Work in process in the shop floor
Throughput	The extent of ground covered by the marching soldiers	Amount produced and sold by the production system
Operating expenses	Amount of energy expounded by the soldiers to complete the march	Cost of transforming the raw material into throughput
Objective	To cover a certain extent of the ground in a given time	To achieve a certain throughput in a given time

12.6.3 Synchronous Manufacturing

The marching-soldiers example shows that the rate at which they can cover ground is a function of the slowest marching soldier in the column. If soldiers ahead of the slowest continue to march at their higher speeds, they will only contribute to increasing the gap between them and the slowest. On the other hand, those behind the slowest will be forced to march only at the speed of the lowest. One way to solve this problem is to first tie a rope connecting all the soldiers (from the first to the last), and provide a drumbeat at a rate equivalent to the slowest and enable the marching soldiers to respond to the rhythm of the drumbeat. By doing this we not only ensure that the gap does not increase with time, but also create a synchronized movement of the soldiers on the ground. Synchronous manufacturing deploys these features in the scheduling and operational control logic. In synchronous manufacturing, the focus is on synchronizing flow rather than on balancing capacities.

This discussion clearly brings out the need to address the scheduling problem in a different context. It is important to plan the scheduling and control of the entire manufacturing system on the basis of availability of resources. The following definitions are common in synchronous manufacturing literature.

- ⇒ *Bottleneck resource*: Any resource whose capacity is equal to or less than the demand placed on it is a **bottleneck resource**. On the contrary, any resource whose capacity is more than the demand placed on it is a non-bottleneck resource.

Any resource whose capacity is equal to or less than the demand placed on it is a **bottleneck resource**.

- ⇒ *Capacity constrained resource (CCR)*: In a manufacturing system it is observed that some resources have the ability to significantly deviate the product flow, from the planned flow, if not properly scheduled. Any resource that can deviate the actual product flow from the planned flow is known as a **capacity constrained resource**. A bottleneck or a non-bottleneck resource may become a CCR.

Any resource that can deviate the actual product flow from the planned flow is known as a **capacity constrained resource (CCR)**.

- ⇒ *Activation of a resource*: Activation merely refers to the deployment of a manufacturing resource to process materials. Merely processing material may not in itself guarantee increase in throughput.
- ⇒ *Utilization of a resource*: Utilization refers to the activation of a manufacturing resource that will lead to improving the throughput.
- ⇒ *Process batch*: A process batch refers to the quantity of a product (or part) processed at a resource before that resource

NOTES

NOTES

changes over to another product (or part). For example, if two parts—X and Y—are processed in Resource R1, then, if 300 units of Part X are processed before changing over to processing Part Y, then the process batch for Part X is 300.

⇒ *Transfer batch*: A transfer batch indicates the quantity moved at the same time from one resource to another resource in a manufacturing system. For example, if 100 units of Part X are moved at a time from Resource R1 to Resource R2 for further processing, then the transfer batch of Part X is 100.

12.6.4 Constraints in a Manufacturing System

Before we address the issue of scheduling with bottleneck resources as the focus, it is helpful to understand how bottlenecks develop in an organization. In a broad sense, bottlenecks are nothing but the manifestation of constraints. In synchronous manufacturing, any element that prevents the system from increasing its throughput is identified as a constraint. There are several constraints in an organization—some of them are hard constraints and others soft. By hard, we mean constraints arising out of capacities, physical infrastructure, resource availability, and so on. On the other hand, soft constraints arise out of policy implications and poor managerial practices. For example, a peculiar and complicated approval process for purchase order release may result in severe material constraints (due to long lead time for procurement). Soft constraints are not obvious to the management. They manifest indirectly in other forms of hard constraints (as our purchase order example shows). It requires special attention to track and remove those constraints.

Market constraints are the limits of production imposed by the market demand. Capacity and material constraints arise out of physical limitations in capacity and material availability. For instance, the short supply of a key ingredient in a manufacturing process puts limits on the system. Similarly, the capacity of available machines and human resources may impose limits on the system. There are also soft constraints arising out of managerial, logistical, and behavioural issues.

Synchronous manufacturing may offer valuable help in scheduling a manufacturing system with constraints. However, in the long run, it is important to invest in methods to remove the constraints. Removing an existing constraint merely shifts the constraint elsewhere and is therefore an ongoing process, as indicated in Figure 12.10.

The guiding principles of synchronous manufacturing are as follows:

1. Do not focus on balancing capacities. Instead, focus on synchronizing the flow.
2. The marginal value of time spent at a non-bottleneck resource is negligible. Do not attempt to reduce time at a non-bottleneck resource.

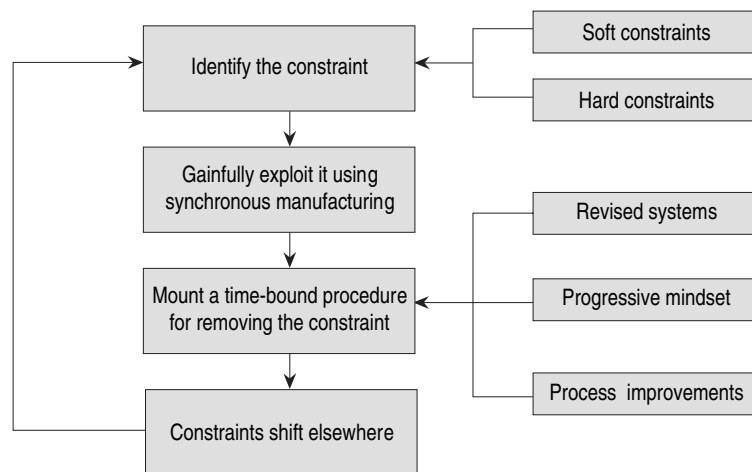


FIGURE 12.10 Constraint management in the long run

NOTES

- 2a. An hour gained at a non-bottleneck resource is a mirage.
- 2b. An hour lost at a non-bottleneck resource is an hour of throughput loss.
3. The level of utilization of a non-bottleneck resource is controlled by other constraints within the system
4. Resources in the system must be utilized, not simply activated.
5. A constraint is any element that prevents the system from achieving the goal of making more money.
6. The transfer batch need not, and many times should not, be equal to the process batch

From the definitions and the guiding principles we have discussed, a few important aspects can be inferred:

- ⇒ The separation of resources as bottleneck and non-bottleneck is very important because while bottleneck resources determine the (planned) output of the system, CCR will ensure that the actual throughput does not deviate from the plan in a manufacturing system. Focusing on maximizing the utilization of a bottleneck resource is instrumental to maximizing the throughput in a manufacturing system. On the other hand, scheduling is done in synchronous manufacturing with reference to CCRs.
- ⇒ Placing buffers in front of bottleneck resources will protect the throughput of the system
- ⇒ Careful choice of the transfer and process batch sizes will ensure the protection of the throughput at the bottleneck as well as proper scheduling of the CCRs
- ⇒ Once these aspects are determined, the rest of the system must work in tandem with these decisions

A drum–buffer–rope (DBR) methodology helps one to achieve these while scheduling a manufacturing system.

NOTES

12.6.5 The Drum–Buffer–Rope (DBR) Methodology

Once a preliminary production plan is established, synchronous manufacturing develops detailed schedules for the CCRs in the plant. On the basis of these schedules, the production plan can be finalized. The resulting modified production plan can become the master production schedule (MPS) and can be used as the basis for making promises to customers. This process of arriving at the MPS on the basis of the detailed schedules for CCRs is the drum in DBR methodology. The MPS, which is based on the processing capabilities and requirements of the CCRs, determines the pace and the sequence of production for the entire plant. The MPS itself is therefore analogous to the drumbeat of the march. The drumbeat ensures that all resources are not merely activated but utilized as per the throughput requirement.

Since statistical fluctuations are inevitable, there could be potential opportunities for the planned flow to deviate from the actual flow. To protect the schedule from disruptions, a time buffer is built into the system. The planned lead time is the sum of the process and set-up times and the time buffer. In effect, the time buffer increases the planned lead time by a sufficient amount so as to accommodate disruptions and fluctuations that are likely to happen in the manufacturing system. Since the bottleneck resource dictates the output of the entire system, an hour of capacity lost at the bottleneck translates directly into an hour of lost throughput. Therefore, the buffer is placed in front of a bottleneck resource. On the contrary, nothing is achieved by placing a buffer ahead of a non-bottleneck resource. It merely increases the inventory in the system but not the throughput.

Once the MPS is finalized, the product structure, routing sheets, and other production policies are used to determine the quantity and timing of planned production at each resource in the plant. Since the MPS is based on the capabilities of CCRs, the entire plant operation is tied through this process to the drumbeat of the MPS. Through this methodology, we have ensured synchronization of schedules at all non-CCRs. Therefore, this is the rope component of the DBR methodology. Figure 12.11 illustrates the DBR methodology in the case of a bottleneck resource in a shop having seven resources. The DBR methodology can be outlined as follows:

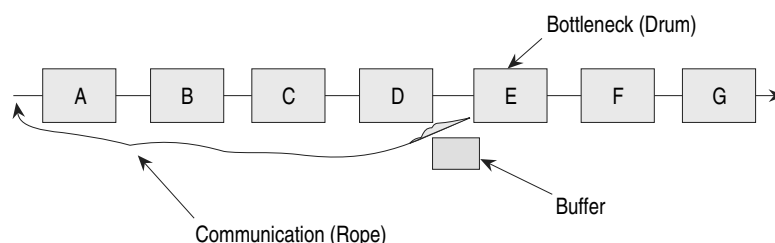


FIGURE 12.11 An illustration of the DBR methodology

- (a) Develop the MPS so that it is consistent with the constraints of the systems (Drum).
- (b) Protect the throughput of the system from statistical fluctuations through the use of buffers at some critical points in the system (Buffer).
- (c) Tie the production at each resource to the drumbeat (Rope).

Several organizations have benefited from the use of the DBR methodology for scheduling operations in a manufacturing system. In response to the demand for easy application of these principles, several computer-based software packages have also been developed to implement the DBR methodology. Synchronous manufacturing principles represent a recent approach to addressing scheduling complexities in organizations. Moreover, the notion of the theory of constraints is also applicable to other operations systems such as management of large projects.

NOTES

12.7 SUMMARY

- The focus shifts from operations planning to operational control in the short term. Scheduling aids operational control in manufacturing and service systems.
- The scheduling context relates to the number of jobs and machines in the system and the physical configuration of the machines. These factors greatly influence the complexity of scheduling.
- Flow shops and job shops are two alternatives for the configuration of a manufacturing system. The scheduling methodology and complexity differ vastly between these two. Job shops are far more complex to schedule than flow shops.
- Johnson's algorithm provides an optimal schedule for a two machine- n job problem using the shortest processing time rule for scheduling. Several variations of Johnson's algorithms are available for flow-shop scheduling.
- Operational control in mass production systems are primarily achieved through use of TAKT time-based scheduling.
- The theory of constraints indicates that scheduling of operations must take into account the existence of bottlenecks and statistical fluctuations in operations.
- Synchronous manufacturing principles apply the theory of constraints and develop alternative schedules using a drum-buffer-rope methodology.

12.8 FORMULA REVIEW

$$\begin{aligned}
 1. \text{ Critical Ratio (CR)} &= \frac{\text{Remaining time}}{\text{Remaining work}} \\
 &= \frac{(\text{Due date} - \text{Current date})}{\text{Remaining processing time}}
 \end{aligned}$$

- 2. Flow time of the job, $F_i = (R_i - C_i)$
- 3. Make span (maximum completion time), $C_{\max} = \max_i \{C_i\}$
- 4. Lateness of the job, $L_i = (C_i - D_i)$
- 5. Tardiness of the job, $T_i = \max(0, L_i)$

12.9 REVIEW QUESTIONS

- How is scheduling linked to other aspects of production planning? Explain with the help of an example.
- What do you mean by the loading of machines? What do you achieve by performing the loading function?
- Classify the following as a job shop or as a flow shop for the purpose of scheduling. Also, identify the relevant parameters of the system that have a bearing on the scheduling complexity.
 - Executive health check-up in a hospital
 - Tirupati temple queue complex having multiple stages
 - Breakfast buffet system at Taj Hotel
 - Final assembly shop of a computer manufacturer
 - Computerized passenger reservation system of Indian Railways
 - Eye camps conducted by a local eye hospital
 - A common heat treatment facility in a manufacturing plant
 - Visa section at the US consulate office at Mumbai
- How would you determine the complexity of the scheduling problem?
- A firm is interested in choosing between the minimization of make span and minimization of flow time as the performance criterion. Prepare a short note advising them on the pros and cons of using these rules.
- What are the implications of using completion-based measures as opposed to due date-based measures for evaluating scheduling rules?
- Explain, by means of a simple example, the difference between lateness and tardiness. Which is a better measure? Why?
- What is a Gantt chart? What is its use in scheduling?
- What is cycle time? What are the practical implications of cycle time measure in mass production scenarios? How do they influence the scheduling of jobs in a manufacturing system?
- An organization has a certain number of workstations in its final assembly for a desired production rate. If the desired production rate has gone up by 10 per cent, how would you respond to the new requirement? Will this change have an effect on cycle time?
- What do you mean by soft and hard constraints in an organization?
- How are synchronous manufacturing principles different from the traditional methods used for scheduling of jobs?
- What is meant by the DBR methodology? What does it seek to achieve?

12.10 PROBLEMS

- Consider a one-machine-six-jobs scheduling problem. All the jobs are available for scheduling at the beginning of the planning horizon itself. The processing times and due dates of the jobs are given in Table 12.13.

TABLE 12.13 PROCESSING TIMES AND DATES OF JOBS

	Job 1	Job 2	Job 3	Job 4	Job 5	Job 6
Processing time (minutes)	12	10	15	13	12	9
Due (minutes from start of planning horizon)	25	45	55	30	90	70

- Use LPT, SPT, FCFS, RAN, EDD, and SR to arrive at a schedule
- For each of the schedules, compute the following performance measures:

- Mean flow time
- Make span
- Mean completion time
- Mean lateness
- Mean tardiness
- Number of tardy jobs

- Identify the rule that performs the best for each of these performance criteria.

- A service provider has two stages in service delivery. Initially, the customers arrive at Stage 1, get the initial part of the service, and then reach Stage 2, where the remaining part of the service is completed. All arriving customers wait in a lounge and are called on some basis into the system once a batch of eight accumulates. All of them go through these two stages, although the time they spend in these two stages may vary depending on their specific requirements. The time spent by a set of eight customers on these two stages is shown in Table 12.14.
 - Use Johnson's rule to schedule customers in the system.
 - Develop a Gantt chart for the schedule.
 - Identify key performance measures for this situation and compute them for the solution obtained using Johnson's rule.
 - If you scheduled the customers using FCFS, what is its impact on the performance criteria?

TABLE 12.14 DURATION SPENT BY CUSTOMERS ON THE TWO STAGES

	Customer Number							
	1	2	3	4	5	6	7	8
Stage 1 (minutes)	11	10	9	13	12	10	9	11
Stage 2 (minutes)	9	14	10	12	14	14	11	10
Arrival time (minutes from start of batch)	0	1	3	4	5	7	8	10

(e) Evaluate and compare the performance of the random rule with Johnson's rule and the FCFS rule.

3. A pure flow shop has three machines and all the jobs visit these machines. The processing time of each job in each machine is given in Table 12.15.

- Use the SPT rule to schedule the jobs in the flow shop and draw a Gantt chart for the schedule.
- Compare the performance of the SPT rule with the EDD rule using the following performance criteria:
 - Mean tardiness
 - Mean flow time
 - Number of tardy jobs
 - Make span
- Compute the utilization of the three machines for both the schedules.

(Hint: Convert it into a two machine problem. First alternative is to combine processing times of m/c1 and m/c2 into a single equivalent machine. Alternative 2 is to combine the processing times m/c2 and m/c3. Solve using Johnson's rule and choose the best alternative)

TABLE 12.15 PROCESSING TIME OF JOBS, MACHINE-WISE

Processing Time	Job Number							
	1	2	3	4	5	6	7	8
Machine 1 (minutes)	14	10	10	13	12	11	9	9
Machine 2 (minutes)	11	14	10	9	14	14	7	10
Machine 3 (minutes)	9	8	10	11	12	7	8	9
Due (minutes from start of planning horizon)	80	45	55	60	90	50	100	70

TABLE 12.16 PROCESSING TIME OF JOBS

Jobs	Machines				
	A	B	C	D	E
1	11	14	17	10	12
2	12	14	16	13	10
3	17	12	11	15	12
4	16	16	12	12	14
5	14	16	10	11	12
6	15	14	13	16	19

- A job shop has five machines for processing. There are six jobs to be processed. However, current capacity constraints indicate that only five jobs can be processed out of the six. The planner needs to know which jobs to assign to the machines. Identify an optimal loading of the shop on the basis of Table 12.16, which contains data on the processing time of each job against each machine.
- A job shop has five jobs for processing. There are four machines in the shop. The jobs visit the machines in a particular order (indicated by each job's routing file). Table 12.17 has details of the machines visited by each job and the processing time of each machine. The table also has information on the due dates.
 - Develop a schedule for the jobs using (i) SPT, (ii) EDD, and (iii) FCFS rules.
 - Use three appropriate performance criteria and evaluate these rules.
 - Compute the machine waiting time and job waiting time for all the schedules obtained using the SPT, EDD and FCFS rules. If you were to select a scheduling rule on the basis of the minimum job waiting time, what will your recommendation be?
 - If there is a penalty for delay at the rate of ₹250 per minute and a reward of ₹100 per minute for completing early, compute the cost of the schedules attributable to these.

TABLE 12.17 PROCESSING TIME OF JOBS AND DUE DATES

Jobs	Machines Visited (Processing Time In Minutes)					Due by (minute)
	1	2	3	4		
1	1 (12)	3 (7)	4 (8)	5 (4)		45
2	2 (7)	4 (6)	3 (5)	5 (6)		35
3	1 (8)	2 (5)	3 (9)	4 (5)		37
4	3 (6)	1 (6)	2 (6)	5 (7)		29
5	2 (9)	4 (8)	5 (6)	3 (7)		40

6. A bulb manufacturer works for 25 days a month on a two-shift basis. Each shift is of 8 hours duration. The marketing demand estimates the monthly demand for the bulbs to be 48,000. If the manufacturing system is a continuous-flow mass-manufacturing type with feeder shops linked to the assembly shops through a material flow control system, then compute the following for the manufacturer:

- (a) The TAKT time for all the shops in the manufacturing system.
- (b) If the successive workstations in the manufacturing plant are uniformly placed at a distance of three metres, at what speed should the material handling system move between the work-stations?

12.11 MINI PROJECTS

1. A vast number of scheduling packages are available in the market today. Scan through the following sources:

- Journals: *OR/MS Today* (a bimonthly magazine from the Operations Research Society) and *IIE Solutions* (a journal of the Institute of Industrial Engineering). These journals periodically bring out a survey of available software for specific areas of production management. Look for surveys pertaining to scheduling software.
- American Production & Inventory Control Society (APICS).
- Search engines: Search with keywords such as “APS”, “advanced production scheduling”, and “scheduling software”.

(a) Based on this exercise, prepare a technical note on a survey of the available software options for scheduling.

(b) Develop a classification scheme to list all the software that you have come across.

2. Visit a health management system (such as a multi-specialty hospital with facilities for in-patients, out-patients, laboratories for diagnostics, operation theatres, and special care systems) and study the scheduling practices with respect to the following:

- The doctors and surgeons visiting the hospital
- The support staff such as nurses, shift doctors, co-workers and laboratory staff
- Major facilities such as operation theatres and laboratories
- The outpatient department

(a) Prepare a note outlining the major findings of the study with respect to the aspects mentioned here. The report should highlight the relevance of scheduling techniques discussed in the chapter, the extent of usage, reasons for non-usage, and recommendations for removing some of the barriers to adopting the known techniques.

(b) Did you notice any significant patterns in the scheduling of personnel in the system that differs from a typical manufacturing system?

(c) If so, identify them and state the reasons for the differences.

3. Read the book *The Goal* by Goldratt and Cox. After reading the book, write a book review. While writing the review, answer the following questions amongst other things that you may like to cover in the book review:

(a) What are the key elements of the alternative method by which Alex Rego tries to solve the problems at his manufacturing set-up? Clearly outline the key conceptual details of his alternative strategy that helped him find a solution.

(b) How do you link his strategies for controlling a manufacturing system to that of the notion of scheduling that we have discussed in the chapter?

(c) What are the key performance measures that Alex Rego uses for controlling his manufacturing system?

(d) Do you think that his concept has a wide applicability? If so, give some examples of situations where this concept can be applied.

12.12 MULTIPLE-CHOICE QUESTIONS

- 12-1.** What system uses a defined framework to address issues associated with the use of available resources and the delivery of products and services as promised to the customers?
A. MRP system
B. Procurement Scheduling
C. Production Scheduling
D. Operations Scheduling
- 12-2.** Which is the process of rank-ordering the jobs in front of each resource to maximize some chosen performance measure?
A. Loading
B. Scheduling
C. Routing
D. Dispatching
- 12-3.** What is the useful method when one job is assigned to one machine and there are as many machines available for assignment as there are jobs?
A. Process Planning
B. Flexible Manufacturing Systems
C. The Assignment Method
D. Production Planning
- 12-4.** What are the parameters adequately describe the scheduling problems that one tries to solve?
A. Number of machines, Number of jobs, Shop configuration
B. Processing time, Due date, Random order
C. Completion measure, Inventory, Cost
D. Due date measure, Lateness, Work-in-Progress
- 12-5.** With how many sets of jobs possible, the floor shop becomes a shop?
A. One
B. Two
C. Three
D. Four
- 12-6.** For Job-shop scheduling which is the very useful graphical method?
A. Gantt charts
B. SPT rules
C. EED rules
D. Operating system
- 12-7.** What is an important element of operational control in manufacturing systems, especially in batch manufacturing and job-shop systems?
A. Production control
B. Input-Output control
C. Time control
D. Operation control
- 12-8.** Operational control issues in mass production systems begin with the computation of:
A. Initial Environment Review (IER)
B. Environment Management System (EMS)
C. Health, Safety and Environment (HSE)
D. TAKT time
- 12-9.** What was developed by Eliyahu Goldratt?
A. The Analogy
B. Synchronous
C. Bottleneck resource
D. The theory of constraints
- 12-10.** Which is the administrative process of authorizing processing of jobs?
A. Dispatching
B. Receiving
C. Allocating
D. Distributing

Answer Keys: 12-1. D 12-2. D 12-3. C 12-4. A 12-5. D 12-6. A 12-7. B 12-8. D 12-9. D 12-10. A

